
MARROW ED8

Pharmacology

Comprehensive Question Bank

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Pharmacokinetics

Question 1:

Fundoscopy evaluation in an HIV patient showed CMV retinitis. He was started on intravenous foscarnet, a highly ionized drug. Which of the following statements is true about the drug?

- a) Excreted mainly by the kidneys
- b) Can cross the blood brain barrier easily
- c) Well absorbed from the intestine
- d) Accumulates in the cellular lipids

Question 2:

Alkalinization of the urine is done in case of toxicity with which of the following substances?

- a) Barbiturate
- b) Amphetamine
- c) Heparin
- d) Morphine

Question 3:

When the $\text{pH} = \text{pK}_a$ of a drug, which of the following statements is true?

- a) Concentration of the drug is 50% ionic and 50 % non-ionic
- b) Concentration of the drug is 90% ionic and 10% non-ionic
- c) Concentration of the drug is 75% ionic and 25 % non-ionic
- d) Concentration of the drug is 25% ionic and 75 % non-ionic

Question 4:

In a patient with nephrotic syndrome, drug toxicity is least likely with which of the following drugs?

- a) Efavirenz

- b) Morphine
- c) Diazepam
- d) Valproate

Question 5:

What does time for peak plasma concentration (T_{max}) indicate?

- a) Rate of elimination
- b) Rate of absorption
- c) Duration of effect
- d) Intensity of effect

Question 6:

The pharmacokinetics of an experimental drug is being studied. If its apparent volume of distribution is 6L/kg, which of the following is true regarding its distribution?

- a) Highly bound to plasma proteins
- b) Confined to vascular compartment
- c) Sequestered in body tissues
- d) Distributed in extracellular fluid volume

Question 7:

Which of the following parameters is used to assess the extent of drug absorption in a plasma concentration vs. time graph?

- a) T_{max}
- b) C_{max}
- c) $T_{1/2}$
- d) Area under the curve

Question 8:

Which of the following statements is true about first-order kinetics?

- a) A constant amount of drug is eliminated in unit time
- b) The half life increases with an increase in dose
- c) The rate of elimination is constant
- d) The rate of elimination is proportional to the plasma concentration

Question 9:

You would consider administering a loading dose for a patient if the drug has

- a) High clearance
- b) High volume of distribution
- c) Long plasma half life
- d) Short plasma half life

Question 10:

Which of the following drugs approach zero-order kinetics at higher therapeutic levels?

- a) Penicillin
- b) Phenytoin
- c) Valproate
- d) Carbamazepine

Question 11:

Which of the following routes of administration will not bypass hepatic first-pass metabolism?

- a) Sublingual
- b) Oral
- c) Intravenous
- d) Transdermal

Question 12:

In the metabolism of xenobiotics, all of the following are phase 1 reactions except?

- a) Oxidation
- b) Reduction
- c) Acetylation
- d) Hydrolysis

Question 13:

Sulphonamides undergo which of the following metabolic reactions?

- a) Acetylation
- b) Methylation
- c) Hydroxylation
- d) Oxidation

Question 14:

All of the following can cause lupus-like syndromes except:

- a) Isoniazid
- b) Penicillin
- c) Hydralazine
- d) Sulphonamide

Question 15:

A patient on long-term anticoagulation therapy with warfarin would need an increased dose when coadministered with which of the following drugs?

- a) Captopril
- b) Erythromycin
- c) Rifampicin
- d) Cimetidine

Question 16:

Contraceptive failure while on oral contraceptive pills is unlikely to occur with which of the following drugs?

- a) Valproate
- b) Rifampicin
- c) Phenytoin
- d) Phenobarbitone

Question 17:

Which of the following microsomal enzymes is important for the activation of procarcinogens?

- a) CYP3A4
- b) CYP2C8/9
- c) CYP2C19
- d) CYP1A1/2

Question 18:

Cytochrome enzymes are not involved in which of the following processes?

- a) Oxidation
- b) Production of bile acids
- c) Synthesis of steroids
- d) Acetylation

Question 19:

A patient is brought to the emergency in an unresponsive state. There is an alleged history of consumption of multiple prescription tablets in an attempt to commit suicide. After the drug was conclusively identified, the doctor recommends dialysis for definitive management. The drug most likely is:

- a) Digoxin
- b) Verapamil
- c) Diazepam
- d) Lithium

Question 20:

Which of the following is not a prodrug?

- a) Enalapril
- b) Imipramine
- c) Sulfasalazine
- d) Cyclophosphamide

Question 21:

Which of the following is not true about P-glycoprotein?

- a) Pgp stands for permeability glycoprotein
- b) Also known as CD243
- c) Also known as ABCB1
- d) Also known as MDR2

Question 22:

For which of the following drugs, is absorption increased after a fatty meal?

- a) Griseofulvin
- b) Amphotericin B
- c) Ampicillin
- d) Aspirin

Question 23:

You prescribe an antibiotic course for a patient following cesarean section. What advice would you give regarding breastfeeding?

- a) No advice as most of drug are not present in breast milk
- b) Avoid breastfeeding during the antibiotic course
- c) Medications to be taken 3-4 hours before next feeding
- d) She can take medications anytime

Question 24:

A 55-year-old patient is on omeprazole for gastroesophageal reflux disease. She is now to be started on dual antiplatelet therapy with aspirin and clopidogrel. You decide to switch to pantoprazole because of the action of omeprazole on which enzyme?

- a) CYP3A4
- b) CYP2D6
- c) CYP2C19
- d) CYP2C20

Question 25:

Subfamily of CYP3A4 is _____

- a) CYP
- b) A
- c) 3
- d) 4

Question 26:

How much drug will remain at the end of 6 hours when a patient is given 200 mg drug, 75 mg excreted in 90 min and if the drug follows first-order kinetics?

- a) 50 mg
- b) 30.5 mg
- c) 12.5 mg
- d) 6.25 mg

Answer Key

Question No.	Correct Option
1	a
2	a
3	a
4	b

5	b
6	c
7	d
8	d
9	b
10	b
11	b
12	c
13	a
14	b
15	c
16	a
17	d
18	d
19	d
20	b
21	d
22	a
23	c
24	c
25	b
26	b

Detailed Explanations

Solution to Question 1:

Highly ionized drugs like foscarnet are mainly excreted by the kidneys.

They are not reabsorbed from the renal tubules and are hence excreted unchanged by glomerular filtration. Dose reduction is necessary in renal impairment to avoid the accumulation of the drug and associated toxicity.

The non-ionized forms of weak acids and bases undergo tubular reabsorption.

Option B: The blood-brain barrier limits the entry of ionized, non-lipid soluble drugs.

Option C: A drug is more readily absorbed from the gastrointestinal tract when it is in its unionized, more lipophilic form.

Option D: As the ionized form has low lipid solubility, it cannot penetrate cell membranes easily.

Solution to Question 2:

Alkalinization of the urine is done in barbiturate poisoning. When the urine is alkaline, weakly acidic drugs like barbiturates are present in highly ionized form and excreted more rapidly.

When the tubular urine is made more alkaline, weak acids like barbiturates become highly ionized and are excreted to a greater extent. Similarly, acidification of urine increases the excretion of weak bases. This property is used in the management of drug overdose.

Drug	Treatment of overdose
Weak acids(e.g., phenobarbital, aspirin)	Alkalinization of urine with sodium bicarbonate
Weak bases(e.g., amphetamine)	Acidification of urine with ammonium chloride

Solution to Question 3:

When the $\text{pH} = \text{pKa}$ of a drug, the concentration of the drug is 50% ionic and 50% non-ionic.

pKa is defined as the pH at which a drug exists as 50% ionized (hydrophilic) and 50% unionized (lipophilic).

For e.g., phenobarbitone is an acidic drug with a pKa of 7.2. At a pH of 7.2, 50% will be unionized. In a more basic plasma ($\text{pH}=7.4$), a lesser fraction (around 40%) will be unionized.

Small changes in pH near pKa of a drug may cause large variations in the degree of ionization.

Solution to Question 4:

In a patient with nephrotic syndrome, drug toxicity is least likely with morphine.

Acidic drugs generally bind to plasma albumin while basic drugs bind to α_1 acid glycoprotein. Nephrotic syndrome reduces plasma albumin levels, therefore there is a decrease in the protein-bound drug fraction and an increase in the free form, resulting in drug toxicity.

This is less likely with drugs having low plasma protein binding such as morphine (only around 35%).

Efavirenz, diazepam, and valproate have a very high plasma protein binding of $>90\%$ and predominantly bind to albumin and are associated with toxicity in hypoalbuminemia.

Solution to Question 5:

The time for peak plasma concentration (T_{max}) indicates the rate at which the drug is absorbed from the administered site. It depends on both the administered site and the drug formulation.

This is used to determine the timing of the administration of hypnotics with respect to sleep timings. A hypnotic with a higher rate of absorption might be preferred to induce sleep at a shorter latency as the target concentration can be achieved earlier.

Solution to Question 6:

In the present case, the volume of distribution of the drug is 6L/kg (420 L in a normal adult with reference weight 70kg) which is very high when compared to the total body water of 42L. Therefore, it can be implied that the drug is having high sequestration in tissues.

The apparent volume of distribution (Vd) is an imaginary volume throughout which the drug is distributed if the plasma concentration remains constant throughout.

$V_d = \text{Dose administered IV} / \text{plasma concentration}$

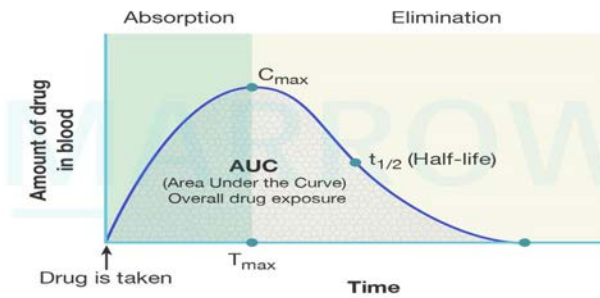
The difference in the volume of distribution based on drug type:

- Low Vd
- Restricted to the intravascular compartment
- Seen in drugs with high molecular weight (e.g., heparin) or high protein binding (e.g., warfarin - 9.8L)
- Medium Vd
- Lipid insoluble drugs (e.g., gentamicin - 20L) do not enter the cells and are distributed in the extracellular fluid volume
- Small water-soluble molecules (e.g., ethanol - 42L) are distributed in the total body water
- High Vd
- Sequestered and accumulated in the tissues
- Small lipophilic molecules (e.g., chloroquine - 13000L)

Solution to Question 7:

The area under the curve of plasma concentration vs. time curve gives the extent of absorption which is calculated by the trapezoidal rule.

The following image shows a plasma concentration vs. time graph:



C_{max}	- Highest concentration of drug in blood
$t_{1/2}$ (Half-life)	- Time at which drug has lost half its maximum concentration

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Two drugs may have the same rate of absorption but a different extent of absorption or bioavailability.

The factors upon which the extent of drug absorption depends are:

- Lipophilicity - If too lipophilic (e.g., acyclovir), the drug is not soluble enough to cross the water layer adjacent to the cell.
- Hydrophilicity - If too hydrophilic (e.g., atenolol), the drug cannot cross the lipid cell membrane.

Solution to Question 8:

The rate of elimination is proportional to plasma concentration in first-order kinetics.

Solution to Question 9:

The volume of distribution of the drug determines the loading dose of the drug.

Loading dose is a relatively large dose of the drug given initially to rapidly achieve the target plasma concentration. It is given by the formula

Loading Dose = Vol. of Distribution $\times$ Desired Steady-state plasma Concentration.

The bioavailability determines the route of administration. Though bioavailability may affect the dose necessary, it will not be the primary factor considered during the calculation of the loading dose.

Option A: Clearance is used to determine the maintenance dose.

Maintenance dose = Clearance $\times$ Desired Steady-state plasma concentration.

Options C and D: Plasma half-life primarily determines the frequency of dosing.

Solution to Question 10:

Phenytoin kinetics changes from first-order to zero-order kinetics at higher therapeutic levels.

True zero-order kinetics is shown only by ethanol.

Some drugs have first-order kinetics at low plasma concentrations while at higher plasma concentration (within therapeutic range), the mechanism for elimination gets saturated and they change over to zero-order kinetics.

Drugs that show early saturation kinetics include:

- Warfarin
- Aspirin
- Theophylline
- Verapamil
- Fluoxetine
- Phenytoin

Some drugs show zero-order kinetics at much higher (non-therapeutic / toxic) levels.

Solution to Question 11:

Hepatic first-pass metabolism occurs in drugs administered via the oral route.

When drugs are given orally, they are absorbed in the intestine and reach the liver through the portal vein. During this process, the drug is acted on by enzymes of the liver and intestine. Additionally, the liver may excrete part of the drug in the bile, further reducing the bioavailability. This is called the first-pass metabolism.

In all the other routes, the drug directly enters systemic circulation without reaching the liver.

Note: Bioavailability is the fraction of an administered dose of unchanged drug that reaches the systemic circulation. Hence, the bioavailability of a drug administered intravenously (I.V) is 100%. It is neither subjected to the first-pass metabolism as in the oral route nor does it depend on local factors as in subcutaneous, intramuscular, or transdermal.

Solution to Question 12:

Acetylation is a phase 2 reaction.

Metabolizing enzymes in hepatic cells:

- Microsomal enzymes - present in the smooth endoplasmic reticulum
- Non-microsomal enzymes - present in mitochondria and cytoplasm

Solution to Question 13:

Sulphonamides undergo acetylation. It is a phase II metabolic reaction where compounds having amino or hydrazine residues are conjugated with the help of acetyl coenzyme-A.

Drugs that undergo acetylation are:

- Sulphonamides
- Hydralazine
- Isoniazid
- PAS
- Dapsone
- Clonazepam
- Procainamide

Solution to Question 14:

Penicillin is not associated with a lupus-like syndrome.

Drug-induced lupus is a consequence of genetic polymorphism of N-acetyl transferase 2 (NAT2) enzyme which results in slow acetylators and fast acetylators:

- Slow acetylators of isoniazid, hydralazine, and sulphonamides can have increased toxicity that includes lupus-like syndrome
- Fast acetylators can have decreased efficacy as a result of faster metabolism of the active drug

Drugs that may result in lupus-like syndrome (The SHIP MeChaniQ has lupus-like syndrome) are:

- Sulphonamides/ Sulfasalazine
- Hydralazine
- Isoniazid
- Procainamide, Para-aminosalicylic acid
- Methyldopa
- Chlorpromazine
- Quinidine

Solution to Question 15:

A patient on long-term anticoagulation therapy with warfarin would need an increased dose when coadministered with rifampicin. This is because rifampicin is a potent microsomal enzyme

inducer and can decrease the effect of warfarin.

Rifampicin induces CYP3A and CYP2D6 enzymes.

Enzyme inducers:

- CYP3A
- Carbamazepine
- Phenobarbitone (CYP2B1)
- Rifampicin (CYP2D6)
- Phenytoin
- Glucocorticoids
- CYP2E1
- Isoniazid
- Chronic alcoholism

Note: Isoniazid is an inducer for CYP2E1 but an inhibitor for CYP3A4

Solution to Question 16:

Contraceptive failure in patients on the oral contraceptive pill is unlikely to occur with valproate. It is an enzyme inhibitor that inhibits the degradation of OCPs and increases their duration of action.

Rifampicin, phenytoin, and phenobarbitone are enzyme inducers and can cause increased metabolism of OCPs leading to contraceptive failure.

Enzyme inducers	Enzyme inhibitors
Mnemonic: CRAP GPSCarba mezepineRifampicinAlcohol (chronic)PhenytoinGriseofulvi nPhenobarbitoneSulfonylure as	Mnemonic: RED VICKRitona virErythromycinDiltiazemVer apamil, ValproateItraconazol eClarithromycin, Cimetidine Ketoconazole

Solution to Question 17:

CYP1A1/2 enzymes are important in the activation of procarcinogens.

They are involved in the metabolism of few drugs like theophylline, caffeine, paracetamol, and carbamazepine.

Inducers of CYP1A1/2:

- Polycyclic hydrocarbons
- Cigarette smoking

- Charbroiled meat
- Rifampicin
- Carbamazepine

Solution to Question 18:

Cytochrome enzymes are not involved in acetylation. Cytochrome enzymes such as CYP450 are involved in phase 1 reactions such as oxidation.

They are also involved in bile acid production from cholesterol; synthesis of steroids and fatty acid signaling molecules like epoxyeicosatrienoic acids.

Solution to Question 19:

The drug most likely is lithium as it is the only one amongst the given drugs that can be removed by dialysis. It is the treatment of choice in case of severe lithium toxicity ($>4\text{mEq/L}$).

The distribution of lithium is mainly restricted to the vascular compartment and it is not bound to any plasma proteins. Dialysis is effective in removing the drugs if they are sufficiently present in the free form in the plasma. Therefore, lithium can be removed by dialysis during an overdose.

Drugs not suitable for dialysis are those with:

- High volume of distribution - digoxin, verapamil, and organophosphates
- High level of plasma protein binding - diazepam

Solution to Question 20:

Imipramine is not a prodrug. It is a tricyclic antidepressant and has therapeutic activity. It also forms an active metabolite, desipramine.

Prodrugs are inactive precursors of active forms. They require metabolic alteration in the body to become active.

Examples of prodrugs:

- Prednisone
- Levodopa
- ACE inhibitors (except lisinopril and captopril)
- Sulfasalazine
- Methyldopa
- Cyclophosphamide
- Clopidogrel

Solution to Question 21:

P-glycoprotein is also known as multidrug resistance protein 1 or MDR1 (not MDR2). It is also denoted by cluster of differentiation 243 (CD243).

P-glycoprotein (P-gp) is one of the first members of the ATP-binding cassette (ABC) transporter, which acts as an efflux membrane transporter. It is responsible for limiting cellular uptake and the distribution of xenobiotics and toxic substances. It is associated with multidrug resistance (MDR).

- Increased P-gp expression: Reduces the absorption of drugs that are substrates for P-gp
- Decreased P-gp expression: Leads to supratherapeutic plasma concentrations and drug toxicity

Solution to Question 22:

The absorption of griseofulvin is increased after a fatty meal.

Note: Fatty meals increase the rate of absorption of efavirenz to such an extent as to cause toxicity. Hence, efavirenz is recommended to be taken on an empty stomach.

Drug absorption reduced /delayed by food	Drug absorption increased by food
Ampicillin Aspirin Atenolol Azithromycin Captopril Indinavir Isoniazid Rifampin Sucralfate Tetracycline Doxycycline	Atovaquone Carbamazepine Chlorthalidone Cefuroxime Diazepam Ganciclovir Lithium Metoprolol Ritonavir Hydralazine Lumefantrine

Solution to Question 23:

If the lactating mother needs to take medications that are deemed safe, the dosing optimally should be 30-60 minutes after nursing and at least 3-4 hours before the next feeding.

A minimum interval of 3-4 hours allows time for partial clearance of the drug. This ensures the infants are not exposed to the peak maternal plasma concentration that may translate to higher levels secreted into milk.

Option A: Most drugs administered are detected in breast milk but most of them are excreted in very small quantities that do not cause any effect in the infants.

Option B: Breastfeeding should not be avoided altogether for all medications. Intake of cytotoxic agents and immunomodulatory agents are contraindications for breastfeeding.

Option D: The timing of medication should preferably be at an interval where the infants are not exposed to higher maternal plasma concentrations.

Solution to Question 24:

Omeprazole is replaced with another proton pump inhibitor (PPI) during coadministration with clopidogrel because of the action on CYP2C19.

Clopidogrel is a prodrug that is activated to its active metabolite by the liver enzyme CYP2C19. Apart from CYP2C19, CYP3A4, and CYP2C9 are also involved to a certain extent.

Omeprazole serves as both substrate and inhibitor for CYP2C19. It may inhibit the activation of clopidogrel and decrease its antiplatelet action. Hence, they are not prescribed together.

Solution to Question 25:

The sub-family of CYP3A4 is A.

In CYP enzymes nomenclature, the root CYP is followed by a number designating the family, a letter denoting the subfamily and another number designating the gene. Therefore, in CYP3A4:

- 3 - family
- A - subfamily A
- 4 - individual gene coding for one specific isoenzyme

There are numerous CYP isoenzymes in humans. Among them, CYP3A4 is the most abundant and carries out the metabolism of nearly 50% of clinically used drugs.

Solution to Question 26:

In first order kinetics, a constant fraction of the drug is excreted in a given time.

75 mg of 200 mg is excreted in 90 minutes

$75/200 = 37.5/100 = 0.375$ is eliminated

After 90 minutes; $200 \text{ mg} - 75 \text{ mg} = 125 \text{ mg}$ remains

The next 90 minutes (after 180 minutes) $= 125 \times 0.375 = 46.875 \text{ mg}$ is eliminated, and $125 - 46.875 = 78.125 \text{ mg}$ remains

The next 90 minutes (after 270 minutes) $= 78.125 \times 0.375 = 29.297 \text{ mg}$ is eliminated, and $78.125 - 29.297 = 48.828 \text{ mg}$ remains

The next 90 minutes (after 360 minutes) $= 48.828 \times 0.375 = 18.310 \text{ mg}$ is eliminated, $48.828 - 18.310 = 30.518 \text{ mg}$ remains after 6 hours.

Simpler way to calculate

$3/8$ fraction is excreted in 90 minutes $\rightarrow 5/8$ remains after 90 minutes.

6 hours (360 min) is 4 times of 90 minutes.

So $200 \times (5/8)(5/8)(5/8)(5/8) = 30.518$ mg remains after 360 minutes

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Pharmacodynamics

Question 1:

Neurotransmitters like noradrenaline, adrenaline, and dopamine act on:

- a) Single-pass transmembrane receptors
- b) Nuclear receptors
- c) Seven-pass transmembrane receptors
- d) Ligand-gated receptors

Question 2:

Which of the following statements is true about G-protein-coupled receptors?

- a) G proteins bind to hormones on the cell surface.
- b) All the three subunits alpha, beta and gamma should bind to each other for G proteins to act.
- c) G proteins act as inhibitory and excitatory because of differences in the alpha subunit.
- d) G protein is bound to GTP in resting state.

Question 3:

Which of the following GPCRs does not act via the K⁺ channel?

- a) Angiotensin 1 receptor
- b) Muscarinic M2 receptor
- c) Dopamine D2 receptor
- d) Serotonin 5 HT1 receptor

Question 4:

Which of the following is an example of a GPCR?

- a) Muscarinic
- b) Nicotinic
- c) GABA-A

d) Insulin

Question 5:

Which of the following is an ion channel receptor?

- a) Nicotinic cholinergic receptor
- b) Muscarinic cholinergic receptor
- c) Beta adrenergic receptors
- d) Opioid receptors

Question 6:

All of the following agents act by intracellular receptors except _____

- a) Thyroid hormones
- b) Vitamin D
- c) Insulin
- d) Steroid

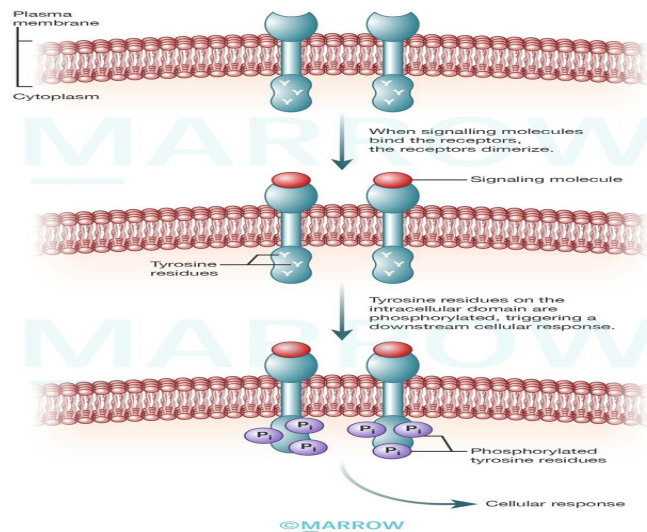
Question 7:

Which among the following is not a second messenger?

- a) Cyclic AMP
- b) Guanylyl cyclase
- c) Diacylglycerol
- d) Inositol triphosphate

Question 8:

Which of the following hormones act through the receptor shown in the image below?



- a) Insulin
- b) Somatostatin
- c) Cholecystokinin
- d) Vasopressin (V1 receptor)

Question 9:

A patient with congestive heart failure is being treated with an intravenous infusion of nesiritide. Which of the following receptors does this drug act through?

- a) G-protein coupled receptors
- b) Tyrosine kinase receptors
- c) JAK-STAT receptors
- d) Ligand-gated guanylyl cyclase receptor

Question 10:

JAK/STAT is a signal transducer for which of the following receptors?

- a) G protein coupled receptors
- b) Ion channels
- c) Transmembrane receptors
- d) Nuclear receptors

Question 11:

Adrenaline is used to treat histamine-induced bronchoconstriction in a patient with brittle asthma. What is this type of antagonism called?

- a) Physiological antagonism
- b) Chemical antagonism
- c) Competitive antagonism
- d) Uncompetitive antagonism

Question 12:

Diamox is given to a patient with glaucoma. It is a non-competitive inhibitor of the carbonic anhydrase enzyme. Which of the following would be the effect of this drug?

- a) Decrease in V_{max}
- b) Decrease in K_m
- c) Decrease in both K_m and V_{max}
- d) No change in V_{max}

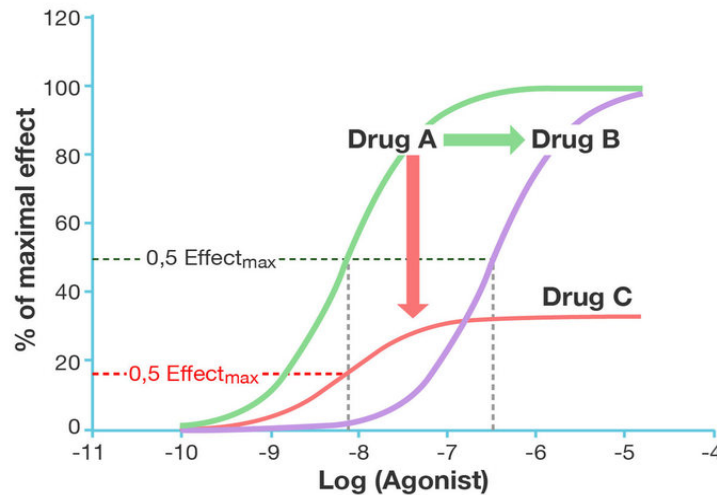
Question 13:

Letrozole, a competitive inhibitor of the enzyme aromatase is prescribed to a patient with breast cancer. Which of the following is true about this type of antagonism?

- a) Antagonism cannot be completely reversed by increased dose of the agonist
- b) An agonist cannot displace an antagonist from the receptor
- c) Agonists and antagonists bind to the same receptor
- d) Dose response curve shifts to the left in the presence of an antagonist

Question 14:

Which of the following statements is true regarding the dose-response curve shown in the image below?



- a) C is competitive antagonist
- b) C is more potent than A
- c) B is more potent than A
- d) B is competitive antagonist

Question 15:

Which of the following is a true statement regarding inverse agonists?

- a) Binds to the receptor and causes maximum action
- b) Binds to the receptor and causes opposite action to that of agonist
- c) Binds to the receptor and causes no action
- d) Binds to the receptor and causes submaximal action

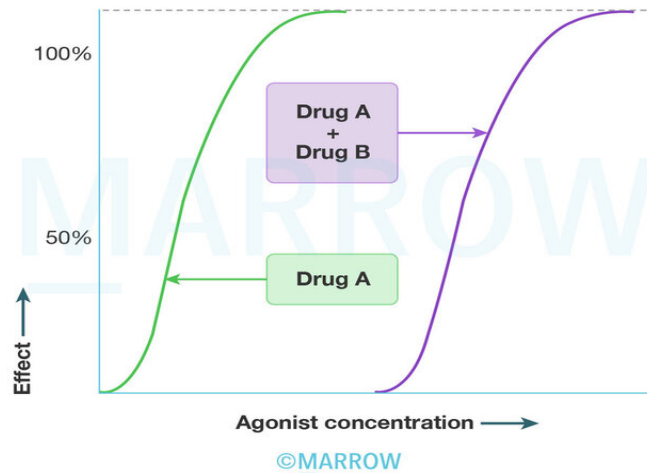
Question 16:

A patient with symptoms of psychosis is prescribed aripiprazole. Which of the following best describes this drug?

- a) No intrinsic activity and high affinity for D2 receptor
- b) High intrinsic activity and no affinity for D2 receptor
- c) Low intrinsic activity and high affinity for D2 receptor
- d) High intrinsic activity and low affinity for D2 receptor

Question 17:

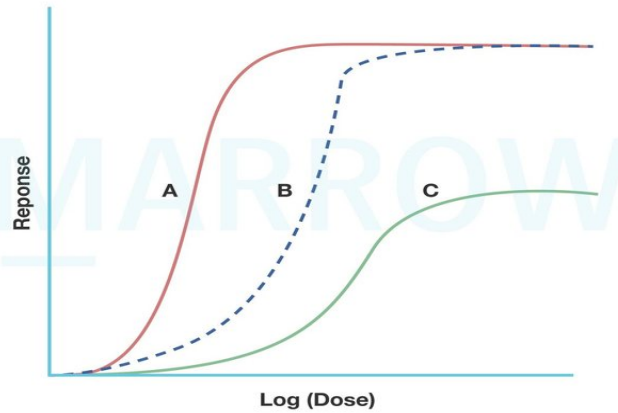
What type of antagonism is shown by drugs A and B in the following dose-response curve?



- a) Competitive antagonism
- b) Noncompetitive antagonism
- c) Uncompetitive antagonism
- d) Cannot be determined

Question 18:

The graph below shows the relationship between the log(dose) and the response of three drugs, A, B, and C. Which of these drugs has the highest potency?

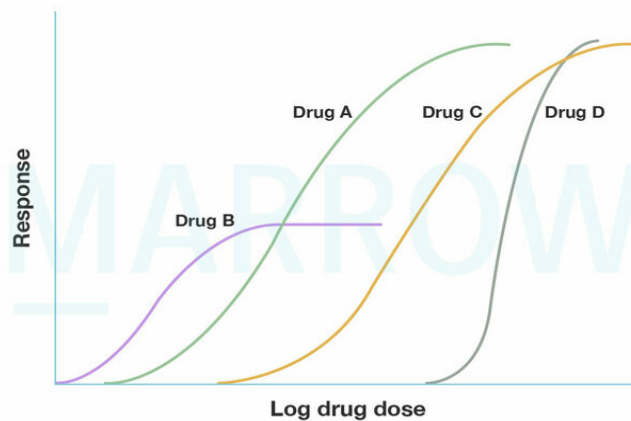


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- a) A
- b) B
- c) Both A and B
- d) C

Question 19:

Which of the following statements is true about the given dose-response curve?



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- a) Drug B is the most potent.
- b) Drug B is the most efficacious.

- c) Drug C is more efficacious than drug D.
- d) Drug A is more potent than drug B.

Question 20:

Which of the following statements regarding the potency and efficacy of a drug is not true?

- a) In a clinical setup, efficacy is more important than potency
- b) In the log dose response curve, the height of the curve corresponds with efficacy
- c) ED₅₀ of the drug corresponds to the efficacy
- d) Drugs that produce a similar pharmacological effect can have different levels of efficacy

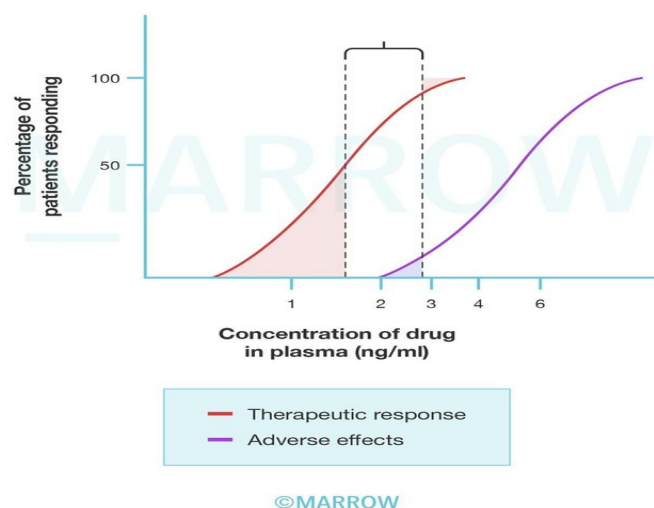
Question 21:

Which of the following shows a U-shaped dose-response curve?

- a) Vitamins
- b) Anti-cancer drugs
- c) Steroids
- d) Chelators

Question 22:

What does the area between the dotted lines in this graph indicate?



- a) Median effective dose
- b) Median lethal dose
- c) Therapeutic index
- d) Therapeutic window

Question 23:

Therapeutic index is used for assessing which of the following factors?

- a) Safety
- b) Efficacy
- c) Potency
- d) Toxicity

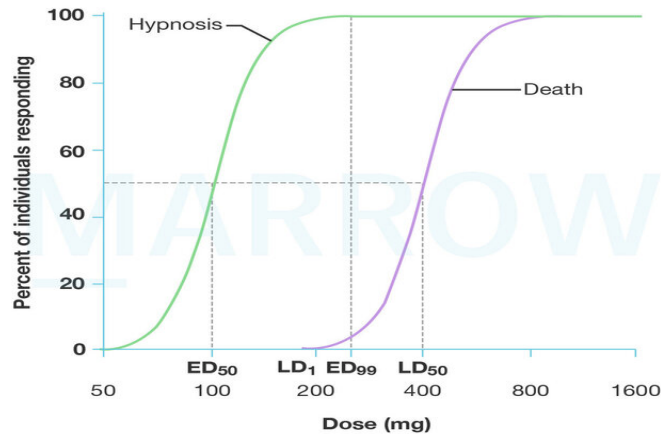
Question 24:

Therapeutic index is calculated with which of the following equations?

- a) ED_{50}/LD_{50}
- b) LD_{50}/ED_{50}
- c) $ED_{50}-LD_{50}$
- d) $ED_{50} \times LD_{50}$

Question 25:

Calculate the therapeutic index of a drug whose dose-response curves are plotted as shown below:



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- a) 1
- b) 2
- c) 4
- d) 8

Question 26:

Therapeutic drug monitoring is required in all except _____.

- a) Phenytoin
- b) Metformin
- c) Tacrolimus
- d) Cyclosporine

Question 27:

Which of the following statements about pharmacogenetics is correct?

- a) It refers to the effect of drugs on genetic expression
- b) It refers to the genetic basis of variability in drug response
- c) It is not synonymous with pharmacogenomics
- d) Drug metabolism is not a highly heritable trait

Question 28:

A 23-year-old male is diagnosed with G6PD deficiency. Which of the following drugs is not contraindicated in his condition?

- a) Cotrimoxazole
- b) Proguanil
- c) Nalidixic acid
- d) Ceftriaxone

Answer Key

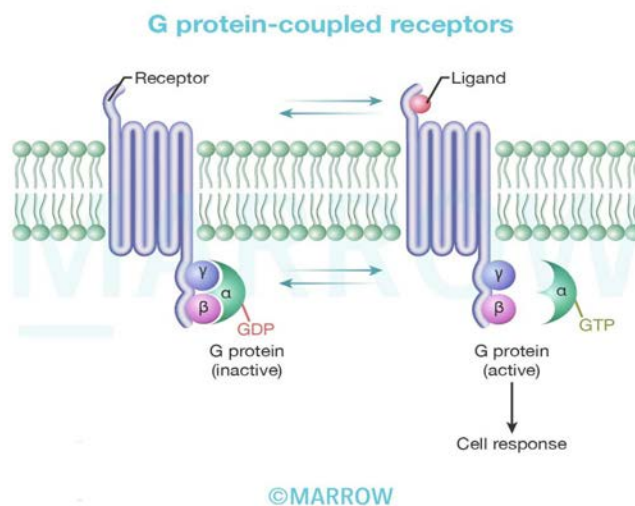
Question No.	Correct Option
1	c
2	c
3	a
4	a
5	a
6	c
7	b
8	a
9	d
10	c
11	a
12	a
13	c
14	d
15	b
16	c
17	a
18	a
19	a
20	c
21	a
22	d
23	a

24	b
25	c
26	b
27	b
28	d

Detailed Explanations

Solution to Question 1:

Noradrenaline, adrenaline, and dopamine act through a bundle of seven alpha-helical trans-membrane receptors which are the G-protein-coupled receptors (GPCRs).



Solution to Question 2:

G proteins act as inhibitory and excitatory because of differences in the alpha subunit.

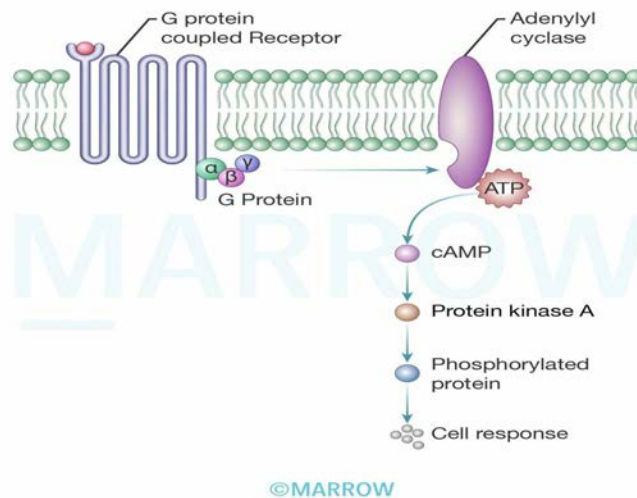
- Excitatory - Gs and Gq
- Inhibitory - Gi

In the resting-state:

- G protein is a heterotrimer consisting of alpha, beta, and gamma subunits. It forms a complex with G-protein-coupled receptors at the intracellular domain. The extracellular domain of GPCR has the binding domain for hormones. G protein subunits do not directly bind them.
- The alpha subunit is bound to GDP.

Binding of ligand results in activation of GPCR, following which:

- Alpha subunit undergoes a conformational change
- Release of GDP and binding of GTP occurs
- GTP bound alpha subunit dissociates from beta-gamma subunit
- Both alpha subunit and beta-gamma subunit bind to and regulate effectors only when they are separated.



Solution to Question 3:

Angiotensin 1 receptor acts via Ca^{2+} channels. It does not act through second messengers via the K^{+} channel.

G-protein coupled receptors activate inwardly rectifying potassium channels and increase the permeability of the potassium ions causing hyperpolarisation of the cell. They are called G protein-coupled inwardly-rectifying potassium channels (GIRKs). These receptors include:

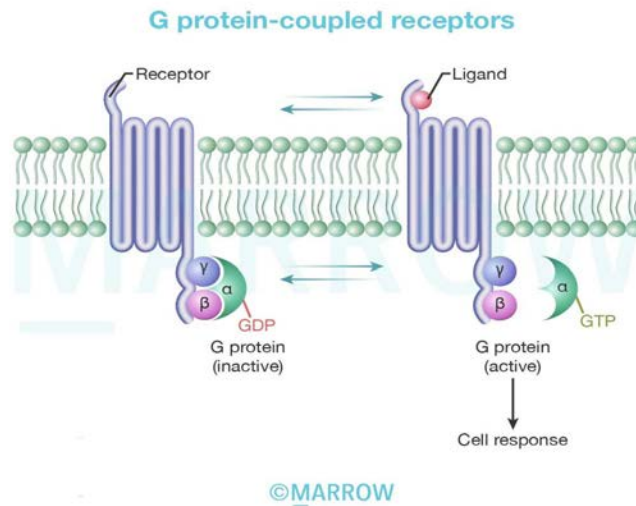
- M2 muscarinic
- A1 adenosine
- α_2 adrenergic
- D2 dopamine
- 5HT1 serotonin
- Somatostatin receptors.

Solution to Question 4:

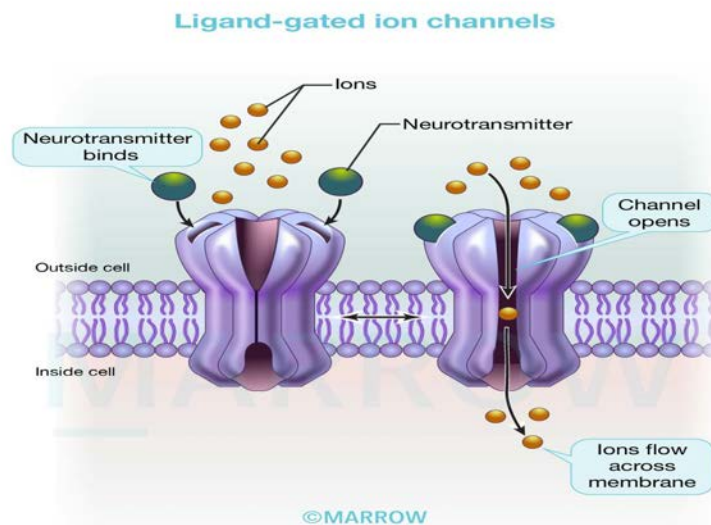
Muscarinic receptors are G-protein-coupled receptors among the given options.

M1, M2, M3, M4, and M5 are the muscarinic Ach receptors.

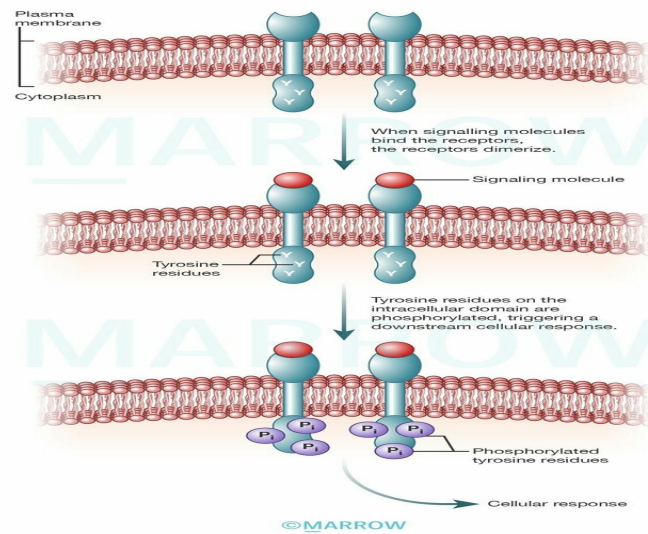
- M1 and M3 act through Gq protein and activate IP3/DAG causing increased cytosolic calcium.
- M2 and M4 act through Gi protein and inhibit adenylyl cyclase causing increased potassium conductance.



Nicotinic receptors and GABA-A are examples of ion channel coupled receptors.



Insulin receptor is a tyrosine kinase receptor.

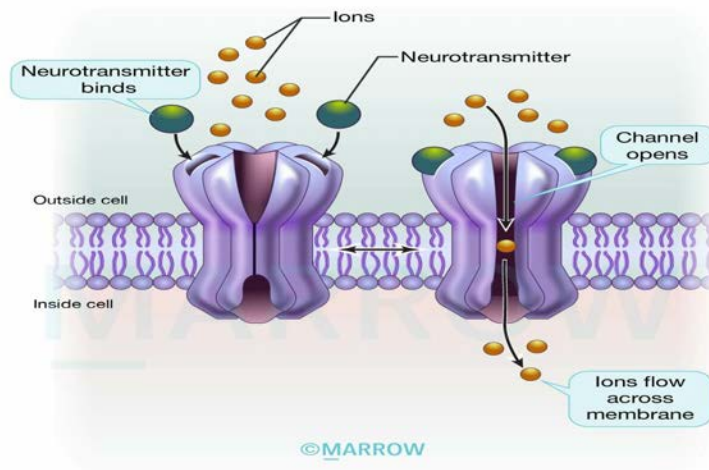


Solution to Question 5:

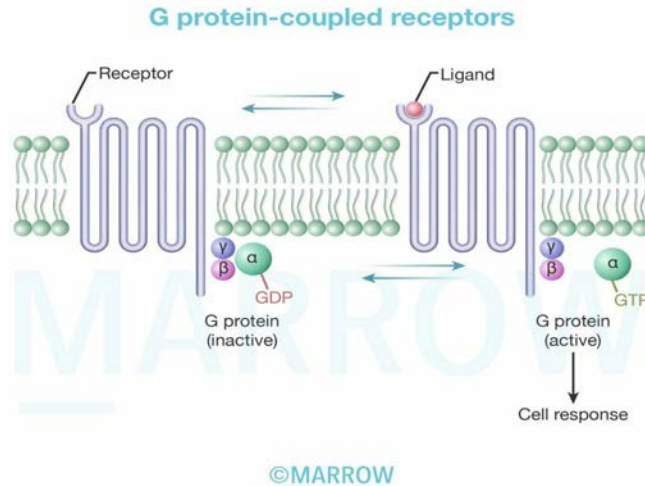
Nicotinic cholinergic receptors (both Nm and Nn) are of ligand-gated ion channel type.

The binding of ligand alters the permeability of ion channels directly and brings about biological action.

Ligand-gated ion channels



Muscarinic cholinergic, beta-adrenergic, and opioid receptors are G protein-coupled receptors.



Solution to Question 6:

Insulin receptors belong to the class of trans-membrane enzyme-linked receptors.

Transmembrane enzyme-linked receptors consist of a large extracellular ligand-binding domain. It is connected to an intracellular subunit having enzymatic property through a single transmembrane helical peptide chain. The enzyme on the cytosolic side is usually a protein kinase (tyrosine) or membrane-bound GC (guanylyl cyclase).

Physiological ligands for transmembrane enzyme-linked receptors are mostly growth factors and chemokines. Other receptors belonging to this class include:

- Receptor tyrosine kinases - Growth factors (PDGF, EGF, VEGF); insulin
- Membrane-bound GC - Natriuretic peptides
- Receptor serine-threonine kinases - TGF- β

Thyroid hormones, vitamin D, and steroids act via intracellular receptors.

Solution to Question 7:

Guanylyl cyclase is an enzyme that converts GTP to cGMP. It is not a second messenger.

Second messengers are:

- Diacylglycerol
- Inositol triphosphate
- Cyclic AMP
- Cyclic GMP
- Calcium

- Nitric oxide

Solution to Question 8:

The receptor in the given image uses tyrosine kinase signaling. Insulin acts through this receptor to exert its functions.

Other hormones acting via tyrosine kinase signaling:

- Growth hormone
- Growth factors like VEGF, PDGF, FGF, NGF
- Prolactin
- Leptin

Solution to Question 9:

Nesiritide is recombinant B-type natriuretic peptide. Natriuretic peptide receptors are transmembrane ligand-gated guanylyl cyclase receptors.

Natriuretic peptides:

- Atrial natriuretic peptide (ANP): released from atrial storage granules in response to an increase in intravascular volume. It decreases blood pressure.
- Brain natriuretic peptide (BNP): released from ventricular tissue in response to volume overload. Along with a decrease in blood pressure, it is known to reduce cardiac hypertrophy and fibrosis.
- C- type natriuretic peptide (CNP): synthesized in the brain and endothelial tissues and functions to stimulate long bone growth.

ANP and BNP act on the NPR-A receptor.

CNP acts on the NPR-B receptor.

Solution to Question 10:

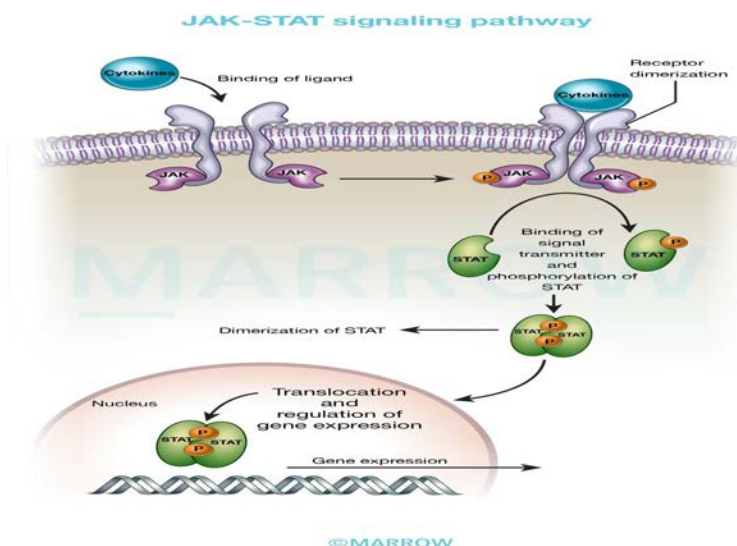
JAK/STAT is a signal transducer for transmembrane receptors lacking intrinsic enzymatic activity.

Interleukin and other cytokines act through such receptors.

On binding of the ligand to such receptors:

- Receptor dimerization occurs
- Increase in affinity for cytosolic JAK (Janus kinase)
- Activation of JAK results in phosphorylation of tyrosine residues of the receptor

- Binding of signal transducer and activator of transcription (STAT) and subsequent phosphorylation of STAT
- STAT dimerizes and translocates to the nucleus to regulate gene expression, resulting in a biological response



Solution to Question 11:

Adrenaline and histamine are physiological antagonists.

Adrenaline and histamine act on bronchial muscles. Histamine acts on H₁ receptors and produces bronchoconstriction. Adrenaline acts on β_2 receptors and produces bronchodilation. Two drugs acting on different receptors and resulting in opposite responses means having pharmacological effects in the opposite direction. This is called physiological antagonism or functional antagonism.

Option B: Chemical antagonism refers to a reaction between two chemicals to neutralize their effects, such as is seen with chelation therapy.

Option C: Competitive antagonism is a type of receptor antagonism that occurs when the agonist and antagonist compete for the same binding site on the receptor.

Option D: In uncompetitive inhibition, inhibitors form a complex with an intermediate complex of the substrate and transporter. This inhibits subsequent translocation.

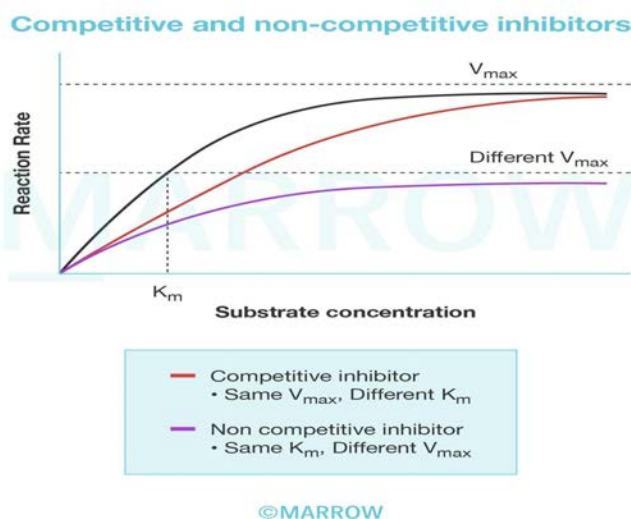
Solution to Question 12:

Non-competitive inhibitors cause a decrease in V_{max} .

In non-competitive inhibition:

- The inhibitor binds to the adjacent site and not the catalytic site of the enzyme
- The catalytic property of the enzyme is reduced and hence V_{max} is decreased

- The affinity of the enzyme is not altered and hence K_m is unaltered



Solution to Question 13:

In competitive antagonism, the drug and the antagonist compete to bind to the same site of the receptor.

Agonists if present in high concentration, can displace the antagonists from the active site and overcome the competitive antagonism completely.

The dose-response curve in the case of competitive antagonism shifts to the right.

Solution to Question 14:

In the above image, drug B is a competitive antagonist.

In competitive antagonism, the DRC is shifted to the right whereas in non-competitive antagonism the DRC is flattened.

Thus in the above image Drug B is the Competitive antagonist while drug C is the non-competitive antagonist.

Potency:

- It is defined as the amount of drug needed to produce a given response.
- A highly potent drug evokes a given response at low concentrations and vice-versa.
- The more the curve is on the left in a dose-response curve (DRC), the greater is its potency.
- Thus, in the above image, the potency is drug A > drug C > drug B.

Efficacy:

- It is defined as the maximum effect produced by the drug.

- It expresses the effect produced by the receptors when bound by a certain drug.
- In the log dose-response curve, the perpendicular height represents the drug efficacy.
- Thus, in the above image, the efficacy is drug A = drug B > drug C.

Solution to Question 15:

Inverse agonists bind to the receptor and cause an action opposite to that of an agonist.

The ability of the drug to bind with a receptor is called affinity.

The capacity of a drug to induce a functional change in the receptor is called intrinsic activity or efficacy.

Type of drugs	Intrinsic activity
Agonists	1
Partial agonists	0 to 1
Competitive antagonists	0
Inverse agonists	0 to -1

Solution to Question 16:

Aripiprazole is a D2 receptor partial agonist. Partial agonists have submaximal or low intrinsic activity and high affinity.

Partial agonists can behave as an antagonist in the presence of an agonist. Partial agonists block the binding of the agonist to the receptor thereby inhibiting a complete response.

Aripiprazole has lower activity than dopamine at D2 receptors. By occupying and blocking the action of dopamine on D2 receptors, it acts similar to an antagonist. It is used in conditions like schizophrenia where dopamine antagonist activity is required.

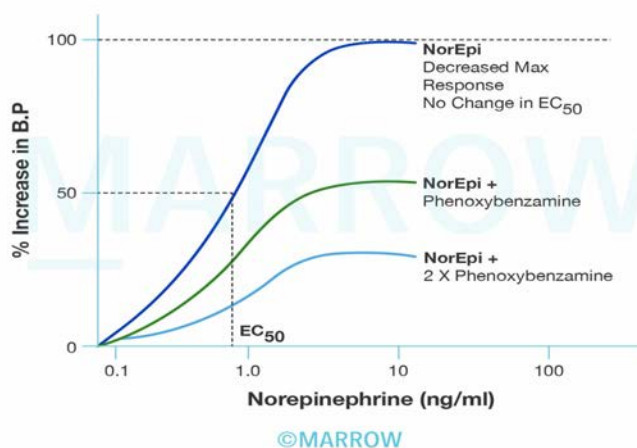
Solution to Question 17:

The parallel rightward shift of the agonist dose-response curve (DRC) is seen with competitive antagonism.

Increasing the dose of the agonist can result in the same maximal response.

In non-competitive antagonism, there will be flattening of the agonist DRC.

Noncompetitive inhibition



Solution to Question 18:

Drug A has the highest potency.

Potency:

- It is defined as the amount of drug needed to produce a given response. A highly potent drug evokes a given response at low concentrations and vice-versa.
- The more the curve is on the left in a dose-response curve (DRC), the greater is its potency.
- Thus, in the above image, the potency is Drug A > Drug B > Drug C.

Efficacy:

- It is defined as the maximum effect produced by the drug. It reflects the capacity of the drug to activate a receptor and produce the response.
- In the log dose-response curve, the perpendicular height represents the drug efficacy.
- Thus, in the above image, the efficacy is Drug A = Drug B > Drug C.
- Drugs having a similar pharmacological effect can have different efficacies e.g furosemide and metolazone are both diuretics but furosemide has greater efficacy than metolazone.

Solution to Question 19:

The more the curve is on the left in a dose-response curve (DRC), the greater is its potency.
Hence, Drug B is the most potent.

Potency:

- It is defined as the amount of drug needed to produce a given response.
- A highly potent drug evokes a given response at low concentrations and vice-versa.

- In the above image, the potency is Drug B > Drug A > Drug C > Drug D.

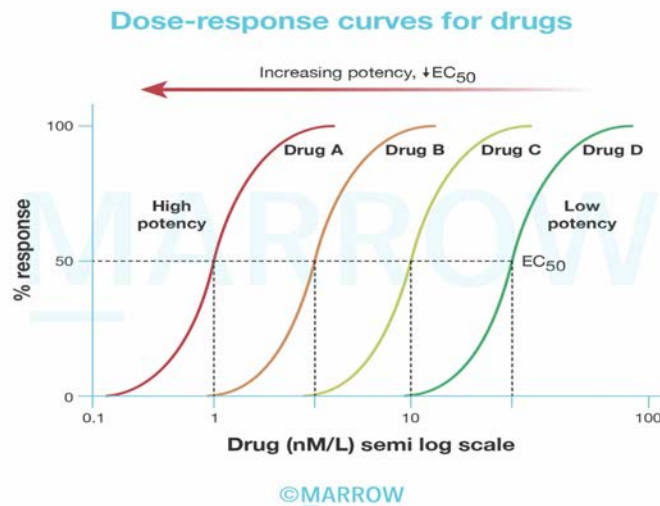
Efficacy:

- It is defined as the maximum effect produced by the drug. It reflects the capacity of the drug to activate a receptor and produce the response.
- In the log dose-response curve, the perpendicular height represents the drug efficacy.
- Thus, in the above image, the efficacy is Drug A = Drug C = Drug D > Drug B.
- Drugs having a similar pharmacological effect can have different efficacies e.g furosemide and metolazone are both diuretics but furosemide has greater efficacy than metolazone.

Solution to Question 20:

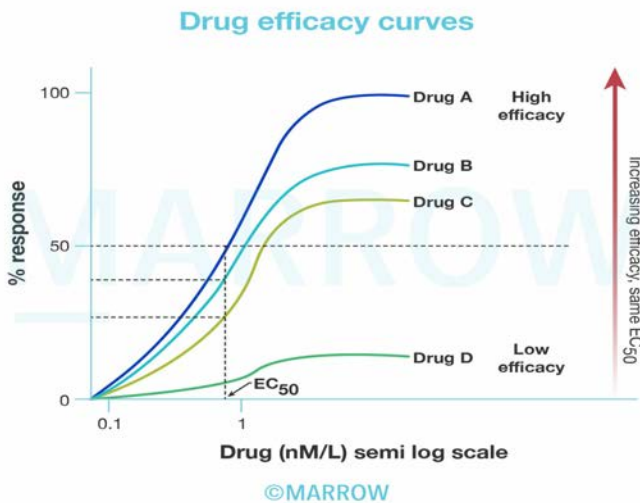
ED₅₀ is a measure of the drug potency, not efficacy.

Potency refers to the amount of the drug required to produce a certain response. The x coordinates (dose axis) on the log dose-response curve (DRC) of a drug represent the potency of the drug. A right shift in the DRC indicates a lower potency and vice versa.



Potency

- It is defined as the amount of drug needed to produce a given response.
- A highly potent drug evokes a given response at low concentrations and vice-versa.
- The efficacy of a drug is a more decisive factor in the choice of a drug in the clinical setting
- The more the curve shifts towards the left in a dose-response curve (DRC), the greater is its potency.
- Thus, in the above image, Drug A is more potent. The potency is Drug A > Drug B > Drug C > Drug D.



Efficacy

- It is defined as the maximum effect produced by the drug.
- It expresses the effect produced by the receptors when bound by a certain drug.
- It is mainly a factor in choosing the dose of a drug.
- In the log dose-response curve, the perpendicular height represents the drug efficacy.
- Thus, in the above image, efficacy is Drug A > B > C > D.

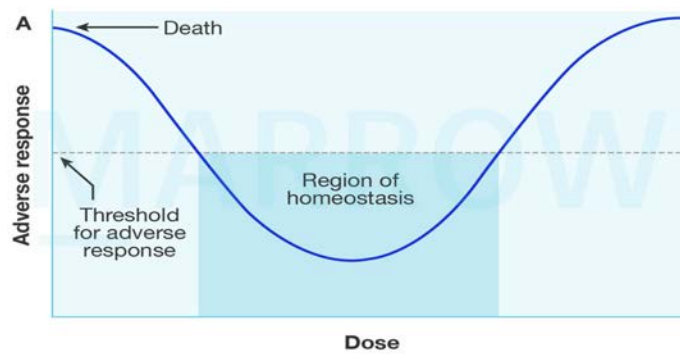
Solution to Question 21:

Vitamins and essential metals show a U-shaped dose-response curve.

At a low dose, adverse effects are observed because there is a deficiency of these nutrients to maintain homeostasis. As dose increases, homeostasis is achieved, and the bottom of the U-shaped dose-response curve is reached. As the dose increases to surpass the amount required to maintain homeostasis, toxicity can ensue.

Thus, adverse effects are seen at both low and high doses.

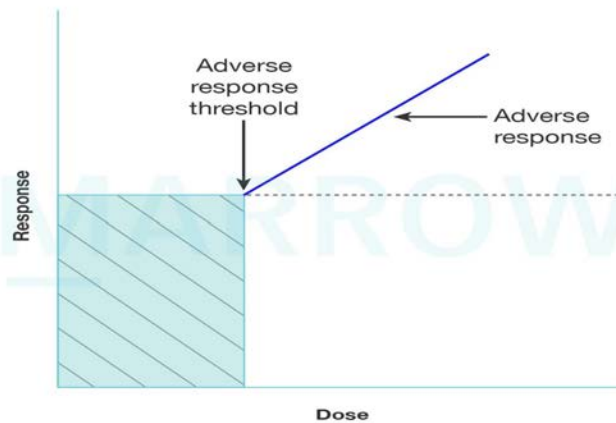
Graph of Adverse response vs Dose



©Marrow

Hockey stick-shaped dose-response curve for toxicants that are also metabolic by-products.

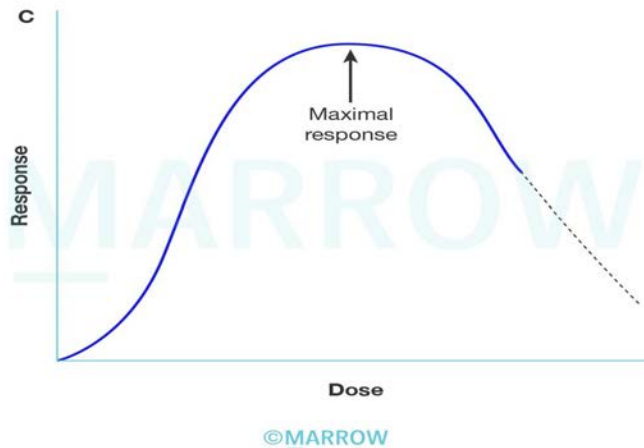
Dose-response curve - Hockey-stick shaped



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Inverted U-shaped dose-response curve for ligands that downregulate their receptors.

Dose-response curve - Inverted U-shaped



Solution to Question 22:

The area between the dotted lines in the graph shows the therapeutic window. It is the range of steady-state concentrations of a drug that provides therapeutic efficacy with minimal toxicity.

The therapeutic index (TI) describes a relationship between the doses of a drug that causes lethal or toxic effects with the dose that causes therapeutic effects. It is also called the therapeutic ratio.

Mathematically, you can calculate TI by following way:

$$\text{Therapeutic Index} = \text{LD}_{50}/\text{ED}_{50} \text{ (or) Therapeutic Index} = \text{TD}_{50}/\text{ED}_{50}$$

Where,

- LD_{50} (Median Lethal dose) is the minimum amount of drug that causes adverse effects in 50% of the population. LD_{50} could also be replaced with Toxic dose (TD_{50}).
- ED_{50} (Median Effective dose) is the quantity of a drug that can produce desired therapeutic effects in 50% of the population.

Solution to Question 23:

Therapeutic index is used to assess the safety of the drug administered.

$$\text{Therapeutic index} = \text{TD}_{50}/\text{ED}_{50}$$

- TD_{50} (Median toxic dose) = Dose required to produce a toxic effect in 50% of animals tested.
- ED_{50} (Median effect dose) = Dose required to produce a therapeutic effect in 50% of the individuals.

Option B: Efficacy is defined as the maximum effect produced by the drug. It expresses the effect produced by the receptors when bound by a certain drug.

Option C: Potency refers to the concentration (EC₅₀) or dose (ED₅₀) required to produce 50% of maximum effect where EC₅₀ is the concentration of agonist resulting in a half-maximal response.

Option D: Median toxic dose is the dose required to produce a toxic effect in 50% of animals tested. Lethal dose is the dose required to kill 50% of animals tested.

Solution to Question 24:

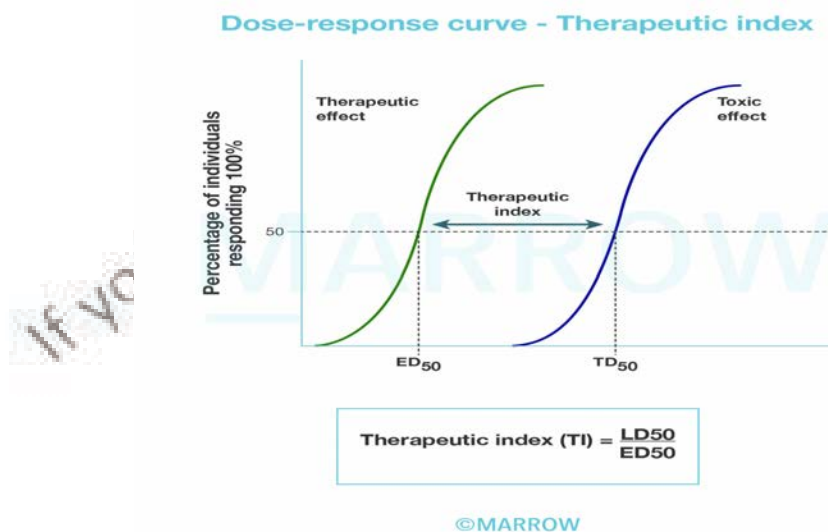
Therapeutic index is calculated with the following equation = LD₅₀/ED₅₀.

Therapeutic index is used to assess the safety of the drug administered.

LD₅₀ (median lethal dose) refers to the dose of the drug at which 50% of the experimental animals die. Larger the LD₅₀, the safer the drug. In humans, TD₅₀ is considered. TD₅₀ refers to the dose of the drug that causes a toxic response in 50% of the population.

ED₅₀ (median effective dose) refers to the dose of the drug at which 50% of the experimental animals show the specific or desired effect

Clinically, the therapeutic range bound by the dose that produces the minimal necessary therapeutic effect and the dose that produces the maximal acceptable adverse effect is more relevant.



Solution to Question 25:

The therapeutic index is 4.

In the above graph,

Median lethal dose (LD₅₀) = 400

Median effective dose (ED₅₀) = 100

Therapeutic index (TI) = LD_{50}/ED_{50} .

Hence,

Therapeutic index (TI) = $400/100 = 4$

Clinically, the therapeutic range bound by the dose that produces the minimal therapeutic effect and the dose that produces the maximal acceptable adverse effect is more relevant.

Solution to Question 26:

Metformin does not require therapeutic monitoring because it does not have a narrow therapeutic index and the effect on blood sugar levels can be easily measured.

Phenytoin, tacrolimus, and cyclosporine have the narrow therapeutic index and therapeutic monitoring is necessary to avoid adverse effects.

Solution to Question 27:

Pharmacogenetics is the study of the genetic basis for variability in drug action.

Genetic differences cause variability in the dose-response curve (DRC) due to alteration in drug action and disposition in the body.

The term is used interchangeably with pharmacogenomics.

Drug metabolism is highly heritable, as assessed using drug exposure studies in monozygotic versus fraternal twins and from other related subjects.

Solution to Question 28:

Ceftriaxone is not contraindicated in G6PD deficient individuals as it does not possess oxidizing property, hence there is no hemolysis.

Drugs having the potential to cause hemolysis in G6PD deficient individuals:

- Anti-malarials - Primaquine, chloroquine, quinine, pyrimethamine, proguanil
- Sulfonamides - Cotrimoxazole
- Fluoroquinolones - Nalidixic acid
- Nitro furan antibacterial - Furazolidone, nitrofurantoin
- Anti-hypertensives - Methyldopa, hydralazine
- Anti-convulsants - Phenytoin
- Anti-arrhythmics - Procainamide

Clinical Trials and Miscellaneous

Question 1:

Pharmacovigilance refers to:

- a) Monitoring of drug safety
- b) Monitoring of unethical trade of drugs
- c) Monitoring of drug potency
- d) Monitoring of drug efficacy

Question 2:

Which of the following organizations regulates the clinical trials and approval of investigational new drugs in India?

- a) CPCSEA
- b) CDSCO
- c) FDA
- d) DCGI

Question 3:

You are working as a research associate on clinical trials of a new drug for migraine. What is the dose of the test drug that is usually not exceeded in phase 0 studies?

- a) 1000 mcg
- b) 1000 mg
- c) 100 mcg
- d) 100 mg

Question 4:

Good clinical practice is not required in which phase of clinical trials?

- a) Preclinical phase
- b) Phase 1 trial

- c) Phase 2 trial
- d) Phase 3 trial

Question 5:

Which of the following is not a primary endpoint in phase 1 studies?

- a) Safety
- b) Tolerability
- c) Pharmacokinetic parameters
- d) Efficacy

Question 6:

Which of the following phases of a clinical trial is also referred to as efficacy trial?

- a) Phase 1
- b) Phase 2
- c) Phase 3
- d) Phase 4

Question 7:

You read a news alert that mentions a novel drug has received marketing approval for diabetes mellitus and that it has shown positive results in a pivotal trial. This corresponds to which phase of clinical trials?

- a) Phase I
- b) Phase II
- c) Phase III
- d) Phase IV

Question 8:

A pharmaceutical company has set up a team of physicians to oversee the Phase IV clinical trial of a new drug for SARS-CoV2 infection, to identify possible drug interactions. Which is the most appropriate statement about this phase of the clinical trial?

- a) Done in a group of 50-500 patients
- b) Done after a drug is approved and marketed
- c) Done for drugs used in rare diseases
- d) Done for formulating guidelines for therapeutic use

Question 9:

Which of the following statements are true?

- a) 1, 3 and 4
- b) 2 and 4
- c) 2, 4 and 5
- d) 1, 2 and 3

Question 10:

A pharmacologically inert substance used along with an active ingredient in a formulation is known as the:

- a) Placebo
- b) Nocebo
- c) Excipient
- d) Inducer

Question 11:

Which one of the following is a placebo?

- a) Herbal medication with no effect known
- b) Physiotherapy
- c) Sham surgery
- d) Cognitive behavioural therapy

Question 12:

Which of the following are not reasons for the alteration of drug dosage in the elderly?

- a) They have decreasing renal function with age
- b) They have loss of lean body mass
- c) They have increased baroreceptor sensitivity
- d) Their total body water is decreased

Question 13:

A patient started on an anti-epileptic for epilepsy developed toxic epidermal necrolysis. It was documented as a type A adverse drug reaction. Which of the following is not true regarding this adverse effect?

- a) These reactions occur due to genetically determined abnormal reactivity to a drug.
- b) These reactions can be predicted from the drug's known pharmacological or toxicological effects
- c) These reactions are generally dose-dependent
- d) These reactions are usually common

Question 14:

Which of the following drugs if taken during pregnancy can lead to this adverse effect?



- a) Tetracycline
- b) Thalidomide
- c) Warfarin

- d) Alcohol

Question 15:

Which of the following drugs can be used in the first trimester of pregnancy?

- a) ACE inhibitors
- b) Angiotensin receptor blockers
- c) Propylthiouracil
- d) Statins

Question 16:

Which of the following drugs cannot cross the placenta and can safely be used in a pregnant woman?

- a) Phenytoin
- b) Diazepam
- c) Morphine
- d) Heparin

Question 17:

'Me-too drugs' do not differ significantly in which of the following pharmacological features as compared to the prototypical drug?

- a) Absorption
- b) Metabolism
- c) Mechanism of action
- d) Elimination

Question 18:

As a medical officer in a PHC, you anticipate a rise in infections following the worsening flood in your area. You write a letter to the district collector requesting to provide an additional supply of essential medicines. Which of the following best describes these medicines?

- a) These are drugs needed to treat emergency conditions

- b) These are drugs needed to treat serious diseases
- c) These are drugs that satisfy the priority health needs of the population
- d) These are drugs that have been in use for a long time

Question 19:

The schedule that deals with prescription drugs under the Drugs and Cosmetics Act is _____.

- a) Schedule P
- b) Schedule H
- c) Schedule Y
- d) Schedule D

Question 20:

Which of the following is a schedule X drug?

- a) Halothane
- b) Thalidomide
- c) Ketamine
- d) Colistin

Question 21:

A first-year resident administers an intramuscular injection to a patient by the Z track technique. What is the advantage of this technique?

- a) Increase absorption rate
- b) Prevent staining of skin
- c) Reduce allergic manifestations
- d) Longer duration of action

Question 22:

Z track technique is used for?

- a) Administration of nicotine patches
- b) Monitoring of carbamazepine therapy
- c) Administration of long acting depot antipsychotics
- d) Monitoring of lithium therapy

Question 23:

A drug label suggests storage in a 'cool' place. What is the temperature at which this drug is to be stored?

- a) - 2 degree Celsius
- b) 0 degree Celsius
- c) 2-6 degree Celsius
- d) 8-15 degree Celsius

Answer Key

Question No.	Correct Option
1	a
2	b
3	c
4	a
5	d
6	b
7	c
8	b
9	b
10	c
11	c
12	c
13	a
14	b
15	c
16	d
17	c

18	c
19	b
20	c
21	b
22	c
23	d

Detailed Explanations

Solution to Question 1:

Pharmacovigilance deals with the monitoring of drug safety.

WHO definition of pharmacovigilance:

"Science and activities related to the detection, assessment, understanding, and prevention of adverse effects or any other drug-related problems".

Solution to Question 2:

Central Drug Standard Control Organisation (CDSCO) regulates the clinical trials and approval of investigational new drugs in India.

The proposal for initiating trials, review of clinical trial results, and recommendations for marketing authorization is done through a committee called the Investigational New Drug (IND) Committee formed by CDSCO.

Preclinical trials are performed on animals, and they are regulated by guidelines prescribed by the Committee for the Purpose of the Control and Supervision of Experiments on Animals (CPCSEA) in India.

The Drugs Controller General of India (DCGI) is the head of the CDSCO, under the Ministry of Health and Family Welfare in India.

Solution to Question 3:

The dose of the test drug in phase 0 studies usually does not exceed 100mcg. Phase 0 studies are also called micro-dosing studies.

Very low doses of drugs, usually 1/100th of the estimated human dose or a maximum of 100 mcg is used.

They are done to obtain pharmacokinetic data in humans at a very early stage of drug development. The micro-dosing stage is not mandatory and not clearly categorized by regulatory authorities yet.

Solution to Question 4:

Good clinical practice is not required in the preclinical phase.

Good clinical practice (GCP) is an international ethical and scientific quality standard for the design, conduct, performance, monitoring, auditing, recording, analyses, and reporting of clinical trials. It is primarily for the conduct of clinical trials in humans.

Preclinical trials are performed on animals and they are regulated by guidelines prescribed by the CPCSEA (Committee for the Purpose of the Control and Supervision of Experiments on Animals) in India. The guidelines are laid down in good laboratory practice (GLP).

Solution to Question 5:

Efficacy is not a primary endpoint in phase 1 studies. Phase 1 studies are generally done in healthy volunteers. Efficacy can be a secondary endpoint in some studies where patients are recruited.

The primary endpoints are:

- Safety
- Tolerability of single/multiple doses
- Pharmacokinetic parameters

The primary endpoint denotes the purpose of the study for which it is primarily designed. The secondary endpoint denotes the measure of the valuable outcome of the study design.

Solution to Question 6:

Efficacy and appropriate dose range is the main aim of phase 2 of clinical trials. It is evaluated on a small group of patients with the target disease.

Phase II of clinical trials involve a small group of patients (50-500). Most of the new drugs will be administered for the first time in patients with the target disease. The study is designed to obtain proof of concept (potential of the drug to treat the disease) and dose-ranging study (provides input for choosing the most appropriate dose for phase 3 studies).

Phase 3 studies are also known as pivotal studies as they establish treatment benefits.

Solution to Question 7:

Phase III trials are usually considered pivotal trials and are necessary for the approval of the majority of drugs. They are also called multi-site trials and are done for 3–5 years.

When the initial Phase III trials are complete, the sponsor (usually a pharmaceutical company) applies to the FDA for approval to market the drug. This application is called either a New Drug Application (NDA) or a Biologics License Application (BLA).

In some drugs related to rare diseases or malignancies, phase II studies are considered as pivotal, and the drug may be approved on the basis of phase II studies.

Solution to Question 8:

Phase IV study is conducted after the drug is approved and marketed by regulatory authorities. It includes post-marketing surveillance and other studies that complement the earlier findings of the drug.

After the drug is marketed for general use, practicing physicians are identified through whom data about the efficacy, acceptability, and adverse effects of the drug are collected on a structured proforma. Patients treated in the normal course form the study population. Therefore the study population is much larger. Uncommon adverse effects or those that occur only after long-term use and unsuspected drug interactions are detected at this stage.

Solution to Question 9:

Orphan drugs are for the treatment or prevention of rare diseases with little commercial value for pharmaceutical companies. Orphan receptors are receptors that are discovered yet no known ligands are identified.

There are two types of drug development. They are:

- Traditional drug development: Initial drug discovery is followed by the characterization of the receptors the drugs are used to act on.
- Contemporary drug development: Due to advances in molecular biology and genome sequencing, receptors are identified initially. The drugs are identified later using chemical screening later resulting in the discovery of orphan receptors.

Special incentives are given by the government to pharmaceutical companies for the development and marketing of orphan drugs.

Few examples of orphan drugs are:

- Liposomal amphotericin B
- Digoxin antibody (Digibind)
- Fomepizole

Solution to Question 10:

An excipient is a pharmacologically inert substance used along with an active ingredient in a formulation. e.g., starch, lactose, FDKP

Role of excipients:

- Add bulk to the active drug if used in small quantities
- Increase the stability of the formulation
- Mask unpleasant taste

Inhaled insulin (Afrezza) consists of 2- to 2.5- μm crystals of the excipient, fumaryl diketopiperazine (FDKP), that provides a large surface area for adsorption of proteins like insulin.

Option A: Placebo is an inert substance given in the form of medication. It is generally not combined with the active ingredient.

- Endorphins may play a role in the placebo effect especially for analgesic action.
- It is used in the controls of clinical trials to elucidate the actual drug effect.
- Not all but only a few patients respond to the placebo.

Option B: Nocebo is the opposite of the placebo effect where the patient experiences a negative psychodynamic effect due to their pessimistic attitude towards the medications or healthcare professionals.

Option C: Inducers refer to agents that increase the expression of cytochrome enzymes involved in the metabolism of drugs.

Solution to Question 11:

Sham surgery (placebo surgery) is a faked surgical intervention that omits the step thought to be therapeutically necessary.

Note: While option A may be considered a placebo due to there being no known effects, it is not certain as to whether all possible effects (known and unknown) have been considered. Hence, weighing options A and C, the better option among the two would be C as sham surgery is a widely used placebo.

Solution to Question 12:

The reason for the alteration of drug dosage in the elderly is decreased baroreceptor sensitivity due to age-related vascular stiffening.

Reasons for the alteration of drug dosage in the elderly are:

- Progressive decline of renal function
- Loss of lean body mass and an increased ratio of fat to muscle, leading to an increased volume of distribution of lipophilic drugs
- Decreased body water - causing volume of distribution (V_d) of hydrophilic drugs decreases.

- Slower absorption due to reduced gut motility.
- Plasma albumin decreases with aging which leads to decreased plasma protein binding.

Solution to Question 13:

Type A (augmented) adverse drug reactions (ADR) are not based upon the qualitatively abnormal responses to the drug. It is based on the pharmacological properties of the drug.

Examples of type A ADR:

- Side effects
- Toxic effects
- Consequences of drug withdrawal.

They are common, reversible, and preventable.

Idiosyncrasy refers to the genetically determined abnormal reactivity to a drug.

Adverse drug reactions can be classified into two types:

•

Type A reactions – sometimes referred to as augmented reactions – are dose-dependent and predictable based on the pharmacology of the drug

•

Type B reactions – bizarre reactions – which are idiosyncratic and not predictable on the basis of drug pharmacology. They can be based on the characteristics of patients.

Solution to Question 14:

The baby has features suggestive of phocomelia and it results due to the consumption of thalidomide during pregnancy.

Phocomelia is associated with underdeveloped limbs and absent pelvic bones. Both upper and lower limbs are affected and a part of the limbs may be missing. Fused fingers and missing thumbs are also seen.

Thalidomide was originally used for the treatment of pregnancy-associated morning sickness but was withdrawn due to this adverse effect.

It re-entered the market for the treatment of erythema nodosum leprosum. It also has antiangiogenic and immunomodulatory effects.

Solution to Question 15:

Propylthiouracil can be used in the first trimester of pregnancy. It is indicated in the management of hyperthyroidism.

Solution to Question 16:

Heparin is a large molecule and cannot cross the placenta. Hence, heparin can be used safely in a pregnant woman.

Unfractionated or low molecular weight heparin is the preferred anticoagulant in pregnancy.

Option A: Phenytoin crosses the placenta and can cause fetal hydantoin syndrome.

Options B and C: Diazepam and morphine are lipophilic molecules and can cross the placenta.

Solution to Question 17:

Me-too drugs are chemically related to the prototype or other chemical compounds that have an identical mechanism of action.

Various statins present in the market can be considered as "Me-too drugs". The pharmacokinetic parameters (absorption, distribution, metabolism, and elimination) vary significantly between them (atorvastatin and rosuvastatin have a very long duration of action), but the core mechanism of action remains the same: HMG CoA reductase inhibition and decrease in cholesterol synthesis.

Solution to Question 18:

Essential medicines are those medicines that satisfy the priority health care needs of the population.

They are selected with due regard to:

- Public health relevance
- Evidence on efficacy and safety
- Comparative cost-effectiveness

WHO publishes a model list of essential medicines that provides a framework for individual nations to draw their own national list. The 21st Essential Medicines List, published in April 2019 brings the total to 460 drugs deemed essential for addressing the most important public health needs globally.

Solution to Question 19:

Schedule H deals with prescription drugs. All the drugs listed under schedule H should be provided to the patient only with the prescription from a registered medical practitioner.

All the drugs under the list must have the warning mentioned below, on their packaging listed prominently.

"Schedule H drug. Warning: To be sold by retail on the prescription of a registered medical practitioner only"

Schedule H1: To be sold on the production of a valid prescription, but the chemist will maintain a separate register where the identity of the patient, contact details of the prescribing doctor, and the name and dispensed quantity of the drug will be recorded. The dispensing quantity should be strictly according to the valid prescription. It is a recently introduced list with the objective of curtailing indiscriminate use of antimicrobials.

Schedule X: To be sold with a valid prescription. Prescription in duplicate, separate license requirements, meticulous storage, and dispensing records must be maintained by the chemist. The drugs that have high abuse liability are included in schedule X.

Solution to Question 20:

Ketamine is a schedule X drug.

According to The Drugs and Cosmetics Act-1945,

Schedule X drugs are those that have high abuse liability and are to be sold with a valid prescription only.

A prescription in duplicate, separate license requirement, meticulous storage, and dispensing records must be maintained by the chemist.

Option B: Thalidomide is a schedule H drug.

Option D: Colistin is a schedule P drug.

Solution to Question 21:

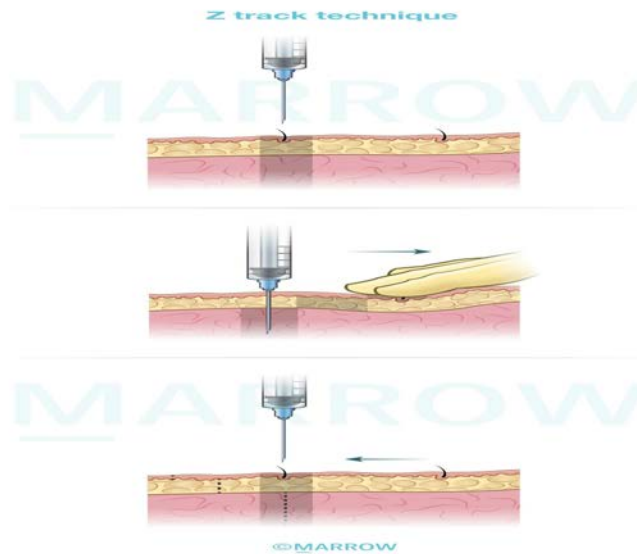
The advantage of the 'Z-track technique' for intramuscular administration is that it prevents skin staining.

Z-track technique procedure:

- Stretch the skin by pulling it laterally away from the intended point of injection.
- Insert the needle into the site at a 90° angle, aspirate by pulling back to see if the needle has a penetrated vessel.
- If blood is aspirated withdraw the needle, and repeat the procedure at another site with new equipment and medication.
- If no blood is aspirated administer the injection.
- Wait for 10 seconds and then withdraw the needle, and release the skin allowing the displaced tissue to seal the needle track.

Advantages:

- The 'Z-track' prevents leakage of the medication into the subcutaneous tissue and skin staining through the tract to the surface of the skin
- Pain intensity is also reduced

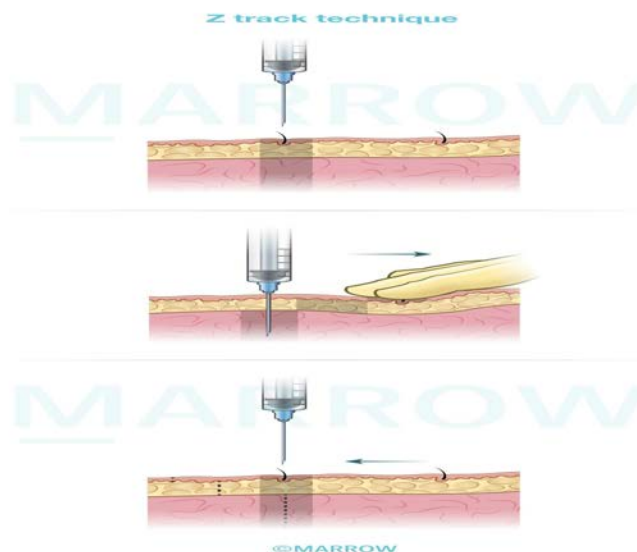


Solution to Question 22:

The Z-track technique is used for the intramuscular administration of long-acting depot antipsychotics.

Advantages:

- The 'Z-track' prevents leakage of the medication into the subcutaneous tissue and skin staining through the tract to the surface of the skin
- Pain intensity is also reduced



Solution to Question 23:

Storing at a cool place means to store at 8-15 degrees C.

Examples for drugs to be stored under cool temperature include insulin, nicorandil.

Label instructions	Temperature
Store frozen	-20oC
Keep cool	8-15oC
Store at room temperature	15-25oC

Pharmacokinetics - Calculations

Question 1:

The elimination after 3 half-lives in first-order kinetics is around:

- a) 12.5%
- b) 75%
- c) 87.5%
- d) 94%

Question 2:

You have to give 180 mg of ceftriaxone to a patient in a 2 ml syringe that has 10 divisions per ml. The concentration of the drug in the vial is 500 mg/5 ml. How many divisions should be filled in 2 ml of the syringe to give 180 mg?

- a) 18
- b) 1.8
- c) 2
- d) 20

Question 3:

500mg of a drug was administered to a patient. 100mg was cleared in 1 hour. How much of the drug would be left in his body after 3 hours if the excretion of the drug follows zero-order kinetics?

- a) 300 mg
- b) 200 mg
- c) 400 mg
- d) 100 mg

Question 4:

200 mg of a drug is administered to a patient, of which 20 mg is excreted in 90 minutes. How much of the drug will remain after 3 hours if it follows first-order kinetics?

- a) 162 mg
- b) 145.8 mg
- c) 50 mg
- d) 40 mg

Question 5:

Calculate the half-life of a drug X administered to a patient if it has a volume of distribution of 50 liters and clearance of 0.1 L/min.

- a) 500 minutes
- b) 440 minutes
- c) 420 minutes
- d) 350 minutes

Question 6:

Calculate hepatic clearance of a drug with hepatic blood flow of 2000 mL/min and hepatic extraction ratio of 0.5.

- a) 4000 mL/min
- b) 2000 mL/min
- c) 1500 mL/min
- d) 1000 mL/min

Question 7:

A patient is administered drug X that follows first-order kinetics. 40 mg intravenous dose results in a plasma concentration of 2mg/L initially and 1mg/L, 4 hours later. What is the clearance of drug X?

- a) 5 L/hr
- b) 10 L/hr
- c) 3.5 L/hr
- d) 7 L/hr

Question 8:

A male patient weighing 50 kg has acute severe asthma, for which theophylline infusion is planned. The target plasma concentration is 10 mg/L. The volume of distribution of theophylline is 0.6 L/kg. What will be the loading dose in this patient for intravenous administration?

- a) 200 mg
- b) 250 mg
- c) 300 mg
- d) 350 mg

Question 9:

Calculate maintenance dose for a drug M with 100% oral bioavailability, which needs a target concentration of 2 mg/L and clearance of 20L/hr and dosed QID.

- a) 300 mg
- b) 160 mg
- c) 250 mg
- d) 240 mg

Question 10:

You have started an infusion of drug Y in a patient at a rate of 200 mg/hr. Drug Y has V_d of 100 liters and clearance of 100 ml/min. What will be the drug concentration at a steady state?

- a) 50 mg/h
- b) 25 mg/h
- c) 36.66 mg/L
- d) 33.33 mg/L

Question 11:

The clearance of a drug which follows first-order kinetics (in one compartment model) is 0.2 L/min and volume of distribution is 40 L. What is the $t_{1/2}$ of that drug?

- a) 140 min
- b) 220 min
- c) 240 min
- d) 180 min

Question 12:

A drug was given IV at a rate of 1.6mg/min. The clearance of the drug is 640ml/min. The half life of the drug is 1.8 hour. Calculate the plasma concentration of the drug at steady state.

- a) 2.88mg/ml
- b) 0.004mg/ml
- c) 3.55mg/ml
- d) 0.002mg/ml

Answer Key

Question No.	Correct Option
1	c
2	a
3	b
4	a
5	d
6	d
7	c
8	c
9	d
10	d
11	a
12	d

Detailed Explanations

Solution to Question 1:

After each half-life, 50% of the drug is eliminated or the plasma concentration becomes half the original value.

Therefore, after,

- 1 half-life = 50% drug is eliminated
- 2 half-lives = 75% drug is eliminated
- 3 half-lives = 87.5% is eliminated
- 4 half-lives = 93% is eliminated

Solution to Question 2:

The syringe should be filled up to 18 divisions to deliver 180 mg ceftriaxone to the patient.

Concentration of drug: 500 mg/5 mL $\rightarrow$ 100 mg/1 mL

The syringe has 10 divisions per mL.

Hence, each division corresponds to $1/10 = 0.1$ mL

The required dosage for the patient is 180 mg that is present in 1.8 mL.

0.1mL = 1 division, 1.8mL = 18 divisions.

Solution to Question 3:

In zero-order kinetics, a constant amount of drug is eliminated in unit time.

In this question, 100 mg of the drug is eliminated in 1 hour.

After 3 hours,

$3 \times 100 = 300$ mg would be eliminated.

$500 - 300 = 200$ mg will be the remaining drug in the body.

Solution to Question 4:

After 3 hours, 162mg of the drug will remain.

In first-order kinetics, a constant fraction of the drug is excreted in a given time.

In the given question,

20mg out of 200mg is excreted in 90 minutes,

$20/200 = 0.1$ fraction is eliminated every 90 minutes.

After 90 minutes, $200\text{mg} - 20\text{ mg} = 180\text{ mg}$ remains

The next 90 minutes (i.e. after 180 minutes) $= 180 \times 0.1 = 18\text{ mg}$ is eliminated.

Therefore, $180 - 18 = 162$ mg remains.

Simpler way:

1/10 fraction of the drug is excreted in 90 minutes $\rightarrow$ 9/10 remains after 90 minutes.

3 hours (180 min) is 2 times 90 minutes.

So $200 \times (9/10)(9/10) = 162$ mg remains after 180 minutes

Solution to Question 5:

The half-life of this drug is 350 minutes.

In first-order kinetics:

$$\text{Plasma half-life (t}_{1/2}\text{)} = 0.693 \times \frac{\text{Volume of distribution (V}_d\text{)}}{\text{Clearance (CL)}}$$

Therefore,

$$t_{1/2} \sim 0.7 \times 50 / 0.1 = 0.7 \times 500$$

$$t_{1/2} \sim 350 \text{ minutes}$$

Where,

- $t_{1/2}$ - Plasma half-life
- V_d - Volume of distribution (50)
- CL - Clearance (0.1)

Solution to Question 6:

The hepatic clearance is 1000 mL/min.

Clearance of a drug by an organ is the difference between the rate of presentation of the drug to the organ and the rate of exit of the drug from the organ. It is calculated by the equation:

$$CL_{\text{organ}} = Q \times E$$

Where,

- Q - Organ blood flow
- E - Extraction ratio (organ-specific)

In the question, it is given that the hepatic blood flow (Q) is 2000 mL/min and the hepatic extraction ratio (E) is 0.5.

Substituting the values in the above equation,

$$CL_{\text{organ}} = 2000 \times 0.5 = 1000 \text{ mL/min}$$

Note: The extraction ratio (E) is the fraction of the drug that is removed from the blood by the eliminating organ.

$$\text{Extraction ratio (E)} = \frac{CA - CV}{CA}$$

Where,

- CA is the concentration of drug presented to the organ
- CA - CV is the drug removed by the organ

Solution to Question 7:

Time taken for the plasma concentration to drop to half,
2 mg/L to 1mg/L = 4 hours (plasma half life- $t_{1/2}$)

$$\text{Volume of distribution (V}_d\text{)} = \frac{\text{Amount of drug in the body}}{\text{Initial plasma concentration}}$$

Amount of drug in the body = 40

Initial plasma concentration = 2

Therefore,

$$V_d = 40/2 = 20 \text{ L}$$

$$\text{Clearance} = 0.693 \times V_d / t_{1/2}$$

$$\text{Clearance} \sim 0.7 \times 20/4$$

$$\text{Clearance} \sim 3.5 \text{ L/hour}$$

Solution to Question 8:

The loading dose in this patient for intravenous administration is 300mg.

Target plasma concentration (C_p) = 10 mg/L

Volume of distribution (V_d) = 0.6 L/kg

In the above patient, total $V_d = 0.6 \times 50 = 30 \text{ L}$

$$\text{Loading dose} = \frac{\text{Target } C_p \times V_d}{F}$$

For intravenous route; $F=1$

Where,

- V_d - Volume of distribution (liters)
- C_p - target serum level (mcg/L)
- F - bioavailability factor

$$\text{Loading dose} = 10 \text{ mg/L} \times 30 \text{ L} = 300 \text{ mg}$$

Note: The loading dose of 300 mg is usually not given as a single intravenous injection; it would be given as an infusion or as divided doses administered over a short period to reduce the toxicity associated with high plasma levels.

Solution to Question 9:

The maintenance dose of the drug is 240 mg.

$$\text{Maintenance dose} = \frac{\text{Target plasma concentration (C}_p\text{)} \times \text{Clearance (CL)} \times \text{Dosing interval}}{\text{Bioavailability (F)}}$$

Target plasma concentration (C_p) = 2 mg/L

Clearance (CL) = 20 L/hr

Assuming dosing interval for QID = every 6th hour = x 6 (since the clearance stated is per hour)

Bioavailability (100%), F = 1

Maintenance dose = 2 x 20 x 6 / 1 = 240 mg

Note: QID corresponds to 4 times a day. Hence, the dosing interval is 24/4 = 6 hours.

Solution to Question 10:

Steady state concentration (C_{ss}) for this drug is 33.33 mg/L.

Steady state concentration (C_{ss}) = Rate of infusion / clearance.

Rate of infusion = 200 mg/h

V_d = 100 L

Clearance = 100 mL/min

Clearance = 100 x 60 mL/hr = 6000 mL/hr = 6 L/h (converted to L/h)

C_{ss} = (200 mg/hr) / (6 L/hr) = 100/3 = 33.33 mg/L

Solution to Question 11:

The t_{1/2} of the drug would be 140 minutes.

$t_{1/2}$ is the time required for the plasma concentration or the amount of a drug in the body to be reduced by 50%. It depends on its clearance and volume of distribution.

$$t_{1/2} = 0.693 \times V_d / Cl$$

where,

- V_d - volume of distribution
- Cl - clearance.

We can take 0.693 approximately as 0.7

Substituting,

$$t_{1/2} = 0.7 \times 40 / 0.2$$

$$t_{1/2} = 0.7 \times 200$$

$$t_{1/2} = 140 \text{ minutes}$$

Clinical implications: Half-life determines the duration of the drug effect. It also indicates whether the accumulation of the drug will occur under multiple dosage regimens thereby making it is essential to decide on the appropriate dosing interval.

Solution to Question 12:

Plasma steady-state concentration is achieved when the amount of drug being administered equals the amount of drug being excreted from the body.

The amount of drug being administered = 1.6mg/min

Amount of drug being excreted = Clearance (CL) x Plasma concentration of the drug (C)

Clearance = 640ml/min

Therefore,

$$1.6\text{mg/min} = 640\text{ml/min (clearance)} \times \text{Plasma concentration of the drug}$$

Hence,

$$\text{Plasma concentration of the drug} = 1.6 \text{ mg/min} / 640 \text{ ml/min} = 0.002 \text{ mg/ml.}$$

ANS Introduction & Parasympathomimetics

Question 1:

What is the principal post-ganglionic neurotransmitter of the parasympathetic system?

- a) Dopamine
- b) Nor-epinephrine
- c) Acetylcholine
- d) GABA

Question 2:

While conducting animal studies, a scientist is experimenting with different properties of drugs acting on the ANS. He identifies a drug that inhibits the rate-limiting step in acetylcholine synthesis in nerve terminals. What is this drug?

- a) Choline
- b) Hemicholinium
- c) Acetylcholine esterase
- d) Botulinium toxin

Question 3:

What is the drug that acts on SNARE?

- a) Hemicholinium
- b) Botulinum toxin
- c) Hexamethonium
- d) Acetylcholinesterase

Question 4:

Which of the following statements regarding acetylcholine is incorrect ?

- a) It has no systemic therapeutic use
- b) It is metabolised by both acetylcholine esterase and butyrylcholine esterase

- c) It is used topically during ophthalmic surgery
- d) It has good CNS penetration

Question 5:

Which of the following causes an increase in ACh release from the presynaptic neuron?

- a) Potassium channel blocker
- b) Sodium channel blocker
- c) Calcium channel blocker
- d) Chlorine channel blocker

Question 6:

You notice your senior performing a vagal maneuver in a stable patient with PSVT. Which of the following muscarinic receptors mediate the therapeutic effect of this maneuver?

- a) M₁
- b) M₂
- c) M₃
- d) M₄

Question 7:

What is the mechanism by which acetylcholine causes a decrease in the heart rate?

- a) Delayed diastolic depolarization
- b) Increase in plateau
- c) Decrease in preload
- d) Increase in afterload

Question 8:

A 40-year-old woman who was diagnosed with Sjogren's syndrome is on treatment with cevimeline. On which receptor does this drug mainly act for the expulsion of secretions from the salivary gland?

- a) M₁

- b) M₂
- c) M₃
- d) M₄

Question 9:

You are asked to prescribe a cholinergic agent to a patient suffering from Alzheimer's disease. Which of the following drugs cannot be chosen?

- a) Galantamine
- b) Donepezil
- c) Tacrine
- d) Memantine

Question 10:

In which of the following conditions should cholinomimetic drugs be avoided?

- a) Closed angle glaucoma
- b) Bradycardia
- c) Cobra bite
- d) Myasthenia gravis

Question 11:

Neostigmine is administered to a post-op patient to reverse neuromuscular blockade. What is its mechanism of action?

- a) It blocks the action of acetylcholine
- b) It interferes with the action of choline transferase
- c) It interferes with the action of choline acetyl transferase
- d) It interferes with the action of acetylcholine esterase

Question 12:

A first-year anesthesia resident administered neostigmine to a patient who was accidentally given a slightly excessive bolus dose of suxamethonium. What is the most probable effect this

is likely to produce?

- a) Complete reversal
- b) Incomplete reversal
- c) No effect
- d) Enhanced relaxation

Question 13:

A 45-year-old woman presented to the medicine OPD with complaints of muscle weakness and blurring of vision following exertion. On examination, the patient had ptosis. What is the drug that can be used in the diagnosis of this condition?

- a) Physostigmine
- b) Edrophonium
- c) Neostigmine
- d) Pyridostigmine

Question 14:

What is the drug used to differentiate myasthenia gravis from cholinergic crisis?

- a) Obidoxime
- b) Edrophonium
- c) Donepezil
- d) Atropine

Question 15:

Which of the following drugs would you use in the treatment of a patient with myasthenia gravis?

- a) d-Tubocurarine
- b) Hexamethonium
- c) Ambenonium
- d) Gallamine

Question 16:

A patient presents to the emergency department with diarrhea, sweating, excessive lacrimation, and salivation. His pulse rate is 55/min and his plasma cholinesterase level was 50U/mL. Which of the following can be used to treat this patient?

- a) Neostigmine
- b) Atropine
- c) Epinephrine
- d) Benztropine

Question 17:

What is the mechanism of action of atropine in the management of organophosphate poisoning?

- a) Reactivates acetylcholinesterase
- b) Competes with acetylcholine release
- c) Binds with both nicotinic and muscarinic acetylcholine receptors
- d) Is a competitive antagonist of acetylcholine

Question 18:

A sanitation worker presented with complaints of blurred vision, breathing difficulty and muscle weakness. He gave a history of spraying malathion this morning. You administer pralidoxime as an antidote. What is the mechanism of action of this drug?

- a) Stimulation of ACh receptors
- b) Inhibition of breakdown of ACh
- c) Blockade of ACh receptors
- d) Reactivation of AChE enzyme

Question 19:

A young man is brought to the ER with a history of ingestion of carbamate-based insecticide. He is agitated with excessive sweating, lacrimation, and salivation. Which of the following should not be used for treatment?

- a) Pralidoxime

- b) Atropine
- c) Magnesium sulfate purgative
- d) Gastric lavage with activated charcoal

Question 20:

A patient with diabetes and COPD developed post-operative urinary retention. Which of the following drugs can be used for short-term treatment of the condition?

- a) Tamsulosin
- b) Bethanechol
- c) Terazosin
- d) Methacoline

Answer Key

Question No.	Correct Option
1	c
2	b
3	b
4	d
5	a
6	b
7	a
8	c
9	d
10	b
11	d
12	d
13	b
14	b
15	c
16	b
17	d
18	d

19	a
20	a

Detailed Explanations

Solution to Question 1:

The principal post-ganglionic neurotransmitter of the parasympathetic system is acetylcholine.

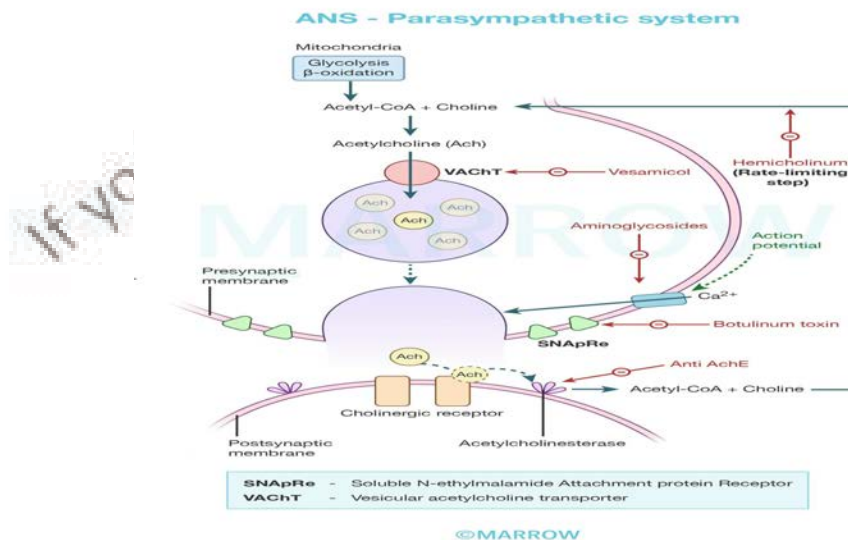
Solution to Question 2:

The rate-limiting step in acetylcholine synthesis is the active uptake of choline into the nerve terminal, which is inhibited by hemicholinium.

Option A: Choline is the precursor of acetylcholine and is taken up from the synaptic cleft into the nerve terminal.

Option C: Acetylcholine esterase is involved in the breakdown of acetylcholine after it is released into the synaptic cleft.

Option D: Botulinum toxin inhibits the release of acetylcholine from nerve terminals.



Solution to Question 3:

The drug which acts on SNARE (Soluble N-ethylmaleimide Sensitive Factor (NSF) Attachment protein Receptor) is botulinum toxin.

SNARE proteins play a critical role in the fusion of the vesicle with the cell membrane and the exocytosis of the neurotransmitter from the pre-synaptic membrane. Inhibition of these SNARE proteins inhibits the release of acetylcholine from synaptic terminals.

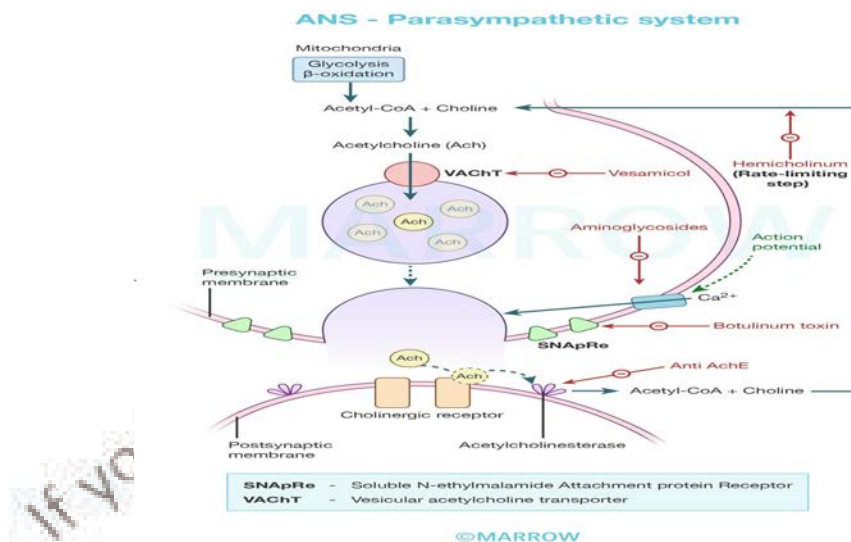
It is useful to treat spastic and neurological conditions associated with increased cholinergic activity like:

- Blepharospasm
- Spastic cerebral palsy
- Torticollis
- Chronic resistant migraine

Option A: Hemicholinium inhibits the uptake of choline into nerve terminals.

Option C: Hexamethonium is a ganglionic blocker.

Option D: Acetylcholine esterase is involved in the breakdown of acetylcholine after it is released into the synaptic cleft.



Solution to Question 4:

Acetylcholine does not have a good CNS penetration as it is a quaternary ammonium compound.

Neostigmine and glycopyrrolate are also quaternary ammonium compounds and their entry into CNS is limited. Tertiary ammonium compounds like physostigmine have good CNS penetration.

Options A and B: Acetylcholine has virtually no systemic use because of its diffuse action and rapid hydrolysis by both acetylcholine esterase and plasma butyrylcholine esterase.

Option C: Acetylcholine is used during ophthalmic surgery for producing rapid miosis.

Solution to Question 5:

Potassium channel blockers cause an increase in ACh release from the presynaptic neuron.

Presynaptic potassium channel block promotes the activation of voltage-gated calcium channels. This leads to a release of acetylcholine in the neuromuscular junction.

Aminopyridines (K⁺ channel blockers) are used in the treatment of Lambert-Eaton myasthenic syndrome which helps to improve muscle strength.

Solution to Question 6:

In vagal maneuver, the parasympathetic nervous system mediates activity on the heart via the M₂ muscarinic receptor (most predominant subtype).

The receptors in the autonomic nervous system (ANS) are of 2 types:

- Muscarinic:
 - 5 Subtypes - M₁, M₂, M₃, M₄, M₅
 - M₁, M₃, M₅ are G_q coupled receptors
 - M₂, M₄ are G_i/G_o coupled receptors
- Nicotinic:
 - 2 Subtypes - Nm (Neuromuscular junction), Nn (CNS, Ganglion, Adrenal glands)
- Nicotinic receptors are ion channels

Receptors of Parasympathetic system	Subtype	Site
Nicotinic	Nm	Neuromuscular junction
Nn	CNS, ganglia, Adrenal gland	
Muscarinic	M ₁	CNS, Ganglia, Salivary, and gastric glands
M ₂	Heart	
M ₃	Smooth muscles, secretory glands, eyes - pupils, and blood vessels	
M ₄	CNS	
M ₅	CNS	

Solution to Question 7:

The mechanism by which acetylcholine causes a decrease in the heart rate is by decreasing the diastolic depolarization.

The acetylcholine released from the vagal nerve acts on the M2 receptor ($\beta\gamma$ subunit of G protein) on the SA node. This opens a special set of potassium channels causing an efflux of potassium (IKAch) which hyperpolarizes the SA node (Ih). The resultant current (Ih) slows the diastolic depolarization thereby causing bradycardia.

Solution to Question 8:

Cevimeline is an M3 receptor agonist resulting in the expulsion of the secretions from the salivary gland. This drug also enhances expulsion of lacrimal secretions in Sjögren's syndrome.

Both M1 and M3 receptors are present in a salivary gland. Activation of M1 increases secretion and M3 causes smooth muscle contraction and expulsion of the secretions.

Receptors of Parasympathetic system	Subtype	Site
Nicotinic	Nm	Neuromuscular junction
Nn	CNS, ganglia, Adrenal gland	
Muscarinic	M1	CNS, Ganglia, Salivary, and gastric glands
M2	Heart	
M3	Smooth muscles, secretory glands, eyes - pupils, and blood vessels	
M4	CNS	
M5	CNS	

Solution to Question 9:

Memantine is an NMDA receptor antagonist (not a cholinergic agent) indicated for the management of moderate to severe Alzheimer's disease.

Donepezil, galantamine, and tacrine are centrally acting cholinesterase inhibitors indicated for Alzheimer's disease.

Note: In this scenario, you are asked to prescribe a cholinergic drug. Tacrine, a cholinergic drug, although hepatotoxic, is still approved for use in Alzheimer's disease. Hence, memantine is the best answer.

Solution to Question 10:

Cholinomimetics cannot be used in bradycardia. They cause bradycardia by its action on M2 receptors on the heart.

In bradycardia, an anticholinergic agent like atropine is indicated.

Option A - Cholinomimetics like pilocarpine are used in both open-angle and closed-angle glaucoma.

Option C - Cobra venom contains a curare-like neurotoxin. Although antivenom is the specific antidote, symptoms can be controlled by a combination of neostigmine and atropine.

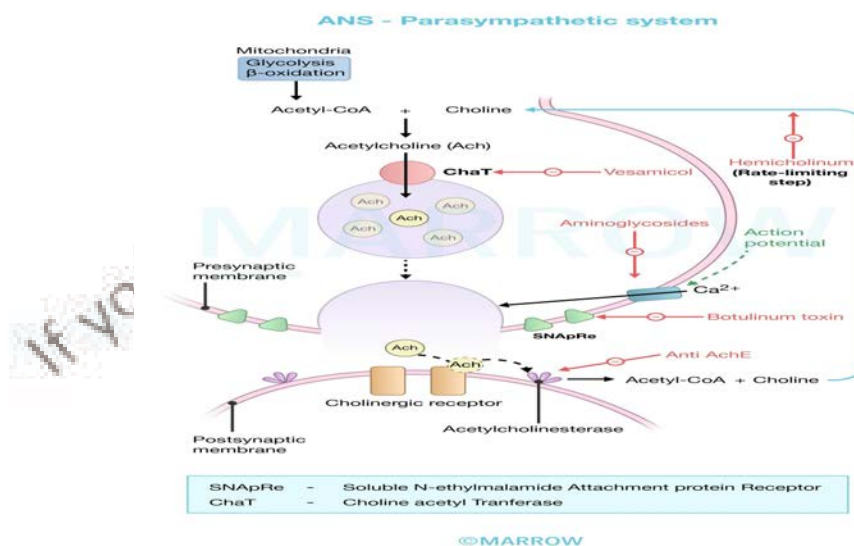
Option D - Cholinesterase inhibitors like neostigmine and pyridostigmine are indicated in myasthenia gravis.

Solution to Question 11:

Muscle weakness is improved by neostigmine as it interferes with the action of acetylcholine esterase.

Neostigmine is a cholinesterase inhibitor used mainly to improve muscle tone in people with myasthenia gravis, and also to reverse the effects of non-depolarizing muscle relaxants such as rocuronium and vecuronium at the end of an operation.

Inhibition of acetylcholine esterase increases the acetylcholine levels in the synapse that can act on receptors over the large surface area and for longer periods of time, reducing the weakness.



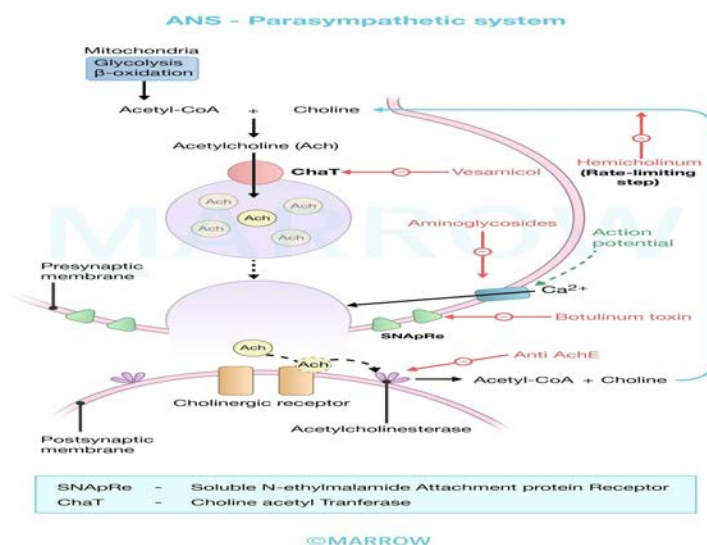
Solution to Question 12:

Neostigmine prolongs the muscle relaxation caused by suxamethonium (succinylcholine) instead of reversing the block.

The contraction of a muscle is achieved by a cycle of ACh binding and its degradation. Depolarising neuromuscular blockers (such as suxamethonium) acts by binding to the receptors much longer than ACh and causing a prolonged depolarization, resulting in an eventual blocking of neuromuscular transmission and flaccid paralysis.

AChE metabolizes ACh and plasma cholinesterase metabolizes suxamethonium (SCh). Neostigmine blocks both enzymes. By inhibiting AChE, neostigmine increases the ACh concentration in the synaptic cleft. But this cannot reverse the block as the depolarized nerves are unresponsive to any stimuli during this period. However, the simultaneous inhibition of plasma cholinesterase decreases the degradation of SCh and therefore potentiates the neuromuscular block of suxamethonium, resulting in prolonged muscle relaxation.

Neostigmine can be used only as a reversal agent for neuromuscular blockade caused by competitive non-depolarising blockers such as vecuronium.



Solution to Question 13:

Edrophonium, a short-acting anticholinesterase agent is used for diagnosing myasthenia gravis. It is also called the tensilon test.

Initially, 2 mg is administered. If improvements of symptoms do not occur within 45 seconds, an additional 8 mg can be administered. Improvement of strength unaccompanied by lingual fasciculations (generally present in non-myasthenics) is a positive test.

Edrophonium is short-acting because:

- The binding site of edrophonium to acetylcholine esterase - choline subsite of the active center
- No reaction with the enzyme - reversible binding only
- Quaternary structure - rapid elimination

Note: An excessive dose of anti-cholinesterase drugs results in a cholinergic crisis.

Solution to Question 14:

The drug used to differentiate myasthenia gravis from a cholinergic crisis is edrophonium.

Muscle weakness is a common presentation in patients with myasthenia gravis and cholinergic crisis (excessive dose of anti-ChE drugs). Intravenous injection of edrophonium improves muscular weakness in a patient with myasthenia gravis and not in a case of cholinergic crisis.

The test used for this differentiation is known as the edrophonium test.

Initially, 2 mg is administered; if improvements of symptoms do not occur within 45 seconds, an additional 8 mg can be administered. Improvement of strength is a positive test for myasthenia gravis.

Solution to Question 15:

Amibenonium, pyridostigmine, and neostigmine are the anti-cholinesterase drugs indicated in the treatment of myasthenia gravis.

Neostigmine and its congeners can improve muscle contractions by allowing released acetylcholine to accumulate and act on the receptors over a large area. They also have an additional direct depolarizing effect on the motor endplate.

Options A and D: d-Tubocurarine and gallamine are non-depolarising neuromuscular blockers and can exacerbate myasthenia.

Option B: Hexamethonium is a ganglionic blocker and not indicated for myasthenia gravis.

Solution to Question 16:

The clinical features are suggestive of organophosphate poisoning and IV atropine is the treatment of choice.

The organophosphates commonly used as pesticides are anticholinesterases. Toxicity occurs by excessive inhalation of vapors, ingestion, or percutaneous absorption. It inhibits the cholinesterase activity which results in an increase of acetylcholine, leading to cholinergic symptoms.

The predominant initial signs are of muscarinic excess which includes: (DUMBBELSS)

- Diarrhea
- Urination
- Miosis
- Bronchospasm
- Bradycardia
- Emesis
- Lacrimation
- Salivation
- Sweating

This may be followed by cognitive disturbances, convulsions, and neuromuscular blockade.

Treatment:

- Maintenance of vital signs particularly respiration
- Removal of clothing and washing of the skin
- IV atropine 2- 5 mg. If IV administration is not possible, then 2 mg IM every 5-10 minutes (till muscarinic symptoms disappear)
- Pralidoxime, a cholinesterase reactivator can also be used for treatment

Solution to Question 17:

Atropine is useful in organophosphate poisoning because it is a competitive antagonist of acetylcholine.

An acetylcholinesterase enzyme has an esteratic and an anionic site. The organophosphates bind to the esteratic (active) site of AChEs (acetylcholine esterases) and deactivate it.

This results in excessive Ach causing increased muscarinic activity like increased tracheobronchial and salivary secretions, bronchoconstriction, miosis, bradycardia, and to a lesser extent peripheral ganglionic and CNS actions. It has relatively insignificant action on nicotinic receptors.

Atropine antagonizes these parasympathetic effects by competitive inhibition of acetylcholine at the muscarinic receptors.

Pralidoxime reactivates the acetylcholinesterase, which had been deactivated in organophosphate poisoning. Pralidoxime attaches to the anionic site of the enzyme that remains unoccupied. Then it reacts with the phosphorus atom of the organophosphorous compound which is attached to the esteratic site to form a phospho-oxime complex. It then dissociates from the enzyme, regenerating the active enzyme.

Solution to Question 18:

Pralidoxime reactivates cholinesterase, which would have been deactivated by phosphorylation caused by malathion.

Pralidoxime attaches to the anionic site of the enzyme that remains unoccupied. Then it reacts with the phosphorus atom of the organophosphorous compound which is attached to the esteratic site to form a phospho-oxime complex. It then dissociates from the enzyme, regenerating the active enzyme.

Solution to Question 19:

Pralidoxime may be hazardous in oral poisoning with carbamate insecticides. It is not effective in carbamate anti-ChE poisoning as the anionic site required for attachment of oximes is occupied.

Oximes are usually contraindicated in carbamate poisoning, as they do not reactivate the enzyme. They also have anti-ChE activity of their own that may worsen the condition.

Option B: Atropine is the mainstay in the management of carbamate poisoning.

Options C and D: Magnesium sulfate purgative and gastric lavage with activated charcoal can be tried in oral carbamate poisoning.

Solution to Question 20:

Tamsulosin is the drug used to relieve the symptoms in cases of postoperative urinary retention in a patient with COPD.

It is also efficacious in the treatment of benign prostatic hyperplasia (BPH) with little effect on blood pressure.

Option B: Bethanechol is the drug of choice to relieve the symptoms in cases of postoperative urinary retention. However, it can cause bronchoconstriction in patients with COPD. Hence tamsulosin is the best answer here.

Option C: Terazosin is an α_1 receptor antagonist, used in the treatment of benign prostatic hyperplasia (BPH).

Option D: Methacholine is a muscarinic agonist but acts specifically on bronchial smooth muscle causing its contraction. It is available as an inhalational agent and is used for the diagnosis of bronchial hyper-reactivity in patients who do not have clinically apparent asthma.

Sympathomimetics

Question 1:

Which of the following inhibits the rate-limiting enzyme in catecholamine synthesis?

- a) Tyrosine
- b) Metyrosine
- c) Propranolol
- d) Valbenazine

Question 2:

An asthmatic patient has been taking a salbutamol inhaler to control his wheezing episodes. Which of the following is not an effect seen on stimulation of the receptor involved?

- a) Inhibition of lipolysis
- b) Increased hepatic gluconeogenesis
- c) Increased muscle glycogenolysis
- d) Smooth muscle relaxation

Question 3:

Which of the following statements most appropriately describes the vasomotor reversal of Dale?

- a) Stimulation of alpha 1 followed by stimulation of beta 2
- b) Blockade of alpha 1 followed by stimulation of beta 2
- c) Stimulation of alpha 1 followed by blockade of beta 2
- d) Stimulation of beta 1 followed by blockade of beta 2

Question 4:

Which of the following drugs when administered would lead to a biphasic reaction on blood pressure?

- a) Adrenaline

- b) Noradrenaline
- c) Dopamine
- d) Dobutamine

Question 5:

Which of the following would not increase with slow IV infusion of adrenaline?

- a) Systolic blood pressure
- b) Diastolic blood pressure
- c) Mean blood pressure
- d) Pulse pressure

Question 6:

Which of the following statements is incorrect?

- a) IV noradrenaline increases systolic and diastolic BP and causes tachycardia
- b) IV adrenaline increases systolic, decreases diastolic BP and causes tachycardia
- c) Dopamine increases peripheral resistance and improves renal perfusion
- d) IV isoproterenol increases systolic and decreases diastolic BP and causes tachycardia

Question 7:

A patient in low output cardiogenic shock is being treated with dobutamine. Which of the following is true about the drug?

- a) Dobutamine decreases peripheral resistance
- b) Acts on D1 and D2 receptors
- c) Decreases renal circulation
- d) Has no effect on coronary circulation

Question 8:

During a rotation in the acute medical ward, you observe a patient in cardiogenic shock being started on a low-dose dopamine infusion. Which of the following would you expect to increase in this patient?

- a) Cardiac output
- b) Urine output
- c) Heart rate
- d) Blood pressure

Question 9:

A stress ECHO has to be done for a 56-year-old COPD patient with high suspicion of coronary artery disease. Which of the following drugs can be used to perform this test?

- a) Thalium
- b) Noradrenaline
- c) Adrenaline
- d) Dobutamine

Question 10:

You notice fenoldopam mentioned on the drug chart of an elderly male patient admitted in the ICU. What condition is he most likely being treated for?

- a) Hypertensive emergency
- b) Congestive heart failure
- c) Uncontrolled migraine
- d) Atrial fibrillation

Question 11:

Which of the following is not an indirectly acting sympathomimetic?

- a) Methamphetamine
- b) Oxymetazoline
- c) Amphetamine
- d) Tyramine

Question 12:

Following spinal anesthesia, the blood pressure of a patient started dropping. Your senior asks you to administer a mixed-acting sympathomimetic. Which of the following loaded syringes will you pick?

- a) Dobutamine
- b) Isoproterenol
- c) Ephedrine
- d) Epinephrine

Question 13:

Which of the following statements is not true about clonidine?

- a) It acts on alpha 2 adrenergic receptors
- b) Can result in rebound hypertension on abrupt withdrawal
- c) Can produce CNS stimulation
- d) Inhibits salivation

Question 14:

You are administering clonidine intravenously to a patient. Which of the following effects would you observe?

- a) Prolonged rapid fall in blood pressure
- b) Transient rise in blood pressure followed by a prolonged fall
- c) Transient fall in blood pressure followed by a prolonged rise
- d) Prolonged rapid rise in blood pressure

Question 15:

In which of the following patients is therapy with an adrenergic-agonist not useful?

- a) 50-year-old woman suffering from acute decompensated heart failure
- b) 35-year-old man in septic shock with persistent hypotension
- c) 40-year-old woman with blood pressure 160/100 mmHg
- d) 55-year-old man with erectile dysfunction

Question 16:

A 22-week pregnant woman presents to the OPD for a routine check-up. On examination, her BP is 150/100 mmHg but she does not have a history of hypertension. You prescribe methyldopa to treat her. On which of the following receptors does this drug act?

- a) α_2
- b) α_1
- c) D₁
- d) β_1

Question 17:

The following table gives BP values before and after administration of drug 'X'. What would be the probable mechanism of action of drug 'X'?

DRUG X		
SBP	120	150
DBP	80	68
HR	72	86

- a) α_1 agonist and β_1 agonist
- b) M₂ agonist and M₃ agonist
- c) α_1 antagonist and β_2 agonist
- d) β_1 agonist and β_2 agonist

Question 18:

Which of the following is not an alpha adrenoreceptor agonist?

- a) Clonidine
- b) Isoproterenol
- c) Guanabenz
- d) Guanfacine

Question 19:

Why is salbutamol preferred over adrenaline in the treatment of an asthmatic patient?

- a) β_1 selectivity
- b) β_2 selectivity
- c) α_1 selectivity
- d) β_3 selectivity

Question 20:

A 56-year-old man diagnosed with chronic bronchitis was prescribed a long-acting beta-2 agonist. Identify the drug.

- a) Albuterol
- b) Formoterol
- c) Pirbuterol
- d) Orciprenaline

Question 21:

Which of the following beta 2 receptor agonists was developed specifically as a uterine relaxant?

- a) Salbutamol
- b) Terbutaline
- c) Ritodrine
- d) Orciprenaline

Question 22:

A child suffering from intermittent bronchial asthma is being treated with β_2 agonists. Which of the following side effects would you not expect?

- a) Tremors
- b) Hypoglycemia
- c) Hypokalemia
- d) Tachycardia

Question 23:

A 4-year-old child on salbutamol is brought to the hospital with complaints of tremors. Which of the following effects of the drug would explain this side effect?

- a) 1, 2, and 5
- b) 2 and 4
- c) 1 and 2
- d) 3 and 5

Question 24:

Which of the following is not an adverse effect of salbutamol?

- a) Tachycardia
- b) Tolerance
- c) Hypokalemia
- d) Hypoglycemia

Question 25:

A 68-year-old man with Parkinson's disease was prescribed mirabegron to control his symptoms of urinary urgency and incontinence. What is the mechanism of action of this drug?

- a) β_2 agonist
- b) β_3 agonist
- c) M3 receptor antagonist
- d) M1 receptor antagonist

Question 26:

What dose of epinephrine is used in anaphylactic shock?

- a) 1:10000 I.V. - 10 ml
- b) 1:1000 I.M. - 0.5 ml
- c) 1:1000 I.M. - 1 ml
- d) 1:1000 I.V. - 0.5 ml

Question 27:

A patient was treated for septicemia with norepinephrine infusion. Which of the following best explains the benefit of this drug?

- a) Bradycardia
- b) Tachycardia
- c) Increases urine output
- d) Increases mean arterial blood pressure

Question 28:

Match the following dilution of adrenaline and its use:

- a) A-1 B-2 C-3
- b) A-2 B-1 C-3
- c) A-3 B-2 C-1
- d) A-1 B-3 C-2

Answer Key

Question No.	Correct Option
1	b
2	a
3	b
4	a
5	b

6	a
7	a
8	b
9	d
10	a
11	b
12	c
13	c
14	b
15	d
16	a
17	d
18	b
19	b
20	b
21	c
22	b
23	c
24	d
25	b
26	b
27	d
28	b

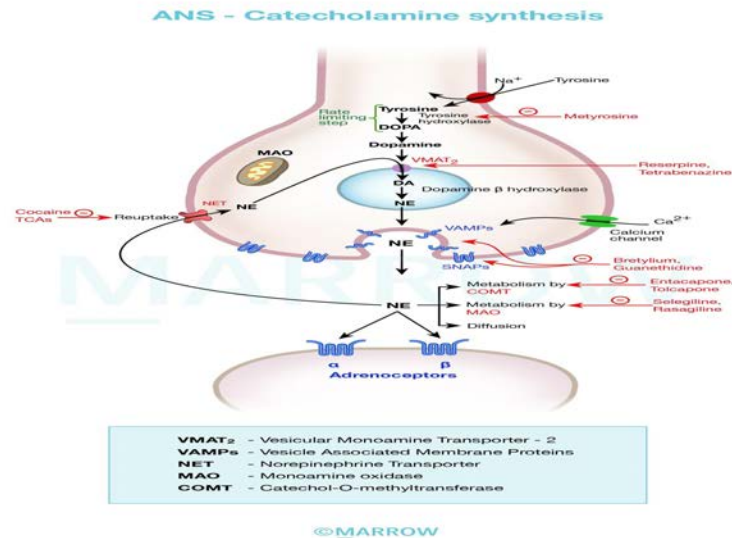
Detailed Explanations

Solution to Question 1:

The rate-limiting enzyme in catecholamine synthesis is tyrosine hydroxylase and is inhibited by metyrosine (alpha-methyl-p-tyrosine).

Tyrosine hydroxylase converts tyrosine to dopa (rate-limiting step). Dopa is then converted to dopamine by dopa decarboxylase.

Metyrosine is indicated preoperatively in pheochromocytoma and in inoperable cases.



Solution to Question 2:

Salbutamol is a short-acting, selective β_2 -adrenergic receptor agonist used in the treatment of asthma and COPD. β_2 receptors mediate the stimulation of lipolysis (not inhibition).

Adrenergic control of adipose tissue:

- Stimulation of lipolysis - α_1 , β_1 , β_2 , and β_3 receptors
- Inhibition of lipolysis - α_2

Adrenergic receptor	Actions
α_1 receptor	Vasoconstriction of large arterioles Genito-urinary smooth muscle - contraction Uterine muscles contraction Mydriasis - by contraction of radial muscles of the iris Increase secretion of glands Gastrointestinal relaxation
α_2 receptor	Autoregulatory in the sympathetic system Vasoconstriction of precapillary arterioles in skeletal muscles Decrease insulin synthesis
β_1 receptor	Increased rhythmicity and force of contraction of the heart Increased renin secretion - JG cells in the kidney

Adrenergic receptor	Actions
β_2 receptor	Smooth muscle relaxation: vasodilation, bronchodilation Uterine muscle relaxation Stimulation of lipolysis Increased insulin secretion Increased hepatic gluconeogenesis Increased muscle glycogenolysis
β_3 receptor	Stimulation of lipolysis Relaxation of detrusor

Solution to Question 3:

Vasomotor reversal of Dale is a blockade of α_1 followed by stimulation of beta (β_2) receptors.

Normally, a high dose of epinephrine injected produces an α_1 -mediated rise in BP followed by a β_2 -mediated fall in BP.

Administration of epinephrine after administration of α -blockers produces only a fall in BP. This is because the α -mediated response is abolished and hence, only the β -mediated response is seen, resulting in a fall in BP. This phenomenon is called the vasomotor reversal of Dale. This was first demonstrated by Sir H.H. Dale.

Solution to Question 4:

Biphasic reaction on blood pressure is seen with the administration of adrenaline. The BP initially rises followed by a fall in BP.

Adrenaline acts on α_1 , α_2 , β_1 , β_2 , and β_3 receptors.

α_1 - present in blood vessels - causes vasoconstriction - increases BP.

β_2 - present in blood vessels - causes vasodilatation - decreases BP.

At higher concentrations of adrenaline, alpha action predominates and hence a rise in BP is seen. Therefore an increase in BP is seen with IV administration of adrenaline.

Due to rapid metabolism and uptake, the concentration of adrenaline decreases, and the β_2 action predominates. Hence a decrease in BP is seen. This is referred to as the biphasic reaction.

Adrenergic drug	Receptors it acts on
Adrenaline	$\alpha_1 + \alpha_2 + \beta_1 + \beta_2 + \text{weak } \beta_3$ action
Noradrenaline	$\alpha_1 + \alpha_2 + \beta_1 + \beta_3$ but no β_2 action

Adrenergic drug	Receptors it acts on
Isoprenaline	$\beta_1 + \beta_2 + \beta_3$ but no α action

Solution to Question 5:

Slow intravenous infusion of adrenaline results in a significant increase in systolic blood pressure with minimal effect on diastolic pressure.

Mean blood pressure and pulse pressure increase. This is because of the increased systolic pressure coupled with decreased, unchanged, or an insignificant increase in diastolic pressure.

At lower concentrations of adrenaline, vascular β_2 receptors are more sensitive than α receptors, hence peripheral resistance decreases.

Solution to Question 6:

IV noradrenaline increases systolic and diastolic BP but is unlikely to cause tachycardia.

IV noradrenaline acts on α_1 , α_2 , and β_1 receptors:

- α_1 - vasoconstriction - Increases DBP.
- β_1 - increases heart rate and contractility - Increases SBP.
- This increases mean arterial pressure which causes a reflex vagal activity that decreases the heart rate leading to bradycardia.

IV adrenaline acts on α_1 , α_2 , β_1 , β_2 , and β_3 receptors:

- β_1 - increases SBP, tachycardia.
- DBP - either β_2 - decrease DBP, α_1 - increase DBP or $\alpha_1 + \beta_2$ - no change.

Dopamine acts on β_1 , α_1 , and D1 receptors:

- D1 - low concentrations - causes vasodilatation thus improving renal perfusion
- β_1 - positive inotropic effect - increases SBP.
- α_1 - higher infusion rates - causes vasoconstriction and increases peripheral resistance

IV isoproterenol is a non-selective β receptor agonist:

- β_1 - increases SBP
- β_2 - vasodilatation - decreases DBP
- β_1 - tachycardia

Solution to Question 7:

Dobutamine decreases peripheral resistance minimally.

Clinically, dobutamine increases cardiac output by selectively increasing myocardial contractility and promotes coronary vasodilation. This is associated with a mild decrease in total peripheral vascular resistance, which is mediated by the reflex withdrawal of sympathetic tone to the vasculature.

Dobutamine has a $t_{1/2}$ of 2 minutes. It resembles dopamine but no action is seen on dopamine receptors.

Renal circulation is variably affected depending upon the sympathetic tone of the patient.

Solution to Question 8:

At lower doses ($< 2 \mu\text{g/kg}$), dopamine only stimulates dopamine receptors in renal vasculature, causing increased urine output.

As the dose increases ($2-5 \mu\text{g/kg}$), additional cardiac stimulant action (β_1) is also seen.

At very high doses ($5-15 \mu\text{g/kg}$), vasoconstriction (α_1) is additionally seen.

Solution to Question 9:

Dobutamine is used to perform a stress test.

Stress echocardiography is primarily done for the evaluation of myocardial ischemia. The basic principle is to increase cardiac stress (either with exercise or with a drug) in a controlled fashion, to aid with the identification and localization of ischemic areas (by detecting new or worsening wall motion abnormalities).

Performing stress echocardiography with a pharmacologic agent such as dobutamine allows for simulation of increased heart rate and increased myocardial physiologic demands in patients who may be unable to exercise due to musculoskeletal or pulmonary comorbidities.

Dobutamine causes cardiac stress by acting on β_1 -adrenergic receptors to increase cardiac contractility and heart rate, and on β_2 -adrenergic receptors to cause peripheral vasodilation. It is preferred over vasodilators used in stress echo tests such as adenosine, dipyridamole, and regadenoson (adenosine derivative).

Solution to Question 10:

Fenoldopam is primarily indicated in the management of hypertensive emergencies. It is a D_1 agonist at peripheral D_1 receptors and acts as a rapid-acting vasodilator.

Drugs used in a hypertensive emergency are:

- Nitroprusside
- Nicardipine

- Labetalol
- Hydralazine
- Esmolol
- Nitroglycerine
- Phentolamine
- Fenoldopam

Solution to Question 11:

Oxymetazoline is a direct-acting non-selective α_1 and α_2 receptor agonist.

Amphetamine, methamphetamine, and tyramine are catecholamine releasing agents and are considered to be indirectly acting sympathomimetic agents.

Solution to Question 12:

Ephedrine is a mixed-acting sympathomimetic agent which can be used to manage hypotension following spinal anesthesia.

Drugs that release norepinephrine and also directly activate receptors are called mixed-acting sympathomimetics.

Ephedrine has a dual action:

- Releasing agent: releases neurotransmitters from nerve endings.
- Direct action on α_1 , α_2 , β_1 , and β_2 receptors.

The adrenergic agents are classified into three sub-groups:

- Directly acting agents:
 - Selective agents - phenylephrine (α_1), clonidine (α_2), dobutamine (β_1), terbutaline (β_2).
 - Non-selective agents - oxymetazoline ($\alpha_1 + \alpha_2$), isoproterenol ($\beta_1 + \beta_2$), epinephrine ($\alpha_1 + \alpha_2 + \beta_1 + \beta_2$) norepinephrine ($\alpha_1 + \alpha_2 + \beta_1$)
 - Mixed acting agents - ephedrine ($\alpha_1 + \alpha_2 + \beta_1 + \beta_2 +$ catecholamine releasing agent)
- Indirectly acting agents:
 - Catecholamine releasing agents - amphetamine and tyramine
 - Uptake inhibitor - cocaine
 - MAO inhibitor - selegiline
 - COMT inhibitor - entacapone

Solution to Question 13:

Clonidine is a centrally acting antihypertensive agent. It is not associated with CNS stimulation. It treats high blood pressure by stimulating presynaptic α_2 receptors in the vasomotor center in the brainstem. This binding has a sympatholytic effect, by suppressing the release of norepinephrine and other sympathomimetics.

Option A - Clonidine is an α_2 receptor agonist that reduces central sympathetic outflow.

Option B - Abrupt withdrawal after long-term therapy can result in rebound hypertension.

Option D - Dry mouth and sedation are the major side effects of clonidine. It can also cause impotency and constipation.

Solution to Question 14:

Intravenous administration of clonidine results in a transient rise in blood pressure followed by a prolonged fall.

α_2 receptors are present on the presynaptic as well as the postsynaptic ganglia. They are the autoreceptors of the sympathetic system.

Clonidine is a directly acting α_2 agonist. When it activates the postsynaptic α_2 receptors in vascular smooth muscle, there is a transient rise in the blood pressure. This is followed by activation of central α_2 receptors, which results in decreases the central sympathetic tone and causes a prolonged fall in blood pressure.

α_2 receptors are present postsynaptically in vascular walls and activation results in the same action as α_1 stimulation i.e., vasoconstriction. But the major receptor that is responsible for vasoconstriction in vascular walls is the postsynaptic α_1 receptor.

Solution to Question 15:

Adrenergic agonists are not useful in the treatment of erectile dysfunction. Adrenergic antagonists (phentolamine) are used locally for vasodilation in erectile dysfunction as PIPE therapy (Phentolamine Induced Penile Erection).

Adrenergic agonists are mainly sympathomimetic due to their action on peripheral vascular or post-synaptic α receptors. However, certain adrenergic agonists (like clonidine) act as a sympatholytic due to their predominant action on pre-synaptic α_2 receptors which inhibits the release of noradrenaline.

Options A and B: In hypotensive states like shock and acute decompensated heart failure, short-term therapy with parenteral sympathomimetic drugs are indicated in treatment.

Option C: Clonidine is an adrenergic agonist which acts on the pre-synaptic α_2 receptor and is used in the treatment of hypertension.

Solution to Question 16:

The given scenario suggests a diagnosis of gestational hypertension, for which methyldopa is used. It acts on α_2 receptors.

Methyldopa is a centrally acting α_2 -adrenergic receptor agonist that is converted into alpha-methyl-norepinephrine. It decreases central sympathetic outflow, decreasing systemic vascular resistance without decreasing cardiac output.

Solution to Question 17:

The probable mechanism of action of drug 'X' is β_1 and β_2 agonism.

According to the given table, systolic BP and heart rate is increased due to β_1 agonistic effect, and diastolic BP is decreased by vasodilatation due to β_2 action.

Option A: Action on α_1 causes $\uparrow$ DBP

Option B: Action on M_2 causes $\downarrow$ HR and action on M_3 causes vasodilatation and reflex increase in HR. Hence there is no net effect on heart rate

Option C: α_1 antagonism causes vasodilatation and $\downarrow$ DBP. β_2 action causes vasodilatation and DBP $\downarrow$

Solution to Question 18:

Isoproterenol is a non-selective β -receptor agonist with no action on α -receptors.

Clonidine, guanabenz, and guanfacine are α_2 receptor agonists.

Solution to Question 19:

Salbutamol is preferred over adrenaline in an asthmatic patient due to β_2 selectivity.

This β_2 selectivity is relative and lost at higher concentrations. Around 40% of the beta receptors are of β_2 type and their activation won't cause cardiac stimulation.

Major adverse effects of non-selective β -agonists are attributed to the stimulation of β_1 receptors in the heart which result in tachyarrhythmias. Selective β_2 receptor agonists are introduced to overcome this major drawback.

Solution to Question 20:

Formoterol is a long-acting β_2 agonist (LABA).

β_2 agonists are classified as:

- Short-acting β_2 receptor agonists (SABA):
- Albuterol (Salbutamol)
- Pirbuterol
- Orciprenaline
- Long-acting β_2 agonists (LABA):
- Salmeterol
- Formoterol
- Ultra-long acting β_2 agonists (VLABA):
- Indacaterol
- Olodaterol
- Vilanterol
- Carmoterol

Solution to Question 21:

Ritodrine was developed specifically as a uterine relaxant.

Salbutamol, orciprenaline, and terbutaline can also be used as uterine relaxants but were initially developed as inhalational agents for asthma. β_2 agonists are associated with increased mortality and are rarely used as uterine relaxants now.

Solution to Question 22:

β_2 agonists are associated with hyperglycemia probably due to β_2 receptor-mediated glycogenolysis and glucagon release.

Adverse effects of beta 2 agonists:

- Hypokalemia - direct effect on skeletal muscle uptake of K^+ ions
- Tachycardia - reflex sympathetic stimulation due to vasodilator action on β_2 receptors and direct effect on cardiac β_1 receptors (β_2 selectivity is lost at high concentrations)
- Muscle tremors - direct effect on β_2 receptors in skeletal muscles and due to hypokalemia
- Hyperglycaemia
- Hyperlipidemia
- Tolerance - desensitization on long-term use

Solution to Question 23:

Tremors due to β_2 agonists can be both due to direct action on muscle spindle and hypokalemia. Initially, it was thought to be due to beta receptor-mediated enhancement of discharge of muscle spindles. However, recently the tremors have also been closely correlated with hypokalemia. Hence both direct action and hypokalemia can explain the tremors associated with β_2 agonist use, although the exact mechanism is still not known.

Solution to Question 24:

Salbutamol in large doses can result in hyperglycemia, not hypoglycemia.

Option A: Tachycardia due to reflex sympathetic stimulation due to vasodilator action on β_2 receptors and direct effect on cardiac β_1 receptors (β_2 selectivity is lost at high concentrations)

Option B: Tolerance occurs due to desensitization on long-term use

Option C: Hypokalemia due to direct effect on skeletal muscle uptake of K^+ ions

Solution to Question 25:

Mirabegron is a β_3 agonist used for the treatment of overactive bladder (OAB). Stimulation of β_3 receptors in detrusor muscle increases the bladder capacity and provides symptomatic relief in overactive bladder.

Antimuscarinic drugs are the mainstay of oral pharmacological treatment for OAB. It acts by inhibiting muscarinic receptor subtypes, M_2 and M_3 in the bladder. Both subtypes are expressed in multiple tissues, resulting in anticholinergic adverse events such as dry mouth, constipation, and cognitive impairment. These adverse effects are not seen with mirabegron as it acts via adrenergic receptors.

Mirabegron is a moderate inhibitor of CYP2D6. Hence, caution has to be exercised when prescribing it along with CYP2D6 substrates such as metoprolol, desipramine, digoxin.

Solution to Question 26:

Epinephrine (1:1000 solution) 0.5 ml via intramuscular route is preferred in the treatment of anaphylactic shock.

For intravenous use, 1:1000 adrenaline must be further diluted to 1:10,000 and should be used very cautiously as slow bolus.

There is a much greater risk of causing harmful side effects by either inappropriate dosage or misdiagnosis of anaphylaxis when using IV adrenaline. Hence, IM injection is preferred over intravenous (IV) bolus because it is faster in many situations and safer (lower risk of cardiovascular complications, such as severe hypertension and ventricular arrhythmias).

For intramuscular, subcutaneous, and inhalational (nebulization) routes of administration, 1:1000 dilution can be used.

In cardiac arrest, the 1:1000 adrenaline must be diluted to 1:10,000 for IV use. The recommended dose of epinephrine is 1mg = 10 mL of 1:10,000 solution IV administered every 3 to 5 minutes during resuscitation as per the CPR guidelines.

Epinephrine is also used in the treatment of hypotension, to induce local vasoconstriction to reduce the diffusion of local anaesthetics and to induce bronchodilation in asthma and COPD.

Solution to Question 27:

The benefit of using noradrenaline in septicemia is that it increases the mean arterial blood pressure. It acts as a powerful peripheral vasoconstrictor and inotrope.

Noradrenaline causes a rise in both systolic BP and diastolic BP, resulting in a rise in the mean arterial blood pressure. This increases the organ perfusion in patients in septic shock or cardiogenic shock.

Another advantage of noradrenaline is that it does so without causing tachycardia, which is often a troublesome event with other inotropes in critical care.

Options A and B: The rise in mean arterial pressure causes reflex bradycardia which is not a major contributor to the treatment of septic shock.

Option C: Low-dose dopamine infusions and furosemide are used to increase urine output in critically ill patients.

Solution to Question 28:

Epinephrine (1:1000 solution) 0.5 ml I.M. is preferred in the treatment of anaphylactic shock.

In cardiac arrest, the 1:1000 adrenaline must be diluted to 1:10,000 for IV route, and used cautiously.

Adrenaline in a concentration of 1: 50,000 to 1: 200,000 is added to local anesthetics for:

- Reducing the systemic absorption
- Prolonging the duration of action
- Providing a more bloodless field for surgery

1:1000 dilution can be used for the following routes of administration:

- Intramuscular
- Subcutaneous
- Inhalational (nebulization)

IM injection is preferred over intravenous (IV) bolus because it is faster in many situations and safer (ie, lower risk of cardiovascular complications, such as severe hypertension and ventricular arrhythmias). There is a much greater risk of causing harmful side effects by inappropriate dosage or misdiagnosis of anaphylaxis when using IV adrenaline.

Epinephrine is also used in the treatment of hypotension, to induce local vasoconstriction to reduce the diffusion of local anaesthetics, to induce bronchodilation in asthma and COPD.

Note: The recommended dose of epinephrine is 1mg = 10 mL of a 1:10,000 solution administered IV every 3 to 5 minutes during resuscitation as per the CPR guidelines. For nebulization, 1:1000 adrenaline is commonly used.

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Parasympatholytics

Question 1:

Which of the following is not an anticholinergic drug?

- a) Ipratropium bromide
- b) Dicyclomine
- c) Atropine
- d) Amphetamine

Question 2:

What is the scientific name of this plant whose roots & leaves were historically used in the treatment of asthma?



- a) Atropa belladonna
- b) Datura stramonium
- c) Calotropis gigantea
- d) Digitalis purpurea

Question 3:

On administration of atropine, which of the following effects is unlikely?

- a) Increased bowel sounds
- b) Mydriasis
- c) Dry mouth
- d) Tachycardia

Question 4:

While treating a patient with acute myocardial infarction, an abnormal rhythm was noted on the ECG. The senior resident decides to administer atropine to control this. What was the likely rhythm observed in the patient?

- a) Sinus tachycardia
- b) Sinus bradycardia
- c) Ventricular arrhythmias
- d) Atrial fibrillation

Question 5:

You administer atropine at a dose of 0.5 mg IV to a patient with symptomatic bradycardia. What would be the effect on heart rate?

- a) Transient tachycardia followed by bradycardia
- b) Transient bradycardia followed by tachycardia
- c) Mild tachycardia
- d) Worsening bradycardia

Question 6:

A delirious child was rushed to the ER with a history of ingestion of an unknown plant. On examination, his pulse rate was 150 beats per minute, and his skin was warm and dry. Pupils were dilated and bowel sounds were sluggish. Which of the following toxidromes is likely to be responsible for the symptoms in this child?

- a) Anticholinergic
- b) Sympathomimetic
- c) Opioid

d) Sedative-Hypnotic

Question 7:

Which of the following would you use in the treatment of atropine toxicity?

- a) Pralidoxime**
- b) Naloxone**
- c) Flumazenil**
- d) Physostigmine**

Question 8:

You are monitoring a patient being treated with atropine for alleged consumption of an organophosphate compound. Suddenly he develops marked excitement and claims to see his late wife sitting next to him. All of the following can be used in the management of this patient except?

- a) Physostigmine**
- b) Lorazepam**
- c) Chlorpromazine**
- d) Diazepam**

Question 9:

Atropine is indicated in all of the following poisonings except _____.

- a) Baygon**
- b) TIK 20**
- c) Parathion**
- d) Endrin**

Question 10:

Which of the following describes the mechanism of mydriasis due to the instillation of a muscarinic antagonist?

- a) Contraction of the sphincter pupillae of the iris**

- b) Paralysis of the dilator pupillae muscles of the iris
- c) Contraction of the dilator pupillae muscles of the iris
- d) Paralysis of the sphincter pupillae of the iris

Question 11:

To achieve which of the following states will you use phenylephrine instead of atropine?

- a) Mydriasis without cycloplegia
- b) Only cycloplegia
- c) Both mydriasis and cycloplegia
- d) Mydriasis and controlled cycloplegia

Question 12:

Which of the following is a short-acting mydriatic that you can use for performing a funduscopy?

- a) Atropine
- b) Homatropine
- c) Cyclopentolate
- d) Tropicamide

Question 13:

A 70-year-old diabetic patient presents to the OPD for diabetic retinopathy screening. Why would you prefer phenylephrine over other anticholinergic drugs as a mydriatic agent before performing funduscopy?

- a) High ciliary tone in elderly patients
- b) Anticholinergics can precipitate glaucoma
- c) Phenylephrine can decrease ciliary tone in elderly patients
- d) All of the above

Question 14:

A young man with a history of motion sickness presents to your OPD with upcoming plans for a road trip. What is the most appropriate treatment for this condition?

- a) Scopolamine patch one night before
- b) Ondansetran one night before and also before trip
- c) Domperidone on the previous night
- d) Dimenhydrinate 1 hour before

Question 15:

Which of the following conditions can be treated with pirenzepine?

- a) Asthma
- b) Peptic ulcer disease
- c) Glaucoma
- d) Hypertension

Question 16:

A 32-year-old woman diagnosed with diarrhoea-predominant irritable bowel syndrome presented to the casualty with severe abdominal pain. She was prescribed a drug that exerts a direct spasmolytic effect on the smooth muscles of the gastrointestinal tract. Which drug could it be?

- a) Dicyclomine
- b) Scopolamine
- c) Galantamine
- d) Rivastigmine

Question 17:

Which of the following agents shows M₃ receptor selectivity?

- a) Oxybutynin
- b) Trospium
- c) Tolterodine
- d) Solifenacin

Question 18:

Among the following antimuscarinic drugs used in the treatment of overactive bladder, which one shows direct spasmolytic activity on smooth muscles?

- a) Tolterodine
- b) Darifenacin
- c) Oxybutynin
- d) Solifenacin

Question 19:

Which of the following is not true regarding the effects of muscarinic antagonists on the urinary bladder?

- a) Lowers intravesicular pressure
- b) Lowers bladder capacity
- c) Lowers the frequency of bladder contractions
- d) Alters bladder sensation during filling

Question 20:

After suffering from a stroke, a middle-aged patient was diagnosed with overactive bladder due to detrusor instability. Which of the following drugs is not useful in this condition?

- a) Solifenacin
- b) Tolterodine
- c) Flavoxate
- d) Duloxetine

Question 21:

In a patient with overactive bladder, what is the most common reason for him to discontinue treatment with muscarinic antagonists?

- a) Xerostomia
- b) Blurred vision
- c) Constipation

d) Dyspepsia

Question 22:

Which of the following has selective action on M1 and M3 receptors?

- a) Ipratropium
- b) Tiotropium
- c) Atropine methonitrate
- d) Umeclidinium

Question 23:

In which of the following patients is the use of trospium contraindicated?

- a) 10-year-old boy with bronchial asthma
- b) 45-year-old woman with hypertension
- c) 60-year-old man with urinary retention
- d) 50-year-old woman with peptic ulcer disease

Question 24:

Which of the following does not produce central anticholinergic syndrome?

- a) Glycopyrrolate
- b) Atropine
- c) Antihistaminics
- d) Tricyclic antidepressants

Question 25:

Which of the following is most commonly used for narcoanalysis?

- a) Atropine sulfate
- b) Pentothal sodium
- c) Phenobarbitone
- d) Scopolamine hydrochloride

Question 26:

Inhibitory effect on mucociliary clearance is minimal with which of the following drugs?

- a) Atropine
- b) Tiotropium
- c) Ipratropium
- d) Both b and c

Answer Key

Question No.	Correct Option
1	d
2	b
3	a
4	b
5	b
6	a
7	d
8	c
9	d
10	d
11	a
12	d
13	b
14	a
15	b
16	a
17	d
18	c
19	b
20	d
21	a
22	b

23	c
24	a
25	b
26	d

Detailed Explanations

Solution to Question 1:

Amphetamine is not an anticholinergic (parasympatholytics) drug. It is an indirectly acting sympathomimetic drug used as a CNS stimulant.

Parasympatholytic drugs are classified as:

Class	Drugs	
Muscarinic antagonists	AtropineScopolamineHomatropineTropicamideIpratropium, TiotropiumDarifenacin, SolifenacinPirenzepine, TelenzepineGlycopyrrolateUmeclidinium	
Nicotinic antagonists	Nn antagonists	TrimethaphanHexamethoniumMecamylamine
Nm antagonists	Depolarizing	SuxamethoniumDecamethonium
Non-depolarizing	PancuroniumVecuroniumRocuroniumd-tubocurarineMivacuriumAtracurium	

Solution to Question 2:

The above plant is Datura stramonium, also called Jamestown or Jimsonweed.

The active ingredient in Datura stramonium includes anticholinergic alkaloids such as atropine, scopolamine, and hyoscyamine. These were helpful in relieving the symptoms associated with asthma via inhibition of M₃ receptors on bronchi, leading to a decrease in bronchospasm and tracheobronchial secretions.

Solution to Question 3:

Atropine does not cause increased bowel sounds. It decreases the amplitude and frequency of peristalsis, resulting in decreased bowel sounds.

Peristalsis suppression due to atropine is incomplete because of the additional control of peristalsis by local reflexes in the enteric nervous system and involvement of other neurotransmitters like 5-HT, enkephalin, dopamine in peristalsis.

Actions of atropine:

- Mydriasis - by blocking the contraction of sphincter pupillae muscle
- Decreased secretions from salivary glands and lacrimal glands - by blocking M1 & M3 receptors
- Tachycardia - due to its vagolytic activity at M2 receptor
- Decreased sweating - by inhibiting sympathetic cholinergic supply to sweat glands which makes the skin hot and dry (atropine fever).

Solution to Question 4:

Atropine is indicated in acute myocardial infarction when it is accompanied by sinus bradycardia due to excessive vagal tone. It is seen commonly in inferior or posterior wall myocardial infarction.

The dosage of atropine has to be carefully adjusted when used in a patient with myocardial infarction. If the dose is too low it will aggravate the bradycardia by action on presynaptic M1 receptors. If the dose is too high it will cause significant tachycardia and aggravate the condition by competitive inhibition of postsynaptic M2 receptors.

Solution to Question 5:

Atropine at average clinical doses of 0.4-0.6 mg administered parenterally results in transient bradycardia followed by tachycardia.

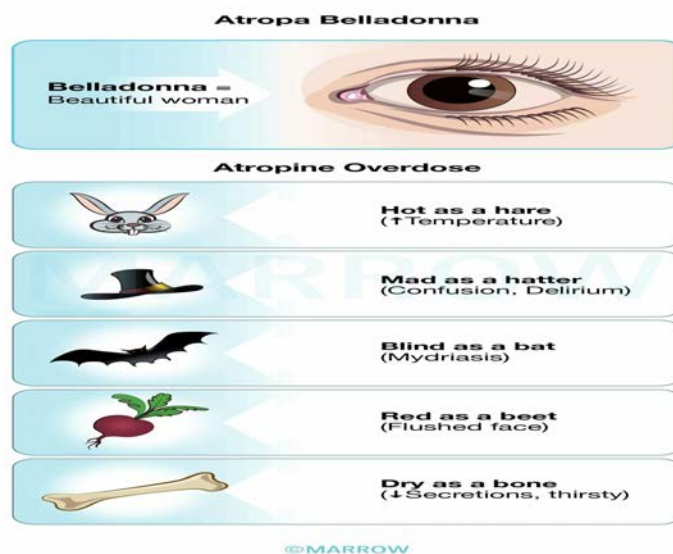
Parasympathetic postganglionic nerve terminals in the SA node contain presynaptic M1 receptors. These receptors normally inhibit the release of ACh. Therefore stimulation of these receptors decreases the release of ACh causing tachycardia. Inhibition of these receptors increases ACh causing bradycardia.

Atropine initially blocks these presynaptic M1 receptors. This increases the amount of ACh released at the SA node and causes bradycardia. Then, the progressive increase in the atropine concentration blocks the postsynaptic M2 receptors on the SA node (decreasing the ACh release) decreasing the parasympathetic tone, thereby causing tachycardia.

Solution to Question 6:

Mydriasis, tachycardia, dry mouth, atropine fever (warm, hot skin) and delirium are classical symptoms of anticholinergic overdose following the ingestion of plants that contain belladonna alkaloids.

Physostigmine is the drug of choice for the treatment of atropine and belladonna poisoning.



Option B: Sympathomimetics can cause mydriasis and tachycardia. Hyperthermia can be seen secondary to increased agitation and aggressive movements. However, delirium is rare and bowel sounds would be normal or increased in sympathomimetic toxidromes.

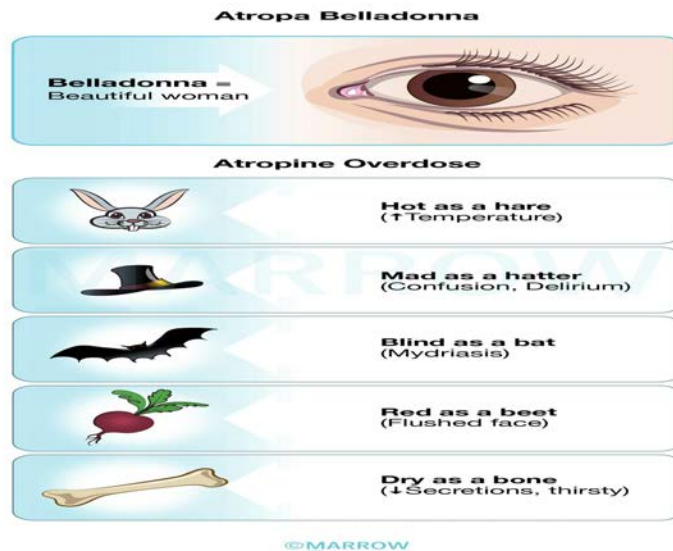
Option C: Opioid overdose results in decreased bowel sounds but other symptoms are not seen.

Option D: In sedative-hypnotic toxidrome, CNS depression and reduced bowel sounds are seen. But it does not cause tachycardia or change in pupil diameter.

Solution to Question 7:

Atropine toxicity is treated with physostigmine, a reversible anticholinesterase agent.

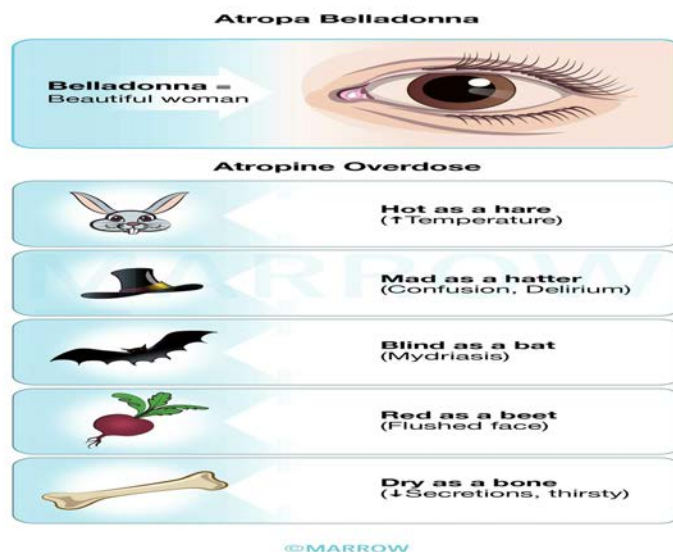
Physostigmine acts by decreasing the breakdown of acetylcholine, and it also overcomes the competitive blockade of atropine in all muscarinic receptors.



Solution to Question 8:

The clinical features of marked excitement and hallucinations are seen in atropine toxicity. Chlorpromazine should be avoided in patients with atropine toxicity.

Chlorpromazine is a low potency typical antipsychotic having high anticholinergic activity. Administering phenothiazines like chlorpromazine will worsen atropine toxicity.



Option A: Physostigmine is indicated as first-line therapy for atropine intoxication

Option B and D: Benzodiazepines can be used when marked excitement is present to calm down the patient

Solution to Question 9:

Endrin is an organochloride pesticide that causes neurotoxicity. There is no antidote available and only symptomatic treatment is done.

Parathion and TIK20 are organophosphates and baygon is a combination of organophosphate and carbamate. For all of these, atropine is the drug of choice.

Solution to Question 10:

Mydriatic action of a muscarinic antagonist is due to the paralysis of the sphincter pupillae (circular) muscle of the iris.

In the iris, miosis occurs due to the constriction of the pupillary sphincter muscles and mydriasis is due to the contraction of the dilator pupillae (radial) muscles. They counter-balance each other.

When muscarinic antagonists are instilled, the pupillary sphincter muscle is paralyzed, causing mydriasis due to the unopposed contraction of the dilator pupillae muscle.

Adrenergic alpha agonists also cause mydriasis, but by mediating the contraction of the dilator pupillae muscle of iris.

Parasympatholytic drugs are classified as:

Class	Drugs	
Muscarinic antagonists	Atropine, Scopolamine, Homatropine, Tropicamide, Ipratropium, Tiotropium, Darifenacin, Solifenacin, Pirenzepine, Telenzepine, Glycopyrrrolate, Umeclidinium	
Nicotinic antagonists	Nicotinic antagonists	Trimethaphan, Hexamethonium, Mecamylamine
Nicotinic antagonists	Depolarizing	Suxamethonium, Decamethonium
Non-depolarizing	Pancuronium, Vecuronium, Rocuronium, d-tubocurarine, Mivacurium, Atracurium	

Solution to Question 11:

Atropine is substituted by phenylephrine in situations where mydriasis is required without cycloplegia, like in fundoscopy.

Phenylephrine, an alpha 1 agonist, acts on the radial muscle (dilator pupillae) of the iris and causes isolated mydriasis. It has no cycloplegic action because sympathetic receptors are present only on the iris and not on the ciliary body.

Atropine has both mydriatic and cycloplegic action, which is required for refraction testing.

In elderly people who have a low ciliary tone, phenylephrine is preferred as a mydriatic. This is because other anticholinergics can precipitate or aggravate glaucoma by relaxation of the ciliary body and altering the forces acting on trabecular meshwork for aqueous outflow.

Atropine, cyclopentolate, and tropicamide are antimuscarinics that are available for ophthalmic use and cause cycloplegia to a variable extent. This cycloplegia/paralysis of accommodation occurs due to their parasympatholytic action, which decreases the tone of the ciliary body.

Solution to Question 12:

Tropicamide is a short-acting mydriatic used in fundoscopy.

It has a rapid onset (20-40 minutes) and a brief duration of action (3-6 hours). The cycloplegia produced by tropicamide is unreliable and insignificant. It is indicated for fundoscopy and refraction testing in adults who have a relatively low ciliary tone when compared to children.

Atropine, which produces complete cycloplegia, is preferred in children as they have a higher ciliary tone. Atropine is usually preferred over other mydriatics for accurate refractory error testing because of reliable cycloplegia. However, it can also precipitate an acute attack of glaucoma and should not be used in patients with glaucoma.

Solution to Question 13:

Phenylephrine is preferred in elderly people as other anticholinergics can precipitate or aggravate glaucoma.

Elderly people have low ciliary tone, and phenylephrine does not alter the tone of the ciliary body. Other anticholinergics cause relaxation of the ciliary body and alter the forces acting on trabecular meshwork for aqueous outflow.

Solution to Question 14:

Scopolamine (Hyoscine) transdermal patch is the most appropriate treatment for motion sickness before a trip.

It should be applied behind the ear about 5-6 hours before the journey, and gives protection for around 72 hours against motion sickness, after which the patch can be replaced if necessary.

Scopolamine is also used for narcoanalysis as it causes sedation and amnesia.

Dimenhydrinate is a first-generation antihistamine that can also be used in motion sickness. Although dimenhydrinate may have equal efficacy it is not preferred because of its sedating properties.

Solution to Question 15:

Pirenzepine is used for treating gastric ulcers. It is a selective M1 anticholinergic drug. It decreases gastric acid secretions at doses that have little effect on salivation or heart rate.

Solution to Question 16:

Dicyclomine is a weak muscarinic receptor antagonist with non-specific direct spasmolytic effects on the smooth muscle of the gastrointestinal tract.

Other options:

Option B: Scopolamine exerts spasmolytic activity indirectly through its antimuscarinic action. It is not preferred due to its adverse effects like drowsiness, dryness, and blurred vision.

Option C and D: Galantamine and rivastigmine are central cholinesterase inhibitors and do not possess spasmolytic activity.

Solution to Question 17:

M3 receptor selectivity is shown by solifenacin and darifenacin.

Oxybutynin and trospium do not show any selectivity. Tolterodine has a more specific action on the urinary bladder but subtype selectivity is absent.

Solution to Question 18:

Oxybutynin is an antimuscarinic drug indicated in the overactive bladder that also exerts a direct spasmolytic action on the smooth muscles of the bladder.

Direct spasmolytic activity is the direct smooth muscle relaxation effect independent of relaxation caused by antimuscarinic action.

The most important adverse effects caused by the use of muscarinic antagonists for an overactive bladder are xerostomia (most common reason for discontinuation), blurred vision, and GI side effects like constipation and dyspepsia (due to blockade of M1, M2 & M3). Drowsiness, dizziness, and confusion can also occur.

Tolterodine, darifenacin, and solifenacin are antimuscarinic agents but they do not have the additional direct antispasmodic effect.

Solution to Question 19:

Muscarinic antagonists increase bladder capacity by blocking M3 receptors on the detrusor. This results in the relaxation of the bladder.

Effects of muscarinic antagonists on the bladder:

- Lowers intravesicular pressure
- Increases bladder capacity
- Lowers the frequency of bladder contractions
- Alters bladder sensation during filling

Solution to Question 20:

Duloxetine is not useful in detrusor instability. It is a selective serotonin and norepinephrine reuptake inhibitor that is efficacious in stress urinary incontinence. It is also indicated in depression, fibromyalgia, and neuropathic pain.

Muscarinic antagonists are useful in this case. They increase bladder capacity by blocking M3 receptors on detrusor. This results in the relaxation of the bladder.

The effects of muscarinic antagonists on the bladder are:

- Lowering of intravesicular pressure
- Increases bladder capacity
- Lower the frequency of bladder contractions
- Alter bladder sensation during filling

Solution to Question 21:

Xerostomia is the most common reason for the discontinuation of muscarinic antagonists in overactive bladder.

The most important adverse effects caused by the use of muscarinic antagonists for an overactive bladder are xerostomia, blurred vision, and GI side effects like constipation and dyspepsia (due to blockade of M1, M2 & M3). Drowsiness, dizziness, and confusion can also occur.

Solution to Question 22:

Tiotropium has a selective action on M1 and M3 receptors.

Tiotropium binds to all M1, M2 and M3 receptors. However, it dissociates slower from M1 and M3 receptors as compared to M2. This shows it has a higher affinity towards M1 and M3 receptors.

Ipratropium and atropine methonitrate do not have any subtype selectivity on muscarinic receptors.

Inhibition of mucociliary clearance by atropine results in the accumulation of lower airway secretions and the formation of mucus plugs. However, tiotropium and ipratropium have minimum interference with mucociliary clearance when compared to atropine, and the incidence

of mucus plugs is also less.

Solution to Question 23:

The use of trospium is contraindicated in urinary retention. It is an antimuscarinic agent and can worsen urinary retention.

Muscarinic antagonists are contraindicated in urinary tract obstruction and in benign prostatic hyperplasia (or used with extreme caution).

Bronchial asthma, hypertension, and peptic ulcer disease are not the contraindications for the use of trospium.

Solution to Question 24:

Glycopyrrolate usually does not produce central anticholinergic syndrome. It is a quaternary amine compound and lipid insoluble. Therefore, glycopyrrolate does not cross the blood-brain barrier.

Non-depolarising skeletal muscle relaxants are used in anesthesia. Neostigmine along with glycopyrrolate are used for reversal of this neuromuscular block. Glycopyrrolate blocks the peripheral muscarinic side effects due to the increased acetylcholine levels after the administration of neostigmine, without causing unwanted central anticholinergic effects.

Peripheral and central anticholinergic syndromes are reactions due to either overdose or any unusual reaction to the usage of drugs like atropine and other muscarinic agents, phenothiazines, antihistamines, and tricyclic antidepressants

Option B: Atropine is a tertiary compound that can cross the blood-brain barrier and cause central anticholinergic syndrome.

Option C and D: Some of the first-generation antihistamines and tricyclic antidepressants have significant anticholinergic activity and can cause central anticholinergic syndrome.

Solution to Question 25:

The most commonly used drug in narcoanalysis is pentothal sodium (thiopentone).

Other drugs used in narcoanalysis are sodium amytal and secobarbital.

Scopolamine hydrochloride was also used in narcoanalysis during World War II. It was popularly known as the 'lie detector'. The amnesic and depressant actions of scopolamine are due to the inhibition of the brain cholinergic system. It would put the subject off guard in case of sustained interrogation and sleep deprivation, eventually forcing the truth out.

Solution to Question 26:

Both ipratropium and tiotropium have a minimal inhibitory effect on mucociliary clearance when compared to atropine. Inhibition of mucociliary clearance by atropine results in accumulation of lower airway secretions and formation of mucus plugs.

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If you purchased this from someone else,
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Sympatholytics

Question 1:

Which of the following is a selective alpha 1 receptor antagonist?

- a) Phenoxybenzamine
- b) Phentolamine
- c) Prazosin
- d) Idazoxan

Question 2:

Which of the following is an incorrect statement regarding prazosin?

- a) Blocks all subtypes of alpha 1 receptors
- b) It is only an arteriolar dilator
- c) It has a beneficial effect on lipid profile
- d) Causes minimal reflex tachycardia

Question 3:

In which of the following patients would prazosin not be used therapeutically?

- a) 45-year-old man with peripheral vascular disease
- b) 25-year-old woman with pheochromocytoma
- c) 30-year-old male with ejaculatory dysfunction
- d) 10-year-old boy stung by a scorpion

Question 4:

A patient presents with complaints of dizziness while getting up from bed. He tells you he was recently diagnosed with BPH and has been taking prazosin. Which of the following drugs would be a good replacement for him?

- a) Terazosin
- b) Silodosin

- c) Doxazosin
- d) Alfuzosin

Question 5:

A 62-year-old male presents with complaints of a weak urine stream and frequent urination in small amounts for the past month. He also mentions that he often feels lightheaded and has low blood pressure. Which of the following alpha-blockers will you prefer to give him?

- a) Alfuzosin
- b) Terazosin
- c) Tamsulosin
- d) Doxazosin

Question 6:

You prescribe tamsulosin to a patient with benign prostatic hyperplasia. What is the main side effect that should be explained to the patient?

- a) Priapism
- b) Postural hypotension
- c) Retrograde ejaculation
- d) Erectile dysfunction

Question 7:

Which of the following is a selective alpha 2 receptor antagonist?

- a) Terazosin
- b) Yasmin
- c) Yohimbine
- d) Alfuzosin

Question 8:

Phenoxybenzamine is indicated in the management of which of the following conditions?

- a) Pheochromocytoma

- b) Essential hypertension
- c) Erectile dysfunction
- d) All of the above

Question 9:

A patient with erectile dysfunction complained of unsatisfactory response to oral medication. He was then administered an intracavernosal injection of phentolamine following which he developed sustained painful erection. Which of the following can be used to reverse his condition?

- a) Phenoxybenzamine
- b) Prazosin
- c) Phenylephrine
- d) Propranolol

Question 10:

In which of the following conditions is propranolol not indicated?

- a) Infantile hemangioma
- b) Variant angina
- c) Essential tremors
- d) Portal hypertension

Question 11:

A 45-year-old man is diagnosed with type 2 diabetes mellitus during a routine health evaluation. He mentions that he is on propranolol for essential tremors. You would caution him against continuing the drug due to what reason?

- a) Causes hypertension
- b) Causes seizures
- c) Masks hypoglycemic symptoms
- d) Causes hypotension

Question 12:

Propranolol is drug of choice for:

- a) Ulcerated infantile hemangioma
- b) Lymphangioma circumscriptum
- c) Capillary malformation
- d) Pyogenic granuloma

Question 13:

Which of the following is a cardioselective beta blocker with antioxidant property?

- a) Sotalol
- b) Metoprolol
- c) Nebivolol
- d) Esmolol

Question 14:

Esmolol is indicated in which of the following conditions?

- a) 1, 3, 5
- b) 3, 5
- c) 3, 4, 5
- d) 1, 2, 3, 5

Question 15:

Which of the following is a false statement?

- a) Metoprolol prolongs survival in congestive cardiac failure
- b) Celiprolol has vasodilatory property
- c) Esmolol is mainly metabolised in the liver
- d) Nebivolol releases nitric oxide

Question 16:

Identify the third-generation beta-blocker that does not cause alpha 1 receptor antagonism.

- a) Labetalol
- b) Carvedilol
- c) Celiprolol
- d) Bucindolol

Question 17:

You are about to add a beta-blocker to the treatment chart of a patient with heart failure. Which of the following is not true about the use of these drugs in this condition?

- a) Should be continued after discharge
- b) Most effective in acute decompensated heart failure
- c) Should be started at a low dose
- d) Carvedilol is widely used in this condition

Question 18:

Which of the following is not true about beta-blockers?

- a) Atenolol is longer acting than metoprolol
- b) Labetalol has both alpha and beta blocking action
- c) Celiprolol undergoes hepatic elimination
- d) Nadolol has longest half life

Question 19:

A nurse in the nephrology ward is managing the drug inventory for patients with severe renal failure. Which of the following beta-blockers is least likely to be present in the drug charts?

- a) Carvedilol
- b) Metoprolol
- c) Sotalol
- d) Labetalol

Question 20:

Beta-blockers are not indicated in which of the following patients?

- a) 65-year-old woman with hypothyroidism
- b) 35-year-old factory worker with alcohol withdrawal
- c) 40-year-old chronic alcoholic with portal hypertension
- d) 22-year-old singer with performance anxiety

Question 21:

Which of the following is a selective beta 2 antagonist?

- a) Betaxolol
- b) Butoxamine
- c) Levobunolol
- d) Celiprolol

Question 22:

Cocaine-like effect of inhibiting neuronal uptake of noradrenaline is seen with which of the following?

- a) Propranolol
- b) Metoprolol
- c) Labetalol
- d) Acebutolol

Question 23:

Match the following beta-blockers with their receptor selectivity and additional actions.

- a) A-2, B-3, C-4, D-1
- b) A-2, B-1, C-4, D-3
- c) A-4, B-2, C-3, D-1
- d) A-1, B-2, C-3, D-4

Answer Key

Question No.	Correct Option
1	c
2	b
3	c
4	b
5	c
6	c
7	c
8	a
9	c
10	b
11	c
12	a
13	c
14	d
15	c
16	c
17	b
18	c
19	c
20	a
21	b
22	c
23	b

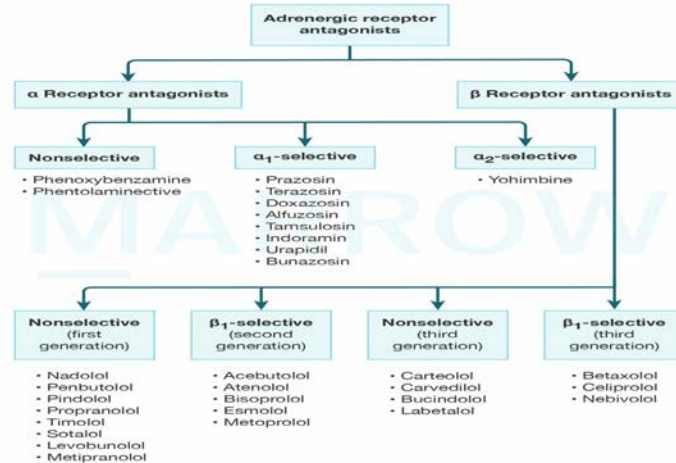
Detailed Explanations

Solution to Question 1:

Prazosin is a sympatholytic drug that is a selective α_1 receptor antagonist.

The sympatholytics are classified into α and β -blockers.

Adrenergic receptor antagonists



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Miscellaneous antiadrenergic drugs that inhibit synthesis or release of neurotransmitters include:

- Metyrosine
- Reserpine
- Tetrabenazine
- Deutetrabenazine

Solution to Question 2:

Prazosin causes dilation of both arterioles and veins.

As a result, both afterload and preload decrease. However, the reflex tachycardia is less pronounced due to the absence of α_2 blockade.

Prazosin blocks all subtypes of the α_1 receptor (α_{1A} , α_{1B} , and α_{1D}). It also has a beneficial effect on lipid profile as it decreases LDL and triglycerides and increases HDL levels.

Solution to Question 3:

Prazosin is not useful to treat a patient with ejaculatory dysfunction. It is indicated in relieving symptoms of benign prostatic hyperplasia (BPH).

Prazosin is an α_1 receptor antagonist and is useful in:

- Relieving vasospasm in peripheral vascular diseases
- Controlling the symptoms of pheochromocytoma preoperatively (as an alternative to phenoxybenzamine)
- Mainstay treatment for scorpion sting along with antivenom

Solution to Question 4:

The given scenario is suggestive of postural hypotension caused by prazosin. These symptoms can be avoided by using silodosin which has a selective antagonist action on α_1A receptors over α_1B receptors.

The prostate stroma has a maximum concentration of α_1A receptors over other α_1 receptor subtypes. α_1B receptors are present on blood vessels which play a role in vasoconstriction. Hence, selective α_1A receptor antagonists have minimal effect on blood pressure as compared to non-selective α_1 antagonists like prazosin.

α_1 receptor antagonists can be classified as:

- Selective antagonists
- Silodosin (α_1A)
- Tamsulosin ($\alpha_1A + \alpha_1D$)
- Non-selective
- Prazosin
- Terazosin
- Doxazosin
- Alfuzosin

Solution to Question 5:

The given scenario is suggestive of benign prostatic hyperplasia (BPH) and tamsulosin should be given to this patient as it has the least effect on blood pressure.

It is an α -blocker that has more selectivity towards $\alpha_1A/1D$ when compared to α_1B receptors. The vasodilatory effects are mainly attributed to the blockade of α_1B receptors present on blood vessels.

The relief of BPH symptoms is mainly due to the blockade of α_1A receptors present predominantly in the prostatic stroma. It lacks the apoptotic effect on prostatic cells that is seen with terazosin and doxazosin.

Solution to Question 6:

Retrograde ejaculation is the main side effect of tamsulosin.

It is a selective $\alpha_1A/1D$ blocker and the relief of BPH symptoms is mainly due to blockade of α_1A receptors. Blockade of α_1D receptors present in the bladder relaxes the urethral sphincter, which is normally closed during ejaculation. This results in the redirection of semen into the urinary bladder.

Sildenafil causes ejaculatory dysfunction due to absence of seminal emission.

Solution to Question 7:

Yohimbine is a competitive antagonist that is selective for α_2 receptors. It was earlier used in the indication of erectile dysfunction but is rarely indicated now due to better results with PDE5 inhibitors.

Terazosin and alfuzosin are selective α_1 receptor antagonists.

Yasmin (drospirenone) is a 4th generation progestin used as an oral contraceptive having mineralocorticoid antagonist action.

Solution to Question 8:

Phenoxybenzamine is indicated in the preoperative preparation of patients with pheochromocytoma. It is also indicated for symptomatic control in inoperable and advanced pheochromocytoma.

Option B: Phenoxybenzamine is not used in the management of essential hypertension because of the irreversible α blockade and increased risk of adverse effects.

Option C: For erectile dysfunction, intracavernous phentolamine, a reversible non-selective α -blocker is indicated.

Solution to Question 9:

The reversal of phentolamine induced priapism (painful erection) can be achieved by intracavernosal injection of phenylephrine.

Intracavernosal phentolamine (α -blocker) is used for the management of impotency. Priapism is a serious adverse effect of this agent. Phenylephrine is an α agonist and counteracts phentolamine induced priapism by producing vasoconstriction.

Phentolamine is useful in:

- Hypertensive crisis due to clonidine withdrawal
- Short term control of intra-operative hypertension in patients with pheochromocytoma
- MAO inhibitor-induced hypertension in cheese reaction

Options A and B: Phenoxybenzamine and prazosin are α -receptor antagonists and can worsen priapism.

Option D: Propranolol is a non-selective β -blocker and is not indicated in this condition.

Solution to Question 10:

Propranolol is not indicated in variant angina. It can cause further constriction of coronary arteries by leaving alpha-mediated coronary vasoconstriction unopposed.

Uses of propranolol include:

- Management of thyrotoxicosis
- Management of hypertension (however, cardioselective beta-blockers are now preferred over non-selective)
- Treatment of infantile hemangioma
- Management of essential tremors and panic attacks
- Prophylaxis of migraine
- Prevention of variceal bleeding in portal hypertension

Solution to Question 11:

Propranolol is generally avoided in diabetic patients because it masks the hypoglycaemic symptoms.

Non-selective β blockers interfere with the counter-regulatory hormones secreted during hypoglycemia, resulting in prolonged hypoglycemia. It also decreases the perception of symptoms such as tremors, tachycardia, and nervousness. These symptoms are manifested due to stimulation of sympathetic beta receptors in response to hypoglycemia. Blunting of sympathetic response results in hypoglycemic unawareness.

Cardioselective beta-blockers have relatively lesser interference on the sympathetic response to hypoglycemia and are preferred in diabetic patients.

Solution to Question 12:

Propranolol is the drug of choice in the management of infantile hemangioma complicated by ulceration or in symptomatic patients.

It acts by causing vasoconstriction, inducing apoptosis and inhibition of angiogenesis.

Solution to Question 13:

Nebivolol is a cardioselective beta-blocker with additional antioxidant property.

Nebivolol:

- Nebivolol is a third-generation, long-acting, and highly selective β_1 adrenergic receptor antagonist

- It stimulates NO-mediated vasodilation via β_3 receptor agonism
- Nebivolol is approved for the treatment of hypertension
- It has potential utility in the treatment of heart failure with reduced ejection fraction
- Nebivolol reduces oxidative stress and may have favorable effects on both carbohydrate and lipid metabolism
- This can be used in metabolic syndrome.

Solution to Question 14:

Esmolol is a short-acting cardioselective beta-blocker that is indicated in:

- Prevention and management of intraoperative tachycardia
- Postoperative hypertension
- Supraventricular tachycardia
- Hypertensive emergency

It is contraindicated in pulmonary hypertension.

Solution to Question 15:

Esmolol contains an ester linkage and it is hydrolyzed rapidly by esterases in erythrocytes and not in the liver. This results in a short half-life of about 8-10 minutes.

It is used in the management of intraoperative and postoperative hypertension as it has a short duration of action and can be discontinued quickly.

Option A: Metoprolol is approved for use in chronic congestive cardiac failure and has shown mortality benefit.

Option B: Celiprolol has shown a relaxant effect on vascular smooth muscle and weak vasodilation due to partial selective β_2 agonism.

Option D: Nebivolol has the additional action of releasing nitric oxide.

Solution to Question 16:

Celiprolol is a third-generation beta blocker having no action on α_1 receptors. It causes vasodilation by partial selective β_2 agonist activity.

Labetalol, carvedilol, and bucindolol are non-selective third-generation beta blockers that possess additional α_1 receptor antagonistic action. This is responsible for their vasodilatory effect.

Note: Third-generation β -blockers are those that possess additional vasodilating actions and include cardioselective and non-selective drugs.

Solution to Question 17:

β -blockers are contraindicated in the treatment of acute decompensated heart failure (ADHF) as they can worsen the condition. They are indicated only in patients with chronic stable heart failure.

Option A: Though contraindicated in ADHF, it is safe to start once the patient is euvolemic. Initiation of β -blocker prior to discharge is associated with improved outcomes and must be continued after discharge.

Option C: β -blockers are initiated at a low dose and slowly titrated to decrease the risk of precipitating heart failure.

Option D: Carvedilol has been shown to reduce mortality in patients with heart failure, and is widely used because of additional benefits like antioxidant properties and peripheral vasodilatory action.

Solution to Question 18:

Celiprolol is excreted predominantly through the renal route. It is excreted unchanged in the urine.

Option A: Atenolol is longer acting with a longer half-life than metoprolol. It is usually administered once daily.

Option C: Labetalol is a non-selective β -blocker with α_1 blocking action similar to carvedilol.

Option D: Nadolol has the longest half-life of 20 hours and is administered once daily.

Solution to Question 19:

Sotalol is least likely to be present in the treatment charts as it is contraindicated in severe renal failure. It has low lipid solubility and is excreted primarily unchanged in the urine.

Carvedilol, metoprolol and labetalol are metabolised by the liver and can be used in renal failure.

Solution to Question 20:

Beta-blockers are not indicated in hypothyroidism. They are indicated in hyperthyroidism and thyrotoxicosis that manifests with increased sympathetic activity.

Options B and C: Beta-blockers are useful in the treatment of alcohol withdrawal and in reducing variceal bleeding in portal hypertension.

Option D: β blockers like propranolol are effective in controlling acute panic symptoms and anxiety associated with performing in public. They reduce symptoms of increased sympathetic

activity such as tachycardia and tremors.

Solution to Question 21:

Butoxamine is a selective β_2 antagonist.

There are no clinical indications currently for selective β_2 antagonists and they are still being researched.

Solution to Question 22:

Cocaine-like effect of inhibiting neuronal uptake of noradrenaline (NE) is seen with labetalol.

It is a third-generation non-selective β blocker with the following actions:

- Non-selective β receptor blockade
- Selective α_1 receptor blockade
- Inhibition of neuronal uptake of NE/cocaine-like effect

Solution to Question 23:

Pindolol is a non-selective first-generation β -blocker with significant intrinsic agonist activity, which may result in a smaller reduction in heart rate and blood pressure. This is useful in patients with diminished cardiac reserve or at risk of bradycardia.

Carvedilol is a non-selective third-generation β -blocker with antioxidant and anti-inflammatory properties. These properties make it useful in the treatment of congestive heart failure.

Betaxolol is a β_1 selective third-generation β -blocker with additional calcium entry blockade activity and a slight membrane-stabilizing property. It does not have agonist activity.

Acebutolol is a β_1 selective second-generation β blocker having both membrane-stabilizing and intrinsic agonist activity. It is useful in the treatment of atrial and ventricular arrhythmias.

Pindolol	Non-selective, intrinsic agonist
Carvedilol	Non-selective, antioxidant
Betaxolol	Beta 1 selective, membrane stabilizing
Acebutolol	Beta 1 selective, intrinsic agonist

Drugs for Glaucoma

Question 1:

You want to prescribe a drug that increases uveo-scleral outflow while treating a patient with open-angle glaucoma. Which of the following drugs would you choose?

- a) Prostaglandin analogues
- b) Carbonic anhydrase inhibitors
- c) Rho-kinase inhibitors
- d) Beta-blockers

Question 2:

Which of the following anti-glaucoma agents has no effect on aqueous outflow?

- a) Netarsudil
- b) Timolol
- c) Brimonidine
- d) Latanoprost

Question 3:

Dipevefrine is a prodrug of _____.

- a) Noradrenaline
- b) Adrenaline
- c) Clonidine
- d) Dopamine

Question 4:

What is the primary mechanism of action of topical carbonic anhydrase inhibitors in glaucoma?

- a) Increasing trabecular outflow
- b) Increasing uveoscleral outflow

- c) Reducing aqueous formation
- d) Relaxation of trabecular meshwork

Question 5:

A 40-year old man who is suffering from open-angle glaucoma reads about netarsudil being approved as a novel agent for his condition. What is the mechanism of action of this drug?

- a) Carbonic anhydrase inhibition
- b) Phosphodiesterase 5 inhibition
- c) Rho kinase inhibition
- d) GPCR inhibition

Question 6:

A patient who was started on treatment for glaucoma comes to the casualty with acute onset breathlessness. On auscultation, bilateral wheeze is heard. Which of the following drugs might have led to this adverse effect?

- a) Timolol
- b) Betaxolol
- c) Latanaprost
- d) Acetazolamide

Question 7:

A patient with diabetic macular edema developed glaucoma. Which of the following drugs would you consider giving last?

- a) Beta blocker
- b) Alpha agonist
- c) Pilocarpine
- d) Prostaglandin analogues

Question 8:

Which of the following is the most appropriate first-line medical therapy in a patient with open-angle glaucoma?

- a) Beta blockers
- b) Miotics
- c) PGF 2 alpha analogs
- d) Carbonic anhydrase inhibitors

Question 9:

You diagnose a patient with primary open-angle glaucoma and decide to prescribe a PGF2 alpha analog. Which of the following drugs would you not choose?

- a) Bimatoprost
- b) Travoprost
- c) Dinoprost
- d) Tafluprost

Question 10:

Which of the following antiglaucoma medications is unsafe in infants?

- a) Timolol
- b) Brimonidine
- c) Latanoprost
- d) Dorzolamide

Question 11:

Which of the following can be administered orally for the treatment of glaucoma?

- a) 1 and 5
- b) 1, 2 and 3
- c) 2, 3 and 4
- d) 1, 2, 3, 4, 5

Question 12:

Intravenous mannitol is used in ____?

- a) Pulmonary edema
- b) Congestive cardiac failure
- c) Acute kidney injury with anuria
- d) Glaucoma

Question 13:

Which of the following drugs should not be given in a patient prone to developing angle-closure glaucoma?

- a) Pilocarpine
- b) Clozapine
- c) Rivastigmine
- d) Neostigmine

Question 14:

While examining a patient using a corneal esthesiometer, you find that corneal sensations are decreased. Which of the following drugs could have led to this adverse effect?

- a) Beta blockers
- b) Carbonic anhydrase inhibitors
- c) Cholinomimetics
- d) Prostaglandin analogs

Question 15:

A patient presents with severe eye pain to the ophthalmology OPD. On examination, there is circumcorneal congestion and the IOP is 40 mm Hg. Which of the following drugs should not be administered immediately to this patient?

- a) Acetazolamide
- b) Timolol
- c) Latanoprost
- d) Pilocarpine

Question 16:

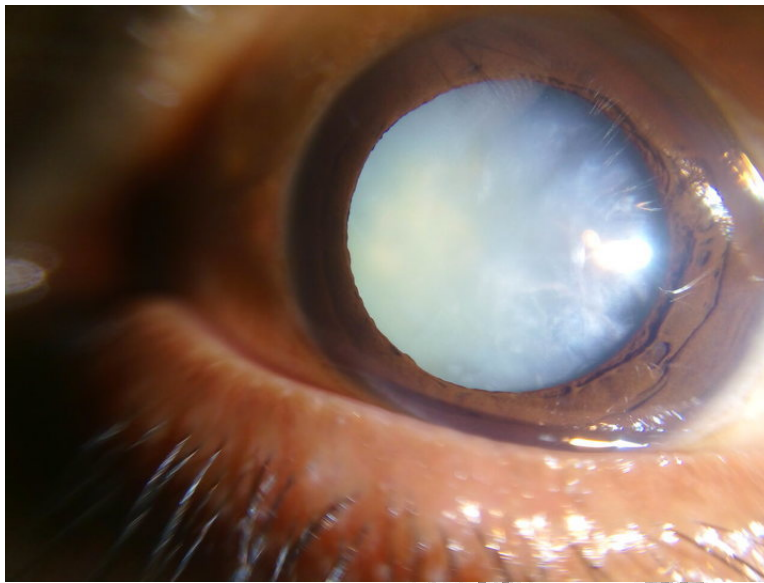
A patient who is on treatment for glaucoma presented with the following finding during his routine follow-up. Which of the following drugs can cause this adverse effect?



- a) Timolol
- b) Latanoprost
- c) Brimonidine
- d) Apraclonidine

Question 17:

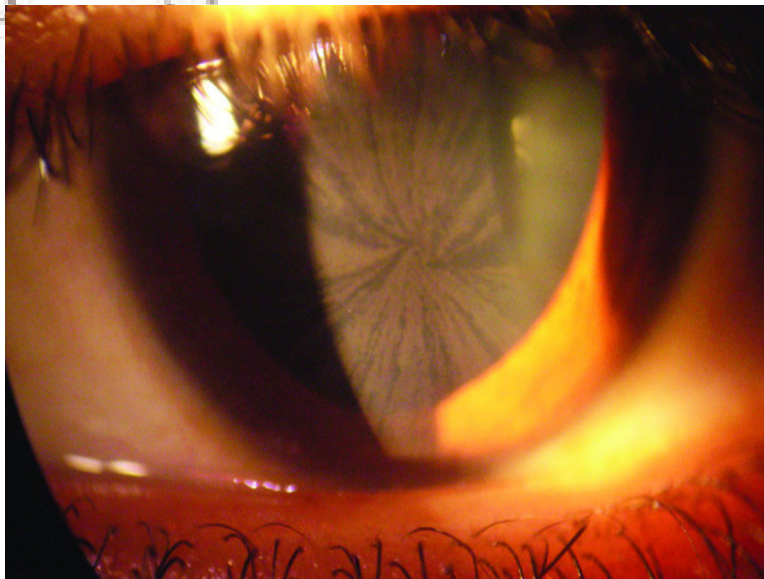
The ophthalmologic examination of a patient who is on anti-glaucoma drugs shows the following finding. Identify the drug that might have resulted in this adverse effect.



- a) Carbonic anhydrase inhibitors
- b) Anticholinesterase agents
- c) Muscarinic antagonists
- d) Sympathomimetic agents

Question 18:

A patient who is on anti-glaucoma therapy presents with corneal findings as shown in the image below. This finding is most commonly observed with which of the following drugs?



- a) Timolol

- b) Apraclonidine
- c) Netarsudil
- d) Brimonidine

Question 19:

A patient on treatment for glaucoma presented with complaints of floaters and flashing lights. He complains of seeing a dark grey curtain move across his field of vision. Which of the following can lead to this condition?

- a) Beta-blockers
- b) Prostaglandin analogs
- c) Cholinergic agonists
- d) Muscarinic antagonists

Question 20:

Your friend who is on treatment for glaucoma tells you that she feels her eyes look prettier than before. On examination, you notice increased iris pigmentation and eyelash hypertrichosis. Which of the following drugs is associated with this effect?

- a) Pilocarpine
- b) Acetazolamide
- c) Brimonidine
- d) Bimatoprost

Question 21:

What is the most probable diagnosis in a patient with a dilated pupil not responsive to 1% pilocarpine?

- a) Diabetic 3rd nerve palsy
- b) Adie's tonic pupil
- c) Uncal herniation
- d) Pharmacological block

Question 22:

Which of the following cannot be used to decrease intraocular pressure in glaucoma?

- a) Dexamethasone
- b) Acetazolamide
- c) Apraclonidine
- d) Mannitol

Answer Key

Question No.	Correct Option
1	a
2	b
3	b
4	c
5	c
6	a
7	d
8	c
9	c
10	b
11	a
12	d
13	b
14	a
15	d
16	d
17	b
18	c
19	c
20	d
21	d
22	a

Detailed Explanations

Solution to Question 1:

The class of anti-glaucoma drugs which act by increasing the uveo-scleral outflow are prostaglandin analogues.

Solution to Question 2:

Timolol is a non-selective beta-blocker. Beta-blockers reduce intra-ocular pressure primarily by reducing aqueous humor production. Another hypothesis states that by reducing ocular blood flow, there may be a decrease in ultrafiltration responsible for aqueous production.

Netarsudil is a rho-kinase inhibitor that acts by increasing trabecular outflow.

Brimonidine is an alpha-agonist that has a small effect on increasing both uveoscleral and trabecular outflow.

Latanoprost is a prostaglandin PGF₂-alpha analog that primarily acts by releasing matrix metalloproteinases, thereby increasing trabecular and uveoscleral outflow.

Solution to Question 3:

Dipivefrine is a prodrug of adrenaline (epinephrine).

Adrenaline has poor corneal penetration. Dipivefrine penetrates the cornea and is converted into adrenaline by the esterases.

Adrenaline lowers intraocular pressure by:

- Increasing trabecular outflow - beta-2 receptor-mediated.
- Increasing both trabecular and uveoscleral outflow, and decreasing aqueous production - alpha receptor-mediated.

Solution to Question 4:

The primary mechanism of action of topical carbonic anhydrase inhibitors in glaucoma is reducing the aqueous formation. Topical carbonic anhydrase inhibitors act by inhibiting type II isoenzyme of carbonic anhydrase in ciliary body epithelium. This leads to a reduction in fluid transport and decreases aqueous formation and IOP.

Solution to Question 5:

Netarsudil is a rho kinase inhibitor that acts by relaxing the trabecular meshwork. This increases the outflow of aqueous humor and reduces intraocular pressure.

Solution to Question 6:

The given clinical scenario is suggestive of acute bronchial asthma which was most likely precipitated by timolol.

Timolol is a non-selective beta-blocker that reduces intraocular pressure by decreasing aqueous formation. It precipitates bronchial asthma in susceptible individuals by beta-2 blockade.

Betaxolol is a beta-1 selective blocker that is used in glaucoma and has a lower propensity to precipitate bronchial asthma.

Latanoprost is a prostaglandin (PGF-2 alpha) analog and does not precipitate bronchial asthma.

Acetazolamide is a carbonic anhydrase inhibitor and does not precipitate bronchial asthma.

Solution to Question 7:

Prostaglandin analogs are the last drugs to be considered in this patient since macular edema is one of their side effects.

Adverse effects of PGF 2alpha analogs are:

- Blurring of vision
- Macular edema (especially in aphakic patients)
- Eyelash hypertrichosis
- Iris pigmentation (Heterochromia iridis)
- Peri-orbital fat depletion

Solution to Question 8:

The most appropriate first-line medical therapy for open-angle glaucoma is PGF-2 alpha analogs.

They act by increasing the aqueous outflow. They have largely replaced beta-blockers and can be considered as first-line agents for open-angle glaucoma because of their following properties:

- Once-daily dosing
- Low incidence of systemic side effects
- Potent IOP-lowering effect

When the use of PGF-2 alpha analogs or beta-blockers is contraindicated, topical carbonic anhydrase inhibitors or alpha-2 agonists are used.

Solution to Question 9:

Dinoprost is a derivative of natural PGF-2 alpha. It is indicated in the termination of pregnancy and post-partum hemorrhage and not for use in glaucoma.

PGF2 alpha analogs that are used topically for the treatment of glaucoma are:

- Latanoprost
- Bimatoprost
- Travoprost
- Tafluprost

Solution to Question 10:

The use of brimonidine is unsafe in infants. It is an alpha-2 agonist that can cross the blood-brain barrier. This may result in CNS depression and apnoea in neonates. Therefore it is contraindicated in children less than 2 years.

° 2 agonists, through the activation of G-protein coupled receptor activation, inhibit the activity of adenylate cyclase. This reduces cAMP and hence reduces aqueous humor production by ciliary action.

Solution to Question 11:

Acetazolamide and methazolamide are carbonic anhydrase inhibitors that can be used orally in the treatment of glaucoma.

Acetazolamide can also be administered intravenously.

Dorzolamide and brinzolamide are topical carbonic anhydrase inhibitors.

Dichlorphenamide is a carbonic anhydrase inhibitor approved for the treatment of familial periodic paralysis.

Solution to Question 12:

Intravenous mannitol is used for the treatment of acute attacks of glaucoma.

Uses of mannitol are:

- To reduce cerebral edema and intracranial pressure
- Enhancement of urinary excretion of salicylates, barbiturates, bromides, and lithium
- Diagnosis of bronchial hyperreactivity (inhalation)
- Antihemolytic urological irrigation during transurethral procedures
- Treatment of acute attacks of glaucoma.

Solution to Question 13:

Clozapine is an atypical antipsychotic having significant anticholinergic action. Drugs having significant anticholinergic action should be avoided or used with extreme caution in patients prone to developing acute angle-closure glaucoma.

The anticholinergic action results in the relaxation of the sphincter pupillae muscle. This results in the iris coming in contact with the lens and thereby obstructing the movement of the aqueous from the posterior to the anterior chamber.

This results in bulging of the iris and further leads to the obliteration of the narrow iridocorneal angle. This blocks the outflow of aqueous precipitating an acute attack of glaucoma.

Solution to Question 14:

Corneal hypoesthesia is a side effect of ocular beta-blockers.

For topical use in glaucoma, beta-blockers with no/minimal membrane-stabilizing action are used. This is the reason why propranolol is not formulated for ocular use. It has the highest membrane-stabilizing action and causes maximum corneal hypoesthesia if used.

Solution to Question 15:

The given scenario is suggestive of acute angle-closure glaucoma. Pilocarpine should not be administered immediately in this condition.

During an acute attack, the elevated pressure in the anterior chamber causes pressure-induced ischemic paralysis of the iris. Pilocarpine would be ineffective in this phase.

Once the IOP is reduced by other agents and reperfusion is achieved, pilocarpine can be used to overcome the pupillary block.

Pilocarpine is avoided in conditions like pseudo-exfoliation syndrome, phacomorphic glaucoma, and aqueous misdirection syndrome as it may exacerbate pupillary block in such cases.

Topical beta-blockers, alpha-2 agonists and prostaglandin analogs can be used to decrease the IOP initially at presentation. Oral or intravenous acetazolamide can also be used.

Solution to Question 16:

The patient has presented with lid retraction which is most likely caused by the use of apraclonidine. It is a congener of clonidine that has primary α_2 agonist action and a subsidiary α_1 agonist action.

Adverse effects like mydriasis and lid retraction are common with apraclonidine.

Brimonidine is also a congener of clonidine with more α_2 selectivity. Mydriasis and lid retraction are uncommon with this agent.

Timolol and latanoprost do not cause lid retraction.

Solution to Question 17:

The ophthalmological examination shows a pearly white opacity of the lens suggestive of cataract. Anticholinesterase agents like echothiophate and cholinergic agonists like pilocarpine are known to be associated with cataract formation.

Solution to Question 18:

The bilateral whorl-like pattern of cream-colored lines on the cornea seen in the image is called cornea verticillata or vortex keratopathy. It is observed with the use of netarsudil. It is usually mild and reversible and not associated with any significant visual abnormalities.

Netarsudil is approved for lowering intraocular pressure in patients with open-angle glaucoma or ocular hypertension.

Other causes of corneal verticillata are:

- Fabry's disease
- Amiodarone
- Chloroquine
- Hydroxychloroquine
- Indomethacin
- Phenothiazines.

Solution to Question 19:

The given scenario is suggestive of retinal detachment. It is associated with the use of miotics i.e., both cholinergic agonists and anticholinesterase agents.

Miotics produce ciliary body contraction which can lead to altered forces at the vitreous base resulting in retinal tears. Hence, they are to be used with extreme caution in patients who are at risk of developing retinal detachment.

Solution to Question 20:

Bimatoprost is a PGF-2 alpha analog. PGF-2 alpha analogs are associated with eyelash hypertrichosis and iris hyperpigmentation.

Bimatoprost is approved for topical application in the treatment of eyelash hypotrichosis.

Solution to Question 21:

The most appropriate answer would be a pharmacological block. Antimuscarinic will occupy the receptors and results in non-responsiveness to pilocarpine.

In diabetic third nerve palsy, the dilated pupil is due to the loss of innervation to constrictor pupillae. Muscarinic receptors on constrictor pupillae will be intact resulting in miosis by pilocarpine.

In Adie's tonic pupil, there is denervation hypersensitivity. Pupil responds to even lower concentrations of pilocarpine.

Uncal herniation causes pressure on the third nerve leading to ipsilateral mydriasis. But muscarinic receptors are intact and thus the pupil responds to pilocarpine.

Solution to Question 22:

Dexamethasone is a long-acting glucocorticoid and is not used to decrease the intraocular pressure in glaucoma.

Acetazolamide, apraclonidine, and mannitol are used in lowering intraocular pressure in glaucoma.

Anti-Anginal Drugs

Question 1:

A 50-year-old male was rushed to the ER with complaints of chest pain for the past 45 minutes and was started on isosorbide mononitrate. How does this drug act?

- a) By decreasing cAMP
- b) By increasing cAMP
- c) By decreasing cGMP
- d) By increasing cGMP

Question 2:

Which of the following drugs is not used in angina?

- a) Fasudil
- b) Sacubitril
- c) Atenolol
- d) Amlodipine

Question 3:

A patient with Prinzmetal angina is administered isosorbide mononitrate and is noticed to have symptomatic relief. What is the mechanism of action of nitrate in this case?

- a) Endothelium independent coronary vasodilation
- b) Reduced cardiac contractility
- c) Increased left ventricular end diastolic volume
- d) Decreased diastolic perfusion pressure

Question 4:

A 45-year-old patient who suffers from Prinzmetal angina requires a formulation of nitroglycerine which has the most rapid onset of action. Which of the following formulations of nitroglycerin can be administered to the patient?

- a) Sublingual spray
- b) Sublingual tablet
- c) Transdermal patch
- d) Topical ointment

Question 5:

In patients with angina, nitroglycerine should not be administered by which of the following routes?

- a) Oral
- b) Sublingual
- c) Intramuscular
- d) Intravenous

Question 6:

A 30-year-old man suspected to have cyanide poisoning is brought to the ER. The medical officer stabilizes the patient and asks the nurse to bring an ampoule of amyl nitrite. By which of the following routes is this drug administered?

- a) Oral
- b) Inhalational
- c) Intravenous
- d) Sublingual

Question 7:

An 18-year-old boy was brought to the ER in an unconscious state. His coworker states he works in a factory manufacturing pesticides. Based on this, the medical officer suspects cyanide poisoning and administers sodium nitrite. Which of the following is the mechanism of action of this drug in this case?

- a) Produces methemoglobinemia
- b) Produces sulfhemoglobinemia
- c) Produces carboxyhemoglobinemia
- d) Produces deoxyhemoglobinemia

Question 8:

A middle-aged man came to the casualty with complaints of abdominal pain. On examination, it is noted that the pain radiates from the loin to the groin of the patient. The resident wants to administer a drug to the patient for symptomatic relief. Which drug is not effective in this condition?

- a) Paracetamol
- b) Aspirin
- c) Morphine
- d) Nitrates

Question 9:

A patient who is on treatment for angina, presents to you with erectile dysfunction. You decide against prescribing sildenafil because the patient is being treated with _____

- a) Calcium channel blockers
- b) Beta blockers
- c) Nitrates
- d) ACE inhibitors

Question 10:

A 30-year-old male, who works in a factory that manufactures explosives, came to the OPD with a peculiar complaint. He says that he gets headaches and feels transient dizziness every Monday after his weekend off. Exposure to which compound is likely to produce his symptoms?

- a) Nitroglycerin
- b) Benzodiazepines
- c) Propranolol
- d) Verapamil

Question 11:

A 60 year old male is on verapamil for angina. Which of the following is not an adverse effect of this medication?

- a) Gingival hyperplasia
- b) Headache
- c) Constipation
- d) Tachycardia

Question 12:

Nifedipine acts by blocking _____ calcium channels.

- a) T- type
- b) R-type
- c) N-type
- d) L - type

Question 13:

Which among the following is a newer dihydropyridine with additional N-type calcium channel blocking activity?

- a) Amlodipine
- b) Nimodipine
- c) Clevidipine
- d) Cilnidipine

Question 14:

Which of the following is more likely to precipitate angina?

- a) Amlodipine
- b) Nifedipine
- c) Diltiazem
- d) Verapamil

Question 15:

A patient with polycystic kidney disease came to the ER with complaints of severe headache. An NCCT brain was taken and it showed subarachnoid hemorrhage. Which of the following

drugs is most appropriate for this patient?

- a) Nifedipine
- b) Amlodipine
- c) Nicardipine
- d) Nimodipine

Question 16:

Peripheral edema seen with the use of calcium channel blockers is primarily due to which of the following mechanisms?

- a) Reduced glomerular filtration rate and renal blood flow
- b) Increased hydrostatic pressure in the capillaries of lower extremities
- c) Decreased oncotic pressure due to loss of plasma proteins in urine
- d) Precapillary constriction and post capillary dilation

Question 17:

A 30-year-old athlete complained of chest pain during his practice sessions. He is diagnosed to be suffering from exertional angina and is prescribed atenolol. Which of the following changes are not caused by this drug?

- a) Decrease in cardiac contractility
- b) Decrease in heart rate
- c) Decrease in blood pressure
- d) Decrease in coronary collateral circulation

Question 18:

Which of the following drugs can be used to balance the undesirable effects of beta-blockers in angina?

- a) Verapamil
- b) Nitrates
- c) Ivabradine
- d) Diltiazem

Question 19:

Which of the following drugs is associated with coronary steal phenomenon?

- a) Dipyridamole
- b) Diltiazem
- c) Propranolol
- d) Verapamil

Question 20:

Which of the following statements about ranolazine is not true?

- a) It is recommended as first-line treatment for acute angina.
- b) Hypotension is uncommon.
- c) It improves glycemic control.
- d) It is metabolised by CYP3A4

Question 21:

Which of the following is a rho-kinase inhibitor?

- a) Fasudil
- b) Ranolazine
- c) Trimetazidine
- d) Nicorandil

Question 22:

Which of the following drugs causes activation of K⁺-ATP channels?

- a) Nicorandil
- b) Molsidomine
- c) Isosorbide dinitrate
- d) Nitroglycerin

Answer Key

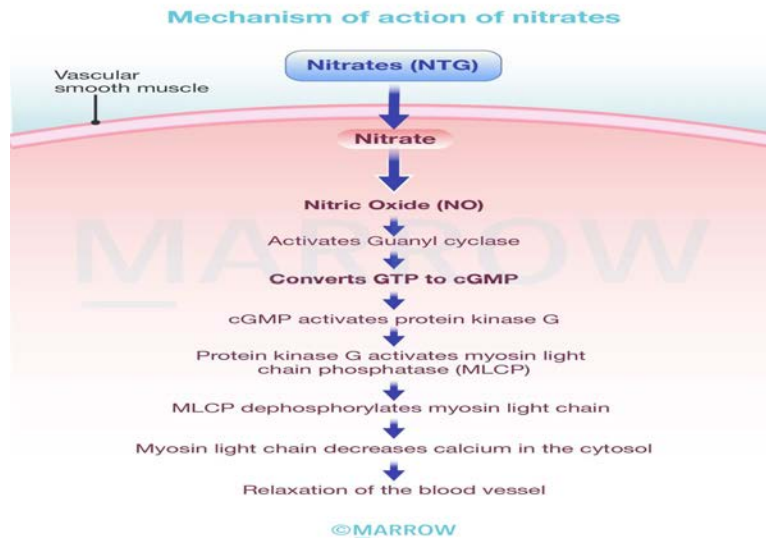
Question No.	Correct Option
1	d
2	b
3	a
4	a
5	c
6	b
7	a
8	d
9	c
10	a
11	d
12	d
13	d
14	b
15	d
16	b
17	d
18	b
19	a
20	a
21	a
22	a

Detailed Explanations

Solution to Question 1:

Nitrates like isosorbide mononitrate act by increasing the concentration of the second messenger cGMP.

They are used to relieve angina and cause vasodilation.



Solution to Question 2:

Sacubitril is not used in angina. Sacubitril is a neprilysin inhibitor, approved for use in the treatment of heart failure.

Solution to Question 3:

The mechanism of action of nitrates in Prinzmetal angina is endothelium-independent coronary vasodilatation.

Prinzmetal angina occurs due to vasospasm of the coronary arteries. Nitrates relax the smooth muscles of epicardial coronary arteries and relieve coronary artery spasm.

In the case of stable angina, the primary mechanism of action of nitrates is preferential dilatation of veins (by increasing cGMP). Venous dilatation leads to a decrease in the venous return to the heart (preload). This decreases the ventricular end-diastolic volume and thereby a decrease in myocardial oxygen consumption. The diastolic perfusion pressure (aortic blood pressure in diastole - left ventricular pressure in diastole) is increased.

Solution to Question 4:

The sublingual route of administration has a very rapid onset of action (1-3 mins). The onset of action is more rapid when administered as a spray when compared to a tablet.

The sublingual route is used for the termination of attacks. Oral extended-release capsule and transdermal patch are used for prophylaxis. The onset of action is approximately ≥ 60 min.

Topical nitroglycerine ointment can be used to provide symptomatic relief in patients with anal fissures.

Solution to Question 5:

Intramuscular nitroglycerin should be avoided due to severe pain in the injection area in patients with angina.

Sublingual and intravenous nitroglycerin can be used for acute effects. Oral extended-release capsules are available for prophylactic use in angina.

Solution to Question 6:

Amyl nitrite is administered via the inhalational route for acute relief in angina and cyanide poisoning.

Amyl nitrite is a highly volatile liquid available in glass ampoules. Its unpleasant odor and very short duration of action have made its use obsolete in angina.

Solution to Question 7:

The mechanism of action of sodium nitrite as an antidote for cyanide poisoning is by the development of methemoglobinemia.

Cyanide binds to the cytochrome oxidase (complex IV) of the electron transport chain (ETC), thereby blocking it. When sodium nitrite is administered, it releases the nitrite ion. This ion binds to hemoglobin to form methemoglobin. The methemoglobin so formed, has a high affinity for the cyanide ion and binds to it, thus, detaching it from the cytochrome oxidase. This regenerates active cytochrome normalizing the function of ETC.

The cyanide ion with methemoglobin forms cyanmethemoglobin. Sodium thiosulphate is administered for the elimination of this compound. The thiosulfate forms thiocyanate ion with the cyanide and this is readily excreted.

Solution to Question 8:

Abdominal pain radiating from loin to groin is suggestive of renal colic. Nitrates are not effective in renal colic.

In renal colic, NSAIDs, and morphine are used for acute pain relief. Due to the short time of action of nitrates, their clinical efficacy as relaxants in genitourinary system is quite low.

Therapeutic Uses of nitrates are:

- Angina pectoris

- Acute coronary syndrome
- Myocardial infarction
- CHF
- Biliary colic
- Esophageal spasm

Solution to Question 9:

The patient would be on nitrates for angina and thus should not be prescribed sildenafil as sildenafil causes dangerous potentiation of nitrate action resulting in severe hypotension.

The combination of the two drugs has a synergistic action on the intracellular cGMP levels that can result in uninhibited vasodilation and a precipitous fall in blood pressure. Hence, the use of sildenafil is a contraindication when on nitrates.

Solution to Question 10:

The patient is most likely suffering from Monday disease which is associated with exposure to nitroglycerin.

Continuous exposure to high levels of nitrates can occur in the chemical industry, especially where explosives are manufactured.

When contamination of the workplace with nitrate compounds is severe, workers find that upon starting their week, they suffer headaches and transient dizziness. After a day or so back at work, tolerance develops. Over the weekend, when exposure to the chemicals is reduced, tolerance disappears, so symptoms recur each Monday.

Solution to Question 11:

Tachycardia is not an adverse effect of verapamil.

Verapamil has a direct negative chronotropic effect on the heart. It is more likely to cause bradycardia because of its significant cardio depressant action. Any reflex stimulation of heart due to peripheral vasodilation is effectively countered by the direct effect.

Common side effects of verapamil include headache, gingival hyperplasia, and constipation.

Solution to Question 12:

Nifedipine acts by blocking L-type calcium channels.

Verapamil, diltiazem, and dihydropyridines block L-type calcium channels by binding to α_1 subunit. Verapamil was the first clinically used calcium channel blocker.

Solution to Question 13:

Cilnidipine is a novel dihydropyridine calcium channel blocker that blocks both L- and N-type calcium channels.

Blockage of N-type calcium channels results in the reduction of sympathetic activity. It provides additional anti-hypertensive action without inducing reflex tachycardia.

Solution to Question 14:

Amongst the given calcium channel blockers, nifedipine is more likely to precipitate angina.

Nifedipine is a dihydropyridine calcium channel blocker having a rapid onset and short duration of action. It results in an abrupt decrease in blood pressure resulting in reflex tachycardia. Increased contractility leads to an increase in O_2 demand which may precipitate angina.

Option A: Amlodipine and extended-release preparations of nifedipine do not cause a steep fall in blood pressure and subsequent activation of sympathetic tone. Therefore, they are less likely to precipitate angina.

Options C and D: Verapamil and diltiazem have relatively less action on peripheral vascular smooth muscles. They have a direct inhibitory effect on cardiac tissue causing negative inotropic effect (verapamil & diltiazem).

Solution to Question 15:

Nimodipine is the most appropriate drug that can be given to this patient.

Nimodipine is indicated in cerebral vasospasm following subarachnoid hemorrhage. It has a high affinity for cerebral blood vessels and appears to reduce mortality after subarachnoid hemorrhage.

Solution to Question 16:

Calcium channel blockers induced peripheral edema is most likely due to increased hydrostatic pressure in the capillaries of the lower extremities.

The increased hydrostatic pressure occurs due to precapillary dilation and reflex postcapillary constriction.

Solution to Question 17:

Atenolol is a beta-blocker which increases the coronary collateral circulation, thereby increasing the blood flow towards the ischaemic region. This provides relief in exertional angina.

Beta-blockers decrease the heart rate, contractility, and arterial blood pressure. These effects are beneficial by decreasing the myocardial O₂ demand. A decrease in heart rate increases the time available for myocardial perfusion.

Solution to Question 18:

Beta-blockers cause an increase in end-diastolic volume and ejection time. These effects can be balanced using nitrates by preferential venous dilatation, thus reducing preload.

Beta-blockers also block the baroreceptor mediated reflex tachycardia caused by nitrates due to vasodilation.

Options A and D: Verapamil and diltiazem are not combined with beta-blockers as they can cause additive suppression of SA node and AV nodal conduction.

Option C: Ivabradine is a pacemaker I(f) "funny" current inhibitor. Its addition does not reverse the undesirable effect but only adds to the bradycardia caused by beta blockers.

Solution to Question 19:

Coronary steal phenomenon is seen with dipyridamole.

In patients with partial obstruction of coronary vasculature, a region of ischemia develops in the myocardium. In the absence of drugs, arterioles in the ischemic areas are usually maximally dilated as a result of local compensatory mechanisms.

When dipyridamole is administered, it acts as a nonselective arteriolar dilator. It can only dilate the vessels of the normal myocardium as the vessels of the ischemic myocardium are already maximally dilated. This diverts (steals) the blood away from the ischemic myocardium towards the normal myocardium. This is termed as coronary steal phenomenon.

Solution to Question 20:

Ranolazine is not recommended as a first-line agent for acute angina. It is approved as a second-line agent for chronic angina.

Ranolazine acts by reducing late sodium current (I_{Na}) that facilitates calcium entry via the sodium-calcium exchanger. The reduction in intracellular calcium concentration results in decreased cardiac contractility. Additionally, it also reduces cardiac fatty acid oxidation and stimulates glucose metabolism. It results in a small decrease in HbA_{1c}, thus improving glycemic control.

It is metabolized by CYP3A4 and hence should not be used with CYP3A4 inhibitors such as macrolides, imidazole antibiotics, and HIV protease inhibitors.

The most frequent side effects are dizziness, headache, nausea, and constipation. Hypotension is uncommon.

Solution to Question 21:

Fasudil is a selective RhoA /rho-kinase inhibitor.

The Rho kinases (ROCK) are a family of enzymes that inhibit vascular relaxation. Their excessive activity has been noted in coronary spasm and pulmonary hypertension. Inhibition of these enzymes results in vascular relaxation.

Option B: Ranolazine acts by reducing late sodium current (INa). This reduces calcium entry via the sodium-calcium exchanger.

Option C: Trimetazidine is a partial fatty acid oxidation (pFOX) inhibitor.

Option D: Nicorandil is an agonist of KATP channels

Solution to Question 22:

Nicorandil causes activation of K⁺-ATP channels. This leads to the hyperpolarization of the smooth muscle cells leading to vasodilatation.

It also has nitrate like cGMP dependent action. Nicorandil dilates both arteries and veins leading to a decrease in afterload and preload of the heart.

Molsidomine, isosorbide dinitrate, and nitroglycerin are nitrates. They do not act on K-ATP channels.

Heart Failure Drugs

Question 1:

What is the half-life of digoxin?

- a) 24 hrs
- b) 40 hrs
- c) 72 hrs
- d) 60 hrs

Question 2:

Digitalis inhibits Na^+/K^+ ATPase pump in cardiac muscle cell membrane. How does this lead to its positive inotropic effect?

- a) By decreasing activity of $\text{Na}^+/\text{Ca}^{++}$ exchanger causing decreased influx of sodium and decreased efflux of Ca^{++}
- b) By decreasing efflux of Na^+ leading to less negative resting membrane potential and opening of voltage gated Ca^{++} channels on the T tubules
- c) By increasing intracellular Na^+ causing increased efflux of Na^+ and increased influx of Ca^{++} through $\text{Na}^+/\text{Ca}^{++}$ exchanger
- d) By increasing intracellular Na^+ decreasing the activity of Ca^{++} pump in the sarcoplasmic reticulum

Question 3:

The action of digoxin will not be affected in which of the following patients?

- a) 43-year-old man with chronic liver disease
- b) 30-year-old woman with electrolyte disturbances
- c) 82-year-old man with renal failure
- d) 60-year-old woman with myocardial infarction

Question 4:

Which of the following does not increase the risk of digoxin toxicity?

- a) Renal impairment
- b) Hyperkalemia
- c) Hypercalcemia
- d) Hypomagnesemia

Question 5:

A patient is to be started on digoxin therapy for heart failure. Despite already being on another drug, dose adjustment of digoxin is not necessary for this patient. Identify this drug.

- a) Amiodarone
- b) Quinidine
- c) Captopril
- d) Levosimendan

Question 6:

What is the most appropriate treatment for a patient with severe digoxin toxicity?

- a) Potassium supplements
- b) Propafenone
- c) Quinidine
- d) Fab fragments of digoxin antibodies

Question 7:

An ECG is done for a patient suspected to have digoxin toxicity. Which of the following would you not expect to see on the ECG?

- a) Ventricular bigeminy
- b) Paroxysmal atrial tachycardia with fast ventricular rate
- c) Regularisation of atrial fibrillation
- d) Bidirectional ventricular tachycardia

Question 8:

A patient with heart disease presents to the emergency with severe vomiting and blurry vision and mentions that he sees a greenish-yellow tinge over all objects. He then develops ventricular arrhythmia. What would be the most appropriate drug for immediate management?

- a) Procainamide
- b) Lignocaine
- c) Atropine
- d) Propranolol

Question 9:

A patient is diagnosed with congestive heart failure. Which of the following is not prescribed to prevent progression and revert cardiac remodelling?

- a) Beta blocker
- b) ACE inhibitor
- c) Digoxin
- d) Aldosterone antagonist

Question 10:

You are starting a treatment regimen containing an angiotensin-converting enzyme inhibitor for a patient with congestive heart failure. Which of the following is true regarding its role in the regimen?

- a) It is the drug of choice unless contraindicated
- b) Used as an alternative to diuretics
- c) It is a substitute for digitalis
- d) Used as adjuncts only in resistant cases

Question 11:

Which of the following drugs can overcome an increase in heart rate and renin release seen in patients with CHF?

- a) Minoxidil
- b) Metoprolol

- c) Metolazone
- d) Milrinone

Question 12:

You see the cardiologist prescribe a non-selective beta-blocker for a patient with congestive heart failure. He mentions that the drug has several beneficial effects. Identify the drug.

- a) Propranolol
- b) Pindolol
- c) Metoprolol
- d) Carvedilol

Question 13:

β blocker therapy is to be started on a patient with congestive heart failure. Which of the following statements is false about the therapy?

- a) The drugs should be started at optimum doses
- b) The dose should be gradually increased over weeks
- c) Special precautions should be taken in cases of NYHA class III and IV
- d) Carvedilol and metoprolol are the preferred drugs

Question 14:

You are going through the treatment chart of a patient with chronic congestive heart failure. Which of the following drugs will you not expect to see?

- a) Ivabradine
- b) Digoxin
- c) Spironolactone
- d) Adrenaline

Question 15:

A 70-year-old man is brought with complaints of severe chest pain. The ECG shows anterolateral wall MI and echocardiography shows an ejection fraction of 38%. Which of the

following is the drug of choice for his condition?

- a) Dopamine
- b) Dobutamine
- c) Epinephrine
- d) Norpinephrine

Question 16:

Which of the following is false about nesiritide?

- a) It is a BNP analogue
- b) It can be used in decompensated CHF
- c) It can be administered orally
- d) It causes loss of Na⁺ in the urine

Question 17:

Which enzyme degrades BNP?

- a) Neutral endopeptidase
- b) Elastase
- c) Pseudocholinesterase
- d) Carboxypeptidase

Question 18:

Isosorbide dinitrate is indicated for congestive heart failure in combination with which of the following vasodilators?

- a) Minoxidil
- b) Hydralazine
- c) Nimodipine
- d) Nitrendipine

Question 19:

A patient with congestive heart failure develops hyponatremia. Which oral vasopressin receptor antagonist will you prescribe for him?

- a) Tolvaptan
- b) Conivaptan
- c) Both a and b
- d) Sacubitril

Question 20:

What is the mechanism of action of milrinone?

- a) Stimulation of PDE3
- b) Inhibition of PDE3
- c) Inhibition of PDE5
- d) Stimulation of PDE5

Question 21:

A trial is being conducted to determine the efficacy of a myofilament calcium sensitizer in patients with heart failure. Which among the following is the drug likely being studied?

- a) Levosimendan
- b) Omapatrilat
- c) Sacubitril
- d) Enoximone

Question 22:

Sacubitril is a new drug approved for chronic heart failure. The mechanism of action of sacubitril is:

- a) Angiotensin II inhibitor
- b) ACE inhibitor
- c) Renin inhibitor
- d) Neutral endopeptidase inhibitor

Question 23:

Which of the following is not used in heart failure?

- a) Trimetazidine
- b) Sacubitril
- c) Metoprolol
- d) Nesiritide

Answer Key

Question No.	Correct Option
1	b
2	a
3	a
4	b
5	d
6	d
7	b
8	b
9	c
10	a
11	b
12	d
13	a
14	d
15	b
16	c
17	a
18	b
19	a
20	b
21	a
22	d
23	a

Detailed Explanations

Solution to Question 1:

The plasma half-life of digoxin is around 40 hours (36-40 hours).

	Digitoxin	Digoxin
Oral absorption	Very good (90-100 %)	Good (60-80%)
Plasma protein binding	95%(t _{1/2} /25-7 days)	25%(t _{1/2} /240 hrs)
Elimination	Hepatic metabolism	Renal excretion (by glomerular filtration)
Dose adjustment in	Liver failure	Kidney failure

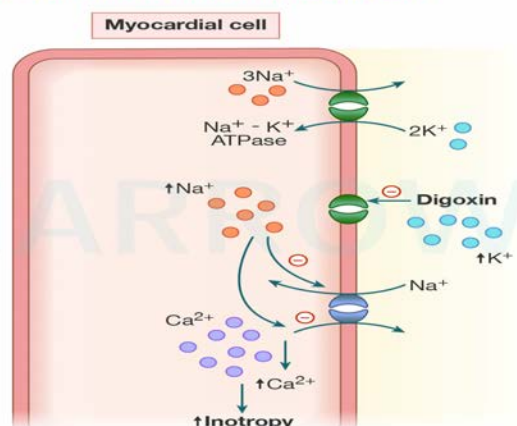
Therapeutic levels of digoxin in the range of 0.5-0.8 ng/ml are beneficial. An increase in levels >1.2 ng/mL is associated with increased mortality.

Solution to Question 2:

The positive inotropic effect of digitalis is due to decreased activity of Na⁺/Ca²⁺ exchanger causing a decreased influx of sodium and decreased efflux of Ca²⁺.

Digitalis selectively binds and inhibits the Na⁺/ K⁺ATPase pump of myocardial fibers which normally pushes Na⁺ outside and brings K⁺ inside. When this pump is inhibited, there is an intracellular accumulation of Na⁺ ions. This results in a relative decrease in the expulsion of Ca²⁺ ions via the Na⁺-Ca²⁺ exchanger (NCX) leading to the accumulation of calcium inside the cells resulting in increased contractility.

Mechanism of action of digoxin



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Solution to Question 3:

The action of digoxin is not affected in chronic liver disease.

Option B: Electrolyte disturbances like hypokalemia, hypomagnesemia, and hypercalcemia increase digitalis toxicity.

Option C: Digoxin is excreted unchanged primarily by kidneys by glomerular filtration. Therefore, dose adjustment of digoxin has to be done in renal failure.

Option D: Chances of arrhythmias increases manifold in post-MI duration. Hence, it is generally avoided post-MI as digitalis itself causes arrhythmias.

	Digitoxin	Digoxin
Oral absorption	Very good (90-100 %)	Good (60-80%) i/v also
Plasma protein binding	95%	25%
Elimination	Hepatic metabolism	Renal excretion (by glomerular filtration)
Dose adjustment in	Liver failure	Kidney failure

Solution to Question 4:

Digoxin toxicity is not increased by hyperkalemia.

In hyperkalemia, the ATPase alpha subunit is dephosphorylated, which is the binding site of digoxin. This reduces its binding and effect. Therefore, hyperkalemia does not increase digoxin toxicity.

Factors predisposing digoxin toxicity:

- Hypokalemia (more digoxin binds to the Na⁺/K⁺ ATPase)
- Hypercalcemia (increases the risk of arrhythmia)
- Hypomagnesemia (Mg⁺⁺ is a co-factor for Na⁺/K⁺ ATPase)
- Renal failure (for digoxin)

Solution to Question 5:

Levosimendan does not increase digitalis toxicity and hence would not require dose adjustment of digoxin.

It is a combined Ca²⁺ channel sensitizer (troponin C binding) and PDE3 inhibitor.

Digoxin is metabolized by P-glycoprotein and drugs that inhibit it increase the plasma levels of digoxin and may result in toxicity. The drugs that increase digoxin toxicity are:

- Amiodarone
- Quinidine
- Propafenone
- Verapamil
- Captopril
- Erythromycin
- Gentamycin
- Ranolazine

Solution to Question 6:

The most appropriate treatment for severe digoxin toxicity is purified Fab fragments of digitalis antibodies (Digibind).

It neutralizes the digoxin action and its dosage is based upon the total dose of digoxin ingested.

Option A: KCl infusion is indicated in mild toxicity as hyperkalemia decreases the binding of digoxin by dephosphorylating Na/K ATPase pump. In acute severe poisoning, potassium is not indicated as it may already be high in plasma. Potassium is also not indicated when a higher degree of AV block is present.

Options B and C: Quinidine and propafenone are not indicated in arrhythmias caused by digoxin.

Solution to Question 7:

Digoxin can cause either paroxysmal or non-paroxysmal atrial tachycardia with variable A-V block but not a fast ventricular rate.

Regularisation of atrial fibrillation with complete A-V block and junctional or ventricular escape rhythm can occur because of the blockade of A-V conduction. Due to the blockade of AV node by digoxin, it is contraindicated in WPW syndrome.

Common arrhythmias associated with digoxin toxicity:

- Ventricular bigeminy
- Atrioventricular junctional rhythm
- Bidirectional ventricular tachycardia
- Premature ventricular depolarizations.

Solution to Question 8:

The given clinical scenario is suggestive of digoxin toxicity. Xanthopsia i.e., yellowing of vision is characteristically seen in this condition. Lignocaine (and phenytoin) has the least effect on AV nodal conduction and is the most appropriate drug for the treatment of digoxin-induced arrhythmias.

Option A: Procainamide and other class IA anti-arrhythmic agents are not indicated in digoxin toxicity because of their inhibitory effects on AV nodal conduction.

Option C: Atropine is used for atrial bradyarrhythmias associated with digoxin toxicity.

Option D: Propranolol is used for atrial tachyarrhythmias associated with digoxin toxicity.

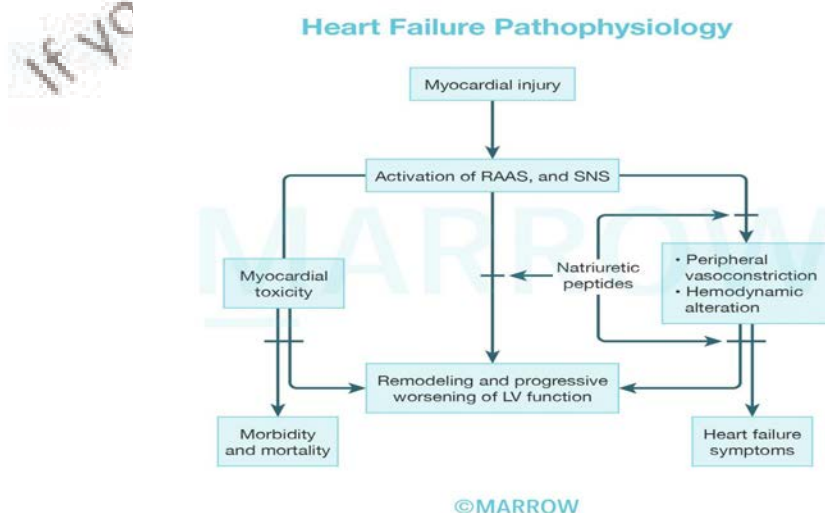
Solution to Question 9:

Digoxin has no role in preventing cardiac remodelling. It increases cardiac contractility and is indicated for symptomatic management of congestive heart failure.

Drugs preventing ventricular remodelling

- ACE inhibitors/ ARBS
- Beta-blockers
- Aldosterone antagonists (spironolactone, eplerenone)

The activation of RAAS and the sympathetic nervous system leads to the remodelling and progressive LV dysfunction. The RAAS activation is prevented by the ACE inhibitors/ARBs and the aldosterone antagonists (spironolactone), while the overactive sympathetic system is inhibited by the beta-blockers.



Solution to Question 10:

ACE inhibitors and ARBs are the first choice drugs in congestive heart failure (CHF) unless contraindicated.

Renin-angiotensin-aldosterone system (RAAS) activation is pivotal in the development of the symptoms and disease progression in CHF.

ACE inhibitors and ARBs afford symptomatic as well as disease-modifying benefits in CHF by causing vasodilation, retarding/preventing ventricular hypertrophy, myocardial cell apoptosis and fibrosis.

Solution to Question 11:

Metoprolol is a selective β_1 blocker. Beta blockade reduces the sympathetic stimulation of the SA node and thus reduces the heart rate. In the same way, β_1 receptors in the JG apparatus are also inhibited by metoprolol, and thus, renin release is reduced.

Option A: Minoxidil is a K^+ channel opener that was earlier used in hypertensive emergencies for its vasodilatory action on blood vessels. This has no effect on renin release and heart rate can increase due to reflex sympathetic stimulation.

Option C: Metolazone is a thiazide diuretic that can be used even in severe renal failure.

Option D: Milrinone is an inodilator. It increases cardiac contractility and causes vasodilation by inhibiting the enzyme PDE_3 .

Solution to Question 12:

The drug most likely being discussed is carvedilol which is a non-selective beta-blocker approved for use in mild to severe CHF.

Additional actions of carvedilol:

- Blockade of alpha receptors
- Blockade of L-type calcium channels
- Antioxidant effect
- Antiproliferative effect

Options A and B: Propranolol and pindolol are non-selective beta-blockers but they are not approved for use in CHF.

Option C: Metoprolol is approved for use in CHF but it is a cardioselective beta-blocker.

Solution to Question 13:

Beta-blockers are not started with optimum doses in patients with heart failure.

Beta-blocker therapy in CHF requires caution, proper patient selection, and observance of several guidelines such as:

- Generally, the starting dose is less than 1/8th (low dose) of the target dose is and then titrated upwards slowly over weeks (doubling every 4 weeks)
- The greatest utility of beta-blocker therapy has been seen in mild to moderate CHF (NYHA I, II)
- There is no place for beta-blockers in decompensated heart disease
- Carvedilol, bisoprolol, and metoprolol are commonly used beta-blockers in CHF.

Solution to Question 14:

Adrenaline is used for the treatment of acute decompensated heart failure, not chronic congestive heart failure.

Drugs used in chronic congestive heart failure include :

- Diuretics
- ACE inhibitors
- Angiotensin receptor blockers
- Angiotensin receptor-neprilysin inhibitor - sacubitril-valsartan
- Hydralazine
- Nitrates
- Digoxin
- Mineralocorticoid receptor antagonists
- Ivabradine - sinus node inhibitor

Solution to Question 15:

The given clinical scenario is suggestive of acute congestive heart failure with systolic dysfunction. Dobutamine is administered as it is the beta-adrenergic agonist of choice for managing this condition.

Dobutamine has a well-balanced action with stimulation of cardiac output. It causes less tachycardia than epinephrine and also has an added advantage of a decrease in pulmonary artery wedge pressure.

At doses that result in a positive inotropic effect, the beta-1 action on the myocardium predominates. In the vasculature, the alpha-1 effects are counterbalanced by beta-2 effects. The net result is an increase in stroke volume, augmented by a small decrease in systemic vascular resistance, thereby resulting in a decrease in afterload.

Solution to Question 16:

Nesiritide is available for intravenous administration only as it is a peptide and is digested by peptidases in the GIT.

Nesiritide is a recombinant brain natriuretic peptide (BNP). It causes natriuresis and vasodilation. It has a half life of 18 minutes. It is approved to relieve dyspnoea and other symptoms in acute refractory decompensated CHF.

Solution to Question 17:

Brain natriuretic peptide (BNP) is degraded by neutral endopeptidase.

Natriuretic peptides:

- Atrial natriuretic peptide (ANP)
- Brain natriuretic peptide
- C-type natriuretic peptide

Nesiritide is a recombinant BNP approved to relieve dyspnoea and other symptoms in acute refractory decompensated CHF.

Solution to Question 18:

Isosorbide dinitrate (ISDN) is indicated in combination with hydralazine for CHF.

Hydralazine is a direct vasodilator and its mechanism of action is not known. It also reduces nitrate tolerance.

The combination of ISDN and hydralazine showed mortality benefits in a specific group of population (African-Americans).

Solution to Question 19:

Tolvaptan is an oral vasopressin receptor antagonist.

It preferentially binds to V₂ receptors over V_{1a} receptors. It has been used in decompensated CHF patients to restore normal sodium levels.

Conivaptan is also a vasopressin receptor antagonist that is administered intravenously and binds to both V₂ and V_{1a} receptors with similar affinity. It is more commonly used for hyponatremia rather than CHF. Oral conivaptan is not yet available.

Solution to Question 20:

Milrinone acts by inhibition of the enzyme phosphodiesterase-3 (PDE 3).

The PDE-3 inhibitors decrease cellular cAMP degradation, resulting in increased levels of cAMP. This results in positive inotropic and chronotropic action of the heart. It also dilates veins and arteries thereby decreasing the preload and afterload.

Solution to Question 21:

Levosimendan is a myofilament calcium sensitizer.

Calcium sensitizers bind and induce a conformational change in thin-filament regulatory protein troponin C. Thus increasing the sensitivity of contractile myofilaments to calcium. The result is an increased force of contraction for a given cytosolic calcium concentration.

Option B: Omapatrilat is a dual ACE and neprilysin inhibitor that failed clinical trials due to increased risk of angioedema.

Option C: Sacubitril is a neprilysin inhibitor approved in combination with valsartan for CHF.

Option D: Enoximone is a PDE-3 inhibitor.

Solution to Question 22:

Sacubitril is a neutral endopeptidase (neprilysin) inhibitor.

Sacubitril is a prodrug that is activated to sacubitrilat. It can inhibit the enzyme neprilysin (neutral endopeptidase), which is responsible for the degradation of atrial and brain natriuretic peptides (reduction of blood volume). Therefore, it increases the levels of atrial and brain natriuretic peptides and lowers blood pressure.

In 2015, the FDA approved the use of sacubitril in combination with valsartan, at a ratio of 1:1, for the treatment of chronic heart failure.

Solution to Question 23:

Trimetazidine is not used in heart failure.

Trimetazidine is a newer antianginal drug that is a pFOX inhibitor which partially inhibits the fatty acid oxidation pathway in the myocardium. This decreases the oxygen requirement of the heart.

Nesiritide (natriuretic peptide), sacubitril (neprilysin inhibitor) and metoprolol (beta-blocker) are used in the treatment of heart failure.

Antihypertensive Drugs

Question 1:

Which of the following is the most appropriate first-line medication for an elderly patient with hypertension?

- a) Hydrochlorothiazide
- b) Furosemide
- c) Chlorthalidone
- d) Spironolactone

Question 2:

Chlorthalidone is being prescribed to an elderly patient for the management of hypertension. Which of the following side effects would not be expected in this patient?

- a) Hyperuricemia
- b) Hypercalcemia
- c) Hyperkalemia
- d) Hyperglycemia

Question 3:

A patient on lithium therapy developed hypertension. He was started on thiazides for hypertension. After few days, he developed coarse tremors and other symptoms suggestive of lithium toxicity. Explain the likely mechanism of this interaction.

- a) Thiazide inhibits the metabolism of lithium
- b) Thiazides act as an add on drug to lithium
- c) Thiazides increase the tubular reabsorption of lithium
- d) Thiazides cause water loss thereby increase lithium levels

Question 4:

A hypertensive patient on losartan requires an add on drug to achieve his blood pressure goal. Which of the following drugs will not increase the risk of hyperkalemia when added to his current medication?

- a) Atenolol
- b) Amiloride
- c) Indapamide
- d) Aliskiren

Question 5:

Long term reduction of blood pressure by thiazide diuretics is mainly by which of the following mechanisms?

- a) Reduction in extracellular volume
- b) Reduction in sympathetic activity
- c) Reduction in peripheral vascular resistance
- d) Increase in extracellular volume

Question 6:

A diabetic patient is diagnosed with hypertension during his annual health check-up. What is the treatment of choice for him?

- a) Chlorthalidone
- b) Carvedilol
- c) Captopril
- d) Nifedipine

Question 7:

A patient is diagnosed with hypertension during routine health evaluation. Which of the following is not recommended as first-line therapy for him?

- a) Verapamil
- b) Doxazosin
- c) Metolazone
- d) Quinapril

Question 8:

Which class of antihypertensive drugs has a favourable effect on lipid profile?

- a) Thiazide diuretics
- b) Beta blockers
- c) Alpha 1 selective blockers
- d) Calcium channel blockers

Question 9:

Methyldopa is primarily used for which of the following conditions?

- a) Hypertension in pregnancy
- b) Renovascular hypertension
- c) Hypertension with gout
- d) Refractory hypertension

Question 10:

A pregnant woman with hypertension, on treatment with methyldopa, comes for routine evaluation. She is found to have a positive Coombs test but is asymptomatic. What is the next course of action for this patient?

- a) Immediately stop methyldopa
- b) Taper the dose of methyldopa
- c) Monitor for signs of hemolysis
- d) Switch to labetalol

Question 11:

Which of the following is not true about clonidine?

- a) Reduction in central sympathetic flow
- b) Prazosin completely antagonizes its action
- c) Sedation and xerostomia are common side effects

- d) It can be combined with vasodilators

Question 12:

A patient presents with complaints of multiple episodes of intermittent palpitations, throbbing headache, and sweating. Blood pressure measured during one of these episodes is 210/120 mm Hg. The results of 24-hour urinary metanephrines were inconclusive. Which of the following could be used for diagnosis?

- a) Phentolamine
- b) Phenoxybenzamine
- c) Clonidine
- d) Prazosin

Question 13:

Which of the following is a selective imidazoline receptor agonist?

- a) Clonidine
- b) Methyldopa
- c) Moxonidine
- d) Brimonidine

Question 14:

Which of the following antihypertensives are not used in pregnancy?

- a) Alpha methyldopa
- b) Hydralazine
- c) Propranolol
- d) Labetalol

Question 15:

Which of the following drugs reduces blood pressure primarily by directly decreasing the heart rate?

- a) Propranolol

- b) Prazosin
- c) Alpha methyl dopa
- d) Nitroprusside sodium

Question 16:

The blood pressure of a patient undergoing a coronary artery bypass surgery suddenly rises to 170/110mm Hg. Which of the following can you administer intravenously for immediate management?

- a) Nifedipine
- b) Amlodipine
- c) Clevidipine
- d) Felodipine

Question 17:

An elderly hypertensive patient is diagnosed with bilateral renal artery stenosis. He is also on treatment for diabetes mellitus. Which among the following drugs would be the most appropriate for him?

- a) Enalapril
- b) Verapamil
- c) Metoprolol
- d) Metolazone

Question 18:

Which of the following is a direct renin inhibitor?

- a) Captopril
- b) Sacubitril
- c) Aliskiren
- d) Losartan

Question 19:

Which of the following statements is incorrect about diazoxide?

- a) It acts by causing prolonged opening of ATP dependent K^+ channels in beta cells
- b) It can cause severe hypoglycaemia
- c) It can be used to treat patients with insulinoma
- d) It is used as an antihypertensive agent

Question 20:

A pregnant woman with preeclampsia and is started on hydralazine. Which of the following statements is incorrect about the drug?

- a) It decreases cerebral vascular resistance
- b) It causes equal dilatation of both arteries and veins
- c) Postural hypotension is not a common problem
- d) It increases plasma renin activity

Question 21:

The efficacy of an antihypertensive agent is being studied in a clinical trial. During the course of the trial, the researchers observed hair regrowth and generalized hypertrichosis in balding patients. What is the mechanism of action of the drug being studied?

- a) Closes K^+ channels in vascular smooth muscle
- b) Activates K^+ channels in vascular smooth muscle
- c) Activates Ca^{2+} channels in vascular smooth muscle
- d) Closes Ca^{2+} channels in vascular smooth muscle

Question 22:

A patient presented to the ER with severe headache and vomiting. On examination, the blood pressure was found to be 220/130 mm Hg and papilledema was also seen. What is the mechanism of the drug of choice for the condition?

- a) Increased cAMP
- b) Calcium channel blockage
- c) Stimulation of guanylyl cyclase
- d) K^+ channel opener

Question 23:

A patient with malignant hypertension is being managed with sodium nitroprusside infusion. He is at risk of developing all of the following adverse effects except?

- a) Methhemoglobinemia
- b) Cyanide toxicity
- c) Drug-induced lupus syndrome
- d) Thiocyanate toxicity

Question 24:

A patient with gout develops hypertension. The uric acid level was found to be elevated. Which of the following is not administered?

- a) Hydrochlorothiazide
- b) Atenolol
- c) Enalapril
- d) Prazosin

Question 25:

You will counsel patients regarding first-dose phenomenon while prescribing which of the following drugs?

- a) Propranolol
- b) Prazosin
- c) Metoprolol
- d) Hydralazine

Question 26:

Which of the following drugs can be given safely in a patient with chronic renal failure?

- a) Aluminium hydroxide
- b) Candesartan
- c) Telmisartan

d) Sucralfate

Answer Key

Question No.	Correct Option
1	c
2	c
3	c
4	c
5	c
6	c
7	b
8	c
9	a
10	c
11	b
12	c
13	c
14	c
15	a
16	c
17	b
18	c
19	b
20	b
21	b
22	b
23	c
24	a
25	b
26	c

Detailed Explanations

Solution to Question 1:

The most appropriate first-line antihypertensive among the given drugs is chlorthalidone.

Thiazide diuretics are recommended as one of the first-line antihypertensive drugs for elderly patients without any comorbidities. Among them, guidelines recommend chlorthalidone over hydrochlorothiazide because of its longer duration of action.

First-choice drugs used in the treatment of hypertension in patients without any comorbidities include:

- ACE inhibitors or ARBs: No evidence of superiority of one class over the other. However, ACE inhibitors have a higher risk of side effects like dry cough and angioedema.

-

Calcium channel blockers: Extended-release, long-acting dihydropyridines are preferred.

-

Thiazide diuretics: Chlorthalidone is usually preferred to hydrochlorothiazide.

Loop diuretics like furosemide and torsemide are less effective antihypertensives than thiazides. They are not indicated for the routine management of hypertension.

In the presence of any associated conditions or comorbidities, the first choice drugs for the treatment of hypertension would be as follows:

Comorbidities	First choice drugs for hypertension
DiabetesChronic kidney disease with proteinuria	ACE inhibitors or ARBs
Angina	Beta-blockers or calcium channel blockers
Heart failure	ACE inhibitors and alpha+beta blockers (e.g., carvedilol)
Benign prostatic hyperplasia	° 1 blockers
Pregnancy	Labetalol
Atrial fibrillation	Diltiazem or verapamil

Solution to Question 2:

Thiazide diuretics cause hypokalemia.

Other side effects of thiazide diuretics include hyperuricemia, hypercalcemia, hyperglycemia and erectile dysfunction. All thiazide diuretics cross the placenta and therefore should be avoided in pregnancy.

Solution to Question 3:

Thiazides act by increasing the tubular reabsorption of lithium.

Thiazide use is accompanied by sodium loss which is compensated in a few days by increased reabsorption of Na^+ in the proximal tubule.

The kidney handles both lithium and Na^+ in the same way. Thus, along with Na^+ reabsorption, there will be an increase in the reabsorption of lithium also, resulting in decreased excretion of lithium.

Solution to Question 4:

Indapamide is a thiazide diuretic that increases K^+ excretion—decreases the risk of hyperkalemia associated with ACE inhibitors or ARBs such as losartan.

Option A: Beta-blockers can contribute to K^+ retention by decreasing renin release and further reducing aldosterone secretion. Non-selective beta-blockers may also interfere with the cellular uptake of K^+ ions.

Option B: K^+ sparing diuretics like amiloride cause a decrease in K^+ excretion and increase the risk of hyperkalemia.

Option D: Aliskiren is a direct renin inhibitor and increases the risk of hyperkalemia when combined with losartan.

Solution to Question 5:

Long term reduction of blood pressure by thiazide diuretics is mainly due to a reduction in peripheral vascular resistance.

Thiazide diuretics initially decrease extracellular volume by their action on NaCl transporter in distal convoluted tubules in the nephrons. However, the compensatory response from activation of RAAS raises the extracellular volume returns almost to pre-treatment levels after a few weeks.

Solution to Question 6:

ACE inhibitors such as captopril or ARBs can be considered as a treatment of choice in hypertensive patients with diabetes because of the renoprotective action. It benefits diabetic patients by reducing the risk of occurrence and progression of diabetic nephropathy.

They are also recommended in patients with CKD irrespective of diabetes status as they can delay the progression of nephropathy.

Option A: Thiazide diuretics worsen glycemic control and are not preferred in diabetic patients.

Option B: Beta-blockers, especially non-selective blockers, worsen glycemic control.

Option D: Calcium channel blockers have a neutral effect on glycemic control and can be used in diabetes if ACE inhibitors or ARBs cannot be given.

Solution to Question 7:

Alpha blockers like doxazosin are not recommended as first-line therapy for uncomplicated essential hypertension.

As per the European Society of Cardiology 2018 guidelines for hypertension, the classes of drugs recommended as first line therapy are:

- ACE inhibitors
- Angiotensin receptor blockers
- Calcium channel blockers
- Thiazide diuretics

Solution to Question 8:

Alpha 1 selective blockers have a favourable effect on lipid profile. They decrease plasma concentrations of triglycerides, total LDL-cholesterol and increase HDL cholesterol.

Option A: Thiazides increase plasma concentrations of triglycerides, LDL-cholesterol and total cholesterol. HDL may decrease or remain unchanged.

Option B: Non-selective beta-blockers increase triglycerides and decrease HDL levels. However, total cholesterol remains unchanged.

Option D: Calcium channel blockers have a neutral effect on the lipid profile.

Solution to Question 9:

Methyldopa is preferred for chronic hypertension in pregnancy based on its safety and efficacy data for both mother and fetus.

Antihypertensives found safe in pregnancy:

- Hydralazine
- Methyldopa
- Dihydropyridine CCBs like nifedipine, amlodipine - however, they should be stopped before labor as they weaken the uterine contractions
- Cardioselective beta-blockers and those with intrinsic sympathomimetic activity

Antihypertensives to be avoided in pregnancy:

- ACE inhibitors, Angiotensin II receptor blockers (ARB)

- Diuretics
- Nonselective beta-blockers
- Nitroprusside

Labetalol is considered the first-line agent for hypertensive emergency in pregnancy.

Solution to Question 10:

The best course of action for the patient is close monitoring for any signs or symptoms of hemolysis.

Coombs test is done to test the presence of antibodies against Rh antigens on the red blood cells, which result in hemolysis. Positive Coombs test is seen in 20% of patients who receive methyldopa.

However, only 1-5% of such patients develop hemolytic anemia and hence, a positive Coombs test is not a necessary indication for stopping methyldopa treatment. They must be monitored closely and any signs that indicate hemolytic anemia needs prompt discontinuation of the drug.

Solution to Question 11:

Prazosin does not antagonize the effect of clonidine. Prazosin is a selective α_1 receptor antagonist whereas clonidine is an α_2 receptor partial agonist.

On oral administration, clonidine stimulates α_{2A} receptors in CNS and results in decreased central sympathetic flow. This causes a decrease in BP.

However, on IV administration postsynaptic α_{2B} receptors are activated at higher doses which cause a transient rise in blood pressure. It also causes significant reductions in HDL lipids, apolipoproteins AI and AII.

It can be combined with vasodilators but has to be used with caution as potentiation of each other's actions can occur.

Solution to Question 12:

The given clinical vignette is suggestive of pheochromocytoma. When first-line investigations like measurement of 24-hour urinary metanephrines are inconclusive, clonidine suppression test is used in the diagnosis of pheochromocytoma.

An oral dose of 0.3 mg clonidine is given, followed by the measurement of norepinephrine (NE) concentration at 3 hours. A concentration of >500 pg/mL of NE indicates a positive test.

Option A: Phentolamine test is not recommended due to its low sensitivity and specificity.

Options B and D: Phenoxybenzamine and prazosin are used in the treatment of pheochromocytoma but have no role in its diagnosis.

Solution to Question 13:

Moxonidine is a selective imidazoline (I₁) receptor agonist.

It is indicated as a second line agent in the long-term management of hypertension. It also has weak alpha 2 agonist action.

Rilmenidine is another selective imidazoline receptor agonist.

Option A: Clonidine is an alpha 2 receptor partial agonist and has additional imidazoline (I₁) receptor agonist action. But the main mode of action is through alpha 2 agonism.

Option B: Methyldopa does not have any action on imidazoline receptors

Option D: Brimonidine is a congener of clonidine.

Solution to Question 14:

Propranolol is a non-selective beta-blocker and is not safe for use during pregnancy.

Antihypertensives to be avoided in pregnancy:

- ACE inhibitors, ARBS
- Diuretics
- Nonselective beta-blockers
- Nitroprusside

Antihypertensives found to be safe in pregnancy:

- Hydralazine
- Methyldopa
- Dihydropyridine CCBs: should be stopped before labor as they weaken the uterine contractions
- Cardioselective beta-blockers and those with intrinsic sympathomimetic activity: may be used as a last choice

Solution to Question 15:

Propranolol reduces blood pressure by directly decreasing the heart rate. It is a nonselective beta-blocker and blocks the beta 1 receptor of the heart (SA node) thus reducing the heart rate and blood pressure.

It also reduces renin secretion by blocking beta 1 receptors in JG cells. Reduction in renin release also contributes to blood pressure reduction.

Option B: Prazosin is a selective alpha 1 blocker and dilates both arteries and veins and may cause reflex tachycardia.

Option C: Alpha methyldopa is a selective alpha 2 receptor agonist. Alpha 2 receptors are autoreceptors present on the presynaptic membrane. On stimulation, central sympathetic discharge is reduced. Alpha methyldopa can indirectly reduce heart rate.

Option D: Nitroprusside is a vasodilator and may cause reflex tachycardia.

Solution to Question 16:

Clevidipine is a dihydropyridine available for intravenous administration and is indicated in the acute management of hypertension. It has a short half-life (1-2 minutes) and a rapid onset of action.

Option A: Nifedipine is a short-acting dihydropyridine that is no longer indicated in the management of hypertension. The rapid reduction in blood pressure and concurrent surges in sympathetic activity are not desirable.

Options B and D: Amlodipine and felodipine are longer-acting dihydropyridines that are available in tablet forms. They are used for long-term treatment of hypertension.

Solution to Question 17:

In the given scenario, verapamil would be the most appropriate drug for the management of hypertension in the patient as it has no adverse effect on glycemic control.

Option A: Efferent arteriolar constriction maintains the glomerular filtration pressure in bilateral renal artery stenosis. ACE inhibitors are contraindicated in bilateral renal stenosis as it can precipitate acute renal failure due to dilation of efferent arterioles and fall in glomerular filtration pressure.

Options C and D: Beta-blockers and thiazides can worsen glycemic control and are not suitable in the given scenario.

Solution to Question 18:

Aliskiren is a direct renin inhibitor. Though its efficacy as an antihypertensive is comparable to that of ACE inhibitors, it is not considered a first-line agent due to higher incidence of side effects.

Option A: Captopril is an angiotensin-converting enzyme inhibitor.

Option B: Sacubitril is a neprilysin inhibitor indicated in combination with valsartan in chronic heart failure patients.

Option D: Losartan is an angiotensin receptor blocker.

Solution to Question 19:

Diazoxide causes hyperglycemia, not hypoglycemia.

Diazoxide is a potent K⁺ channel opener and opens up the ATP-sensitive K⁺ channel and thus prevents depolarization of pancreatic beta cells and subsequent insulin release. It is used in some cases of insulinoma to reduce insulin secretion and prevent hypoglycemia.

It was earlier used as an intravenous agent for hypertensive emergencies due to its strong vasodilatory action.

Solution to Question 20:

Hydralazine causes preferential dilatation of arteries over veins.

Hydralazine acts directly on vascular smooth muscles. The precise mechanism is unknown but involves a decrease in intracellular Ca²⁺ concentrations.

Option A: It causes a selective decrease in vascular resistance in coronary, cerebral and renal circulation, with a smaller effect in skin and muscle.

Option C: Because of preferential dilatation of arterioles over veins, postural hypotension is not a common problem.

Option D: Hydralazine-induced vasodilation causes reflex activation of the sympathetic system through baroreceptor-mediated reflexes resulting in increased plasma renin activity.

Solution to Question 21:

The antihypertensive drug with the side effect of excess hair growth is minoxidil and it acts by activating ATP-sensitive K⁺ channels in smooth muscles. This causes hyperpolarization that results in the relaxation of smooth muscles of arterioles.

Topical minoxidil is also used for the treatment of androgenetic alopecia.

Solution to Question 22:

The given scenario of severe hypertension (blood pressure $\geq 180/\geq 120$ mmHg) with fundoscopic changes (flame haemorrhages and/or papilloedema) is suggestive of malignant hypertension. The drug of choice is IV nicardipine which acts by calcium channel blockage.

Nicardipine blocks the influx of extracellular calcium across the vascular and myocardial smooth muscle cell membranes by blocking the voltage-sensitive calcium channels during depolarization. The decrease in intracellular calcium causes dilation of the coronary and systemic arteries.

Other preferred agents for malignant hypertension are clevidipine with labetalol or esmolol.

Note: Sodium nitroprusside is no longer the treatment of choice for acute hypertensive problems

Solution to Question 23:

Drug-induced lupus is not an adverse effect of sodium nitroprusside infusion.

When nitroprusside is administered at a high rate, it can result in the production of methemoglobin and cyanide. Cyanide is further metabolized to thiocyanate. On prolonged infusion, thiocyanate toxicity can also occur especially in patients with renal failure.

Solution to Question 24:

Hydrochlorothiazides are not given to this patient.

Gout is a metabolic disease associated with hyperuricemia. Thiazides increase direct urate reabsorption in the proximal renal tubules and are associated with elevated serum uric acid levels and may precipitate gout. Hence, hydrochlorothiazides are not used to treat hypertension in gout patients.

Solution to Question 25:

First-dose phenomenon is characteristic of prazosin and patients must be counselled regarding the effects.

Within 30-90 minutes of the initial dose of prazosin or when the dosage is increased, there is marked orthostatic hypotension or syncope. This is known as the first-dose phenomenon. After a few initial doses, tolerance develops to this side effect. Initiating low doses at bedtime is recommended to minimize this phenomenon.

Solution to Question 26:

Telmisartan can be given safely in a patient with chronic renal failure (CRF) because it is mainly cleared from circulation by biliary excretion.

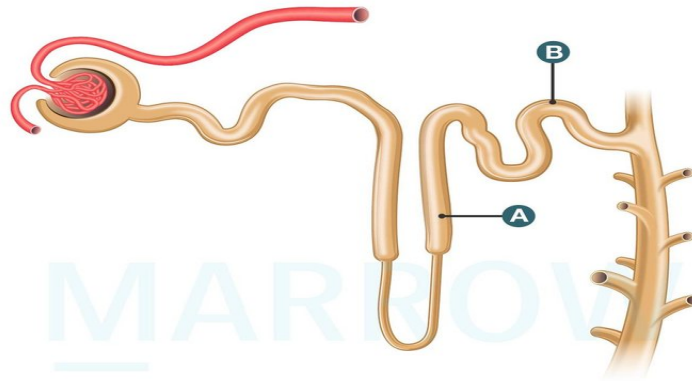
Options A and D: Sucralfate is a basic aluminum salt of sucrose sulfate. Sucralfate, aluminum hydroxide, and magnesium hydroxide in CRF patients can cause the accumulation of Al^{3+} and Mg^{2+} ions as they are preferentially excreted via the kidney after systemic absorption. Absorbed Al^{3+} can contribute to osteoporosis, encephalopathy and proximal myopathy in CRF patients.

Option B: Candesartan is cleared from circulation by renal and biliary excretion. Hence, plasma clearance is affected by renal insufficiency.

Diuretics

Question 1:

Choose the correct pair matching with the diagram:



- a) A: Loop diuretics, B: Carbonic anhydrase inhibitors
- b) A: Loop diuretics, B: Thiazides
- c) A: Thiazides, B: Loop diuretics
- d) A: Thiazides, B: Carbonic anhydrase inhibitors

Question 2:

Diuretics acting on which part of the nephron have maximum efficacy?

- a) Proximal tubule
- b) Thick ascending limb of the loop of Henle
- c) Collecting tubule
- d) Distal tubule

Question 3:

Which of the following statements about furosemide is true?

- a) Acute pulmonary edema is an indication
- b) Acts on the proximal convoluted tubules
- c) Causes mild diuresis
- d) Given only by parenteral route

Question 4:

A 70-year-old man with pulmonary edema is started on furosemide. Which of the following would you not expect after starting the drug?

- a) Hyperglycemia
- b) Hypomagnesemia
- c) Hypokalemia
- d) Acidosis

Question 5:

High ceiling diuretics are used in the treatment of all of the following patients except:

- a) 40-year-old man with anasarca
- b) 35-year-old woman with a head injury and cerebral edema
- c) 56-year-old woman with acute pulmonary edema
- d) 48-year-old man with pulmonary hypertension

Question 6:

A 65-year-old woman with chronic kidney disease was given a diuretic intravenously. Following administration of the drug, she developed tinnitus and diminished hearing. Which of the following drugs was most likely given?

- a) Thiazide
- b) Spironolactone
- c) Ethacrynic acid
- d) Triamterene

Question 7:

Carbonic anhydrase inhibitors do not cause:

- a) Increased bicarbonate excretion
- b) Urinary alkalization
- c) Metabolic alkalosis
- d) Potassium loss

Question 8:

Which of the following can be administered intravenously?

- a) Brinzolamide
- b) Dorzolamide
- c) Acetazolamide
- d) Methazolamide

Question 9:

A 30-year-old man joins a pilgrimage to a shrine at a mountain peak. At the summit, he develops tightness in the chest, associated with breathlessness and altered sensorium. Which of the following can be given for symptomatic relief?

- a) Mannitol
- b) Digoxin
- c) Diltiazem
- d) Acetazolamide

Question 10:

A patient with a glioblastoma multiforme is started on mannitol as palliative therapy. Which of the following statements is false regarding its diuretic action?

- a) Renal blood flow is increased
- b) Reduction in medullary tonicity
- c) Extracellular fluid compartment is expanded
- d) GFR is decreased

Question 11:

A 45-year-old woman presents with complaints of swelling of both feet and of breathlessness on exertion. After evaluation, she was diagnosed with congestive cardiac failure. Which of the following drugs is contraindicated in this patient?

- a) Mannitol
- b) Spironolactone
- c) Furosemide
- d) Hydrochlorothiazide

Question 12:

A patient with hypertension is prescribed chlorthalidone. What would be the effect of this drug on the excretion of electrolytes?

- a) A
- b) B
- c) C
- d) D

Question 13:

Thiazide diuretics are not used in the treatment of which of the following conditions?

- a) Idiopathic hypercalciuria with nephrocalcinosis
- b) Hyperlipidemia
- c) Congestive heart failure
- d) Nephrogenic diabetes insipidus

Question 14:

A 45-year-old man with hypertension is prescribed chlorthalidone. Which of the following is unlikely to occur in this patient?

- a) Hyponatremia
- b) Hypokalemia
- c) Erectile dysfunction

d) Hypocalcemia

Question 15:

A 42-year-old man in the nephrology ward has a GFR of less than 30mL/min, suggestive of severe renal failure. What is the best drug that can be given to him?

- a) Chlorthalidone**
- b) Metolazone**
- c) Xipamide**
- d) Hydrochlorothiazide**

Question 16:

A patient has been on lisinopril since being diagnosed with hypertension last year. His blood pressure readings are quite high, and the doctor considers adding a diuretic. Which amongst the following would be the ideal add on drug?

- a) Spironolactone**
- b) Eplerenone**
- c) Hydrochlorthiazide**
- d) Amiloride**

Question 17:

A 10-year-old boy is brought with complaints of vomiting and lethargy after swallowing several pills of his father's medication. Serum electrolytes indicate hyperkalemia. Which of the following drugs can you rule out?

- a) Triamterene**
- b) Spironolactone**
- c) Amiloride**
- d) Indapamide**

Question 18:

Which of the following diuretics do not need to be present in the tubular lumen for pharmacological action?

- a) Thiazide diuretics
- b) Loop diuretics
- c) Carbonic anhydrase inhibitors
- d) Aldosterone antagonists

Question 19:

Canrenone is a metabolite of:

- a) Acetazolamide
- b) Spironolactone
- c) Eplerenone
- d) Metolazone

Question 20:

A 52-year-old man with alcoholic cirrhosis develops ascites. What is the drug of choice for management?

- a) Chlorothiazide
- b) Acetazolamide
- c) Spironolactone
- d) Mannitol

Question 21:

Which of the following is not a side effect seen in a patient receiving spironolactone?

- a) Gynaecomastia
- b) Hyperkalemia
- c) Metabolic alkalosis
- d) Renal insufficiency

Question 22:

Which of the following is not an indication for amiloride use?

- a) Lithium induced polyuria
- b) Cystic fibrosis
- c) Calcium nephrolithiasis
- d) Liddle syndrome

Question 23:

A 35-year-old man with asthma presents with complaints of increased urination. The doctor checks that he has been taking theophylline and counsels that it could be a side effect. What is the mechanism of action causing diuresis?

- a) Adenosine antagonism
- b) PDE-4 inhibition
- c) Beta-2 agonism
- d) PDE-3 inhibition

Answer Key

Question No.	Correct Option
1	b
2	b
3	a
4	d
5	d
6	c
7	c
8	c
9	d
10	d
11	a
12	c
13	b
14	d

15	b
16	c
17	d
18	d
19	b
20	c
21	c
22	c
23	a

Detailed Explanations

Solution to Question 1:

A is the thick ascending limb of the loop of Henle and loop diuretics act on this part of the nephron.

B is the distal convoluted tubule and thiazides act on this part of the nephron.

Site of action of other diuretics:

- Proximal tubule - carbonic anhydrase inhibitors
- Collecting ducts and late distal tubule - epithelial sodium channel (ENaC) inhibitors and aldosterone antagonists

Solution to Question 2:

Loop diuretics have the maximum efficacy. They act on the thick ascending loop of Henle by blocking $\text{Na}^+ - \text{K}^+ - 2\text{Cl}^-$ symporter.

Option A: Although 65% of Na^+ is reabsorbed in the proximal tubule, diuretics acting here do not have maximum efficacy. This is because the thick ascending limb of the loop of Henle has a great reabsorptive capacity and it absorbs most of the filtrate from the proximal tubule.

Options C and D: Diuretics acting distal to the thick ascending limb of the loop of Henle have limited efficacy because most of the reabsorption of Na^+ (>90%) would have already occurred by the time the tubular fluid reaches there.

Solution to Question 3:

Furosemide is indicated in acute pulmonary edema. It is a loop diuretic that acts on the thick ascending loop of Henle and inhibits $\text{Na}^+ - \text{K}^+ - 2\text{Cl}^-$ symporter.

Loop diuretics have maximum efficacy amongst all the diuretics and are therefore called high ceiling diuretics. They can be given by both the oral and parenteral routes.

Solution to Question 4:

Furosemide causes metabolic alkalosis.

The adverse effects of furosemide (loop diuretics) are:

- Hypovolemia
- Hyponatremia
- Hypokalemia
- Hypocalcemia
- Hypomagnesemia
- Hyperuricemia
- Increased LDL-C and decreased HDL-C
- Glucose intolerance - hyperglycemia
- Ototoxicity

Solution to Question 5:

High ceiling diuretics are not used in pulmonary hypertension.

High ceiling or loop diuretics have diuretic properties as well as vasodilatory action.

- Diuretic property - removing excess fluid in hepatic, renal, and cardiac diseases
 - Vasodilatory action - quick symptomatic relief of symptoms in LVF and acute pulmonary edema.
- They are also used for the treatment of chronic CHF.

Pulmonary artery hypertension is a condition where there is increased pulmonary vascular resistance. Since nitric oxide is an important regulator of pulmonary vascular resistance, PDE5 inhibitors like sildenafil, vardenafil or tadalafil are primarily used in the treatment of pulmonary arterial hypertension.

Solution to Question 6:

Ethacrynic acid is the diuretic that was most likely given as it is commonly associated with ototoxicity.

Ototoxicity occurs with rapid intravenous infusion and the hearing impairment and deafness are usually reversible.

Ethacrynic acid should be used only in patients who are not able to tolerate other loop diuretics.

Solution to Question 7:

Carbonic anhydrase inhibitors do not cause metabolic alkalosis. They cause metabolic acidosis due to increased bicarbonate excretion

Option B: Increased bicarbonate excretion causes urinary alkalization.

Option D: Increased delivery of Na^+ to distal tubules causes increased K^+ excretion.

Solution to Question 8:

Acetazolamide is a carbonic anhydrase inhibitor that can be administered intravenously. It is indicated in the acute management of angle-closure glaucoma for the rapid reduction of IOP before surgery. The oral formulation is given for open-angle glaucoma.

Options A and B: Brinzolamide and dorzolamide are available only as ophthalmic drops and used in open-angle glaucoma.

Option D: Methazolamide is available as an oral formulation and used for open-angle glaucoma.

Solution to Question 9:

The symptoms are suggestive of acute mountain sickness and acetazolamide can be given for symptomatic relief.

It can also be used for prophylaxis.

Solution to Question 10:

Mannitol is an osmotic diuretic. These drugs do not cause much change in the GFR.

Osmotic diuretics cause extraction of fluid from the intracellular compartment resulting in the expansion of the extracellular compartment. The renal blood flow increases and renin release is inhibited. An increase in renal medullary blood flow removes NaCl and urea from the renal medulla thus reducing medullary tonicity. There is not much change in the total GFR.

Solution to Question 11:

Mannitol is contraindicated in patients with cardiac failure.

Mannitol is an osmotic diuretic. It acts by extracting water from the intracellular compartments, which results in the expansion of the extracellular fluid. This results in volume overload and frank pulmonary edema in patients with cardiac failure.

Solution to Question 12:

Chlorthalidone is a thiazide diuretic which increases Na^+ and K^+ excretion and decreases Ca^{2+} excretion.

Solution to Question 13:

Thiazide diuretics are not used in the treatment of hyperlipidemia. They result in an increase in the levels of LDL, total cholesterol, and triglycerides.

Option A: Thiazide reduces Ca^{2+} excretion and is used in the treatment of idiopathic hypercalciuria with nephrocalcinosis.

Option C: They are also used in the treatment of edema associated with CHF, hepatic cirrhosis, and nephrotic syndrome.

Option D: Thiazides are the mainstay for the treatment of nephrogenic diabetes insipidus. They reduce the kidney's ability to excrete free water. They increase proximal tubular water absorption and block the ability of DCT to form dilute urine.

Solution to Question 14:

Hypocalcemia is unlikely to occur in a patient on chlorthalidone, a thiazide diuretic.

Thiazide diuretics do not cause hypocalcemia. They result in hypercalcemia.

Thiazide and loop diuretics share similar side effects except that loop diuretics cause Ca^{2+} excretion (loop loses calcium) whereas thiazides retain Ca^{2+} and cause hypercalcemia.

Other side effects of thiazide diuretics are:

- Hypokalemia
- Hyponatremia
- Hypomagnesemia
- Metabolic alkalosis
- Hypercalcemia
- Hyperuricemia

Solution to Question 15:

Metolazone is effective in severe renal failure ($\text{GFR} < 30 \text{ ml/min}$).

Metolazone works primarily in the distal convoluted tubule, rather than the proximal convoluted tubule, and has little effect on GFR and can hence be used in patients with a reduced GFR. Indapamide is another thiazide diuretic that may be effective in severe renal failure.

Solution to Question 16:

Hydrochlorothiazide would be the ideal add-on drug. It is a thiazide diuretic and causes hypokalemia, which can counteract the hyperkalemia caused by lisinopril, an ACE inhibitor.

Aldosterone causes Na^+ absorption in the collecting duct and the distal tubule in exchange for H^+ and K^+ . ACE inhibitors decrease angiotensin II levels and thereby, decrease aldosterone levels. Decreased aldosterone level causes K^+ retention by reducing the exchange. Therefore, ACE inhibitors cause retention of K^+ .

Diuretics that cause elevation of plasma K^+ levels are not suitable for combining with ACE inhibitors.

Potassium-sparing diuretics like spironolactone, eplerenone, and amiloride are contraindicated with ACE inhibitors.

Solution to Question 17:

Indapamide can be ruled out as it is a thiazide diuretic that causes hypokalemia. The other drugs are potassium-sparing diuretics that cause hyperkalemia.

K^+ sparing diuretics work by 2 mechanisms:

- Blocking the ENaC (epithelial sodium channel)
- Triamterene
- Amiloride
- Aldosterone receptor antagonist
- Spironolactone
- Eplerenone
- Canrenone

Solution to Question 18:

Aldosterone antagonist acts from the interstitial side of the tubular cells. They do not have to be present in the tubular lumen for their action.

Aldosterone antagonists enter the tubular cells from the basolateral membrane. They competitively inhibit the binding of aldosterone to mineralocorticoid receptors.

Thiazide diuretics, loop diuretics, ENaC inhibitors, and carbonic anhydrase inhibitors have to be present in the lumen for its action. They are all secreted from the proximal tubule into the tubular fluid through the organic anion/ cation transport process.

Osmotic diuretics reach the lumen via glomerular filtration.

Solution to Question 19:

Canrenone is a metabolite of spironolactone.

Spironolactone is metabolized completely in the liver. It is also converted to an active metabolite, canrenone that is responsible for more than 50% of the spironolactone action. Spironolactone has a half-life of 1-2 hours while canrenone has a half-life of around 18 hours.

Eplerenone is an aldosterone receptor antagonist similar to spironolactone and metabolized primarily by CYP3A4, but no active metabolites have been identified.

Solution to Question 20:

The diuretic of choice in a patient with cirrhotic ascites is spironolactone.

Spironolactone is a competitive inhibitor of the mineralocorticoid receptor in the late distal tubule and collecting duct of the kidneys. It causes a decrease in sodium reabsorption and potassium excretion in the distal tubule. As a result, spironolactone promotes sodium diuresis but maintains body potassium levels.

Spironolactone is particularly helpful in edematous states caused or exacerbated by hyperaldosteronism such as in cirrhotic ascites.

Solution to Question 21:

Aldosterone antagonists cause hyperchloremic metabolic acidosis and not metabolic alkalosis. They inhibit H⁺ secretion in parallel with K⁺ secretion thus resulting in acidosis.

Other side effects are:

- Hyperkalemia
- Gynecomastia
- Renal insufficiency.

Solution to Question 22:

Amiloride is not used in the treatment of calcium nephrolithiasis. The condition is sometimes treated with thiazide diuretics but not amiloride.

Amiloride is a K⁺ sparing diuretic that acts by inhibiting the epithelial Na⁺ channels. The various uses of amiloride are:

- Edema/hypertension - usually in combination with thiazides
- Lithium-induced polyuria - blocks lithium transport into collecting tubules
- Cystic fibrosis - aerosolized amiloride enhances mucociliary clearance; acts by inhibiting Na⁺ absorption from epithelial surfaces and increasing hydration of secretions
- Liddle syndrome - hyperactive epithelial sodium channels (ENaC) activity is reduced

Solution to Question 23:

The diuretic effect of theophylline is due to the blockade of adenosine (A₁) receptors.

Theophylline is a methylxanthine drug that acts by -

- Inhibition of phosphodiesterase enzyme (PDE)
- PDE3 inhibition - relaxation of bronchial smooth muscle
- PDE4 inhibition - inhibition cytokines and chemokine release, which leads to a decrease in immune cell migration and activation
- Blockade of adenosine receptors inhibits bronchoconstriction, however, blockade of A₁ receptors in the glomerular afferent arteriole and PCT causes diuresis.
- Activating enzyme histone deacetylase-2, which is responsible for its anti-inflammatory action

Anti-Arrhythmic Drugs

Question 1:

A patient diagnosed with ventricular tachycardia was started on a class 1B anti-arrhythmic drug. Identify the drug.

- a) Mexiletine
- b) Quinidine
- c) Flecainide
- d) Propafenone

Question 2:

The additional mechanism by which class 1A drugs additionally prolong action potential duration is:

- a) Blocking Na⁺ channels
- b) Opening Na⁺ channels
- c) Blocking K⁺ channels
- d) Opening K⁺ channels

Question 3:

Which of the following antiarrhythmics has the fastest rate of recovery on Na⁺ channel blockade?

- a) 1, 3 and 5
- b) 2 and 4
- c) 4 and 5
- d) 2 only

Question 4:

Dofetilide belongs to which class of anti-arrhythmic drugs?

- a) Class I

- b) Class II
- c) Class III
- d) Class IV

Question 5:

An athlete diagnosed with atrial fibrillation was administered an antiarrhythmic drug. Several hours later, he complains of lower abdominal pain and not having passed urine since admission. Examination reveals a palpable bladder. Which was the most probable drug administered?

- a) Quinidine
- b) Disopyramide
- c) Procainamide
- d) Quinine

Question 6:

Which of the following drugs is associated with QT prolongation?

- a) Lignocaine
- b) Quinidine
- c) Magnesium sulfate
- d) All of the above

Question 7:

A patient presents with sinus bradycardia and bronchospasm. She was previously diagnosed with AV nodal reentrant tachycardia and had been started on an antiarrhythmic drug. Which of the following drugs is most likely being used?

- a) Flecainide
- b) Propafenone
- c) Mexiletine
- d) Lidocaine

Question 8:

Blurred vision in a patient being treated for paroxysmal supraventricular tachycardia is most likely due to:

- a) Esmolol
- b) Dronedarone
- c) Dofetilide
- d) Flecainide

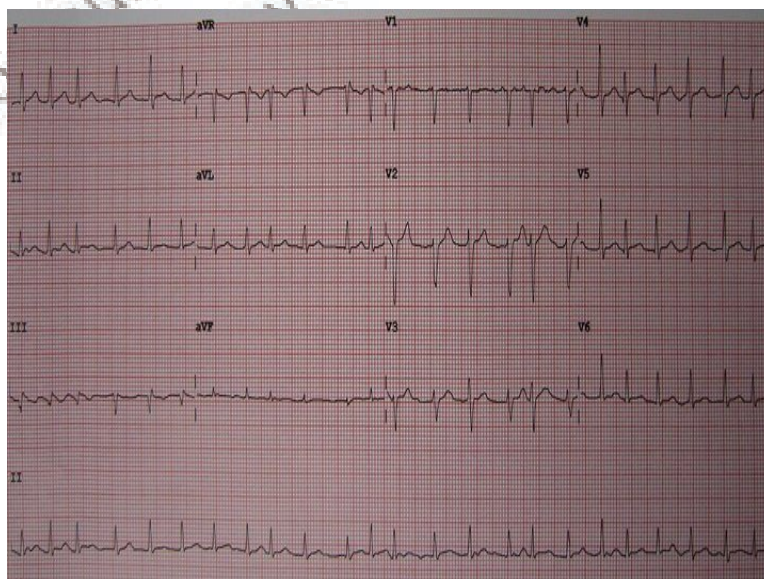
Question 9:

Which of the following antiarrhythmics has both class II and class III action?

- a) Propranolol
- b) Esmolol
- c) Sotalol
- d) Nadolol

Question 10:

A patient presented to the ER with dizziness and palpitations. ECG is shown below. Which of the following drugs cannot be used in this condition?



- a) Digoxin
- b) Verapamil
- c) Quinidine

d) Lignocaine

Question 11:

Which of the following is incorrect regarding lignocaine?

- a) Gastrointestinal absorption is good
- b) High volume of distribution
- c) In heart failure patients, both loading and maintenance doses are decreased
- d) Indicated for prophylaxis of arrhythmias in all cases of suspected MI

Question 12:

Which of the following is a correct statement regarding antiarrhythmics?

- a) Sotalol has additional class II action
- b) Esmolol has additional class IV action
- c) Propranolol has additional class I action
- d) Both A and C

Question 13:

A 60-year-old man presents to the OPD with complaints of bluish discoloration of the skin on the sun-exposed areas. He has been on medication for monomorphic ventricular tachycardia for the past 3 years. Which drug is most likely implicated in this condition?

- a) Lidocaine
- b) Adenosine
- c) Amiodarone
- d) Mexiletine

Question 14:

Chronic usage of an antiarrhythmic drug leads to optic neuritis and smurf skin or ceruloderma in a patient. Which of the following is not a mechanism of action of this drug?

- a) K⁺ and Na⁺ channel blockade
- b) Ca²⁺ channel blockade

- c) Adrenergic blockade
- d) Cl⁻ channel blockade

Question 15:

Which of the following adverse effects is least likely in a patient who is on chronic amiodarone therapy?

- a) Dry cough
- b) Corneal microdeposits
- c) Hepatic dysfunction
- d) Nephrotoxicity

Question 16:

Which of the following antiarrhythmic drugs is contraindicated in a patient with interstitial lung disease?

- a) Amiodarone
- b) Sotalol
- c) Quinidine
- d) Lignocaine

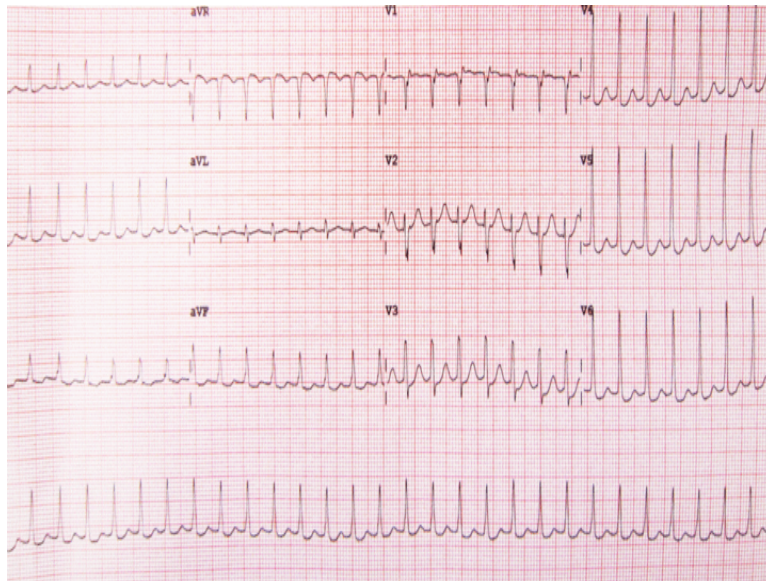
Question 17:

Which of the following statements about dronedarone is incorrect?

- a) It does not contain iodine moiety
- b) Has stronger antiadrenergic effects than amiodarone
- c) Less effective antiarrhythmic when compared to amiodarone
- d) Dose adjustment has to be done in renal failure

Question 18:

Given below is the ECG of a patient who presented to the emergency room with sudden onset palpitations and breathlessness. Which of the following drugs can be used in this condition?



- a) Atropine
- b) Mexiletine
- c) Verapamil
- d) Lidocaine

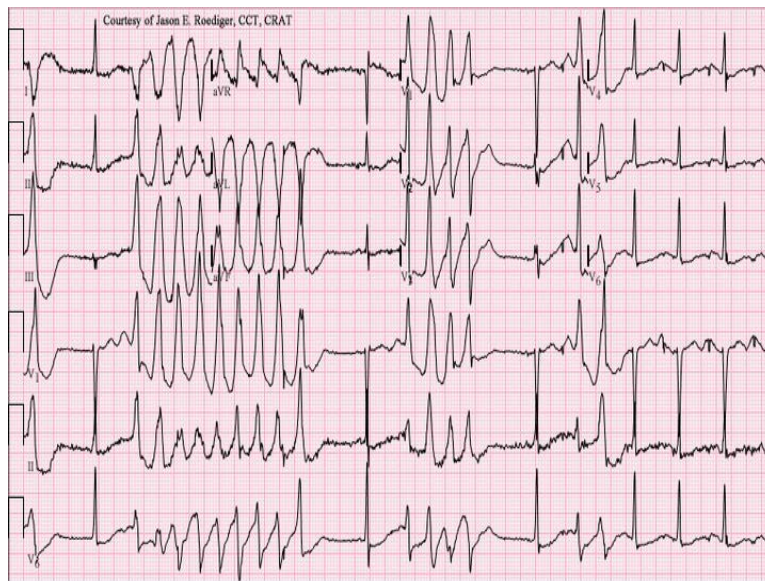
Question 19:

Vernakalant is indicated in which of the following types of arrhythmias?

- a) Ventricular fibrillation
- b) Atrial fibrillation
- c) Atrial flutter
- d) Ventricular flutter

Question 20:

A patient presented with palpitations and light-headedness. His ECG is shown below. Which of the following drugs is least likely to cause this condition?



- a) Quinidine
- b) Disopyramide
- c) Procainamide
- d) Lidocaine

Question 21:

Which of the following is the drug of choice for the immediate management of a patient with paroxysmal supraventricular tachycardia?

- a) Verapamil
- b) Diltiazem
- c) Digoxin
- d) Adenosine

Question 22:

A patient who was recently diagnosed with a seizure disorder presents to the casualty with altered mental status. His ECG shows a prolonged QTc. There is history of sudden cardiac deaths in his family at an early age. On further evaluation, a diagnosis of congenital long QT syndrome is made. Which is the most suitable pharmacological agent to treat this condition?

- a) Metoprolol
- b) Mexilitine
- c) Nicorandil

d) Isoprenaline

Question 23:

An elderly patient with heart failure and atrial fibrillation is stabilized with digoxin. By what mechanism does it control the arrhythmia in this patient?

- a) Vagolytic action
- b) Sympatholytic action
- c) Increasing vagal tone
- d) Decreasing atrial repolarization by blocking Na⁺ channels

Question 24:

A male patient who is suffering from a psychotic disorder is on antipsychotic, antidepressant and antihypertensive treatment. He suddenly developed tachycardia and chest tightness. On evaluation he had metabolic acidosis. ECG showed widened QRS complex and prolonged QT interval. Which of the following drugs is responsible for this?

- a) Amitriptyline
- b) Amlodipine
- c) Chlorthalidone
- d) Enalapril

Answer Key

Question No.	Correct Option
1	a
2	c
3	b
4	c
5	b
6	b
7	b
8	d

9	c
10	d
11	d
12	d
13	c
14	d
15	d
16	a
17	d
18	c
19	b
20	d
21	d
22	a
23	c
24	a

Detailed Explanations

Solution to Question 1:

Mexiletine is a class 1B antiarrhythmic drug. It is a lidocaine analog.

Solution to Question 2:

Class IA antiarrhythmics block K⁺ channels leading to prolongation of action potential duration.

The main action of class IA antiarrhythmics is blocking Na⁺ channels causing:

- Slowing the upstroke of the action potential
- Slowing conduction
- Prolonging the QRS duration

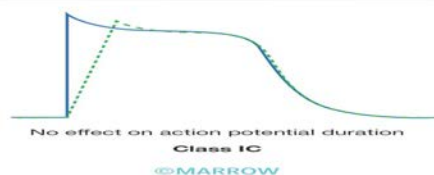
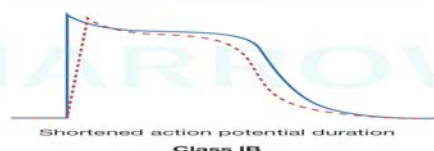
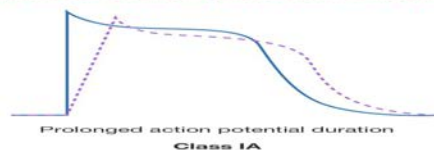
Solution to Question 3:

Phenytoin and mexiletine which are class IB antiarrhythmic drugs have the fastest rate of recovery on Na⁺ channel blockade.

They show the fastest dissociation kinetics to Na⁺ channel blockade (<1 sec).

	Na ⁺ channel blockade	Potassium channel	Action potential duration
Class IA	Intermediate- 1-10 sec	Significant K ⁺ channel blocker	Prolonged
Class IB	Rapid- <1 sec	K ⁺ channel opener	Shortened or no effect
Class IC	Slow- >10 sec	Negligible K ⁺ channel blocker	Variable or no effect

Class I Antiarrhythmic drugs – Effect on action potential duration



Solution to Question 4:

Dofetilide belongs to class III of anti-arrhythmic drugs.

Solution to Question 5:

The given scenario is suggestive of urinary retention. Disopyramide, a class 1A antiarrhythmic is the drug most likely to cause these symptoms in this patient.

It causes prominent anticholinergic side effects like urinary retention, dry mouth, blurred vision, precipitation of glaucoma, and constipation.

It is the drug of choice for vagally mediated atrial fibrillation such as sleep-induced or atrial fibrillation in athletes.

Solution to Question 6:

QT prolongation is seen with quinidine.

It causes torsades de pointes, a life-threatening form of polymorphic ventricular tachycardia.



Solution to Question 7:

Propafenone is the most likely drug being used in this patient as sinus bradycardia and bronchospasm are adverse effects seen with the use of this drug.

These side effects are due to its weak beta-blocking activity.

Solution to Question 8:

Blurred vision is the most common dose-related non-cardiac adverse effect of flecainide therapy.

Flecainide is a class 1C antiarrhythmic agent. It acts mainly by sodium channel blockade. It does not cause any change in the action potential duration.

Flecainide is used in paroxysmal supraventricular tachycardia (PSVTs):

- Atrioventricular nodal re-entrant tachycardia (AVNRT)
- AV re-entrant tachycardia (AVRT)
- Atrial fibrillation/atrial flutter (without structural heart disease)

Solution to Question 9:

Sotalol is a class III antiarrhythmic agent with additional beta-adrenergic blocking effects. (class II antiarrhythmic action)

Propranolol is a class II antiarrhythmic agent that has an additional Na⁺ channel blocking effect. (class I antiarrhythmic action)

Esmolol is a short-acting beta-blocker with minimal/no action on cardiac Na⁺ and K⁺ channels.

Solution to Question 10:

The scenario and ECG findings (absent P waves and RR intervals that are irregularly irregular) are suggestive of atrial fibrillation. Lignocaine is not used in atrial arrhythmias.

Lignocaine is a class 1B anti-arrhythmic agent. It blocks Na⁺ channels mostly in the closed/inactivated state. It is not used in atrial arrhythmias because the atrial action potential is so short that the Na⁺ channels are in a closed/ inactivated state for only a brief amount of time.

Lignocaine is used in the acute treatment of ventricular tachycardia and ventricular fibrillation.

Drugs mostly used for supraventricular arrhythmia are:

- Adenosine
- Verapamil
- Diltiazem
- Digoxin

Drugs used for ventricular arrhythmia are:

- Lidocaine
- Mexiletine

Drugs used in both conditions are:

- Amiodarone
- Beta-blockers
- Class 1a / 1c antiarrhythmics

Solution to Question 11:

Lignocaine decreases the incidence of arrhythmias in patients with suspected MI. However, it increases mortality probably by precipitating heart block or congestive heart failure. Hence it is not routinely used for prophylaxis in MI patients but it is used only when indicated.

Gastrointestinal absorption of lignocaine is good but bioavailability is very less due to high first-pass hepatic metabolism. Therefore, it is not used through the oral route.

Lignocaine has a high volume of distribution. The loading dose is required to achieve rapid action.

In heart failure, both volumes of distribution and clearance are reduced. Both the loading dose and maintenance dose have to be reduced.

Solution to Question 12:

Sotalol has both class II (beta-blocker) and class III action (K⁺ channel blockade). Though it is a nonselective beta-blocker, as an antiarrhythmic, it is classified under class III.

Propranolol is a non-selective beta-blocker with additional membrane-stabilizing action especially at higher doses: Na⁺ channel blockade/ class I action.

Esmolol is a beta 1 selective blocker that does not have any additional class effect.

Solution to Question 13:

The scenario is suggestive of blue man syndrome which is most likely caused by amiodarone.

One of the side effects of amiodarone is photosensitivity. In some cases, bluish discoloration of the skin is seen, predominantly in the sun-exposed areas. This could be due to the immune response in dermal layers by amiodarone or the accumulation of amiodarone in superficial layers of skin.

The symptoms are reversible on drug discontinuation. The syndrome is related to both dose and photosensitivity.

The side effects of amiodarone include:

- Pulmonary fibrosis
- Hypotension, depressed ventricular function
- Torsades de pointes
- Corneal microdeposits
- Hepatotoxicity
- Neuropathies
- Photosensitivity
- Thyroid dysfunction

Other such syndromes associated with drugs are:

- Red man syndrome - Vancomycin
- Gray baby syndrome - Chloramphenicol

Solution to Question 14:

Optic neuritis and smurf skin (blue man syndrome or ceruloderma) are caused by amiodarone. It acts by blocking K⁺, Na⁺, and Ca²⁺ channels. It also blocks adrenergic effects. It does not block Cl⁻ channels.

Amiodarone is classified as a class III antiarrhythmic agent. It exerts its function through the following mechanisms:

- Blockade of transient outward delayed rectifier and inward rectifier K⁺ currents
- Blockade of inactivated Na⁺ channels with a rapid rate of recovery
- Decreases Ca²⁺ current
- Non-competitive adrenergic blockade

It is the drug of choice for acute treatment of VT and VF.

The side effects of amiodarone include:

- Pulmonary fibrosis
- Hypotension, depressed ventricular function
- Torsades de pointes
- Corneal microdeposits
- Hepatotoxicity
- Neuropathies
- Photosensitivity
- Thyroid dysfunction

Solution to Question 15:

Nephrotoxicity is not an adverse effect of chronic amiodarone therapy. Pulmonary fibrosis associated with dry cough is the most serious side effect of chronic amiodarone therapy. It can rapidly progress and become fatal.

The side effects of amiodarone include:

- Pulmonary fibrosis
- Hypotension, depressed ventricular function
- Torsades de pointes
- Corneal microdeposits
- Hepatotoxicity
- Neuropathies
- Photosensitivity
- Thyroid dysfunction

Solution to Question 16:

Amiodarone is contraindicated in a patient with interstitial lung disease.

The most serious adverse effect during chronic amiodarone therapy is pulmonary fibrosis, which can be rapidly progressive and fatal. Underlying lung disease, doses of 400 mg/day or more, and recent pulmonary insults such as pneumonia appear to be risk factors for this condition.

Amiodarone is a class III antiarrhythmic drug.

The side effects of amiodarone include:

- Pulmonary fibrosis
- Hypotension, depressed ventricular function
- Torsades de pointes
- Corneal microdeposits
- Hepatotoxicity
- Neuropathies
- Photosensitivity
- Thyroid dysfunction

Solution to Question 17:

Dronedarone undergoes hepatic metabolism and hence dose adjustment is not required in renal failure as it is not excreted through the kidneys.

Dronedarone is a non-iodinated congener of amiodarone. Like amiodarone, it is a blocker of multiple ion channels. It has stronger antiadrenergic effects than amiodarone. It is less effective in maintaining sinus rhythm and is considered less antiarrhythmic when compared to amiodarone.

Unlike amiodarone, it does not cause thyroid dysfunction or pulmonary fibrosis. However, it causes hepatotoxicity and increases the risk of stroke and heart failure.

Solution to Question 18:

The clinical scenario and the ECG findings are suggestive of paroxysmal supraventricular tachycardia (PSVT). Verapamil and diltiazem are used in both acute treatment and prophylaxis of PSVT.

Their mechanism of action is by slowing SA node automaticity and AV nodal conduction velocity. Additionally, they also decrease cardiac contractility and reduce blood pressure.

Dihydropyridines like nifedipine and amlodipine do not have any antiarrhythmic property.

Drugs used for supraventricular arrhythmias are:

- Adenosine
- Verapamil
- Diltiazem

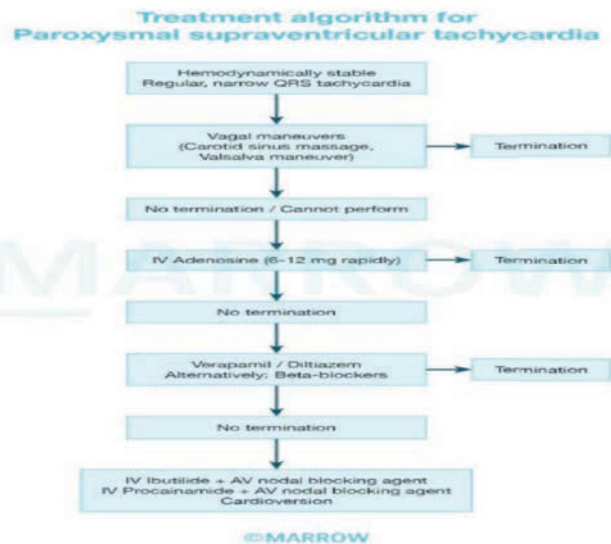
- Digoxin

Drugs used for ventricular arrhythmias are:

- Lidocaine
- Mexiletine

Drugs used in both conditions are:

- Amiodarone
- Beta-blockers
- Class 1A / 1C antiarrhythmics



Solution to Question 19:

Vernakalant is a newer antiarrhythmic agent indicated in atrial fibrillation.

Its mechanisms of action include:

- Frequency and voltage-dependent block of the early and late components of the sodium current
- The muscarinic potassium current I_{KACH} , which is constitutively activated in atrial fibrillation is also blocked

It is indicated for rapid termination of atrial fibrillation in patients with no or minimal structural heart disease.

Solution to Question 20:

The given scenario and the ECG findings (rapid asynchronous complexes and an undulating baseline) point towards a diagnosis of torsades de pointes. Lidocaine use is unlikely to cause this condition.

Lidocaine belongs to class IB antiarrhythmics. It is considered to have the least arrhythmogenic potential among antiarrhythmics.

Among the antiarrhythmics, class Ia and class III drugs prolong action potential duration and thus prolong QT interval. They are also prone to cause torsades de pointes.

Solution to Question 21:

The drug of choice for the termination of PSVT is adenosine.

Drugs used for supraventricular arrhythmias are:

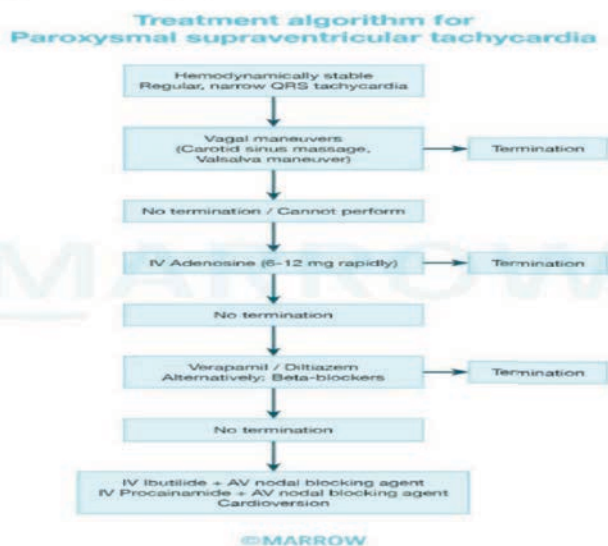
- Adenosine
- Verapamil
- Diltiazem
- Digoxin

Drugs used for ventricular arrhythmias are:

- Lidocaine
- Mexiletine

Drugs used in both conditions are:

- Amiodarone
- Beta-blockers
- Class 1a / 1c antiarrhythmics



Solution to Question 22:

Beta-blockers like metoprolol are the mainstay of pharmacological therapy in both asymptomatic and symptomatic patients with congenital LQTS. They reduce both syncope and sudden cardiac death (SCD).

Long QT syndrome (LQTS) is a disorder of ventricular myocardial repolarization characterized by a prolonged QT interval on the electrocardiogram that can lead to symptomatic ventricular arrhythmias and an increased risk of sudden cardiac death.

Solution to Question 23:

Digoxin results in an increased vagal tone which helps in controlling the ventricular rate in atrial fibrillation.

Increased vagal tone increases the AV nodal refractoriness due to blockade of Ca^{2+} channels. It also results in hyperpolarisation and shortening of atrial action potentials due to activation of ACh-mediated K^{+} channels in the atria.

Solution to Question 24:

The given clinical scenario is suggestive of torsades de pointes. Only amitriptyline causes this condition, among the above drugs.

Torsades de pointes (French: twisting of points) is a life-threatening form of polymorphic ventricular tachycardia with rapid asynchronous complexes and an undulating baseline on ECG.



Hypolipidemic Drugs

Question 1:

A 58-year-old patient with coronary artery disease has a deranged lipid profile and is started on statins. Which enzyme does this drug act on?

- a) Acyl coA synthase
- b) Acyl coA reductase
- c) HMG coA synthetase
- d) HMG coA reductase

Question 2:

Hepatic cholesterol synthesis is maximal between midnight and 2 am. So most of the statins are preferably administered in the evening for maximal effect. Which statin can be taken irrespective of the time of the day?

- a) Simvastatin
- b) Pravastatin
- c) Atorvastatin
- d) Lovastatin

Question 3:

A patient with hypercholesterolemia on statin therapy presents to your OPD for follow-up. You notice his cholesterol is still high and decide to add on another drug that directly inhibits the absorption of cholesterol. Which of the following will you prescribe?

- a) Ezetimibe
- b) Orlistat
- c) Cholestyramine
- d) Niacin

Question 4:

Which drug decreases Lipoprotein(a)?

- a) Fenofibrate
- b) Niacin
- c) Gemfibrozil
- d) Colestipol

Question 5:

Which of the following drugs would you warn a patient taking statins against, due to the highest risk of myopathy?

- a) Fenofibrate
- b) Gemfibrozil
- c) Colestipol
- d) Niacin

Question 6:

A patient who is on statin therapy asks about a new drug called evolocumab that his friend is taking for hyperlipidemia. What is the route of administration of this drug?

- a) Intramuscular
- b) Intradermal
- c) Subcutaneous
- d) Intravenous

Question 7:

Which among the following is a microsomal triglyceride transfer protein inhibitor?

- a) Ticlopidine
- b) Gugulipid
- c) Lomitapide
- d) Evinacumab

Question 8:

A patient with hereditary hyperlipoproteinemia is prescribed a fibrate. What is the mechanism of action of this drug?

- a) Increased lipolysis of triglycerides
- b) Inhibit lipolysis in adipose tissues
- c) Inhibit HMG CoA synthetase
- d) Binds bile acids and bile salts in the intestine

Question 9:

A 45-year-old obese man presented to your OPD. His lipid profile showed elevated levels of cholesterol and triglycerides. On examination, you notice palmar xanthomas. What is the drug of choice for his condition?

- a) Statins
- b) Niacin
- c) Fibrates
- d) Alirocumab

Question 10:

What is the drug of choice for type IIa and IIb hyperlipoproteinemias?

- a) Statins
- b) Fibrates
- c) Niacin
- d) Bile acid sequestrants

Question 11:

A patient with homozygous familial hypercholesterolemia has mipomersen as part of his treatment regimen. By which of the following mechanisms does this drug act?

- a) Inhibition of microsomal triglyceride transfer
- b) Inhibition of PCSK9
- c) Inhibition of apo B-100 synthesis
- d) Inhibition of cholesterol absorption

Answer Key

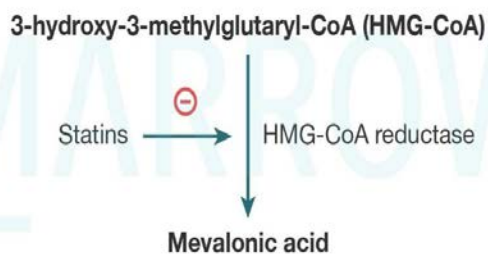
Question No.	Correct Option
1	d
2	c
3	a
4	b
5	b
6	c
7	c
8	a
9	c
10	a
11	c

Detailed Explanations

Solution to Question 1:

Statins act on the HMG-CoA reductase enzyme, thus inhibiting cholesterol synthesis.

Mechanism of action of Statins



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Note: Necrotizing myopathy is a newly defined form of idiopathic inflammatory myopathy. Patients who have anti-HMG Co-A reductase antibodies should not be given statins to control

high cholesterol as they may develop this type of statin-induced necrotizing myopathy.

Solution to Question 2:

Atorvastatin has a longer half-life (19 ± 9 hours) and may be taken at any time of the day.

Other drugs that can be taken irrespective of the time of day:

- Rosuvastatin - has a half-life of 20 ± 6 hours
- Pitavastatin - has a half-life of 12 hours

All other statins, which have a plasma half-life of around 4 hours or less, should preferably be administered in the evening.

Solution to Question 3:

Ezetimibe is the drug that prevents hypercholesterolemia by directly inhibiting the absorption of cholesterol.

Ezetimibe acts by interfering with a specific cholesterol transport protein NPC1L1 in the intestinal mucosa and reduces the absorption of both dietary and biliary cholesterol. It causes a compensatory increase in hepatic cholesterol synthesis but LDL cholesterol is decreased. Statins block this enhanced cholesterol synthesis (cholesterol synthesis inhibitors) and so their combination has a synergistic effect.

Option B: Orlistat is an anti-obesity drug that acts by inhibiting pancreatic and gastric lipase and thus interferes with the digestion and absorption of dietary triglycerides.

Option C: Cholestyramine is a bile acid sequestrant. The bile-acid sequestrants are highly positively charged and bind negatively charged bile acids. Because of their large size, the resins are not absorbed, and the bound bile acids are excreted in the stool.

Cholestyramine primarily reduces LDL levels by increasing hepatic bile acid synthesis; which results in increased LDL uptake and diversion of cholesterol to bile acid production.

Option D: Niacin reduces lipolysis by hormone-sensitive lipase (HSL) and reduces hepatic triglyceride synthesis.

Solution to Question 4:

The drug that decreases Lipoprotein(a) is niacin.

Niacin is the only lipid-lowering agent that lowers lipoprotein(a) levels significantly. It also results in a decrease in LDL, VLDL, and triglycerides along with an increase in HDL cholesterol.

The main adverse effect of niacin is cutaneous flushing and pruritus which can be controlled by pretreatment with aspirin. Niacin can also cause hepatotoxicity, GI toxicity, and hyperuricemia.

Solution to Question 5:

Myopathy risk is highest when statins are used along with gemfibrozil.

Gemfibrozil is a fibrate that inhibits the uptake of statins into the hepatocytes and prevents its metabolism, thereby increasing its bioavailability. Thus, the risk of toxicity of statins increases.

Other drugs interacting similarly with statins include:

- Cyclosporine
- Digoxin
- Warfarin
- Macrolide antibiotics
- Azole antifungals.

Option A: Fenofibrate does not have much interaction with statins.

Option C: Colestipol is a bile acid sequestrant and does not have much effect on statin metabolism.

Option D: Niacin can also rarely increase the risk of statin-induced myopathy but the risk is much lower than that with gemfibrozil. So here gemfibrozil is the best answer.

Solution to Question 6:

Evolocumab is a PCSK-9 inhibitor that is administered as a subcutaneous injection.

PCSK-9 is a protease that binds to the LDL receptors on the surface of hepatocyte and enhances the lysosomal degradation of the LDL receptor. This results in higher plasma LDL levels. inhibition of PCSK-9 results in a reduction of plasma LDL levels.

Two approved PCSK-9 inhibitors are alirocumab and evolocumab.

Solution to Question 7:

Microsomal triglyceride transfer protein (MTP) inhibitor is lomitapide.

MTP adds triglycerides to nascent VLDL in the liver and to chylomicrons in the intestine. Its inhibition decreases VLDL secretion and thus decreases the accumulation of LDL in the plasma. It is approved for use in homozygous familial hypercholesterolemia.

Options A: Ticlopidine is an antiplatelet agent.

Option B: Gugulipid is a plant extract that is commonly used for hyperlipidemia but has very little/no evidence to support its use.

Option D: Evinacumab is a recently approved angiopoietin-like 3 (ANGPTL3) inhibitor. It is indicated as an adjunct to other LDL-cholesterol lowering therapies for homozygous familial hypercholesterolemia in adults and pediatric patients aged 12 years and older.

Solution to Question 8:

Fibrates increase the lipoprotein lipase activity through PPAR- α and cause increased lipolysis of triglycerides.

Lipoprotein lipase (LPL) is a key enzyme in the degradation of VLDL. Fibrates bind to PPAR alpha and cause stimulation of lipoprotein lipase enzyme. Increased LPL synthesis enhances the clearance of triglyceride-rich lipoproteins. Additionally, fibrates also stimulate fatty acid oxidation and reduce the expression of apo C-III (inhibitor of lipolysis).

Class	HMG- CoA reductase inhibitors (statins)	Fibrates	Niacin	Bile acid-binding resins	Ezetimibe
Important examples	Lovastatin, atorvastatin, fluvastatin, pravastatin, simvastatin, rosuvastatin, pitavastatin	Gemfibrozil and fenofibrate	Niacin	Colestipol, cholestyramine, colesevelam	Ezetimibe
Mechanism of action	Inhibits HMG-CoA reductase	PPAR- α mediated increased lipolysis	Decreases the production of VLDLs and decreases the catabolism of HDLs	Binds to bile acids in the intestinal lumen and increases excretion. This results in increased conversion of hepatic cholesterol to bile acids, and reduction in hepatic cholesterol content.	Inhibits intestinal sterol absorption
Major effects on lipoprotein levels	Reduces LDL levels	Reduces triglyceride levelsA moderate increase in HDL levels	Reduces triglycerides, LDL, and lipoprotein a levelsSignificant increase HDL levels	Reduces LDL levels	Reduces LDL cholesterolMinimal increase in HDL levelAlso decreases the absorption of phytosterols

Class	HMG- CoA reductase inhibitors (statins)	Fibrates	Niacin	Bile acid-binding resins	Ezetimibe
Important side effects	MyopathyElevations of serum aminotransferase activityElevation of creatine kinase activity	Rashes, gastrointestinal symptoms, myopathy, arrhythmiasHypokalemia, elevated aminotransferases, or alkaline phosphatase	Cutaneous vasodilationElevations in aminotransferases levels	ConstipationBloating	Reversible hepatic function impairmentRarely myositis

Solution to Question 9:

The given scenario of elevated cholesterol and triglycerides along with the presence of palmar xanthomas (pathognomonic sign) is indicative of type III hyperlipoproteinemia. The drugs of choice for this condition are fibrates.

Fibrates are also preferred in patients having severe hypertriglyceridemia (TG >1000 mg/dl) who are at risk for pancreatitis.

The response to fibrate therapy in Type III hyperlipoproteinemia patients include:

- Elevated cholesterol and triglyceride levels are dramatically lowered.
- Tuberoeruptive and palmar xanthomas may regress completely.
- Angina and intermittent claudication also improve.

The image below shows palmar xanthoma.



Solution to Question 10:

The drugs of choice for type IIa and IIb hyperlipoproteinemias are statins.

In type IIa or IIb hyperlipoproteinemias, there is increased LDL cholesterol with +/- increase in TG. Statins have the maximum LDL lowering property among the hypolipidemic drugs. Therefore, they are the drugs of choice in this case.

In type III/IV hyperlipoproteinemias, there is mainly an increase in triglyceride(TG) level. Fibrates reduce the TGs contained in VLDL and increase the HDL. So they are the drugs of choice for these conditions.

Solution to Question 11:

Mipomersen is an antisense oligonucleotide that acts as an inhibitor of apo B-100 synthesis.

Mipomersen binds to the mRNA of apo B-100 resulting in degradation of apo B-100 mRNA and eventually reduction of apo B-100 synthesis.

Renin-Angiotensin-Aldosterone System

Question 1:

Which of the following ACE inhibitors is not a prodrug?

- a) Fosinopril
- b) Enalapril
- c) Ramipril
- d) Lisinopril

Question 2:

A patient with BP 225/145 presents with acute pulmonary edema and is diagnosed with hypertensive emergency. The active metabolite of which ACE inhibitor is available for intravenous use in this patient?

- a) Ramipril
- b) Perindopril
- c) Enalapril
- d) Fosinopril

Question 3:

Which of the following statements regarding ACE inhibitors is true?

- a) They inhibit the conversion of angiotensinogen to angiotensin I
- b) Omission of prior diuretic dose decreases the risk of postural hypotension
- c) Lisinopril is shorter acting than enalapril
- d) They are contraindicated in diabetic patients

Question 4:

All of the following are indications for the use of ACE inhibitors except:

- a) Hypertensive emergency
- b) Myocardial infarction

- c) Left ventricular dysfunction
- d) Pheochromocytoma

Question 5:

In which of the following patients are ACE inhibitors not contraindicated?

- a) 1 and 3
- b) 1,2 and 3
- c) 2,4 and 5
- d) 3, 4 and 5

Question 6:

A 53-year-old hypertensive patient has been on ACE inhibitors for the last 12 years. Which of the following effects would you expect to see in this patient?

- a) Rightward shift in renal pressure natriuresis curve
- b) Reduction in filtration fraction
- c) Significant increase in heart rate
- d) Decrease in compliance of large arteries

Question 7:

A patient who is started on lisinopril for hypertension is concerned about its adverse effects. Which among the following is not an adverse effect of this drug?

- a) Dysgeusia
- b) Hypokalemia
- c) Neutropenia
- d) Skin rash

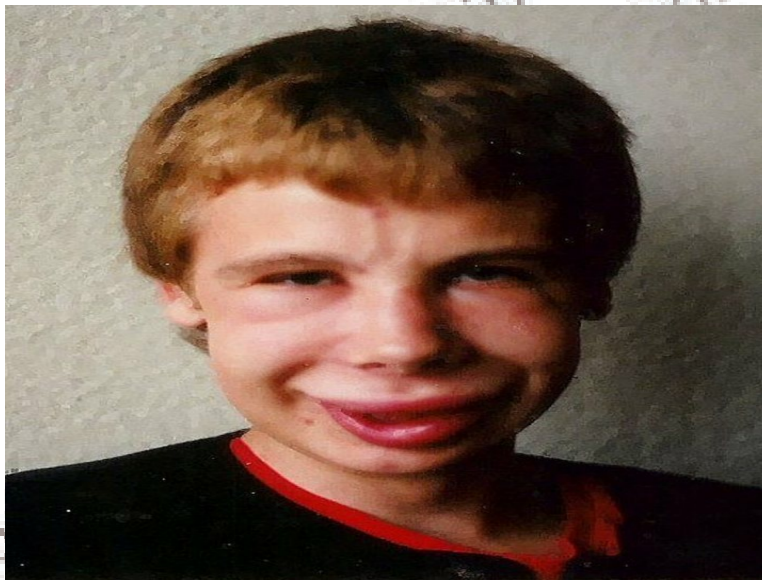
Question 8:

A patient presents to the casualty with cough, difficulty in breathing, and non-indurating edema of his face and lips. He had recently been started on ramipril for hypertension. Which among the following mediates these adverse effects?

- a) Bradykinin and renin
- b) Angiotensin II and bradykinin
- c) Only bradykinin
- d) Histamine

Question 9:

An 18-year-old boy is brought to the casualty with complaints of swelling of his face and lips as shown in the image below. On further enquiry, it is noted that he has suffered from similar episodes in the past whenever he becomes emotional. He is prescribed icatibant. What is the likely mechanism of action of this drug?



- a) Bradykinin B2 receptor agonist
- b) Bradykinin B2 receptor antagonist
- c) Bradykinin B1 receptor agonist
- d) Bradykinin B1 receptor antagonist

Question 10:

A 40-year-old diabetic male is on ACE inhibitors for hypertension. Which among the following is the most serious adverse effect you would expect to see in this patient?

- a) Hypotension
- b) Hypokalemia
- c) Hypocalcemia

d) Hypercalcemia

Question 11:

A 40-year-old man is diagnosed with hypertension and is prescribed losartan. Which of the following statements about the drug is incorrect?

- a) It is a competitive angiotensin receptor antagonist
- b) It attenuates platelet aggregation
- c) Associated with negligible cough
- d) Causes hyperuricemia

Question 12:

Combined use of ACE inhibitors and ARBs are not routinely indicated primarily due to the increased risk of which of the following?

- a) Hyperkalemia
- b) Hypokalemia
- c) Angioedema
- d) Heart failure

Answer Key

Question No.	Correct Option
1	d
2	c
3	b
4	d
5	c
6	b
7	b
8	c
9	b
10	a

11	d
12	a

Detailed Explanations

Solution to Question 1:

Lisinopril is not a prodrug.

All ACE inhibitors are prodrugs except lisinopril and captopril.

Solution to Question 2:

Enalapril is available in its active form as enalaprilat for intravenous use.

Enalapril is a prodrug of enalaprilat. It is indicated in conditions where oral drugs cannot be used or rapid onset of action is desired like in a case of hypertensive emergency.

Solution to Question 3:

The omission of prior diuretic dose decreases the risk of postural hypotension.

ACE inhibitors cause arteriolar dilation and increase the compliance of larger arteries. ACE inhibitors induced hypotension is thereby due to a decrease in total peripheral resistance. In diuretic-treated patients, the renin-angiotensin-aldosterone system is more active to compensate for volume loss. Hence, ACE inhibitors increase the risk of postural hypotension in these patients.

Option A: ACE inhibitors inhibit the conversion of angiotensin I to angiotensin II.

Option C: Lisinopril is longer acting than enalapril. Lisinopril is usually administered once daily in all patients. Enalapril requires twice-daily administration in some patients.

Option D: ACE inhibitors and ARBs are first-line agents in diabetic patients because of their renoprotective effects.

Solution to Question 4:

ACE inhibitors are usually avoided in conditions like pheochromocytoma which is associated with high renin activity. Abrupt inhibition of the RAAS system can result in severe hypotension (usually postural).

The majority of pheochromocytomas are treated surgically. Phenoxybenzamine is often used in preparing the patient for surgery and also to manage hypertension in inoperable or malignant

pheochromocytoma. Phentolamine can also be used in the short-term control of hypertension in these patients.

ACE inhibitors are indicated in hypertensive emergency, myocardial infarction, and left ventricular dysfunction.

Solution to Question 5:

ACE inhibitors are not contraindicated in unilateral renal artery stenosis, left ventricular systolic dysfunction, and scleroderma renal crisis.

Patient 1: ACE inhibitors are teratogenic and can cause hypoplasia of organs, renal failure, anuria, and growth retardation in the fetus. Therefore, they are contraindicated in pregnancy.

Patient 2: ACE inhibitors can be used with caution in unilateral renal artery stenosis if the other kidney is functioning normally.

Patient 3: Efferent arteriolar constriction maintains the glomerular filtration pressure in bilateral renal artery stenosis. ACE inhibitors are contraindicated in bilateral renal stenosis as they can precipitate acute renal failure due to dilation of efferent arterioles and fall in glomerular filtration pressure.

Patient 4: ACE inhibition in patients with systolic dysfunction prevents or delays the progression of heart failure, decreases the incidence of sudden death and myocardial infarction, and improves the quality of life. ACE inhibition reduces afterload and systolic wall stress, and both cardiac output and cardiac index increase.

Patient 5: The use of ACE inhibitors considerably improves the survival of patients with scleroderma renal crisis.

Solution to Question 6:

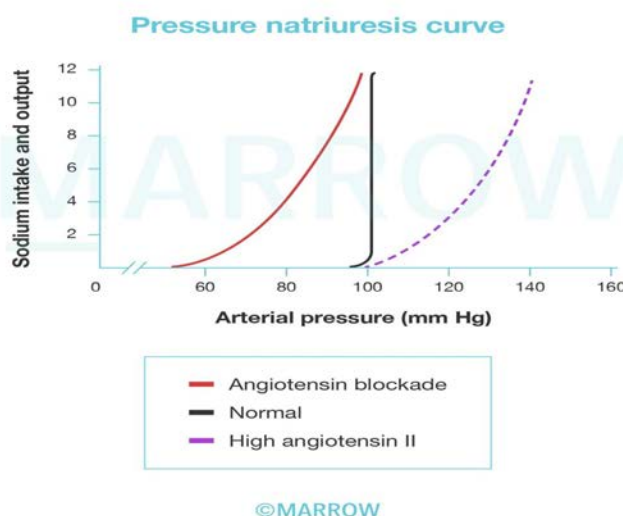
ACE inhibitors, when used for a long time in patients with hypertension, cause a reduction in filtration fraction.

Filtration fraction is the ratio of GFR to that of renal plasma flow. ACE inhibitors increase renal blood flow without altering GFR and thus result in reduction of filtration fraction.

Option A: The graph plotted between urinary sodium excretion and mean blood pressure is known as the pressure natriuresis curve. For example, when dietary sodium intake is low, renin release is stimulated. Angiotensin II acts on the kidneys to shift the renal pressure – natriuresis curve to the right. This increases sodium and water retention.

Conversely, when dietary Na^+ is high or when ACE inhibitors are used, there is the withdrawal of angiotensin II. The renal pressure natriuresis curve shifts to the left.

The following image depicts a pressure natriuresis curve:



Options C and D: ACE inhibitors do not cause a significant increase in heart rate and can have a favorable change in the compliance of large arteries on long-term use.

Solution to Question 7:

ACE inhibitors are associated with the development of hyperkalemia.

Hyperkalemia occurs due to the inhibition of angiotensin II-mediated aldosterone secretion. Decreased aldosterone levels result in decreased tubular excretion of potassium in the kidneys leading to hyperkalemia.

The adverse effects of ACE inhibitors are:

- Hypotension
- Hyperkalemia
- Cough
- Angioedema
- Skin rash
- Teratogenicity
- Dysgeusia (alteration in or loss of taste)

Extremely rare but reversible side effects include neutropenia, glycosuria in the absence of hyperglycemia, anemia, and hepatotoxicity.

Solution to Question 8:

The given scenario is suggestive of ACE inhibitor-induced angioedema which occurs due to bradykinin.

Angioedema is most commonly mediated by histamine. However, bradykinin-mediated angioedema, though less common, can result due to hereditary or acquired defects in C1 esterase inhibitor and as a side effect of ACE inhibitors.

ACE has structural similarity to another enzyme called kininase. Kininase normally metabolizes bradykinin, substance P, and prostaglandins. ACE inhibitors when administered also block kininase enzyme which results in an increase in the concentration of these substrates. An increase in the level of bradykinin/ substance P levels results in persistent brassy cough and angioedema in the patients.

Solution to Question 9:

The given clinical scenario and image showing swelling of the face and lips is suggestive of hereditary angioedema. Icatibant is a selective bradykinin B2 receptor competitive antagonist indicated for the treatment of acute attacks of hereditary angioedema.

The other drug which is approved for the treatment of acute attacks of hereditary angioedema is ecallantide which is a synthetic plasma kallikrein inhibitor. Both these drugs are administered subcutaneously.

Solution to Question 10:

The most significant adverse effect seen with the use of ACE inhibitors is hypotension.

ACE inhibitors are used in the management of hypertension and are one of the first-line drugs for this condition.

Option B: ACE inhibitors may cause hyperkalemia in patients with renal insufficiency or diabetes.

Options C and D: ACE inhibitors do not affect calcium levels significantly.

Solution to Question 11:

Losartan use is not associated with hyperuricemia. In fact, losartan is uricosuric and can also be used in patients with gout.

Losartan is a competitive angiotensin II receptor antagonist. It binds to the AT₁ receptor with high affinity. It is also a competitive antagonist of the thromboxane A₂ receptor and attenuates platelet aggregation. Losartan has no effect on ACE and it does not inhibit bradykinin breakdown. This results in negligible incidences of cough.

Solution to Question 12:

The combined use of ACE inhibitors and ARBs are not commonly indicated as they increase the risk of hyperkalemia.

They also increase the risk of hypotension and renal injury. Any such use should be done under the supervision of specialists with frequent monitoring.

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Anti-diuretics

Question 1:

A 50-year-old man was rushed to the ER with impaired consciousness following a hunger strike for a day. His wife reported, he had a head injury 1 month back and since then had complaints of polyuria. On examination, he was found to be severely dehydrated and MRI brain showed an absence of the characteristic bright spot in the pituitary. Which of the following is the drug of choice for his condition?

- a) Desmopressin
- b) Amiloride
- c) Thiazide diuretics
- d) Triamterene

Question 2:

A patient having polyuria and polydipsia is diagnosed with central diabetes insipidus. He is started on intranasal desmopressin. Which of the following is not a reason for choosing this over intravenous vasopressin?

- a) Higher potency
- b) Higher selectivity for V1 receptor
- c) Has lower vasoconstrictor activity
- d) Longer acting

Question 3:

A 30-year-old woman with pituitary adenoma is non-compliant with her treatment with cabergoline. She has now presented with bitemporal hemianopia and features suggestive of diabetes insipidus. Which of the following drugs cannot be used to manage her condition?

- a) Desmopressin
- b) Furosemide
- c) Chlorpropamide
- d) Carbamazepine

Question 4:

Tachyphylaxis is a concern with the use of desmopressin in which of the following patients?

- a) A 50-year-old man with central DI post meningitis
- b) A 25-year-old woman on lithium therapy with nephrogenic DI
- c) A 17-year-old girl with von willebrand's disease
- d) A 7-year-old child with primary nocturnal enuresis

Question 5:

A 50-year-old man who is on treatment for small cell carcinoma of the lung presents with acute onset confusion, nausea, and vomiting. He appears to be euvolemic. Laboratory investigations reveal serum sodium levels to be 120 mEq/l and urine osmolality is 300 mOsm/kg . Which of the following is not used as a treatment option in his condition?

- a) Demeclocycline
- b) Desmopressin
- c) Restriction of free water intake
- d) 3% NaCl

Question 6:

The best approach in the treatment of chronic SIADH is _____

- a) Tolvaptan
- b) Reducing fluid intake to <500 ml per day
- c) Demeclocycline
- d) Fludrocortisone

Question 7:

A patient with liver cirrhosis presenting with ascites and pedal edema is administered conivaptan. Which of the following statements regarding the drug is true?

- a) Vasopressin receptor antagonist
- b) V2 selective action
- c) Given orally

- d) Used in hypernatremia

Question 8:

A 50-year-old alcoholic with deranged liver function tests is to be started on treatment for chronic SIADH. Which of the following drugs should not be used in his treatment regimen?

- a) Conivaptan
- b) Tolvaptan
- c) Demeclocycline
- d) Lixivaptan

Question 9:

A patient who was diagnosed with ADPKD was prescribed tolvaptan. After a few weeks, he presented with dry mouth and increased thirst. What is the likely cause?

- a) Increased free water clearance
- b) V₂ receptor agonism
- c) Increased renal cAMP levels
- d) Increase in urine osmolality

Answer Key

Question No.	Correct Option
1	a
2	b
3	b
4	c
5	b
6	a
7	a
8	b
9	a

Detailed Explanations

Solution to Question 1:

The given scenario of polyuria following a head injury and an absence of a pituitary bright spot on the MRI brain is suggestive of central diabetes insipidus. The drug of choice for this condition is desmopressin, a V₂ receptor agonist.

Mechanism of action of desmopressin is: It stimulates the V₂ receptors on the basolateral membrane of the principal cells in the collecting tubules. This increases the insertion of water channels (aquaporin 2) into the apical membrane. This leads to increased water reabsorption in the collecting ducts. Activation of V₂ receptors also increase the permeability of urea.

Option B: Amiloride is a potassium-sparing diuretic. It is indicated for lithium-induced nephrogenic diabetes insipidus.

Option C: Thiazides paradoxically exert an antidiuretic effect in DI but are not the drugs of choice. They are generally used in nephrogenic DI.

Option D: Triamterene is also a potassium-sparing diuretic that can be used to treat nephrogenic DI.

Note: Patients with DI usually do not develop overt signs of dehydration due to increased intake of water to compensate for polyuria. But when water intake is impaired they can present with dehydration.

Solution to Question 2:

Desmopressin is preferred in the treatment of central diabetes insipidus as it is a selective agonist of the V₂ receptor.

Desmopressin has minimal or no action on V₁ receptors resulting in little vasoconstrictor activity. It is more potent than vasopressin. It is longer acting than vasopressin as enzymatic degradation of desmopressin is slow.

Lypressin acts on both V₁ and V₂ receptors. Terlipressin is a synthetic vasopressin analog that has more selectivity for the V₁ receptor and produces fewer adverse effects than lypressin.

Solution to Question 3:

Furosemide is not used for the treatment of diabetes insipidus. It is a loop diuretic that might further worsen diabetes insipidus.

Option A: The drug of choice for central diabetes insipidus is desmopressin. Desmopressin is a V₂ receptor agonist and synthetic analog vasopressin.

Option C: Chlorpropamide enhances the action of circulating vasopressin and reduces urine formation in central DI.

Option D: Carbamazepine also reduces urine formation in central DI. The exact mechanism is unknown.

Solution to Question 4:

Tachyphylaxis refers to the rapid development of tolerance when doses of a drug are repeated in quick succession. It is a concern when desmopressin is used in bleeding disorders because long-term use of desmopressin can cause depletion of factor VIII and von Willebrand factor stores.

The usefulness of desmopressin in bleeding disorders is limited to preoperative preparation, postoperative bleeding, excessive menstrual bleeding, and emergency situations. In patients with type I von Willebrand disease and in some with type II vWD, desmopressin elevates von Willebrand factor and shortens the bleeding time.

Option A: It does not cause tachyphylaxis in central DI.

Option B: Desmopressin is not indicated in nephrogenic DI.

Option D: Bedtime administration of desmopressin for primary nocturnal enuresis provides a high response rate that is sustained with long-term use. However, intranasal desmopressin is no longer recommended for the treatment of primary nocturnal enuresis because of the increased risk of hyponatremia.

Solution to Question 5:

The given clinical scenario is suggestive of syndrome of inappropriate ADH secretion (SIADH) which can occur as a paraneoplastic syndrome of small cancer of the lung. Desmopressin is not used in the treatment of SIADH.

Since excess vasopressin is the cause for SIADH, giving a vasopressin analog like desmopressin would only worsen the condition.

Management of SIADH:

- Fluid restriction
- Hypertonic saline(3%) + loop diuretics is the treatment of choice in severe symptomatic hyponatremia.
- Tolvaptan and conivaptan (vasopressin receptor antagonists) are preferred for long-term use.
- Demeclocycline (a tetracycline drug) is used as an alternative.

Solution to Question 6:

The best approach in the treatment of chronic SIADH is the oral administration of a vaptan like tolvaptan, which is a selective V2 receptor antagonist. This increases urinary water

excretion by blocking the antidiuretic effect of AVP.

Other treatment options for chronic SIADH include:

- Fluid intake < 500 ml/day
- Demeclocycline
- Fludrocortisone

Increased thirst is a frequently observed characteristic of hyponatremia in SIADH. Fluid restriction is usually a burden to the patient. It is unreliable, impractical, and slow to work. As thirst is particularly difficult to suppress, water restriction in SIADH frequently leads to non-compliance in the long run. Vaptans would be especially suited for the treatment of these patients. Serum sodium concentrations are also corrected by vaptans and hence, vaptans substantially improve the quality of life in these patients.

Solution to Question 7:

Conivaptan is a nonselective vasopressin receptor antagonist.

Conivaptan antagonizes both V1a and V2 receptors. It is administered intravenously and is used in the treatment of euvoletic and hypervolemic hyponatremia. Vaptans have been proven to mobilize ascites and improve hyponatremia in patients with cirrhosis.

Vasopressin receptor blockers are classified as:

Classification of receptor blockers	Drugs
Nonselective (mixed V1/V2)	Conivaptan
V1 selective(V1 RA)	Relcovaptan
V1b selective(V3 RA)	Nelivaptan
V2 selective(V2 RA)	LixivaptanMozavaptanSatavaptanTolvaptan

Solution to Question 8:

Tolvaptan should not be used in patients with liver disease as it has been associated with liver damage. Administration of tolvaptan is generally limited to 30 days to limit the risk of hepatotoxicity.

Solution to Question 9:

Tolvaptan causes increased free water clearance, which causes dehydration. Hence, the patient may present with dry mouth and increased thirst.

Tolvaptan is a vasopressin V₂ receptor antagonist. Decreased binding of vasopressin to the V₂ receptor in the kidney lowers the adenylyl cyclase activity leading to decreased intracellular cAMP. Decreased cAMP concentrations prevent the fusion of aquaporin 2 containing vesicles with the plasma membrane, leading to an increase in urinary water excretion (free water clearance), and a decrease in urine osmolality.

The common side effects of tolvaptan include:

- dehydration
- hypernatremia
- hypovolemia
- liver injury

Tolvaptan is used to delay disease progression in patients with ADPKD.

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Sedatives and Hypnotics

Question 1:

Which of the following is a benzodiazepine?

- a) Flurazepam
- b) Flumazenil
- c) Amobarbital
- d) Zaleplon

Question 2:

A physician prescribes clonazepam to a patient with generalized anxiety disorder. Choose the false statement regarding its mechanism of action.

- a) Bind at the α/γ interface of GABA-A receptor
- b) Increase the affinity of the GABA-A receptor for GABA
- c) Activate GABA-A receptors directly opening the Cl channel
- d) Increase the frequency of opening of GABA receptor Cl channel

Question 3:

Which is the shortest acting benzodiazepine?

- a) Remimazolam
- b) Midazolam
- c) Lorazepam
- d) Temazepam

Question 4:

Which of the following does not act via GABA potentiation?

- a) Zopiclone
- b) Midazolam

- c) Thiopentone
- d) Promethazine

Question 5:

A patient diagnosed with insomnia is taking a drug belonging to the Z-compounds family. Which of the following does not belong to this family?

- a) Zopiclone
- b) Zaleplon
- c) Zonisamide
- d) Eszopiclone

Question 6:

Which of the following agents does not bind to GABA receptor chloride channel complex?

- a) Ethanol
- b) Alphaxolone
- c) Zolpidem
- d) Buspirone

Question 7:

Which of the following is an inverse agonist at the benzodiazepine receptor?

- a) Phenobarbitone
- b) Beta-carboline
- c) Flumazenil
- d) Picrotoxin

Question 8:

A 33-year-old woman with suspected diazepam overdose was administered the antidote for benzodiazepine poisoning. Which of the following statements is false about this antidote?

- a) It is a specific antagonist of benzodiazepines

- b) It may be used to treat barbiturate poisoning
- c) It is administered intravenously
- d) It acts at the same site on GABA channel where benzodiazepines bind

Question 9:

Which of the following is a competitive antagonist of GABA at the GABA-A receptor?

- a) Flumazenil
- b) Picrotoxin
- c) Bicuculline
- d) Propofol

Question 10:

In which of the following patients is the administration of barbiturates contraindicated?

- a) 29-year-old lady with history of anxiety disorder
- b) 33-year-old man with history of acute intermittent porphyria
- c) 50-year-old man with refractory status epilepticus
- d) 11-day-old neonate with kernicterus

Question 11:

A 28-year-old man with insomnia is prescribed a melatonin analog by his physician. Which of the following drugs would have been prescribed?

- a) Doxepin
- b) Suvorexant
- c) Ramelteon
- d) Trazodone

Question 12:

A young woman, who is blind, presented with complaints of night-time insomnia. What is the drug of choice for her condition?

- a) Agomelatine
- b) Zolpidem
- c) Tasimelteon
- d) Ramelteon

Question 13:

Your senior resident asks you to prescribe an orexin receptor inhibitor to a patient with insomnia. Which of the following drugs would you prescribe?

- a) Suvorexant
- b) Doxepin
- c) Ritanserin
- d) Agomelatine

Question 14:

A 68-year-old man with alcoholic liver disease is scheduled to undergo surgery. Which of the following benzodiazepines should be avoided in him?

- a) Lorazepam
- b) Oxazepam
- c) Triazolam
- d) Diazepam

Answer Key

Question No.	Correct Option
1	a
2	c
3	a
4	d
5	c
6	d
7	b

8	b
9	c
10	b
11	c
12	c
13	a
14	d

Detailed Explanations

Solution to Question 1:

Flurazepam is a benzodiazepine.

Class of drugs	Mechanism of action	Examples
Benzodiazepines	Facilitate the action of GABA by increasing the frequency of opening of GABA mediated chloride channel	Alprazolam, Clonazepam, Lorazepam, Midazolam, Oxazepam, Temazepam, Estazolam, Flurazepam
Barbiturates	Facilitate the action of GABA by increasing the opening duration of GABA mediated chloride channel	Amobarbital, Phenobarbital, Butobarbital, Pentobarbital
Z drugs	Act like benzodiazepines	Zaleplon, Zolpidem, Zopiclone, Eszopiclone
Melatonin receptor agonists	Activates MT ₁ and MT ₂ receptors	Ramelteon, Tasimelteon
Orexin antagonist	Blocks orexins, neuropeptides that promote wakefulness	Suvorexant, Daridorexant
5HT receptor agonist	Affinity for 5HT > D ₂ receptor	Buspirone

Solution to Question 2:

Clonazepam is a benzodiazepine. These are allosteric modulators of the GABA_A receptor. They do not directly activate GABA_A receptors to open the Cl⁻ channel.

GABA_A receptor is a transmembrane inhibitory receptor in the CNS that functions as a chloride ion channel and has 5 subunits.

Benzodiazepines are allosteric modulators of the GABA_A receptor that bind at the α/γ interface of the receptor. This increases the affinity of the GABA_A receptor for GABA thus increasing the

frequency of the opening of the channel.

Barbiturates, on the other hand, have a different binding site to the receptor and they increase the duration of the GABA gated chloride channel opening.

Solution to Question 3:

Remimazolam is the shortest-acting benzodiazepine. Its half-life is just around 37-53 mins.

Classification of benzodiazepines:

- Long-acting:

- Flurazepam

- Diazepam

- Quazepam

- Chlordiazepoxide

- Intermediate-acting:

- Alprazolam

- Temazepam

- Lorazepam

- Short-acting:

- Triazolam

- Midazolam

- Ultra-short acting

- Remimazolam

Note: Remimazolam was approved by FDA in 2020.

Solution to Question 4:

Promethazine does not act via GABA potentiation. It is a first-generation H1 receptor antagonist.

Zopiclone, midazolam, and thiopentone act via GABA potentiation.

Solution to Question 5:

Zonisamide is a sulfonamide anticonvulsant drug, that has no activity on GABA receptors. It does not belong to the Z-compound drugs. It is an adjunct drug for Parkinson's disease, partial-onset seizures, infantile spasm, mixed seizure types of Lennox–Gastaut syndrome, myoclonic and generalized tonic-clonic seizure.

Z compounds are benzodiazepine receptor agonists. They have less anticonvulsant and muscle relaxant properties when compared to benzodiazepines.

They are used for the treatment of insomnia. They include:

- Zolpidem
- Zaleplon
- Zopiclone
- Eszopiclone

Solution to Question 6:

Buspirone does not bind to the GABA receptor chloride channel complex. It is a partial agonist at 5-HT_{1A} receptor used in the treatment of anxiety.

Option A: Ethanol releases GABA and also has a binding site on the GABA receptor chloride channel complex.

Option B: Alphaxolone is a neurosteroid and it acts through activation of GABA_A receptors. These receptors are BZD insensitive.

Option C: Zolpidem belongs to novel benzodiazepine receptor agonists called Z compounds. It binds to the BZD receptor present on the GABA receptor chloride channel complex.

Solution to Question 7:

The inverse agonist at the benzodiazepine receptor is beta-carboline.

It impedes the action of GABA and can cause anxiogenic, convulsive, and memory-enhancing effects.

Option A: Barbiturates (phenobarbitone) are agonists at an allosteric site on the GABA_A receptor that facilitate the action of GABA.

Option C: Flumazenil is a competitive antagonist at the benzodiazepine receptor.

Option D: Picrotoxin blocks Cl⁻ channel non-competitively.

Solution to Question 8:

Flumazenil is the antidote used in benzodiazepine poisoning. It is not used to treat barbiturate poisoning.

Flumazenil is a benzodiazepine (BZD) receptor-specific antagonist. It has no affinity for barbiturate binding sites on the GABA receptor Cl⁻ channel complex. Flumazenil competes with BZD agonists and inverse agonists for the BZD receptors.

Flumazenil is available only for intravenous use. It is used to treat benzodiazepine overdose and for the reversal of sedative effects produced by benzodiazepines administered during general anesthesia.

Solution to Question 9:

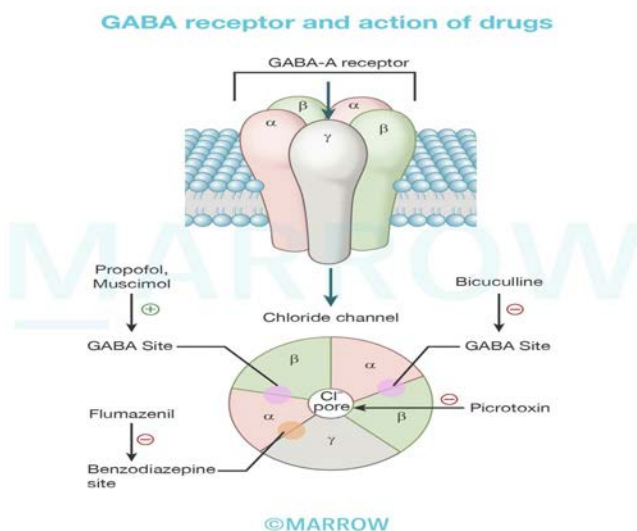
Bicuculline is a competitive antagonist of GABA at the GABA-A receptor.

Option A: Flumazenil is a competitive antagonist at the alpha/gamma interface of the benzodiazepine site. This is different from the binding site of GABA which is the alpha/beta interface.

Option B: Picrotoxin blocks Cl⁻ channel non-competitively.

Option D: Propofol is an agonist at the GABA-A receptor.

Note: Muscimol is an agonist at the GABA-A receptor.



Solution to Question 10:

The administration of barbiturates is contraindicated in acute intermittent porphyria.

The rate-limiting enzyme in porphyria synthesis is δ -aminolaevulinic acid (ALA)-synthetase. Barbiturates are enzyme inducers and they result in the increased synthesis of porphyrins and subsequent precipitation of acute intermittent porphyria.

Options A and C: Barbiturates are used to treat anxiety and refractory status epilepticus.

Option D: Phenobarbitone can be used to treat kernicterus in newborns as it increases hepatic glucuronosyltransferase (UGT) enzyme activity and the conjugation of bilirubin.

Solution to Question 11:

Ramelteon is a melatonin analog approved for the treatment of insomnia, particularly with difficulties of sleep onset.

It acts on both MT1 and MT2 receptors.

- MT1 receptor - promotes the onset of sleep (hence used for induction of sleep)
- MT2 receptor - shifts the timing of the circadian system.

The other melatonin agonists are:

- Tasimelteon - used in insomnia in blind people
- Agomelatine - used in major depressive disorder

Option A: Doxepin is a tricyclic antidepressant and functions as a serotonin-norepinephrine reuptake inhibitor. It is used in depression, anxiety, and some forms of insomnia.

Option B: Suvorexant is a dual orexin receptor antagonist used in the treatment of insomnia.

Option D: Trazodone is a mixed agonist and antagonist at various serotonin receptors used as an antidepressant and in alcohol dependence.

Solution to Question 12:

The drug of choice for insomnia in blind people is tasimelteon. It is a selective agonist of the melatonin (MT1, MT2) receptors.

Tasimelteon is approved for the treatment of non-24-h sleep-wake syndrome (nighttime insomnia and daytime sleepiness, alternating with asymptomatic periods) in totally blind patients experiencing circadian rhythm disorder.

Solution to Question 13:

Suvorexant is an inhibitor of both orexin1 and orexin2 receptors. It is approved for the treatment of insomnia.

Orexins are produced in the lateral hypothalamus neurons and play a major role in maintaining the sleep cycle. Orexins promote wakefulness, while antagonists at orexin receptors enhance REM and non-REM sleep.

Option B: Doxepin is a tricyclic antidepressant and has H1 receptor antagonism effect. It is used for insomnia at lower doses.

Option C: Ritanserin is a 5HT_{2A/2C} receptor antagonist indicated in insomnia.

Option D: Agomelatine is a melatonin receptor agonist and a 5HT_{2C} receptor antagonist indicated in insomnia.

Solution to Question 14:

Diazepam and its long-acting metabolites can accumulate significantly in elderly patients with liver disease and therefore should not be used in such cases.

Diazepam is a long-acting benzodiazepine. It is metabolized in the liver to active metabolites like nordazepam which are long-acting. Nordazepam is further converted to oxazepam which is conjugated with glucuronic acid to form inactive metabolites.

Lorazepam, oxazepam, and triazolam are also metabolized in the liver but do not form long-acting active metabolites. As such, they can be used in elderly patients with liver disease but with caution.

Oxazepam and lorazepam are intermediate-acting and undergo glucuronide conjugation directly to form inactive metabolites.

Triazolam is a short acting benzodiazepine. It forms alpha-hydroxy active metabolites but they are short-acting and undergo glucuronide conjugation to form inactive metabolites.

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Anti-epileptics I

Question 1:

What is the most important mechanism of antiepileptic activity of phenytoin?

- a) Prolongation of Na⁺ channel inactivation
- b) Reduction in Ca²⁺ influx
- c) Facilitation of GABA response
- d) Increasing the K⁺ influx

Question 2:

A known case of seizure disorder has been on long-term phenytoin therapy. Which of the following statements about this drug would be incorrect?

- a) Extensively bound to plasma proteins
- b) Metabolism follows only zero order kinetics
- c) Requires therapeutic drug monitoring
- d) Metabolized extensively by the CYP enzymes in the liver

Question 3:

Which of the following is an incorrect statement regarding fosphenytoin?

- a) Water soluble prodrug of phenytoin
- b) Fosphenytoin displaces phenytoin from protein-binding sites
- c) Causes hyperphosphatemia in renal failure patients
- d) ECG monitoring is not required when administering IV

Question 4:

A patient presenting to the ER with generalized tonic-clonic seizures is started on phenytoin therapy. All of the following adverse effects can be expected in this patient except?

- a) 1, 3 and 4
- b) 2 and 4

- c) 2 only
- d) 3 and 5

Question 5:

Which of the following features would you not expect to find in a baby born to a woman on phenytoin therapy?

- a) Digital hypoplasia
- b) Gum hyperplasia
- c) Microcephaly
- d) Cleft lip

Question 6:

A patient with recent-onset primary generalized epilepsy develops a skin reaction due to phenytoin sodium. What is the most appropriate course of action?

- a) Switch to clonazepam
- b) Stop and restart phenytoin sodium after two weeks
- c) Switch to sodium valproate
- d) Switch to ethosuximide

Question 7:

A 51-year-old male complains of severe, sharp, piercing intermittent electrical shock-like pain sensation on the left side of his face. The pain was triggered by talking, washing his face, eating, and brushing his teeth. Which of the following drugs is most effective to treat this patient?

- a) Phenytoin
- b) Carbamazepine
- c) Sodium valproate
- d) Lamotrigine

Question 8:

Which of the following statements is true about carbamazepine?

- a) It is indicated in focal seizures with impaired awareness
- b) It is an enzyme inhibitor
- c) It causes hypernatremia
- d) It is the drug of choice for status epilepticus

Question 9:

If carbamazepine is added to ongoing valproate therapy, what adverse effect is risked?

- a) Pancreatitis
- b) Thrombocytopenia
- c) Hepatotoxicity
- d) Hyperammonemia

Question 10:

In patients receiving carbamazepine, which of the following idiosyncratic reactions are not expected to occur?

- a) Rash
- b) Agranulocytosis
- c) Peripheral neuropathy
- d) Steven Johnson syndrome

Question 11:

A 70-year-old man is on carbamazepine therapy. Which of the following is expected to occur in this patient?

- a) Hypernatremia
- b) Hyponatremia
- c) Hypocalcemia
- d) Hypokalemia

Question 12:

An 8-year-old child diagnosed to have focal seizures was initiated on oxcarbazepine. Which of the following would be incorrect regarding the agent being used?

- a) Keto analog of carbamazepine
- b) Enzyme induction potency is less when compared to carbamazepine
- c) Metabolised by glucuronide conjugation
- d) Hyponatremia risk is less when compared to carbamazepine

Question 13:

A neonate diagnosed with rhabdomyosarcoma was started on vincristine, cyclophosphamide, and cisplatin. A week after initiation of treatment, the baby had multiple episodes of seizures, refractory to diazepam and lidocaine. Workup revealed an elevated ADH level and euvoletic hyponatremia. Which of the following drugs can be used to treat this patient?

- a) Carbamazepine
- b) Phenytoin
- c) Oxcarbazepine
- d) Phenobarbital

Question 14:

A 5th-grade student, after receiving multiple complaints from teachers at school for repeated episodes of inattention was brought to the OPD. The mother gave a history that her child had brief spells of staring associated with lip-smacking movements. Which anti-epileptic drug is the most effective for this child?

- a) Carbamazepine
- b) Ethosuximide
- c) Phenytoin
- d) Gabapentin

Question 15:

A child with absence seizures improves after being treated with ethosuximide. Which of the following is the mechanism of action of this drug?

- a) Inhibits L type Ca^{2+} channels
- b) Inhibits N type Ca^{2+} channels

- c) Inhibits T type Ca^{2+} channels
- d) Inhibits R type Ca^{2+} channels

Question 16:

A 10-year-old child is brought to the OPD with a history of vacant stares lasting for about half a minute. The EEG shows a characteristic 3 Hz spike and wave pattern. Which of the following drugs can be used for this condition?

- a) 1, 2 and 5
- b) 2 and 3
- c) 2 only
- d) 3, 4 and 5

Question 17:

Which of the following is not an enzyme inducer?

- a) Phenytoin
- b) Phenobarbital
- c) Carbamazepine
- d) Sodium valproate

Question 18:

Which of the following is not the mechanism of action of sodium valproate?

- a) Prolongation of Na^{+} channel inactivation
- b) Inhibition of T type Ca^{2+} channels
- c) Inhibition of GABA transaminase
- d) Inhibition of GABA transporter

Question 19:

Which of the following are not adverse effects of sodium valproate?

- a) Weight gain

- b) Alopecia
- c) Liver damage
- d) Osteomalacia

Question 20:

Which of the following antiepileptic drugs does not cause fulminant hepatitis as a serious complication?

- a) Sodium valproate
- b) Felbamate
- c) Phenytoin
- d) Vigabatrin

Question 21:

A 23-year-old female patient is admitted to the neurology ward for evaluation of GTCS. She reports that she has been on oral contraceptive pills for the past 6 months. Which of the following drugs would you prescribe to this patient?

- a) Carbamazepine
- b) Phenobarbital
- c) Valproate
- d) Phenytoin

Question 22:

A 16-year-old patient was brought to the ER with bilateral jerking movements of arms. EEG revealed diffuse, symmetric, bilateral 4-6 Hz polyspike and wave discharges with a frontocentral predominance. What is the most appropriate drug for this patient?

- a) Lamotrigine
- b) Levetiracetam
- c) Topiramate
- d) Sodium valproate

Question 23:

A 55-year-old man visits the neurology OPD with complaints of tremors in both his hands which improve on alcohol intake. On examination, there is a significant worsening of tremors towards the end of finger-to-nose testing and head-nodding movements. Which antiepileptic is approved for use in this condition?

- a) Phenobarbital
- b) Phenytoin
- c) Primidone
- d) Valproate

Question 24:

Pseudolymphoma may occur in patients on long-term therapy with:

- a) Carbamazepine
- b) Phenobarbital
- c) Phenytoin
- d) Primidone

Question 25:

What is the most common adverse effect of phenytoin in children and young adolescents on chronic use?

- a) Hirsutism
- b) Osteomalacia
- c) Gingival hyperplasia
- d) Megaloblastic anemia

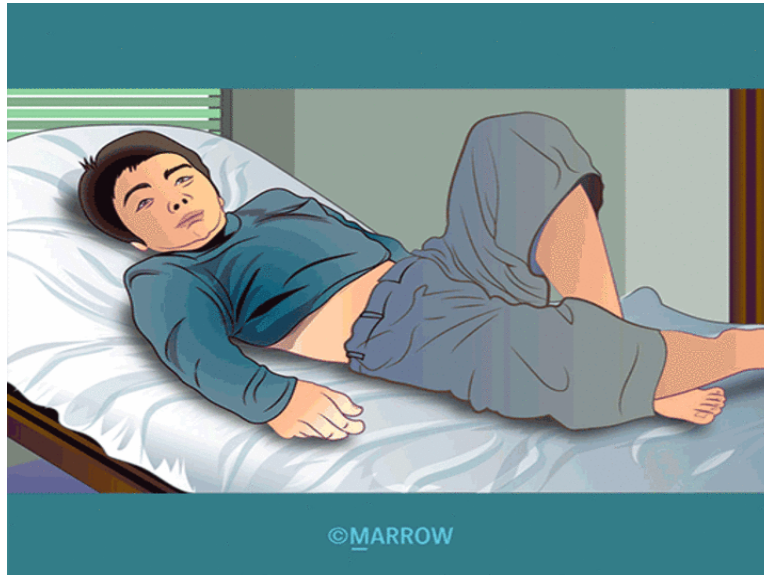
Question 26:

Which of the following is a carbamazepine analogue that has an advantage of once daily dosing?

- a) Oxcarbazepine
- b) Eslicarbazepine acetate
- c) Dacarbazine
- d) Licarbazepine

Question 27:

A patient with a history of mitral stenosis presents with the given intractable finding that is refractory to conventional therapy. What is the drug of choice?



- a) Probenecid
- b) Diazepam
- c) Haloperidol
- d) Sodium valproate

Question 28:

A 10-year-old child is brought to the OPD with a history of vacant stares lasting for about half a minute. The EEG shows ≈ 3 Hz spike and wave pattern. The first-line drug for the management of this condition is:

- a) Sodium valproate
- b) Lamotrigine
- c) Levitiracetam
- d) Felbamate

Question 29:

An 18-year-old girl was brought to the OPD with complaints of excessive growth of hair on her face and chest. She is a known case of epilepsy and is on treatment with phenytoin. Which of the following drugs would not cause this condition?

- a) Metoclopramide
- b) Methyldopa
- c) Amantadine
- d) Phenothiazines

Answer Key

Question No.	Correct Option
1	a
2	b
3	d
4	d
5	b
6	c
7	b
8	a
9	c
10	c
11	b
12	d
13	b
14	b
15	c
16	b
17	d
18	d
19	d
20	d
21	c
22	d
23	c
24	c

25	c
26	b
27	d
28	a
29	c

Detailed Explanations

Solution to Question 1:

The most important mechanism of antiepileptic activity of phenytoin is the prolongation of Na^+ ion channel inactivation. It slows down the rate of recovery of Na^+ channels and thus limiting the propagation of the action potential.

Phenytoin also inhibits voltage-gated Ca^{2+} channels. This reduces the influx of calcium inside the membrane and reduces the inflow of potassium ions during repolarization.

At higher or toxic doses, phenytoin facilitates GABA response and inhibits the action of glutamate.

Solution to Question 2:

Phenytoin follows first-order kinetics at lower concentrations and as the concentration increases, the elimination changes to zero-order kinetics. The rate of elimination of phenytoin varies with the serum concentration.

As the blood levels rise, the maximal capacity of the liver to metabolize phenytoin is reached. Thus even a small increase in the dosage can result in disproportionally high plasma levels. Therefore, it requires therapeutic drug monitoring.

About 90% of phenytoin is bound to plasma proteins, mainly albumin. Phenytoin is metabolized by the CYP enzymes in the liver.

Solution to Question 3:

Fosphenytoin like phenytoin requires continuous ECG monitoring during IV infusion due to the risk of arrhythmias.

Fosphenytoin is a water-soluble prodrug of phenytoin. Fosphenytoin is also extensively protein bound, mainly albumin. This binding is saturable and fosphenytoin displaces phenytoin from protein-binding sites.

Hydrolysis of fosphenytoin yields phosphate, which increases the risk of hyperphosphatemia in patients with renal failure. It is contraindicated in patients with sinus bradycardia, sino-atrial block, second and third-degree A-V block.

Solution to Question 4:

Phenytoin causes hypocalcemia and hyperglycemia.

It interferes with the metabolism of vitamin D and intestinal calcium absorption and results in hypocalcemia. Hyperglycemia and glycosuria can be seen due to the inhibition of insulin secretion.

Long-term usage of phenytoin can also cause cutaneous pseudolymphoma which is mainly T-cell predominant.

Solution to Question 5:

The newborn baby is unlikely to have gum hyperplasia. A baby born to a pregnant mother who was on phenytoin therapy will be suffering from fetal hydantoin syndrome.

Features of fetal hydantoin syndrome are:

- Microcephaly
- Digital hypoplasia
- Cleft lip/ cleft palate

Gum hyperplasia is the most common adverse effect of phenytoin in children and young adolescents on chronic use.

Solution to Question 6:

Hypersensitivity reaction to phenytoin warrants discontinuation of the drug. Switching to sodium valproate is the most appropriate course of action for this patient.

Sodium valproate is one of the first-line agents indicated for GTCS. Clonazepam and ethosuximide are primarily used in absence seizures and are not indicated for GTCS.

Clonazepam is indicated as adjunctive therapy in the absence seizures when first-line agents (ethosuximide, sodium valproate, lamotrigine) fail to control the seizures adequately.

Solution to Question 7:

The clinical features in this patient are suggestive of a diagnosis of trigeminal neuralgia and carbamazepine is the most effective agent for its treatment.

If carbamazepine is ineffective, then second-line agents like lamotrigine and phenytoin are used.

Sodium valproate although effective for trigeminal neuralgia lacks enough evidence for use.

Solution to Question 8:

Carbamazepine is indicated in focal seizures with impaired awareness.

Carbamazepine is an antiepileptic drug that acts by prolonging the inactivated state of the Na⁺ channel similar to phenytoin. It is used in:

- Focal seizures with impaired awareness
- Focal seizures with intact awareness
- Generalized tonic-clonic seizures
- Trigeminal and related neuralgias
- Manic depressive psychosis.

Option B: Carbamazepine is an enzyme inducer. It is a substrate as well as an inducer of CYP3A4 and CYP2C9.

Option C: Carbamazepine causes hyponatremia and not hypernatremia. Other side effects of carbamazepine include agranulocytosis or aplastic anemia, hepatotoxicity, water retention and hyponatremia (by potentiating ADH action), and dose-related neurotoxicity.

Option D: It is not used in status epilepticus. Lorazepam is the drug of choice for status epilepticus.

Solution to Question 9:

If carbamazepine is added to ongoing valproate therapy, it can result in hepatotoxicity.

A combination of valproate and carbamazepine increases hepatotoxicity. Valproic acid may inhibit carbamazepine clearance and increase steady-state carbamazepine blood levels.

Solution to Question 10:

Peripheral neuropathy is not an idiosyncratic reaction that can occur after carbamazepine use. It is a dose-dependent CNS adverse effect. It can be minimized by gradually increasing the dose from initiation or reducing the maintenance dose.

Idiosyncratic reactions are hypersensitivity reactions like:

- Rash
- Steven-Johnson syndrome
- Eosinophilia
- Lymphadenopathy.
- Diplopia

Serious hematological toxicity includes aplastic anemia and agranulocytosis.

Solution to Question 11:

Carbamazepine is expected to cause hyponatremia in this patient.

Carbamazepine potentiates the action of antidiuretic hormone (ADH). It can result in water retention with decreased osmolarity and hyponatremia, especially in elderly patients with cardiac disease.

Solution to Question 12:

The risk of hyponatremia with oxcarbazepine is higher when compared to carbamazepine.

Option A: Oxcarbazepine is a keto analog of carbamazepine. It is a prodrug. The mechanism of action of oxcarbazepine is by blocking voltage-gated sodium channels (same as carbamazepine).

Option C: Active form is metabolized by glucuronide conjugation.

Option B: Oxcarbazepine does not cause auto-induction of metabolizing enzymes. It can induce CYP enzymes but less potent when compared to carbamazepine.

Solution to Question 13:

The given clinical scenario is suggestive of refractory seizures induced by hyponatremia caused by SIADH (due to anticancer drugs). Phenytoin is used to treat this patient. It is known to decrease ADH secretion and stop seizures by correcting severe hyponatremia.

Options A and C: Carbamazepine and oxcarbazepine increase the antidiuretic hormone (ADH) levels. This leads to increased sensitivity of renal tubules to ADH levels and hyponatremia. Carbamazepine cannot be used in SIADH because of its antidiuretic effects.

Option D: Phenobarbital does not affect ADH secretion or action and thus will be ineffective in stopping the seizures.

Solution to Question 14:

The clinical features are suggestive of absence seizures and ethosuximide is the most effective agent for this condition.

First-line drugs in:

- Typical absence seizures: Ethosuximide (drug of choice), valproate, lamotrigine
- Atypical absence seizures: Sodium valproate

Solution to Question 15:

The mechanism of action of ethosuximide is by inhibition of T-type Ca^{2+} channels in thalamic neurons at clinically relevant concentrations.

It also affects other ion channels affected include voltage-gated sodium channels and potassium channels.

Solution to Question 16:

The given clinical scenario is suggestive of absence seizures. Ethosuximide and valproate are the first-line drugs in this condition.

Absence seizures are characterized by sudden, brief lapses of consciousness without loss of postural control. There is no postictal confusion. The following subtle motor signs may accompany the episodes

- Rapid blinking of the eyelids
- Chewing movements
- Small-amplitude clonic movements of the hands

Hyperventilation has a tendency to provoke absence seizures. First-line drugs include:

- Typical absence seizures: Ethosuximide (drug of choice), valproate, lamotrigine
- Atypical absence seizures: Valproate

Solution to Question 17:

Sodium valproate is not an enzyme inducer. It is metabolized by CYP2C9/19 and is an inhibitor of CYP2C9.

Phenytoin, phenobarbital, and carbamazepine are inducers of CYP and UGT enzymes.

Solution to Question 18:

Sodium valproate has no action on the GABA transporter.

Mechanism of action of sodium valproate:

- Prolongation of sodium channel inactivation
- Inhibition of T-type Ca^{2+} channels
- Inhibition of GABA transaminase

- Increases GABA synthesis from glutamic acid.

Solution to Question 19:

Osteomalacia is not an adverse effect of sodium valproate.

The side effects of valproate (mnemonic: VALPROATE)

- Vomiting
- Alopecia
- Liver toxicity
- Pancreatitis
- Rash
- Obesity
- Anorexia
- Teratogenicity (most common: spina bifida), thrombocytopenia, tremors
- Edema

Solution to Question 20:

Vigabatrin is not associated with fulminant hepatitis. It is largely excreted unchanged in urine without undergoing any hepatic metabolism.

Sodium valproate, felbamate and phenytoin are associated with fulminant hepatitis.

Solution to Question 21:

Valproate is the drug used for this patient as it does not cause contraceptive failure.

Valproate is an enzyme inhibitor and it inhibits the degradation of OCPs. OCPs thus have a longer duration of action and less incidence of contraceptive failure with valproate.

However, other antiepileptic drugs like carbamazepine, phenytoin, phenobarbital, primidone, oxcarbazepine, topiramate are enzyme inducers. They cause increased degradation of OCPs and hence lead to contraceptive failure.

Solution to Question 22:

The clinical features and EEG findings are suggestive of juvenile myoclonic epilepsy. The most appropriate drug for management is sodium valproate.

In patients with myoclonic seizures which are refractory to valproate, clonazepam or clobazam may be added.

Solution to Question 23:

The clinical features are suggestive of a diagnosis of essential tremors in this patient. Primidone is considered to be the first-line therapy for the same along with propranolol.

Bilateral, symmetric tremors involving the hands associated with head nodding (titubation) and worsening of tremors towards the end of goal-directed movements such as drinking from a glass or finger-to-nose testing is suggestive of essential tremors.

Primidone is a barbiturate and functions as a GABA enhancer. It is an enzyme inducer. It is used in the treatment of generalized tonic-clonic seizures and partial seizures.

It has side effects like sedation, ataxia, and hyperactivity.

Solution to Question 24:

Phenytoin on long-term use can result in pseudolymphoma.

This lymphadenopathy can be difficult to distinguish from malignant lymphoma. Although some studies suggest a causal relationship between phenytoin use and Hodgkin's disease, the data are far from conclusive. Almost all lesions are benign and termed as pseudolymphoma.

Solution to Question 25:

Gingival hyperplasia occurs in patients on chronic phenytoin therapy. It is the most common manifestation in children and young adolescents.

Gingival hyperplasia due to phenytoin therapy occurs due to tissue overgrowth secondary to altered collagen metabolism. The toothless portion of gums is not involved. It does not warrant discontinuation of the drug. It can be minimized by maintaining good oral hygiene.

Solution to Question 26:

Eslicarbazepine acetate is an analog of carbamazepine that has the advantage of once-daily dosing. It is a prodrug and is rapidly converted to active licarbazepine.

Oxcarbazepine is an analog of carbamazepine that is also converted to licarbazepine.

Oxcarbazepine does not have the advantage of once-daily dosing like eslicarbazepine acetate.

Dacarbazine is not a carbamazepine analog like the other drugs and is a chemotherapy medication used in the treatment of melanoma and Hodgkin's lymphoma.

Solution to Question 27:

The drug used for refractory intractable rheumatic chorea is sodium valproate, as it doesn't have extrapyramidal side effects.

Sydenham's chorea is characterized by involuntary, purposeless, jerky movements which mainly affect the head and upper limbs. In the tongue, it is visible as the characteristic darting movements.

Medications used to treat Sydenham chorea are:

- Mild cases - Providing a calm environment.
- Severe cases:
 - Dopamine antagonists - haloperidol, pimozide
 - Antiepileptic medications - sodium valproate, carbamazepine.
 - Severe, refractory cases - Steroids (thought to be immune-mediated)
 - IV immunoglobulins, plasmapheresis - limited studies, decreased efficacy (not currently recommended)

Solution to Question 28:

The given clinical scenario and ≈ 3 Hz spike and wave pattern on EEG is suggestive of atypical absence seizures. The first-line drug in the treatment of atypical absence seizures is sodium valproate.

Symmetric 3 Hz spike and wave pattern on EEG is seen in typical absence seizures. First-line drugs in its treatment include:

- Ethosuximide (drug of choice)
- Valproate
- Lamotrigine

Solution to Question 29:

The given clinical vignette of excessive hair growth secondary to treatment with phenytoin is suggestive of hirsutism. Amantadine does not cause hirsutism or hypertrichosis but its characteristic side effect is livedo reticularis (bluish discoloration of the skin).

Following are the drugs causing hirsutism:

- Phenytoin
- Anabolic steroids

- Danazol
- Metoclopramide
- Methyldopa
- Phenothiazines
- Progestins
- Reserpine
- Testosterone

Following are the drugs causing hypertrichosis:

- Cyclosporine
- Diazoxide
- Hydrocortisone
- Minoxidil
- Penicillamine
- Phenytoin
- Psoralens
- Streptomycin

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Anti-epileptics II

Question 1:

Which of the following anti-epileptics is an AMPA glutamate receptor antagonist?

- a) Levetiracetam
- b) Brivaracetam
- c) Perampanel
- d) Rufinamide

Question 2:

Which of the following mechanisms of action is observed with levetiracetam?

- a) GABA transaminase inhibition
- b) GABA transporter inhibition
- c) Binding to synaptic vesicular protein SV2A
- d) Inhibition of neuronal sodium channel

Question 3:

A known epileptic patient on levetiracetam has been seizure-free for 2 years. But he develops agitation and anger issues interfering with his day-to-day activities as a result of the drug intake. Which of the following measures is to be taken next?

- a) Slowly taper the drug over next 6 months
- b) Continue levetiracetam since a 5 year seizure free interval is needed
- c) Stop levetiracetam and start on a different anti-epileptic
- d) Discontinue the drug as he is seizure free

Question 4:

Tiagabine has been initiated as an adjunctive treatment for focal seizures in a patient. What is the mechanism of action of this drug?

- a) Inhibiting GABA transaminase

- b) Increasing GABA synthesis
- c) Inhibiting GABA transporter
- d) Inhibiting glutamate release

Question 5:

A 25-year-old patient diagnosed with Lennox-Gastaut syndrome was prescribed a drug with NMDA antagonist activity. Identify the drug.

- a) Memantine
- b) Riluzole
- c) Ketamine
- d) Felbamate

Question 6:

An antiepileptic agent which acts by opening neuronal potassium channels was prescribed to a patient admitted to the neurology ward. On discharge, the patient was asked to have a regular ophthalmological examination every 6 months. Which drug is most likely prescribed?

- a) Ezogabine
- b) Rufinamide
- c) Vigabatrin
- d) Gabaculine

Question 7:

All of the following statements are true about zonisamide except:

- a) Indicated in focal seizures
- b) Nephrolithiasis is not a side effect
- c) Inhibits T type calcium channel
- d) Blocks low voltage gated Na⁺ channel

Question 8:

Patients on which of the following antiepileptic drugs will experience weight loss?

- a) Sodium valproate
- b) Vigabatrin
- c) Topiramate
- d) Carbamazepine

Question 9:

A 55-year-old patient presented to the OPD with complaints of pain over a healed dermatomal rash on his lateral chest wall. Gabapentin was prescribed to this patient. The analgesic activity of this drug is mainly exerted by which of the following?

- a) Increasing the duration of GABA Cl⁻ channel opening
- b) Inhibiting the GABA transaminase enzyme
- c) Inhibiting the uptake of GABA at the presynaptic membrane
- d) Inhibiting the Ca²⁺ channel current

Question 10:

An infant presents with sudden jerky movements of the head, neck and arms. On EEG evaluation, a characteristic pattern of hypsarrhythmia is observed. Which is the GABA transaminase inhibitor approved for the treatment of this condition?

- a) Vigabatrin
- b) Tiagabine
- c) Progabide
- d) Pregabalin

Question 11:

What is the drug of choice in the treatment of infantile spasms?

- a) ACTH
- b) Phenobarbitone
- c) Carbamazepine
- d) Vigabatrin

Question 12:

Which antiepileptic drug is associated with permanent visual loss?

- a) Topiramate
- b) Ethosuximide
- c) Vigabatrin
- d) Felbamate

Question 13:

Which of the following antiepileptic drugs are approved for prophylaxis in patients suffering from migraines?

- a) 2, 3 and 4
- b) 3 and 5
- c) 1, 4 and 5
- d) 1 and 3

Question 14:

A 22-year-old female is brought to the casualty with seizures for the past 10 minutes. History from relatives suggests no seizure-free intervals in the past 10 minutes. After stabilizing the vitals of the patient, which is the first drug you administer to the patient?

- a) Lorazepam
- b) Diazepam
- c) Phenobarbitone
- d) Thiopentone sodium

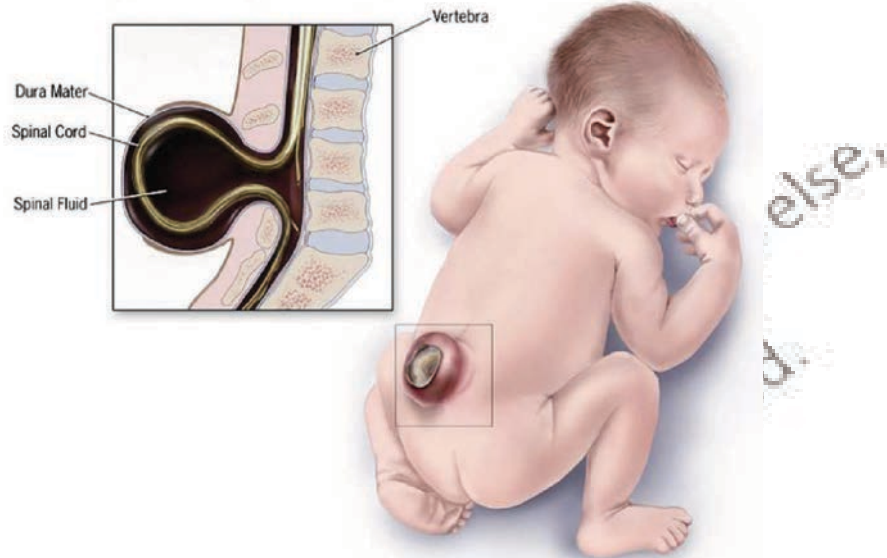
Question 15:

A 25-year-old newly married female comes to the OPD with a positive urine pregnancy test. Her history is suggestive of antiepileptic use since 8 years. Which of the following statements is incorrect about further management in this patient?

- a) Anti epileptic agents has to be continued at stabilised doses
- b) Immediately stop antiepileptics once pregnancy is detected
- c) Folate supplementation is a must
- d) No antiepileptic is deemed safe in pregnancy

Question 16:

An infant is born with a birth defect as shown in the image below. It is the most common teratogenic effect of:



- a) Phenytoin
- b) Topiramate
- c) Sodium valproate
- d) Phenobarbital

Question 17:

Which is the treatment of choice for juvenile myoclonic epilepsy in pregnancy?

- a) Levetiracetam
- b) Carbamazepine
- c) Vigabatrin
- d) Phenytoin

Question 18:

A 29-year-old primigravida, at 4-months period of gestation and history of juvenile myoclonic epilepsy, visits your clinic. She has been regularly taking sodium valproate and is now

concerned about the medication possibly affecting her child. What would you suggest to this patient?

- a) Immediately taper valproate and start lamotrigine
- b) Continue valproate with serial drug monitoring
- c) Switch to carbamazepine
- d) Add lamotrigine to valproate

Question 19:

Which of the following is recommended for prophylaxis in a pregnant woman who is on anti-epileptic drugs with an enzyme-inducing property?

- a) Vitamin K
- b) Anti D immunoglobulin
- c) Indomethacin
- d) Prostacyclin

Question 20:

Which of the following antiepileptics is effective in Lennox-Gastaut syndrome, but shows the highest propensity to cause rashes in young children warranting its withdrawal?

- a) Sodium valproate
- b) Levetiracetam
- c) Lamotrigine
- d) Vigabatrin

Question 21:

Patients receiving which of the following anti-microbial drugs are least likely to develop seizures?

- a) Beta lactam compounds
- b) Quinolones
- c) Macrolides
- d) Isoniazid

Question 22:

Which of the following drugs of abuse is least likely to be associated with seizures?

- a) Amphetamine
- b) Cocaine
- c) Cannabinoids
- d) Phencyclidine

Question 23:

A patient presented to the OPD with a skin rash for 3 days. He gives a history of antiepileptic drug use for 2 months and is ataxic on examination. He is diagnosed to have Stevens-Johnson syndrome. Which of the following drugs are implicated in these adverse effects?

- a) Phenobarbital
- b) Lamotrigine
- c) Topiramate
- d) Ethosuximide

Question 24:

Which of the following drugs if used can lead to behavioral effects such as aggression?

- a) Phenytoin
- b) Phenobarbitone
- c) Levetiracetam
- d) Pregabalin

Question 25:

Which of the following drug is approved for use in fibromyalgia?

- a) Gabapentin
- b) Pregabalin
- c) Tiagabine
- d) Topiramate

Question 26:

What is the first-line treatment for infantile spasms associated with tuberous sclerosis?

- a) ACTH
- b) Vigabatrin
- c) Glucocorticoids
- d) Sodium valproate

Answer Key

Question No.	Correct Option
1	c
2	c
3	a
4	c
5	d
6	a
7	b
8	c
9	d
10	a
11	a
12	c
13	d
14	a
15	b
16	c
17	a
18	b
19	a
20	c
21	c
22	c
23	b
24	c

25	b
26	b

Detailed Explanations

Solution to Question 1:

Perampanel is a selective non-competitive antagonist of the AMPA-type glutamate receptor. It is approved for adjunctive therapy of focal-onset seizures.

Options A and B: Levetiracetam and brivaracetam act on SV2A proteins and modulates GABA and glutamate release.

Option D: Rufinamide acts on voltage-gated sodium channels.

Solution to Question 2:

Levetiracetam has a binding affinity to synaptic vesicular protein SV2A. SV2A plays a role in vesicle recycling following exocytosis of neurotransmitter.

When levetiracetam bind to this synaptic vesicular protein, it decreases the release of excitatory neurotransmitter glutamate thus preventing seizures.

Levetiracetam also inhibits N-type Ca^{2+} channels and Ca^{2+} release from intracellular stores. Thus, it is considered a broad-spectrum antiepileptic as it has an effect on multiple channels.

Solution to Question 3:

The mood changes observed in the above patient must be due to an adverse effect of levetiracetam. Abrupt stoppage of the drug can result in rebound seizures. So the best management will be to slowly taper the drug over the next 6 months.

Adverse effects of levetiracetam:

- Somnolence
- Asthenia
- Ataxia
- Dizziness.
- Mood and behavioral changes
- Psychotic reactions (rare)

Solution to Question 4:

Tiagabine acts by inhibiting the GAT1 GABA transporter, thereby reducing its reuptake into neurons. It also increases the duration of inhibitory impulses in the synapse by prolonging the action of GABA.

Tiagabine is approved as an add-on therapy for refractory partial seizures with or without generalization.

Other antiepileptic drugs that have GABA enhancing activity are vigabatrin and stiripentol.

Solution to Question 5:

Felbamate is the NMDA receptor antagonist prescribed to this patient. It also potentiates GABA action. It is indicated in refractory primary or secondary generalized seizures and Lennox-Gastaut syndrome.

Felbamate is associated with aplastic anemia and liver failure.

Option A: Memantine is used in Alzheimer's disease.

Option B: Riluzole is used in amyotrophic lateral sclerosis.

Option C: Ketamine causes dissociative anesthesia and used in general anesthesia.

Solution to Question 6:

Ezogabine (retigabine) acts by opening neuronal potassium channels. By opening the K⁺ channels, stabilizes the resting membrane potential and reduces the excitability of the nerve membrane.

FDA recommends that patients on chronic ezogabine therapy need visual acuity and fundus examination every 6 months due to the possibility of retinal abnormalities.

Option B: Rufinamide acts by prolonging an inactive state of the Na⁺ channel.

Option C: Vigabatrin is a GABA transaminase inhibitor.

Option D: Gabaculine is an irreversible GABA reuptake inhibitor, that has only experimental value, due to high toxicity.

Solution to Question 7:

Nephrolithiasis is a side effect of zonisamide.

Zonisamide is a broad-spectrum antiepileptic, that acts by blocking low voltage Na⁺ channels and T type calcium channels. It also has carbonic anhydrase inhibiting activity that may result in metabolic acidosis and the formation of renal stones.

It is used in the treatment of focal seizures.

Solution to Question 8:

Topiramate causes weight loss as one of its common side effects.

Topiramate has mixed mechanisms of action:

- Reduces voltage gate Na⁺ currents
- Activates hyperpolarizing K⁺ currents
- Enhances postsynaptic GABAA current
- Limits the activation of AMPA subtype of glutamate receptors.

Topiramate is used in focal seizures, GTCS and Lennox-Gastaut syndrome.

The side effects:

- Cognitive impairment
- Weight loss
- Fatigue
- Renal calculi

Sodium valproate, vigabatrin (highest incidence), and carbamazepine cause weight gain.

Solution to Question 9:

Gabapentin exerts its analgesic activity by inhibiting the Ca²⁺ channel current in this patient who is most likely suffering from post-herpetic neuralgia.

Gabapentin and pregabalin have close structural similarity to GABA. Still, they do not act through the effects of GABA receptors or any other mechanism related to GABA-mediated neurotransmission.

They inhibit the calcium channel current which decreases the entry of calcium in the presynaptic neuron. When calcium entry in the neuron is reduced, there is a decrease in the release of glutamate. This lowers the excitability of the neuron, thus reducing seizure activity.

Therapeutic use of these drugs:

Gabapentin: Focal seizures, postherpetic neuralgia pain

Pregabalin: Focal seizures, postherpetic neuralgia pain, diabetic neuropathy pain, spinal cord injury pain, fibromyalgia pain.

Solution to Question 10:

The given clinical scenario along with hypsarrhythmia on EEG is suggestive of West syndrome. Vigabatrin is an inhibitor of the GABA transaminase that can be used in this

condition.

Inhibition of the enzyme GABA transaminase leads to an increase in the concentration of GABA in the synaptic cleft which is an inhibitory neurotransmitter. This contributes to its anticonvulsant property of vigabatrin.

West syndrome is a rare type of epilepsy syndrome that begins in infancy. It is characterized by a triad of:

- Infantile seizures (also called salaam spells)
- Developmental regression
- Hypsarrhythmia i.e Himalayan wave pattern on EEG

Treatment of West syndrome includes:

- ACTH
- Vigabatrin if associated with tuberous sclerosis

Solution to Question 11:

The drug of choice in the treatment of infantile spasms or West syndrome is adrenocorticotrophic hormone (ACTH.)

The mechanism of action of ACTH or glucocorticoids is unknown but may involve a reduction in inflammatory processes. Vigabatrin is used if the spasms are associated with tuberous sclerosis.

Other antiseizure medications that may be helpful are valproate, topiramate, zonisamide, or benzodiazepines.

Solution to Question 12:

Vigabatrin is associated with progressive and permanent visual loss.

It is a GABA transaminase inhibitor indicated in:

- West syndrome associated with tuberous sclerosis
- Refractory focal seizures with impaired awareness.

Solution to Question 13:

Sodium valproate, topiramate and gabapentin are approved anti-epileptics for use in prophylaxis of migraine.

Carbamazepine, phenytoin and rufinamide are not used for prophylaxis of a migraine.

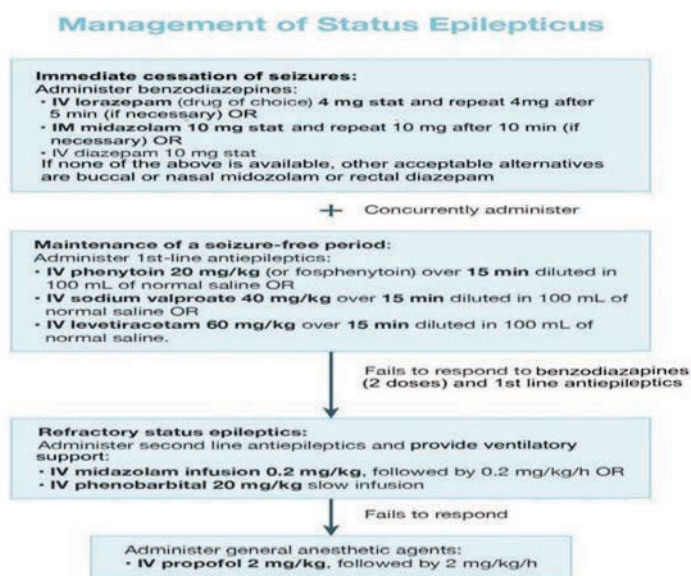
Solution to Question 14:

This scenario described above is of status epilepticus. The first drug used in the management is lorazepam.

Status epilepticus refers to continuous seizures or repetitive, discrete seizures with impaired consciousness in the inter-ictal period. For practical purposes, it is defined as any one of the following:

- Continuing seizures ≥ 5 minutes or
- No regaining of consciousness between 2 seizures.

Management of status epilepticus:



Solution to Question 15:

Stopping antiepileptics abruptly once pregnancy is detected in this patient is not recommended. If this is done, then there is a lot of risk to the fetus by uncontrolled seizures and can result in anoxia and intellectual disability. Antiepileptics have to be continued and then the dose can be tapered slowly if there are no episodes of seizures.

Management of seizures in pregnancy:

- If the patient is a known case of seizure and is on any antiepileptic drug: The same prescribed drug is continued
- If the first episode occurs during pregnancy: The drug preferred is either lamotrigine or levetiracetam
- Folic acid (1-4 mg/day) should be given to all

No antiepileptic is deemed safe in pregnancy. So, it is important to have the patient on monotherapy at the lowest effective dose, especially during the first trimester.

Solution to Question 16:

The above illustration is suggestive of an infant with spina bifida of the open type which is the most common teratogenic effect of sodium valproate.

Option A: Phenytoin is usually associated with fetal hydantoin syndrome.

Option B: Topiramate has shown growth defects in animal studies and hypospadias have been shown in the male fetus but the definite relationship is lacking.

Option D: Phenobarbital has teratogenic effects similar to phenytoin.

Solution to Question 17:

The drug of choice for juvenile myoclonic epilepsy (JME) in pregnancy is levetiracetam. Lamotrigine or levetiracetam are alternative first-line in pregnancy as they have fewer teratogenic effects.

The drug of choice for JME is valproate. However, in pregnant women or women of childbearing age, valproate should be avoided because of the increased risks of fetal malformations and neurodevelopmental delay.

Carbamazepine, vigabatrin, and phenytoin are not indicated in juvenile myoclonic epilepsy as they may worsen myoclonic jerks and absence seizures.

Solution to Question 18:

In the given scenario, the patient should be advised to continue valproate, and serial drug monitoring should be done.

Valproate has a higher teratogenic potential. It is usually avoided before conception if other antiepileptics with less teratogenicity (lamotrigine, levetiracetam) can be used. But for a woman who is pregnant and is already taking valproate, changing the drug is not recommended.

Option A - The patient is presenting in the second trimester when organogenesis would have been complete. Therefore, tapering the drug and introducing lamotrigine is not an option as precipitating seizures has a greater risk of fetal harm at this stage.

Option C: Carbamazepine is not indicated in JME as it may worsen the myoclonic jerks and absence seizures.

Option D- Adding additional drugs to stabilized doses may increase teratogenicity. So adding lamotrigine to valproate is also not a viable option.

Solution to Question 19:

Vitamin K is recommended for prophylaxis in pregnant women who are on antiepileptic drugs having enzyme-inducing properties.

Antiepileptic drugs with enzyme inducing properties:

- Phenytoin
- Carbamazepine
- Oxcarbazepine
- Topiramate
- Phenobarbital
- Primidone

They induce CYP enzymes that interfere with the metabolism of vitamin K. Deficiency of vitamin K-dependent clotting factors can occur in the newborn leading to coagulopathy and intracerebral hemorrhage.

Although hemorrhage is uncommon, the mother should be treated with oral vitamin K, 20 mg/day in the last two weeks of pregnancy. The infant should receive an intramuscular injection of vitamin K (1 mg) at birth.

Solution to Question 20:

Lamotrigine is effective in Lennox-Gastaut syndrome and has the highest propensity to cause rashes in young children.

Lamotrigine is also associated with toxic epidermal necrosis and Stevens-Johnson Syndrome in some patients. The incidence of serious rash and other serious skin conditions is higher with lamotrigine than other anti-epileptics provided in the options. The occurrence of rash warrants the stoppage of the drug.

Option A: Sodium valproate is the drug of choice in Lennox-Gastaut syndrome. It can cause fulminant hepatitis especially in children < 2 years.

Option B: Levetiracetam is also effective in most of the seizures present in Lennox-Gastaut syndrome. It can increase blood pressure in children < 4 years.

Option D: Vigabatrin is also effective in Lennox-Gastaut syndrome, especially as an add-on therapy. It can cause permanent visual field defects.

Solution to Question 21:

Macrolides are the least likely anti-microbial out of the given options to cause seizures.

Antimicrobials associated with causing seizures:

- Beta-lactam and related compounds
- Quinolones

- Acyclovir
- Isoniazid
- Ganciclovir

Other groups of drugs that are implicated in causing epilepsy:

- Alkylating agents - busulfan, chlorambucil
- Antimalarials - chloroquine, mefloquine
- Anesthetics/ analgesics - meperidine, fentanyl, tramadol, local anesthetics
- Immunomodulatory drugs - cyclosporine, tacrolimus, interferons
- Psychotropics:
 - Antidepressants - bupropion
 - Antipsychotics - clozapine
 - Anti-manic drugs - lithium
 - Drugs of abuse - amphetamine, cocaine, phencyclidine, methylphenidate
 - Miscellaneous - Theophylline, memantine.

Solution to Question 22:

Cannabinoids are being investigated for the treatment of various types of epilepsy and are least likely to cause seizures.

Drugs of abuse that are implicated in causing seizures are:

- Amphetamine
- Cocaine
- Phencyclidine
- Methylphenidate

Solution to Question 23:

Lamotrigine is known to cause such adverse effects in patients.

Adverse effects of lamotrigine:

- Skin rash
- Stevens-Johnson syndrome
- Dizziness
- Diplopia
- Sedation

- Ataxia
- Headache

Solution to Question 24:

Children who are on levetiracetam have a higher risk of developing several behavioral side-effects such as aggression, hostility, and nervousness compared to children who do not use levetiracetam.

These side effects are seen in less frequency with brivaracetam.

Solution to Question 25:

Pregabalin is approved for use in fibromyalgia. It is also approved for use in diabetic peripheral neuropathy.

Mechanism of analgesic action:

- Interacts at the binding site of the alpha, delta subunit of voltage-dependent Ca^{2+} channels
- Correlates with decreased Ca^{2+} channel function and release of neurotransmitters
- Decreased neurotransmitter release is associated with reduced neuronal hyperexcitability.

Options A and C: Gabapentin and tiagabine are also useful in fibromyalgia but are not approved for this indication and are used off-label.

Option D: Topiramate's efficacy is not encouraging in fibromyalgia and neuropathic pain.

Solution to Question 26:

Vigabatrin is particularly effective in infantile spasms associated with tuberous sclerosis and is considered the first-line treatment in this condition.

ACTH is considered to be the drug of choice for infantile spasms that are not associated with tuberous sclerosis.

Anti-psychotics

Question 1:

What is the primary mechanism of action of typical antipsychotics?

- a) D2 receptor antagonism
- b) 5HT₂ receptor antagonism
- c) D2 receptor agonism
- d) 5HT₂ receptor agonism

Question 2:

A patient with psychosis was prescribed an antipsychotic medication. Which one of the following side effects cannot be explained by the antagonistic action of the drug on D₂ receptors?

- a) Constipation
- b) Hyperprolactinemia
- c) Affective flattening
- d) Neuroleptic malignant syndrome

Question 3:

A woman presented with sudden onset high-grade fever, muscle rigidity, and altered sensorium. She had been on antipsychotic therapy with chlorpromazine. What is the most probable diagnosis?

- a) Malignant hyperthermia
- b) Neuroleptic malignant syndrome
- c) Anticholinergic syndrome
- d) Serotonin syndrome

Question 4:

A 28-year-old patient presents to the OPD with complaints of a brownish hue to her vision since 2 weeks. She has a history of being on antipsychotic medication for the past 10 weeks. Complete workup reveals pigmentary retinopathy in this patient. Which of the following drugs is implicated in this condition?

- a) Ziprasidone
- b) Thioridazine
- c) Quetiapine
- d) Chlorpromazine

Question 5:

A patient having both positive and negative symptoms of schizophrenia was prescribed an atypical antipsychotic with partial D2 agonism. Identify the drug.

- a) Clozapine
- b) Fluphenazine
- c) Aripiprazole
- d) Risperidone

Question 6:

Which of the following atypical antipsychotics is a preferential D3 receptor partial agonist?

- a) Aripiprazole
- b) Cariprazine
- c) Clozapine
- d) Amisulpride

Question 7:

Which of the following antipsychotics should be given sublingually?

- a) Quetiapine
- b) Asenapine
- c) Olanzapine
- d) Amisulpride

Question 8:

A schizophrenic patient was initiated on haloperidol therapy. 2 days later he presented to the OPD with deviation of his eyes, asymmetric neck posture and inability to open his mouth completely. What is the most probable diagnosis?

- a) Acute dystonia
- b) Tardive dyskinesia
- c) Parkinsonism
- d) Akathisia

Question 9:

A 50-year-old man complains of feeling restless and irritable with an irresistible urge to move. He had recently been started on antipsychotic therapy with haloperidol. He appears agitated and paces up and down the room. What is the diagnosis?

- a) Ekbom disease
- b) Magnan syndrome
- c) Restless leg syndrome
- d) Akathisia

Question 10:

Which of the following drugs does not cause neuroleptic malignant syndrome?

- a) Amantadine
- b) Domperidone
- c) Quetiapine
- d) Metoclopramide

Question 11:

Which of the following effects of chlorpromazine is not due to its antidopaminergic activity?

- a) Hyperprolactinemia
- b) Extrapyramidal side effects
- c) Antiemetic

- d) Orthostatic hypotension

Question 12:

Which of the following atypical antipsychotics has the highest risk of galactorrhea?

- a) Iloperidone
- b) Risperidone
- c) Clozapine
- d) Aripiprazole

Question 13:

A 24-year-old with schizophrenia has not responded to adequate treatment with two different antipsychotics. Which of the following would be the drug of choice for his condition?

- a) Olanzapine
- b) Clozapine
- c) Risperidone
- d) Paliperidone

Question 14:

A 50-year-old man is brought to the casualty following an episode of seizure. He says he was recently switched to a higher dose of his antipsychotic medication. Which drug is he most likely on?

- a) Olanzapine
- b) Clozapine
- c) Aripiprazole
- d) Risperidone

Question 15:

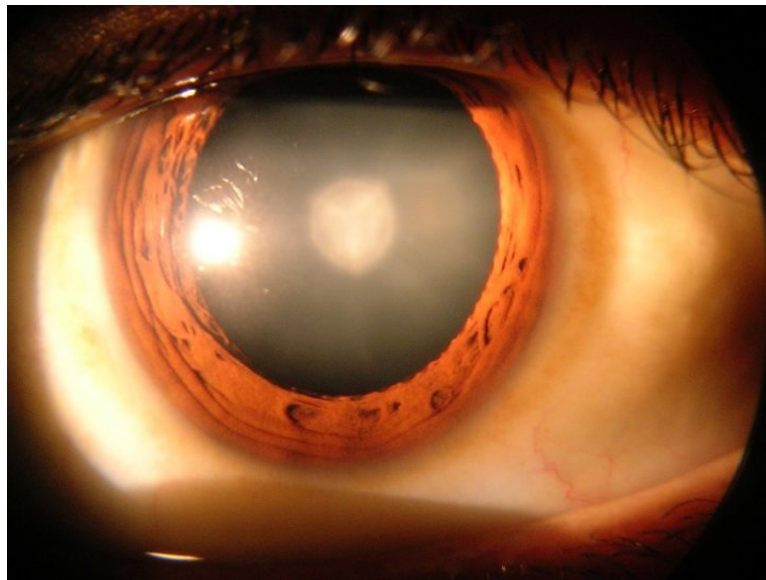
Which of the following is not true regarding clozapine?

- a) More potently blocks D2 as compared to D1 receptors
- b) Drug concentration is kept below 350ng/ml to avoid agranulocytosis

- c) Should not be used along with carbamazepine
- d) Should be discontinued if the neutrophil count is below 500/microL

Question 16:

A 34-year-old woman presented to the ophthalmology OPD with complaints of diminution of vision. She has bipolar disorder and has been on antipsychotic therapy for the past 8 months. Ophthalmic examination revealed the finding seen in the below image . What drug was she most likely on?



- a) Clozapine
- b) Quetiapine
- c) Aripiprazole
- d) Risperidone

Question 17:

Which of the following antipsychotics cause QT prolongation at therapeutic doses?

- a) 1, 2, 5
- b) 3, 4
- c) 1, 3, 4
- d) 1, 2

Question 18:

Patient X and patient Y are on different antipsychotics. X has gained a lot of weight and has been feeling dizzy when she stands. Y has not gained weight, but is very restless and keeps pacing around the room. The drugs they are most likely taking are:

- a) X- Haloperidol, Y- Chlorpromazine
- b) X- Haloperidol, Y- Risperidone
- c) X- Clozapine, Y- Quetiapine
- d) X- Olanzapine, Y- Haloperidol

Question 19:

Which of the following is a side effect of ziprasidone?

- a) Sialorrhea
- b) Lenticular deposits
- c) Prolongation of QT interval
- d) Agranulocytosis

Question 20:

A 45-year-old woman suffering from Huntington's chorea is on haloperidol. Which of the following adverse drug effects is best attributed to the supersensitivity of dopamine receptors?

- a) Muscle rigidity, labile blood pressure, hyperreflexia, hyperthermia, tachycardia, tremors
- b) Lip smacking, facial dyskinesia, protrusion and rolling of the tongue, involuntary movements of the extremities
- c) Facial grimacing, involuntary upward eye movement, laryngeal spasm and arching of the trunk forward
- d) Stooped posture, rigidity, shuffling gait, bradykinesia, and pill rolling tremors

Question 21:

A 24-year-old man is brought by his family with complaints of mental instability. On questioning, he says that he sees his grandparents in the room and talks to them often but the family confirms that it's been 10 years since his grandparents passed away. He was then prescribed an antipsychotic drug following which he developed parkinsonism. Which of the

following antiviral drugs can be used for treating his condition?

- a) Foscarnet
- b) Acyclovir
- c) Amantadine
- d) Interferons

Question 22:

Which of the following statement is false with respect to antipsychotics and pregnancy?

- a) Olanzapine crosses the placenta
- b) Olanzapine is unlikely to increase the risk of birth defects
- c) There is a high risk of gestational diabetes and neonatal seizures with clozapine
- d) Guidelines recommend switching from clozapine to another drug during pregnancy

Answer Key

Question No.	Correct Option
1	a
2	a
3	b
4	b
5	c
6	b
7	b
8	a
9	d
10	a
11	d
12	b
13	b
14	b
15	b
16	b

17	c
18	d
19	c
20	b
21	c
22	d

Detailed Explanations

Solution to Question 1:

Typical or first-generation antipsychotics primarily act by postsynaptic D₂ receptor blockade.

The second-generation or atypical antipsychotics mainly act by blocking 5HT_{2A} receptors but they also have weak D₂ blocking effect.

Histaminic H₁ and muscarinic M₁ receptors are blocked by most of the first and second generation agents. This contributes to sedation and anticholinergic effects of antipsychotics.

Option C: Ropinirole and pramipexole are D₂ agonists used in restless leg syndrome and Parkinson's disease.

Option D: Lorcaserin is a 5HT_{2C} agonist that was used to treat obesity. It was withdrawn in 2020 as it was linked to increased cancer risk.

Antipsychotics	Drugs
First-generation	D ₂ antagonists Chlorpromazine Haloperidol Fluphenazine, Trifluoperazine Thiothixene
Second-generation	5HT _{2A} and D ₂ antagonists Clozapine Loperidone Olanzapine Quetiapine Risperidone Ziprasidone D ₂ partial agonists Aripiprazole Cariprazine D ₂ and D ₃ antagonist Amisulpride 5HT _{2A} inverse agonist without D ₂ binding Pimavanserin

Solution to Question 2:

Constipation and dry mouth are side effects of antipsychotics that have potent antimuscarinic action.

Hyperprolactinaemia, affective flattening, and rigidity in neuroleptic malignant syndrome occur due to D₂ receptor blockade.

Apart from their action on D2 and 5-HT receptors, antipsychotic drugs also exert the following effects:

Action of drug	Example	Side effect
Anti-muscarinic	ChlorpromazineThioridazine Olanzapine	Dry mouth, constipation, dilated pupil, urinary retention
Alpha blocker	ChlorpromazineClozapine	Hypotension, sedation
H1 blocker	Risperidone	Antiemetic, sedation

Solution to Question 3:

The features are suggestive of neuroleptic malignant syndrome (NMS) that is associated with the use of typical antipsychotics like chlorpromazine.

Malignant neuroleptic syndrome is characterized by extreme muscle rigidity, fever, unstable BP and myoglobinemia. It can even lead to death. It can occur weeks to months after initiation of antipsychotic therapy and can persist for up to a week after discontinuation of the drug.

The probable mechanism of this condition is dopamine antagonism.

It is treated by stopping the offending antipsychotic immediately, supportive care, dantrolene, and bromocriptine.

Solution to Question 4:

The pigmentary retinopathy seen in this patient is an adverse effect seen with the use of thioridazine which is a typical antipsychotic.

In advanced cases, the deposits may resemble retinitis pigmentosa and the deposits are associated with 'browning of vision'. The drug is rarely used as it causes prolongation of QTc interval.

Solution to Question 5:

Aripiprazole is an atypical antipsychotic showing D2 partial agonism. It has a higher affinity for D2 receptors than for 5HT2A receptors.

As a partial agonist, aripiprazole has a dual-action which makes it useful in patients with both positive and negative symptoms of schizophrenia.

Options A & D: Clozapine and risperidone are atypical antipsychotics with predominant 5HT2A antagonistic activity over D2 antagonism.

Option B: Fluphenazine is a typical antipsychotic. They have predominant D2 antagonism property and such action on the mesocortical pathway may increase the negative symptoms of

schizophrenia.

Solution to Question 6:

Cariprazine is an atypical antipsychotic which is also a preferential D₃ receptor partial agonist. Like other atypical antipsychotics, it also blocks D₂ and 5-HT₂ receptors.

Option A: Aripiprazole is a dual D₂/D₃ partial agonist but it has no preference for the D₃ receptor.

Option C: Clozapine is a D₂ receptor antagonist.

Option D: Amisulpride is a D₂ and D₃ receptor antagonist.

Solution to Question 7:

Asenapine is only available as an orally disintegrating tablet preparation that has to be administered sublingually.

It undergoes very high first-pass metabolism and so bioavailability is very low via the oral route. Hence, sublingual administration is preferred.

Solution to Question 8:

The given scenario of deviation of the eyes (oculogyric crisis), asymmetric position of the head or neck (torticollis) and inability to open the mouth (trismus) are pointing towards a diagnosis of haloperidol-induced acute dystonia.

These extrapyramidal symptoms (EPS) occur due to the antagonism of dopamine D₂ receptors.

Acute dystonia is more commonly associated with high potency typical antipsychotics like haloperidol and occurs within 1 to 5 days of initiation of therapy. Benztropine or diphenhydramine may be used for treatment.

Solution to Question 9:

The features mentioned are suggestive of akathisia that is commonly associated with the use of high potency typical antipsychotics like haloperidol.

Akathisia is described as subjective complaints of restlessness accompanied by observed movements such as movement of the legs, rocking from foot to foot, pacing, or inability to sit or stand still. It can occur at any time after 5-60 days from starting the drug.

Treatment of akathisia:

- Following a slower dose titration schedule, dose reduction, or switching to a less potent antipsychotic

- First-line drugs - beta-blockers
- Second line drugs - benzodiazepines like lorazepam
- Third line drugs - anticholinergic agents like trihexyphenidyl, benztropine

Solution to Question 10:

Amantadine does not cause neuroleptic malignant syndrome.

Neuroleptic malignant syndrome resembles very severe parkinsonism and is caused by D2 receptor antagonists. Amantadine increases the presynaptic release of dopamine and has agonist and modulatory action at postsynaptic D2 receptors.

Option C: Quetiapine is an atypical antipsychotic. Although this class of antipsychotics has a lower chance of causing NMS compared to typical antipsychotics, but cases of fatal NMS following the administration of quetiapine have been recorded.

Options B and D: Metoclopramide and domperidone are gastro kinetic agents with D2 antagonism. Although domperidone has restricted entry across blood-brain barrier when compared to metoclopramide, there are reports of NMS with this agent.

Solution to Question 11:

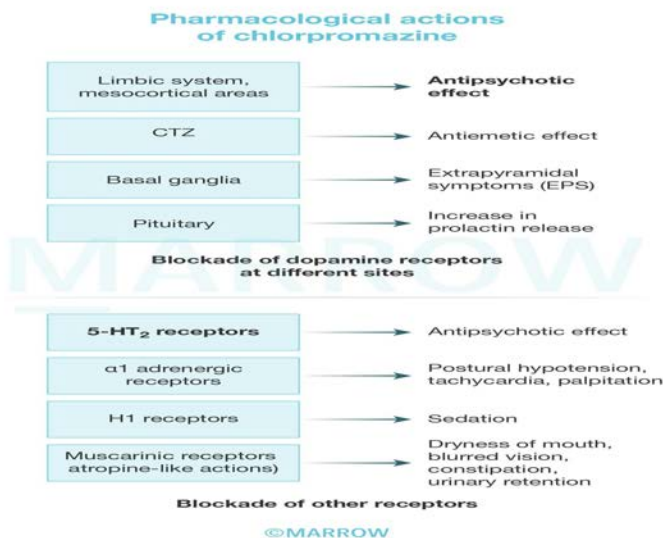
Orthostatic hypotension is due to the alpha-receptor blocking the action of chlorpromazine.

Chlorpromazine is a typical or first-generation antipsychotic with D2 blocking and additional alpha-receptor-blocking and anticholinergic activity.

Option A: Dopamine inhibits prolactin release causing hypoprolactinemia and hence antidopaminergic action of chlorpromazine causes hyperprolactinemia.

Option B: Extrapyramidal side effects occur due to D2 antagonism in the nigrostriatal region.

Option C: The antiemetic action is due to D2 antagonism in the chemoreceptor trigger zone (CTZ).



Solution to Question 12:

Among atypical antipsychotics, risperidone and paliperidone have the highest risk of galactorrhea. The adverse effect occurs due to their higher D₂ blocking potency.

Solution to Question 13:

The drug of choice for treatment-resistant schizophrenia is clozapine.

Treatment-resistant schizophrenia is the lack of response to two different antipsychotics given at adequate dosages for a minimum duration of 4-6 weeks. If patients demonstrate even a mild amount of improvement during this period, it may be reasonable to wait before changing the medication, since patients may improve at a steady rate for 3-6 months.

Solution to Question 14:

The drug most likely implicated here is clozapine as it has the highest risk of precipitating seizures. It causes dose-dependent lowering of the seizure threshold.

Clozapine-associated seizures can be treated with sodium valproate.

Clozapine side effects: (Mnemonic - "Some say Clozapine has too many nasty adverse effects")

- Seizures
- Sedation
- Constipation

- Hypotension/hypersalivation
- Tachycardia
- Mass increase: weight gain and increased appetite
- Nausea and vomiting
- Agranulocytosis (neutropenia)
- Enuresis and polyuria

Solution to Question 15:

Agranulocytosis is an immune-mediated adverse effect and unlike seizures, it is not dependent on dose. Thus, agranulocytosis can happen at any plasma concentration of clozapine and concentration < 350 ng/mL does not lower the risk.

Option A: Clozapine is an atypical antipsychotic, which acts by blocking the 5HT₂, α_2 and D₄ receptors. The potency of block is D₄ $>$ D₂ $>$ D₁.

Option C: Concomitant administration of carbamazepine with clozapine is contraindicated because of the increased risk of leukopenia. In addition, carbamazepine is an enzyme inducer and reduces the half-life of clozapine.

Option D: Administration of clozapine requires regular blood count monitoring due to the risk of agranulocytosis. Monitoring is performed weekly for the first 6 months and once in two weeks for the next 6 months. Patients must be asked to report for evaluation in case of symptoms such as fever or sore throat. Clozapine should be discontinued if the neutrophil count drops below 500 cells/ μ L.

Solution to Question 16:

The image shows a whitish opacity in the pupillary region and this in combination with the presenting features are suggestive of cataract. This is an adverse effect seen with the use of quetiapine which is an atypical antipsychotic.

Solution to Question 17:

Haloperidol, quetiapine and ziprasidone cause QT prolongation at therapeutic doses.

Solution to Question 18:

The most likely option amongst the given options is:

Patient X is taking olanzapine which has side effects of weight gain and hypotension.

Olanzapine and clozapine account for most cases of weight gain and drug-induced diabetes mellitus.

Patient Y is taking haloperidol and has developed the side effect of akathisia. This is an extrapyramidal effect in which the patient is restless, irritable, and feels an urge to keep moving.

Haloperidol has a high risk of developing extrapyramidal side effects such as akathisia, dystonia, and tardive dyskinesia.

Solution to Question 19:

Ziprasidone causes prolongation of QT interval. It should be avoided in patients with a history of cardiac arrhythmias and should not be given in combination with other drugs known to prolong the QT interval.

The other options mentioned are side effects of clozapine.

Solution to Question 20:

Lip-smacking, facial dyskinesia, protrusion and rolling of the tongue, and involuntary movements of the extremities are characteristic of tardive dyskinesia. This is caused by supersensitivity of dopamine receptors.

It occurs as a consequence of chronic postsynaptic D2 receptor blockade by antipsychotics, which results in upregulation and supersensitivity of postsynaptic dopamine receptors.

Tardive dyskinesia is usually seen in patients treated with typical antipsychotics (first generation) for more than a year. This may be reversible with early recognition and drug discontinuation. Treatment is with valbenazine, deutetrabenazine (VMAT2 inhibitors).

Solution to Question 21:

Amantadine is an antiviral agent that is the most commonly used medication for antipsychotic-induced parkinsonism.

It increases the concentration of dopamine (DA) at the synapse and appears to act by certain mechanisms like:

- presynaptic DA reuptake blockade
- facilitation of DA release
- postsynaptic DA agonism
- receptor modulation

Solution to Question 22:

The guidelines recommend continuing the same antipsychotic, as switching the drug during pregnancy would cause a higher rate of relapse.

Antipsychotic drugs and pregnancy:

- Crossing the placenta: Olanzapine > Haloperidol > Risperidone > Quetiapine.
- Olanzapine is unlikely to increase the risk of birth defects above the background population risk. However, it can cause low birth weight and increased NICU admissions.
- Clozapine is associated with gestational diabetes and neonatal seizures.

Note: Although antipsychotic drugs appear to be relatively safe in pregnancy, the available data does not provide adequate guidance on the choice of agent. Because of limited experience in humans, olanzapine should be used in pregnancy only when the potential benefit justifies the potential risk to the fetus.

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Anti-manic Drugs

Question 1:

Which is a mood stabiliser with an anti-suicidal role?

- a) Carbamazepine
- b) Lamotrigine
- c) Lithium
- d) Valproate

Question 2:

A 25-year-old man is brought to the psychiatry OPD by his mother with complaints that he has been behaving aggressively for the last 10 days. He believes he owns the biggest automobile company in the world and talks about how his made-up company will put satellites into space someday. She also says that he sleeps for only 3 hours a day. Which of the following drugs cannot be used to treat this patient?

- a) Sodium valproate
- b) Lamotrigine
- c) Carbamazepine
- d) Lithium

Question 3:

A patient is being treated with lithium for manic-depressive psychosis. Which of the following is true about the use of this drug?

- a) Can be given alone for acute episodes
- b) Serum concentration monitoring is not used for dose adjustment
- c) Cause benign and reversible flattening of T wave on ECG
- d) Causes significant weight loss

Question 4:

A patient on lithium therapy is diagnosed to be hypertensive. Which of the following antihypertensive drugs is contraindicated here?

- a) Clonidine
- b) Beta blockers
- c) Calcium channel blockers
- d) Thiazide diuretics

Question 5:

A patient with a history of mania, on lithium therapy, is brought with complaints of vomiting and multiple episodes of diarrhea. On examination, coarse tremors and an ataxic gait are noted. What is the expected level of serum lithium in this patient?

- a) > 1.2 mEq/L
- b) > 0.8 mEq/L
- c) > 1.5 mEq/L
- d) > 0.5 mEq/L

Question 6:

At what serum lithium levels would you consider dialysis in acute lithium toxicity?

- a) > 2.0 mEq/L
- b) > 4.0 mEq/L
- c) > 3.0 mEq/L
- d) > 1.5 mEq/L

Question 7:

Patients with which of the following conditions do not warrant lithium therapy?

- a) Chemotherapy induced neutropenia
- b) Major depression
- c) Vasculogenic headache
- d) Generalised anxiety disorder

Question 8:

Which of the following must be administered initially in patients with acute mania with psychotic features?

- a) Lithium
- b) Olanzapine
- c) Carbamazepine
- d) Sodium valproate

Question 9:

A patient with mania has been initiated on lithium therapy. While counseling the patient about the medication, which of the following effects do you mention as the most common CNS adverse effect?

- a) Ataxia
- b) Slurred speech
- c) Fine tremors
- d) Coarse tremors

Question 10:

A patient on chronic lithium therapy is diagnosed to have hypothyroidism. What should be the next course of action?

- a) Stop lithium immediately and start exogenous thyroid replacement
- b) Stop lithium immediately and wait for normalisation of thyroid status
- c) Continue lithium along with exogenous thyroid replacement
- d) Decrease the dose of lithium and wait for normalisation of thyroid status

Question 11:

A 34-year-old woman presents with complaints of increased frequency of urination and fatigue. She gives a history of lithium intake for bipolar mood disorder for the past 6 years. On investigations, her serum osmolality was 300 mOsm/L and her urine osmolality was 250 mOsm/L. Which of the following drugs is used to treat this patient?

- a) Amiloride

- b) Demeclocycline
- c) Thiazide diuretics
- d) Indomethacin

Question 12:

A female patient is on first-line pharmacological therapy for bipolar disorder for 6 months. She is fasting for a few days due to religious reasons and presented with seizures, coarse tremors, confusion, and weakness of limbs. Which of the following investigations should be done to help diagnose her condition?

- a) Serum potassium level
- b) Serum lithium level
- c) ECG
- d) MRI

Question 13:

A 28-year-old woman on lithium therapy now presents with a positive urine pregnancy test. Which of the following teratogenic effects can be expected in the fetus?

- a) Cardiac defects
- b) Facial defects
- c) CNS defects
- d) Urogenital defects

Question 14:

Which drug, when given to a mother in her third trimester of pregnancy, would result in the following finding in her newborn?



- a) Venlafaxine
- b) Diazepam
- c) Promethazine
- d) Clozapine

Answer Key

Question No.	Correct Option
1	c
2	b
3	c
4	d
5	c
6	b
7	d
8	b
9	c
10	c
11	a
12	b
13	a
14	b

Detailed Explanations

Solution to Question 1:

Lithium is a mood stabilizer with an antisuicidal role.

Maintenance treatment with lithium has shown to decrease the rate of suicide in mood disorders. Lithium was found to be better than other mood stabilizers such as carbamazepine, lamotrigine, and valproate.

Lithium is also used in recurrent depression with a cyclic pattern is controlled by either lithium or imipramine. It is also used to augment standard antidepressant response in acute major depression in those patients who have had an inadequate response to monotherapy.

Solution to Question 2:

The clinical features given are suggestive of acute mania in this patient and lamotrigine is not indicated in this condition.

It requires a slow, extended titration to minimize the risk of Stevens-Johnson syndrome and is used for maintenance therapy in bipolar disorder.

Lithium, sodium valproate, carbamazepine, and atypical antipsychotics are indicated in acute mania.

Lithium is indicated in both acute mania and maintenance therapy.

Solution to Question 3:

Prolonged use of lithium causes a benign and reversible flattening of T-wave.

Option A: Lithium is usually not given alone in acute episodes as the onset of antimanic action is slow. Benzodiazepines, anticonvulsants, or atypical antipsychotics are administered along with lithium in the early phases of treatment.

Option B: Lithium has a very narrow therapeutic index and monitoring of serum concentration is used for guiding dose adjustment.

Option D: Lithium causes significant weight gain.

Solution to Question 4:

Thiazides are contraindicated in patients who are on lithium therapy.

Thiazide diuretics cause significant reductions in Li^+ clearance that result in toxic levels.

The K⁺ sparing diuretics have modest effects on lithium excretion, with a mild increase in serum levels.

Loop diuretics have a limited impact on Li⁺ serum levels.

Solution to Question 5:

Clinical features are suggestive of lithium toxicity, where serum levels ≥ 1.5 mEq/L are considered toxic.

Lithium has a narrow therapeutic index (0.8–1.2 mEq/L) and therefore requires monitoring to suspect toxicity. The toxic effects of lithium usually start manifesting when the serum level crosses 1.5 mEq/L.

Acute intoxication is characterized by:

- Vomiting
- Profuse diarrhea
- Coarse tremors
- Ataxia
- Coma
- Convulsions.

A fine action tremor is common in patients taking lithium at therapeutic doses and can develop at any time during treatment. Therapeutic lithium tremor is generally symmetric, limited to the arms, and nonprogressive. Tremor from lithium toxicity is coarser and more disabling, may also affect the legs, and occurs with other symptoms of toxicity.

Serious effects of lithium toxicity will involve confusion, hyperreflexia, and seizures. Motor and cognitive side effects can become irreversible with persistent cerebellar tremor being the most common.

Solution to Question 6:

In acute overdoses, hemodialysis is indicated in:

- Serum lithium level is ≥ 4.0 mEq/L
- Patients exhibiting symptoms of toxicity.

Note: While hemodialysis is considered when serum lithium level is 4.0 mEq/L and above, signs of lithium toxicity usually start showing when the serum level is ≥ 1.5 mEq/L. One should not get confused between the two.

Solution to Question 7:

Lithium is not indicated in generalized anxiety disorder.

Bipolar disorder is the primary indication for lithium.

Option A: Effect of leukocytosis may hasten recovery from cancer chemotherapy-induced leukopenia and agranulocytosis.

Option B: Unipolar depression can be treated with lithium as an adjuvant in resistant cases.

Option C: Vasculogenic headaches like cluster headaches.

Solution to Question 8:

In acute mania, even in patients without overt psychotic manifestations, atypical antipsychotics like olanzapine is used initially for rapid control of psychosis. Atypical antipsychotics are also effective as monotherapy.

The therapeutic effects of lithium, carbamazepine, and valproic acid are seen only after a few days. Therefore they are effective only with daily dosing with adequate serum monitoring.

Solution to Question 9:

The most common CNS adverse effect of lithium in the therapeutic dose range is fine postural hand tremors. Therapeutic lithium tremor is generally symmetric, limited to the arms, and nonprogressive.

Options A and B: Ataxia, slurred speech, and/or incoordination are adverse effects seen during peak CNS/serum concentrations. Single bedtime dosing can avoid these adverse effects.

Option D: Coarse tremors are seen in case of lithium toxicity and is more disabling and may also affect the legs.

Solution to Question 10:

Hypothyroidism developed in a patient on chronic lithium therapy is treated by exogenous replacement of thyroid hormones.

It is not a reason for the discontinuation of lithium therapy.

Solution to Question 11:

The given clinical scenario with a history of lithium intake and investigations showing $\uparrow$ serum osmolality and $\downarrow$ urine osmolality is suggestive of nephrogenic diabetes insipidus secondary to lithium toxicity. Amiloride is the drug of choice for this condition.

Amiloride is a potassium-sparing diuretic. It inhibits the renal epithelial Na⁺ channel (ENaC channel) in collecting duct cells. It blocks the entry of Li⁺ ions through the ENaC channels and thus mitigates diabetes insipidus induced by lithium.

Thiazide diuretics cause significant reductions in Li⁺ clearance that result in toxic levels. Thiazides are contraindicated in patients who are on lithium therapy.

Solution to Question 12:

The clinical features given in the question are most likely suggestive of lithium toxicity, and hence serum lithium levels should be measured to confirm it.

Lithium is a mood stabilizer with a very narrow therapeutic index and is the first line of management in bipolar disorder.

The kidney handles both lithium and Na⁺ in the same way via ENaC channel. Thus during fasting/dehydration, along with Na⁺ reabsorption in renal tubules, there will be an increase in the reabsorption of lithium leading to lithium toxicity.

Lithium has a very narrow therapeutic index of 0.8-1.2 mEq/L. The severity of lithium toxicity is often divided into the following three grades:

A fine tremor is common in patients taking lithium at therapeutic doses. Therapeutic lithium tremor is generally symmetric, limited to the arms, and nonprogressive. The tremor from lithium toxicity is coarser and more disabling, may also affect the legs, and occurs with other symptoms of toxicity. Other adverse effects include mental confusion, hypothyroidism, nephrogenic diabetes insipidus, and edema.

As lithium is readily dialyzed, lithium overdose or toxicity is treated with hemodialysis.

*Note - Serum Na⁺ levels are significant with respect to lithium toxicity, whereas serum potassium levels become significant in digoxin toxicity. Hypokalemia increases the risk of digoxin toxicity.

Grade	Sr. lithium concentration(in mEq/L)	Symptoms
Mild symptoms	1.5-2.5	nausea, vomiting, tremor, and fatigue
Moderate intoxication	2.5-3.5	confusion, agitation, delirium, tachycardia, and hypertonia
Severe intoxication	>3.5	coma, seizures, hyperthermia, and hypotension

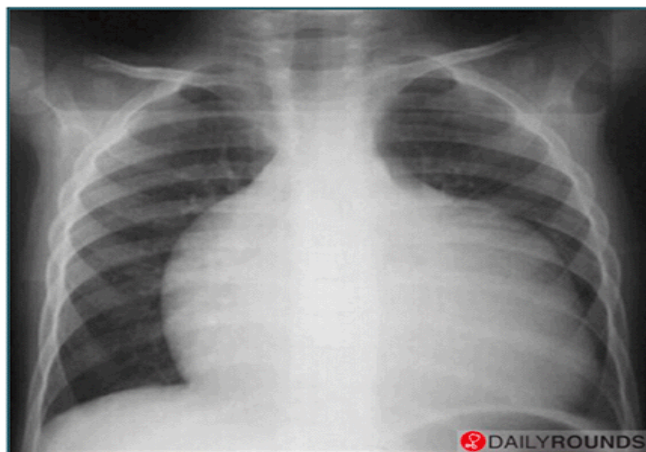
Solution to Question 13:

Lithium use in pregnancy can lead to cardiac defects.

Ebstein's anomaly is the specific cardiac defect with lithium use in pregnancy.

In Ebstein's anomaly, the radiological sign seen is a box-shaped heart.

Ebstein's Anomaly - box-shaped heart



Solution to Question 14:

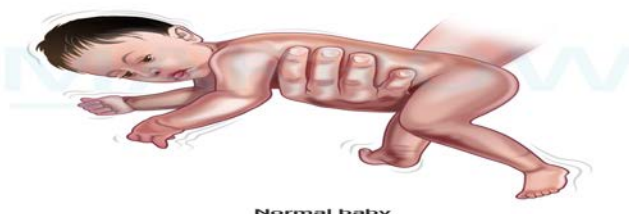
Benzodiazepines, when given during the third trimester of pregnancy can lead to floppy baby syndrome.

Benzodiazepines in pregnancy:

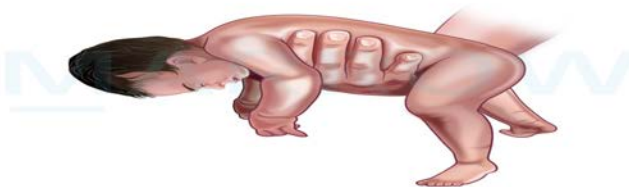
- First trimester - cleft palate, GI anomalies
- Third trimester - floppy baby syndrome with neonatal hypotonia, as a withdrawal effect.

Note: Fetal lithium exposure is also associated with floppy baby syndrome.

Normal baby vs Floppy baby



Normal baby



Floppy baby

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Drugs for Parkinson's Disease

Question 1:

Which of the following drugs crosses the blood-brain barrier?

- a) Dopamine
- b) Levodopa
- c) Both A and B
- d) Carbidopa

Question 2:

A 55-year-old is diagnosed with Parkinson's disease and is started on levodopa therapy. Which of the following statements is incorrect about the drug?

- a) In parkinsonism, phenothiazines reduce its efficacy
- b) It is a prodrug
- c) Pyridoxine reduces effect of levodopa in parkinsonism
- d) Domperidone reduces levodopa induced emesis and its therapeutic potential

Question 3:

After initiation of levodopa therapy, a 55-year-old patient starts experiencing adverse effects. Which of these side effects is not decreased by the addition of carbidopa?

- a) Nausea and vomiting
- b) On and off effect
- c) Cardiac arrhythmias
- d) Hallucinations

Question 4:

A 60-year-old male is brought to the casualty in a confused state. On examination, he has severe muscle rigidity. His son says that the patient is on treatment for Parkinson's disease but has forgotten to take his medication as they went on a holiday trip. What is the likely

diagnosis in this patient?

- a) Wearing off phenomenon
- b) On / Off phenomenon
- c) Neuroleptic malignant syndrome
- d) Serotonin syndrome

Question 5:

A patient on long-term treatment with levodopa and carbidopa for Parkinson's disease is diagnosed with on/off phenomenon. Which of the following drugs is used to treat this condition?

- a) Safinamide
- b) Amantadine
- c) Benztropine
- d) Domperidone

Question 6:

A patient who is on levodopa therapy presents with confusion, delusions of grandeur, and hallucinations. What is the most appropriate treatment for his condition?

- a) Haloperidol
- b) Chlorpromazine
- c) Trifluoperazine
- d) Quetiapine

Question 7:

While discharging a patient with Parkinson's disease, the treating physician writes a prescription of levodopa along with another drug that can increase the bioavailability of the former. Identify the drug.

- a) Entacapone
- b) Ropinirole
- c) Amantadine
- d) Apomorphine

Question 8:

Which of the following anti-Parkinsonism drugs is correctly matched with its mechanism of action?

- a) Tolcapone - COMT inhibition
- b) Rasagiline - Muscarinic antagonism
- c) Amantidine - MAO-B inhibition
- d) Apomorphine - Inhibition of dopa

Question 9:

Which of the following statements is false regarding the use of selegiline in Parkinsonism?

- a) It acts as a selective MAO A inhibitor
- b) Incidence of cheese reaction is minimal
- c) May be used for on / off phenomenon
- d) It forms amphetamines as metabolites

Question 10:

Which of the following dopamine agonists is an ergot derivative?

- a) 3, 4 and 5
- b) 1, 2 and 3
- c) 1 and 2
- d) 4 and 5

Question 11:

A 50-year-old female patient visits the OPD with complaints of burning sensation associated with an irresistible urge to move her legs since 2 weeks. It occurs especially at night and the discomfort gets relieved on moving her legs. Which is the first-line drug that can be prescribed for this patient?

- a) Benztropine
- b) Gabapentin

- c) Riluzole
- d) Clonazepam

Question 12:

Which of the following dopamine receptor agonists has antioxidant properties?

- a) Bromocriptine
- b) Pergolide
- c) Pramipexole
- d) Ropinirole

Question 13:

Which of the following drugs would you prescribe as a transdermal patch to patients with Parkinson's disease?

- a) Selegiline
- b) Rotigotine
- c) Entacapone
- d) Apomorphine

Question 14:

A 30-year-old female is on treatment with ropinirole for restless leg syndrome. Which of the following adverse effects are not expected to be seen in this patient?

- a) Drowsiness
- b) Postural hypotension
- c) Cardiac fibrosis
- d) Hallucination

Question 15:

Which anti-Parkinsonism drug is an NMDA-glutamate receptor antagonist?

- a) Amantadine

- b) Rotigotine
- c) Benzhexol
- d) Trihexyphenidyl

Question 16:

A 55-year-old patient presents to the dermatology OPD with findings as shown in the image below. Which of the following drugs can result in this adverse effect?



- a) Pergolide
- b) Memantine
- c) Amantadine
- d) Trihexyphenidyl

Question 17:

Which is the 5HT_{2A} inverse agonist approved for use in psychosis associated with Parkinsonism?

- a) Lorcaserin
- b) Pimavanserin
- c) Pitolisant
- d) Mianserin

Question 18:

A 30-year-old male patient presented to the casualty department with bradykinesia, rigidity, and tremors. His attender reported that he had been diagnosed with schizophrenia and had recently started treatment. What is the drug of choice to treat the patient's presenting symptoms?

- a) Levodopa
- b) Benzhexol
- c) Amantadine
- d) Selegiline

Answer Key

Question No.	Correct Option
1	b
2	d
3	d
4	c
5	a
6	d
7	a
8	a
9	a
10	c
11	b
12	c
13	b
14	c
15	a
16	c
17	b
18	b

Detailed Explanations

Solution to Question 1:

Levodopa crosses the blood-brain barrier.

It is a naturally occurring aromatic amino acid and an immediate precursor of dopamine that can cross the blood-brain barrier. Dopamine does not cross the blood-brain barrier because of the polar nature of the molecule. Levodopa is also a polar compound but it can cross the blood-brain barrier through an aromatic amino acid transporter.

Therefore, levodopa is used therapeutically when dopamine is required for central action. Levodopa is converted to dopamine by the decarboxylase enzyme.

Carbidopa is a peripheral decarboxylase inhibitor and does not penetrate the blood-brain barrier.

Solution to Question 2:

Domperidone is an antidopaminergic antiemetic that can reverse levodopa-induced emesis without affecting the therapeutic potential as it cannot cross the blood-brain barrier.

The antiemetic effect is through its action on CTZ which is outside the blood-brain barrier.

Option A: Typical antipsychotics like phenothiazines can reduce the efficacy of levodopa because of its central D2 blocking property.

Option B: Levodopa is a prodrug. It is converted into dopamine by the decarboxylase enzyme in CNS.

Option C: Pyridoxine can reduce the effect of levodopa by increasing its peripheral conversion to dopamine by potentiating peripheral dopa decarboxylase enzyme.

Solution to Question 3:

Behavioral adverse effects like confusion, hallucinations, nightmares, delusions, postural hypotension are not reduced after the addition of carbidopa.

In fact, they are increased in patients who are taking levodopa with peripheral decarboxylase inhibitors. This is because higher levels of dopamine are formed in CNS.

The combination of levodopa with peripheral decarboxylase inhibitors like carbidopa inhibits peripheral adverse effects of dopamine. This decreases the adverse effects like:

- Anorexia, nausea, and vomiting
- Cardiac arrhythmias
- On and off effect of motor performance

Solution to Question 4:

Abrupt withdrawal of levodopa or dopaminergic agonists can precipitate neuroleptic malignant syndrome. It is characterized by extreme rigidity, confusion, and hyperthermia.

It is an emergency condition that can turn fatal if immediate intervention is not done. Neuroleptic malignant syndrome is commonly associated with the use of typical antipsychotics.

Option A: Wearing off phenomenon - Each dose of levodopa improves mobility for a period of 1-2 hours. After that, the effect of the drug wears off, and rigidity and akinesia return. This can be surpassed by increasing the dosing and frequency.

Option B: On/off phenomenon - Rapid fluctuation between on and off periods.

- On period: When the patient has a beneficial effect of the medications but with disabling dyskinesias
- Off period: No beneficial effects from anti-Parkinsonism medications.

'Wearing off' and 'On / Off' phenomenon occur with long-term treatment of levodopa.

Option D: Serotonin syndrome occurs with co-prescription of MAO-inhibitors and SSRIs.

Solution to Question 5:

Safinamide is used for the treatment of the on/off phenomenon.

Management of On/Off phenomenon:

- To prolong the 'On' phase:
- COMT inhibitor - entacapone, tolcapone
- Irreversible MAO-B inhibitor - selegiline, rasagiline
- Reversible MAO-B inhibitor - safinamide.
- Rescue from 'Off' phase: injection apomorphine

Dopaminergic agonists like pramipexole, ropinirole, and rotigotine are also effective in the treatment of the on/off phenomenon.

Solution to Question 6:

The given clinical scenario is suggestive of levodopa-induced psychosis for which quetiapine is the most appropriate treatment.

Quetiapine and clozapine are the two atypical antipsychotics that are most effective and best tolerated in levodopa-induced psychosis.

Typical antipsychotics are effective against levodopa-induced psychosis but may cause marked worsening of parkinsonism. Hence, atypical antipsychotics are deemed safer.

Solution to Question 7:

Entacapone is a peripheral COMT inhibitor that increases the bioavailability of levodopa. It inhibits the peripheral metabolism of levodopa thereby increasing the bioavailability of levodopa.

Options B and D: Ropinirole and apomorphine are dopamine receptor agonists and do not have any significant effect on levodopa bioavailability.

Option C: Amantadine which is an antiviral drug is also anti-Parkinsonian due to its varied actions including increased release and decreased uptake of dopamine. It has no significant effect on the bioavailability of levodopa.

Solution to Question 8:

Tolcapone and entacapone are selective, potent, and reversible COMT (catechol-O-methyl transferase) inhibitors.

COMT metabolizes levodopa and blockade of this pathway prolongs the half-life of levodopa and allows a larger fraction of the administered dose to reach the brain.

Option B: Rasagiline is a selective MAO-B inhibitor.

Option C: Amantadine acts by antagonism of the NMDA type of glutamate receptors.

Option D: Apomorphine is a dopaminergic agonist that is FDA-approved as a 'rescue therapy' for the acute intermittent treatment of 'off' episodes.

Solution to Question 9:

Selegiline acts as a selective MAO-B inhibitor.

At higher doses, MAO-A can also be inhibited. Higher doses are used for depression and administered in the form of the transdermal patch.

Option B: Selective MAO-B inhibitors do not interfere significantly with the peripheral metabolism of dietary amines and do not cause cheese reaction.

Option C: It has lower efficacy than levodopa but when added with levodopa it prolongs its action and also decreases the wearing-off effect of levodopa.

Option D: Selegiline forms metabolites like amphetamine and methamphetamine that can result in anxiety, insomnia, and other undesirable effects. Rasagiline, unlike selegiline, does not form amphetamines as metabolites.

Solution to Question 10:

Bromocriptine and pergolide are ergot derivatives that act as potent agonists at the D2 receptor.

Pramipexole and ropinirole are non-ergot derivatives having D2/D3 receptor agonistic action. They are better tolerated and have fewer GI symptoms than bromocriptine.

Dopamine agonists can be classified as follows:

- Ergot derivatives - Bromocriptine, pergolide, cabergoline
- Non-ergot derivatives - Pramipexole, ropinirole, apomorphine, rotigotine

Dopaminergic agonists exert subtype-selective activation of dopaminergic receptors involved in the Parkinsonism circuit of the striatum.

Solution to Question 11:

The symptoms in this patient are suggestive of restless leg syndrome and gabapentin is the first-line choice of treatment for this condition.

Restless leg syndrome is a typical sensory-motor disorder that generally affects the legs during rest periods, especially sleep. There is an irresistible urge to shake the legs associated with burning sensation, itching, and cramps of the legs. In idiopathic cases, dopaminergic hypofunction is postulated.

In patients with restless legs syndrome, iron stores are initially repleted, and the aggravating factors are eliminated. If RLS still persists, initial therapy with a gabapentinoid (pregabalin, gabapentin, gabapentin enacarbil) are preferred, rather than a dopamine agonist (pramipexole, ropinirole, rotigotine). This is due to the high frequency of intermediate and long-term augmentation with dopamine agonists, as well as their risk of impulse control disorders.

FDA-approved drugs for the treatment of restless leg syndrome:

- Gabapentin
- Ropinirole
- Pramipexole
- Rotigotine

A dopamine agonist is used as an alternative in patients with increased risk for side effects from gabapentinoids, such as those with obesity and its complications, a past or present history of moderate to severe depression, a disorder causing gait instability or respiratory failure, and a history of substance use disorder.

Option A: Benztropine is an anticholinergic that can be used for drug-induced parkinsonism.

Option C: Riluzole is a benzothiazole derivative that is used in amyotrophic lateral sclerosis.

Option D: Benzodiazepines like clonazepam can be used as co-adjuvant drugs in the treatment of restless leg syndrome but should not be used as first-line therapy.

Solution to Question 12:

Pramipexole is a dopamine receptor agonist that has an additional antioxidant effect.

Pramipexole has a possible neuroprotective effect by its ability to scavenge hydrogen peroxide and enhance neurotrophic activity.

Solution to Question 13:

Rotigotine is a dopamine agonist used in Parkinson's disease as a transdermal patch.

A transdermal patch can lead to various local reactions at the application site. Rotigotine was withdrawn from some markets due to crystal formation on the patch but later reintroduced with a newer formulation.

Selegiline transdermal patch is approved for depression and not for Parkinson's disease.

Solution to Question 14:

Cardiac fibrosis is a side effect of pergolide, not ropinirole.

The adverse effects of ropinirole include:

- Nausea
- Drowsiness
- Postural hypotension
- Hallucinations

Cardiac and pleural fibrosis are seen with pergolide which is also a dopamine agonist. Pergolide was withdrawn from the market due to reports of serious valve damage.

Solution to Question 15:

Amantadine which is an antiviral agent, also used in parkinsonism has NMDA glutamate receptor antagonist action.

This action is responsible for its antidyskinetic properties. It also promotes presynaptic synthesis and release of dopamine.

Option B: Rotigotine is a dopamine agonist available as a transdermal patch.

Options C and D: Benzhexol and trihexyphenidyl are central anticholinergics.

Solution to Question 16:

The above presentation is of livedo reticularis, which is an adverse effect of amantadine.

The patient presents with a common skin finding consisting of a mottled vesicular pattern that appears as a lace-like purplish discoloration of the skin which is due to the local release of catecholamines resulting in post-capillary vasoconstriction.

Adverse effects of amantadine are:

- Insomnia, restlessness, confusion
- Nightmares
- Anticholinergic effects
- Hallucinations
- Livedo reticularis
- Edema of ankles

Solution to Question 17:

Pimavanserin is a selective 5-HT_{2A} receptor inverse agonist that is approved for the management of delirium and hallucinations associated with Parkinson's disease.

Option A: Lorcaserin is a 5-HT_{2C} agonist indicated as an anti-obesity agent.

Option C: Pitolisant is an H₃ receptor inverse agonist approved for the treatment of narcolepsy.

Option D: Mianserin is an anti-depressant.

Solution to Question 18:

The given clinical scenario is suggestive of drug-induced parkinsonism, for which the drug of choice is benzhexol, also known as trihexiphenidyl.

Drug-induced parkinsonism is caused by certain drugs that have D₂ antagonistic action, such as typical antipsychotics. Due to the decrease in dopaminergic activity, they result in an unbalanced cholinergic activity. Benzhexol is a centrally acting antimuscarinic agent. It acts by restoring the cholinergic balance in the striatal interneurons.

The following drugs are implicated in drug-induced parkinsonism:

Drug class	Examples
Typical antipsychotics	Phenothiazines, haloperidol
Atypical antipsychotics	Risperidone, olanzapine
Dopamine depleters	Reserpine, tetrabenazine
Antiemetics	Metoclopramide, levosulpiride
Calcium channel blockers	Flunarizine, cinnarizine

Antidepressant and Antianxiety Drugs

Question 1:

Which of the following best describes the mechanism of action of antidepressants?

- a) Change in neurotransmitter receptor sensitivity
- b) Decrease in level of neurotransmitters
- c) Change in efficacy of neurotransmitters
- d) Increase the reuptake of neurotransmitters

Question 2:

Moclobemide belongs to which class of drugs?

- a) SSRI
- b) SNRI
- c) MAO inhibitor
- d) Tricyclic antidepressant

Question 3:

Which of the following is a selective MAO-B inhibitor?

- a) Moclobemide
- b) Isocarboxazid
- c) Phenelzine
- d) Selegiline

Question 4:

In which of the following patients is transdermal selegiline approved for use?

- a) 30-year-old man with obsessive compulsive disorder
- b) 64-year-old man with Parkinson's disease
- c) 29-year-old woman with depression

d) 33-year-old woman with generalized anxiety disorder

Question 5:

A college student on MAO inhibitor therapy for depression consumed two bottles of beer during a party. The next day he started experiencing severe headache, nausea, and vomiting. He was taken to the casualty by his friends where his BP was recorded to be 180/110mmHg. Which of the following compounds would be responsible for this complication?

- a) Tyramine
- b) Tyrttophan
- c) Threonine
- d) Thiamine

Question 6:

A patient presents to the medicine OPD with hypertension. He is already on imipramine for depression. Which of the following antihypertensive drugs will not be effective in him?

- a) Enalapril
- b) Atenolol
- c) Diltiazem
- d) Clonidine

Question 7:

Which tricyclic antidepressant has an additional dopaminergic receptor antagonist action?

- a) Clomipramine
- b) Doxepin
- c) Imipramine
- d) Amoxapine

Question 8:

Which of the following antidepressants can be used for the treatment of a 6-year-old child diagnosed with nocturnal enuresis?

- a) Imipramine
- b) Amitriptyline
- c) Desipramine
- d) Clomipramine

Question 9:

Which of the following side effects of tricyclic antidepressants are correctly matched to the receptor involved in their pathogenesis?

- a) 1, 4 and 5
- b) 1, 2 and 3
- c) 3, 4 and 5
- d) 4 and 5 only

Question 10:

A 30-year-old woman having depression took multiple tablets of amitriptyline in an attempt to commit suicide. Which of the following statements regarding the management is incorrect?

- a) Physostigmine is not recommended
- b) Sodium bicarbonate should be administered to treat acidosis
- c) Atropine sulphate should be administered as an antidote
- d) Diazepam should be injected to control seizures

Question 11:

A psychiatrist plans to prescribe an SSRI to a patient diagnosed with generalized anxiety disorder. Which of the following was not likely prescribed?

- a) Fluoxetine
- b) Citalopram
- c) Paroxetine
- d) Duloxetine

Question 12:

In a middle-aged man with a history of closed-angle glaucoma, which of the following SNRIs should be avoided?

- a) Venlafaxine
- b) Duloxetine
- c) Desvenlafaxine
- d) Fluoxetine

Question 13:

A 27-year-old man on antidepressant therapy with TCAs for the past five years is being switched to SSRIs. Which of the following would not be an advantage of this change?

- a) Minimal lowering of seizure threshold
- b) Minimal anticholinergic side effects
- c) Minimal sexual dysfunction
- d) Minimal sedation

Question 14:

Selective serotonin reuptake inhibitors are not the drug of choice in which of the following patients?

- a) A college student with an acute panic attack
- b) An adolescent girl with social phobia
- c) A young lady with post traumatic stress disorder
- d) A middle-aged man with generalized anxiety disorder

Question 15:

A patient diagnosed with depression has improved symptomatically after initiating SSRIs. On discontinuing the drug, he became irritable and complained of insomnia, nausea, and flu-like symptoms. Which of the following drugs has the least risk of causing these symptoms?

- a) Paroxetine
- b) Fluoxetine
- c) Sertraline
- d) Fluvoxamine

Question 16:

A young man with obsessive-compulsive disorder is taking SSRIs. Which of the following adverse effects would you not expect in him?

- a) Diarrhea
- b) Nausea
- c) Weight loss
- d) Loss of libido

Question 17:

Which of the following drugs is a serotonin receptor antagonist?

- a) Milnacipran
- b) Trazodone
- c) Citalopram
- d) Fluoxetine

Question 18:

Which of the following drugs inhibit dopamine and noradrenaline reuptake?

- a) Clozapine
- b) Bupropion
- c) Zolpidem
- d) Mirtazapine

Question 19:

A young woman diagnosed with major depressive disorder is refractory to first-line antidepressant therapy. The psychiatrist now decides to prescribe a recently approved NMDA antagonist. Which of the following was most likely prescribed?

- a) Brexanolone
- b) Esketamine
- c) Lurasidone

d) Lemborexant

Question 20:

A 30-year-old married man complains of low energy and lack of motivation. He mentions that he feels useless and has no hope for the future. He is willing to take medication but is apprehensive about its side effects on his sexual life. Which of the following drugs would you prescribe to him?

- a) Fluoxetine**
- b) Venlafaxine**
- c) Mirtazapine**
- d) Imipramine**

Question 21:

Which group does the antianxiety drug buspirone belong to?

- a) Partial antagonist of 5-HT 1A**
- b) Partial agonist of 5-HT 1A**
- c) Partial antagonist of 5-HT 2A**
- d) Partial agonist of 5-HT 2A**

Question 22:

A psychiatry resident should not prescribe a tricyclic antidepressant to a patient having which of the following co-morbidities?

- a) Essential hypertension**
- b) Asthma**
- c) Narrow angle glaucoma**
- d) Increased intracranial tension**

Question 23:

A 35-year-old man was admitted to the emergency ward with priapism lasting for about 7 hours. A detailed history revealed the initiation of a drug for a mood disorder by a

psychiatrist 3 weeks ago. Which of the following drugs is responsible for his condition?

- a) Trazodone
- b) Venlafaxine
- c) Bupropion
- d) Mirtazapine

Question 24:

A man feels frustrated because of his wife's decreased sexual desire. She was recently diagnosed with depression and started on a drug for the same. Which of the following drugs was most likely prescribed?

- a) Fluoxetine
- b) Mirtazapine
- c) Bupropion
- d) Trazodone

Question 25:

Which of the following is not an adverse effect of escitalopram?

- a) Nausea
- b) Vivid dreams
- c) Anorgasmia
- d) Sialorrhoea

Question 26:

In children and adolescents treated with an SSRI, there appears to be an increased risk of particular ideations. Clinicians are therefore advised to remain vigilant of which of the following?

- a) Nihilistic ideations
- b) Schizotypal ideations
- c) Suicidal ideations
- d) Homicidal ideations

Answer Key

Question No.	Correct Option
1	a
2	c
3	d
4	c
5	a
6	d
7	d
8	a
9	a
10	c
11	d
12	b
13	c
14	a
15	b
16	c
17	b
18	b
19	b
20	c
21	b
22	c
23	a
24	a
25	d
26	c

Detailed Explanations

Solution to Question 1:

Antidepressants increase levels of neurotransmitters at the synapse by preventing their reuptake. Due to a prolonged increase in the levels of neurotransmitters, there is a change in neurotransmitter receptor sensitivity which is thought to be the major mechanism responsible for the treatment of depression.

Chronic treatment with a number of antidepressants desensitizes presynaptic autoreceptors and heteroreceptors, producing long-lasting changes in monoaminergic neurotransmission. Postreceptor effects of antidepressant treatment, including modulation of GPCR signaling and activation of protein kinases and ion channels, are involved in the mediation of the long-term effects of antidepressant drugs.

Solution to Question 2:

Moclobemide is a reversible and selective inhibitor of the MAO-A enzyme.

Monoamine hypothesis states that in depression, there is a decrease in monoamines like serotonin (5HT) and norepinephrine. These monoamines are metabolized by the enzyme monoamine oxidase (MAO). MAO inhibitors, therefore, increase these monoamines by inhibiting the enzyme.

The various classes of drugs used in depression and anxiety are listed below:

Class of drugs	Examples
Selective serotonin reuptake inhibitors (SSRI)	Fluoxetine Fluvoxamine Paroxetine Sertraline Citalopram
Serotonin-norepinephrine reuptake inhibitors	Venlafaxine Desvenlafaxine Duloxetine Milnacipran
Tricyclic antidepressants	Amitriptyline Imipramine Nortriptyline Desipramine Amoxapine
Atypical antipsychotics	Aripiprazole Olanzapine Quetiapine Risperidone
Monoamine oxidase inhibitors (MAOI)	Isocarboxazid Selegiline Phenelzine Moclobemide
Atypical antidepressants	Bupropion Trazodone Mirtazapine Mianserin

Solution to Question 3:

Selegiline is a selective MAO-B inhibitor.

Nonselective MAO inhibitors	Selective MAO inhibitors
Phenelzine Isocarboxazid Tranylcypromine	MAO-A inhibitors - Moclobemide MAO-B inhibitors - Selegiline, Rasagiline

Solution to Question 4:

Selegiline transdermal patch is approved for depression. The transdermal patch provides the high concentration required for antidepressant action. At higher doses, MAO-A is also inhibited and is highly effective in depression.

Selegiline is a selective MAO-B inhibitor at low doses. It is used in Parkinson's disease at low doses, administered orally.

Solution to Question 5:

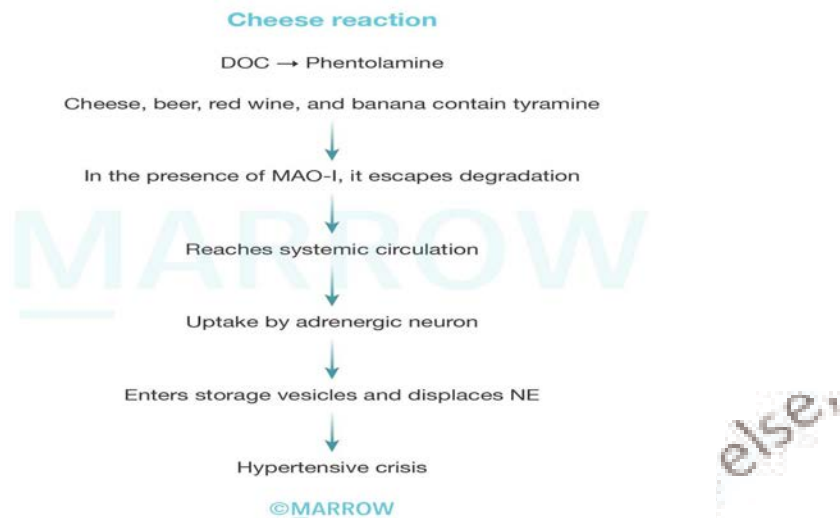
The given clinical presentation of a hypertensive crisis in a patient on MAO inhibitor therapy is suggestive of cheese reaction, which is seen after consumption of foods rich in tyramine (beer).

Tyramine is an indirect sympathomimetic. It induces the release of catecholamines from the neurons. Tyramine can be produced in high concentrations in protein-rich foods by decarboxylation of tyrosine during fermentation.

Normally tyramine is inactivated when taken orally because it is readily metabolized by MAO in the liver. In patients who are on MAO inhibitor therapy, the tyramine taken in the diet is not inactivated and results in its increased sympathomimetic effect resulting in elevated blood pressure, termed as hypertensive crisis or cheese reaction. Drug of choice for cheese reaction is IV phentolamine.

When MAO inhibitors are taken along with SSRI or TCA, it results in serotonin syndrome due to the overstimulation of 5HT receptors. The clinical features are a triad of:

- Cognitive effect - delirium, coma
- Autonomic effects- hypertension, tachycardia
- Somatic effects- myoclonus, hyperreflexia, tremors



Solution to Question 6:

Clonidine will not be effective in this patient.

Imipramine, a tricyclic antidepressant would abolish the antihypertensive effect of clonidine, by inhibiting its transport into the adrenergic neurons.

Side effects of tricyclic antidepressants:

Receptor blocked	Side effects
Muscarinic receptor	Anticholinergic - Dry mouth, dilated pupils, tachycardia, urinary retention
Alpha 1 adrenergic receptor	Drowsiness, postural hypotension, sexual dysfunction
H1 receptor	Weight gain, drowsiness

Solution to Question 7:

The tricyclic antidepressant having an additional dopaminergic receptor antagonist action is amoxapine.

Tricyclic antidepressants block the transporter of norepinephrine (NET) and serotonin (SERT) thereby preventing their uptake. Some of the TCA additionally block other receptors like alpha 1, H1, and M1.

- TCA blocking NET selectively - desipramine, nortriptyline, amoxapine.
- TCA blocking both NET and SERT equally - imipramine, amitriptyline.

Clomipramine, doxepin, and imipramine are tertiary amines and do not possess any dopaminergic antagonist effect.

Solution to Question 8:

Imipramine is a tricyclic antidepressant that can be used for the treatment of nocturnal enuresis.

In children above 5 years, imipramine 25 mg at bedtime is effective for nocturnal enuresis.

Intranasal desmopressin is also effective in the treatment of nocturnal enuresis.

Option B: Amitriptyline is used in postherpetic neuralgia. It is also used in neuropathic pain.

Option C: Desipramine is used in the treatment of cocaine dependence.

Option D: Clomipramine is the best TCA for OCD.

Solution to Question 9:

The blockade of H₁, α₁, and muscarinic receptors causes sedation, postural hypotension, and constipation respectively.

Tricyclic antidepressants (TCAs) are drugs that act by inhibiting the reuptake of noradrenaline and serotonin from the synaptic clefts.

The side effects of TCAs are due to blockade of other receptors:

- Muscarinic (M) receptors: constipation, dry mouth (not sialorrhea), arrhythmia
- Histamine (H₁) receptors: sedation
- 5HT receptors: weight gain due to increased appetite
- Alpha₁ receptors: postural hypotension

Solution to Question 10:

Tricyclic antidepressants like amitriptyline are competitive antagonists of muscarinic receptors. Atropine being an antimuscarinic drug should not be administered.

Tricyclic antidepressant overdose presents with anticholinergic effects. Hypotension occurs because of α₁ receptor blockade.

TCAs act on Na⁺ channels and cause slowed conduction and depressed cardiac contractility resulting in serious arrhythmias. Seizures and agitation may occur followed by depression.

Management of TCA overdose:

- Initial resuscitation by securing airway, breathing, and circulation.
- Hypotension - IV fluids and norepinephrine infusion.

- Cardiac-depressant effects - bolus dose of sodium bicarbonate helps overcome Na^+ blockade by increasing extracellular Na^+ . It also addresses metabolic acidosis which is often present.
- Arrhythmias - lignocaine for ventricular arrhythmias.
- Seizures - benzodiazepines like diazepam/lorazepam.

Physostigmine can reverse anticholinergic effects but can aggravate cardiac depression and cause seizures. Therefore physostigmine is not recommended.

Solution to Question 11:

Duloxetine is not an SSRI but a serotonin-norepinephrine reuptake inhibitor (SNRI). It blocks the uptake of both serotonin and norepinephrine.

The selective serotonin reuptake inhibitors (SSRIs) are:

- Fluoxetine
- Fluvoxamine
- Citalopram
- Escitalopram
- Sertraline
- Dapoxetine
- Paroxetine

Solution to Question 12:

Duloxetine should be avoided as it is contraindicated in patients with closed-angle glaucoma. SNRIs like duloxetine and milnacipran have anticholinergic side effects like mydriasis which can precipitate closed-angle glaucoma attacks.

Duloxetine is the longest-acting SNRI with no withdrawal symptoms. In addition to the treatment of depression and anxiety, it is also used in fibromyalgia, peripheral neuropathy, and stress incontinence.

Option A: Venlafaxine is the shortest acting SNRI with the maximum risk of withdrawal symptoms. It is approved for use in PTSD, panic disorder, and depression associated with hypotension. It is contraindicated in pregnancy.

Option C: Desvenlafaxine is a longer-acting SNRI and hence has lesser risk of withdrawal symptoms.

Option D: Fluoxetine is an SSRI, not an SNRI. It is avoided as it can cause mydriasis and narrow the angle.

Solution to Question 13:

SSRIs have a higher incidence of sexual dysfunction when compared to TCAs.

SSRI-induced sexual dysfunction can result in loss of libido, delayed ejaculation, and interference with orgasm.

Option A: The potential of SSRIs to lower the seizure threshold is less than that of TCAs.

Options B and D: SSRIs have a selective action on serotonin reuptake with no effect on cholinergic or histaminergic receptors.

Solution to Question 14:

In acute panic attacks, benzodiazepines or propranolol are the preferred anxiolytics. SSRIs are not the drug of choice as they have a long latent period before beneficial effects are seen.

SSRIs are considered as the drug of choice for:

- Social anxiety disorder
- Post-traumatic stress disorder
- Generalized anxiety disorder
- Obsessive-compulsive disorder

Solution to Question 15:

The given clinical presentation is suggestive of drug discontinuation or withdrawal symptoms which are seen after sudden discontinuation of SSRIs. Fluoxetine has the least risk of causing drug-discontinuation symptoms.

Drug discontinuation symptoms of SSRIs are:

- Nausea
- Paresthesia
- Flu-like symptoms
- Insomnia
- Irritability
- Fatigue.

Withdrawal is least with fluoxetine as its metabolite norfluoxetine has the longest half-life. Thus the plasma level of norfluoxetine is maintained for a long time even after stopping the drug, minimizing withdrawal symptoms. Withdrawal is seen maximum with paroxetine.

Solution to Question 16:

Weight loss is not an adverse effect of SSRIs.

Side effects of SSRIs:

- Nausea and diarrhea (early)
- Loss of libido
- Erectile dysfunction
- Headache
- Sleep disturbances - insomnia or hypersomnia.

Solution to Question 17:

Trazodone is a serotonin receptor antagonist.

Trazodone blocks postsynaptic 5HT_{2A} receptors and is considered an atypical antidepressant. It has weak serotonin reuptake inhibitory effect.

Nefazodone is another serotonin receptor antagonist.

Option A: Milnacipran is a serotonin and norepinephrine reuptake inhibitor.

Options C and D: Citalopram and fluoxetine are selective serotonin reuptake inhibitors.

Solution to Question 18:

Bupropion is an atypical antidepressant that inhibits the reuptake of noradrenaline and dopamine.

Indications of bupropion are:

- Prevention of a seasonal depressive disorder
- Smoking cessation
- ADHD
- Peripheral neuropathy

Option A : Clozapine is an atypical antipsychotic that selectively blocks dopamine (D₄) receptors in the brain.

Option C : Zolpidem - It is a nonbenzodiazepine hypnotic. It acts as a selective agonist in benzodiazepine receptor.

Option D : Mirtazapine - block histamine (H₁) receptors.

Solution to Question 19:

Esketamine is an NMDA antagonist recently approved for the treatment of treatment-resistant depression.

Esketamine is a non-selective, non-competitive antagonist of the NMDA receptor, an ionotropic glutamate receptor. Though the exact mechanism by which it acts as an antidepressant is unknown, a single dose of intranasal Esketamine has shown to have short-term antidepressant efficacy suggesting a possible use in suicidality.

Option A: Brexanolone has GABAA modulatory action. It is given via intravenous route for post-partum depression.

Option C: Lurasidone is a second-generation antipsychotic that is an antagonist at dopamine (D2) and serotonin (5-HT2A) receptors. It is approved for use in depressive episodes in bipolar disorder.

Option D: Lemborexant is an orexin receptor antagonist indicated for the treatment of adult patients with insomnia.

Solution to Question 20:

The given clinical presentation is suggestive of depression. Mirtazapine can be prescribed to this patient as sexual dysfunction is not a side effect associated with its use.

Mirtazapine and mianserin block histamine (H1) receptors, and also have high antagonistic activity against 5HT2A, 5HT2C, and 5HT3 receptors.

Due to its sedative effect, mirtazapine is also used in depression with insomnia.

Option A: Fluoxetine is an SSRI that can cause side effects like GI disturbances, risk of suicidal behavior, sexual dysfunction, and weight gain. Vilazodone is an SSRI that is not associated with sexual dysfunction or weight gain.

Option B: SNRIs like venlafaxine have side effects that include nausea, dizziness, sexual dysfunction, and risk of suicidal thoughts or behavior.

Option D: TCAs like imipramine have side effects like orthostatic hypotension, weight gain, GI disturbances, sexual dysfunction, seizures, and irregular heartbeats.

The various classes of drugs used in depression and anxiety are listed below

Class of drugs	Examples
Selective serotonin reuptake inhibitors (SSRI)	FluoxetineFluvoxamineParoxetineSertralineCitalopram
Serotonin-norepinephrine reuptake inhibitors	VenlafaxineDesvenlafaxineDuloxetineMilnacipran
Tricyclic antidepressants	AmitriptylineImipramineNortriptylineDesipramineAmoxapine
Atypical antipsychotics	AripiprazoleOlanzapineQuetiapineRisperidone

Class of drugs	Examples
Monoamine oxidase inhibitors (MAO)	Isocarboxazid Selegiline Phenelzine
Atypical antidepressants	Bupropion Trazodone Mirtazapine Mianserin

Solution to Question 21:

The antianxiety drug buspirone is a partial agonist of 5-HT_{1A} receptors. The most important locations of the 5-HT_{1A} receptor are raphe nuclei of brainstem and hippocampus.

It mimics the antianxiety properties of benzodiazepines but does not interact with GABA_A receptors or display the sedative and anticonvulsant properties of benzodiazepines. Buspirone is primarily used in the treatment of generalized anxiety disorder.

5-HT_{1A} partial agonists are:

- Buspirone
- Gepirone
- Ipsapirone

Solution to Question 22:

Tricyclic antidepressants produce anticholinergic effects of pupillary dilatation and are thus, contraindicated in narrow-angle glaucoma.

Side effects of tricyclic antidepressants:

Receptor blocked	Side effect
Muscarinic receptor	Anticholinergic- dry mouth, dilated pupils, tachycardia, urinary retention
Alpha 1 adrenergic receptor	Drowsiness, postural hypotension, sexual dysfunction
H ₁ receptor	Weight gain, drowsiness

Solution to Question 23:

Trazodone causes priapism. Intracavernous injection of phenylephrine can be used to treat priapism.

Solution to Question 24:

Fluoxetine is a selective serotonin reuptake inhibitor (SSRI) that is associated with the side effect of sexual dysfunction.

Fluoxetine is used in the treatment of major depression, obsessive-compulsive disorder, post-traumatic stress disorder, bulimia and anorexia nervosa, generalized anxiety disorders, and premenstrual dysphoric disorder.

Other adverse effects include dry mouth, urinary retention, insomnia, GI disturbances, risk of suicidal behavior, and weight gain.

Solution to Question 25:

Sialorrhoea is not an adverse effect of escitalopram.

Escitalopram is a selective serotonin reuptake inhibitor (SSRI) mainly used to treat major depressive disorder or generalized anxiety disorder.

Nausea is the commonest side effect of SSRIs as they target the 5HT₃ receptor.

Vivid dreams may occur because the drug suppresses REM sleep (the stage of dream), hence when patients eventually fall into REM sleep it may be more intense.

Anorgasmia is the persistent inability to achieve orgasm despite responding to sexual stimulation. A proposed mechanism of escitalopram causing sexual dysfunction involves stimulation of the 5HT_{2C} receptors.

Solution to Question 26:

Selective serotonin reuptake inhibitors (SSRIs) should be used carefully in the young for the management of depression due to increased suicidal ideation. In children and adolescents, there appears to be a slightly increased risk of suicidal ideations and attempts, but not of completed suicide.

Somatic symptoms of depression improve faster than psychological symptoms when a patient is started on SSRIs and hence patients have the energy to act on their thoughts of deliberate self-harm and suicide (paradoxical suicide).

Opioids - Functions and Classification

Question 1:

A middle-aged woman with resistant intractable back pain is planned to receive neuraxial opioids for analgesia. Which of the following is the site of action of this drug in the spinal cord?

- a) Ventral horn of spinal cord
- b) Dorsal horn of spinal cord
- c) Lateral horn of spinal cord
- d) No spinal component is seen with opioid analgesia

Question 2:

A primigravida in labor is administered opioids for epidural analgesia. Which of the following statements is false regarding the drug administered?

- a) Act on dorsal horn substantia gelatinosa
- b) Can cause itching
- c) Function of the GIT is not affected
- d) Can cause respiratory depression

Question 3:

A patient is receiving morphine for pain management. Which of the following findings would you not expect?

- a) Urinary retention
- b) Vomiting
- c) Pruritus
- d) Mydriasis

Question 4:

An experimental drug that acts only as an agonist at the mu-opioid receptor is being developed. Which of the following effects would be seen with this drug?

- a) 1 & 4 only
- b) 1, & 3 only
- c) 1, 2 & 3 only
- d) 2 & 5 only

Question 5:

Dysphoria is mediated by which opioid receptor?

- a) Mu
- b) Kappa
- c) Delta
- d) Zeta

Question 6:

A patient is administered intravenous opioids for postoperative pain management. This drug elicits its analgesic action through which of the following receptors?

- a) 1, 2 & 3
- b) 1 & 3 only
- c) 1, 2, 4 & 5
- d) 1, 2, 3 & 4

Question 7:

Following surgery, a woman received multiple doses of intravenous opioids for pain management. Later, her respiratory rate became 8 per minute. Which receptor mediates this adverse effect?

- a) μ
- b) κ
- c) δ
- d) Both μ & κ

Question 8:

Which of the following drugs does not aggravate opioid-induced respiratory depression?

- a) Alcohol
- b) Barbiturate
- c) Diazepam
- d) Naloxone

Question 9:

Which of the following is an opiate?

- a) Pentazocine
- b) Heroin
- c) Fentanyl
- d) Morphine

Question 10:

What is the other name for diacetylmorphine?

- a) Codeine
- b) Heroin
- c) Methadone
- d) Meperidine

Question 11:

A patient who has a history of substance abuse with a semisynthetic opioid is now referred to a rehabilitation center. Which of the following drugs is he likely addicted to?

- a) Morphine
- b) Heroin
- c) Codeine
- d) Thebaine

Question 12:

A 24-year-old pregnant woman is in labor. Your senior administered her a synthetic opioid to provide labor analgesia. Which of the following would be the drug administered to her?

- a) Heroin
- b) Dihydromorphone
- c) Butorphanol
- d) Buprenorphine

Question 13:

A 54-year-old woman is receiving acupuncture treatment for her back pain. Which of the following opioid peptides does not mediate analgesia in her?

- a) Endorphin
- b) Nociceptin
- c) Enkephalin
- d) Morphine

Question 14:

A patient presented with biliary colic. The intern has administered an analgesic, following which the pain got aggravated. Which of the following could be responsible for the aggravation of the patient's symptoms?

- a) Morphine
- b) Diclofenac
- c) Nefopam
- d) Etoricoxib

Question 15:

You decide to administer morphine for analgesia in your patients. In which of the following patients would you avoid morphine?

- a) A 50 year old man with hypotension
- b) A 32 year old woman with head injury

- c) A 44 year old woman with hypothyroidism
- d) A 21 year old man with type 1 diabetes mellitus

Question 16:

Which of the following statements are true?

- a) 1, 3, 4 & 5
- b) 1, 2, 3 & 5
- c) 1, 3 & 5
- d) 3 & 5 only

Question 17:

A patient presented with pelvic fracture and head trauma after a road traffic accident. He was given IV morphine for pain relief following which he developed respiratory difficulty. What could be the most likely cause?

- a) Pelvic Fracture
- b) CO₂ retention due to opioid use
- c) Cerebral Edema
- d) Pulmonary embolism

Question 18:

Which of the following is not a contraindication to the use of morphine?

- a) Bronchial asthma
- b) Renal failure
- c) Emphysema
- d) Acute myocardial infarction

Question 19:

Which of the following is true about the WHO analgesic ladder for chronic pain in adults?

- a) Analgesics are given 'on demand'

- b) Adjuvants are indicated only for mild pain
- c) Step 3 includes use of Morphine
- d) Intravenous route is preferred

Question 20:

You prescribe a cough syrup to a patient that has an opioid cough suppressant as one of the constituents. Which would be the opioid present in the syrup?

- a) Guaifenesin
- b) Diphenoxylate
- c) Codeine
- d) Naltrindole

Question 21:

How does dextromethorphan differ from codeine?

- a) Its antitussive action can be more easily blocked by naloxone
- b) It depresses mucociliary function of the airway mucosa
- c) Addiction is more common
- d) It causes less constipation

Answer Key

Question No.	Correct Option
1	b
2	c
3	d
4	a
5	b
6	d
7	a
8	d
9	d

10	b
11	b
12	c
13	d
14	a
15	b
16	b
17	b
18	d
19	c
20	c
21	d

Detailed Explanations

Solution to Question 1:

The spinal component of opioid analgesia is seen due to the action in the dorsal horn of the spinal cord.

Opioid analgesia:

- Spinal component: opioids act in the substantia gelatinosa of the dorsal horn to inhibit the release of excitatory transmitters like substance P from primary afferents carrying pain impulses. This action is through interneurons that are involved in the gating of pain impulses.
- Supraspinal component: action at the medulla, periaqueductal gray matter, limbic, and cortical areas alter the processing and interpretation of pain impulses. It also sends inhibitory impulses through descending pathways to the spinal cord.
- Peripheral action: acts on small primary afferent terminals in the skin or deeper structures, decreasing their sensitization following tissue injury.

Solution to Question 2:

Gastrointestinal motility is decreased by epidural opioids.

Option A: The epidural opioids provide more direct access to the first pain-processing synapse i.e., substantia gelatinosa by acting on the dorsal horn of the spinal cord.

Option B: Symptoms such as pruritus, nausea, and vomiting are also common with the epidural route.

Option D: Respiratory depression is seen with epidural opioids but the incidence is lesser than the parenteral route.

Solution to Question 3:

Miosis (not mydriasis) would be expected in a patient receiving morphine.

Opioids stimulate the Edinger-Westphal nucleus of the III cranial nerve resulting in miosis.

Urinary retention, vomiting, and pruritus are actions of opiates.

Solution to Question 4:

Mu opioid receptor activation results in miosis, nausea, and vomiting.

Mu receptor actions (mnemonic- MUSCARINE):

- Miosis
- Urine retention
- Sedation
- Constipation
- Analgesia
- Respiratory depression
- Increased muscular rigidity
- Negative bile flow (contraction of the sphincter of Oddi)
- Euphoria

Solution to Question 5:

Dysphoria is mediated by kappa opioid receptor.

Kappa receptor actions:

- Dysphoria, psychotomimetic
- Analgesia (Spinal and supraspinal)
- Miosis
- Sedation
- Physical dependence
- Reduced GI motility

Solution to Question 6:

Analgesia produced by opioids is mediated by μ (mu), κ (kappa), and δ (delta) receptors. Opioid receptors are part of the G protein-coupled receptors (GPCRs) family.

Supraspinal analgesia is predominantly mediated by the μ_1 receptor and spinal analgesia by the κ and δ receptor.

Solution to Question 7:

The given features suggest opioid-induced respiratory depression, which is mediated by the action of opioids on μ receptors (μ_2) in the brainstem respiratory centers.

Drug of choice for opioid-induced respiratory depression is naloxone.

Solution to Question 8:

Naloxone is an opioid antagonist, which can reverse the respiratory depression induced by opioids.

Respiratory depression can be prolonged when opioids are taken with CNS depressant substances like alcohol, barbiturates, benzodiazepines, inhalational anesthetics, and other sedative hypnotics.

Solution to Question 9:

Morphine is an opiate.

The term opiate specifically describes the naturally occurring alkaloids: morphine, codeine, thebaine, and papaverine.

Options A and C: Pentazocine and fentanyl are synthetic opioids.

Option B: Heroin is a semisynthetic opioid.

Solution to Question 10:

Heroin is structurally diacetylmorphine.

It is more potent than morphine and causes more euphoric effects. It is banned in some countries.

Diacetylmorphine is converted into monoacetylmorphine and morphine, to which the pharmacological actions of heroin are mainly attributed. Monoacetylmorphine is more lipid-soluble than morphine.

Codeine is also known as methyl morphine.

Meperidine is also known as pethidine.

Solution to Question 11:

He is likely addicted to heroin as it is the only semisynthetic opioid among the given options. Morphine, codeine, and thebaine are natural opioids derived from poppy *Papaver somniferum*.

Solution to Question 12:

Butorphanol would be the drug administered to her as it is the only synthetic opioid among the given options.

Heroin, dihydromorphone, and buprenorphine are semisynthetic opioids.

Solution to Question 13:

Morphine does not mediate analgesia in acupuncture as it is not an endogenous opioid. It is a naturally occurring opioid derived from poppy *Papaver somniferum*.

One of the most accepted mechanisms of action of acupuncture is that it works by releasing endogenous opioid peptides resulting in analgesia. Endogenous opioid peptides are produced physiologically in the body and act through an opioid receptor.

Endogenous opioid peptides:

- Enkephalins (more potent at δ receptor)
- β -endorphins (more potent at μ receptor)
- Dynorphins (more potent at κ receptor)
- Orphanin FQ or Nociceptin
- Neoendorphins
- Endomorphins

Solution to Question 14:

Morphine is the analgesic which could be responsible for the aggravation of the patient's symptoms.

It causes constriction of the sphincter of Oddi causing an increase in fluid pressure in the common bile duct and gall bladder.

Note: Pethidine causes lesser spasm of the sphincter of Oddi and can be used in biliary colic.

Solution to Question 15:

Morphine should be avoided in a patient with head injury because:

- Morphine retains CO₂ that increases intracranial pressure (ICP) and adds to the already increased ICP.
- Therapeutic doses can result in marked respiratory depression.
- Interferes with the assessment of progress by causing miosis, vomiting, and altered mentation.

Among the above options, head injury seems to be the most appropriate answer and some authorities consider head injury as a contraindication.

Option A: Hypotension can increase the susceptibility to vasodilating effects of morphine. Caution needs to be exercised.

Option C: Sensitivity to opioids is increased in hypothyroidism. Caution needs to be exercised.

Option D: Diabetes is not a contraindication for morphine use.

Solution to Question 16:

Statements 1, 2, 3 & 5 are true.

Option 1: Morphine is used in pulmonary edema due to its vasodilatory and anxiolytic actions.

Option 2: Morphine should be avoided as it causes constriction of the sphincter of Oddi causing an increase in fluid pressure in the common bile duct and gall bladder. Pethidine causes lesser spasm of the sphincter of Oddi and can be used in biliary colic.

Option 3: Indomethacin is an NSAID used in headache.

Option 4: The use of aspirin can worsen peptic ulcer disease.

Option 5: Single dose of pethidine (meperidine) can be effective in the treatment of postanesthetic shivering.

Solution to Question 17:

The most likely cause of respiratory difficulty in the given clinical scenario is CO₂ retention due to opioid use, as the patient was administered morphine after head trauma.

Head injury is considered as a contraindication to morphine use because:

- Morphine retains CO₂ that increases intracranial pressure (ICP) and adds to the already increased ICP.
- Therapeutic doses can result in marked respiratory depression.
- Interferes with the assessment of progress by causing miosis, vomiting, and altered mentation.

Solution to Question 18:

Acute myocardial infarction is not a contraindication for morphine use.

Morphine is the opioid of choice for myocardial infarction as it relieves anxiety, chest pain, and myocardial oxygen demand.

Contraindications to the use of Morphine:

- Bronchial asthma: Histamine release can cause bronchoconstriction and precipitate attacks of asthma.
- Renal failure: Morphine-6-glucuronide (metabolite) can accumulate and lead to prolonged respiratory depression.
- Emphysema: Respiratory depression and decreased response to hypoxia can cause hypercapnia in patients with COPD
- Head injury: Hypercapnia due to respiratory depression can increase ICP. Miosis, vomiting, and effect on wakefulness make interpreting the signs of head injury difficult.
- Biliary colic: Constriction of sphincter of Oddi can worsen biliary spasm
- Hypotension and shock

Solution to Question 19:

The true statement is the use of morphine falls under step 3 of WHO analgesic ladder for chronic pain in adults.

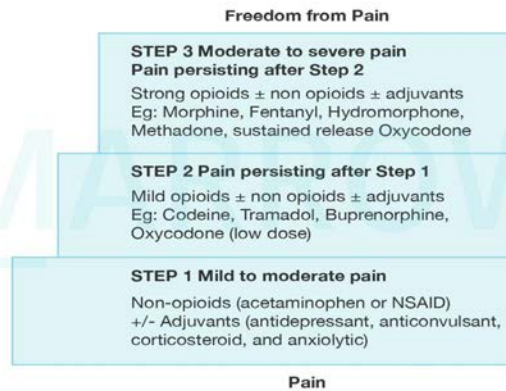
WHO has developed 3 step ladder for chronic/cancer pain relief in adults.

The 3 main principles are as follows:

- By the clock: Drugs should be given “by the clock” or “around the clock” rather than “on-demand”.
- By the mouth: The oral route is preferred. If not feasible, the least invasive route should be considered.
- By the ladder: Drugs are given in the following order, Non-opioids → Mild opioids → Strong opioids. Adjuvants are added at any stage of the WHO ladder to relieve anxiety.

Note: This 3-step ladder approach is inexpensive and 80-90% effective when utilized optimally.

Step-ladder for chronic or cancer pain relief



Solution to Question 20:

Codeine (methyl morphine) is an opioid used in the treatment of cough.

It is a more selective cough suppressant than morphine and widely used as an antitussive agent. It has antitussive actions at subanalgesic doses.

The analgesic action of codeine is mainly due to its conversion to morphine.

Solution to Question 21:

Dextromethorphan causes fewer gastrointestinal side effects like constipation when compared with codeine.

It is a synthetic central NMDA (N-methyl D-aspartate) receptor antagonist.

It is used as an antitussive agent and has no analgesic property.

Option A: The antitussive action of dextromethorphan is not blocked by naloxone.

Option B: Dextromethorphan does not depress the mucociliary function of the airway mucosa.

Option C: It has no addictive property.

Synthetic Opioids

Question 1:

Which of the following opioids has maximum plasma protein binding capacity?

- a) Morphine
- b) Sufentanil
- c) Fentanyl
- d) Pethidine

Question 2:

Which among the following is the most potent opioid analgesic?

- a) Fentanyl
- b) Sufentanil
- c) Remifentanyl
- d) Morphine

Question 3:

Which of the following has the most rapid analgesic action?

- a) Fentanyl
- b) Remifentanyl
- c) Sufentanil
- d) Meperidine

Question 4:

A middle-aged man with liver metastasis and altered LFT is posted for a surgery. What would be the opioid of choice in him?

- a) Fentanyl
- b) Alfentanil

- c) Sufentanil
- d) Remifentanil

Question 5:

A 48-year-old woman with CKD is scheduled for a surgery. Her serum creatinine is 2.4mg/dL and eGFR is 13. Which of the following opioids can be safely used in her?

- a) Pethidine
- b) Remifentanil
- c) Alfentanil
- d) Morphine

Question 6:

A patient who underwent an intra-abdominal surgery is about to receive intraspinal opioid injection for analgesia. Which of the following drugs should not be used for this purpose?

- a) Sufentanyl
- b) Alfentanyl
- c) Fentanyl
- d) Remifentanil

Question 7:

Which of the following is the shortest acting opioid?

- a) Fentanyl
- b) Remifentanil
- c) Alfentanil
- d) Sufentanil

Question 8:

Which among the following is not included in neuroleptic anesthesia?

- a) Nitrous oxide

- b) Propofol
- c) Droperidol
- d) Fentanyl

Question 9:

You are planning to use a transdermal patch for one of your patients who require around-the-clock opioids for his cancer pain. Which of the following drugs will you prescribe?

- a) Fentanyl
- b) Remifentanyl
- c) Sufentanyl
- d) Alfentanyl

Question 10:

Which of the following is the drug of choice for rescue analgesia?

- a) Morphine
- b) Codeine
- c) Oxycodone
- d) Fentanyl

Question 11:

A 60-year-old man was posted for bronchoscopy under procedural sedation. After the administration of anesthetic drugs, he developed chest wall rigidity and ineffective spontaneous ventilation. Attempts at manual ventilation were unsuccessful due to the non-compliant chest wall. The opioid that is most commonly associated with this condition is?

- a) Morphine
- b) Fentanyl
- c) Pethidine
- d) Tramadol

Question 12:

Upon recovering from anesthesia, a patient started feeling cold accompanied by shivering. Which is the opioid indicated to treat this condition?

- a) Morphine
- b) Meperidine
- c) Methylnorphine
- d) Methadone

Question 13:

A 50-year-old housewife with diabetic nephropathy and receiving dialysis weekly comes to your clinic with an accidental burn on her left hand. On examination, 2nd-degree burns are seen. Which of the following opioids should not be prescribed to her for analgesia?

- a) Pethidine
- b) Methadone
- c) Fentanyl
- d) Alfentanil

Question 14:

A woman was administered an intravenous opioid for managing postoperative pain. Within 5 minutes, she started developing severe redness and pruritus along the distribution of the vein into which the drug was injected. Which of the following drugs would be responsible for this?

- a) Meperidine
- b) Alfentanil
- c) Remifentanil
- d) Fentanyl

Question 15:

A 16-year-old boy received an intravenous opioid for analgesia. A few minutes later, his heart rate decreased from 72/min to 58/min. Which among the following drugs is least likely to be associated with this finding?

- a) Fentanyl
- b) Sufentanil

- c) Alfentanil
- d) Meperidine

Question 16:

Which of the following opioids has local anesthetic property?

- a) Morphine
- b) Methadone
- c) Pethidine
- d) Codeine

Question 17:

Apart from mu-opioid receptors, on which other receptor does tramadol act?

- a) Anticholinergic
- b) 5HT_{1a} inhibition
- c) Serotonergic and norepinephrine reuptake inhibition
- d) Antihistaminic

Question 18:

Which of the following statements is false regarding tramadol?

- a) Safe in epileptics
- b) Less potent than morphine
- c) Less respiratory depression
- d) Synthetic analogue of codeine

Question 19:

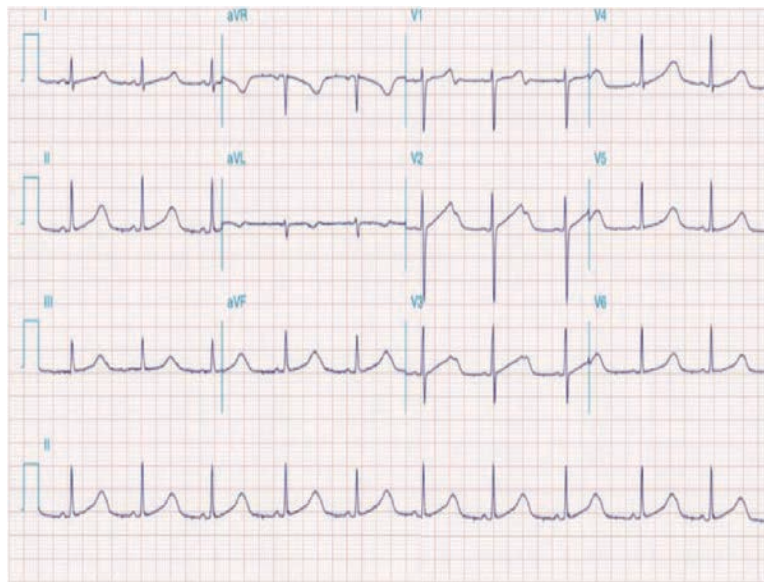
A 24-year-old woman requires an intranasal opioid spray for her migraine. Which of the following would you prescribe her?

- a) Morphine
- b) Buprenorphine

- c) Butorphanol
- d) Methadone

Question 20:

A 40-year-old man, on treatment with fluoxetine for depression, was started on an opioid for severe neuropathic pain. A few days later he had an episode of syncope and was brought to the casualty. ECG taken is given below, with QTc = 477 ms. Which of the following drugs is most likely the cause?



- a) Morphine
- b) Meperidine
- c) Methadone
- d) Propoxyphene

Question 21:

Which of the following statements is false regarding methadone?

- a) Long acting mu receptor agonist
- b) Used for chronic pain relief
- c) Rapidly absorbed from GIT and appears in plasma within 30 min of oral administration
- d) Onset of analgesia is 30-60 min after parenteral and 1-2 h after oral administration

Question 22:

A young man is scheduled for laparoscopic cholecystectomy. His medical history is significant for depression and he is on monoamine oxidase inhibitors for the same. Which of the following opioids can be safely administered to him?

- a) Tramadol
- b) Meperidine
- c) Dextromethorphan
- d) Morphine

Question 23:

A woman sustained multiple fractures in a motorcycle accident. Her physician prescribed an oral opioid for analgesia which is a partial agonist at mu and a full agonist at kappa receptors. Identify the drug given.

- a) Pentazocine
- b) Buprenorphine
- c) Tramadol
- d) Fentanyl

Question 24:

A cancer patient is dependent on morphine for pain management. His physician decides to switch him to buprenorphine. Due to which property he chooses this new drug?

- a) Partial agonism at mu receptor
- b) Partial agonism at kappa receptor
- c) Partial antagonism at mu receptor
- d) Partial antagonism at kappa receptor

Question 25:

An elderly man with bone metastasis is receiving intravenous opioids for pain management. Which of the following drugs can be safely administered to him without a risk of developing respiratory depression?

- a) Morphine

- b) Buprenorphine
- c) Fentanyl
- d) Alfentanyl

Question 26:

Which of the following statements is false?

- a) Naltrexone is an antagonist at μ receptor
- b) Meperidine is a pure agonist at μ receptor
- c) Nalbuphine is an agonist at κ receptor
- d) Butorphanol is an antagonist at κ receptor

Answer Key

Question No.	Correct Option
1	b
2	b
3	b
4	d
5	b
6	d
7	b
8	b
9	a
10	d
11	b
12	b
13	a
14	a
15	d
16	c
17	c
18	a
19	c

20	c
21	d
22	d
23	a
24	a
25	b
26	d

Detailed Explanations

Solution to Question 1:

Sufentanil has maximum plasma protein binding capacity.

More than 90% of sufentanil is bound to alpha acid glycoprotein in plasma.

Plasma protein binding capacity of opioids:

- Sufentanil: 90%
- Fentanyl: 80-85%
- Pethidine (meperidine): 65-75%
- Morphine: 36%

Solution to Question 2:

Sufentanil is the most potent opioid. It is 1000 times more potent than morphine.

On the other hand, fentanyl is around 100 times more potent than morphine.

Solution to Question 3:

The most rapid analgesic action is achieved by remifentanyl.

Analgesic action (peak) is achieved within 1 to 1.5 min following intravenous administration.

Analgesic action- duration for peak action:

- Remifentanyl: 1 to 1.5 min
- Fentanyl: 5 min
- Sufentanil: 5 min
- Meperidine: 15 min

The rapid onset of action and the rapid recovery (within 15 min) owing to metabolism by plasma esterases of remifentanyl make it an ideal analgesic for short painful procedures.

Solution to Question 4:

Opioid of choice in patients with compromised hepatic function is remifentanyl.

Remifentanyl is hydrolyzed by plasma and tissue esterases and hence its metabolism is independent of hepatic function.

Other opioids like fentanyl, alfentanil, and sufentanil are metabolized by the liver.

Solution to Question 5:

The given lab values in a CKD patient are suggestive of renal failure and remifentanyl can be safely used in renal failure patients.

Remifentanyl is hydrolyzed by plasma and tissue esterases. The pharmacokinetics and the pharmacodynamics of remifentanyl are not altered by renal dysfunction.

Option A: Pethidine is metabolized to norpethidine, accumulation of which can lead to CNS excitatory effects (seizures).

Option C: High levels of alfentanil can lead to prolonged action.

Option D: Morphine is metabolized to morphine-6-glucuronide which can accumulate leading to respiratory depression.

Note: Fentanyl clearance is not altered by renal failure but its plasma protein binding is decreased which can alter the concentration of the free drug. Hence, fentanyl is the second drug of choice after remifentanyl.

Solution to Question 6:

Remifentanyl should not be used for intraspinal injection.

Remifentanyl is formulated in combination with glycine. Glycine acts as an inhibitory neurotransmitter in the spinal dorsal horn that causes a reversible motor weakness when it is injected intrathecally.

For this reason, remifentanyl is not approved for spinal or epidural use.

Solution to Question 7:

The shortest acting opioid is remifentanyl with $t_{1/2}$ of 8-20 minutes.

The rapid metabolism by tissue esterases and rapid clearance results in its ultra short-acting nature.

Remifentanyl has the fastest onset of action and recovery, making it the opioid of choice in daycare surgery.

Solution to Question 8:

Propofol is not included in neuroleptic anesthesia.

Neuroleptic anesthesia includes fentanyl, droperidol, and nitrous oxide.

Neuroleptic analgesia includes fentanyl and droperidol.

Neuroleptic anesthesia is characterized by analgesia, the absence of clinically apparent motor activity, suppression of autonomic reflexes, maintenance of cardiovascular stability, and amnesia.

QT interval prolongation and torsades de pointes due to droperidol have made this technique obsolete.

Solution to Question 9:

Fentanyl is available as a transdermal patch and can be prescribed to this patient. It is mainly used for chronic pain.

Various routes of administration of fentanyl are: intravenous, epidural, spinal, transdermal therapeutic system (transdermal patch and transdermal iontophoresis), oral (buccal tablets, films, lozenges).

Solution to Question 10:

Fentanyl is the drug of choice for providing rescue analgesia.

Rescue analgesia involves the use of medications to minimize post-operative pain. Mild pain can be controlled with oral analgesics but severe pain requires opioids like fentanyl.

Fentanyl is preferred over morphine or oxycodone due to its rapid onset of action, lesser incidence of postoperative nausea, vomiting, and other adverse effects.

Solution to Question 11:

Chest wall rigidity and ineffective ventilation is a classic presentation of 'wooden chest syndrome'. This opioid-induced muscle rigidity is most commonly associated with the use of lipophilic opioids such as fentanyl and its congeners (alfentanil, sufentanil, and remifentanyl).

Risk factors for wooden chest syndrome:

- Rapid administration of a large dose of fentanyl
- Concomitant use of nitrous oxide
- Old age

The clinical presentation can vary from mild hoarseness to chest wall rigidity severe enough to prevent adequate bag-and-mask ventilation.

Management of wooden chest syndrome:

- Discontinuation of the opioid
- Replacement of the opioid with a non-lipophilic version (such as hydromorphone or morphine) for analgesia
- Consider naloxone for reversal (you do not want to reverse analgesia on an intubated patient unless absolutely necessary)
- Addition of a non-depolarizing neuromuscular blocker for paralysis
- Supportive ventilatory care

Solution to Question 12:

Meperidine is commonly indicated for post-anesthetic shivering.

It prevents post-anesthetic shivering by lowering the temperature threshold at which shivering is triggered by the hypothalamus.

A single dose of 12.5 to 35mg meperidine (pethidine) is effective in the treatment of postanesthetic shivering.

Infusion-related chills and rigors associated with amphotericin B, aldesleukin, trastuzumab, and alemtuzumab are also managed with meperidine in combination with other agents.

Solution to Question 13:

Pethidine (meperidine) should not be prescribed to her as it is contraindicated in patients with renal failure.

In renal failure, there is an accumulation of its metabolite norpethidine which causes excitatory syndrome consisting of hallucinations, tremors, muscle twitches, and convulsions. Therefore, pethidine is avoided in renal failure patients.

Methadone, alfentanil, and fentanyl primarily undergo hepatic metabolism.

Solution to Question 14:

The clinical findings are suggestive of histamine release, which is common with the use of meperidine (pethidine) due to mast cell activation.

Codeine, morphine, and hydromorphone are also associated with histamine release.

Fentanyl, alfentanil, and remifentanyl do not lead to histamine release.

Solution to Question 15:

Meperidine (pethidine) is least likely to be associated with bradycardia.

Intravenous meperidine produces tachycardia as it has a structure similar to atropine. Unlike other opioids, meperidine causes mydriasis due to its anticholinergic effect.

Pentazocine, which is an opioid agonist-antagonist, also produces tachycardia, hypertension, and mydriasis.

Solution to Question 16:

The opioid with local anesthetic property is pethidine (meperidine), which is especially seen after intrathecal administration.

Intrathecal pethidine can provide spinal anesthesia for endoscopic urological procedures and surgery of the lower limb, perineum, and lower abdomen including cesarean section.

However, due to low potency and systemic opioid side effects, it is usually not used for this purpose.

Solution to Question 17:

Tramadol inhibits the reuptake of serotonin and norepinephrine, apart from its action on the mu-opioid receptor.

It is a synthetic codeine analog that is a mu-opioid receptor agonist.

Tramadol is a centrally acting analgesic used in the treatment of mild to moderate pain.

Solution to Question 18:

Tramadol is contraindicated in epileptics.

Tramadol is epileptogenic and can cause seizures in predisposed individuals and in those taking medications like monoamine oxidase inhibitors and neuroleptic drugs.

It is a synthetic analogue of codeine and a weak opioid agonist with 1/6000 affinity for μ receptors as that of morphine.

Apart from opioid action, it also inhibits the reuptake of 5HT and NE. Due to non-opioid receptor-mediated actions, it produces less respiratory depression and less constipation.

Solution to Question 19:

Butorphanol is a synthetic opioid available as a nasal spray for use in migraine headaches and painful conditions.

It is used parenterally as an analgesic for moderate to severe pain or as an adjunct to general anesthesia.

Opioids available for intranasal administration are butorphanol and fentanyl.

Solution to Question 20:

The given ECG findings show QT prolongation which is associated primarily with the use of methadone.

In the ECG QTc (corrected QT interval = $QT/\sqrt{RR}$) is 477ms. Normal QTc is about 440ms. Values of more than 440ms in males and 460ms in females are considered to be prolonged. QT prolongation can present as syncope or cardiac arrest.

Methadone has additive action with agents known to cause QT prolongation. Serious cardiac arrhythmias including torsades de pointes have been noted with methadone use.

Solution to Question 21:

Methadone's onset of analgesia is 10-20 minutes after the parenteral route, and 30-60 minutes after the oral route of administration.

Option A: Methadone is a long-acting mu receptor agonist with a plasma half-life of 15-40 hours.

Option B: It is an analgesic and used for chronic pain relief.

Option C: Absorption from GIT is rapid and appears in plasma within 30 minutes and reaches peak concentrations at around 4 hours.

Solution to Question 22:

Morphine can be safely administered as it does not inhibit the neuronal reuptake of serotonin and hence does not trigger serotonin syndrome in patients on MAO inhibitors.

Phenylpiperidine series of opioids such as meperidine, tramadol, dextromethorphan, and methadone inhibit the neuronal reuptake of serotonin leading to serotonergic overactivity. This can cause serotonin syndrome in patients receiving monoamine oxidase inhibitors.

Solution to Question 23:

Pentazocine is a partial agonist at mu and a full agonist at kappa receptors.

Analgesia produced by pentazocine is predominantly mediated by the κ receptor.

Pentazocine is a mixed agonist-antagonist opioid agent. It is a full agonist at the κ receptor and a partial agonist or weak antagonist at the μ receptor.

Buprenorphine is a partial agonist at mu receptors but an antagonist at kappa receptors.

Solution to Question 24:

Buprenorphine is a partial agonist at the mu receptor with a lower intrinsic activity compared to morphine and hence does not produce withdrawal symptoms in patients with morphine dependence.

Buprenorphine is a synthetic thebaine congener and also acts as an antagonist at kappa receptor.

It is more potent than morphine at mu receptor but with lower intrinsic activity and ceiling effect. Due to lower intrinsic activity, it has no withdrawal symptoms and can be used in the treatment of morphine dependence.

Solution to Question 25:

Buprenorphine can be safely administered to this patient because of its ceiling effect on respiratory depression.

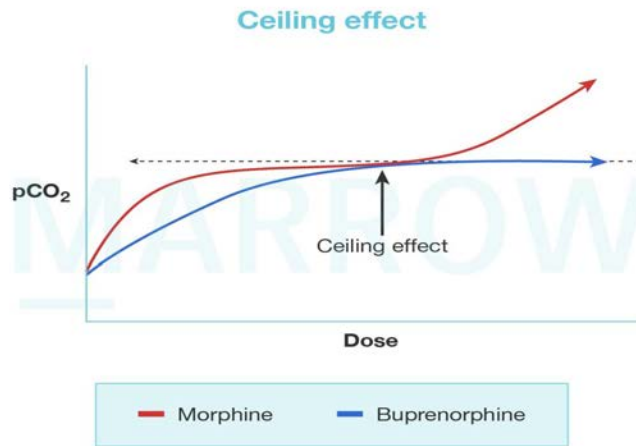
Ceiling effect is defined as no further increase in the effect of the drug with a further increment in dose, once the ceiling level is reached.

Opioid agonist-antagonist compounds produce respiratory depression (and analgesia) with a ceiling effect which means increasing the dose further does not produce any further respiratory depression and may actually result in increased ventilation (due to the predominance of antagonist actions).

Other drugs that show the ceiling effect are:

- Butorphanol
- Nalbuphine
- Pentazocine.

Respiratory depression with morphine, fentanyl & alfentanil does not have a ceiling effect because they are pure opioid agonists. As the dose increases, respiratory depression also increases progressing to respiratory arrest.



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Solution to Question 26:

Butorphanol is an agonist at κ receptor. It may be a partial agonist or antagonist at μ receptor.

Pentazocine, butorphanol, buprenorphine, meptazinol, nalorphine, nalbuphine, and dezocine are mixed agonist-antagonist opioids.

All opioid agonist-antagonists are μ antagonists and κ agonists, except buprenorphine, pentazocine, and dezocine. Buprenorphine, pentazocine, and dezocine are partial agonists at the μ receptor.

Option A: Naltrexone is a pure antagonist at μ , κ , and δ receptors.

Option B: Meperidine is a pure agonist of both μ and κ receptors.

Opioid Antagonists

Question 1:

Which of the following is a pure opioid antagonist?

- a) Naloxone
- b) Naltrexone
- c) Nalorphine
- d) Both a and b

Question 2:

Which of the following statements is incorrect about opioid antagonists?

- a) Naloxone is a short acting drug
- b) Naltrexone lowers craving in alcoholics
- c) Nalmefene can be used for treatment of opioid overdose
- d) Nalmefene is shorter acting than naloxone

Question 3:

Which of the following is false about naltrexone?

- a) Is orally active
- b) Is used in the treatment of alcohol dependence
- c) Is less potent than naloxone
- d) Is a pure opioid antagonist

Question 4:

A patient is on chronic opioid therapy for pain management. He would not develop tolerance against which of the following actions of opioids?

- a) Miosis
- b) Analgesia

- c) Euphoria
- d) Nausea and vomiting

Question 5:

An elderly man, who is on chronic morphine therapy for pain relief, visits your clinic with complaints of not passing stools for the last 4 days. Which of the following drugs is best suited to treat his condition?

- a) Naltrexone
- b) Methylnaltrexone
- c) Dextromethorphan
- d) Nalmefene

Question 6:

Which of the following is not an antagonist at μ receptor?

- a) Naltrindole
- b) Naloxone
- c) Nalbuphine
- d) Naltrexone

Question 7:

A 27-year-old man presented to the casualty with slurred speech and shallow breathing. On examination, you find he has bradycardia, hypotension, and pin-point pupils. Which drug will you choose to treat him?

- a) Nalorphine
- b) Naltrexone
- c) Naloxone
- d) Nalmefene

Question 8:

A college student who is addicted to intravenous heroin abruptly stopped abusing the drug during exams. Which of the following would you not expect as a part of withdrawal?

- a) Lacrimation
- b) Constipation
- c) Increased salivation and running nose
- d) Yawning

Question 9:

An elderly woman who was dependent on morphine for her palliative care recently got switched to another opioid medication. A few days later she started developing withdrawal symptoms. Which of the following opioids would have been prescribed to her?

- a) Pethidine
- b) Pentazocine
- c) Morphine
- d) Methadone

Question 10:

A 33-year-old man with a history of heroin addiction is referred to a rehabilitation center. Which of the following would you use as a substitute drug for maintenance therapy?

- a) Methyldorphine
- b) Methadone
- c) Meperidine
- d) Methylphenidate

Question 11:

A physician prescribes naltrexone to a patient undergoing treatment for opioid addiction. What is the function of this drug in opioid addiction?

- a) Prevention of relapse
- b) Treatment of withdrawal
- c) Treatment of overdose
- d) Prevention of withdrawal

Question 12:

Which of the following is incorrect about the use of naltrexone in the treatment of opioid addiction?

- a) Discontinuation of naltrexone does not cause withdrawal symptoms
- b) It satisfies craving in opioid addicts
- c) It blocks the effects of opioids if they are used
- d) It is administered only when the patient has completed detoxification from opioids

Question 13:

You are treating a 31-year-old man with heroin dependence. Which of the following drugs would you not prescribe to him?

- a) Disulfiram
- b) Buprenorphine
- c) Clonidine
- d) Lofexidine

Answer Key

Question No.	Correct Option
1	d
2	d
3	c
4	a
5	b
6	a
7	c
8	b
9	b
10	b
11	a

12	b
13	a

Detailed Explanations

Solution to Question 1:

Both naloxone and naltrexone are pure opioid antagonists.

Nalorphine is an agonist-antagonist. It has antagonistic activity at the mu receptor and additional agonistic activity at the kappa receptor.

Note: Nalmefene is sometimes classified as a pure antagonist but it has additional kappa receptor partial agonist activity.

Solution to Question 2:

Nalmefene is longer-acting (not shorter-acting) than naloxone.

Solution to Question 3:

Naltrexone is more potent than naloxone.

Naltrexone is a pure opioid antagonist which is orally active.

Alcohol craving is reduced by naltrexone. It is also approved for the prevention of relapse in opioid addiction treatment.

Solution to Question 4:

Tolerance is developed to all the above actions of opioids except for miosis.

Exceptions to opioid tolerance (3C's):

- Constricted pupil (miosis)
- Constipation
- Convulsions

Solution to Question 5:

The given clinical picture points towards opioid-induced constipation. Methylnaltrexone, an opioid antagonist with predominant peripheral action is best suited for treating opioid-induced constipation.

Methylnaltrexone is a peripheral μ receptor antagonist. It was introduced in the treatment of opioid-induced constipation refractory to other measures.

It is a quaternary ammonium compound and hence cannot cross the blood-brain barrier. It can reverse the effects mediated by peripheral opioid receptors and spares CNS effects like analgesia.

The peripherally acting opioid antagonists include:

- Methylnaltrexone
- Naloxegol
- Naldemedine
- Alvimopan

Solution to Question 6:

Naltrindole is a new non-peptide drug which has a selective antagonistic activity on the δ receptor. This drug is still in the trial phase and is not yet in use.

Option B: Naloxone is considered a pure opioid receptor antagonist but at high doses, it is found to have partial agonistic activity on the μ - and κ -opioid receptors.

Option C: Nalbuphine is an agonist-antagonist opioid which acts as an antagonist at the μ receptor and an agonist at the κ receptor.

Option D: Naltrexone is a μ -, δ - and κ -opioid receptor antagonist.

Solution to Question 7:

The given clinical features are suggestive of opioid overdose. Naloxone is the drug of choice for opioid overdose.

Opioid overdose symptoms:

- Triad of coma, respiratory depression, and pinpoint pupils (miosis)
- Hypotension, bradycardia
- Shallow breathing
- Hypothermia

Naloxone is also used for reversing neonatal asphyxia due to opioid usage in labor. Naloxone is a pure opioid antagonist which reverses all effects except sedation.

Note: Naloxone is contraindicated in infants born to mothers with opiate addiction as it may precipitate acute neonatal withdrawal with seizures.

Solution to Question 8:

Diarrhea (not constipation) is seen in opioid withdrawal.

Opioid withdrawal symptoms (mnemonic- O MY WITHDRAWALS):

- Mydriasis
- Yawning
- Weakness
- Insomnia, irritability
- Tremors, tachycardia
- Hypertension, hyperactive
- Diarrhea
- Rhinorrhoea
- Apnoeic attacks
- Weight loss
- Anxiety
- Lacrimation
- Sweating

Solution to Question 9:

Pentazocine can precipitate withdrawal symptoms in opioid-dependent patients.

It acts as a weak antagonist or partial agonist at mu-opioid receptors which may precipitate withdrawal symptoms.

Options A, C, and D: Pethidine, morphine, and methadone are pure agonists at mu-receptors and do not precipitate withdrawal symptoms.

Solution to Question 10:

Methadone is used as a substitute drug for the maintenance therapy of opioid (heroin) dependence. This is because of its good oral bioavailability and long half-life.

Methadone is substituted for opioids during detoxification to avoid withdrawal symptoms. Once stabilized, the dose is gradually reduced and the drug is stopped slowly over days.

Due to its long half-life and accumulation in tissues, discontinuing the drug still results in low concentrations of the drug in plasma resulting in protracted withdrawal symptoms.

Maintenance therapy for opioid dependence:

- Long-acting opioid agonist - a safer substitute like methadone is given in a supervised manner to prevent withdrawal symptoms
- Partial agonist - buprenorphine can also be used
- Clonidine, lofexidine - alpha-2 agonists act as sympatholytic and reduce the withdrawal symptoms
- LAAM(L-Alpha-Acetyl-Methadol) - opioid agonist which reduces craving
- Naltrexone - is an opioid antagonist used in maintenance therapy to prevent relapses

Solution to Question 11:

Naltrexone is used for the prevention of relapse in opioid addiction.

It is given only after initial opioid detoxification. The patient should be opioid-free for 7-10 days and have a negative naloxone challenge (no withdrawal after naloxone administration).

Naltrexone is a pure opioid antagonist which is more potent than naloxone. It is orally active and has a long duration of action which makes it suitable for opioid blockade therapy in post-addicts.

Solution to Question 12:

Naltrexone does not satisfy the craving of opioid addicts. It is an opioid antagonist with a high affinity for μ opioid receptors.

Naltrexone does not suppress craving in opioid addiction but it suppresses craving in chronic alcoholics.

Opioid agonists such as methadone and buprenorphine suppress craving and withdrawal symptoms and are therefore used in the first stage of detoxification in opioid addiction.

Option A: Discontinuation of naltrexone ingestion has no negative physical consequences, as opposed to discontinuation of agonist maintenance therapies, which results in the emergence of opioid withdrawal symptoms. This is a major reason for noncompliance and treatment failure.

Option C: It blocks the effects of opioids and prevents the user from experiencing subjective symptoms of opioid intoxication such as euphoria on subsequent use. As the user does not experience these symptoms, he does not develop a craving for the substance. Thus, naltrexone prevents relapse.

Option D: It is only administered when the patient has completed a medically supervised detoxification from opioids because it can cause immediate withdrawal symptoms.

Solution to Question 13:

Disulfiram cannot be used to treat heroin dependence.

It is an aldehyde dehydrogenase inhibitor used in the maintenance therapy of alcohol abstinence.

Maintenance therapy for opioid dependence:

- Long-acting opioid agonist - a safer substitute like methadone is given in a supervised manner to prevent withdrawal symptoms
- Partial agonist - buprenorphine can also be used
- Clonidine, lofexidine - alpha-2 agonists act as sympatholytic and reduce the withdrawal symptoms
- LAAM(L-Alpha-Acetyl-Methadol) - opioid agonist which reduces craving
- Naltrexone - is an opioid antagonist used in maintenance therapy to prevent relapses

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Alcohols

Question 1:

Effect of acute ethanol consumption is inhibitory on which of the following ligand-gated ion channels?

- a) GABA-A receptor
- b) Glycine receptor
- c) NMDA receptor
- d) Nicotinic ACh receptor

Question 2:

The black-outs associated with high levels of ethanol can be attributed to its action on which of the following receptors?

- a) GABA-A receptor
- b) GABA-B receptor
- c) NMDA receptor
- d) Glycine receptor

Question 3:

Ethanol ingestion results in enhanced diuresis. What is the most probable mechanism causing this?

- a) Inhibition of ADH action in collecting tubules
- b) Stimulation of ADH action in collecting tubules
- c) Stimulation of ADH release from pituitary gland
- d) Inhibition of ADH release from pituitary gland

Question 4:

What effect does acute consumption of ethanol have on cytochrome enzymes?

- a) Induction

- b) Inhibition
- c) Stimulation
- d) No effect

Question 5:

Which of the following is an aldehyde dehydrogenase inhibitor that you would prescribe for the pharmacological management of a patient addicted to alcohol?

- a) 1 and 2
- b) 2 only
- c) 1, 2 and 5
- d) 1, 2, 3 and 4

Question 6:

A 51-year-old diabetic patient presents to your OPD with severe restlessness, palpitations, flushing, and sweating. His medication history includes OHAs and an antibiotic that was started 3 days back. On enquiring, he admits that he consumed two shots of whiskey after taking his medications. Which of the following drugs is not likely to cause similar effects?

- a) Metronidazole
- b) Chlorpropamide
- c) Cefoperazone
- d) Metformin

Question 7:

A 40-year-old man with alcohol dependence is referred to a rehabilitation center for deaddiction. Which of the following drugs would you not prescribe for maintenance of deaddiction?

- a) Morphine
- b) Naltrexone
- c) Acamprosate
- d) Disulfiram

Question 8:

Which of the following anti-epileptics is effective in maintaining abstinence in a patient undergoing treatment for alcohol deaddiction?

- a) Vigabatrin
- b) Topiramate
- c) Phenytoin
- d) Zonisamide

Question 9:

A young man presented to the ER with complaints of headache, vomiting, and blurring of vision after he consumed a bottle of illicit liquor. His ABG shows a high anion gap metabolic acidosis. Which of the following statements would be incorrect regarding his management?

- a) Fomepizole is used intravenously
- b) Intravenous ethanol is an alternative to fomepizole
- c) Hemodialysis is not effective
- d) Folinic acid is also used in treatment

Question 10:

How is the administration of ethanol going to help in the management of methanol poisoning?

- a) Ethanol accelerates metabolism of methanol
- b) Ethanol slows down metabolism of methanol
- c) Ethanol reacts with methanol to form an inactive metabolite
- d) Ethanol reverses the actions of methanol

Question 11:

A middle-aged man undergoing treatment for alcohol deaddiction was started on disulfiram. Which of the following statements is true regarding this drug?

- a) When used with alcohol it causes respiratory difficulties
- b) It is an anticraving drug

- c) It can be initiated even in minimally motivated persons
- d) Its action persists up to 2-3 days after stopping the drug

Answer Key

Question No.	Correct Option
1	c
2	c
3	d
4	b
5	b
6	d
7	a
8	b
9	c
10	b
11	a

Detailed Explanations

Solution to Question 1:

The effect of acute ethanol consumption is inhibitory on NMDA receptor-operated channels.

Effects of acute ethanol on ion channels:

Enhancement

- GABA-A receptor-operated channels
- Glycine receptor-operated channels
- Nicotinic ACh receptor-operated channels
- 5HT₃ receptor-operated channels
- G protein-coupled inwardly rectifying K⁺ channels
- Large conductance, Ca²⁺/ voltage-activated K⁺ channels

Inhibition

- NMDA receptor-operated channels

- Voltage-gated Ca^{2+} channels

Solution to Question 2:

Blackouts are periods of memory loss that occur with high ethanol consumption and may result from the ability of ethanol to inhibit the action of glutamate on NMDA receptors.

NMDA receptor is implicated in many aspects of cognitive function including learning and memory.

Solution to Question 3:

Ethanol inhibits the release of antidiuretic hormone (ADH)/ vasopressin from the posterior pituitary gland. This results in enhanced diuresis observed with ethanol ingestion.

Inhibition of ADH action in collecting tubules is not the mechanism by which alcohol induces diuresis. Stimulation of ADH release or action will have an antidiuretic action.

Solution to Question 4:

Acute consumption of ethanol inhibits the function of several cytochrome enzymes that are involved in the metabolism of drugs. CYP2E1 is one of the isoenzymes that is specifically inhibited.

Chronic consumption of alcohol causes induction of cytochrome enzymes, particularly CYP2E1, increasing the rate of metabolism of drugs dependent on CYP2E1 for metabolism.

Solution to Question 5:

Disulfiram is an irreversible inhibitor of both cytosolic and mitochondrial forms of aldehyde dehydrogenase.

Ethanol consumption during disulfiram therapy can lead to increased concentrations of acetaldehyde. Increased acetaldehyde levels can cause flushing, pulsating headache, throbbing pain in the head and neck, and other aversive reactions.

Disulfiram is indicated in alcohol deaddiction maintenance therapy.

Option 1: Fomepizole is an inhibitor of alcohol dehydrogenase.

Option 3: Acamprosate is an NMDA receptor antagonist with modest GABA-A receptor agonistic activity.

Option 4: Naltrexone is an opioid antagonist that suppresses craving in chronic alcoholics.

Option 5: Chlordiazepoxide is a benzodiazepine that can be used to treat acute alcohol withdrawal and anxiety.

Solution to Question 6:

The given clinical presentation is suggestive of a disulfiram-like reaction. Metformin is not associated with a disulfiram-like reaction following alcohol ingestion.

Disulfiram-like reaction with alcohol ingestion is seen with the following drugs:

- Metronidazole
- 1st generation sulfonylureas like chlorpropamide
- Cephalosporins like cefoperazone, cefamandole
- Griseofulvin
- Procarbazine

Solution to Question 7:

Morphine, an opioid receptor agonist, is not indicated for alcohol deaddiction maintenance. Opioid receptor antagonist is indicated for this condition.

Drugs approved for the maintenance therapy and to prevent relapse:

- Naltrexone: It is an orally active opioid receptor antagonist.
- Acamprosate: An NMDA receptor antagonist with modest GABA-A receptor agonistic activity.
- Disulfiram: A deterrent drug that works by inhibiting aldehyde dehydrogenase enzyme.

Solution to Question 8:

Topiramate has shown efficacy in maintaining abstinence and reducing craving in chronic alcoholism.

Though effective, topiramate is not yet approved for use in alcohol deaddiction maintenance. It is used off-label for this indication.

Ondansetron and baclofen are other drugs that are found to be effective but not yet approved for this use.

Solution to Question 9:

The given clinical picture and labs are suggestive of accidental methanol toxicity. Hemodialysis is used in severe cases of methanol toxicity.

A serum methanol concentration greater than 50 mg/dl is serious enough to consider hemodialysis. It can be used to remove both methanol and its toxic metabolite formic acid from the blood.

Option A: Fomepizole, an alcohol dehydrogenase inhibitor, is administered intravenously in methanol toxicity. It acts by inhibiting the formation of formic acid.

Option B: Intravenous ethanol is an alternative to fomepizole. It has a higher affinity to alcohol dehydrogenase than methanol. Thus, saturation of ADH enzymes results in less formation of toxic formic acid.

Option D: Folate-dependent systems are responsible for the oxidation of formic acid to CO₂. Both folinic and folic acid are often administered during methanol toxicity but the exact benefit is yet to be determined.

Note: Ethanol is given intravenously in the management of methanol and ethylene glycol poisoning.

Solution to Question 10:

Administration of ethanol slows down the metabolism of methanol.

Ethanol has a higher affinity for alcohol dehydrogenase than methanol.

Formic acid is the metabolite of methanol which is highly toxic. Therefore, slowing down the metabolism of methanol reduces its toxic effects.

Solution to Question 11:

Alcohol consumption during disulfiram therapy can lead to increased concentrations of acetaldehyde. This can cause aversive reactions resulting in flushing, pulsating headache, throbbing pain in the head and neck, nausea, vomiting, rashes, respiratory difficulties, chest pain, and autonomic manifestations.

Disulfiram is an irreversible inhibitor of both cytosolic and mitochondrial forms of aldehyde dehydrogenase. It is used in behavioral modification for alcohol de-addiction.

Option B: The principle of action is aversive conditioning (not anti-craving).

Option C: Disulfiram can be given only to highly motivated individuals. It should not be administered until the person has abstained from alcohol for at least 12 hours.

Option D: The disulfiram-alcohol reaction may persist as long as 14 days after the last dose of disulfiram.

Drugs for Alzheimer's Disease and Other Neurodegenerative Disorders

Question 1:

A 70-year-old male patient is brought to the OPD by his son, with complaints of increasing forgetfulness. Neuroimaging shows marked cortical atrophy of temporal and parietal lobes. What is the most striking neurochemical disturbance in this condition?

- a) Deficiency of acetylcholine
- b) Deficiency of dopamine
- c) Deficiency of serotonin
- d) Deficiency of glutamate

Question 2:

Which of the following is not a reversible cholinesterase inhibitor indicated in a patient with Alzheimer's disease?

- a) Donepezil
- b) Rivastigmine
- c) Galantamine
- d) Memantine

Question 3:

A neurology resident changed the current oral medication of a patient with Alzheimer's disease to a transdermal patch of a cholinesterase inhibitor. What must be the new drug that was prescribed?

- a) Tacrine
- b) Donepezil
- c) Rivastigmine
- d) Galantamine

Question 4:

Which of the following NMDA receptor antagonist would you prescribe to a patient with severe Alzheimer's disease?

- a) Memantine
- b) Ketamine
- c) Piracetam
- d) Amantadine

Question 5:

Which of the following statements is incorrect regarding the management of Alzheimer's disease?

- a) Monoclonal antibodies can be used in the treatment
- b) Cholinesterase inhibitors are commonly used
- c) Memantine is approved only in moderate to severe disease
- d) Cholinesterase inhibitors and memantine have no effect on behavioral and psychiatric symptoms

Question 6:

A 35-year-old lady came to the OPD with complaints of occasional shock-like sensations down her spine and muscle spasms. T2 weighted MRI shows typical Dawson fingers and oligoclonal bands were detected in the CSF study. Which of the following is approved for the treatment of her condition?

- a) Interferon alpha
- b) Interferon beta
- c) Interferon gamma
- d) Infliximab

Question 7:

You plan to start disease-modifying therapy in a 40-year-old patient with relapsing-remitting multiple sclerosis. Which of the following drugs would you not include in the treatment regimen?

- a) Glatiramer acetate
- b) Methylprednisolone

- c) Fingolimod
- d) Cladribine

Question 8:

John Cunningham (JC) virus antibody titer is being measured in a patient with multiple sclerosis. During treatment with which of the following drugs are the titres mandatory?

- a) Alemtuzumab
- b) Natalizumab
- c) Glatiramer
- d) Interferon

Question 9:

A 30-year-old lady presents with complaints of sudden, involuntary, and uncontrollable dance-like movements. Her mother had also developed similar symptoms at the age of 40. Which of the following drugs is indicated for the treatment of the involuntary movements associated with her condition?

- a) Benzhexol
- b) Tetraabenazine
- c) Tetrahydrazoline
- d) Benztropine

Question 10:

Which of the following is not a mechanism of action of Riluzole?

- a) Inhibition of glutamate release
- b) Inhibition of GABA reuptake
- c) NMDA glutamate receptor antagonism
- d) Kainate glutamate receptor antagonism

Question 11:

Which of the following new drugs has been approved for the treatment of amyotrophic lateral sclerosis?

- a) Edaravone
- b) Piracetam
- c) Dabrafenib
- d) Trametinib

Answer Key

Question No.	Correct Option
1	a
2	d
3	c
4	a
5	d
6	b
7	b
8	b
9	b
10	b
11	a

Detailed Explanations

Solution to Question 1:

The given clinical scenario is suggestive of Alzheimer's disease (AD). The most striking neurochemical disturbance in AD is the deficiency of acetylcholine in the subcortical area (especially in the nucleus basalis of Meynert).

Neurotransmitters like glutamate and 5-HT and other areas of the brain like the hippocampus are also involved in the pathogenesis of this disease.

The drugs used in Alzheimer's disease are:

- Central acetylcholine esterase inhibitors targeting acetylcholine deficiency
- NMDA antagonists targeting glutamate excitotoxicity

Solution to Question 2:

Memantine is an NMDA receptor antagonist used to treat moderate to severe dementia in Alzheimer's.

Donepezil, rivastigmine, and galantamine are reversible cholinesterase inhibitors indicated in Alzheimer's disease. Irreversible antagonists are not indicated in Alzheimer's disease.

Solution to Question 3:

Rivastigmine is a cholinesterase inhibitor that is available as a transdermal patch for use in Alzheimer's disease. It can be applied for over 24 hours. Transdermal delivery reduces the fluctuation and avoids direct gastric irritation effects associated with oral formulations.

Option A: Tacrine is an anticholinesterase that is obsolete because of its association with hepatotoxicity.

Options B and D: Donepezil and galantamine transdermal formulations are under development and yet to be available in the market.

Solution to Question 4:

Memantine is a noncompetitive N-methyl-D-aspartate (NMDA) receptor antagonist indicated in Alzheimer's disease. It slows the deterioration of patients with moderate and severe Alzheimer's disease. It is claimed to reduce the excitotoxicity of glutamate.

Option B - Ketamine is an NMDA receptor antagonist indicated as an anesthetic agent.

Option C - Piracetam is a nootropic agent (a drug that enhances cognitive function) that is used off-label to enhance memory.

Option D - Amantadine is an antiviral agent with NMDA receptor antagonistic action which is also indicated in Parkinson's disease due to its action on dopamine release and reuptake.

Solution to Question 5:

Cholinesterase inhibitors and memantine reduce some of the behavioral and psychiatric symptoms in Alzheimer's disease.

SSRIs like citalopram or benzodiazepines for agitation and atypical antipsychotics for psychiatric symptoms may be considered in severe cases. But atypical antipsychotics in dementia-related psychosis have shown higher mortality.

Option A - Amyloid beta-directed monoclonal antibodies approved for the treatment of Alzheimer's disease are - lecanemab, donanemab, aducanumab

Option B - Cholinesterase inhibitors are commonly used and are considered as first-line in early or mild to moderate stages of Alzheimer's disease.

Option C - Memantine is approved only in moderate-severe Alzheimer's disease. There is not enough data to support its use in mild to moderate stages.

Note- Aducanumab has been discontinued in the treatment of Alzheimer's

Solution to Question 6:

The given scenario is suggestive of multiple sclerosis. Interferon beta is approved for use in this condition. It reduces the frequency of attacks in relapsing-remitting multiple sclerosis.

The probable mechanisms of action of interferon-beta are:

- Immune modulation and a decrease in T cell proliferation.
- Limiting the entry of T cells into CNS.
- Reducing pro-inflammatory and increasing regulatory cytokines.

Solution to Question 7:

The disease-modifying therapy of relapsing-remitting multiple sclerosis (MS) does not include methylprednisolone. It is used in acute attacks and exacerbations for MS symptom control. The long-term value or disease-modifying potential of glucocorticoid therapy is unknown.

FDA approved disease-modifying therapy of relapsing-remitting multiple sclerosis are:

- Interferon-beta 1a
- Interferon-beta 1b
- Glatiramer
- Dimethyl fumarate
- Teriflunomide
- Alemtuzumab
- Natalizumab
- Fingolimod
- Mitoxantrone
- Cladribine

Solution to Question 8:

John Cunningham (JC) virus antibody titre measurement is mandatory during treatment with monoclonal antibodies like natalizumab. Progressive multifocal leukoencephalopathy (PML) is a life-threatening disease caused by the JC virus in patients with multiple sclerosis.

It is more common when the patient is immunocompromised during their treatment with natalizumab. The current recommendation is to start natalizumab only in a patient who tests negative for JC antibodies.

Seroconversion can occur anytime during treatment. Regular monitoring for JC antibody every 6 months is recommended in JC antibody-negative patients on natalizumab.

Solution to Question 9:

This patient is most likely suffering from chorea associated with Huntington's disease, for which tetrabenazine is indicated. It is a reversible inhibitor of vesicular monoamine transporter-2 (VMAT-2).

Other drugs used in Huntington's disease are:

- Reserpine, an irreversible inhibitor of VMAT-2 and a longer-acting agent as compared to tetrabenazine.
- Deutetrabenazine refers to deuterium attached to carbon in place of hydrogen, in tetrabenazine (deuterated tetrabenazine). It is a newer form of tetrabenazine approved for the treatment of chorea associated with Huntington's disease. It slows down its metabolism, hence it is longer acting than tetrabenazine. It is associated with better control of chorea.

Option A - Benzhexol is a central anticholinergic indicated in parkinsonism.

Option C - Tetrahydrozoline is a topical alpha-adrenergic agonist indicated for the relief of conjunctival congestion.

Option D - Bzotropine is a muscarinic receptor antagonist that can be used for the management of acute dystonia.

Solution to Question 10:

Riluzole does not act by causing inhibition of GABA reuptake.

Riluzole is the only drug known to improve survival in amyotrophic lateral sclerosis (ALS), also known as Lou Gehrig's disease. The mechanism of action involves the reduction of glutamate-induced excitotoxicity by:

- Inhibition of glutamate release
- Antagonism of NMDA and kainate-type glutamate receptor
- Direct action on the voltage-dependent sodium channels.

Solution to Question 11:

Edaravone was approved by the FDA for the treatment of amyotrophic lateral sclerosis (ALS).

It has free radical scavenging properties that may reduce oxidative stress. The exact mechanism of action is unknown. It is to be administered as an IV infusion with an initial treatment cycle of daily dosing for 14 days, followed by a 14-day drug-free period. Subsequent treatment cycles consist of dosing on 10 of 14 days, followed by a 14 day drug-free period.

The major adverse effect of edaravone is hypersensitivity because the infusion contains sodium bisulfite. Other adverse effects include bruising, gait disorder, and headache.

Tafinlar (dabrafenib) and Mekinist (trametinib), are both indicated as monotherapy in BRAF V600E/K (BRAF V600E mutation-positive) melanoma. They are approved for combined administration, for the treatment of anaplastic thyroid cancer (ATC) that cannot be removed by surgery or that has spread to other parts of the body (metastatic).

Piracetam is a nootropic agent (a drug that enhances cognitive function) that is used off-label to enhance memory. It is also used to treat myoclonus.

General Principles of Antimicrobial Therapy

Question 1:

A young man who collapsed at a concert was suspected to have bacterial meningitis. The other attendees of the concert were administered ciprofloxacin by the medical staff on duty. What is this kind of antimicrobial therapy called?

- a) Preemptive therapy
- b) Prophylactic therapy
- c) Empirical therapy
- d) Suppressive therapy

Question 2:

The total microbial population was found to be susceptible to antimicrobials on testing but a subset of the same population was resistant which was undetected. What is this phenomenon called?

- a) Cross resistance
- b) Heteroresistance
- c) Idioresistance
- d) Alloresistance

Question 3:

Which of the following drugs shows concentration-dependent killing?

- a) Penicillins
- b) 5-fluorocytosine
- c) Aminoglycosides
- d) Vancomycin

Question 4:

A patient having pneumococcal pneumonia could not be treated using penicillin G as the microbe was found to be resistant to it. How is this resistance mainly acquired?

- a) Conjugation
- b) Transduction
- c) Transformation
- d) Lysogenic conversion

Question 5:

What is the extrachromosomal DNA present in bacteria that transmits drug resistance called?

- a) Cytochrome
- b) Plasmid
- c) Introns
- d) Exons

Question 6:

How is multidrug resistance transferred among bacteria?

- a) Conjugation
- b) Transformation
- c) Transduction
- d) Transfection

Question 7:

An example of a bacteriostatic drug is _____

- a) Aminoglycoside
- b) Vancomycin
- c) Metronidazole
- d) Linezolid

Question 8:

You want to prescribe a bactericidal drug to a patient having an infection. Which of the following drugs will you not choose?

- a) Isoniazid
- b) Tigecycline
- c) Daptomycin
- d) Ciprofloxacin

Question 9:

A patient developed a surgical site infection with *Klebsiella*, which is found to be sensitive to a cell wall synthesis inhibitor. Which of the following antibiotics is this likely to be?

- a) Aminoglycosides
- b) Cefepime
- c) Doxycycline
- d) Erythromycin

Question 10:

A patient has presented to your OPD complaining of vomiting and rice water stools. You decide to treat him with a protein synthesis inhibitor. Which antibiotic do you choose?

- a) Cefotetan
- b) Doxycycline
- c) Ciprofloxacin
- d) Oxacillin

Question 11:

Which of the following antibacterial agents does not act by inhibiting cell wall synthesis?

- a) Carbapenems
- b) Monobactams
- c) Cephalosporins
- d) Nitrofurantoin

Question 12:

You are studying the susceptibility of Streptococci to bacitracin in the lab. What is the site of action of this drug?

- a) Cell wall
- b) Cell membrane
- c) Nucleic acid
- d) Ribosome

Answer Key

Question No.	Correct Option
1	b
2	b
3	c
4	c
5	b
6	a
7	d
8	b
9	b
10	b
11	d
12	a

Detailed Explanations

Solution to Question 1:

The antimicrobial treatment initiated in a patient who is not yet infected is called prophylactic therapy. The goal is to prevent infection in some patients and to prevent dangerous manifestations of infections in other patients.

Option A: Preemptive therapy is therapy in an asymptomatic individual with evidence of infection (high-risk).

Option C: Empirical therapy is broad-spectrum therapy in a patient with symptoms of infection but no microbiological confirmation of the causative agent.

Option D: Suppressive therapy is given to patients when the infection is controlled but not completely eradicated by the initial round of antimicrobial treatment. This is usually indicated for immunosuppressed patients.

Solution to Question 2:

Heteroresistance occurs when a subset of the total microbial population is resistant, despite the total microbial population being susceptible on testing. The resistant subpopulation is usually well below the assay-detection limits.

Heteroresistant Staphylococci and *M. tuberculosis* have led to therapeutic failures and mortality. *C. albicans* and *C. neoformans* with heteroresistance to fluconazole have resulted in increased treatment failures.

Note: For viruses, the resistant viral species occurring due to the above phenomenon is termed as 'quasi-species'. Quasi-species resistant to antiretroviral agents have resulted in the failure of antiretroviral therapy.

Solution to Question 3:

Aminoglycosides exhibit concentration-dependent killing and are preferably dosed once daily for maximum effect.

Peak concentration above MIC (minimum inhibitory concentration) matters for certain classes of drugs for their antimicrobial action. It is referred to as concentration-dependent killing. The period during which the concentration remains above MIC is not relevant for such drugs. Aminoglycosides and quinolones have this property.

Time-dependent killing: The antimicrobial action is dependent on the period for which the concentration of drug remains above the MIC. Penicillins, 5-fluorocytosine and vancomycin have this property.

Solution to Question 4:

Transformation is the molecular basis of penicillin G resistance in pneumococci and meningococci.

Solution to Question 5:

Drug resistance transmitting extrachromosomal DNA present in bacteria is a plasmid.

They are usually involved in the horizontal transfer of drug resistance gene in bacteria. Chromosomal DNA can also be involved in the transfer of resistance.

Solution to Question 6:

Multidrug resistance is transferred through conjugation.

Multiple genes, each encoding for resistance to a drug can be present in a single bacterial cell. Typically such genes are present on resistance (R) plasmids. The main mode of transfer of resistant gene-containing plasmids is through conjugation.

Solution to Question 7:

Linezolid is an example of a bacteriostatic drug.

Primarily bacteriostatic drugs	Primarily bactericidal drugs
Sulfonamides Erythromycin Tetracyclines Clindamycin Chloramphenicol Linezolid Ethambutol	Penicillins, Cephalosporins Aminoglycosides Vancomycin Polypeptides Ciprofloxacin Rifampin Metronidazole Isoniazid Cotrimoxazole Pyrazinamide

Solution to Question 8:

Tigecycline is not a bactericidal drug. Tigecycline belongs to a new class of glycylcyclines which is a congener of tetracycline. It is a protein synthesis inhibitor and generally considered as a bacteriostatic.

Isoniazid, daptomycin and ciprofloxacin are all bactericidal agents.

Solution to Question 9:

Cefepime is an antibiotic that acts by inhibiting cell wall synthesis.

Cefepime is a fourth-generation cephalosporin. Cephalosporins act by inhibiting cell wall synthesis.

Aminoglycosides, doxycycline, and erythromycin are protein synthesis inhibitors.

Solution to Question 10:

Doxycycline is a tetracycline that acts by inhibiting protein synthesis. It is the drug of choice for cholera.

Options A & D: Cefotetan and oxacillin are beta-lactam antibiotics that act by inhibiting cell wall synthesis.

Option C: Ciprofloxacin is a fluoroquinolone that acts by inhibiting DNA gyrase.

Solution to Question 11:

Nitrofurantoin is an antibacterial drug that does not act by inhibiting cell wall synthesis.

The antibacterial activity of nitrofurantoin appears to correlate with the rapid intracellular conversion of nitrofurantoin to highly reactive intermediates by bacterial reductases. These intermediates react nonspecifically with many ribosomal proteins and disrupt the synthesis of proteins, RNA, DNA, and metabolic processes.

Carbapenems, monobactams, and cephalosporins are beta-lactam antibiotics that act by inhibiting cell wall synthesis.

Solution to Question 12:

Bacitracin acts on the cell wall by inhibiting the synthesis of the cell wall.

It interferes with the dephosphorylation of bactoprenol (lipid carrier) which carries peptidoglycan subunits outside the cell membrane to the growing cell wall. It can also alter the integrity of the cell membrane.

Bacitracin is mainly active against gram-positive cocci and bacilli.

Antimalarial Drugs

Question 1:

Which of the following antimalarial drugs is a slow acting schizonticide?

- a) Artemether
- b) Mefloquine
- c) Pyrimethamine
- d) Quinine

Question 2:

Which of the following is gametocidal to all the species causing malaria?

- a) Primaquine
- b) Chloroquine
- c) Quinine
- d) None

Question 3:

A researcher who was studying plasmodium species decided to visit a malaria-endemic place in Odisha to do his clinical study for 5 weeks. He is planning to leave after 2 days. Which of the following drugs can be taken as prophylaxis before visiting the malaria-endemic place?

- a) Artesunate
- b) Mefloquine
- c) Doxycycline
- d) Chloramphenicol

Question 4:

A 35-year-old woman presented with fever and chills. Peripheral blood smear revealed findings suggestive of *P. vivax* malaria. Which among the following is the most appropriate drug for this patient?

- a) Chloroquine
- b) Primaquine
- c) Sulfadoxine-pyrimethamine
- d) Artesunate

Question 5:

Radical cure is required for malaria caused by which of the following species?

- a) *P. falciparum* and *P. vivax*
- b) *P. falciparum* and *P. malariae*
- c) *P. vivax* and *P. malariae*
- d) *P. vivax* and *P. ovale*

Question 6:

Tafenoquine is used in which of the following conditions?

- a) Radical cure of *Plasmodium vivax*
- b) Prophylaxis of malaria in pregnancy
- c) Treatment of severe *falciparum* malaria
- d) Treatment of malaria in children <2 years

Question 7:

Chloroquine is not useful in the treatment of which of the following conditions?

- a) Systemic lupus erythematosus
- b) Porphyria cutanea tarda
- c) Giardiasis
- d) Sjogren's Syndrome

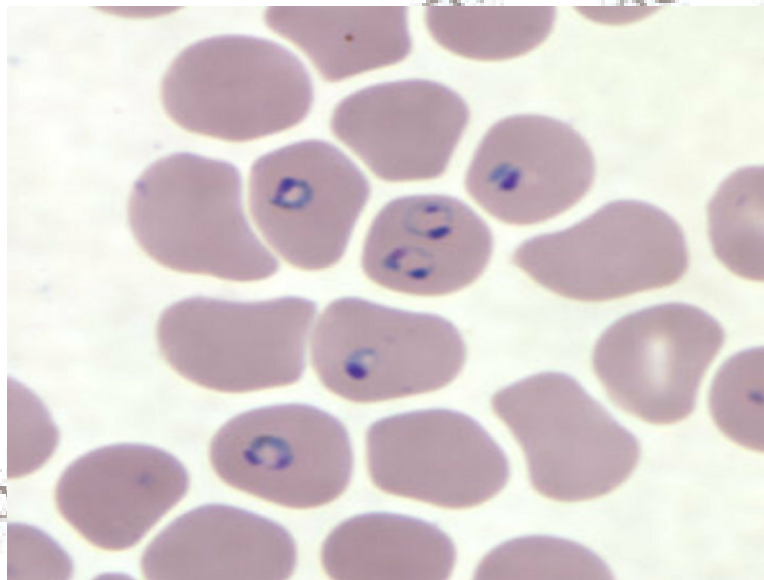
Question 8:

A 26-year-old woman who is 10 weeks pregnant, comes with history of fever and chills. She is diagnosed with *falciparum* malaria. Which of the following drugs will you choose for her treatment?

- a) Quinine plus clindamycin
- b) Quinine plus doxycycline
- c) Artemether plus lumefantrine
- d) Artesunate plus pyrimethamine-sulfadoxine

Question 9:

A 32-year-old man presents with fever with chills and seizures. On examination, his blood pressure was 70/50 mm of Hg and labored breathing was present. Routine blood investigations revealed Hb of 4.8 g/dl. ABG was done and the pH was 7.21. Findings in peripheral blood smear are shown below. Which of the following is the drug of choice in this condition?



- a) Chloroquine
- b) Quinine
- c) Artesunate
- d) Artemether

Question 10:

Which of the following is not a WHO recommended artemisinin-based combination therapy?

- a) Artesunate plus amodiaquine
- b) Artesunate plus quinine
- c) Artesunate plus pyrimethamine-sulfadoxine

- d) Artesunate plus mefloquine

Question 11:

You are posted in a health centre in Nagaland as part of your rural service and a patient is diagnosed with uncomplicated falciparum malaria. Which of the following artemisinin-based combination therapy is recommended by NVBDCP for this patient?

- a) Artesunate - sulfadoxine+pyrimethamine
- b) Artesunate - chloroquine
- c) Artesunate - mefloquine
- d) Artemether - lumefantrine

Question 12:

Loading dose is essential for chloroquine because of which of the following properties?

- a) Undergoes fast hofmann degradation
- b) Rapid hepatic elimination
- c) High volume of distribution
- d) High first-pass metabolism

Question 13:

While writing a prescription for a patient diagnosed with malaria, the medical officer instructed the patient to take a particular drug along with a high fatty meal. Identify the drug.

- a) Mefloquine
- b) Lumefantrine
- c) Chloroquine
- d) Amodiaquine

Question 14:

A patient who was diagnosed with malaria was initiated on treatment about 3 days ago. Now she presents with seizures and confusion. Which of the following drugs is most likely to have caused this condition?

- a) Artesunate
- b) Arteether
- c) Lumefantrine
- d) Mefloquine

Question 15:

The finding shown in the image below is seen as a side effect of which of the following drugs?



- a) 1, 4 and 5 only
- b) 1, 3 and 5 only
- c) 1 and 3 only
- d) 2 and 4 only

Question 16:

A pregnant patient with severe malaria is undergoing treatment with quinine. Which of the following adverse effects is not associated with this drug?

- a) Tinnitus
- b) Hypoglycemia
- c) Hypertension
- d) Blackwater fever

Question 17:

A patient who was initiated on treatment for malaria two days ago, now presents with complaints of passage of dark-colored urine. Hemolysis due to G6PD deficiency is suspected. With which of the following drugs is this condition more common?

- a) Quinine
- b) Chloroquine
- c) Hydroxychloroquine
- d) Primaquine

Answer Key

Question No.	Correct Option
1	c
2	a
3	c
4	a
5	d
6	a
7	d
8	a
9	c
10	b
11	d
12	c
13	b
14	d
15	c
16	c
17	d

Detailed Explanations

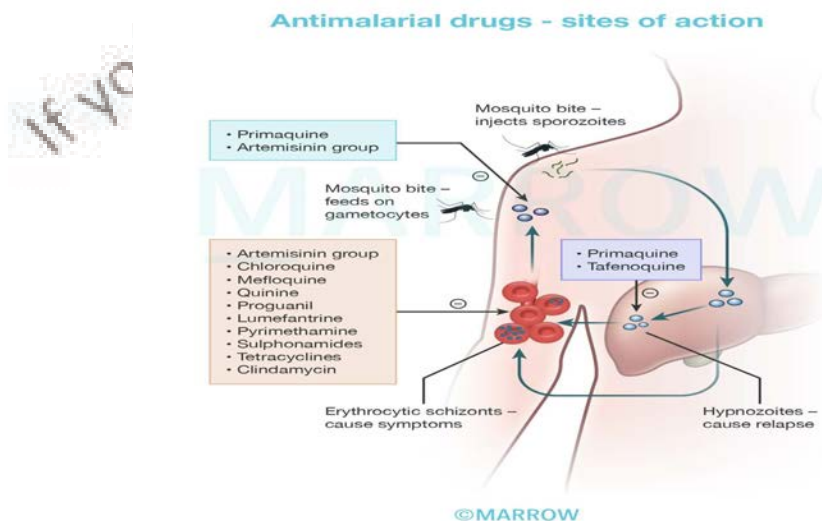
Solution to Question 1:

Pyrimethamine is a slow-acting schizonticide.

Female Anopheles mosquito inoculates Plasmodium species in the form of sporozoites into the human host during blood meal. Sporozoites migrate to the liver and mature into hepatic schizonts. Hepatic schizonts are released into the blood in the form of merozoites. Merozoites in the erythrocyte transform into ring form and erythrocytic schizonts. Some merozoites in the blood transform into gametocytes.

Antimalarial drugs are classified based on their onset of action of schizonticidal function as below:

- Fast onset of action
- Artemisinin derivatives like artemether, artesunate and artemisinate are fastest acting
- Chloroquine
- Intermediate onset of action
- Mefloquine
- Quinine
- Proguanil
- Lumefantrine
- Slow onset of action
- Pyrimethamine
- Sulfonamides
- Tetracyclines
- Clindamycin



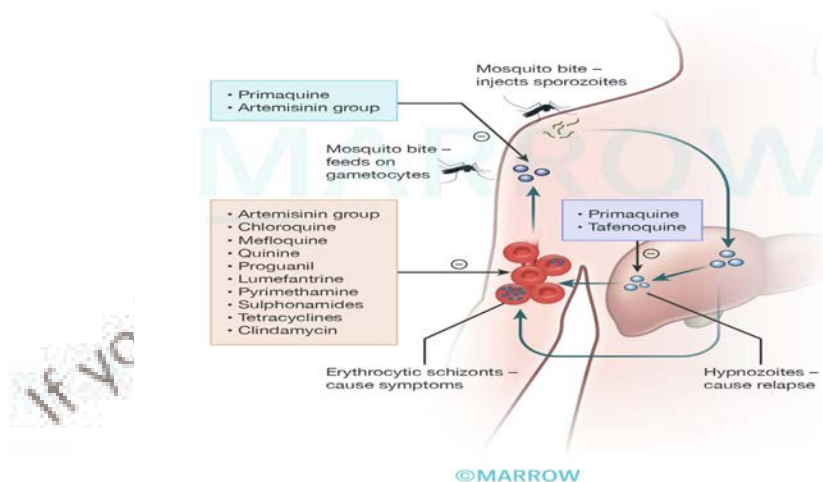
Solution to Question 2:

Primaquine is gametocidal to all species causing malaria. It inhibits transmission of malaria.

The drugs with clinically important gametocidal activity (against all species) are:

- High activity:
- Artemisinin
- Primaquine
- Low to moderate activity:
- Chloroquine
- Quinine
- Atovaquone/proguanil
- No activity:
- Mefloquine
- Pyrimethamine
- Sulfadoxine
- Tetracycline

Antimalarial drugs - sites of action



Solution to Question 3:

Doxycycline can be taken as prophylaxis by the researcher before going to malaria-endemic places.

Chemoprophylaxis against malaria for travelers:

- If the duration of travel is less than 6 weeks then doxycycline 100mg once a day is to be started 2 days before travel and continued atleast 4 weeks after travel.
- If the duration of travel is more than 6 weeks then mefloquine 250mg once a week is to be started 2 weeks before travel and continued atleast 4 weeks after travel.

Solution to Question 4:

Most appropriate drug for the treatment of malaria due to *P. vivax* is chloroquine.

P. vivax treatment recommendations are:

- Chloroquine-susceptible infections: Chloroquine or artemisinin-based combination therapy (ACT).
- Chloroquine-resistant infections: ACT is the drug of choice but quinine is used during 1st trimester of pregnancy.

In the absence of resistance data in the question, chloroquine appears to be the most appropriate answer.

Solution to Question 5:

Radical cure is required for malaria caused by *P. vivax* and *P. ovale*.

In case of *Plasmodium vivax* and *Plasmodium ovale*, malarial parasites have an exoerythrocytic stage (hypnozoites). In the exoerythrocytic phase malarial parasites are found in the liver that can cause relapse of malaria after clinical cure.

Therefore, they require radical cure by drugs acting on the exoerythrocytic stage. Primaquine is effective against the exoerythrocytic stages of *Plasmodium vivax* and *Plasmodium ovale* in the liver. Primaquine is administered for 14 days for radical cure.

Solution to Question 6:

Tafenoquine is used in the radical cure of *Plasmodium vivax*. Total eradication of plasmodium species which includes clinical curative and hypnozoites phase are called radical cure.

Primaquine and tafenoquine (derivative of primaquine) have very high activity against the latent tissue forms (hypnozoites) of the species. It is primarily used for the chemoprophylaxis and radical cure of *P. vivax* and *P. ovale* infections.

Solution to Question 7:

Chloroquine is used in the treatment of systemic lupus erythematosus, porphyria cutanea tarda and giardiasis, due to its anti-inflammatory effect; but it is not used in the treatment of Sjogren's Syndrome.

Chloroquine and hydroxychloroquine concentrate in lysosomes and have an anti-inflammatory effect. It also has antihistaminic and antiarrhythmic properties.

Non-antimalarial uses of chloroquine are:

- Hepatic amoebiasis
- Giardiasis
- Rheumatoid arthritis
- Systemic lupus erythematosus
- Discoid lupus erythematosus
- Porphyria cutanea tarda
- Severe polymorphous light eruption

Solution to Question 8:

The woman is in the 1st trimester of pregnancy and falciparum malaria during 1st trimester of pregnancy is treated with quinine plus clindamycin.

Worldwide, most species of Falciparum malaria are resistant to chloroquine. In all trimesters, pregnant women diagnosed with uncomplicated malaria caused by chloroquine-resistant P. falciparum infection can be treated with a combination of quinine and clindamycin.

Quinine is considered the drug of choice in the first trimester. In the second and third trimesters artemether-lumefantrine can be used, but not in the first trimester.

Vivax malaria during pregnancy is treated with chloroquine as chloroquine is safe in pregnancy and children.

Clindamycin can be used in children for chloroquine-resistant cases. It is safe to use in children and pregnant women.

Option B: Doxycycline is useful in the treatment of malaria but not used in pregnancy.

Options C and D: Artemisinin-based combination therapy (ACT) is not recommended during the 1st trimester of pregnancy because of a lack of safety data.

Solution to Question 9:

This scenario depicts a case of severe complicated falciparum malaria, and the drug of choice in this condition is artesunate. The image shows ring forms/early trophozoites in erythrocyte, which are binucleate (double chromatin) and this is characteristic of P. falciparum malaria.

Severe complicated malaria is caused by P. falciparum and is characterized by hyperparasitemia, hyperpyrexia, acidosis, respiratory distress, bleeding manifestation, cardiovascular collapse, renal failure and cerebral malaria.

As per WHO guidelines, all patients with severe malaria should be treated with intravenous or intramuscular artesunate for at least 24 hours. Once the patient recovers and can tolerate oral medications, oral ACT for 3 days is recommended.

Artemether is used only if parenteral artesunate is not available.

Solution to Question 10:

WHO recommended artemisinin-based combination therapies are:

- Artemether plus lumefantrine
- Artesunate plus pyrimethamine-sulfadoxine
- Artesunate plus mefloquine
- Artesunate plus amodiaquine
- Dihydroartemisinin plus piperaquine

Solution to Question 11:

The artemisinin-based combination therapy (ACT) recommended by NVBDCP in the north-eastern states is a combination of artemether and lumefantrine.

NVBDCP guidelines for the treatment of uncomplicated falciparum cases:

- North-eastern states: Artemether-lumefantrine based ACT
- Rest of the country: Artesunate-sulfadoxine+pyrimethamine

In the north-eastern states (Arunachal Pradesh, Assam, Manipur, Meghalaya, Mizoram, Nagaland, and Tripura), the presently recommended ACT for *P. falciparum* malaria is a fixed-dose combination of artemether-lumefantrine due to the recent reports of late treatment failures with the current combination of AS+SP.

Solution to Question 12:

Loading dose is essential for chloroquine because of its property of high volume of distribution.

Chloroquine is distributed extensively in the liver, spleen, kidney, lungs, skin and, to a lesser extent, brain and spinal cord. Because of extensive tissue binding, a loading dose is required for achieving effective plasma concentrations earlier in the therapy.

Solution to Question 13:

The drug described here is lumefantrine and its absorption increases when it is administered along with fatty food or milk because it is highly lipophilic.

Solution to Question 14:

The case depicts neuropsychiatric adverse effects which are commonly associated with the use of mefloquine. The US FDA has put a "black box warning" for this drug due to permanent and serious neuropsychiatric side effects.

The CNS toxicity caused by mefloquine includes:

- Seizures
- Acute psychosis
- Severe vertigo
- Confusion or decreased sensorium

Solution to Question 15:

The finding shown in the image is bulls-eye maculopathy, and chloroquine and hydroxychloroquine use are associated with this side effect.

Bull's-eye maculopathy refers to the ophthalmoscopic appearance of a central area of retinal pigment epithelial depigmentation surrounded by relatively normal retinal pigment epithelium, giving a bull's-eye appearance to the macula.

Solution to Question 16:

Quinine is not associated with hypertension. It is associated with hypotension.

Quinine is used in the treatment of drug-resistant and severe *P. falciparum* malaria. It acts against asexual erythrocytic forms but has no significant effect on hepatic forms of malarial parasites.

Option A: Quinine use is associated with cinchonism which manifests as tinnitus, visual and auditory disturbances, headache, nausea, and vomiting.

Option B: Quinine has a stimulatory effect on pancreatic beta cells and can cause hypoglycemia.

Option D: Blackwater fever is a rare type of hypersensitivity reaction to quinine. It presents with massive hemolysis, hemoglobinemia, and hemoglobinuria that may lead to anuria, renal failure, and death.

Solution to Question 17:

Hemolysis in a G6PD deficient patient is more common with primaquine.

G6PD deficient patients are highly sensitive to oxidative stress. They are at increased risk of hemolysis due to the oxidant property of primaquine. Therefore, before initiating primaquine therapy, the G6PD status of the patient has to be determined. Passage of dark color urine is an indication of hemolysis and the drug has to be stopped if it occurs.

Quinine, chloroquine and hydroxychloroquine can also cause hemolysis in G6PD deficient patients, but the incidence is less compared to primaquine.

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you may have been scammed.

Sulfonamides , Quinolones and Urinary antiseptics

Question 1:

Sulfonamides are structural analogs of which of the following compounds?

- a) Para-amino benzoic acid
- b) Pteridine
- c) Dihydropteroic acid
- d) Dihydrofolic acid

Question 2:

Which of the following is not a topically used sulfonamide?

- a) Sulfacetamide
- b) Sulfasalazine
- c) Silver sulfadiazine
- d) Mafenide

Question 3:

Trimethoprim is an inhibitor of which of the following enzymes?

- a) Dihydropteroate synthase
- b) Dihydrofolate reductase
- c) Dihydropteroate reductase
- d) Dihydrofolate synthase

Question 4:

Which of the following combinations is indicated for the prophylaxis of malaria?

- a) Sulfamethoxazole-trimethoprim
- b) Sulfadoxine-pyrimethamine
- c) Sulfadiazine-pyrimethamine

d) Sulfamethoxazole-pyrimethamine

Question 5:

Which of the following options correctly describes the dose ratio of trimethoprim and sulfamethoxazole in co-trimoxazole?

- a) 1 : 20 (trimethoprim : sulfamethoxazole)
- b) 1 : 5 (trimethoprim : sulfamethoxazole)
- c) 20 : 1 (trimethoprim : sulfamethoxazole)
- d) 5 : 1 (trimethoprim : sulfamethoxazole)

Question 6:

Which of the following is the drug of choice for *Pneumocystis jiroveci* pneumonia prophylaxis in a patient who has undergone organ transplantation?

- a) Aerosolized pentamidine
- b) Cotrimoxazole
- c) Dapsone
- d) Dapsone plus pyrimethamine

Question 7:

Co-trimoxazole is not prescribed in which of the following conditions?

- a) Chancroid
- b) Lower urinary tract infection
- c) Prostatitis
- d) Nocardiosis

Question 8:

An infant had a serum bilirubin level of 33 mg/dL. The mother might have taken which of the following drugs during the third trimester for this to occur?

- a) Ampicillin
- b) Co-trimoxazole

- c) Azithromycin
- d) Chloramphenicol

Question 9:

A girl presents to the emergency room with severe abdominal pain, dark-colored urine, and seizures. You find out that she was recently prescribed co-trimoxazole for a UTI. What is the probable diagnosis?

- a) Acute intermittent porphyria
- b) Porphyria cutanea tarda
- c) Congenital erythropoietic porphyria
- d) Hepatoerythropoietic porphyria

Question 10:

What is the mechanism of action of fluoroquinolones in gram-negative bacteria?

- a) Inhibits bacterial DNA helicase enzyme
- b) Inhibits bacterial primase enzyme
- c) Inhibits bacterial DNA gyrase enzyme
- d) Inhibits bacterial topoisomerase IV

Question 11:

Which of the following fluoroquinolones has maximum bioavailability?

- a) Ciprofloxacin
- b) Levofloxacin
- c) Gatifloxacin
- d) Norfloxacin

Question 12:

A patient who is a known case of cystic fibrosis presents to the OPD with complaints of fever and increased cough with expectoration. Sputum culture is positive for *Pseudomonas aeruginosa*. Which of the following fluoroquinolones can be used in this condition?

- a) 1 and 5 only
- b) 1, 3 and 5 only
- c) 1 and 3 only
- d) 2 and 4 only

Question 13:

Which of the following quinolones is not approved for use in a patient diagnosed with urinary tract infection?

- a) Nalidixic acid
- b) Ciprofloxacin
- c) Moxifloxacin
- d) Norfloxacin

Question 14:

A patient presents with a wound on his leg with purulent discharge. Pus culture was positive for MRSA. Which of the following drugs will you prescribe?

- a) Ciprofloxacin
- b) Sparfloxacin
- c) Delafloxacin
- d) Levofloxacin

Question 15:

Which of the following fluoroquinolones can be used in a patient with rifampicin-resistant leprosy?

- a) Ciprofloxacin
- b) Norfloxacin
- c) Ofloxacin
- d) Pefloxacin

Question 16:

A patient presents to the OPD with sudden onset abdominal cramps, fever, and diarrhea. He is a foreigner who is on a tour in India and is a known asthmatic on theophylline. You decide to start antibiotics. However, you would avoid ciprofloxacin in this patient because _____.

- a) Theophylline inhibits ciprofloxacin metabolism
- b) Ciprofloxacin inhibits theophylline metabolism
- c) Ciprofloxacin decreases the effect of theophylline
- d) Theophylline induces metabolism of ciprofloxacin

Question 17:

A 52-year-old man diagnosed with multi-drug resistant tuberculosis requires second-line agents for his therapy. However, he is found to have a creatinine clearance of 50 ml/min. Which of the following fluoroquinolones can be started in this patient without dosage adjustment?

- a) Ciprofloxacin
- b) Levofloxacin
- c) Moxifloxacin
- d) Ofloxacin

Question 18:

A 65-year-old man presents with symptoms suggestive of Achilles tendon rupture. He had undergone renal transplantation and is now being treated with an antimicrobial agent. Which of the following antimicrobials might he be on?

- a) Cotrimoxazole
- b) Ciprofloxacin
- c) Capreomycin
- d) Clotrimazole

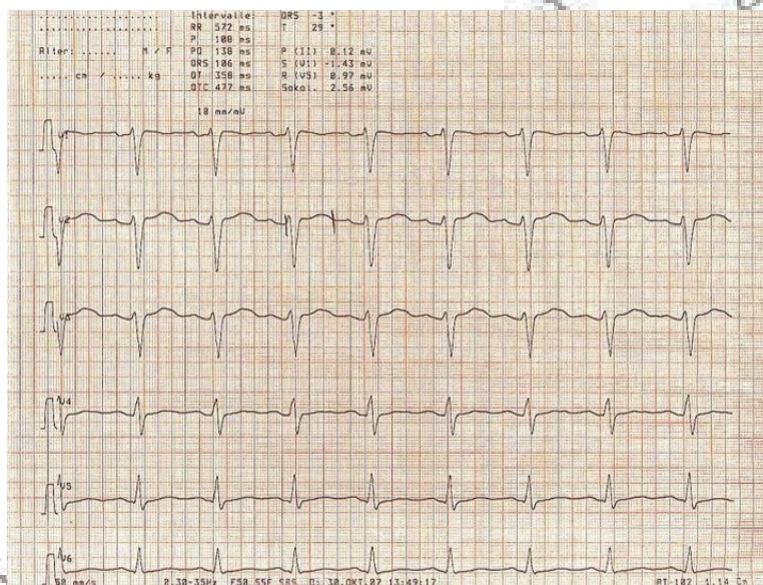
Question 19:

A 60-year-old woman undergoing treatment with moxifloxacin was brought to the casualty with delirium and seizures. According to her son, she had been taking diclofenac tablets over the counter to relieve her ankle pain, which recently started. Which of the following best describes the reason for her current ER admission?

- a) Inhibition of GABA metabolism
- b) Enhanced GABA binding to its receptor
- c) Displacement of GABA from receptor
- d) Inhibition of glycine metabolism

Question 20:

The ECG of a patient undergoing treatment with a quinolone is given below and you notice that $QT_c = 477$ ms. Which of the following drugs has the highest risk of causing this condition?



- a) Moxifloxacin
- b) Ciprofloxacin
- c) Gemifloxacin
- d) Ofloxacin

Question 21:

Which of the following is a urinary antiseptic?

- a) Galantamine
- b) Methenamine
- c) Mecamylamine
- d) Pyrimethamine

Question 22:

Which of the following drugs would you prescribe to a patient in order to relieve pain due to urinary tract infection?

- a) Methenamine
- b) Nitrofurantoin
- c) Phenazopyridine
- d) Phenacetin

Question 23:

A patient presents to the casualty with fever with chills and chest pain. He also gives history of passing brown-colored urine. He was prescribed an antimicrobial drug for urinary tract infection three days ago and his symptoms began after he had started this medication. Which of the following antimicrobial agents had most likely been prescribed?

- a) Methenamine
- b) Nitrofurantoin
- c) Norfloxacin
- d) Gatifloxacin

Question 24:

Which of the following fluoroquinolone is not given in liver disease?

- a) Ofloxacin
- b) Levofloxacin
- c) Pefloxacin
- d) Lomefloxacin

Answer Key

Question No.	Correct Option
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1	a
2	b
3	b
4	b
5	b
6	b
7	a
8	b
9	a
10	c
11	b
12	d
13	c
14	c
15	c
16	b
17	c
18	b
19	c
20	a
21	b
22	c
23	b
24	c

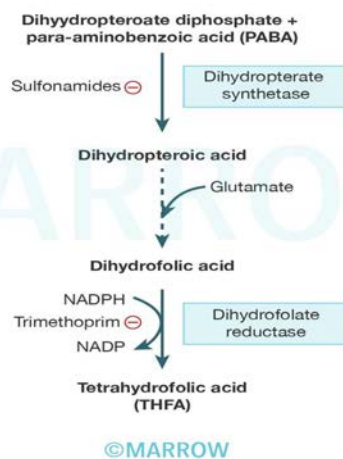
Detailed Explanations

Solution to Question 1:

Sulfonamides are structural analogs of para-aminobenzoic acid (PABA) that interfere in the synthesis of folic acid.

Sulfonamides competitively inhibit dihydropteroate synthase, the enzyme that is involved in the incorporation of PABA into pteridine (dihydropteroate diphosphate) to form dihydropteroic acid, which is an immediate precursor of folic acid.

Sulfonamides - Mechanism of action



Solution to Question 2:

Sulfasalazine is a sulfonamide which is not used topically. It is used orally in the management of ulcerative colitis, rheumatoid arthritis, and regional enteritis.

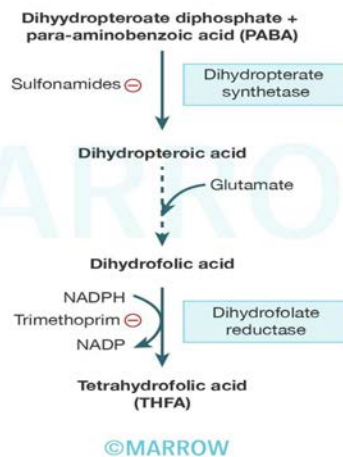
Sulfacetamide is used topically as eye drops for ophthalmic infections.

Silver sulfadiazine and mafenide are used topically for the prevention of burn infections.

Solution to Question 3:

Trimethoprim is a selective inhibitor of microbial dihydrofolate reductase. Dihydrofolate reductase is involved in the reduction of dihydrofolate to tetrahydrofolate.

Sulfonamides - Mechanism of action



Solution to Question 4:

Sulfadoxine-pyrimethamine combination is used for prophylaxis and treatment of malaria caused by mefloquine-resistant strains of *Plasmodium falciparum*.

Sulfamethoxazole-trimethoprim is used in the treatment of *Pneumocystis jiroveci* pneumonia.

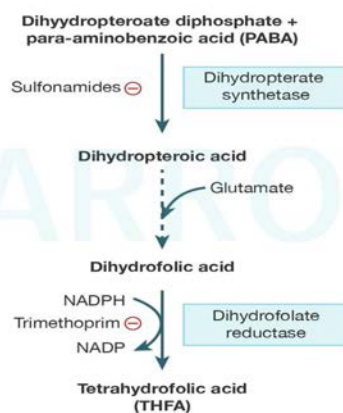
Sulfadiazine-pyrimethamine is used in the treatment of toxoplasmosis.

Solution to Question 5:

Co-trimoxazole is a combination of trimethoprim with sulfamethoxazole in the dose ratio of 1:5 (trimethoprim : sulfamethoxazole).

The antimicrobial activity of this combination is because of its actions on sequential steps in the pathway of synthesis of tetrahydrofolic acid as shown in the image.

Sulfonamides - Mechanism of action



Each agent alone usually exerts bacteriostatic activity. But when an organism is sensitive to both agents, bactericidal activity may be achieved.

Because trimethoprim is more lipid-soluble than sulfamethoxazole, it has a larger volume of distribution than the latter drug.

Therefore, when 1 part of trimethoprim is given with 5 parts of sulfamethoxazole (dose ratio is 1:5; trimethoprim:sulfamethoxazole), the peak plasma concentrations are in the ratio of 1:20, which is optimal for the combined effects of these drugs in vitro.

Solution to Question 6:

Cotrimoxazole is considered the drug of choice for *Pneumocystis jiroveci* pneumonia prophylaxis in immunocompromised patients like patients who have undergone organ transplantation and HIV-positive patients.

Other drugs that can be used for prophylaxis are:

- Aerosolized pentamidine
- Dapsone
- Dapsone plus pyrimethamine
- Atovaquone

Solution to Question 7:

Chancroid was earlier treated with cotrimoxazole but it is not used anymore because better alternatives like ceftriaxone, azithromycin, erythromycin, and ciprofloxacin exist.

Options B & C: Cotrimoxazole is indicated in both upper and lower urinary tract infections and also in prostatitis.

Option D: Patients with severe nocardiosis are initially treated with a combination including cotrimoxazole, amikacin, and ceftriaxone or imipenem.

Solution to Question 8:

Cotrimoxazole is a combination drug consisting of trimethoprim and sulfamethoxazole, of which sulfamethoxazole is a sulfonamide. Sulphonamides displace bilirubin from plasma albumin, leading to elevation of serum bilirubin levels.

These drugs can cross the placenta and are secreted in milk and can therefore cause kernicterus in newborn infants if they are given to pregnant women near term or during lactation.

Solution to Question 9:

The given neurovisceral symptoms following the use of sulfonamides point towards a diagnosis of acute intermittent porphyria (AIP).

AIP is a rare heme biosynthesis disorder due to a partial deficiency of the enzyme uroporphobilinogen deaminase (PBGD). Diagnosis is made based on elevated urine PBG levels. Alcohol, infections, pregnancy, and certain medications can trigger attacks by inducing heme synthesis. This results in the accumulation of porphyrin precursors, causing symptoms.

Side effects of sulfonamides are:

- Acute intermittent porphyria - sulphonamides precipitate attacks
- Hypersensitivity reaction - urticaria, erythema nodosum, erythema multiforme of the Stevens-Johnson type, exfoliative dermatitis, and photosensitivity

- Crystalluria - due to low solubility. It is treated with good hydration and alkalization of urine
- Hemolysis in G-6-PD deficiency (sulphonamides are oxidizers)
- Kernicterus in the newborn (displace bilirubin)

Solution to Question 10:

Fluoroquinolones inhibit two essential type II topoisomerases, DNA gyrase (in gram-negative bacteria) and topoisomerase IV (in gram-positive bacteria).

DNA gyrase introduces negative supercoils and thereby reduces the torsional strain that occurs when double-stranded DNA is being unwound for replication or translation. Topoisomerases IV is responsible for unlinking daughter chromosomes after replication. Quinolone antibiotics form a complex with the topoisomerase enzyme and the DNA, and thereby block DNA replication and bacterial cell division.

Solution to Question 11:

Levofloxacin and ofloxacin have the maximum bioavailability among fluoroquinolones. Hence, intravenous and oral doses are the same for these drugs.

Solution to Question 12:

Ciprofloxacin and levofloxacin have substantial activity against *Pseudomonas aeruginosa* and can be used in this patient.

Mild to moderate respiratory exacerbations owing to *Pseudomonas aeruginosa* in patients with cystic fibrosis respond to oral ciprofloxacin or levofloxacin therapy.

Activity against streptococci is significantly greater with newer quinolones like levofloxacin, gemifloxacin, and moxifloxacin.

Moxifloxacin has clinically significant activity against anaerobes including *B. fragilis*. It is reported to have similar efficacy to piperacillin-tazobactam for complicated intra-abdominal infection.

Solution to Question 13:

Moxifloxacin does not accumulate significantly in urine and is not approved for the treatment of urinary tract infections (UTIs).

UTIs are the only indication for the use of nalidixic acid. Ciprofloxacin and norfloxacin can be used in UTIs caused by susceptible organisms.

Solution to Question 14:

Delafloxacin is a newer fluoroquinolone approved for use in acute bacterial skin and skin structural infections (ABSSSI) caused by susceptible bacteria including MRSA.

Compared to other fluoroquinolones which are zwitterionic in nature, delafloxacin is anionic. This enables the molecule to achieve increased intracellular penetration and concentration, especially under acidic conditions (skin wounds). This results in increased antimicrobial activity and a decreased adverse event profile at clinically potent doses.

Solution to Question 15:

Ofloxacin can be used in this patient as it is the fluoroquinolone that has high activity against *M. leprae* and is indicated in alternative multidrug therapy regimens for leprosy. It also has high activity against *Mycobacterium tuberculosis*.

Solution to Question 16:

Ciprofloxacin should not be given to this patient because it is a cytochrome P450 inhibitor and inhibits the metabolism of theophylline and can potentially lead to theophylline toxicity.

Pefloxacin is another drug that inhibits the metabolism of theophylline and results in its toxicity.

Solution to Question 17:

Moxifloxacin is metabolized predominantly by the liver and hence can be used safely in patients with decreased renal function without dose adjustment.

Fluoroquinolones, especially moxifloxacin, may be used as part of multiple-drug regimens for the treatment of multidrug-resistant tuberculosis.

Most of the available fluoroquinolones are cleared predominantly by kidneys and dosage needs to be adjusted in patients with renal failure. But fluoroquinolones such as moxifloxacin and pefloxacin undergo hepatic elimination and so should be used carefully in patients with liver disease.

Solution to Question 18:

Among the following options, the most likely drug to have caused this condition is ciprofloxacin, since Achilles tendon rupture and tendinitis are side effects of quinolones. Elderly patients (>60 years), patients on corticosteroids and solid organ transplant recipients are at

higher risk.

Solution to Question 19:

The given clinical scenario is suggestive of a fluoroquinolones drug interaction, where NSAIDs augmented the displacement of GABA from its receptors, leading to CNS excitation symptoms.

Fluoroquinolones act as selective antagonists of GABAA receptors and thereby have the potential to cause CNS excitation. NSAIDs such as diclofenac cause the displacement of GABA from its receptors, further enhancing the neuropsychiatric adverse effects caused by fluoroquinolones.

Solution to Question 20:

The given scenario describes QT interval prolongation (Normal QTc = 440 ms). Among the given options, moxifloxacin has the highest risk for causing this condition.

Gemifloxacin, ofloxacin, and ciprofloxacin have a relatively lower risk than moxifloxacin. Ciprofloxacin has the least risk for QT interval prolongation and torsades de pointes. Sparfloxacin has the highest overall risk of QT prolongation and has been withdrawn from many countries due to its propensity to cause torsades de pointes or polymorphic ventricular tachycardia.

Withdrawn Fluoroquinolones	Associated Adverse Events
Sparfloxacin and Gatifloxacin	Phototoxicity, QTc prolongation
Trovafloxacin	Hepatotoxicity
Gatifloxacin	Hypoglycemia
Grepafloxacin	Cardiotoxicity
Temafloxacin	Immune hemolytic anemia
Clinafloxacin	Phototoxicity

Solution to Question 21:

Methenamine is a urinary antiseptic and is indicated in urinary tract infections. At pH<5.5, methenamine releases formaldehyde that inhibits bacteria.

Option A: Galantamine is a central anticholinesterase approved for the treatment of Alzheimer's disease.

Option C: Mecamylamine is a ganglionic blocker that can be used as an antihypertensive agent.

Option D: Pyrimethamine is an antimalarial agent.

Solution to Question 22:

Phenazopyridine can be prescribed to this patient since it is a urinary anesthetic. It is used as a local anesthetic in urinary tract disorders to relieve the pain of lower urinary tract irritation, as in cystitis, urethritis, or prostatitis.

It does not have any antibacterial action but affords symptomatic relief of burning sensation, dysuria and urgency due to cystitis.

Solution to Question 23:

The patient had most likely been prescribed nitrofurantoin as the scenario depicts side effects of this drug.

Nitrofurantoin is a urinary antiseptic used against lower UTI caused by *E. coli* and enterococci. Its use results in brown urine. It is also associated with acute pneumonitis with fever, chills, cough, dyspnea, chest pain, pulmonary infiltration, and eosinophilia within hours to days of initiation of therapy. The symptoms resolve quickly after discontinuation of the drug.

The antibacterial activity of nitrofurantoin is due to its intracellular conversion to highly reactive intermediates by bacterial reductases. These intermediates react nonspecifically with ribosomal proteins and disrupt the synthesis of proteins, RNA, DNA, and other metabolic processes.

Solution to Question 24:

Pefloxacin is mainly metabolized in the liver and hence avoided in patients with liver disease.

Quinolones with primarily hepatic metabolism include moxifloxacin, pefloxacin, and trovafloxacin (withdrawn from the market due to hepatotoxicity).

Quinolones primarily excreted by the kidneys include ciprofloxacin, levofloxacin, norfloxacin, lomefloxacin, and ofloxacin.

Antimicrobials Acting on 30s Subunit

Question 1:

A patient diagnosed with tuberculosis is undergoing treatment with streptomycin. Which of the following is the primary intracellular site of action of this drug?

- a) 30S ribosomal subunit
- b) DNA helicase enzyme
- c) 50S ribosomal subunit
- d) Topoisomerase IV enzyme

Question 2:

Which of the following statements about aminoglycosides is false?

- a) They are teratogenic
- b) They are excreted unchanged in urine
- c) They are bacteriostatic
- d) They are distributed only extracellularly

Question 3:

A young man working in a rabbit farm presents with fever, lymphadenopathy and an ulcer on the hand and is diagnosed with tularemia. What is the drug of choice for this condition?

- a) Amikacin
- b) Neomycin
- c) Kanamycin
- d) Streptomycin

Question 4:

A child presented with recurrent episodes of cough with expectoration. Investigations revealed a positive sweat chloride test. Inhalational formulation of which of the following aminoglycosides is approved for use in this condition?

- a) Amikacin
- b) Gentamicin
- c) Tobramycin
- d) Streptomycin

Question 5:

The organisms detected in culture and sensitivity reports of various specimens obtained from the patients from a medical ward are given below. In which of the following infections would amikacin not be indicated?

- a) 4 only
- b) 1 and 4 only
- c) 5 only
- d) 2 and 3 only

Question 6:

Oral neomycin will not be effective for treating which of the following patients?

- a) A patient requiring pre-operative antisepsis for bowel surgery
- b) A elderly man with hepatic encephalopathy
- c) A child with enteropathogenic E. coli associated diarrhea
- d) A woman with a cut wound on the finger

Question 7:

A patient with hospital-acquired pneumonia is being treated with an aminoglycoside antibiotic by a single daily dose administration method. This method is preferred over the multiple daily dosage method because of which of the following properties of aminoglycosides?

- a) Decreased CYP450 mediated metabolism
- b) Long post antibiotic effect
- c) Increased binding to plasma proteins
- d) Time dependent killing

Question 8:

A patient who was diagnosed with urinary tract infection receives a single intramuscular dose of gentamicin. Within how many minutes of administration is the peak concentration of this drug achieved?

- a) 0-30 minutes
- b) 30-90 minutes
- c) 90-120 minutes
- d) 120-180 minutes

Question 9:

The urine culture and sensitivity report of a patient with urinary tract infection is positive for aminoglycoside-resistant *Enterococcus faecalis*. What is the most common mode of this resistance?

- a) Inhibition of transport across bacterial membranes
- b) Production of inactivating enzymes
- c) Low affinity of bacterial ribosome
- d) Drug efflux mechanism through the membrane

Question 10:

In which of the following patients will you prescribe aminoglycosides at a lower dose?

- a) A urosepsis patient with associated AKI
- b) A cystic fibrosis patient with acute pulmonary exacerbation
- c) A diabetic patient with infected burn wounds
- d) A CLD patient with osteomyelitis

Question 11:

A patient who was on aminoglycoside therapy presents with nausea and vomiting. Investigations revealed rising serum creatinine levels, and granular casts were seen on urine microscopy. Which of the following aminoglycosides is least likely to cause this condition?

- a) Amikacin
- b) Gentamicin

- c) Streptomycin
- d) Tobramycin

Question 12:

Vestibular toxicity is predominantly seen with which of the following aminoglycosides?

- a) Amikacin
- b) Kanamycin
- c) Streptomycin
- d) Neomycin

Question 13:

A patient who was recently started on i.v. gentamicin for enterococcal endocarditis presented to the OPD with complaints of headache and dizziness. By which of the following mechanisms does the drug cause this adverse effect?

- a) Direct hair cell toxicity
- b) Inhibition of hair cell Na^+/K^+ ATPase
- c) Non-cumulative toxicity
- d) Interference with Ca^{2+} channels

Question 14:

An elderly patient who was taken up for an emergency laparotomy showed prolonged respiratory depression which did not resolve with neostigmine. On inquiry, the attendant confirmed that the patient was on aminoglycoside therapy. Which of the following is least likely to cause this effect?

- a) Amikacin
- b) Kanamycin
- c) Gentamicin
- d) Tobramycin

Question 15:

Which of the following is the drug of choice for scrub typhus?

- a) Doxycycline
- b) Azithromycin
- c) Chloramphenicol
- d) Ciprofloxacin

Question 16:

By which of the following mechanisms do tetracyclines inhibit protein synthesis?

- a) Inhibition of initiation of translation and misreading of mRNA
- b) Binding to 30S subunit and inhibiting the binding of aminoacyl-tRNA to A site
- c) Binding to 30S subunit and inhibiting peptide bond formation
- d) Inhibits binding of aminoacyl-t-RNA to 50S subunit of ribosome

Question 17:

Which of the following organisms is resistant to tetracyclines?

- a) MRSA
- b) Pseudomonas aeruginosa
- c) Rickettsia
- d) Mycoplasma

Question 18:

In patients with which of the following conditions is doxycycline not used?

- a) Leptospirosis
- b) Lymphogranuloma venerum
- c) Borreliosis
- d) Typhoid fever

Question 19:

Tetracycline is used for the prophylaxis of which of the following conditions?

- a) Cholera
- b) Brucellosis
- c) Leptospirosis
- d) Meningitis

Question 20:

The urine culture and sensitivity report of a patient is positive for tetracycline-resistant *Enterococcus faecalis*. Which among the following is a minocycline derivative that can be used in this condition?

- a) Methacycline
- b) Lymecycline
- c) Doxycycline
- d) Tigecycline

Question 21:

A patient diagnosed with the syndrome of inappropriate secretion of antidiuretic hormone is prescribed a drug known to cause nephrogenic diabetes insipidus. Identify this drug.

- a) Minocycline
- b) Demeclocycline
- c) Tetracycline
- d) Doxycycline

Question 22:

A patient develops dizziness, ataxia, and nausea after he was started on an antimicrobial drug. Which of the following drugs can be implicated?

- a) Tetracycline
- b) Oxytetracycline
- c) Doxycycline
- d) Minocycline

Question 23:

An infant on antibiotics becomes very irritable. Investigations reveal increased intracranial pressure. However, the condition resolves on discontinuation of the drug. Which drug had the baby likely been on?

- a) Aminoglycosides
- b) Tetracyclines
- c) Chloramphenicol
- d) Macrolides

Question 24:

Which of the following drugs does not require dose reduction in patients with renal failure?

- a) Gentamicin
- b) Vancomycin
- c) Amphotericin B
- d) Doxycycline

Answer Key

Question No.	Correct Option
1	a
2	c
3	d
4	c
5	c
6	d
7	b
8	b
9	b
10	a
11	c
12	c
13	a
14	d

15	a
16	b
17	b
18	d
19	a
20	d
21	b
22	d
23	b
24	d

Detailed Explanations

Solution to Question 1:

Streptomycin is an aminoglycoside antibiotic and the primary intracellular site of action of aminoglycosides is the 30S ribosomal subunit.

Mechanism of action of aminoglycosides:

- Aminoglycosides are transported actively across the cell membrane into the cytoplasm (oxygen-dependent process)
- Binding of aminoglycosides to 30S ribosomes cause misreading of mRNA template and result in incorporation of incorrect amino acids in the growing polypeptide chain
- The resulting aberrant proteins may be inserted into the cell membrane leading to altered permeability of the cell membrane of bacteria causing bacterial death (bactericidal)

Solution to Question 2:

Aminoglycosides are rapid bactericidal agents.

Aminoglycosides like streptomycin and tobramycin cause hearing loss in a newborn child when taken during pregnancy. They are polar compounds and are excreted usually unchanged in the urine. They are distributed mainly in the extracellular compartment.

Primarily bacteriostatic drugs are:

- Sulfonamides
- Erythromycin
- Tetracyclines
- Clindamycin
- Chloramphenicol

- Linezolid
- Ethambutol

Primarily bactericidal drugs are:

- Penicillins
- Cephalosporins
- Aminoglycosides
- Vancomycin
- Polypeptides
- Ciprofloxacin
- Rifampin
- Metronidazole
- Isoniazid
- Cotrimoxazole
- Pyrazinamide

Solution to Question 3:

Streptomycin (if not available, gentamicin) is considered the drug of choice for the treatment of tularemia.

Tularemia is caused by *Francisella tularensis*, an aerobic gram-negative bacteria.

Solution to Question 4:

The given clinical scenario is suggestive of cystic fibrosis. Inhalational formulation of tobramycin is approved for use in cystic fibrosis patients with *Pseudomonas aeruginosa* infections.

Option A: Injectable formulation of amikacin is used in cystic fibrosis.

Options B & D: Inhalational formulations of gentamicin and streptomycin are available but they are usually not used.

Note: The question is about approved inhalational formulation.

Solution to Question 5:

Amikacin has less activity against enterococci and hence is not indicated in infections caused by this organism. In enterococcal infections, gentamicin along with penicillin (or ampicillin) is given.

Amikacin is used against serious nosocomial gram-negative bacillary infections which are resistant to gentamicin and tobramycin. It has good activity against most strains of the following organisms:

- *Pseudomonas aeruginosa*
- *Serratia*
- *Proteus*
- *Klebsiella*
- *Enterobacter*
- *Escherichia coli*

Solution to Question 6:

Topical neomycin is used widely for a variety of infections of the skin and mucous membranes and to prevent infection of minor cuts and wounds.

Oral neomycin can be used for the eradication of bacteria confined to GIT in the following conditions:

- Pre-operative antisepsis for bowel surgery
- Hepatic encephalopathy: To reduce deleterious ammonia producing bacteria
- Enteropathogenic *E. coli* associated diarrhea

Orally administered neomycin is poorly absorbed from the GI tract and is eliminated unchanged in the feces.

Neomycin is highly nephrotoxic in human beings and hence should not be administered systemically.

Solution to Question 7:

Single daily dose aminoglycoside administration is more preferable to multiple daily dose administration because of the long post-antibiotic effect.

For certain classes of drugs such as aminoglycosides and quinolones, its antimicrobial action is dependent on the peak concentration above MIC (Minimum Inhibitory Concentration). It is referred to as concentration-dependent killing.

High-dose, extended-interval dosing schemes take advantage of the concentration-dependent activity of aminoglycosides to achieve maximal initial bacterial killing. And then because of the postantibiotic effect of aminoglycosides, good therapeutic response can be attained even when concentrations fall below MIC for a substantial fraction of the dosing interval.

Therefore, a single daily dose would be equal to or more effective than multiple 8 hourly doses. This dosing strategy may also reduce the ototoxicity and nephrotoxicity of these drugs.

Time-dependent killing is an effect seen when the antimicrobial action is dependent on the duration, the drug concentration remains above the MIC. Beta-lactams, macrolides, and vancomycin have this property.

Solution to Question 8:

Gentamicin is an aminoglycoside antibiotic and the peak concentration of aminoglycosides following intramuscular administration is achieved after 30-90 minutes.

Aminoglycosides need therapeutic dose monitoring because of their low therapeutic index especially in patients in whom renal function is compromised. Both trough and peak levels are calculated for aminoglycosides.

Solution to Question 9:

Clinically, the most common cause of aminoglycoside resistance is drug inactivation by enzymes. The inactivating enzymes are commonly acquired through conjugation and transfer of plasmids containing resistance genes.

Causes of aminoglycoside resistance are:

- Drug inactivation by enzymes
- Increased drug efflux
- Decreased permeability to the drug
- Modifications of the 30S ribosomal subunit that interferes with binding of the aminoglycosides

Solution to Question 10:

The dose of aminoglycosides is to be reduced in patients with renal failure (AKI) because clearance of the drug is reduced.

Aminoglycosides undergo minimum metabolism and are primarily excreted almost entirely by glomerular filtration. Therefore, renal failure decreases the clearance of the drug. Because the incidence of nephrotoxicity and ototoxicity is likely related to the overall exposure to aminoglycosides, it is critical to reduce the maintenance dose and dosing interval of these drugs in patients with impaired renal function.

Option B: Renal clearances are increased and the half-life is reduced in cystic fibrosis patients necessitating higher doses.

Option C: Burn patients have a more rapid drug clearance probably due to increased drug loss through burnt tissue and require higher doses in them.

Option D: Aminoglycoside dose adjustment is not required in liver disease

Solution to Question 11:

The scenario depicts a case of aminoglycoside-induced nephrotoxicity and the least nephrotoxic among aminoglycosides is streptomycin because it does not concentrate in the renal cortex.

Among aminoglycosides, neomycin has the highest and streptomycin has the least nephrotoxicity.

Adverse effects of aminoglycosides are:

- Nephrotoxicity
- Ototoxicity
- Acute neuromuscular blockade and apnea
- Hypersensitivity reactions (rare)

Solution to Question 12:

Vestibular toxicity is predominantly seen with streptomycin and gentamicin.

Ototoxicity seen with aminoglycosides:

- Cochlear toxicity (high-pitched tinnitus) - amikacin, kanamycin, neomycin
- Vestibular toxicity (moderately intense headache and vertigo) - streptomycin & gentamicin
- Both vestibular toxicity and cochlear toxicity - tobramycin

Solution to Question 13:

The scenario depicts a case of gentamicin-induced ototoxicity which is caused due to direct hair cell toxicity. Degeneration of vestibular hair cells and neurons leads to vertigo and associated headache.

Streptomycin and gentamicin produce predominantly vestibular effects. Amikacin, kanamycin, and neomycin primarily affect auditory function. Tobramycin affects both equally.

Solution to Question 14:

The given scenario depicts the neuromuscular blockade effect of aminoglycosides, and tobramycin has the least potency for blockade in this group.

Decreasing order of potency for the neuromuscular blockade is:

Neomycin & Kanamycin & Amikacin & Gentamicin & Tobramycin

It is a recognized clinical side effect of aminoglycoside antibiotics, and this effect becomes significant (prolonged respiratory depression) when the drug is used in conjunction with neuromuscular blocking agents or calcium channel blockers. They are avoided in patients with myasthenia gravis.

Aminoglycosides cause neuromuscular blockade by interfering with calcium channels and thereby inhibiting acetylcholine's pre-junctional release and post-junctional sensitivity. The neuromuscular blockade produced by these antibiotics is not reversed by neostigmine but is reversed by IV calcium gluconate.

Solution to Question 15:

Doxycycline is the drug of choice for scrub typhus.

Azithromycin, chloramphenicol, or rifampin can be administered as alternatives in patients allergic to doxycycline and in pregnancy.

Solution to Question 16:

Tetracyclines inhibit protein synthesis by binding to 30S subunit and inhibiting the binding of aminoacyl-tRNA to the A (acceptor) site of the mRNA – ribosome complex.

Solution to Question 17:

All strains of *Pseudomonas aeruginosa* are resistant to the action of tetracyclines.

Tetracyclines are active against MRSA and MSSA. Doxycycline and minocycline can be used in cutaneous MRSA infections.

Rickettsia and chlamydia are highly susceptible to tetracyclines. Tetracyclines are considered first line agents for chlamydial infections like non-specific urethritis/ endocervicitis and rickettsial infections like typhus, rocky mountain spotted fever, Q fever.

They are also second-line drugs in patients with syphilis who are allergic to penicillin.

Tetracyclines are bacteriostatic drugs which are more active against gram-positive than gram-negative. They are effective against spirochetes, *Haemophilus ducreyi*, *Vibrio cholera*, *Mycoplasma pneumoniae*.

Solution to Question 18:

Doxycycline is not used in typhoid infection. Enterobacteriaceae and gram-negative anaerobes are resistant to tetracyclines, hence doxycycline is not used in intraabdominal infections.

Third-generation cephalosporins and quinolones are used in treating typhoid fever.

Doxycycline treatment for 21 days is the first-line therapy for lymphogranuloma venereum. It is also indicated as one of the first-line treatments for borreliosis.

Doxycycline is the alternative to penicillin for the treatment of leptospirosis.

Solution to Question 19:

Tetracycline is used for the prophylaxis of cholera. It is recommended for close contacts of a case of cholera. It is supposed to be effective in preventing spread.

Doxycycline in combination with rifampicin is used for prophylaxis of brucellosis. It is also used in the prophylaxis of leptospirosis.

Note: The question is about the drug tetracycline and not about the class of tetracyclines as a whole.

Solution to Question 20:

Tigecycline is a glycylcycline and a derivative of minocycline that is active against most bacteria that are resistant to classical tetracyclines, and hence can be used in this condition.

The mechanism of action is similar to tetracyclines. It is used against enterobacteriaceae, gram-negative anaerobes, and enterococcus which are resistant to tetracyclines.

Solution to Question 21:

The drug described here is demeclocycline. Nephrogenic diabetes insipidus is associated with the use of this drug. This side effect of the drug is utilized in the treatment of SIADH (syndrome of inappropriate secretion of antidiuretic hormone).

Solution to Question 22:

The drug described here is minocycline as it is associated with vestibular toxicity which manifests as dizziness, ataxia, nausea, and vomiting. Symptoms usually occur soon after the initial dose is given and disappears 24-48 hours after drug administration is stopped.

Uses of minocycline are:

- Cutaneous MRSA infections
- Along with sulfonamides for nocardiosis (alternative therapy)
- Subgingival administration in dentistry (extended-release preparations)

Solution to Question 23:

The given scenario depicts a case of pseudotumor cerebri and tetracyclines are associated with this condition in infants, even when given in normal doses. Only elevated intracranial pressure is seen without any abnormality in the spinal fluid. Increased pressure returns to normal after therapy is discontinued.

Other side effects of tetracyclines are:

- Teratogenic
- Deposit in developing teeth and bones, cause brownish discoloration of teeth (chelating property)
- Marked suppression of normal flora of intestine causing superinfection. Pseudomembranous enterocolitis is a rare side effect

Solution to Question 24:

Doxycycline does not require dose reduction in renal failure. Doxycycline, minocycline, and glycylcyclines like tigecycline have fewer renal side effects.

Antibiotics causing acute kidney injury are as follows:

- Acute tubular necrosis:
- Amphotericin B
- Vancomycin
- Acyclovir
- Foscarnet
- Tenofovir
- Acute interstitial nephritis:
- Penicillin
- Cephalosporins
- Quinolones
- Rifampicin

Antimicrobials Acting on 50s Subunit

Question 1:

Gray baby syndrome is associated with the use of which of the following drugs?

- a) Clindamycin
- b) Chloramphenicol
- c) Tetracycline
- d) Cotrimoxazole

Question 2:

Chloramphenicol acts by binding to:

- a) 50s ribosomal subunit
- b) Topoisomerase IV enzyme
- c) 30s ribosomal subunit
- d) DNA helicase enzyme

Question 3:

A patient who was diagnosed with meningitis is on chloramphenicol therapy. Which of the following blood dyscrasias as a side effect of this drug?

- a) Hypoplastic anemia
- b) Agranulocytosis
- c) Aplastic anemia
- d) Thrombocytopenia

Question 4:

Which of the following antibiotics structurally resembles aminoacyl t-RNA and inhibits the protein synthesis by premature termination?

- a) Tetracycline
- b) Chloramphenicol

- c) Puromycin
- d) Erythromycin

Question 5:

Which of the following groups of antibiotics has additional anti-inflammatory and immunomodulatory activities?

- a) Tetracyclines
- b) Polypeptides
- c) Fluoroquinolones
- d) Macrolides

Question 6:

An unvaccinated child diagnosed with diphtheria is initiated erythromycin therapy. This drug acts by interfering with:

- a) Translocation at 30s ribosome
- b) Translocation at 50s ribosome
- c) Transcription at 50s ribosome
- d) Signal transduction of 50s ribosome

Question 7:

What is the drug of choice for scrub typhus during pregnancy and otherwise respectively?

- a) Chloramphenicol, doxycycline
- b) Doxycycline, doxycycline
- c) Doxycycline, azithromycin
- d) Azithromycin, doxycycline

Question 8:

What is the drug of choice for diphtheria carrier state?

- a) Amoxicillin

- b) Erythromycin
- c) Rifampicin
- d) Tetracycline

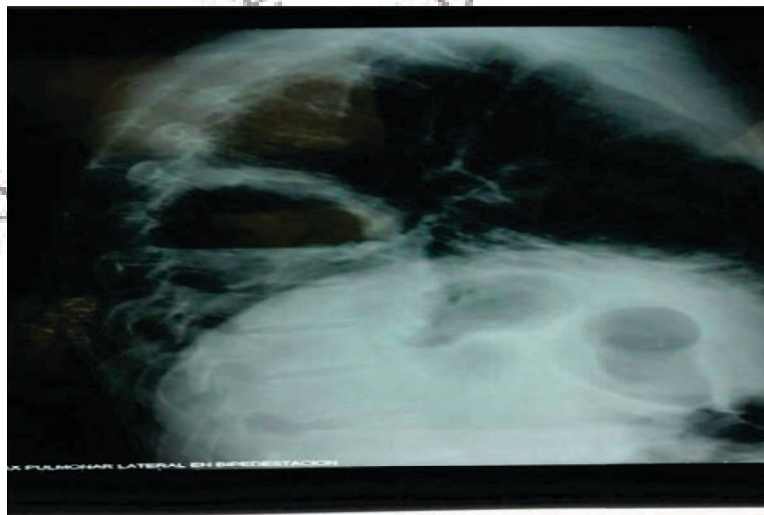
Question 9:

What is the rationale for the use of erythromycin in a patient with diabetic gastroparesis?

- a) It increases cAMP and increases intestinal motility
- b) It decreases bacterial count in intestinal hypomotility
- c) It binds to adenyl cyclase and increases intestinal motility
- d) It binds to motilin receptors and stimulates intestinal motility

Question 10:

The chest x-ray of a patient who presented with fever and cough with expectoration is shown below. Following discussion, a lincosamide antibiotic is prescribed. Which of the following will you choose?



- a) Telithromycin
- b) Spectinomycin
- c) Clindamycin
- d) Streptomycin

Question 11:

Which of the following is a serious adverse effect you would consider as a possibility in a patient receiving clindamycin?

- a) Nephrotoxicity
- b) Pseudotumor cerebri
- c) Pseudomembranous colitis
- d) Aplastic anemia

Question 12:

Which of the following statements about streptogramins is true?

- a) Quinupristin-dalfopristin combination is not active against gram-positive cocci
- b) Synergistic activity of quinupristine and dalfopristine has bactericidal action
- c) Quinupristin binds to the 30S subunit of ribosomes
- d) Dalfopristin binds to the 30S subunit of ribosomes

Question 13:

Which of the following binds to the P site of 50S ribosome and inhibits bacterial protein synthesis?

- a) Daptomycin
- b) Tetracycline
- c) Linezolid
- d) Meropenem

Question 14:

Linezolid is approved for use in which of the following patients?

- a) 1 and 5 only
- b) 1, 2 and 4 only
- c) 1, 3 and 5 only
- d) 1, 2, 3 and 4 only

Question 15:

Which of the following is true about tedizolid?

- a) Active against gram positive organisms
- b) Poor oral bioavailability
- c) Active against anaerobes
- d) Causes peripheral neuropathy

Answer Key

Question No.	Correct Option
1	b
2	a
3	c
4	c
5	d
6	b
7	d
8	b
9	d
10	c
11	c
12	b
13	c
14	d
15	a

Detailed Explanations**Solution to Question 1:**

Gray baby syndrome is associated with the use of chloramphenicol, especially in premature infants.

It develops due to the developmental deficiency of the enzyme glucoronyl transferase. This enzyme metabolizes chloramphenicol in neonates. Its deficiency leads to an inability to excrete

unconjugated drugs adequately through renal excretion.

The initial symptoms are:

- Vomiting
- Refusal to suck
- Irregularly rapid respiration
- Periods of cyanosis.

After 1-2 days:

- Flaccid baby
- Hypothermia
- Ashen-gray color skin discoloration

If there is a recovery, usually there are no sequelae.

Solution to Question 2:

Chloramphenicol acts by binding to the 50s ribosomal subunit.

It inhibits the binding of amino acid containing tRNA to acceptor site on 50s subunit of the ribosome. This prevents peptide bond formation by inhibiting the interaction between peptidyltransferase enzyme and the amino acid.

Solution to Question 3:

The patient is most likely to have aplastic anemia, as it is the most common blood dyscrasia associated with the use of chloramphenicol. It can lead to fatal pancytopenia that has a high rate of mortality.

Hematological toxicities of chloramphenicol include:

- Dose-related toxicity - pancytopenia (anemia, thrombocytopenia, leukopenia)
- Idiosyncratic response - aplastic anemia

Blood dyscrasias occurring with chloramphenicol:

- Aplastic anemia (70%)
- Hypoplastic anemia
- Agranulocytosis
- Thrombocytopenia

Chloramphenicol remains an alternative drug for the treatment of meningitis caused by *H. influenzae*, *N. meningitidis*, and *S. pneumoniae* in developing countries and also in patients who have a severe allergy to β -lactams.

Solution to Question 4:

Puromycin structurally resembles aminoacyl t-RNA and inhibits the protein synthesis by premature termination. It is used as an antibiotic in cell cultures.

Option A: Tetracycline binds to the 30s ribosome and prevents attachment of aminoacyl t-RNA to the acceptor site.

Option B: Chloramphenicol binds to the 50s subunit of the ribosome and inhibits the binding of amino acid containing tRNA to the acceptor site on 50s subunit of the ribosome.

Option D: Erythromycin acts by interfering with translocation at the 50s ribosome.

Solution to Question 5:

Macrolides are a group of antibiotics that have additional anti-inflammatory and immunomodulatory activities.

They are being investigated and used in pulmonary diseases for their immunomodulatory properties.

Examples of macrolide antibiotics:

- Erythromycin
- Azithromycin
- Clarithromycin

Solution to Question 6:

Erythromycin acts by interfering with translocation at the 50s ribosome.

It does not inhibit the peptide bond formation but the translocation step is inhibited where the peptidyl t-RNA molecule moves from the acceptor site on the ribosome to the peptidyl donor site.

Solution to Question 7:

The drug of choice for a case of scrub typhus during pregnancy is azithromycin. For other patients the drug of choice is doxycycline.

An alternative treatment option in pregnant women and patients allergic to doxycycline is chloramphenicol.

Solution to Question 8:

The drug of choice for diphtheria carrier state is erythromycin.

Erythromycin for 7 days is very effective for eradicating the diphtheria carrier state or acute infections.

Solution to Question 9:

Erythromycin can be used in diabetic gastroparesis as it binds to motilin receptors and stimulates intestinal motility.

It has been used in conditions with hypomotility of the gastrointestinal tract like in postoperative ileus to promote peristalsis and to increase gastric emptying.

Erythromycin is used off-label as a prokinetic agent in the intensive care setting and in patients with diabetic gastroparesis.

Solution to Question 10:

Clindamycin, a lincosamide is used in the management of lung abscess and anaerobic lung and pleural space infections.

Clindamycin is a congener of lincomycin and binds to the 50S subunit of bacterial ribosomes and inhibits protein synthesis.

Option A: Telithromycin is a ketolide antibiotic.

Option B: Spectinomycin is an aminocyclitol antibiotic that binds to the 30S subunit of ribosomes and inhibits protein synthesis.

Option D: Streptomycin is an aminoglycoside antibiotic.

Solution to Question 11:

Clindamycin use is to be limited because of the risk of pseudomembranous colitis.

Pseudomembranous colitis is a lethal syndrome caused by *Clostridium difficile* due to alteration in normal intestinal microbiota. It is characterized by fever, watery diarrhea and increased peripheral white blood cell count. Metronidazole and oral vancomycin are used in the treatment of this condition.

Drugs causing pseudomembranous colitis are:

- Penicillins
- Clindamycin
- Rifaximin

- Tetracycline (rare side effect)
- Erythromycin

Solution to Question 12:

Quinupristin and dalfopristin are a combination of streptogramins and the synergistic activity of these drugs has bactericidal action.

Quinupristin binds to the 50s subunit of ribosomes and acts similar to macrolides by interfering in translocation and inhibiting protein synthesis.

Dalfopristin binds to a site nearby, resulting in conformational changes in the 50S ribosome thereby increasing the activity of quinupristin. The synergic activity of these drugs can have bactericidal action in certain susceptible bacteria.

Quinupristin/dalfopristin is active against gram-positive cocci and organisms responsible for atypical pneumonia (e.g., *M. pneumoniae*, *Legionella* spp., and *C. pneumoniae*).

Quinupristin is a streptogramin B molecule whereas dalfopristin is a streptogramin A molecule.

Solution to Question 13:

Linezolid binds to the P site of the 50s ribosomal subunit and inhibits bacterial protein synthesis. It is an oxazolidinone that is active against various resistant organisms including MRSA.

Linezolid is well absorbed orally and the bioavailability approaches 100%, hence intravenous and oral doses are the same for this drug.

Linezolid is associated with the side effect of myelosuppression (anemia, leukopenia, pancytopenia, thrombocytopenia). Thrombocytopenia is the most common side effect. Platelet count monitoring is done in a patient with linezolid who is at risk of myelosuppression and blood dyscrasias. Linezolid also causes serotonin syndrome due to monoamine oxidase(MAO) inhibition.

Solution to Question 14:

Linezolid is approved for use in the following conditions:

- Vancomycin-resistant *E. faecium* infections
- Methicillin resistance/susceptible *S. aureus* infections
- Community-acquired pneumonia due to *S.pneumoniae*
- Nosocomial pneumonia due to *S. aureus*.

Solution to Question 15:

Tedizolid is active against gram-positive organisms.

Tedizolid belongs to the class of oxazolidinones. It has poor activity against most gram-negative bacteria and anaerobes. It inhibits protein synthesis by binding to the P site of the 50S ribosomal subunit.

It is used for the treatment of skin and soft tissue infections, respiratory tract infections caused by streptococci, and *S. aureus* including MRSA.

Common adverse effects include myelosuppression leading to anemia, leukopenia, thrombocytopenia, and pancytopenia.

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Antiretroviral Drugs

Question 1:

Which class of antiretroviral drugs does nevirapine belong to?

- a) Nucleoside reverse transcriptase inhibitors
- b) Non-nucleoside reverse transcriptase inhibitors
- c) Nucleotide reverse transcriptase inhibitors
- d) Protease inhibitors

Question 2:

Resistance to zidovudine develops due to:

- a) Increased efflux of the drug from nucleus
- b) Mutations of reverse transcriptase
- c) Increased efflux of the drug from inside the cell
- d) Decreased zidovudine triphosphate formation

Question 3:

A patient diagnosed with AIDS is planned to be started on a lamivudine-based-antiretroviral therapy. Which of the following drugs will you not add to her regimen?

- a) Zidovudine
- b) Stavudine
- c) Emtricitabine
- d) Abacavir

Question 4:

Which of the following statements about lamivudine is false?

- a) It is an effective agent for patients with chronic hepatitis B
- b) Dose is lower for blocking HIV replication than HBV replication

- c) Should not be used as monotherapy in HBV/HIV infected patients
- d) Anti-HBe seroconversion occurs in a minority of the patients

Question 5:

A patient on antiretroviral therapy presents with severe tingling and numbness in the legs. On examination, you find that his tendon reflexes are reduced. Which of the following drugs has the highest propensity to cause this condition?

- a) Zidovudine
- b) Lamivudine
- c) Stavudine
- d) Didanosine

Question 6:

Which of the following drugs are not protease inhibitors?

- a) 1 and 3
- b) 2 and 4
- c) 3 and 5
- d) 1 and 5

Question 7:

During a clinical trial, it was noted that the bioavailability of some drugs was substantially improved on ritonavir administration. Which of the following drugs does not exhibit this property?

- a) Saquinavir
- b) Lopinavir
- c) Nelfinavir
- d) Darunavir

Question 8:

Which of the following is a pharmacological booster with no antiretroviral activity?

- a) Ritonavir
- b) Vorinostat
- c) Belinostat
- d) Cobicistat

Question 9:

The antiviral drug having dual antiviral activity against HIV and HBV is:

- a) Enfuvirtide
- b) Abacavir
- c) Emtricitabine
- d) Ritonavir

Question 10:

Which of the following is a subcutaneously administered drug, that will be helpful in a patient showing high viral load despite good compliance with ART?

- a) Maraviroc
- b) Enfuvirtide
- c) Raltegravir
- d) Emtricitabine

Question 11:

A patient showing treatment failure with multiple antiretroviral drugs is now started on maraviroc. This drug acts against:

- a) CCR5
- b) CXCR4
- c) CD4
- d) CCR5 and CD4

Question 12:

The husband of an AIDS patient noticed that his wife had become forgetful and confused lately. She complained of dizziness and lack of sleep. Eventually, she started behaving bizarrely, paranoid that everybody is trying to kill her. Yesterday she was found in her room talking to somebody who did not exist in reality. Which of the following drugs is most commonly associated with this side effect?

- a) Nevirapine
- b) Efavirenz
- c) Etravirine
- d) Delavirdine

Question 13:

A patient presenting with low-grade fever and weight loss of 3 months is diagnosed to have HIV-2 infection. Which of the following drugs is not indicated here?

- a) Ritonavir
- b) Tenofovir
- c) Efavirenz
- d) Zalcitabine

Question 14:

A 52-year-old man who has been on ART for the past two months presents with abdominal pain. On evaluation, he was found to have acute pancreatitis. Which of the following drugs is associated with the highest risk for causing this condition?

- a) Didanosine
- b) Lamivudine
- c) Zidovudine
- d) Abacavir

Question 15:

A patient diagnosed with AIDS is initiated on antiretroviral therapy with zidovudine. Which of the following side effects is he most likely to develop?

- a) Anemia
- b) Peripheral neuropathy

- c) Lactic acidosis
- d) Hepatotoxicity

Question 16:

Of all the nucleoside reverse transcriptase inhibitors, the use of which of the following is most strongly associated with fat wasting?

- a) Zidovudine
- b) Lamivudine
- c) Stavudine
- d) Abacavir

Question 17:

A patient on antiretroviral therapy comes to you for follow-up. His lipid profile reveals high cholesterol levels. Which of the following drugs is not associated with this condition?

- a) Atazanavir
- b) Saquinavir
- c) Amprenavir
- d) Nelfinavir

Question 18:

A 48-year-old known hemophiliac on antiretroviral therapy suddenly developed headache, vomiting and altered consciousness and was rushed to a hospital. Which of the following protease inhibitors is implicated here?

- a) Saquinavir
- b) Ritonavir
- c) Tipranavir
- d) Indinavir

Question 19:

A patient on antiretroviral therapy comes for regular follow-up. You note nail changes as shown below. Which of the following drugs causes this condition?



- a) Abacavir
- b) Ritonavir
- c) Zidovudine
- d) Didanosine

Question 20:

Which is the only antiretroviral drug that may have teratogenic properties?

- a) Didanosine
- b) Emtricitabine
- c) Efavirenz
- d) Nevirapine

Question 21:

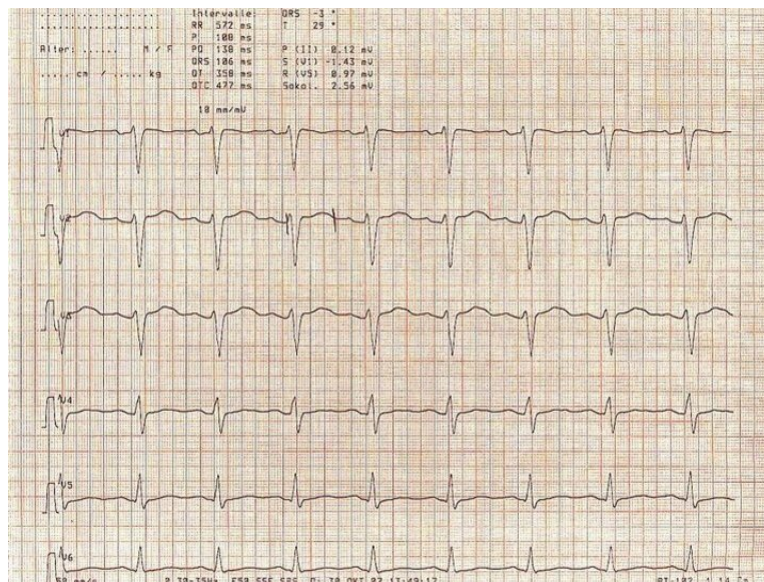
A patient undergoing treatment for HIV complains of dizziness. On examination, you note his supine BP to be 140/80 mmHg and standing BP to be 110/70 mmHg. Which of the following drugs would have caused this condition?

- a) Maraviroc
- b) Raltegravir

- c) Enfuvirtide
- d) Nelfinavir

Question 22:

The ECG of a patient on antiretroviral therapy is shown below. You observe that the QTc = 477 ms. This can be attributed to:



- a) Raltegravir
- b) Rilpivirine
- c) Etravirine
- d) Efavirenz

Question 23:

A 45-year-old HIV-positive man was started on HAART. On follow-up testing, his CD4 cell count had decreased below 200/mm³. Subsequently, he developed diarrhea and redistribution of body fat. What is false about the causative drug class?

- a) Atazanavir has the lowest propensity to increase total cholesterol levels.
- b) Tipranavir is used in sub-therapeutic doses as a pharmacodynamic booster in cART.
- c) Indinavir can cause nephrolithiasis when adequate hydration is not maintained.
- d) Ritonavir is a potent inhibitor of the CYP3A4 isoform of P450 & may alter levels of drugs.

Question 24:

An AIDS patient on treatment with AZT, lamivudine, indinavir, ketoconazole, and cotrimoxazole, develops breast hypertrophy, central adiposity, hyperlipidemia, insulin resistance, and nephrolithiasis. Which of the following drugs could have caused these?

- a) Azidothymidine
- b) Indinavir
- c) Ketoconazole
- d) Lamivudine

Question 25:

Which of the following is a monoclonal antibody approved for use in the treatment of HIV?

- a) Bevacizumab
- b) Ibalizumab
- c) Panitumumab
- d) Pembrolizumab

Question 26:

Which of the following is an integrase inhibitor indicated in the treatment of patients with HIV?

- a) Lopinavir
- b) Abacavir
- c) Raltegravir
- d) Ritonavir

Answer Key

Question No.	Correct Option
1	b
2	b
3	c

4	b
5	c
6	c
7	c
8	d
9	c
10	b
11	a
12	b
13	c
14	a
15	a
16	c
17	a
18	c
19	c
20	c
21	a
22	b
23	b
24	b
25	b
26	c

Detailed Explanations

Solution to Question 1:

Nevirapine is a non-nucleoside reverse transcriptase inhibitor (NNRTI)

US FDA has approved doravirine, a new non-nucleoside reverse transcriptase inhibitor (NNRTI) for the treatment of adults with HIV-1 infection given in combination with other ARTs.

Solution to Question 2:

Resistance to zidovudine occurs due to mutations in the reverse transcriptase enzyme.

The mutations are referred to as thymidine analog mutations (TAM) because of their ability to confer cross-resistance to other thymidine analogs such as stavudine.

M184V mutation, which is associated with the use of lamivudine or emtricitabine, restores zidovudine susceptibility in zidovudine-resistant HIV.

Solution to Question 3:

Lamivudine and emtricitabine are not combined because of similar actions and no added benefits. Emtricitabine is a cytidine analog. It is chemically related to lamivudine and shares most of its pharmacodynamic characteristics.

The same holds good for the zidovudine and stavudine combination. Zidovudine and stavudine are thymidine analogues. In fact, efficacy may even decrease because of competition for a similar activation process.

Solution to Question 4:

The daily dose of lamivudine indicated for HIV (300 mg) is higher than that of the dose indicated for HBV (100 mg).

Lamivudine has both anti-HIV and anti-HBV activity and is an effective agent for patients with chronic hepatitis B. It should not be used alone because resistance can rapidly emerge in HIV viruses.

Anti-HBe seroconversion occurs in only a minority of the patients taking lamivudine.

Drugs that cause anti-HBe seroconversion:

- Lamivudine
- Entecavir
- Adefovir
- Tenofovir
- Telbivudine
- Pegylated interferon

Solution to Question 5:

The scenario is suggestive of peripheral neuropathy. The drug causing maximum peripheral neuropathy is stavudine.

The incidence of peripheral neuropathy is very rare with both zidovudine and lamivudine. Lamivudine is considered one of the least toxic antiretroviral agents.

Didanosine is frequently associated with peripheral neuropathy, but stavudine is associated with the highest risk of peripheral neuropathy.

Note: Zalcitabine was associated with a higher incidence of peripheral neuropathy. It is no longer marketed.

Solution to Question 6:

Abacavir is a guanosine analogue and is a nucleoside reverse transcriptase inhibitor.

Raltegravir is an integrase inhibitor.

Nelfinavir, saquinavir and ritonavir are protease inhibitors.

Solution to Question 7:

The bioavailability of nelfinavir is not substantially improved with ritonavir administration because it undergoes metabolism primarily by CYP2C19 and ritonavir is a weak inhibitor of CYP2C19.

All other protease inhibitors undergo hepatic oxidative metabolism predominantly by the CYP3A4 enzyme. Ritonavir, being a potent inhibitor of CYP3A4 acts as a pharmacological booster in combination with other protease inhibitors.

Solution to Question 8:

Cobicistat is a pharmacological booster with no antiretroviral activity.

Cobicistat is a ritonavir analog developed as an exclusive pharmacological booster that inhibits the CYP3A4 enzyme and has no antiviral activity. It is available only in combination with atazanavir or darunavir or elvitegravir for HIV infections.

Ritonavir is also used as a pharmacological booster but it has antiretroviral activity.

Vorinostat and belinostat are histone deacetylase inhibitors. Vorinostat is approved for the treatment of patients with cutaneous T-cell lymphoma with persistent or recurrent disease after two systemic therapies.

Cobicistat	Ritonavir
Specific CYP3A4 inhibition	Potent inhibitor of CYP3A4 and weak inhibitor of CYP2D6
Better tolerated	Less tolerated
Reduced impact on plasma lipids	Causes dose-dependent elevation in serum total cholesterol and triglycerides
Lacks P450 enzyme induction	Moderate inducer of P450 enzymes

Solution to Question 9:

Antiviral drug having dual antiviral activity against HIV and HBV is emtricitabine.

It is a nucleoside reverse transcriptase inhibitor similar to lamivudine. It has activity against HIV-1, HIV-2, and HBV.

Solution to Question 10:

The drug described in this case is enfuvirtide, which is an anti-HIV drug that is administered subcutaneously.

It is used only in the treatment of those who have evidence of HIV replication (high viral load) despite ongoing antiretroviral therapy.

It binds to the gp41 subunit of the viral envelope and prevents the fusion of the virus to the cellular membranes thereby preventing entry of the virus.

Enfuvirtide is active against HIV-1 only.

Solution to Question 11:

Maraviroc is a chemokine receptor antagonist acting against the CCR5 receptor.

It binds to the host cell CCR-5 receptor and prevents the binding of viral GP-120 to host cells. It is indicated in patients with evidence of infection with predominantly CCR-5 tropic virus.

Solution to Question 12:

The given scenario describes psychiatric or CNS adverse effects, which are commonly seen with efavirenz.

The CNS/ psychiatric adverse effects seen with the use of efavirenz are:

- Frank psychosis (depression, hallucinations, and/or mania)
- Impaired concentration
- Dysphoria
- Dizziness
- Vivid dreams
- Insomnia

Solution to Question 13:

Efavirenz is not indicated for HIV-2 infection. It is a non-nucleoside reverse transcriptase inhibitor (NNRTI).

NNRTIs are HIV-1 specific and have no activity against HIV-2. The binding site for NNRTI is virus strain-specific which only inhibits the HIV-1 reverse transcriptase. It is approved only for HIV-1 infections and should not be used for HIV-2 infections.

Ritonavir, tenofovir, and zalcitabine are indicated for HIV-2 infections.

Solution to Question 14:

The maximum risk of pancreatitis is seen with didanosine.

It should, therefore not be combined with hydroxyurea, alcohol, or pentamidine.

Solution to Question 15:

This patient is most likely to develop anemia since it is the most common side effect of zidovudine.

Higher dosages of zidovudine lead to bone marrow suppression resulting in anemia and granulocytopenia.

Peripheral neuropathy, hepatotoxicity, and lactic acidosis are rare side effects of zidovudine.

Solution to Question 16:

Of all the nucleoside reverse transcriptase inhibitors, the use of stavudine is most strongly associated with fat wasting (lipoatrophy).

Solution to Question 17:

Atazanavir is not associated with dyslipidemia, fat redistribution, or metabolic syndrome. This is in contrast to all the other protease inhibitors, which causes side effects like hypertriglyceridemia, hypercholesterolemia and other signs of lipodystrophy.

Patients treated with atazanavir have significantly lower fasting triglyceride and cholesterol concentrations than patients treated with nelfinavir and lopinavir. Clinically, when abacavir (Nucleoside Reverse Transcriptase Inhibitors) is substituted for protease inhibitors in patients having dyslipidemia, it leads to an improvement in lipid profile.

Solution to Question 18:

The given scenario is suggestive of intracranial hemorrhage and the protease inhibitor that is associated with this condition is tipranavir. It is a non-peptidic protease inhibitor

Tipranavir use has been associated rarely with intracranial hemorrhage (including fatalities) and bleeding episodes in patients with hemophilia.

Solution to Question 19:

The image shows nail hyperpigmentation and it is commonly encountered with the use of zidovudine. This is seen with chronic zidovudine administration.

The adverse effects of zidovudine are:

- Fatigue, malaise, myalgia, and anorexia
- Headache and insomnia
- Erythrocytic macrocytosis
- Nail hyperpigmentation
- Myopathy
- Bone marrow suppression - anemia and granulocytopenia

Solution to Question 20:

The only antiretroviral drug that may have teratogenic properties is efavirenz.

It has shown teratogenicity in primates. However, clinical trials have suggested that there is no significantly increased risk of teratogenicity in humans as compared to other antiretroviral drugs.

Solution to Question 21:

The given clinical scenario depicts a case of orthostatic hypotension and maraviroc is associated with dose-dependent and concentration-dependent orthostatic hypotension.

Maraviroc is a chemokine receptor antagonist acting against the CCR5 receptor. It binds to the host cell CCR-5 receptor and prevents the binding of viral GP-120 to host cells. It is indicated in patients with evidence of infection with predominantly CCR-5 tropic virus.

Solution to Question 22:

The given ECG indicates QT interval prolongation (normal QTc = 440ms) and this is observed with the use of rilpivirine. It is a non-nucleoside reverse transcriptase inhibitor (NNRTI).

Qtc interval prolongation is observed with higher concentrations of the drug. Caution has to be exercised when rilpivirine is administered along with CYP3A4 inhibitors that can increase the concentration of rilpivirine.

Solution to Question 23:

The given clinical scenario is suggestive of adverse effects of gastrointestinal tolerance and fat redistribution due to protease inhibitors. HAART therapy includes this class of drugs along with 2 nucleoside reverse transcriptase inhibitors +/- NNRTIs. Ritonavir, not tipranavir, is used in sub-therapeutic doses as a pharmacodynamic booster of other protease inhibitors in cART.

This is because ritonavir is a potent inhibitor of the CYP3A4 isoform of P450. It inhibits their CYP3A-mediated metabolism and permits the use of lower doses of other protease inhibitors. Nelfinavir is the only protease inhibitor not boosted by ritonavir.

Indinavir is notable for its tendency to precipitate in the urinary tract, causing nephrolithiasis, unless the patient maintains a high state of hydration.

Solution to Question 24:

Indinavir, a protease inhibitor, is the most likely cause for these adverse effects.

Patients being treated with protease inhibitors (e.g., indinavir) develop a syndrome involving the derangement of lipid metabolism. This causes changes in body fat distribution. Type 2 diabetes mellitus has also been reported. This is called HIV/HAART-associated lipodystrophy.

Indinavir can also precipitate in the urinary tract, causing nephrolithiasis, unless the patient is maintained in a high state of hydration.

Indinavir binds to the site of HIV-1 protease activity and inhibits cleavage of viral gag-pol polyprotein precursors into individual functional proteins required for infectious HIV. This results in the formation of immature, non-infectious viral particles.

Option A: Azidothymidine or zidovudine, is a nucleoside reverse transcriptase inhibitor. Chronic zidovudine administration has been associated with nail hyperpigmentation.

Option C: Ketoconazole is an antifungal agent that inhibits fungal ergosterol synthesis. Its adverse effects include liver dysfunction and testosterone synthesis inhibition leading to gynecomastia.

Option D: Lamivudine is a nucleoside reverse transcriptase inhibitor. It is one of the least toxic antiretroviral drugs. Neutropenia, headache, and nausea have been reported at higher-than-recommended doses.

Solution to Question 25:

Ibalizumab is an anti-CD4 monoclonal antibody approved for use in the treatment of multidrug-resistant HIV-1. It is administered intravenously.

It acts as an entry inhibitor. It blocks HIV-1 from infecting CD4+ T cells by binding to domain 2 of CD4 cells. It also interferes with post-attachment steps required for the entry of HIV-1 virus particles into host cells and thus prevents the viral transmission that occurs via cell-cell fusion.

Solution to Question 26:

The integrase inhibitor indicated in the treatment of HIV is raltegravir.

It inhibits the catalytic activity of the HIV-encoded integrase and prevents the integration of virus DNA into the host chromosome. It is eliminated primarily by glucuronidation by UGT1A1.

The other two FDA-approved integrase inhibitors are elvitegravir (EVG) and dolutegravir (DTG).

Options A and D: Lopinavir and ritonavir are protease inhibitors.

Option B: Abacavir is a nucleoside reverse transcriptase inhibitor.

Note: Dolutegravir (DTG) is recommended as the preferred third drug for HIV post-exposure prophylaxis as per the recent MOHFW Jan 2020 guideline, in line with WHO.

The FDA approved dolutegravir tablets and suspension for the treatment of HIV-1 infection in pediatric patients at least four weeks old and weighing at least 3 kg who have never been treated for HIV or who have been treated, but not with an integrase inhibitor class of drug.

WHO recently recommended long-acting Cabotegravir, an HIV integrase strand transfer inhibitor may be offered as a HIV prevention option for people at substantial risk of HIV infection. It is an IM injectable form of pre-exposure prophylaxis that has been shown to be highly effective at reducing the risk of HIV acquisition. The first 2 injections were administered 4 weeks apart, followed thereafter by an injection every 8 weeks.

Penicillins

Question 1:

Which of the following organisms is most sensitive to penicillin?

- a) Staphylococcus
- b) Streptococcus
- c) Pneumococcus
- d) Treponema pallidum

Question 2:

Which of the following statements about penicillin-binding proteins is false?

- a) Present within cell membrane of bacteria
- b) Mutation in the PBPs gives rise to resistance
- c) PBPs are involved in protein synthesis
- d) These are targeted by imipenem

Question 3:

In which of the following patients is penicillin G is not used as treatment?

- a) 38-year-old man with bacterial meningitis
- b) 23-year-old trekker with scrub typhus
- c) 28-year-old pregnant lady with syphilis
- d) 30-year-old police officer infected with anthrax

Question 4:

In which of the following conditions is Penicillin G not the drug of choice?

- a) Actinomycosis
- b) Syphilis
- c) Lyme disease

d) Rat-bite fever

Question 5:

Which of the following is false about penicillins?

- a) Penicillin V is absorbed orally
- b) Benzathine penicillin is short acting penicillin
- c) Cloxacillin is beta lactamase and acid resistant
- d) Ampicillin is not resistant to beta lactamases

Question 6:

A patient is prescribed probenecid along with penicillin. What is the rationale behind this?

- a) It is a beta lactamase inhibitor
- b) It increases the duration of action of penicillins
- c) It increases the excretion of penicillins in kidney
- d) It decreases the metabolism of penicillins in liver

Question 7:

The culture and sensitivity report from a wound swab is positive for penicillinase-producing *Staphylococcus aureus*. Which of the following antimicrobials can be administered?

- a) Amoxicillin
- b) Penicillin G
- c) Piperacillin
- d) Cloxacillin

Question 8:

Which of the following is not an extended spectrum penicillin?

- a) Ampicillin
- b) Cloxacillin
- c) Carbenicillin

d) Piperacillin

Question 9:

A 22-year-old woman presented with dysuria and high-grade fever. Urine culture sensitivity shows the growth of carbenicillin-sensitive E. coli. Which of the following is true about the antibiotic?

- a) Is effective in Pseudomonas infection**
- b) Has no effect in Proteus infection**
- c) Is a ureidopenicillins antibiotic**
- d) Is administered orally**

Question 10:

Which among the following is not a beta-lactamase enzyme inhibitor?

- a) Tazobactam**
- b) Clavulanic acid**
- c) Piperacillin**
- d) Avibactam**

Question 11:

Which of the following antimicrobials is effective against an organism producing extended-spectrum beta-lactamases?

- a) Amoxicillin - clavulanic acid**
- b) Penicillin**
- c) Piperacillin - tazobactam**
- d) Ceftriaxone**

Question 12:

Which of the following is an acid-susceptible penicillin?

- a) Methicillin**
- b) Ampicillin**

- c) Amoxicillin
- d) Cloxacillin

Question 13:

A patient who is being treated with an antimicrobial agent presents with neutropenia. Which of the following drugs could be the most likely cause?

- a) Nafcillin
- b) Methicillin
- c) Carbenicillin
- d) Ampicillin

Question 14:

A 28-year-old man undergoing treatment with penicillin presents with fever, myalgia, and exacerbation of pre-existing cutaneous lesions. Which of the following diseases is he most probably being treated for?

- a) Rheumatic endocarditis
- b) Gonorrhoea
- c) Syphilis
- d) Staphylococcal scalded skin syndrome

Question 15:

A patient who is undergoing treatment with antibiotics presents with diarrhea, fever, and leukocytosis. Clostridium difficile associated diarrhea is suspected. This condition is most commonly associated with the use of which of the following group of drugs?

- a) Aminopenicillins
- b) Tetracyclines
- c) Carbapenems
- d) Macrolide

Question 16:

A patient being treated with ampicillin presents with diarrhea, abdominal pain, and fever. *C. difficile* infection is suspected. Which of the following monoclonal antibodies is approved for use in this patient?

- a) Actoxumab
- b) Bezlotoxumab
- c) Bevacizumab
- d) Adalimumab

Question 17:

Systemic use of bacitracin is restricted due to:

- a) Ototoxicity
- b) Nephrotoxicity
- c) Cardiotoxicity
- d) Pulmonary toxicity

Question 18:

Daptomycin is not indicated in which of the following conditions?

- a) Complicated skin and soft-tissue infections
- b) Complicated bacteremia
- c) Community acquired pneumonia
- d) Right-sided endocarditis

Question 19:

Polymyxins are effective against:

- a) Gram-positive bacteria
- b) Gram-negative bacteria
- c) Acid-fast bacilli
- d) Anaerobic bacteria

Question 20:

What is the mechanism of resistance by lactamases to β -lactam antibiotics?

- a) Alteration of penicillin binding protein
- b) Degradation of antibiotics
- c) Drug efflux
- d) Alteration in 50s ribosome structure

Question 21:

A known asthmatic patient with fever and sore throat was prescribed an antibiotic. Two days later he presents with a maculopapular rash. Which of the following drugs might have caused this?

- a) Prednisolone
- b) Salbutamol
- c) Fexofenadine
- d) Amoxicillin

Question 22:

Which of the following statements is correct for a patient on high dose of penicillin and gentamicin?

- a) TDM is done with penicillin because of narrow therapeutic index
- b) TDM is done with gentamicin because of narrow therapeutic index
- c) TDM is done with both because of dangerous side effects
- d) Both drugs do not need monitoring

Answer Key

Question No.	Correct Option
1	d
2	c
3	b
4	c

5	b
6	b
7	d
8	b
9	a
10	c
11	c
12	a
13	a
14	c
15	a
16	b
17	b
18	c
19	b
20	b
21	d
22	b

Detailed Explanations

Solution to Question 1:

Among the given organisms, *Treponema pallidum* is most sensitive to penicillin. Penicillin G remains the drug of choice for the treatment of syphilis.

Staphylococcus, *Streptococcus*, and *Pneumococcus* have developed variable resistance to penicillin G and penicillin G is no longer considered the drug of choice for most of the infections caused by them.

Solution to Question 2:

The targets for the action of β -lactam antibiotics are collectively termed penicillin-binding proteins (PBPs) and they are not involved in protein synthesis.

Transpeptidase enzyme and related proteins constitute the penicillin-binding proteins (PBPs) which are present in the bacterial cell membrane. Transpeptidase is involved in the synthesis of peptidoglycan of the bacterial cell wall. All β -lactam groups of antibiotics bind to PBPs and inhibit the last step in peptidoglycan synthesis.

Mutation in the different PBPs confers resistance to the bacteria against the beta-lactams. Methicillin resistance in *Staphylococcus aureus* (MRSA) occurs due to alteration in the PBPs.

Imipenem targets the PBPs, disrupting cell wall synthesis and causing death of the micro-organism.

Beta-lactam group of antibiotics include:

- Penicillins
- Cephalosporins
- Carbapenems
- Monobactam

Solution to Question 3:

Penicillin G is not used against rickettsial infections like scrub typhus, and tetracyclines are the drug of choice in this condition.

Penicillins are ineffective against amebae, plasmodia, rickettsiae, fungi or viruses.

Uses of penicillin are:

- Prophylaxis of rheumatic carditis (benzathine penicillin)
- Syphilis
- Leptospirosis
- Cervicofacial actinomycosis
- Rat-bite fever
- Meningococcal meningitis
- Anthrax
- Listeria

Solution to Question 4:

Penicillin G is not the drug of choice for Lyme disease, but is an alternative to doxycycline, amoxicillin and cefuroxime in the treatment of Lyme disease.

Penicillin G is the drug of choice for all forms of actinomycosis, syphilis and rat-bite fever.

Solution to Question 5:

Benzathine penicillin G and procaine penicillin G are repository preparations of penicillin G and are long-acting.

They are given as deep I.M. injection. Benzathine penicillin has the longest half-life and is administered every 2-4 weeks.

Penicillin V is an acid-resistant penicillin preparation that is administered orally.

Ampicillin has an extended spectrum but is not resistant to beta-lactamases.

Penicillinase-resistant penicillins are:

- Cloxacillin
- Dicloxacillin
- Flucloxacillin
- Methicillin
- Nafcillin

Solution to Question 6:

Probenecid is used along with penicillin because it increases the duration of action of penicillin.

Penicillin is excreted by tubular secretion in the kidney. Probenecid is a uricosuric drug used in gout that competes with the same transporter (OATP: organic anion transporter) used by penicillin for tubular secretion. Therefore, it decreases the rate of excretion and increases the duration of action of penicillin.

Solution to Question 7:

The drug that can be used here is cloxacillin since it is not penicillinase susceptible. It is penicillinase-resistant penicillin.

Bacteria can inactivate β -lactam antibiotics by the action of the β -lactamase enzyme. The β -lactamase enzyme activity can be narrow or broad spectrum. In general, gram-positive bacteria produce and secrete a large amount of β -lactamase. Most of these enzymes are penicillinases.

Penicillinase-resistant penicillins are:

- Cloxacillin
- Dicloxacillin
- Flucloxacillin
- Methicillin
- Nafcillin

Option B: Penicillin G encompasses the natural preparations and salts of penicillin. They have a narrow spectrum and are susceptible to penicillinase.

Options A and C: Amoxicillin and piperacillin are extended-spectrum penicillins but are susceptible to penicillinase.

Solution to Question 8:

Cloxacillin is not extended-spectrum penicillin. It is penicillinase-resistant penicillin.

Extended-spectrum penicillins have greater activity than penicillin against gram-negative bacteria because of their enhanced ability to penetrate the Gram-negative outer membrane.

Extended-spectrum penicillins are:

- Aminopenicillins - ampicillin, amoxicillin
- Carboxypenicillins - carbenicillin, ticarcillin
- Ureidopenicillins - piperacillin, mezlocillin

Solution to Question 9:

Carbenicillin is effective in Pseudomonas infection.

It is extended-spectrum penicillin belonging to the carboxypenicillins group. It has activity against Pseudomonas aeruginosa and indole-positive Proteus. It is acid labile and is not administered orally.

Antipseudomonal penicillins are:

- Carboxypenicillins - carbenicillin, ticarcillin
- Ureidopenicillins - mezlocillin, piperacillin

Solution to Question 10:

Piperacillin is not a beta-lactamase enzyme inhibitor. It is an extended-spectrum beta-lactam (ureidopenicillin) drug having significant antipseudomonal action.

Beta-lactamases are a family of enzymes produced by many gram-positive and gram-negative bacteria that inactivate beta-lactam antibiotics by opening the beta-lactam ring.

Beta-lactamase inhibitors available for clinical use are:

- Clavulanic acid
- Sulbactam
- Tazobactam
- Avibactam

Solution to Question 11:

Piperacillin-tazobactam is an antimicrobial that is effective against organisms producing extended-spectrum beta-lactamases.

Extended-spectrum beta-lactamase (ESBL) enzyme produced by bacteria causes resistance to most of the penicillins, cephalosporins (third and fourth generation), and monobactams. ESBL is most prevalent among gram-negative organisms like Klebsiella and E. coli.

ESBL can be inhibited by combining beta-lactamase inhibitors like clavulanic acid or tazobactam. Carbapenems and quinolones have inherent activity against ESBL.

Note: As piperacillin has greater activity against Klebsiella and other gram-negative organisms than amoxicillin, piperacillin-tazobactam can be considered as a better option in this question.

Solution to Question 12:

Methicillin and nafcillin are penicillinase-resistant penicillins that are acid susceptible and are administered parenterally.

Methicillin use is associated with higher incidence of interstitial nephritis.

Ampicillin, amoxicillin and cloxacillin are acid-resistant penicillins.

Solution to Question 13:

Neutropenia is seen with nafcillin. It is penicillinase-resistant parenteral penicillin.

It is associated with side effects like neutropenia and interstitial nephritis. In patients with renal failure, it is associated with seizures.

Methicillin is also a penicillinase-resistant penicillin. Its use is associated with a higher incidence of interstitial nephritis than other penicillins, hence it is no longer used.

Solution to Question 14:

The given scenario is suggestive of Jarisch-Herxheimer reaction and it is associated with the use of penicillin in syphilis.

Jarisch-Herxheimer reaction is a host immune reaction due to the release of spirochetal antigen by the dying T.pallidum in patients on treatment with penicillin. This occurs in primary and most cases of secondary syphilis and is characterized by fever, shivering, myalgia, exacerbation of lesions and vascular collapse. Symptomatic treatment suffices and penicillin treatment need not be interrupted.

Solution to Question 15:

Clostridium difficile associated diarrhea is mostly due to aminopenicillins.

Antibiotic-associated colitis is caused by *Clostridium difficile* infection due to the destruction of the normal gut microbial flora by the use of antibiotics.

Although all antibiotics can cause the condition, the most commonly implicated are:

- Ampicillin (aminopenicillin)
- Clindamycin
- Third-generation cephalosporins
- Fluoroquinolones.

Solution to Question 16:

Bezlotoxumab is a monoclonal antibody that binds to *Clostridium difficile* toxin B, indicated to reduce the recurrence of *Clostridium difficile* infection (CDI).

Actoxumab is a monoclonal antibody that binds to *Clostridium difficile* toxin A but it is not approved because of lack of efficacy.

Adalimumab is an anti-TNF alpha antibody used in rheumatoid arthritis and various other conditions.

Solution to Question 17:

The systemic use of bacitracin is restricted due to nephrotoxicity.

Bacitracin is a polypeptide antibiotic and it inhibits the synthesis of the bacterial cell wall.

Currently, it is used topically for dermal infections and for ophthalmic infections. Parenteral use of this antibiotic leads to nephrotoxicity.

Solution to Question 18:

Daptomycin is not indicated in community-acquired pneumonia because even though it penetrates adequately into the lung, it is inactivated by pulmonary surfactant and thus is not useful.

Daptomycin is a cyclic lipopeptide antibiotic. It acts by depolarising bacterial membranes resulting in loss of membrane potential and cell death.

Daptomycin is indicated in the following infections:

- Complicated skin and soft-tissue infections
- Complicated bacteremia caused by staphylococcus and streptococcus
- Right-sided endocarditis
- Vancomycin-resistant enterococcal infections

Its efficacy in the above-mentioned conditions is comparable to that of vancomycin.

Solution to Question 19:

Polymyxins are effective against gram-negative bacteria.

Polymyxin group of antibiotics includes polymyxin B and colistin which have their antimicrobial spectrum limited primarily to gram-negative aerobic bacteria. They interact with the phospholipid of the bacterial cell membranes and disrupt it.

Polymyxins are used in various topical formulations. Parenteral formulations are used only in serious systemic infections that are resistant to other antimicrobials. Nephrotoxicity limits widespread systemic use.

Solution to Question 20:

Resistance by lactamases is through the degradation of antibiotics.

The bacterial cell wall is made of heteropolymeric peptidoglycan that confers mechanical stability. The β -lactam antibiotics inhibit the last step in peptidoglycan synthesis by acylating the transpeptidase via cleavage of the $-\text{CO}-\text{N}-$ bond of the β -lactam ring. These transpeptidase targets are collectively called Penicillin Binding Proteins (PBPs)

Beta-lactamases are enzymes produced by certain bacteria that inactivates β -lactam antibiotics by hydrolysis of the β -lactam ring in their structure.

The resistance to penicillins and other beta-lactam antibiotics can be conferred through one of the four mechanisms:

- Inactivation of antibiotic by lactamases
- Modification of target PBPs
- Impaired penetration of drug to target PBPs
- Antibiotic efflux

Solution to Question 21:

Skin rashes of all types may be caused by allergy to penicillin.

Amoxicillin and ampicillin are aminopenicillins. The incidence of skin rashes appears to be highest following the use of ampicillin, at about 9%.

Amoxicillin is a close congener of ampicillin with the following features:

- Better oral absorption; food does not interfere with the absorption
- Incidence of diarrhea is less
- More active against penicillin-resistant pneumococci

Solution to Question 22:

Gentamicin is an aminoglycoside antibiotic that requires therapeutic drug monitoring (TDM) as it has a narrow therapeutic index. It is nephrotoxic and ototoxic (predominantly vestibular effects).

The threshold for toxicity for gentamicin is the plasma concentration of 2 µg/mL, the maximum recommended for prolonged exposure.

Aminoglycoside antibiotics act by binding to the 30s ribosome and inhibiting protein synthesis. Their ability to cause misreading of DNA templates and the formation of aberrant proteins that get incorporated into bacterial cell membranes makes them bactericidal agents.

Common ototoxic drugs are:

- Antibiotics - aminoglycosides (gentamicin & tobramycin & amikacin & neomycin), erythromycin, tetracyclines.
- Chemotherapeutic agents - cisplatin, bleomycin, nitrogen mustards.
- High dose aspirin, acetaminophen, NSAIDs
- Antimalarials - quinine, quinidine
- Phosphodiesterase-5 inhibitors - sildenafil

Cephalosporins, Vancomycin and Carbapenems

Question 1:

Which of the following is not a first-generation cephalosporin?

- a) Cefadroxil
- b) Cefazolin
- c) Cephalexin
- d) Cefaclor

Question 2:

A patient is undergoing orthopedic surgery. Which of the following first-generation cephalosporins can you use for surgical prophylaxis?

- a) Cephalexin
- b) Cefadroxil
- c) Cephadrine
- d) Cefazolin

Question 3:

You want to prescribe an oral second-generation cephalosporin to a patient with pneumonia. Which of the following drugs will you choose?

- a) Cefepime
- b) Cephadrine
- c) Cefaclor
- d) Cefadroxil

Question 4:

In comparison with first-generation cephalosporins, second-generation cephalosporins have which of the following properties?

- a) Greater activity against gram positive bacteria

- b) Lesser activity against gram positive bacteria
- c) Lesser activity against anaerobes
- d) Lesser hepatotoxicity

Question 5:

Which of the following is a third-generation cephalosporin that can be given orally?

- a) Cefixime
- b) Cefotaxime
- c) Ceftriaxone
- d) Cefoperazone

Question 6:

Cephalosporins are not useful in which of the following infections?

- a) Gonorrhea
- b) Lyme disease
- c) Listeriosis
- d) Pseudomonas infections

Question 7:

The pus culture and sensitivity report of a hospitalized patient with diabetic ulcer is positive for pseudomonas. Which of the following cephalosporins can be used in this patient?

- a) Cefpodoxime proxetil
- b) Cephadrine
- c) Cefotetan
- d) Ceftazidime

Question 8:

Which of the following statements about cefepime is false?

- a) Fourth generation cephalosporin

- b) Dose should be reduced in renal failure
- c) Possesses antipseudomonal activity
- d) Has good activity against MRSA

Question 9:

A 34-year-old man presents with a discharging wound post-operatively. Cultures reveal infection with MRSA. Which of the following cephalosporins can be used for this patient?

- a) Cefepime
- b) Cefpirome
- c) Ceftolozane
- d) Ceftaroline

Question 10:

A patient is to be started on a cephalosporin antibiotic. His serum creatinine levels yesterday at the time of admission was 1.1 mg/dl and now it is 1.5 mg/dl. Which of the following cephalosporins can be started without dose reduction in this patient?

- a) Cefuroxime
- b) Cefoperazone
- c) Ceftazidime
- d) Cefotaxime

Question 11:

What is the drug of choice for a patient who presents with an initial episode of *Clostridium difficile* infection?

- a) Metronidazole
- b) Vancomycin
- c) Fidaxomicin
- d) Loracarbef

Question 12:

What is the mechanism of action of vancomycin?

- a) Cell wall synthesis inhibition
- b) Cell membrane inhibition
- c) Peptide synthesis inhibition
- d) 50s ribosome inhibition

Question 13:

Oral vancomycin is used in the treatment of :

- a) 1, 4 and 5
- b) 1, 2 and 4
- c) 1 and 3
- d) 3 and 5

Question 14:

A patient on intravenous vancomycin develops flushing and erythematous lesions. On examination, tachycardia, and hypotension are noted. By what mechanism does vancomycin cause these effects?

- a) Allergic reaction
- b) Direct toxic effect on mast cells
- c) Histamine antagonist action
- d) Immune-complex deposition

Question 15:

Match the following bacteria with the drug of choice for each.

- a) A-1, B-3, C-4, D-2
- b) A-2, B-1, C-4, D-3
- c) A-3, B-4, C-1, D-2
- d) A-4, B-1, C-2, D-3

Question 16:

A patient allergic to penicillin requires antibiotic therapy. Which of the following can you safely prescribe?

- a) Aztreonam
- b) Cefepime
- c) Imipenem
- d) Ceftriaxone

Question 17:

Which of the following carbapenems has increased risk of causing seizures?

- a) Meropenem
- b) Imipenem
- c) Etrapanem
- d) Doripenem

Question 18:

Which of the following statements is false about extended-spectrum beta-lactamases?

- a) They can hydrolyse penicillins, cephalosporins as well as monobactam
- b) Carbapenems are hydrolysed by ESBL
- c) Ambler classification of ESBL is based on amino acid sequence
- d) Third and fourth generation cephalosporins are used for detection of ESBL

Question 19:

Cilastatin is administered to prolong the action of which of the following drugs?

- a) Meropenem
- b) Imipenem
- c) Cefazolin
- d) Piperacillin

Question 20:

A patient with a complicated urinary tract infection is put on a newly approved combination of meropenem and a non-suicidal beta-lactamase inhibitor. Identify this drug.

- a) Sulbactam
- b) Tazobactam
- c) Vaborbactam
- d) Clavulanic acid

Answer Key

Question No.	Correct Option
1	d
2	d
3	c
4	b
5	a
6	c
7	d
8	d
9	d
10	b
11	c
12	a
13	c
14	b
15	b
16	a
17	b
18	b
19	b
20	c

Detailed Explanations

Solution to Question 1:

Cefaclor is not a first-generation cephalosporin. It is a second-generation oral cephalosporin.

First-generation cephalosporins are :

- Oral formulation
- Cefadroxil
- Cephalexin
- Cephadrine
- Parenteral formulation - cefazolin

Solution to Question 2:

The first-generation cephalosporin indicated for surgical prophylaxis is Cefazolin. It is the preferred agent for surgical prophylaxis when skin flora is the likely pathogen.

Cephalexin, cefadroxil, and cephradine are first-generation oral cephalosporins.

Solution to Question 3:

Cefaclor is a second-generation oral cephalosporin.

Option A: Cefepime is a fourth-generation cephalosporin.

Options B and D: Cefadroxil and Cephradine are oral first-generation cephalosporins.

Solution to Question 4:

Second-generation cephalosporins have lesser activity against gram-positive bacteria when compared with first-generation cephalosporins.

Properties of second-generation cephalosporins, when compared to first-generation cephalosporins, are:

- Lesser activity against gram-positive bacteria
- Broader gram-negative activity against Hemophilus species, indole-positive Proteus, and Klebsiella species
- Anaerobes like B.fragilis are susceptible to second-generation cephalosporins like cefoxitin and cefotetan but not first-generation cephalosporins

Solution to Question 5:

Cefixime is a third-generation oral cephalosporin.

Solution to Question 6:

Ampicillin is the drug of choice for listeriosis, which is caused by *Listeria monocytogenes*, and cephalosporins are not useful in treating this condition.

Ceftriaxone is a third-generation cephalosporin, which is considered as a treatment of choice for gonorrhea and severe forms of Lyme disease.

Cefoperazone and ceftazidime are third-generation cephalosporins that have clinically significant activity against *Pseudomonas*.

Cephalosporins	Therapeutic uses
First-generation	Skin and soft tissue infection Serious infections due to MSSA Perioperative surgical prophylaxis
Second generation	Upper respiratory tract infections (sinusitis, otitis media) Cefoxitin/cefotetan - gynecologic infections, perioperative surgical prophylaxis
Third generation	Community-acquired pneumonia, meningitis, urinary tract infections Streptococcal endocarditis Gonorrhea Severe Lyme disease

Solution to Question 7:

The drug that can be used in this patient is Ceftazidime, which is a third-generation parenteral cephalosporin having significant anti-pseudomonal action.

Cefpodoxime proxetil is a third-generation oral cephalosporin. It does not have significant antipseudomonal action.

Cephadrine and cefotetan are first and second-generation cephalosporins with no antipseudomonal action.

Antipseudomonal cell wall synthesis inhibitors are:

- Cephalosporins:
- Third generation - Ceftazidime, Cefoperazone
- Fourth generation - Cefepime and ceftipime. They are also used against enterobacter
- Fifth generation - Ceftolozane

- Extended-spectrum penicillins:
- Carboxypenicillins - carbenicillin, ticarcillin
- Ureidopenicillins - piperacillin, mezlocillin

Solution to Question 8:

Cefepime, which is a fourth-generation cephalosporin, has good activity against *Pseudomonas*, *Enterobacter*, and *S. aureus*, but not against MRSA. It is effective in serious infections like hospital-acquired pneumonia, septicemia and febrile neutropenia.

Cefepime is excreted primarily through kidneys and dose reduction is necessary for renal failure. Encephalopathy and non-convulsive status epilepticus have been reported with the use of cefepime.

Solution to Question 9:

Ceftaroline is the cephalosporin that is indicated for MRSA infections and hence can be used in this patient. It is considered as anti-MRSA cephalosporin and along with ceftobiprole is grouped as fifth-generation cephalosporins.

Ceftaroline has gram-negative and also enhanced gram-positive coverage. It is effective against MRSA and penicillin-resistant *S. pneumoniae*.

Cefepime and ceftiofame are fourth-generation cephalosporins with no significant action on MRSA.

Ceftolozane is a fifth-generation antipseudomonal cephalosporin with no action on MRSA.

Solution to Question 10:

The given scenario is suggestive of acute kidney injury and the cephalosporin that does not require dose reduction in this patient is cefoperazone.

All cephalosporins are usually excreted by glomerular filtration and tubular secretion. They need a dose reduction in renal failure, except cefpiramide and cefoperazone that have predominant biliary excretion.

Dose reduction is necessary for cefoperazone and cefpiramide in patients with hepatic impairment.

Solution to Question 11:

The drug of choice for the treatment of initial episode of *Clostridium difficile* infection (CDI) is fidaxomicin. It is a macrocyclic antibiotic.

Fidaxomicin is preferred because of its safety profile and higher beneficial effects. But in resource-limited settings where it is not available, vancomycin can be used for the treatment of this condition.

Treatment of *Clostridium difficile* infection:

- Fidaxomicin
- Vancomycin
- Metronidazole

Vancomycin and fidaxomicin are more effective than metronidazole. All of the above drugs are given orally.

Loracarbef is a second-generation cephalosporin and is not indicated in pseudomembranous colitis caused by *Clostridium difficile*.

NOTE:

As per the Infectious Disease Society of America (IDSA) management guidelines 2021, Fidaxomicin is now preferred over Vancomycin for the treatment of initial episode of *Clostridium difficile* infection. Vancomycin can be used as an alternative drug.

Solution to Question 12:

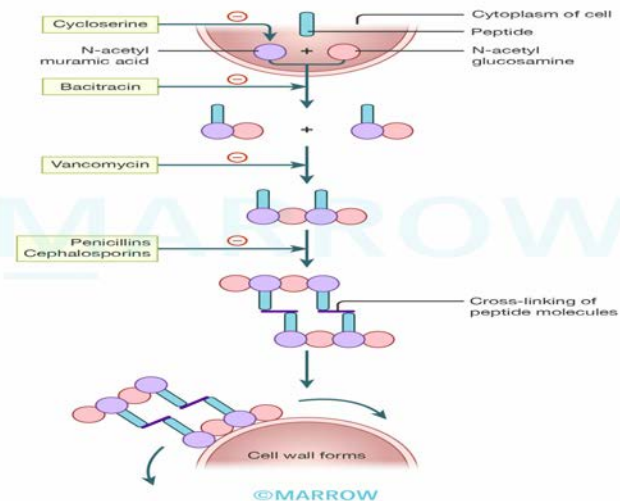
The mechanism of action of vancomycin is cell wall synthesis inhibition.

The bacterial cell wall has long chains of peptidoglycan and is cross-linked to each other by the action of the enzyme transglycosylase. Vancomycin is a glycopeptide group of antibiotics that binds to the D-alanyl-D-alanine terminus of the synthesized peptidoglycan. This inhibits the transglycosylase enzyme and prevents elongation and cross-linking of peptidoglycan. This leads to the weakening of the cell wall causing lysis.

It exhibits a time-dependant or concentration-independent killing of bacteria. It primarily acts on Gram +ve organisms which are resistant to other antibiotics such as methicillin-resistant staphylococcus and ampicillin resistant enterococci.

Lipoglycopeptides are a new generation of glycopeptide congeners which include telavancin, dalbavancin, and oritavancin. Dalbavancin and Oritavancin are administered as a single intravenous dose for skin and soft-tissue infections.

Cell wall synthesis inhibitors-Site of action



Solution to Question 13:

Oral vancomycin is used in the treatment of *Clostridium difficile* associated diarrhea and staphylococcal enterocolitis.

Its oral formulation is not absorbed systemically and hence is used for local action in the gastrointestinal tract.

Solution to Question 14:

The given scenario depicts red man syndrome associated with vancomycin. It occurs due to a direct toxic effect on mast cells resulting in the release of histamine.

Red man or red neck syndrome is a non-allergic reaction due to rapid intravenous infusion of vancomycin characterized by:

- Flushing
- Erythematous, urticarial reactions
- Tachycardia
- Hypotension

Slow infusion, prophylactic use of antihistamines can reduce the incidence.

Solution to Question 15:

Vancomycin is the drug of choice for methicillin-resistant *Staphylococcus aureus* (MRSA). It should be used only if the organism is resistant to other drugs.

Metronidazole is the drug of choice for anaerobic infections caused by Bacteroides.

Metronidazole is used in the treatment of the following conditions:

- Giardiasis
- Trichomoniasis
- Amoebiasis
- Bacterial vaginosis
- Pseudomembranous colitis

Meropenem is used in infections caused by ESBL producing bacteria like Klebsiella.

Carbapenems are not hydrolyzed by ESBL (extended-spectrum beta-lactamases). Therefore, carbapenems like imipenem, meropenem and ertapenem are resistant to hydrolysis by ESBLs. They are indicated in the treatment of infections caused by ESBL producing bacteria, like Klebsiella and E. coli.

Penicillins like amoxicillin, ampicillin can be used in H. influenzae infections.

Solution to Question 16:

Aztreonam can be safely given to a patient with a history of allergy to penicillin. It is a monobactam group of β -Lactam antibiotics resistant to narrow-spectrum beta-lactamase produced by bacteria.

It lacks cross-sensitivity with the other beta-lactams, except for ceftazidime. It has activity only against gram-negative bacteria. It has no activity against gram-positive bacteria and anaerobes.

Solution to Question 17:

Imipenem is the carbapenem that is associated with an increased risk of causing seizures, especially when high doses are given to patients with CNS lesions.

Excretion of imipenem is mainly renal and dosage modification is required to avoid seizures in renal failure patients.

Solution to Question 18:

Carbapenems like imipenem, meropenem, ertapenem are resistant to hydrolysis by extended-spectrum beta-lactamases (ESBL) and hence are indicated in the treatment of infections caused by ESBL producing bacteria.

Beta-lactamases are enzymes produced by bacteria that are able to hydrolyze antibiotics and make them ineffective, thus making the bacteria resistant to these antibiotics. They confer resistance against:

- Penicillins
- Cephalosporins
- Monobactams

Beta lactamases hydrolyze first and second-generation cephalosporins but are ineffective against third and fourth-generation.

Beta lactamases are classified into four categories based on their amino acid sequences, by the Ambler classification.

ESBLs also hydrolyze third and fourth-generation cephalosporins - cefotaxime, ceftazidime, ceftriaxone and cefepime.

Solution to Question 19:

Cilastatin is administered to prolong the action of imipenem.

Imipenem is rapidly hydrolyzed by the enzyme renal dehydropeptidase. It is administered with cilastatin, a reversible inhibitor of dehydropeptidase to inhibit the breakdown and prolong the action of imipenem.

Meropenem is not hydrolyzed by renal dehydropeptidase. Cilastatin administration is not necessary with meropenem.

Solution to Question 20:

Vaborbactam is a recently approved non-suicidal beta-lactamase inhibitor. It is indicated in combination with meropenem for complicated urinary tract infections.

Sulbactam, tazobactam and clavulanic acid are suicidal inhibitors of beta-lactamase enzyme.

Suicidal inhibitors are irreversible inhibitors that can permanently inactivate the β -lactamase through secondary chemical reactions in the enzyme active site.

Miscellaneous Antimicrobials

Question 1:

Which of the following best describes pyronaridine?

- a) Antimalarial
- b) Antiretroviral
- c) Antifungal
- d) Antibacterial

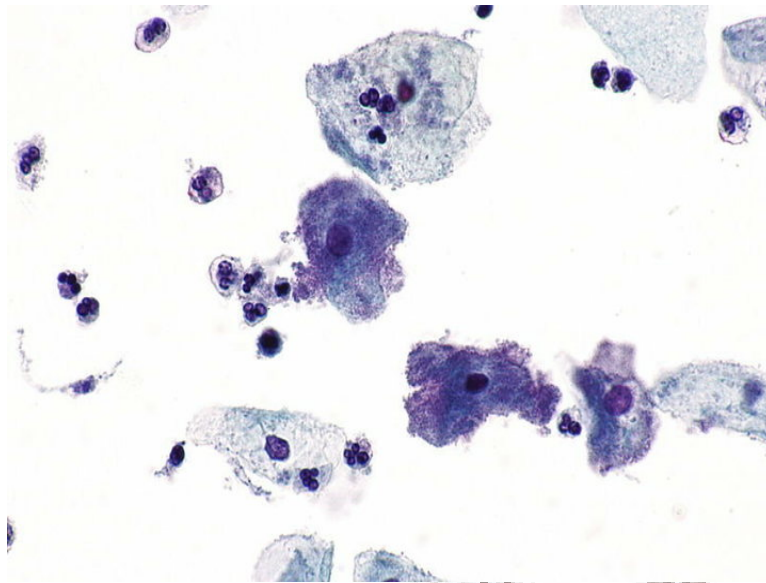
Question 2:

Which of the following drugs gets deposited in the retina?

- a) Chlortetracycline
- b) Chloramphenicol
- c) Chloroquine
- d) Chlordiazepoxide

Question 3:

A 30-year-old woman presented with itching and malodorous discharge in the vagina. Whiff's test was positive and microscopy of the discharge is shown below. What is the drug of choice for this condition?



- a) Ampicillin
- b) Ciprofloxacin
- c) Metronidazole
- d) Clotrimazole

Question 4:

Which of the following is best associated with lumefantrine?

- a) Antimycobacterial
- b) Antifungal
- c) Antimalarial
- d) Antiameobic

Question 5:

Suramin is indicated in which of the following conditions?

- a) African trypanosomiasis
- b) American trypanosomiasis
- c) Cutaneous leishmaniasis
- d) HIV infection

Question 6:

A patient presents with watery diarrhea, fever and abdominal pain. Investigations reveal a WBC count of 16,000 cells/microL. The patient has been on treatment with ampicillin for a few days. Which of the following antibiotics is approved for use in this condition?

- a) Azithromycin
- b) Roxithromycin
- c) Fidaxomicin
- d) Clindamycin

Question 7:

Which of the following best describes tolinaftate?

- a) Antifungal agent
- b) Antiviral agent
- c) Antimalarial agent
- d) Antileprotic agent

Question 8:

A class of antibiotics that is clinically indicated in combination with penicillins is inactivated if combined with penicillins in the same solution before administration. Identify this class of drugs.

- a) Tetracyclines
- b) Aminoglycosides
- c) Macrolides
- d) Glycopeptides

Question 9:

Which of the following antibiotics is not excreted through bile?

- a) Erythromycin
- b) Cefoperazone
- c) Rifampicin

d) Gentamicin

Question 10:

What is the mechanism of action of oseltamivir and zanamivir?

- a) DNA polymerase inhibition**
- b) Protein synthesis inhibition**
- c) Nucleotide analogue**
- d) Neuraminidase inhibition**

Question 11:

Rifampicin can be used as first-line therapy in which of the following patients?

- a) 1, 3 and 5**
- b) 1, 2, 3 and 4**
- c) 1, 2 and 3**
- d) 1, 2 and 4**

Question 12:

Which of the following drugs gets deposited in cornea?

- a) Leflunamide**
- b) Chloroquine**
- c) Methotrexate**
- d) Sulfasalazine**

Question 13:

Which of the following is an inhibitor of RNA synthesis?

- a) 5-Fluorouracil**
- b) Rifampicin**
- c) Nitrofurantoin**
- d) Novobiocin**

Answer Key

Question No.	Correct Option
1	a
2	c
3	c
4	c
5	a
6	c
7	a
8	b
9	d
10	d
11	d
12	b
13	b

Detailed Explanations

Solution to Question 1:

Pyronaridine is an antimalarial agent. It is an erythrocyte schizonticide with high efficacy and mechanism of action similar to chloroquine.

It is used in combination with artesunate.

Solution to Question 2:

Chloroquine has a large volume of distribution and is deposited in the retina.

Long-term use of chloroquine results in retinopathy. The classical finding is 'bull's-eye retinopathy'

Solution to Question 3:

The clinical scenario, a positive Whiff test and clue cells on microscopy point towards a diagnosis of bacterial vaginosis. Metronidazole (oral or intravaginal) or intravaginal clindamycin are considered the drugs of choice.

The male partner should be treated in case of recurrent infections of bacterial vaginosis.

Solution to Question 4:

Lumefantrine is an antimalarial agent.

Lumefantrine is an orally active, high efficacy; long-acting erythrocytic schizonticide related structurally and acts similar to halofantrine.

Lumefantrine is used only in combination with artemether.

Solution to Question 5:

Suramin is primarily indicated in the treatment of African trypanosomiasis. It is indicated only in the early stages of African trypanosomiasis because of its limited entry across the blood-brain barrier.

It has no clinical utility in American trypanosomiasis. Suramin has in vitro activity against retroviral reverse transcriptase but is ineffective in HIV infection.

Solution to Question 6:

The scenario is suggestive of pseudomembranous colitis caused by *Clostridium difficile* and fidaxomicin is a macrocyclic antibiotic that is indicated as an alternative to vancomycin in severe and recurring *Clostridium difficile* infections.

It acts by inhibiting protein synthesis through the blockade of RNA polymerase.

Solution to Question 7:

Tolnaftate is a topical synthetic antifungal agent.

It distorts the mycelial growth of yeast-like fungi and molds. It is effective in the treatment of dermatomycosis and tinea.

Solution to Question 8:

Aminoglycosides can be inactivated by various penicillins in vitro and thus should not be admixed in solution before administration.

Aminoglycosides and penicillin combination is indicated in several infections and they should not be admixed in the same solution before administration.

Solution to Question 9:

Gentamicin is not excreted through bile. It is an aminoglycoside antibiotic and aminoglycosides undergo minimum metabolism and are excreted almost entirely by the renal route.

Erythromycin, cefoperazone, and rifampicin are excreted predominantly in bile.

Solution to Question 10:

Oseltamivir and zanamivir are neuraminidase inhibitors.

They competitively and reversibly interact with the enzyme sites to inhibit viral neuraminidase activity. This interferes with the release of the progeny from infected host cells, halting the spread of infection.

Early administration hastens the resolution of symptoms if given within 48 h of infection.

Oseltamivir is given at a dose of 75 mg, BD for 5 days for the treatment of Influenza A and B.

For prophylaxis, it is given at a dose of 75mg, OD and continued for the duration of exposure.

Solution to Question 11:

Rifampicin is the first-line therapy and is included in the regimens for both tuberculosis and leprosy.

Doxycycline combined with rifampicin is considered a first-line agent for brucellosis.

Solution to Question 12:

Chloroquine, an anti-malarial drug may get deposited in the cornea leading to corneal opacity.

Ocular toxicity of chloroquine:

- Short-term therapy - blurring of vision and diplopia, reversible on stoppage of drug.
- Long-term use (high-dose):
- Keratopathy
- Pigmentary retinopathy
- Optic atrophy

Hydroxychloroquine has lower ocular toxicity compared to chloroquine.

Drugs causing corneal opacity:

- Amiodarone
- Phenothiazine
- Indomethacin
- Chloroquine

Long-term treatment with these drugs can cause a whorl-like opacity in the corneal epithelium called cornea verticillata. It is also seen in Fabry's disease.

Solution to Question 13:

Rifampicin is an inhibitor of RNA synthesis. It binds to the beta subunit of DNA dependent RNA-polymerase (rpoB) to form a stable drug-enzyme complex. The binding suppresses the chain formation in RNA synthesis.

Option A: 5-Fluorouracil inhibits thymidylate synthesis.

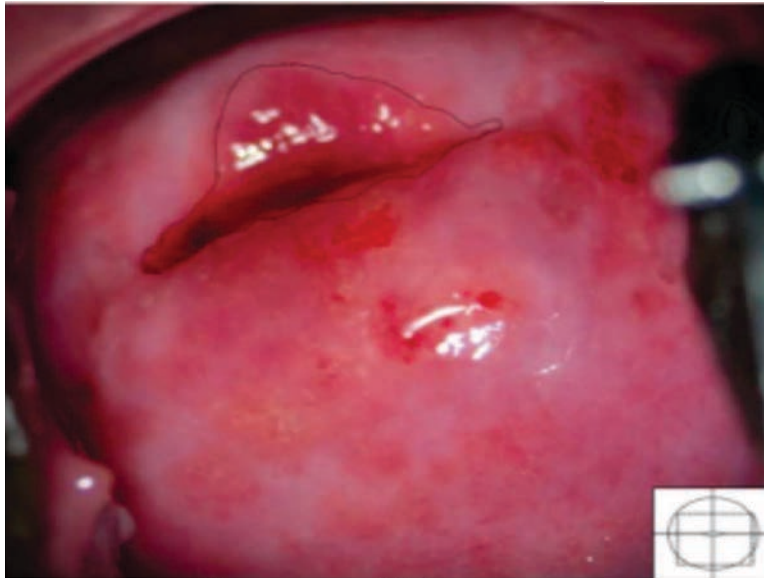
Option C: Nitrofurantoin produces highly reactive intermediates that cause DNA damage

Option D: Novobiocin is a potent inhibitor of bacterial DNA gyrase.

Anti-Protozoal agents and Anthelmintic drugs

Question 1:

A 25-year-old woman presents with malodorous vaginal discharge and dysuria. The examination finding is shown below. What is the drug of choice for this condition?



- a) Tinidazole
- b) Mebendazole
- c) Metronidazole
- d) Secnidazole

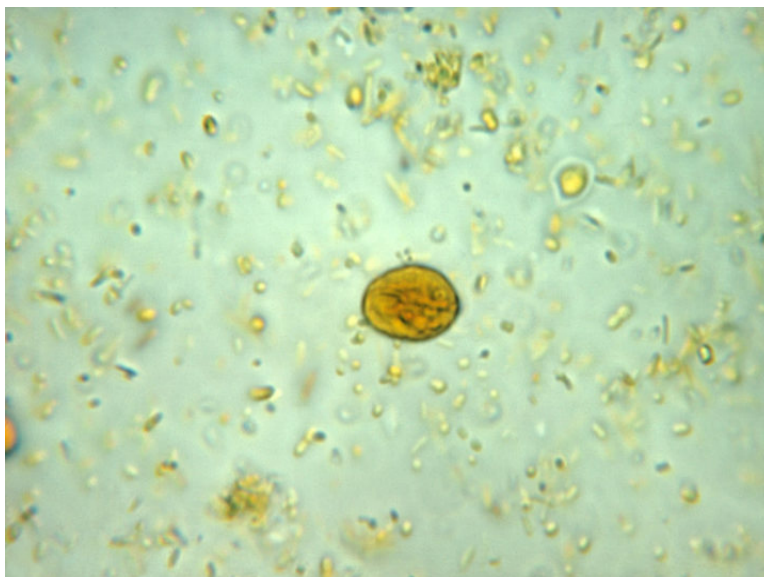
Question 2:

Metronidazole is selectively active against anaerobic and microaerophilic organisms because:

- a) Aerobes have oxygen gradient inside the cell which deactivates metronidazole moiety
- b) Only anaerobes and microaerophilic organisms reduce it to generate reactive nitro radicals
- c) Aerobes have efflux drug pumping mechanism for metronidazole
- d) Metronidazole inserts pore in the cell membrane of anaerobes and microaerophilic bacteria

Question 3:

A child presents with diarrhea, weight loss, and abdominal pain. His mother reports stools to be greasy and foul-smelling. A wet preparation of the stool sample is shown below. What is first-line therapy for this condition?



- a) Metronidazole
- b) Tinidazole
- c) Albendazole
- d) Mebendazole

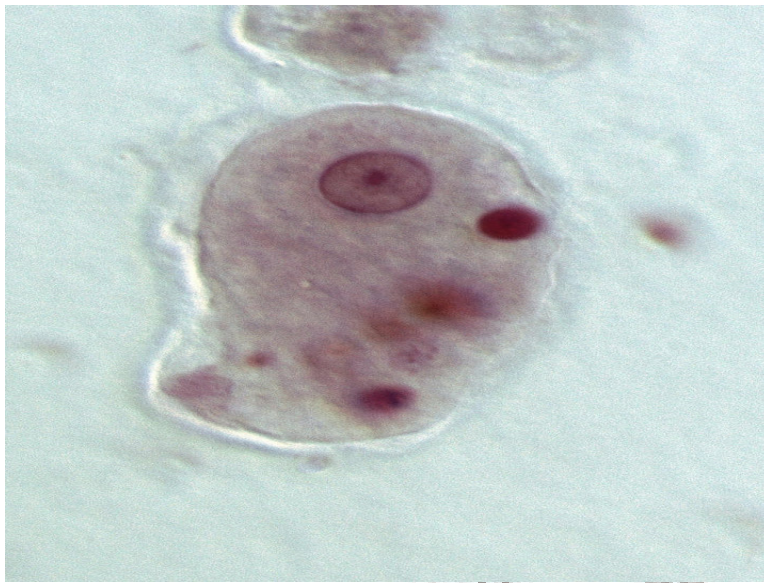
Question 4:

Metronidazole is least likely to be effective in the treatment of:

- a) Hepatic amebiasis
- b) B. fragilis infection
- c) Pseudomembranous colitis
- d) Pneumocystosis

Question 5:

A 6-year-old boy presents with lower abdominal pain and diarrhea. The stool examination finding is shown below. Which of the following drugs cannot be used for the eradication of intestinal cysts in this condition?



- a) Diloxanide furoate
- b) Nitazoxanide
- c) Metronidazole
- d) Iodoquinol

Question 6:

Which of the following is used in the treatment of a patient with asymptomatic intestinal amoebiasis?

- a) Metronidazole
- b) Tinidazole
- c) Paromomycin
- d) Tetracycline

Question 7:

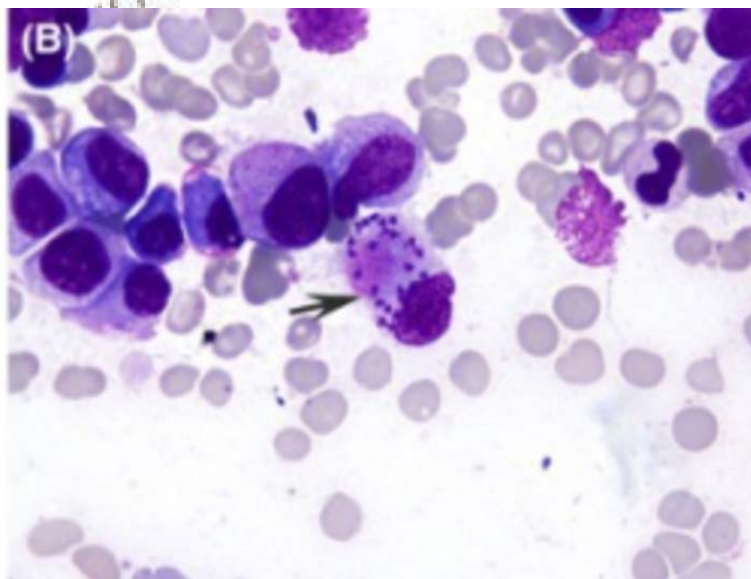
A 21-year-old male presented to the OPD complaining of blood-streaked stools. Sigmoidoscopy was done and it showed ulcers in the sigmoid. Biopsy was taken from this area and the histopathological image is given below. Which of the following is the most appropriate treatment?



- a) Intravenous ceftriaxone
- b) Intravenous metronidazole
- c) Intravenous paromomycin
- d) Intravenous chloroquine

Question 8:

A patient presents with sudden onset fever with chills and malaise. On examination splenomegaly was present. Bone marrow picture is shown in the image. Which of the following drugs is not used in the treatment of this condition?



- a) Paromomycin

- b) Liposomal amphotericin B
- c) Hydroxychloroquine
- d) Miltefosine

Question 9:

A 32-year-old woman presents with the finding as shown in the image. She gives history of travel to South America and was bitten by a reduvid bug. Which of the following is the drug of choice for this condition?



- a) Nifurtimox
- b) Albendazole
- c) Benznidazole
- d) Metronidazole

Question 10:

A patient presents in the late stage of West African sleeping sickness. What is the drug of choice in this condition?

- a) Nifurtimox
- b) Nifurtimox - eflornithine combination
- c) Nifurtimox - melarsoprol combination
- d) Melarsoprol

Question 11:

Which of the following drugs can be used as first-line treatment in patients with early-stage of East African trypanosomiasis?

- a) Suramin
- b) Pentamidine
- c) Melarsoprol
- d) Eflornithine

Question 12:

The microscopy of fecal sample of a patient showed presence of *Ascaris lumbricoides* eggs. This condition can most effectively be treated with which of the following drugs?

- a) Metronidazole
- b) Mebendazole
- c) Albendazole
- d) Pyrantel pamoate

Question 13:

What is the primary mechanism of action of benzimidazole?

- a) Inhibition of microtubule polymerization
- b) Stabilisation of microtubule polymers
- c) Inhibition of protein synthesis
- d) Paralysis of respiratory muscles

Question 14:

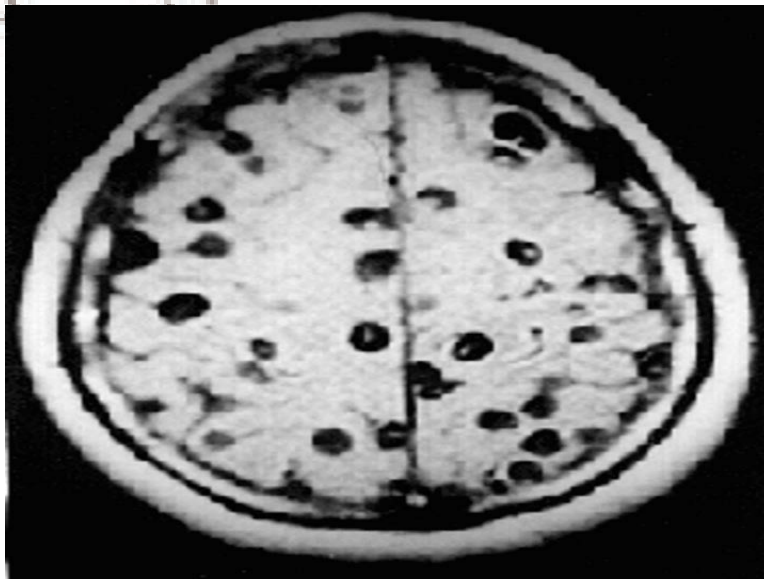
A young man working at a dairy farm presents with fever and abdominal pain. Examination of his stool samples revealed the finding shown in the image. Which of the following benzimidazoles is indicated in this condition?



- a) Albendazole
- b) Mebendazole
- c) Thiabendazole
- d) Triclabendazole

Question 15:

The MRI of a patient who presented with new-onset seizures is shown below. What is the drug of choice?



- a) Praziquantel

- b) Albendazole
- c) Levamisole
- d) Pentamidine

Question 16:

A 25-year-old man hospitalized with a hydatid cyst refused to undergo surgical removal of the cyst. He was treated with high dose albendazole for 3 months. This patient should be monitored for toxicity to the:

- a) Liver
- b) Kidney
- c) Pancreas
- d) Peripheral nerves

Question 17:

What is the drug of choice in patients with onchocerciasis?

- a) Albendazole
- b) Praziquantel
- c) Ivermectin
- d) Diethylcarbamazine

Question 18:

A patient presents with itchy, serpiginous eruptions as shown in the image. He says that these lesions advance rapidly to different sites. What is the primary mechanism of action of the drug preferred for this disease?



- a) Activation of glutamate gated chloride channels
- b) Inhibition of glutamate gated chloride channels
- c) Inhibition of nicotinic acetylcholine receptor
- d) Activation of nicotinic acetylcholine receptor

Question 19:

What is the first-line treatment for a patient presenting with the finding shown below?



- a) DEC
- b) Pyrantel pamoate

- c) Mebendazole
- d) Ivermectin

Question 20:

A 17-year-old boy presents with painless hematuria. He has a history of travel to Kenya. Urine microscopy revealed the finding shown below. What is the drug of choice in this condition?



- a) Albendazole
- b) Mebendazole
- c) Praziquantel
- d) Triclabendazole

Question 21:

Which of the following drugs causes flaccid paralysis of *Ascaris lumbricoides*?

- a) Metrifonate
- b) Levamisole
- c) Piperazine
- d) Tribendimidine

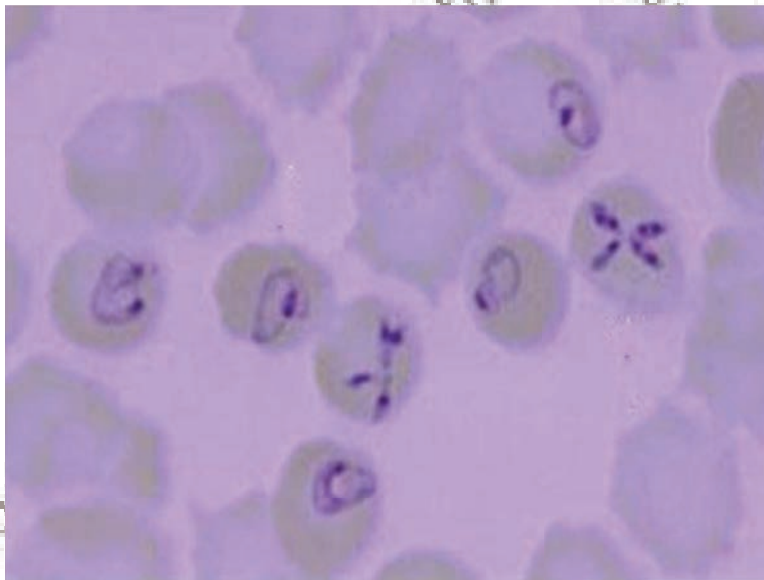
Question 22:

Diethylcarbamazine is used for the treatment of:

- a) Filariasis
- b) Dracunculiasis
- c) Schistosomiasis
- d) Ascariasis

Question 23:

A patient presents with fever, fatigue, and malaise. Investigations reveal anemia and thrombocytopenia. Thin blood smear is shown below. What is the recommended therapy for this condition?



- a) Clindamycin plus atovaquone
- b) Azithromycin plus atovaquone
- c) Azithromycin plus clindamycin
- d) Clindamycin plus quinine

Question 24:

A patient presented with fever and cervical lymphadenopathy. Investigations revealed IgM toxoplasma antibodies. Which of the following agents is most effective in treating this condition?

- a) Trimethoprim

- b) Pyrimethamine
- c) Pentamidine
- d) Pyridoxine

Answer Key

Question No.	Correct Option
1	c
2	b
3	b
4	d
5	c
6	c
7	b
8	c
9	c
10	b
11	a
12	c
13	a
14	d
15	b
16	a
17	c
18	a
19	d
20	c
21	c
22	a
23	b
24	b

Detailed Explanations

Solution to Question 1:

The given scenario and image showing strawberry vagina and cervix are suggestive of trichomoniasis. The drug of choice is metronidazole. It is given as a single dose of 2g.

Both metronidazole and tinidazole are effective in the management of trichomonas vaginitis. Tinidazole has been reported to have fewer GI-related side effects and less resistance compared to metronidazole. However, it is much more expensive and is preferred in cases refractory to metronidazole.

Solution to Question 2:

Metronidazole is selectively active against anaerobic and microaerophilic organisms because only these organisms reduce it to generate reactive nitro radicals.

They contain electron transport components that donate an electron to metronidazole. This results in the formation of a highly reactive nitro radical anion that targets DNA and kills the organism.

Solution to Question 3:

The clinical scenario and the presence of cysts of *Giardia lamblia* in stool point towards a diagnosis of giardiasis, and tinidazole is the treatment of choice for this condition.

Both metronidazole and tinidazole are considered first-line drugs however, tinidazole is considered superior in its efficacy in comparison to metronidazole.

Drug dosage:

- Metronidazole: 250- 500mg TID orally for 7 days
- Tinidazole: 2g single dose orally

The cyst of *Giardia* is small and oval, and is surrounded by a hyaline cyst wall. Its internal structure includes two pairs of nuclei grouped at one end. A young cyst contains one pair of nuclei. The axostyle lies diagonally, forming a dividing line within the cyst wall.

Solution to Question 4:

Metronidazole is not effective in the management of pneumocystosis caused by *Pneumocystis jirovecii*. Co-trimoxazole is used in the management of this condition.

Metronidazole is used in the following conditions:

- Giardiasis
- Trichomoniasis
- Amoebiasis

- Bacterial vaginosis
- Anaerobic infections - bacteroides and clostridium.
- Pseudomembranous colitis (alternative to vancomycin & fidaxomicin)

Solution to Question 5:

The image shows hematophagous trophozoite of *Entamoeba histolytica* and metronidazole is not an effective agent against cysts of amoeba in the colon.

Metronidazole is used in the treatment of amoebiasis (amoebic colitis, amoebic liver abscess). It is well absorbed from the small intestine and only a small amount of the unabsorbed drug reaches the colon, which is insufficient to kill the cysts.

Hence, luminal agents like diloxanide furoate, nitazoxanide or iodoquinol are required for the eradication of cysts and trophozoites in the intestine.

Solution to Question 6:

The drug of choice for the treatment of asymptomatic intestinal luminal amoebiasis due to *E. histolytica* is paromomycin.

Paromomycin (aminosidine) is an aminoglycoside that has poor oral absorption and hence it acts only in the intestinal lumen as a luminal amebicide. It has no effect against extraintestinal organisms.

Solution to Question 7:

The given clinical picture and the histopathological image showing flask-shaped ulcers are suggestive of intestinal amoebiasis. The most appropriate treatment is intravenous metronidazole.

The amoebic ulcer which is flask-shaped is the typical lesion seen in intestinal amoebiasis. The ulcers are multiple and are confined to the colon. Sometimes, in secondary intestinal lesions, the ulcers are numerous in the cecum, followed by the sigmoido-rectal region.

Three classes of drugs are used in the treatment of amebiasis:

- Luminal amoebicides- diloxanide furoate, iodoquinol, paromomycin, and tetracycline. They act in the intestinal lumen but not in tissues.
- Tissue amoebicides- emetine and chloroquine. They are effective in systemic infection but less effective in the intestine.
- Both luminal and tissue amebicides- Metronidazole, tinidazole and imidazole. They act on both sites and are the drug of choice for treating amebic colitis and amebic liver abscess.

Solution to Question 8:

The given scenario and the bone marrow picture showing macrophages infected with intracellular Leishman-Donovan bodies point towards a diagnosis of visceral leishmaniasis. Hydroxychloroquine is not indicated in the treatment of this condition.

The drugs effective in visceral leishmaniasis are:

- Liposomal amphotericin B (preferred)
- Sodium stibogluconate (not in India)
- Miltefosine
- Paromomycin
- Pentamidine (rarely used)

Hydroxychloroquine is an antimalarial agent and is also used in the treatment of dermatological conditions like:

- Cutaneous lupus erythematosus
- Cutaneous dermatomyositis
- Polymorphous light eruption
- Porphyria cutanea tarda
- Sarcoidosis

Solution to Question 9:

The scenario and the image showing Romaña sign point towards a diagnosis of Chagas disease (American trypanosomiasis) and the drug of choice in this condition is benznidazole.

American trypanosomiasis is caused by *Trypanosoma cruzi* and can be managed by either benznidazole or nifurtimox. Due to its favorable safety profile, benznidazole is considered the drug of choice.

Romaña sign is the unilateral painless periorbital swelling associated with the acute stage of Chagas disease.

Albendazole and metronidazole are not indicated for American trypanosomiasis. Albendazole is used in infections with *Ascaris lumbricoides*, *Trichuris trichiura* and hookworms. Metronidazole is used in the treatment of trichomoniasis and amebiasis.

Solution to Question 10:

Nifurtimox in combination with eflornithine is the drug of choice for late-stage West African sleeping sickness. It is caused by *Trypanosoma brucei gambiense*.

Nifurtimox alone is indicated in American trypanosomiasis.

Melarsoprol is the drug of choice for late-stage East African (Rhodesian) trypanosomiasis. It is not indicated in combination with nifurtimox.

Solution to Question 11:

Suramin is the first-line therapy for early-stage East African trypanosomiasis caused by *Trypanosoma brucei rhodesiense*.

It is also active against *Trypanosoma brucei gambiense* but only indicated as a second-line agent when pentamidine fails or is contraindicated.

Melarsoprol is the only effective drug available for the treatment of the late encephalopathic stage of East African sleeping sickness. It is caused by *Trypanosoma brucei rhodesiense*. Melarsoprol is effective in earlier stages also but is avoided because of its toxicity.

Solution to Question 12:

Roundworm infestation is most effectively treated with albendazole. It is considered as the drug of choice for the treatment of roundworm (*Ascaris lumbricoides*) infestation.

Albendazole is considered as the drug of choice for treatment of infestation by:

- Whipworm - *Trichuris trichura* causing trichuriasis
- Hookworm - *Ancylostoma duodenale*
- Roundworm - *Ascaris lumbricoides*
- Dog tapeworm - *Echinococcus granulosus* causing hydatid disease

Albendazole is administered as a single oral 400-mg dose to adults and children more than 2 years of age. Safety is not established for children less than 2 years. However, according to WHO, if the benefit outweighs the risk, albendazole at 200 mg can be given to children between the ages of 1-2 years.

Solution to Question 13:

The primary mechanism of action of benzimidazoles is by inhibition of microtubule polymerization. Benzimidazoles bind to beta-tubulin and cause inhibition of microtubule polymerization.

Drugs belonging to the benzimidazole group are:

- Albendazole
- Mebendazole
- Triclabendazole
- Thiabendazole.

Mechanisms of action:

- Inhibition of mitochondrial fumarate reductase
- Reduced glucose transport
- Uncoupling of oxidative phosphorylation

Solution to Question 14:

The given image showing operculated egg (egg having a lid-like structure on its end) is suggestive of *Fasciola hepatica* and the benzimidazole indicated for infection with this organism is triclabendazole.

Solution to Question 15:

The MRI findings (multiple cysticerci within the brain) are suggestive of neurocysticercosis and albendazole is the drug of choice in this condition.

Glucocorticoid therapy is usually begun before initiating albendazole and continued for several days after the commencement of therapy. It reduces the inflammatory reaction to the dead and dying cysticerci of *Taenia solium*.

Note: Even though Albendazole and Praziquantel are used as a combination, the single best answer choice to this question is Albendazole.

Solution to Question 16:

The patient described above should be monitored for toxicity to the liver. Liver dysfunction is one of the common adverse effects of albendazole therapy and liver function tests should be routinely conducted when long-term therapy is initiated.

Long-term albendazole therapy is also associated with bone marrow toxicity. Therefore, blood counts should also be regularly monitored.

Solution to Question 17:

Ivermectin is the drug of choice for onchocerciasis caused by *Onchocerca volvulus* in adults and children aged more than 5 years.

Ivermectin is the drug of choice for:

- Strongyloidiasis
- Onchocerciasis

It is associated with Mazzotti reaction (fever, pruritic rash, lymphadenopathy) as a side effect when given for filariasis due to immune system activation by the dying parasite. This reaction is also seen with the use of diethylcarbamazine.

Solution to Question 18:

The clinical scenario and the image showing larva currens, points towards a diagnosis of strongyloidiasis, and the drug of choice in this condition is ivermectin. The mechanism of action of ivermectin primarily involves activation of glutamate-gated chloride channels found only in invertebrates.

Ivermectin activates glutamate-activated chloride channels in the nerves and muscles of nematode and results in the influx of chloride into the cell. This causes hyperpolarization of cells and results in paralysis of nematodes.

Solution to Question 19:

The above image shows cutaneous larva migrans. Ivermectin is the first-line drug for the treatment of cutaneous larva migrans.

The first-line drugs used in cutaneous larva migrans include:

- Ivermectin - 200µg/kg orally single dose
- Albendazole - 400mg orally for 3 days

Solution to Question 20:

The clinical scenario and the image showing the egg of *Schistosoma hematobium* (egg with terminal spine), point towards the diagnosis of schistosomiasis. Praziquantel is the drug of choice for schistosomiasis caused by all *Schistosoma* species that can infect humans.

A single oral dose of 40 mg/kg or three doses of 20 mg/kg each, given 4–6 h apart, generally produce cure rates of 70%–95% and consistently high reductions (>85%) in egg counts.

Solution to Question 21:

Piperazine causes flaccid paralysis of *Ascaris lumbricoides*.

It acts as a GABA receptor agonist and increases chloride ion conductance across the *Ascaris* muscle membrane. This results in hyperpolarisation and reduced excitability, which leads to muscle relaxation and flaccid paralysis.

Metrifonate, levamisole and tribendimidine are anthelmintic drugs that cause spastic paralysis.

Solution to Question 22:

Diethylcarbamazine (DEC) is used for the control and treatment of lymphatic filariasis caused by *W. bancrofti*, *B. malayi* and *B. timori*.

Other indications for the use of DEC are:

- Loiasis
- Tropical pulmonary eosinophilia

Solution to Question 23:

The given scenario and the microscopic image showing tetrads or Maltese cross, point towards a diagnosis of babesiosis and the recommended therapy for this condition is azithromycin plus atovaquone.

Babesiosis is a zoonotic infection that resembles malaria. It presents with febrile illness, hemolysis and hemoglobinuria.

For mild or moderate disease: Azithromycin plus atovaquone.

For severe disease:

- Recommended regimen - azithromycin plus atovaquone.
- Alternative regimen - clindamycin plus quinine.

Severe babesiosis typically occurs in patients with one or more of the following characteristics: age >50 years, neonatal prematurity, asplenia/hyposplenism, HIV/AIDS, malignancy, and immunosuppressive therapy.

Solution to Question 24:

Pyrimethamine is the most effective agent against toxoplasmosis and is present in most of the regimens. It is combined with sulfadiazine or clindamycin for toxoplasmosis.

Trimethoprim plus sulfamethoxazole (cotrimoxazole) is an alternative.

Antifungal Agents

Question 1:

Which of the following is a polyene antifungal agent?

- a) Caspofungin
- b) Griseofulvin
- c) Amphotericin B
- d) Terbinafine

Question 2:

What is the site of action of amphotericin B in fungi?

- a) Nucleic acid
- b) Ribosome
- c) Cell membrane
- d) Cytoplasm

Question 3:

Which formulation of amphotericin B has the highest risk of nephrotoxicity?

- a) Liposomal amphotericin B
- b) Conventional amphotericin B
- c) Amphotericin B lipid complex
- d) Amphotericin B colloidal dispersion

Question 4:

A diabetic patient presents with periorbital swelling as shown below. She complains of significant visual impairment and a fungal etiology is suspected. What is the most appropriate drug for this patient?



- a) Itraconazole
- b) Liposomal amphotericin B
- c) Ketoconazole
- d) Broad spectrum antibiotics

Question 5:

A patient diagnosed with mucormycosis is being treated with intravenous amphotericin B. Following administration of the drug, he starts experiencing chills, fever and dyspnea. These symptoms occur most frequently with which formulation?

- a) Liposomal amphotericin B
- b) Conventional amphotericin B
- c) Amphotericin B lipid complex
- d) Amphotericin B colloidal dispersion

Question 6:

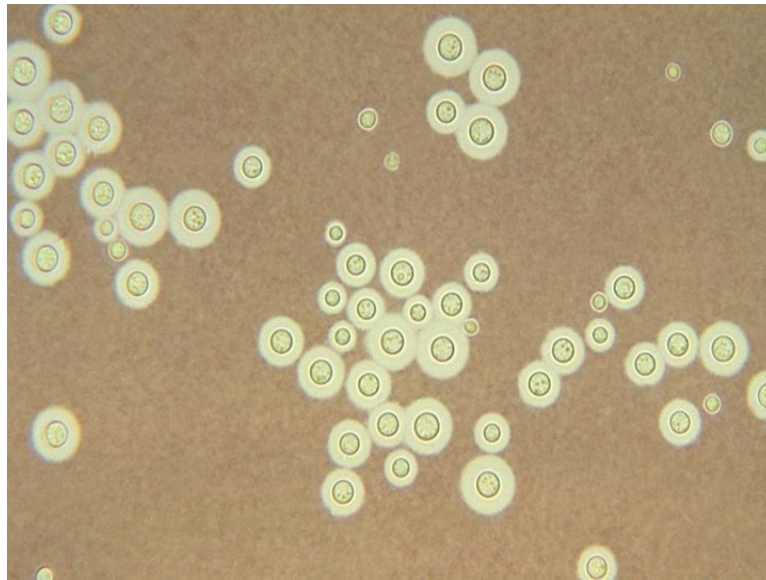
An immunocompromised patient with invasive mucormycosis is administered intravenous amphotericin B. Which frequently associated electrolyte disturbance would you be concerned about?

- a) Hypokalemia
- b) Hyperkalemia
- c) Hypermagnesemia

d) Hyponatremia

Question 7:

An HIV-positive patient presents with headache and nuchal stiffness. The CD4 count is 90 and CSF sample with India-ink staining shows the presence of encapsulated yeast cells. Which of the following is the treatment of choice for the patient?



- a) Fluconazole
- b) Amphotericin B
- c) Flucytosine
- d) Amphotericin B plus flucytosine

Question 8:

Identify the wrongly matched pair of antifungals with its mechanism of action:

- a) Ketoconazole : Lanosterol-14-demethylase inhibition
- b) Flucytosine : Inhibits microtubule synthesis and thus hinders mitosis
- c) Terbinafine : Squalene epoxidase inhibition
- d) Amphotericin B : Sequesters ergosterol

Question 9:

A washerman presents with itchy, annular scaly plaques in the groin. He is prescribed with an antifungal that acts by interfering with mitosis. Identify the drug.

- a) Caspofungin
- b) Ciclopirox
- c) Natamycin
- d) Griseofulvin

Question 10:

To which of the following patients will you not prescribe griseofulvin?

- a) A middle aged man with tinea capitis
- b) A student with tinea cruris
- c) A child with tinea versicolor
- d) An athlete with tinea pedis

Question 11:

Which of the following can cause contraceptive failure in a lady on oral contraceptive pills?

- a) Ketoconazole
- b) Fluconazole
- c) Griseofulvin
- d) Flucytosine

Question 12:

Fungal cell wall synthesis is altered by which of the following drugs?

- a) Caspofungin
- b) Amphotericin B
- c) Nystatin
- d) Flucytosine

Question 13:

What is the mechanism of triazoles?

- a) Interferes with DNA and RNA synthesis
- b) Inhibition of ergosterol synthesis
- c) Forms pores in fungal membranes
- d) Inhibits epoxidation of squalene in fungi

Question 14:

A patient was diagnosed with coccidioidal meningitis. What is the drug of choice for the condition?

- a) Itraconazole
- b) Fluconazole
- c) Amphotericin B
- d) Griseofulvin

Question 15:

A middle aged man presents with the condition shown in the image. Which of the following drugs can cause this condition?



- a) Fluconazole
- b) Itraconazole
- c) Ketoconazole

d) Voriconazole

Question 16:

Which of the following is not an indication for ketoconazole use?

- a) Cushing's syndrome**
- b) Dermal leishmaniasis**
- c) Addison's disease**
- d) Kala azar**

Question 17:

While administering which of the following drugs will you counsel patients regarding transient visual and auditory hallucinations?

- a) Oral voriconazole**
- b) Intravenous voriconazole**
- c) Oral itraconazole**
- d) Intravenous itraconazole**

Question 18:

Omeprazole dose is reduced when given along with voriconazole because of what action of the latter drug?

- a) CYP3A4 inhibition**
- b) CYP3A4 induction**
- c) CYP2C19 inhibition**
- d) CYP2C19 induction**

Question 19:

In a patient with athlete's foot, which keratolytic ointment would you prescribe to enhance treatment outcome?

- a) Benzoic acid**
- b) Salicylic acid**

- c) Undecylenic acid
- d) Ciclopirox olamine

Question 20:

A patient presented with multiple annular itchy skin lesions with central clearing over his back. You may prescribe any of the following except?

- a) Itraconazole
- b) Terbinafine
- c) Amphotericin B
- d) Griseofulvin

Question 21:

Which of the following drugs is not used to treat candida infections?

- a) Ketoconazole
- b) Nystatin
- c) Amphotericin B
- d) Griseofulvin

Question 22:

Which of the following is not true about antifungal drugs?

- a) Amphotericin B is more effective parenterally
- b) Griseofulvin is effective orally
- c) Ciclopirox olamine is effective in systemic mycoses
- d) Fluconazole is effective orally as well as intravenously

Question 23:

Topical antifungal agents have no role in the treatment of patients with which of the following conditions?

- a) Dermatophytosis

- b) Tinea versicolor
- c) Tinea nigra
- d) Chromoblastomycosis

Question 24:

Which of the following antifungal drugs are used topically?

- a) 1, 2, 3
- b) 1, 3, 5
- c) 1, 2, 3, 4, 5
- d) 1, 3, 4, 5

Question 25:

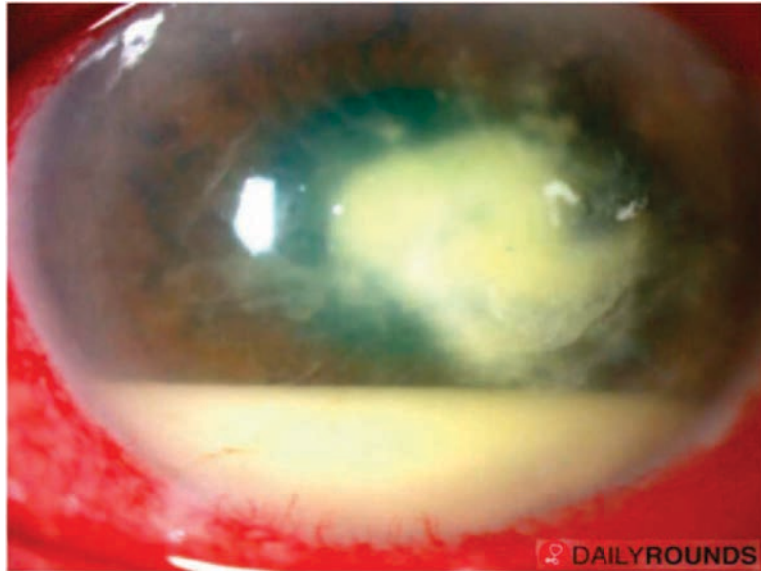
A 48-year-old woman presented with yellowish discolouration of toenails. The discolouration spread proximally as a streak in the nail and later, the entire nail separated from the nailbed. What is the mechanism of action of the drug of choice for her?



- a) Inhibition of microtubule formation
- b) Binding to ergosterol and altering membrane functions
- c) Formation of pore in cell membrane
- d) Inhibition of ergosterol synthesis

Question 26:

A farmer presented with complaints of redness and excessive watering from the right eye. Slit-lamp examination reveals the following finding. Microscopy of the corneal ulcer scraping shows branched septate hyphae. Which of the following drugs can be used in the treatment of this patient?



- a) Nystatin 3.5%
- b) Fluconazole 0.2%
- c) Natamycin 5%
- d) Oral voriconazole

Question 27:

Voriconazole is not effective against which of the following?

- a) Candida albicans
- b) Mucormycosis
- c) Candida tropicalis
- d) Aspergillosis

Answer Key

Question No.	Correct Option
1	c
2	c
3	b
4	b
5	d
6	a
7	d
8	b
9	d
10	c
11	c
12	a
13	b
14	b
15	c
16	c
17	b
18	c
19	b
20	c
21	d
22	c
23	d
24	d
25	d
26	c
27	b

Detailed Explanations

Solution to Question 1:

Amphotericin B is a polyene antifungal.

Other polyene antifungals are:

- Amphotericin B

- Nystatin
- Natamycin
- Hamycin

Amphotericin B is the least toxic among polyenes and is used systemically for fungal infections. Nystatin and hamycin have high systemic toxicity and are used only for local and topical actions.

Solution to Question 2:

The site of action of amphotericin B in fungi is the cell membrane.

Amphotericin B binds to the ergosterol in the cell membrane of fungi and forms pores or channels. They increase the permeability of the membrane to ions and cause leakage of cytosolic molecules. This leads to a loss of integrity of the cell membrane and causes fungicidal effects.

Recent evidence suggests that the amphotericin B sequesters ergosterol from lipid layers, much like a sponge, resulting in fungal cell death.

Solution to Question 3:

Nephrotoxicity risk is highest with conventional amphotericin B.

Conventional amphotericin B (C-AMB) is not a lipid-based formulation of amphotericin B. Amphotericin B is insoluble in water, hence C-AMB is formulated by complexing with bile salt deoxycholate to make it suitable for intravenous infusion.

Nephrotoxicity risk with amphotericin B formulations in decreasing order:

- C-AMB > Amphotericin B lipid complex (ABLC) > Liposomal amphotericin B (L-AMB)
- All lipid formulations of amphotericin B (ABLC, L-AMB, ABCD) are less nephrotoxic than conventional amphotericin B

Note: Amphotericin B colloidal dispersion (ABCD) has not been compared with other lipid-based formulations

Solution to Question 4:

The given scenario is suggestive of invasive mucormycosis in the form of orbital cellulitis.

Intravenous administration of amphotericin B is the treatment of choice for invasive mucormycosis.

Mucormycosis is an opportunistic infection seen in patients with diabetes mellitus, leukaemia, lymphoma and other immunocompromised states. The most prevalent species is *Rhizopus Oryza*.

Solution to Question 5:

Development of fever and chills following intravenous administration is suggestive of infusion-related reactions which occur most frequently with amphotericin B colloidal dispersions.

Infusion-related adverse events are commonly seen with amphotericin B and it has been nicknamed 'shake and bake syndrome' due to the associated chills and fever.

Decreasing order of frequency of infusion-related reactions with amphotericin B formulations:

Amphotericin B colloidal dispersions (ABCD) > conventional amphotericin B (C-AMB) > amphotericin B lipid complex (ABLC) > liposomal amphotericin B (L-AMB).

Solution to Question 6:

Amphotericin B causes hypokalemia and hence potassium levels should be monitored in this patient.

Amphotericin B can cause renal tubular acidosis and renal wasting of K^+ and Mg^{2+} leading to hypokalemia and hypomagnesemia during and several weeks after therapy. Potassium supplements are required in around one-third of patients on prolonged amphotericin B therapy.

Solution to Question 7:

The given clinical scenario and the presence of encapsulated yeast cells in CSF with India ink staining are suggestive of cryptococcal meningitis. Amphotericin B plus flucytosine is the treatment of choice for this patient.

Cryptococcal meningitis is an AIDS-defining opportunistic fungal CNS infection that generally occurs in untreated patients, with $CD4^+$ T cell counts $<100/\mu L$.

In cryptococcal meningitis, amphotericin B combined with flucytosine is the current gold standard treatment in immunocompromised patients, especially in AIDS patients.

In non-immunocompetent patients, the role of additional flucytosine along with amphotericin B is less clear.

Oral fluconazole is used for consolidation and maintenance therapy after two weeks of induction therapy with amphotericin B.

Solution to Question 8:

Flucytosine acts by inhibition of fungal DNA synthesis.

It is a pyrimidine antimetabolite. After uptake into fungal cells, fungal cytosine deaminase activates it to 5-fluorouracil, which is an anticancer drug. This is further converted into metabolites that inhibit thymidylate synthase which is necessary for DNA synthesis. Thus fungal DNA synthesis is inhibited. Mammalian cells are spared as they lack cytosine deaminase enzyme.

Flucytosine has clinically useful activity against (3C):

- Cryptococcus
- Candida species
- Agents causing chromoblastomycosis

Solution to Question 9:

Itchy annular plaques in the groin are suggestive of tinea cruris and the drug most likely given was griseofulvin which acts by interfering with fungal mitosis.

Microtubule function is necessary for the formation of the mitotic spindle during cell division. Griseofulvin inhibits microtubule function and thereby disrupts mitotic spindle formation and inhibits cell division.

Tinea cruris is a dermatophytosis involving the groin and is also known as jock itch or dhobi's itch.

Solution to Question 10:

Griseofulvin is not useful in the treatment of tinea versicolor. It is treated with topical azoles.

Tinea capitis, tinea cruris and tinea pedis are dermatophytoses. Griseofulvin is indicated only for dermatophytoses caused by species of Microsporum, Epidermophyton, and Trichophyton. It is considered the drug of choice for tinea capitis in children. It is reserved only for large topical tinea infections when topical therapy is inadequate.

Tinea versicolor is not a dermatophytosis. It is a superficial mycosis caused by lipophilic yeast Malassezia.

Solution to Question 11:

Griseofulvin is a cytochrome enzyme inducer and can cause contraceptive failure when taken with oral contraceptive pills.

It induces hepatic CYP enzymes and can increase the metabolism of drugs. Warfarin metabolism is increased and the dose has to be adjusted according to the change in response. Estrogen metabolism is also increased thereby reducing the efficacy of low-dose estrogen-based oral contraceptives.

Note: Ketoconazole is a cytochrome enzyme inhibitor.

Solution to Question 12:

Fungal cell wall synthesis is altered by caspofungin.

1,3- β -D glucan is an essential component of the fungal cell wall which maintains cell integrity. Echinocandin group of antifungals like caspofungin, anidulafungin, and micafungin acts by inhibiting the synthesis of 1,3- β -D-glucan and thereby weakens the fungal cell wall and increases the osmotic susceptibility.

Echinocandins are fungicidal against *Candida* species. They show fungistatic activity against *Aspergillus* species.

Solution to Question 13:

Triazoles act by inhibition of ergosterol synthesis.

The azole group inhibits the fungal cytochrome P450 enzyme called 14- α -sterol demethylase and results in an impairment of conversion of lanosterol to ergosterol. This results in the depletion of membrane ergosterol.

Azole antifungals are of two types:

- Imidazoles - clotrimazole, ketoconazole, luliconazole
- Triazoles - itraconazole, fluconazole, voriconazole, posaconazole and isavuconazole

Solution to Question 14:

Fluconazole is the drug of choice for coccidioidal meningitis caused by *Coccidioides immitis* because of better CSF penetration.

Itraconazole is also effective in this condition but it is not the drug of choice. Amphotericin B is indicated in patients in whom triazoles have failed.

Note: Amphotericin B is the drug of choice for initial treatment of cryptococcal meningitis caused by *Cryptococcus neoformans* and fluconazole is used for maintenance therapy.

Solution to Question 15:

The image above depicts gynecomastia and can be caused by ketoconazole due to its antiandrogenic action.

Ketoconazole use results in various antiandrogenic effects. It decreases testosterone production and displaces testosterone from protein binding sites in plasma. It also decreases hydrocortisone production, especially at higher doses. Adverse effects that can occur are gynecomastia, loss of hair, libido, and oligozoospermia.

Fluconazole, itraconazole and voriconazole do not have antiandrogenic action.

Solution to Question 16:

Ketoconazole is not indicated for use in Addison's disease.

It is a potent nonselective inhibitor of adrenal and gonadal steroidogenesis by inhibiting 17 α -hydroxylase enzymes which are involved in steroid hormone synthesis and it can hence worsen Addison's disease. However, this property also makes it useful in the treatment of Cushing's syndrome.

Azoles interfere with leishmanial cell membrane biosynthesis and are therefore used in dermal leishmaniasis and kala-azar.

Solution to Question 17:

Transient visual or auditory hallucinations are commonly seen with the use of intravenous voriconazole. These usually occur at night.

Other side effects of voriconazole are:

- Hepatotoxicity
- Prolonged QTc interval and increased risk of torsades de pointes
- Anaphylactoid reaction when given intravenously

Solution to Question 18:

Voriconazole is a strong inhibitor and substrate of CYP2C19. Concomitant omeprazole administration can result in increased omeprazole levels because of inhibition of metabolism.

In a patient on omeprazole 40 mg/day, the dose of omeprazole should be reduced to half when voriconazole is initiated. Additionally, higher levels of omeprazole can also interfere in voriconazole metabolism as omeprazole also inhibits CYP2C19.

Solution to Question 19:

Salicylic acid is a keratolytic agent that can be prescribed to enhance the treatment outcome in patients with athlete's foot (i.e. tinea pedis).

Salicylic acid accelerates the desquamation of the stratum corneum. It is used in combination with a fungistatic drug called benzoic acid because eradication of infection is possible only if the infected stratum corneum is shed. This combination is called Whitfield's ointment and is used in the treatment of tinea pedis.

Solution to Question 20:

Pruritic annular skin lesions with central clearing are suggestive of dermatophytosis for which amphotericin B must not be prescribed. Azoles, terbinafine, and griseofulvin are indicated for the treatment of dermatophytosis.

Dermatophytes are inhibited by amphotericin B but drug concentrations attained in the skin are very low and ineffective for the treatment of dermatophytosis.

The uses of amphotericin B are:

- Treatment of choice in invasive mucormycosis
- Candida albicans infection
- Cryptococcus neoformans infection
- Endemic mycosis

Solution to Question 21:

Griseofulvin is not used to treat candida infection. It is active against most dermatophytes but not against candida and other fungi causing deep mycosis.

Ketoconazole, nystatin, and amphotericin B are used in the treatment of candida infections.

Solution to Question 22:

Ciclopirox olamine is a topical antifungal agent and is not effective in systemic mycoses.

Ciclopirox olamine is a broad spectrum fungicidal agent that acts by chelating trivalent metal ions which leads to inhibition of the metal-dependent enzyme that degrades free radicals. It is available as a nail lacquer formulation in the treatment of onychomycosis.

Option A: Amphotericin B is more effective parenterally for systemic mycoses. However, local or topical administration of amphotericin B drops is used in mycotic corneal ulcers.

Option B: The absorption of griseofulvin is less when administered orally but microfining the drug particles increases the absorption.

Option D: Fluconazole can be administered orally as well as intravenously.

Solution to Question 23:

Topical antifungal agents have no role in the treatment of subcutaneous mycoses like chromoblastomycosis. Itraconazole is the first-line antifungal indicated in this condition.

Topical antifungal agents are used in the treatment of:

- Dermatophytosis
- Candidiasis

- Piedra
- Tinea versicolor
- Tinea nigra
- Fungal keratitis

Solution to Question 24:

Tolnaftate, butenafine, naftifine, and clotrimazole are used topically.

Allylamine antifungals available for topical application are:

- Terbinafine
- Naftifine
- Butenafine

Azoles that are used topically are:

- Clotrimazole
- Miconazole
- Econazole

Tolnaftate is a topical synthetic antifungal agent. It distorts the mycelial growth of yeast-like fungi and molds. It is effective in the treatment of dermatomycosis and tinea.

Solution to Question 25:

The clinical features are suggestive of tinea unguium or onychomycosis. The drug of choice is oral terbinafine which acts by inhibition of ergosterol synthesis.

Terbinafine acts as a non-competitive inhibitor of squalene epoxidase, an early step enzyme in the synthesis of fungal ergosterol. This results in the accumulation of sterol squalene which is toxic to the organism.

The efficacy of terbinafine in onychomycosis is improved when combined with amorolfine 5% nail lacquer.

Solution to Question 26:

Slit-lamp examination showing a corneal ulcer with satellite lesions and a thick hypopyon are suggestive of mycotic keratitis (fungal corneal ulcer) while branched septate hyphae are seen with *Aspergillus* and *Fusarium*. Natamycin 5% suspension is the drug of choice in this condition.

Treatment of fungal corneal ulcer:

- Topical antifungals: Aspergillus and Fusarium infections are treated with natamycin, amphotericin B and voriconazole while Candida infections are treated with nystatin.
- Systemic antifungals: Ketoconazole or voriconazole may be needed if there is a severe corneal ulcer, perforated ulcer, or suspicion of endophthalmitis.

Solution to Question 27:

Voriconazole is not effective against mucormycosis.

Mucormycosis is treated by amphotericin B. Posaconazole can also be used. However, other azoles are not effective against this fungus.

Voriconazole is the drug of choice for the treatment of aspergillosis. It can also be used for Candida species (including glabrata and krusei), Scedosporium and Fusarium.

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First Line Drugs for Tuberculosis

Question 1:

A 35-year-old lady presents with pulmonary TB is started on anti-tubercular therapy. What is the rationale for administering multi-drug therapy to her?

- a) To delay development of resistance
- b) To reduce toxicity
- c) To broaden antimicrobial spectrum
- d) To prevent toxin release from the organism

Question 2:

Which is the most effective drug against extracellular mycobacteria?

- a) Isoniazid
- b) Rifampicin
- c) Pyrazinamide
- d) Ethambutol

Question 3:

A patient was diagnosed with pulmonary tuberculosis and was started on first-line antitubercular drugs. Which of the following drug acts by inhibiting the synthesis of the marked layer shown in the image?

The diagram illustrates the complex cell wall structure of *Mycobacterium tuberculosis*. It shows several layers: the outermost layer consists of mycolic acid chains (pink wavy lines) and lipoteichoic acids (purple wavy lines) embedded in a layer of arabinogalactan (green wavy lines). Below this is the peptidoglycan layer (orange wavy lines). The innermost layer is the cytoplasmic membrane (blue wavy lines). The diagram also shows the presence of capsular proteins and carbohydrates (pink spheres) and Lipoarabinomannan (LAM) (green spheres) on the surface. A red box labeled 'Text' is present in the center of the diagram.

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- ### Question 4:

- a) Failure of therapy
- b) Peripheral neuropathy
- c) Hepatotoxicity
- d) Allergic reactions

Question 5:

- Rifampicin
- Ethionamide
- Cycloserine

d) Ethambutol

Question 6:

A 4-month-old boy was brought to the ER with recurrent seizures. His mother informs you that he is receiving isoniazid prophylaxis as she was diagnosed with TB during pregnancy. ABG revealed a high anion gap metabolic acidosis. What would be the mainstay of treatment?

- a) Benzodiazepines**
- b) Barbiturates**
- c) Pyridoxine**
- d) Thiamine**

Question 7:

A 30-year-old man undergoing treatment for TB visits your OPD for follow-up. His wife states that recently he is forgetting the names of his colleagues and neighbors, but later recollects them correctly. Which drug is implicated to cause this condition?

- a) Ethionamide**
- b) Isoniazid**
- c) Ethambutol**
- d) Pyrazinamide**

Question 8:

A 66-year-old woman, on anti-tubercular therapy, presented with pain in the right hand and shoulder for the past 1 month. On physical examination, her wrist was swollen and her shoulder was extremely painful with a limited range of motion. Which drug is most likely associated with this condition?

- a) Levofloxacin**
- b) Isoniazid**
- c) Rifampin**
- d) Ethambutol**

Question 9:

Before initiating treatment for a sputum-positive TB patient, the physician orders for Cartridge Based Nucleic Acid Amplification Test in order to test for drug resistance. Which anti-tubercular drug resistance is he checking for?

- a) Isoniazid
- b) Rifampicin
- c) Pyrazinamide
- d) Ethambutol

Question 10:

Which of the following rifamycin drugs would you advise a patient to take on an empty stomach?

- a) Rifabutin and rifapentine
- b) Rifabutin and rifampicin
- c) Rifampicin only
- d) Rifapentine only

Question 11:

An HIV-positive patient on ART with zidovudine, lamivudine, and indinavir develops pulmonary tuberculosis. Which of the following anti-tubercular drugs should be avoided in him?

- a) Isoniazid
- b) Rifampicin
- c) Streptomycin
- d) Ethambutol

Question 12:

Which of the following rifamycins is not used for mycobacterial infections?

- a) Rifampicin
- b) Rifaximin
- c) Rifapentine
- d) Rifabutin

Question 13:

A Gene Xpert test was conducted on the sputum sample of a patient with suspected TB. The results were positive and showed resistance to rifampicin. What is the primary mechanism of resistance against this drug?

- a) Alteration in target enzyme
- b) Induction of efflux pumps
- c) Alteration in activation pathway
- d) Drug inactivating enzymes

Question 14:

In which of the following conditions can rifampicin be used?

- a) 1, 3 and 5
- b) 2 and 4 only
- c) 1, 3, 4 and 5
- d) 1, 2, 3 and 5

Question 15:

You are prescribing first-line anti-tubercular therapy to a patient newly diagnosed with TB. Which of the following is incorrect about his therapy?

- a) Red green color impairment is an early sign of ethambutol induced optic neuritis
- b) Ethambutol accumulates in renal failure
- c) Flu-like syndrome is usually seen if rifampicin is taken on a daily basis
- d) Hyperuricemia is a recognised side effect of pyrazinamide

Question 16:

A young woman undergoing treatment for tuberculosis visits your OPD for a routine checkup. She informs you that recently she has noticed an orange-colored discoloration of her contact lenses. Which of the following drugs is most likely to be responsible for this?

- a) Isoniazid

- b) Pyrazinamide
- c) Ethambutol
- d) Rifampicin

Question 17:

You are planning to start treatment in a middle-aged man diagnosed with pulmonary tuberculosis. He has a history of chronic kidney disease and his current creatinine clearance is 25 mL/min. Which of the following drugs is least likely to require dose modification?

- a) Ethambutol
- b) Streptomycin
- c) Rifampicin
- d) Isoniazid

Question 18:

Pseudojaundice is an adverse effect of which of the following drugs?

- a) Phenytoin
- b) Rifabutin
- c) Pyrazinamide
- d) Isoniazid

Question 19:

Mycobacterium tuberculosis nicotinamidase is involved in the activation of which of the following antitubercular drug?

- a) Isoniazid
- b) Ethambutol
- c) Pyrazinamide
- d) Cycloserine

Question 20:

A 47-year-old man on treatment for TB presents with complaints of severe pain and swelling in his left great toe since today morning, as shown in the image below. Which of the following drugs is most likely to cause this?



- a) 1 and 2
- b) 2, 3 and 4
- c) 2 and 4
- d) 1, 3 and 5

Question 21:

A 33-year-old woman receiving treatment for TB presents to your OPD with complaints of blurry vision and pain on movement of the left eye for the past 2 days. Examination showed Marcus Gunn pupil. Which anti-tubercular drug is most likely associated with this condition?

- a) Ethambutol
- b) Isoniazid
- c) Ethionamide
- d) Cycloserine

Answer Key

Question No.	Correct Option
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1	a
2	b
3	a
4	b
5	b
6	c
7	b
8	b
9	b
10	c
11	b
12	b
13	a
14	d
15	c
16	d
17	c
18	b
19	c
20	c
21	a

Detailed Explanations

Solution to Question 1:

The reason for administering multi-drug therapy in tuberculosis is to prevent the development of resistance.

Resistance to individual drugs in *Mycobacterium tuberculosis* occurs by spontaneous point mutations that occur at a low but predictable rate (10^{-7} to 10^{-10}). Development of this drug resistance is almost invariably the result of monotherapy, i.e., the failure of the health care provider to prescribe at least 2 drugs to which the tubercle bacilli are susceptible. Hence, multi-drug therapy is recommended for the treatment of tuberculosis.

Solution to Question 2:

The most effective drug against extracellular mycobacteria is Rifampicin.

Option A: Isoniazid and Rifampicin act against both intracellular and extracellular mycobacterium. But, Rifampicin inhibits the extracellular mycobacteria at much lower drug concentrations than isoniazid.

Option B: Pyrazinamide acts only against intracellular mycobacterium.

Option C: Ethambutol acts against only extracellular mycobacteria but is not as effective as rifampicin. It is a bacteriostatic drug. Fast-multiplying bacteria are highly susceptible to its action.

Note: Streptomycin is primarily active against only the extracellular tubercle bacilli as its penetration into the cells is very poor.

Solution to Question 3:

Isoniazid is the first-line antitubercular drug that acts by inhibiting the synthesis of mycolic acid.

Isoniazid is a first-line bactericidal drug that is used for the treatment of tuberculosis. This drug enters bacilli by passive diffusion. It then gets activated to its toxic form by the enzyme catalase-peroxidase (the enzyme encoded by the Kat gene).

Normally, InhA and KasA are involved in the synthesis of mycolic acid. Upon activation, the reactive metabolite of isoniazid forms complexes with NAD that inhibit InhA and KasA. Hence, isoniazid inhibits the production of mycolic acid.

The reactive isoniazid metabolite also forms complexes with NADP. This complex inhibits mycobacterial dihydrofolate reductase which results in the interruption of DNA synthesis.

Side effects of isoniazid include:

- Peripheral neuritis due to interference with the production of active coenzyme pyridoxal phosphate from pyridoxine. Therefore, pyridoxine 10mg/day (vitamin B6) is given prophylactically along with isoniazid to prevent neurotoxicity.
- Hepatitis - isoniazid must be stopped at the first sign of hepatotoxicity.
- Rashes
- Anaemia
- Arthralgia

Toxicity of isoniazid results in metabolic acidosis, seizures, and coma.

Other options:

Option B: Rifampin acts by binding to DNA-dependent RNA polymerase to form a stable drug-enzyme complex. This complex suppresses RNA synthesis. Side effects of rifampin are hepatitis, flu-like illness, orange-red colored urine, and body secretions.

Option C: Pyrazinamide is a prodrug, it is converted into an active metabolite pyrazinoic acid which makes the intracellular environment acidic and collapses the transmembrane proton motive force, thereby killing the bacteria. Side effects include hyperuricemia and hepatotoxicity.

Option D: Ethambutol inhibits arabinose transferase III, thereby decreasing arabinogalactan biosynthesis. This results in disruption in mycobacterial cell wall synthesis. Optic neuritis is an

important side effect of ethambutol. It results in decreased visual acuity and loss of red-green discrimination.

Solution to Question 4:

Isoniazid, a key agent in the ATT, is metabolized primarily through N-acetylation by the genetically polymorphic N-acetyltransferase-2 (NAT2) enzyme. Slow acetylators (group A) of isoniazid are prone to develop peripheral neuropathy.

The rate of metabolism of isoniazid is determined by the genetic polymorphism of NAT2 enzyme and classified broadly into two groups - slow and fast acetylators.

In slow acetylators, the drug accumulates due to slow metabolism resulting in an increased risk of side effects, and hence they should undergo periodic monitoring. Peripheral neuropathy is a more frequent side effect of INH when compared to hepatotoxicity.

Fast acetylators metabolize the drug rapidly and show less response to standard doses and the therapy may be less effective.

Solution to Question 5:

Over-expression of the gene for *InhA* leads to cross-resistance of isoniazid (INH) with ethionamide.

Mechanisms of INH resistance	
Mutation of KatG	most common mechanism confers a high level of resistance
Overexpression of <i>InhA</i>	confers low level of resistance cross-resistance with ethionamide
Overexpression of <i>ahpC</i>	increases the bacterial survival under oxidative stress
Induction of efflux pumps	cross-resistance with ethambutol

Solution to Question 6:

The given clinical scenario is suggestive of isoniazid overdose for which the mainstay of treatment is intravenous pyridoxine.

Isoniazid overdose results in decreased pyridoxine availability to neurons. This decreases the levels of GABA, which in turn lowers the seizure threshold. Benzodiazepines can be used to

control seizures.

Prolonged treatment with isoniazid leads to the deficiency of pyridoxine which can lead to peripheral neuropathy and other neurological manifestations like ataxia, paresthesia, and numbness.

Clinical triad of isoniazid overdose:

- Seizures refractory to treatment with phenytoin and barbiturates
- High anion gap metabolic acidosis (HAGMA) resistant to treatment with sodium bicarbonate
- Coma

Solution to Question 7:

Isoniazid is the antitubercular drug that is implicated in the causation of transient memory loss.

Neurological toxicities like peripheral neuropathy, optic neuritis, and seizures can occur with isoniazid therapy. It is also associated with abnormalities like:

- Euphoria
- Transient memory loss
- Separation of ideas and reality
- Loss of self-control
- Florid psychosis

Pyridoxine therapy can prevent and reverse most of the neurological toxicity and mental abnormalities.

Solution to Question 8:

The given clinical presentation is suggestive of shoulder-hand syndrome which is associated with the use of isoniazid.

This syndrome is characterized by painful arthropathy in the upper extremity joints. It can also present as generalized arthropathy.

Other drugs that can cause shoulder-hand syndrome are phenobarbitone and phenytoin.

Solution to Question 9:

The Cartridge Based-Nucleic Acid Amplification Test (CBNAAT) is a molecular test for tuberculosis that detects the DNA of tubercle bacilli within 2 hours and mutations in the *rpoB* gene responsible for rifampicin resistance.

Solution to Question 10:

Rifampicin should be advised to be taken on an empty stomach because food decreases its absorption.

Rifabutin absorption is not altered by food.

Rifapentine absorption is enhanced by fatty food and hence it is recommended to be taken with food.

Solution to Question 11:

Rifampicin should be avoided as it is a potent inducer of CYP enzymes including CYPs 1A2, 3A4, 2C9, and 2C19, which in turn can induce the metabolism of indinavir and results in therapeutic failure of ART.

Rifabutin causes the least induction of cytochrome enzymes and is therefore preferred in HIV patients who are on protease inhibitors/NNRTIs.

Cytochrome enzyme-inducing potency of rifamycins is: rifampicin > rifapentine > rifabutin.

Note:

Cytochrome p-450 inducers	Cytochrome p-450 inhibitors
Barbiturates Carbamazepine Rifampin Phenytoin Aminoglutethimide Cigarette smoking Chronic alcoholism	Macrolide antibiotics Azole antifungals Chloramphenicol Omeprazole Selective serotonin reuptake inhibitors (SSRIs) HIV protease inhibitors Cimetidine Ciprofloxacin Metronidazole

Solution to Question 12:

Rifaximin is not used for mycobacterial infections. It is a synthetic rifamycin that is poorly absorbed from the GIT and is indicated in the management of GI conditions like:

- Traveler's diarrhea
- Irritable bowel syndrome with diarrhea
- Hepatic encephalopathy

Rifamycins indicated in mycobacterial infections are rifampicin, rifapentine, and rifabutin.

Solution to Question 13:

Rifampicin resistance is primarily due to alteration in the target enzyme.

Rifampicin acts by binding to the β subunit of DNA-dependent RNA polymerase (rpoB) and inhibits chain formation in RNA synthesis. Rifampicin is a potent inducer of cytochrome enzymes CYP3A4, CYP1A2, CYP2C9, and CYP2C19. Alteration of the target enzyme (rpoB) by a mutation in the gene coding for rpoB is the primary mechanism of resistance to rifampicin. Gene Xpert identifies mutations in the rpoB gene to detect rifampicin resistance.

Efflux pump induction has also been demonstrated as a mechanism of rifampicin resistance.

Solution to Question 14:

Besides its anti-tubercular action, rifampicin also has certain non-tubercular uses which include:

- Prophylaxis of meningococcal and H. influenzae meningitis
- Treatment of MRSA, diphtheroid, and legionella infections
- Eradication of staphylococcal nasal carrier state in patients with chronic furunculosis
- A combination of doxycycline and rifampicin is used as the first-line therapy for brucellosis
- A part of the WHO regimen for standard therapy of both paucibacillary and multibacillary leprosy

Rifampicin is not used in the treatment of Nocardia species. Cotrimoxazole is considered the drug of choice for pleuropulmonary nocardiosis.

Solution to Question 15:

Flu-like syndrome is usually seen when rifampicin is administered less than twice weekly (high dose) and it is uncommon when administered daily (lower dose).

Flu-like syndrome, characterized by fevers, chills, malaise, and arthralgia is a known dose-dependent adverse effect of rifampicin therapy and tends to occur late into the patient's treatment course. It resolves with either discontinuation of rifampin or switching to a daily-dose regimen. It can often be confused with occult infections or treatment failure.

Options A and B: Red-green colour impairment is one of the early signs of ethambutol-induced optic neuritis. Ethambutol is primarily excreted unchanged by kidneys and accumulation occurs in renal failure. Ethambutol does not undergo hepatic metabolism and can be safely prescribed for the treatment of TB in patients with hepatitis.

Option D: Pyrazinamide interferes with urate excretion and can result in hyperuricemia.

Solution to Question 16:

Rifampicin is the drug most likely to cause orange-colored discoloration of contact lenses in this case.

Drugs like rifampicin and rifabutin are known to cause reddish-orange discoloration of the skin, urine, feces, saliva, sweat, and tears (which can stain the patient's contact lenses).

Solution to Question 17:

Rifampicin is the least likely drug to require dose modification in renal failure patients as it is metabolized in the liver and primarily excreted in bile.

In patients with renal failure, the half-life of isoniazid will be prolonged, but its adverse effects are not significantly increased. Hence dose modification is not usually done for isoniazid in clinical practice as a reduction in dose may render the regimen ineffective and increase the chances of resistance.

Ethambutol and streptomycin are primarily excreted unchanged through the renal system and need dose modification in renal failure.

Solution to Question 18:

Pseudojaundice is an adverse effect of rifabutin. It resolves on discontinuation of the drug.

Unique adverse effects of rifabutin are:

- Polymyalgia/arthritis
- Pseudojaundice
- Anterior uveitis

Solution to Question 19:

Mycobacterium tuberculosis nicotinamidase is involved in the activation of pyrazinamide.

Nicotinamidase which is produced by M. tuberculosis is also called pyrazinamidase. It is encoded by the pncA gene and its mutations are associated with pyrazinamide resistance. It converts pyrazinamide to active pyrazinoic acid.

Pyrazinoic acid has the following actions:

- Inhibition of fatty acid synthetase type 1 leads to interference in mycolic acid synthesis
- Reduction of intracellular pH
- Disruption of membrane transport

Solution to Question 20:

The given clinical history and findings of first metatarsophalangeal joint swelling are suggestive of an acute episode of gout. Antitubercular drugs that are most likely to cause hyperuricemia and precipitate gout are pyrazinamide and ethambutol.

Both pyrazinamide and ethambutol inhibit uric acid excretion in the kidney causing hyperuricemia.

Solution to Question 21:

The given clinical features along with Marcus Gunn pupil are suggestive of optic neuritis, which is a dose-dependent adverse effect associated with ethambutol.

Optic neuritis resulting in decreased visual acuity and loss of red-green discrimination is the most serious adverse effect of ethambutol. It can be unilateral or bilateral and usually recovers when the drug is stopped.

Tests of visual acuity and red-green discrimination before the start of therapy and periodically thereafter are recommended.

Second Line Drugs for Tuberculosis

Question 1:

Which of the following drugs is not hepatotoxic?

- a) Pyrazinamide
- b) Isoniazid
- c) Cycloserine
- d) Ethionamide

Question 2:

A 24-year-old woman who was started on treatment for MDR-TB 6 months back, presented with complaints of fatigue, amenorrhea, and constipation since the past 1 month. On examination, a swelling was noticed on the anterior aspect of her neck. Her lab reports show raised TSH levels with low free T₃ and T₄ levels. Which anti-tubercular drug is likely to cause this adverse effect?

- a) Streptomycin
- b) Pyrazinamide
- c) Cycloserine
- d) Ethionamide

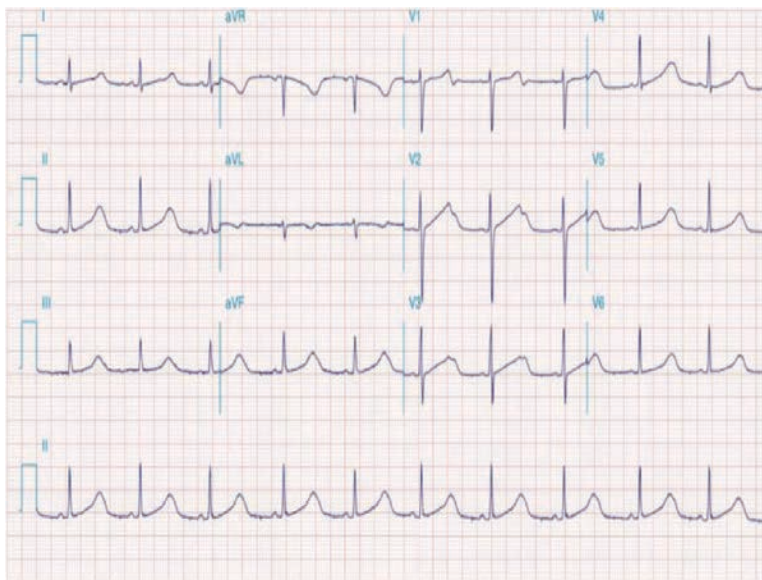
Question 3:

What is the mechanism of anti-mycobacterial action of bedaquiline?

- a) Inhibition of mycolic acid synthesis
- b) Inhibition of protein synthesis
- c) Inhibition of ATP synthase
- d) Inhibition of RNA synthesis

Question 4:

A 68-year-old man with hypertension was diagnosed with drug-resistant pulmonary TB. One week after starting ATT he presented to the casualty following an episode of syncope. ECG showed a corrected QT interval of 477ms, which is shown below. Which drug used in his anti-tubercular therapy is responsible for this adverse effect?



- a) Clofazimine
- b) Isoniazid
- c) Bedaquiline
- d) Para-amino salicylic acid

Question 5:

What is the mechanism of action of delamanid?

- a) Inhibition of ATP synthase
- b) Inhibition of mycolic acid synthesis
- c) Inhibition of DNA gyrase
- d) Inhibition of alanine racemase

Question 6:

A 30-year-old pregnant woman was diagnosed with tuberculosis. Which of the following antitubercular drugs should not be used in her?

- a) Isoniazid

- b) Streptomycin
- c) Rifampicin
- d) Ethambutol

Question 7:

An HIV-positive patient with a CD4 count of 40/mm³ requires prophylaxis against *M. avium* complex bacteremia. Which of the following drugs would you prefer?

- a) Azithromycin
- b) Dapsone
- c) Isoniazid
- d) Rifabutin

Question 8:

An elderly woman on ART with a CD4 count of 35/ μ L and a high viral load presents with a 2 week history of cough, fever, and night sweats. She was diagnosed with pulmonary *M. avium* complex infection. Which of the following is not included in the triple-drug therapy for the management of her condition?

- a) Rifabutin
- b) Isoniazid
- c) Ethambutol
- d) Clarithromycin

Question 9:

A patient undergoing treatment for TB developed resistance to rifampicin and isoniazid after 3 months. He can be treated with all the following drugs except _____.

- a) Tobramycin
- b) Amikacin
- c) Moxifloxacin
- d) Ofloxacin

Question 10:

Which of the following beta lactam class has shown activity against MDR-TB?

- a) Aminopenicillins
- b) Monobactams
- c) Carboxypenicillins
- d) Carbapenems

Answer Key

Question No.	Correct Option
1	c
2	d
3	c
4	c
5	b
6	b
7	a
8	b
9	a
10	d

Detailed Explanations

Solution to Question 1:

Cycloserine is not a hepatotoxic drug. The adverse effects of cycloserine are mainly neurological.

Half of the patients on cycloserine (1g/day) experience neuropsychiatric symptoms. It ranges from headaches to psychosis, seizures, and suicidal ideas. Therefore, cycloserine is contraindicated in patients with epilepsy.

Ethionamide, rifampicin, isoniazid and pyrazinamide are hepatotoxic agents. They are either contraindicated or used with extreme caution in hepatitis.

Solution to Question 2:

The given clinical features and lab reports are suggestive of hypothyroidism with goiter, which is a known adverse effect of ethionamide.

MDR-TB is treated with 2nd line agents that include ethionamide (ETH). ETH is not a first-line agent due to its gastrointestinal side effects. It is structurally related to thionamides (propylthiouracil and methimazole) and therefore causes hypothyroidism by preventing organification of iodine in thyroid hormone synthesis. Another antitubercular drug that can cause hypothyroidism is Para-amino salicylic acid (PAS).

ETH/PAS-induced hypothyroidism should be treated with thyroxine replacement therapy, instead of changing the regimen. Thyroid function usually optimizes within 2 months after completing MDR therapy.

Solution to Question 3:

The mechanism of anti-mycobacterial action of bedaquiline is by inhibition of ATP synthase.

Bedaquiline is a diarylquinoline that targets the subunit c of the ATP synthase of *M. tuberculosis*. This leads to inhibition of the proton pump activity of the ATP synthase and causes a failure in the bacillary energy metabolism.

Solution to Question 4:

The given clinical scenario along with a corrected QT (QTc) interval of 477ms is suggestive of QTc prolongation which is an adverse effect associated with bedaquiline.

In the ECG QTc (corrected QT interval = $QT/\sqrt{RR}$) is 477ms. Normal QTc is about 440ms. Values of more than 440ms in males and 460ms in females are considered to be prolonged. QTc prolongation usually presents as syncope or cardiac arrest.

Although QT prolongation can cause Torsades de pointes, no cases have been reported with bedaquiline. A 5-fold higher rate of death has been reported in patients on bedaquiline versus those on placebo. Therefore, the World Health Organization (WHO) advises caution when administering this medication. Bedaquiline is contraindicated in the age group of <18 years.

Solution to Question 5:

Delamanid acts by inhibiting mycolic acid synthesis.

Pretomanid and delamanid are bicyclic nitroimidazoles that act by inhibiting mycolic acid (a component of the mycobacterial cell wall) and protein synthesis in the mycobacteria.

Pretomanid in combination with bedaquiline and linezolid was approved by FDA in 2019 for the treatment of adult patients with extensively drug-resistant (XDR) TB and for patients who are treatment-intolerant or nonresponsive to multidrug-resistant pulmonary (MDR) TB treatment.

In India, delamanid is only approved for the management of MDR-TB.

Isoniazid also acts by inhibiting mycolic acid synthesis.

Other options:

Option A: Bedaquiline acts by inhibition of ATP synthase. It is a diarylquinoline that targets the subunit c of the ATP synthase of *M. tuberculosis*. This leads to inhibition of the proton pump activity of the ATP synthase and causes a failure in the bacillary energy metabolism.

Option C: Fluoroquinolones act by inhibition of DNA gyrase. They are used as second-line agents in the treatment of tuberculosis.

Option D: Cycloserine is a D-alanine analogue used as a second-line drug for MDR-TB that acts by inhibition of alanine racemase. Alanine racemase converts L-alanine to D-alanine which is needed for the mycobacterial cell wall.

Solution to Question 6:

Streptomycin should not be used for the treatment of TB in a pregnant patient as it can cause hearing loss in the newborn.

First-line treatment for tuberculosis in pregnancy is done with isoniazid (along with pyridoxine), rifampicin, ethambutol, and pyrazinamide.

Solution to Question 7:

Azithromycin 1200 mg given once a week, is preferred for prophylactic therapy against bacteremia caused by *Mycobacterium avium* complex (MAC) in people living with AIDS.

Clarithromycin 500 mg twice daily, may also be administered for prophylaxis. Rifabutin 300 mg once daily is used for patients who are intolerant to macrolides.

Prophylaxis is indicated when CD4 counts are less than 50/mm³ and should be discontinued once the CD4 count is more than 100/mm³.

Dapsone is used as an alternative agent for prophylaxis of *pneumocystis jirovecii* infection, whereas isoniazid is used for prophylaxis of *Mycobacterium tuberculosis* infection.

Solution to Question 8:

Triple drug therapy for pulmonary *Mycobacterium avium* complex (MAC) infections does not include isoniazid.

Treatment for pulmonary MAC infection is triple-drug therapy with:

- A rifamycin, like rifampicin or rifabutin (In HIV patients on ART, rifabutin is preferred)
- Ethambutol

- A macrolide, like clarithromycin or azithromycin

MAC infection is an AIDS-defining opportunistic infection, occurring predominantly at CD4+ T cell counts of $<50/\mu\text{L}$ and high viral load. The average CD4+ T cell count at the time of diagnosis is $10/\mu\text{L}$. The most common presentation is disseminated disease with fever, weight loss, and night sweats.

Solution to Question 9:

Tobramycin cannot be used in the treatment of MDR-TB (rifampicin and isoniazid resistant) as it is not considered a second-line antitubercular drug.

Injectable aminoglycosides like amikacin, kanamycin, capreomycin, and fluoroquinolones like moxifloxacin and ofloxacin are second-line drugs used for treating MDR-TB.

Solution to Question 10:

Carbapenems have shown activity against *Mycobacterium tuberculosis*.

Note: Until recently beta-lactams were assumed to lack action against *Mycobacterium tuberculosis*. However, carbapenems in conjunction with clavulanate have demonstrated efficacy against *Mycobacterium tuberculosis*.

Anti-leprosy drugs

Question 1:

What is the mechanism of action of dapsone, considering that it is used in the treatment of leprosy, fungal infections and, also for immunomodulatory actions?

- a) Cell wall synthesis inhibitor
- b) Folic acid synthesis inhibitor
- c) Protein synthesis inhibitor
- d) Cell membrane inhibitor

Question 2:

Dapsone would not be helpful in which of the following patients?

- a) HIV positive patient with $CD4 < 100$ and IgG positive for Toxoplasma
- b) An elderly man with pemphigoid
- c) HIV positive patient with *Pneumocystis jirovecii* pneumonia
- d) A child with schistosomiasis

Question 3:

What is the treatment of choice for the patient's condition depicted below?



- a) Rifampicin + dapsone + cycloserine, for 12 months
- b) Rifampicin + dapsone + clofazimine, for 6 months
- c) Rifampicin + ofloxacin + minocycline, for 6 months
- d) Rifampicin + dapsone + clofazimine, for 12 months

Question 4:

All of the following are administered under supervision in India except?

- a) Clofazimine
- b) Pyrazinamide
- c) Rifampicin
- d) Dapsone

Question 5:

A patient presented with multiple hypopigmented skin lesions and loss of sensation over the lesions. What is the most bactericidal drug for the condition?

- a) Clofazimine
- b) Dapsone
- c) Rifampicin
- d) Ethionamide

Question 6:

What is the most bactericidal anti leprotic drug amongst the given drugs?

- a) Amoxicillin
- b) Ciprofloxacin
- c) Ofloxacin
- d) Erythromycin

Question 7:

Thalidomide is useful in the treatment of which of the following conditions?

- a) Downgrading reactions
- b) Reversal reactions
- c) Erythema nodosum leprosum
- d) All of the above

Question 8:

A patient with borderline leprosy develops severe type 1 lepra reaction during the course of his multidrug therapy. What would be the most appropriate treatment for him?

- a) Glucocorticoids
- b) Clofazimine
- c) Thalidomide
- d) Plasmapheresis

Question 9:

A patient with leprosy, who is on multidrug therapy, presents with painful raised lesions on his forehead as shown in the image. Which of the following must be avoided in his condition?



- a) Clofazimine
- b) Corticosteroids
- c) Thalidomide
- d) Rifampicin

Question 10:

A patient presents with complaints of reddish-black discolouration of skin and body secretions. Which anti-mycobacterial agent can cause this condition?

- a) Clofazimine
- b) Cycloserine
- c) Capreomycin
- d) Ciprofloxacin

Question 11:

Rifampin is not active against which of the following Mycobacterial species?

- a) Mycobacterium avium complex
- b) Mycobacterium leprae
- c) Mycobacterium abscessus
- d) Mycobacterium kansasii

Answer Key

Question No.	Correct Option
1	b
2	d
3	d
4	d
5	c
6	c
7	c
8	a
9	d
10	a
11	c

Detailed Explanations

Solution to Question 1:

The mechanism of action of dapsone is inhibition of folic acid synthesis.

Dapsone [4,4'-diaminodiphenylsulphone (DDS)] is a structural analog of para-aminobenzoic acid (PABA) and a competitive inhibitor of dihydropteroate synthase in the folate pathway. It is weakly bactericidal and can cause mild hemolysis, and rarely anemia or psychosis.

DDS syndrome is a rare side effect of dapsone therapy in leprosy patients which starts after 5-6 weeks of starting dapsone and leads to exfoliative dermatitis, hepatosplenomegaly, lymphadenopathy, fever and hepatitis and may be fatal. It is also called 'five-week dermatitis'.

Solution to Question 2:

Dapsone is not used in the treatment of schistosomiasis. Praziquantel is the drug recommended for this condition.

Dapsone with pyrimethamine and leucovorin can be given for prophylaxis of Toxoplasma encephalitis in HIV patients with CD4 < 100. Along with trimethoprim, it can be given for prophylaxis and treatment of Pneumocystis pneumonia. However, cotrimoxazole remains the drug of choice in both conditions.

The uses of dapsone are:

Infections

- Leprosy
- Pneumocystis jirovecii infections (prophylaxis and treatment)
- Malaria
- Toxoplasma gondi infections (prophylaxis)

Inflammatory conditions

- Dermatitis herpetiformis
- Pemphigoid
- Relapsing chondritis
- Linear IgA bullous disease

Solution to Question 3:

The given image of a man with nodular lesions on his face depicts lepromatous leprosy and is treated by standard multidrug regimen (MDR-WHO recommended) for 12 months.

It consists of:

- Rifampicin: 600 mg once a month
- Dapsone: 100 mg daily
- Clofazimine: 300 mg once a month + 50 mg once daily

Solution to Question 4:

Dapsone is administered daily in both multibacillary and paucibacillary leprosy without supervision.

Clofazimine is indicated as a once-monthly supervised dose (300 mg) and daily unsupervised doses (50 mg) in multibacillary leprosy and paucibacillary leprosy.

Pyrazinamide is an ATT drug which is administered under direct supervision.

Rifampicin is indicated as a once-monthly supervised dose for both multibacillary and paucibacillary leprosy.

Solution to Question 5:

The given clinical scenario is suggestive of leprosy and rifampicin is the most bactericidal anti-leprotic drug.

Clofazimine and dapsone are bacteriostatic agents, while ethionamide is not indicated in leprosy.

Solution to Question 6:

Ofloxacin is the most bactericidal anti leprotic drug amongst the given drugs.

It exhibits high activity against *M. leprae*, and it is being used in alternative multidrug therapy (MDT) regimens in rifampicin-resistant cases. As a component of MDT, it has been found to hasten the bacteriological and clinical response.

Among the macrolides, clarithromycin has activity against leprosy bacilli and is used for alternative MDT in cases resistant to both rifampicin and fluoroquinolones. Azithromycin shows activity against *Mycobacterium avium* complex but not lepra bacilli.

Note: Rifampicin is the most potent bactericidal anti leprotic drug.

Solution to Question 7:

Thalidomide is useful in the treatment of erythema nodosum leprosum (ENL) or type 2 lepra reaction. It is indicated in recurring and persistent ENL lesions despite the completion of at least two courses of glucocorticoid therapy.

Thalidomide is not effective in type 1 lepra reaction. If type 1 reactions occur after initiation of antimycobacterial therapy, they are termed reversal reactions and it is histologically more tuberculoid. When they occur before the initiation, they are termed downgrading reactions and it is histologically more lepromatous.

Solution to Question 8:

The most appropriate therapy for severe type 1 lepra reactions is glucocorticoids.

Clofazimine, though effective in type 1 lepra reaction, is less efficacious than glucocorticoids.

Thalidomide and plasmapheresis are not indicated in the treatment of type 1 lepra reaction.

Solution to Question 9:

The presence of tender, erythematous, subcutaneous nodules as seen in the image is suggestive of Type II lepra reaction in which rifampicin must be avoided.

Rifampicin is highly bactericidal to *M. leprae* and results in high kill rates and massive release of bacterial antigens and may aggravate type 2 lepra reactions (also called erythema nodosum leprosum).

Options A and B: Clofazimine and corticosteroids are indicated in both type 1 and type 2 lepra reactions.

Option C: Thalidomide is indicated in type 2 lepra reactions.

Solution to Question 10:

Reddish-black discolouration of skin and body secretions is observed with the use of clofazimine.

Clofazimine has good penetration into many tissues and results in reddish-black discolouration of skin and body secretions. This may take a long time to resolve.

Solution to Question 11:

Rifampin is not active against *Mycobacterium abscessus*. It is innately resistant to rifampicin.

First-line therapy for *Mycobacterium abscessus* includes cefoxitin (or imipenem) + amikacin + clarithromycin.

Rifampicin is effective against *Mycobacterium avium* complex, *Mycobacterium leprae*, and *Mycobacterium kansasii*. (Options A, B, D)

Anti-virals (Non-retroviral)

Question 1:

A 54-year-old man was brought to the casualty with fever, headache, focal seizures, and right hemiparesis. MRI brain showed hyperintensity in the left frontotemporal region. PCR of his CSF sample was positive for HSV DNA. What would be the drug of choice for this condition?

- a) Valacyclovir
- b) Acyclovir
- c) Gancyclovir
- d) Foscarnet

Question 2:

A patient diagnosed with shingles was prescribed acyclovir. Which of the following statements is false regarding this drug?

- a) Acyclovir is a nucleoside congener
- b) Nephrotoxicity is a dose limiting side effect of acyclovir
- c) Valacyclovir has a higher bioavailability than acyclovir
- d) Acyclovir does not enter the CSF

Question 3:

What is the most common cause of resistance to acyclovir in HSV-1?

- a) Induction of efflux pumps
- b) Mutation of viral DNA polymerase
- c) Deficient viral thymidine kinase activity
- d) Expression of inactivating enzymes

Question 4:

In which of the following patients can valacyclovir be used as the drug of choice?

- a) An elderly woman with avian influenza

- b) An infant with RSV bronchiolitis
- c) An adolescent with chickenpox
- d) An immunocompromised patient with CMV retinitis

Question 5:

A patient showed persistence of genital herpes lesions even after 7 days of high-dose oral acyclovir therapy. Which of the following antivirals can you use to manage this case?

- a) Ganciclovir
- b) Abacavir
- c) Foscarnet
- d) Valacyclovir

Question 6:

Which of the following is a true statement about amantadine?

- a) The most common side effect is CNS toxicity
- b) It is effective only against Influenza B
- c) It is routinely used for the treatment of influenza
- d) It is largely metabolised by the liver

Question 7:

A young woman with a recent travel history presented with fever and respiratory difficulty. She was diagnosed with Influenza A and was started on oseltamivir. What is the mechanism of action of this drug?

- a) DNA polymerase inhibition
- b) Protein synthesis inhibition
- c) Nucleoside analogue
- d) Neuraminidase inhibition

Question 8:

Zanamivir is to be given for a patient with an influenza B infection. How will you administer this drug?

- a) Oral route
- b) Inhalational route
- c) Intravenous route
- d) Intramuscular route

Question 9:

A 22-year-old man is admitted with acute H1N1 influenza pneumonia. He is having severe nausea and vomiting and is not able to take any oral medications. Which of the following drugs can be administered intravenously to him?

- a) Oseltamivir
- b) Zanamivir
- c) Peramivir
- d) Entecavir

Question 10:

A 40-year-old man, who has history of contact with a known case of influenza, is having fever, cough, and runny nose since today morning. Rapid antigen test turns out to be positive. The endonuclease inhibitor that can be used for his treatment is _____.

- a) Oseltamivir
- b) Baloxavir
- c) Zanamivir
- d) Favipravir

Question 11:

Which of the following drugs is approved for prophylaxis against respiratory syncytial virus (RSV) infection?

- a) Omalizumab
- b) Palivizumab
- c) Denosumab

d) Ribavirin

Question 12:

A hepatitis B infected patient awaiting liver transplantation is admitted with ascites and encephalopathy. Which of the following is the drug of choice for him?

- a) Interferon alpha
- b) Lamivudine
- c) Tenofovir
- d) Telbivudine

Question 13:

Sofosbuvir is indicated in the treatment of which of the following patients?

- a) 1 and 2 only
- b) 1 and 3 only
- c) 2 and 5 only
- d) 2, 3 and 4 only

Question 14:

Which of the following is not a direct-acting agent (DAAs) used in the treatment of HCV infections?

- a) NS5B polymerase inhibitors
- b) NS5A inhibitors
- c) NS3/4A protease inhibitors
- d) NS4B inhibitors

Question 15:

A 52-year-old woman with history of hepatitis B infection presents with fatigue, vomiting, and weight loss. Her serological markers are given in the table below. What is the treatment of choice for her condition?

- a) Paritaprevir booster with ritonavir

- b) Pegylated interferon-alpha
- c) Velpatasvir + Voxilaprevir + Sofosbuvir
- d) Tenofovir + Entecavir

Question 16:

A young woman presents with painless warty lesions on the vulva as shown in the image below. Which of the following is a topical agent that can be prescribed for her condition?



- a) Docosanol
- b) Nystatin
- c) Imiquimod
- d) Acyclovir

Question 17:

Which of the following drugs is currently approved for the treatment of small pox?

- a) Tecovirimat
- b) Lumacaftor
- c) Pegvaliase
- d) Apalutamide

Answer Key

Question No.	Correct Option
1	b
2	d
3	c
4	c
5	c
6	a
7	d
8	b
9	c
10	b
11	b
12	c
13	c
14	d
15	b
16	c
17	a

Detailed Explanations

Solution to Question 1:

The given clinical scenario and investigations are suggestive of HSV encephalitis for which the drug of choice is intravenous acyclovir. The treatment is effective only if it is started early.

The clinical hallmark of HSV encephalitis is the acute onset of fever and focal neurologic signs with hemorrhagic necrosis of especially the temporal lobe. The most sensitive investigation is HSV DNA demonstration in CSF by PCR.

Solution to Question 2:

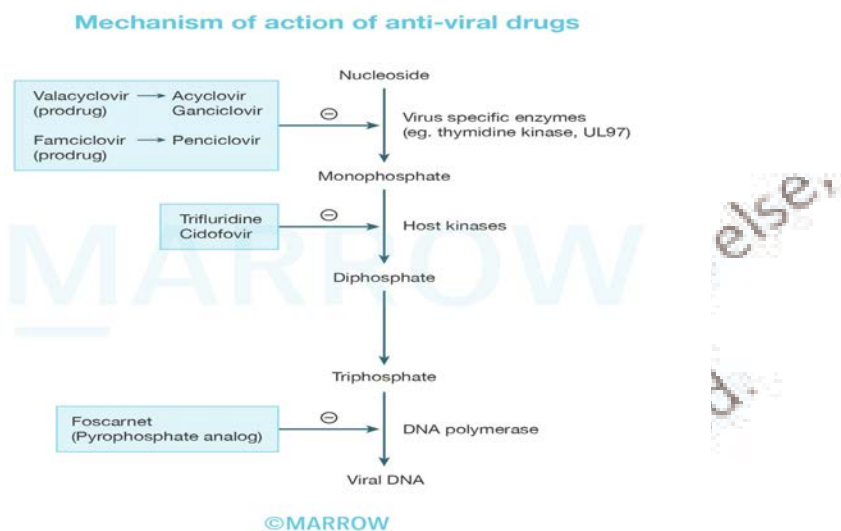
Acyclovir can enter CSF as it is widely distributed into all body fluid compartments, including vesicular fluid and aqueous humor. It can be used for treating shingles caused by HZV.

Option A: Acyclovir is a guanosine nucleoside congener. The other nucleoside congeners are cidofovir, ganciclovir, and famciclovir.

Option B: Nephrotoxicity and neurotoxicity, though rare and reversible, are dose-limiting side effects of acyclovir.

Option C: Acyclovir has a low bioavailability which is overcome by administering valacyclovir, a prodrug with higher bioavailability.

Mechanism of action of anti-viral drugs:



Solution to Question 3:

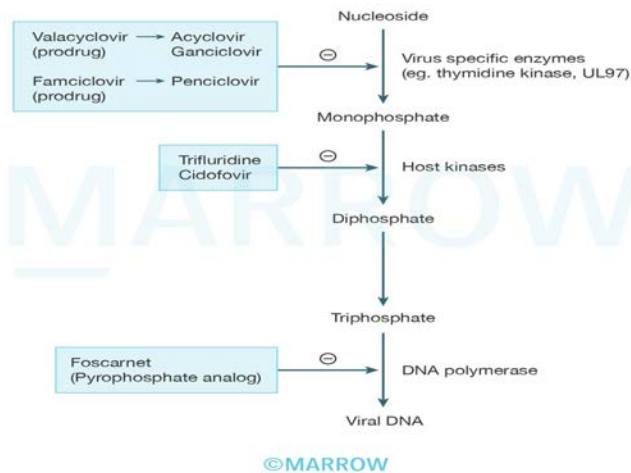
The most common cause of resistance to acyclovir in HSV-1 is deficient or absent viral thymidine kinase activity.

Viral thymidine kinase is needed in the conversion of acyclovir to acyclovir triphosphate which competitively inhibits the viral DNA polymerase activity.

Mutation of viral DNA polymerase is rare.

Mechanism of action of anti-viral drugs:

Mechanism of action of anti-viral drugs



Solution to Question 4:

Oral valacyclovir 1g three times daily for 5-7 days is considered the drug of choice for chickenpox (VZV) in an immunocompetent patient. Treatment should be started ideally within 24 hours of rash onset.

Oral acyclovir (800 mg five times daily) is an alternative drug, but valacyclovir is preferred due to its higher bioavailability and less frequent dosing. In immunocompromised patients, varicella infection is treated with IV acyclovir.

Option A: Oseltamivir is used for the treatment of avian influenza.

Option B: Ribavirin is used for treating RSV bronchiolitis.

Option D: Ganciclovir, foscarnet, and cidofovir are approved for CMV retinitis.

Solution to Question 5:

Intravenous foscarnet is used to treat acyclovir-resistant HSV (genital herpes) and VZV infections.

It inhibits viral nucleic acid synthesis by directly inhibiting HSV DNA polymerase (or HIV reverse transcriptase). It is active against herpes simplex, CMV, VZV, and HIV.

Metabolic abnormalities are very common with foscarnet. Symptomatic hypocalcemia and nephrotoxicity are major dose-limiting toxicities of foscarnet.

Solution to Question 6:

The most common side effect of amantadine is minor CNS toxicity which includes nervousness, difficulty concentrating, and insomnia. It is effective only against Influenza A.

Most strains of influenza are now resistant to amantadine and hence, it is not routinely used for the treatment or prophylaxis of influenza infections. Amantadine is largely excreted unmetabolized in the urine.

Solution to Question 7:

Oseltamivir is a neuraminidase inhibitor.

Drugs like oseltamivir and zanamivir competitively and reversibly interact with the enzyme sites to inhibit viral neuraminidase activity. This interferes with the release of the progeny from infected host cells, halting the spread of infection.

Early administration hastens the resolution of symptoms if given within 48 h of infection.

Oseltamivir is given at a dose of 75 mg, twice a day for 5 days for the treatment of influenza A and B. For prophylaxis, it is given at a dose of 75mg, once per day, and continued for the duration of exposure.

Solution to Question 8:

Zanamivir is delivered by oral inhalation of dry powder in a lactose carrier. Its oral bioavailability is very low (<5%).

Zanamivir is a neuraminidase inhibitor similar to oseltamivir and is active against both influenza A and B.

The different routes of administration of neuraminidase inhibitors are as follows:

- Zanamivir: Inhalational
- Oseltamivir: Oral
- Peramivir: Intravenous

Solution to Question 9:

Peramivir is a neuraminidase inhibitor available for intravenous administration for acute influenza (H1N1).

It is indicated in acute influenza infections where the patient is not able to take drugs via the oral route or if inhalational (zanamivir) formulation is unavailable.

Solution to Question 10:

The endonuclease inhibitor which can be used for the treatment of acute uncomplicated influenza in adults is baloxavir marboxil.

It is a novel drug that was approved by the FDA in 2018. It is used to treat acute uncomplicated influenza in patients 12 years of age and older who have been symptomatic for less than 48 hours.

It is a selective inhibitor of influenza cap-dependent endonuclease which blocks proliferation by inhibiting mRNA synthesis.

Solution to Question 11:

Palivizumab is a monoclonal antibody approved for prophylaxis against respiratory syncytial virus (RSV) infection.

It is a humanized monoclonal antibody against F-glycoprotein present on the surface of RSV. It is used in high-risk infants (prematurity and immunosuppressed conditions) and children (with cardiovascular and pulmonary diseases).

The treatment of RSV infections is done with inhaled ribavirin.

Option A: Omalizumab is a monoclonal antibody against IgE indicated in asthma.

Option C: Denosumab is a monoclonal antibody against RANKL indicated in osteoporosis.

Option D: Ribavirin is used for treatment of RSV infections but is not approved for prophylaxis.

Solution to Question 12:

The given clinical scenario of ascites and encephalopathy in a hepatitis B infected patient is suggestive of decompensated liver disease. Tenofovir is the drug of choice for decompensated cirrhosis due to hepatitis B infection.

Tenofovir is one of the first-line drugs for chronic hepatitis B infections. It is available as two prodrug formulations tenofovir disoproxil fumarate (TDF) and tenofovir alafenamide (TAF).

Option A: Interferon-alpha is one of the first-line agents in individuals with ongoing HBV-DNA replication and liver inflammation. But it is contraindicated in decompensated cirrhosis with HBV infection as they can precipitate clinical deterioration and increase the risk of bacterial infections.

Options B and D: Lamivudine and Telbivudine are not considered first-line agents for HBV infections because of the increased frequency of resistance associated with their use.

Solution to Question 13:

Sofosbuvir is indicated in the treatment of hepatitis C infection only. It is a directly acting antiviral, NS5B nucleotide polymerase inhibitor.

The goal of treatment in hepatitis C is a sustained virologic response (SVR). It is defined as the absence of a virus in the blood (undetectable HCV RNA), 12 weeks after completion of treatment. This can be achieved by the use of pan-genotypic directly acting antiviral drugs, irrespective of the disease stage.

Solution to Question 14:

NS4B inhibitors are not direct-acting agents (DAAs).

Direct-acting agents (DAAs) are drugs that target specific viral proteins which are vital to the replication and lifecycle of HCV.

The currently available DAAs inhibit 3 specific non-structural (NS) proteins:

-

NS3/4A protease inhibitors

-

NS5A inhibitors

-

NS5B polymerase inhibitors

Solution to Question 15:

The given scenario and serology are suggestive of chronic hepatitis D infection in a hepatitis B carrier for which the treatment of choice is pegylated interferon-alpha (pegIFN- α).

It is the only drug currently available to treat chronic hepatitis D infection.

Solution to Question 16:

Warty lesions on the vulva as seen in the given image is suggestive of condyloma acuminata caused by HPV (6, 11). Imiquimod is a topical agent with antiviral and immunomodulatory properties indicated in the treatment of external genital and perianal warts.

Other topical agents that can be used include podophyllotoxin and trichloroacetic acid.

Solution to Question 17:

Tecovirimat has been recently approved as a treatment of smallpox.

It was approved under the FDA's animal rule, which allows findings from animal studies to support an FDA approval when it is not feasible or ethical to conduct efficacy trials in

humans. Cidofovir has also been approved by the FDA for smallpox under this rule.

Both these drugs are currently stockpiled by CDC's strategic national stockpile, which has the medical supplies required to protect the public in case of a public health emergency.

Tecovirimat could also be used under an investigational new drug (IND) protocol to treat adverse reactions from vaccinia vaccination.

Option B: Lumacaftor is a drug approved for cystic fibrosis.

Option C: Pegvaliase has been approved for the treatment of phenylketonuria in adults.

Option D: Apalutamide is a nonsteroidal antiandrogen medication which is used in the treatment of prostate cancer. It is specifically indicated for use in conjunction with castration in the treatment of non-metastatic castration-resistant prostate cancer.

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Hypothalamus and Pituitary

Question 1:

You decide to start recombinant growth hormone therapy in a child with idiopathic short stature. The treatment primarily works by the activation of which of the following?

- a) G protein coupled receptors
- b) JAK/STAT pathway
- c) Nuclear receptors
- d) PI3/Akt pathway

Question 2:

In which of the following conditions is growth hormone therapy not indicated?

- a) Turner syndrome
- b) Prader Willi syndrome
- c) Sotos syndrome
- d) Noonan syndrome

Question 3:

All of the following are sound advice to the respective patients except:

- a) 1, 2 and 5
- b) only 5
- c) 2 and 4
- d) 3, 4 and 5

Question 4:

A middle-aged man with tall stature, frontal bossing, prognathism, large hands, and feet is started on a growth hormone receptor antagonist. Identify the drug.

- a) Octreotide
- b) Lanreotide

- c) Pegvisomant
- d) Cabergoline

Question 5:

A 23-year-old lady presented to the gynecology OPD with complaints of amenorrhea and milky discharge from her breasts. Her urine pregnancy test is negative and her serum prolactin level is 175 µg/L. The resident prescribes a long-acting D2 agonist. Which of the following drugs is she likely to have been prescribed?

- a) Bromocriptine
- b) Lisuride
- c) Cabergoline
- d) Apomorphine

Question 6:

In which of the following conditions is octreotide not used?

- a) Glucagonoma
- b) Insulinoma
- c) Short bowel syndrome
- d) Glioma

Question 7:

Valvular heart disease is not associated with the use of which of the following drugs?

- a) Bromocriptine
- b) Quinagolide
- c) Cabergoline
- d) Pergolide

Question 8:

In which of the following patients is bromocriptine not indicated?

- a) A 35-year-old lady with galactorrhea-amenorrhea complex

- b) A 30-year-old male patient with acromegaly
- c) A 23-year old lady with endogenous depression
- d) A 45-year old parkinson's patient with type 2 diabetes mellitus

Question 9:

A 60-year-old chronic alcoholic patient with a history of coronary bypass surgery presents with haematemesis to the casualty. An upper GI endoscopy shows the following finding. Which of the following drugs will you administer to this patient?



- a) Desmopressin
- b) Propranolol
- c) Octreotide
- d) Terlipressin

Question 10:

Pulsatile dosing of GnRH agonist is used for the treatment of _____.

- a) Male infertility
- b) Endometriosis
- c) Precocious puberty
- d) Prostate cancer

Question 11:

A 30-year-old lady who is suffering from dysmenorrhea and infertility is diagnosed with endometriosis. She has been prescribed an intranasal spray containing a GnRH agonist. What is the likely drug?

- a) Leuprolide
- b) Goserelin
- c) Histrelin
- d) Nafarelin

Answer Key

Question No.	Correct Option
1	b
2	c
3	b
4	c
5	c
6	d
7	b
8	c
9	c
10	a
11	d

Detailed Explanations**Solution to Question 1:**

Growth hormone primarily acts through activation of the JAK-STAT signaling pathway.

Cytokines such as interferon-gamma and prolactin hormone also act by the JAK STAT pathway.

Solution to Question 2:

Growth hormone therapy is not indicated in Sotos syndrome.

Indications for GH therapy are:

- GH Deficiency in children & adults
- AIDS-associated wasting
- Malabsorption associated with the short-bowel syndrome (As GH stimulates the adaptation of GI epithelial cells).
- Children with other conditions associated with short stature despite adequate GH production:
 - Turner syndrome
 - Noonan syndrome
 - Prader-Willi syndrome
 - Chronic renal insufficiency
 - Small for gestational age babies
 - Idiopathic short stature

Sotos syndrome (cerebral gigantism) is characterized by excessive physical growth during the first few years of life resulting in a disproportionately large and long head with a slightly protrusive forehead, large hands and feet, hypertelorism, and down-slanting eyes. It is accompanied by mild cognitive impairment, developmental abnormalities, hypotonia, and speech abnormalities.

The image given below shows a large head, large hands, and characteristic eye features in a child with Sotos syndrome.



Solution to Question 3:

GH deferral is only recommended for the first year after treatment of leukemia.

Because of the anti-insulin action of GH, there is an increased incidence of type 2 diabetes mellitus and therefore, fasting glucose levels should be monitored periodically during GH therapy. Due to the rare potential for developing intracranial hypertension, fundoscopic examination is recommended at the initiation of therapy and at periodic intervals thereafter. Rapid growth induced by GH can cause slipped epiphyses or scoliosis in children.

Estrogens (OCs and estrogen supplements) inhibit GH action so that a larger dose is needed to maintain the same IGF-1 level.

The adverse effects of growth hormone are:

Children	Adults
Intracranial hypertension Increased risk of type 2 diabetes Slipped epiphysis Scoliosis	Peripheral edema Carpal tunnel syndrome Arthralgias Myalgia

Solution to Question 4:

The given clinical scenario depicts acromegaly. Pegvisomant is a growth hormone receptor antagonist that is indicated for the treatment of acromegaly.

Octreotide and lanreotide are synthetic somatostatin analogs that can be used in the management of acromegaly.

Cabergoline is a dopamine agonist used primarily to treat hyperprolactinemia also has an indication in the management of acromegaly.

Solution to Question 5:

The given clinical scenario depicts hyperprolactinemia. Cabergoline is a long-acting D₂ agonist. Its t_{1/2} is >60 hours. It is indicated for the treatment of hyperprolactinemia and acromegaly.

Bromocriptine, lisuride, and apomorphine are short-acting when compared with cabergoline.

Bromocriptine is an ergot alkaloid derivative used in the management of type 2 diabetes mellitus, hyperprolactinemia, acromegaly, and Parkinson's disease. Cabergoline is preferred over bromocriptine for hyperprolactinemia due to lower adverse effects such as nausea and also due to dosing advantage (twice a week instead of daily dosing).

Lisuride is indicated in early Parkinson's disease but is also known to decrease prolactin levels and prevent migraine attacks. Apomorphine is a morphine derivative used to treat hypomobile "off" episodes of advanced Parkinson's disease.

Solution to Question 6:

Octreotide is not used in glioma.

Uses of octreotide are :

- Secretory diarrhea in carcinoid syndrome and VIPoma
- Short bowel syndrome, AIDS
- Pituitary tumors causing acromegaly
- Islet cell tumors like glucagonoma, insulinoma
- Acute bleeding in esophageal varices
- Thyrotrope adenoma

Adverse effects include gall stones, sinus bradycardia, and conduction abnormalities. Vit B12 deficiency can occur on long term use.

Solution to Question 7:

Quinagolide is not associated with valvular heart disease.

Quinagolide is a non-ergot D2 receptor agonist.

Cabergoline, Bromocriptine, and Pergolide are ergot derivatives and are associated with valvular heart disease. The risk of valvular heart disease from ergot therapy is associated with higher doses of the drug (necessary for Parkinson's disease treatment) but not with the lower doses used in treating hyperprolactinemia.

Note: Pergolide was withdrawn from the market due to reports of serious valve damage.

Solution to Question 8:

Bromocriptine is not indicated in endogenous depression.

It is a semisynthetic ergot D2 receptor agonist.

Uses of bromocriptine:

- Hyperprolactinemia (causing galactorrhoea, amenorrhea, or infertility in women and gynecomastia or infertility in men)
- Parkinsonism
- Acromegaly
- Type 2 diabetes mellitus
- Hepatic coma
- Suppression of lactation

Solution to Question 9:

The given clinical history of the patient and the endoscopic image showing prominent cherry-red spots are suggestive of bleeding esophageal varices.

Octreotide is indicated in the treatment of acute variceal bleeding. It is a somatostatin analog that reduces splanchnic blood flow significantly and decreases variceal bleeding.

Other drugs that can be used for the management of bleeding esophageal varices:

- Terlipressin
- Vasopressin
- Propranolol

Option A: Desmopressin is a selective V₂ receptor agonist used for central diabetes insipidus and nocturnal enuresis.

Option B: Propranolol (non-selective beta-blocker) is used in the prophylaxis of variceal bleeding and not in its acute management.

Option D: Terlipressin is a vasoconstrictor used for bleeding esophageal varices but may compromise cardiac function and induce ischemia and hence can't be used in this case.

Note: Although terlipressin is a vasoconstrictor and is used for bleeding oesophageal varices, the patient in this scenario has a history of MI (undergone coronary bypass surgery). Terlipressin can compromise cardiac function (reduces cardiac output and ejection fraction) and induce ischemia. Hence, it is not preferred in this case.

Solution to Question 10:

Pulsatile administration of GnRH agonists results in increased FSH and LH secretion and is therefore used to treat male and female infertility.

Non-pulsatile or continuous dose of GnRH agonist results in suppression and decreased concentrations of gonadotropins LH and FSH. They are used in:

- Endometriosis
- Uterine fibroids
- Central precocious puberty
- Ca prostate
- Ca ovary and breast
- Controlled ovary stimulation

Solution to Question 11:

Nafarelin is a GnRH agonist administered via nasal spray.

It is approved for use in endometriosis and central precocious puberty.

Routes of administration of other GnRH agonists:

- Intranasal and subcutaneous - Buserelin
- Intramuscular and subcutaneous - Leuprolide
- Subcutaneous implants - Histrelin and Goserelin

Some GnRH antagonists are:

- Degarelix
- Ganirelix
- Cetrorelix (Synthetic)

All GnRH antagonists are administered subcutaneously.

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Thyroid and Antithyroid Agents

Question 1:

Levothyroxine/liothyronine acts by binding to thyroid receptors. Thyroid hormone receptors belong to which of the following class of receptors?

- a) Transmembrane receptor linked to tyrosine kinase
- b) Nuclear receptor
- c) Ion channel receptor
- d) G protein coupled receptor

Question 2:

In comparison to thyroxine, liothyronine has _____.

- a) Longer half life
- b) Better absorption
- c) Slower onset of action
- d) Less potency

Question 3:

A 23-year-old hypothyroid patient taking levothyroxine regularly has now become pregnant. What would your advice be regarding levothyroxine?

- a) Switch to liothyronine as levothyroxine is teratogenic
- b) Increase the dose as pregnancy causes an increased demand for levothyroxine
- c) Continue the same dose but monitor the serum levels on a fortnightly basis
- d) Stop levothyroxine till the end of the first trimester as it has teratogenic potential

Question 4:

A patient on levothyroxine supplementation would need to increase her dose if she is taking _____.

- a) Methylprednisolone

- b) Aspirin
- c) Estradiol
- d) Furosemide

Question 5:

A 45-year-old patient suffering from Graves disease has been on treatment for the past three weeks. She has now developed fever and sore throat with an absolute neutrophil count of 80 neutrophils per microlitre of blood. The use of which of the following drugs can result in such a clinical picture?

- a) Only 1
- b) 1, 2 or 3
- c) 1 or 2
- d) 3, 4 or 5

Question 6:

A patient developed thyroid storm in the post-operative ward and was immediately started on IV propranolol and a loading dose of propylthiouracil. The antithyroid drug acts by inhibiting which of the following deiodinases?

- a) D1 only
- b) D2 only
- c) D1 and D2
- d) D3

Question 7:

A 40-year-old man presented with complaints of increased sweating and weight loss over the past two months. On examination, he is tachycardic, has exophthalmos, and a diffusely enlarged thyroid gland. Which of the following is the drug of choice for the management of this condition?

- a) Methimazole
- b) Propylthiouracil
- c) Levothyroxine
- d) Lugol's iodine

Question 8:

A 23-year-old primiparous woman presented with features suggestive of thyrotoxicosis, at 7 weeks of gestation. Which of the following is the drug of choice for the treatment of her condition?

- a) Methimazole
- b) Carbimazole
- c) Propylthiouracil
- d) Radioactive iodine

Question 9:

Which of the following statements best describes the Wolff-Chaikoff effect?

- a) Decrease in thyroid hormone release due to low doses of iodine
- b) Increase in thyroid hormone release due to high doses of iodine
- c) Decrease in thyroid hormone synthesis due to high doses of iodine
- d) Increase in thyroid hormone synthesis due to low iodine intake

Question 10:

A patient suffering from Graves disease is posted for thyroidectomy. He is administered iodides preoperatively. What is the primary benefit of administering iodides in this patient?

- a) To render the patient euthyroid
- b) To make the gland firm and less vascular
- c) To prevent hypothyroidism
- d) To augment radioactive iodine uptake

Question 11:

A patient suffering from hyperthyroidism is administered a drug that acts by inhibiting the release of thyroid hormone. Identify the drug.

- a) Propylthiouracil
- b) Radioactive iodine

- c) Thiocyanate
- d) Potassium iodide

Question 12:

A 43-year-old lady with papillary thyroid cancer is being treated with radioactive iodine 131. What is the half-life of this isotope?

- a) 8 hours
- b) 8 days
- c) 8 weeks
- d) 8 years

Question 13:

Which of the following statements about Lugol's iodine is incorrect?

- a) Used to make thyroid gland less vascular
- b) Used to block thyroid hormone release rapidly
- c) Tolerance is an issue
- d) Augments radioactive iodine uptake into thyroid gland

Question 14:

A 40-year-old lady who is a known case of Graves disease was posted for thyroidectomy. During the surgery, she developed hyperthermia, agitation and the monitors showed an arrhythmia. She was managed with aggressive IV fluid therapy, ice packs, and multiple drugs. Which of the following drugs that were used, do not inhibit the peripheral conversion of T₄ to T₃?

- a) Propranolol
- b) Glucocorticoids
- c) Iopanoic acid
- d) Potassium iodide

Answer Key

Question No.	Correct Option
1	b
2	b
3	b
4	c
5	c
6	a
7	a
8	c
9	c
10	b
11	d
12	b
13	d
14	d

Detailed Explanations

Solution to Question 1:

Thyroid hormone receptors belong to the class of nuclear receptors.

Nuclear receptors are a family of ligand-regulated transcription factors that are activated by various lipid-soluble signals, including retinoic acid, thyroid hormone, and steroid hormones. They primarily function as transcription factors and directly interact with the genomic DNA to regulate gene expression.

Nuclear receptors can either remain anchored in the cytoplasm via chaperone proteins until freed by ligand-binding (e.g., steroid and vitamin D receptors) or can reside in the nucleus bound to specific DNA response elements even in the absence of ligand (e.g., thyroid hormone, PPAR γ , and retinoic acid receptors).

Solution to Question 2:

Liothyronine has better absorption compared to levothyroxine. It is almost completely absorbed (100%) when compared to thyroxine (80%).

Note: Liotrix is a 4:1 mixture of Levothyroxine sodium (LT₄): Liothyronine sodium (LT₃)

	Liothyronine(T3)	Levothyroxine(T4)
Onset of action	Rapid	Slow
Potency	More potent	Less potent
Half-life	T _{1/2} -18-24 hr	T _{1/2} - 7-8 days
Source	25% from gland; 75% from peripheral conversion	Solely produced by the gland
Plasma protein binding	0.2% unbound	0.02% unbound

Solution to Question 3:

The dose of levothyroxine in patients who become pregnant should usually be increased. This is because demand for levothyroxine increases in pregnancy and low thyroxine levels are associated with adverse fetal outcomes such as mental retardation.

The dose of levothyroxine is increased by 30-50% by asking the patient to take 2 extra pills per week. Increased demand for levothyroxine in pregnancy is due to:

- The increased serum concentration of thyroxine-binding globulin (TBG) induced by estrogen
- Increased expression of type 3 deiodinase in the placenta
- A small amount of passage of levothyroxine to the fetus

Thyroid hormones are not teratogenic. Liothyronine is not preferred in pregnancy because T₃ has a short half-life and requires frequent dosing to limit large fluctuations in levels. In addition, excess T₃ often acts to normalize TSH while lowering T₄, thereby risking inadequate T₄ availability to the fetus.

Solution to Question 4:

Patients who start estrogen therapy will need to increase their thyroid supplementation doses.

Estrogens raise the circulating levels of thyroxine-binding globulin (TBG), thereby increasing the bound fraction and decreasing the free (bioactive) fraction of circulating thyroxine (T₄). As a consequence, estrogen therapy increases the T₄ supplementation requirements of women being treated for primary hypothyroidism.

Drugs increasing TBG levels:

- Estrogens
- Tamoxifen
- Methadone
- 5-fluorouracil

Drugs decreasing TBG levels:

- Androgens
- Corticosteroids
- Niacin

Drugs decreasing binding of T₄ to TBG (by displacement):

- Salicylates (aspirin, mefenamic acid)
- Heparin
- Furosemide

Solution to Question 5:

This patient is suffering from agranulocytosis which is a serious adverse effect that can be seen with the use of either methimazole or propylthiouracil.

Agranulocytosis usually occurs in the initial weeks or months of therapy. It occurs rapidly and periodic monitoring of granulocyte levels is generally ineffective.

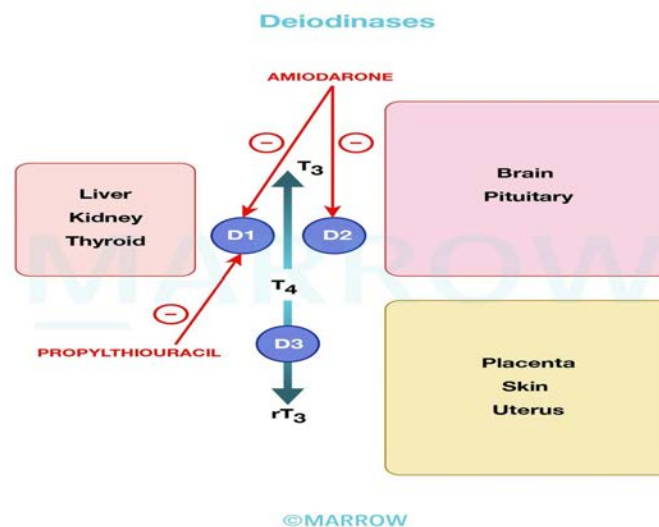
However, it is reversible on discontinuation of the drug. Therefore, patients who are on these drugs should be instructed to report immediately if they develop fever or sore throat.

Solution to Question 6:

Propylthiouracil inhibits only type I deiodinase (D1).

Deiodinases are the enzymes that are involved in the peripheral conversion of T₄ to either T₃ or reverse T₃. Type I deiodinase (D1) and type II deiodinase (D2) convert T₄ to T₃. Type III (D3) is involved in the conversion of T₄ to reverse T₃.

Note: Amiodarone inhibits both types I and type II deiodinases.



Solution to Question 7:

The given clinical scenario is suggestive of Graves disease and the drug of choice is methimazole.

Option A: Methimazole is preferred over propylthiouracil due to fewer side effects and single-daily dosing (compared to PTU which is given TID).

Option C: Levothyroxine is used in hypothyroidism.

Option D: Lugol's iodine is a thyroid hormone-release inhibitor but not used as a first-line drug.

Solution to Question 8:

The drug of choice for the treatment of thyrotoxicosis during the first trimester of pregnancy is propylthiouracil (PTU). It is because it has lower teratogenic effects than carbimazole or methimazole.

Carbimazole and methimazole are associated with teratogenic effects such as:

- Aplasia cutis congenita
- Choanal atresia
- Treacheosophageal fistula
- Fetal goiter

However, propylthiouracil use is limited to the first trimester due to the risk of hepatotoxicity. It is replaced with carbimazole or methimazole in the 2nd and 3rd trimesters of the pregnancy.

The lowest effective antithyroid drug dose should be used throughout gestation. This is to maintain the maternal serum-free T₄ level in the upper limit of the non-pregnant normal reference range.

Radioiodine therapy is contraindicated during pregnancy and lactation.

Solution to Question 9:

The Wolff-Chaikoff effect is best described as iodine-induced hypothyroidism.

It is a normal, protective downregulation of thyroid hormone synthesis and release via transient inhibition of iodide organification in the presence of an iodine load (exogenous supplementation, iodinated contrast dye, drugs like amiodarone). This occurs in the thyroid gland which is under pituitary control.

With persistent high iodine load, an escape phenomenon occurs in normal individuals after around 10 days as the iodide levels fall inside the follicle. In autoimmune diseases (Hashimoto's or Grave's), however, the suppression may persist.

The Jod-Basedow phenomenon is iodine-induced hyperthyroidism. It is also seen after an iodine load. However, it is seen in thyroid glands where autoregulation and pituitary control have failed. When exposed to an iodine load in excess, instead of blocking the organification process, the thyroid gland (areas of the autonomously functioning gland) continues to uptake the iodine, producing more thyroid hormone independent of normal regulatory mechanisms. It is seen in endemic goiter, toxic multinodular goiter, thyroid adenoma, and sometimes in Grave's disease.

Solution to Question 10:

The primary benefit of the preoperative use of iodides in Grave's disease is to make the gland firm and less vascular. This makes manipulation and surgical resection easier.

Antithyroid drugs like methimazole or carbimazole are indicated to render the patient euthyroid.

Solution to Question 11:

Potassium iodide inhibits the release of the thyroid hormone.

Iodides of sodium and potassium and Lugol's iodine act by inhibiting thyroid hormone release. Hence, they are also called thyroid constipating agents. They are the fastest-acting agents.

The beneficial effect of iodides peaks over 10-15 days and gradually disappears. Tolerance is hence an issue with the use of iodides.

Solution to Question 12:

The half-life of radioactive I¹³¹ is 8 days

I¹³¹ emits both beta and gamma rays. It is used both for diagnostic and therapeutic purposes for thyroid disorders.

I¹²⁷ is the stable isotope of iodine.

I¹²³ is the radioactive isotope of iodine that emits only gamma rays. The action is short-lived with a half-life of around 13 hours. It is indicated for diagnostic indications only.

Solution to Question 13:

Lugol's iodine does not augment the uptake of radioactive iodine in the thyroid gland.

Radioactive iodine uptake into the thyroid gland is reduced by Lugol's iodine as the uptake is inversely proportional to the serum concentration of stable iodine.

Lugol's iodine makes the thyroid gland less vascular and firmer. It blocks the release of thyroid hormone. The beneficial effect peaks over 10-15 days and gradually disappears. Therefore tolerance is an issue.

Solution to Question 14:

The given clinical picture is suggestive of a thyrotoxic crisis/thyroid storm. Potassium iodide does not inhibit the peripheral conversion of T₄ to T₃.

The major action of potassium iodide is the inhibition of hormone release. It is usually given 1 hour after PTU administration to avoid iodine stimulating thyroid hormone synthesis.

Drugs inhibiting the peripheral conversion of T₄ to T₃ are:

- Propranolol
- Glucocorticoids
- Propylthiouracil
- Amiodarone
- Iopanoic acid

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Corticosteroids

Question 1:

Glucocorticoids act on which of the following?

- a) G protein couple receptor
- b) Intracellular cytoplasmic receptor
- c) Ligand gated ion channel
- d) JAK-STAT pathway

Question 2:

Glucocorticoids do not have permissive action on which of the following?

- a) Angiotensin
- b) Adrenaline
- c) Noradrenaline
- d) Bradykinin

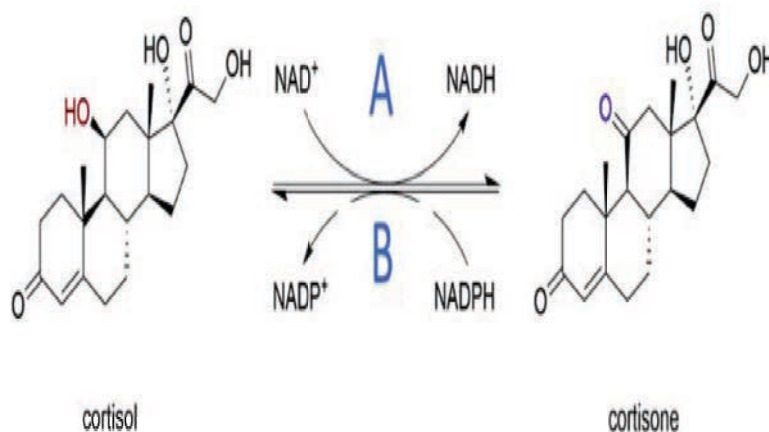
Question 3:

A pregnant woman diagnosed with Cushing's syndrome does not want surgery and requests medical treatment. Which of the following drugs can be recommended?

- a) Mifepristone
- b) Flutamide
- c) Finasteride
- d) Metyrapone

Question 4:

The enzyme A in the following image corresponds to?



- a) 17 β -Hydroxysteroid dehydrogenase
- b) 3 β -Hydroxysteroid dehydrogenase
- c) 11 β -Hydroxysteroid dehydrogenase1
- d) 11 β -Hydroxysteroid dehydrogenase2

Question 5:

Which of the following is a prodrug?

- a) Prednisolone
- b) Methylprednisolone
- c) Hydrocortisone
- d) Prednisone

Question 6:

A patient with moderate COVID-19 was started on a long-acting glucocorticoid. Which of the following drugs could it be?

- a) Triamcinolone
- b) Prednisolone
- c) Dexamethasone
- d) Hydrocortisone

Question 7:

Highest mineralocorticoid activity is seen with which of the following?

- a) Aldosterone
- b) DOCA
- c) Fludrocortisone
- d) Triamcinolone

Question 8:

Which of the following is not a mineralocorticoid receptor antagonist?

- a) Spironolactone
- b) Eplerenone
- c) Drospirenone
- d) Metyrapone

Question 9:

A woman diagnosed with Sheehan's syndrome is planned for the initiation of hormone replacement. Glucocorticoids should be administered prior to thyroxine supplementation due to what concern?

- a) Interference in absorption of glucocorticoids
- b) Precipitation acute adrenal insufficiency
- c) Decrease in the metabolism of glucocorticoids
- d) Suppression the HPA axis

Question 10:

A pregnant woman came for routine antenatal screening. During prenatal testing, the fetus was suspected to have congenital adrenal hyperplasia. What is the drug of choice for treating the fetus in-utero?

- a) Dexamethasone
- b) Betamethasone
- c) Hydrocortisone

d) Prednisolone

Question 11:

A patient is started on hormone replacement for chronic adrenal insufficiency. What is the replacement dose of hydrocortisone that has to be given (in mg/day)?

- a) 10
- b) 15
- c) 20
- d) 5

Question 12:

A pregnant woman at 33 weeks of gestation presents to your antenatal clinic with preterm labour. Which corticosteroid will you administer for stimulation of fetal lung maturation?

- a) Hydrocortisone
- b) Triamcinolone
- c) Methylprednisolone
- d) Betamethasone

Question 13:

A patient with chronic adrenal insufficiency is on long term corticosteroid therapy. Which of the following side effects is he unlikely to experience?

- a) Muscular hypertrophy
- b) Peptic ulceration
- c) Psychosis
- d) Hypertension

Question 14:

You are planning to add oral steroids to manage a case of severe asthma. Presence of which of the following comorbidities is not a relative contraindication for starting the medication?

- a) Diabetes mellitus

- b) Hypertension
- c) Eczematous skin disease
- d) Peptic ulcer disease

Question 15:

A 54-year-old woman is being evaluated for chronic back pain. She mentions that she had been on corticosteroids for asthma since childhood. Which of the following actions of the drug would not cause her condition?

- a) Inhibition of gonadal steroidal hormones
- b) Reduced gastrointestinal absorption of calcium
- c) Suppressive action on osteoblasts
- d) Inhibition of action of osteoclasts

Question 16:

Osteonecrosis following glucocorticoid therapy is most commonly associated with which site?

- a) Femoral head
- b) Distal femur
- c) Jaw bone
- d) Humeral head

Question 17:

A patient with Cushing's syndrome has very high levels of ACTH. Which of the following is a CRH preparation that can be used to determine if it is a pituitary source or an ectopic source?

- a) Gonadorelin
- b) Corticorelin
- c) Ganirelix
- d) Cosyntropin

Question 18:

Which of the following is a steroid receptor antagonist that can be given to a patient with Cushing's syndrome due to ectopic ACTH?

- a) Ketoconazole
- b) Metyrapone
- c) Mifepristone
- d) Aminoglutethimide

Question 19:

Which of the following intravenous anaesthetics can be used for acute control of hypercortisolism?

- a) Thiopentone
- b) Propofol
- c) Ketamine
- d) Etomidate

Question 20:

A patient is diagnosed with inoperable adrenocortical carcinoma and needs medical adrenalectomy. Which of the following drugs could be used for him?

- a) Mitotane
- b) Methotrexate
- c) Doxorubicin
- d) 5-Fluorouracil

Question 21:

Which of the following antifungal drugs can be used in the treatment of a patient with Cushing syndrome?

- a) Ketoconazole
- b) Fluconazole
- c) Itraconazole
- d) Miconazole

Answer Key

Question No.	Correct Option
1	b
2	d
3	d
4	d
5	d
6	c
7	a
8	d
9	b
10	a
11	c
12	d
13	a
14	c
15	d
16	a
17	b
18	c
19	d
20	a
21	a

Detailed Explanations

Solution to Question 1:

Glucocorticoids act on the intracellular cytoplasmic receptors.

They diffuse into cytosol and bind to the cytoplasmic receptors and influence nuclear changes by activation of the receptor.

Solution to Question 2:

Glucocorticoids do not have permissive action on bradykinin. They inhibit inflammatory mediators such as histamine, bradykinin, and eicosanoids.

Glucocorticoids possess permissive action on the vasoconstrictor effect of angiotensin and lipolytic action of adrenaline and noradrenaline. In the absence of glucocorticoids, these hormones have low activity but the presence enhances their activity several folds.

Solution to Question 3:

Metyrapone can be recommended as it is approved in the treatment of pregnant women with Cushing's syndrome. It inhibits the synthesis of cortisol by inhibiting steroid 11 β -hydroxylase reversibly.

Cushing's syndrome (CS) is a manifestation of symptoms that are precipitated by excessive cortisol production. Transsphenoidal surgery, radiation, and pharmacotherapy constitute the standard treatment for tumor-related causes of CS (e.g. pituitary adenoma).

The three main pharmacological approaches are:

- Cortisol synthesis inhibitors: Ketoconazole, metyrapone, mitotane, etomidate, aminoglutethimide
- ACTH secretion inhibitors: Cyproheptadine, pasireotide, octreotide, cabergoline
- Glucocorticoid receptor antagonist: Mifepristone

Note: Metyrapone is also used to test the hypothalamic-pituitary feedback pathway in the diagnosis of Cushing syndrome.

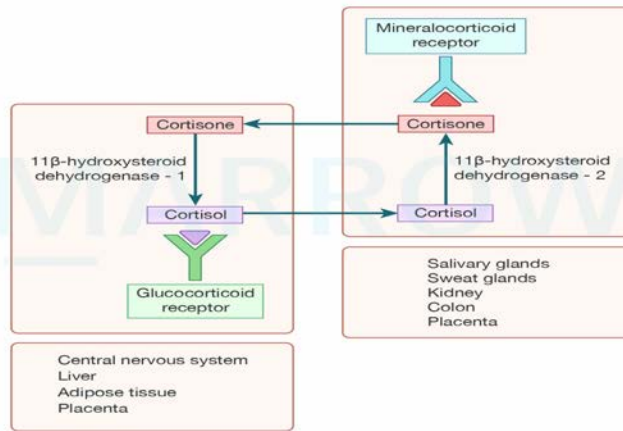
Solution to Question 4:

"A" here corresponds to 11-beta-HSD2 (11 β -Hydroxysteroid dehydrogenase2). It converts active cortisol to inactive cortisone.

Cortisol is active and can bind to both glucocorticoids and mineralocorticoid receptors whereas cortisone is inactive and does not bind to either. The selective inactivation of cortisol is required in certain sites such as the distal nephron so that the specific autoregulatory responses to aldosterone can be achieved.

Therefore, cortisol (active) is metabolized to cortisone (inactive) by 11-beta-HSD2 in the kidney, so as to protect renal mineralocorticoid receptors from cortisol binding. Some tissues, like the liver, can convert inactive cortisone back to active cortisol through an enzyme called 11-beta-HSD1 ("B" in the image). This completes the "cortisone–cortisol shuttle" because the kidney can inactivate cortisol where it is not needed and the liver can reactivate it afterward.

The Cortisol - Cortisone Shunt



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However, when circulating cortisol levels are very high, as in severe Cushing syndrome, they can overwhelm this protective mechanism, allowing cortisol to bind to and activate the mineralocorticoid receptor. This leads to volume expansion, hypertension, and hypokalemia often observed in hypercortisolism. A mutation of 11-beta-HSD2 also causes a similar clinical picture and is the reason for the syndrome of apparent mineralocorticoid excess.

17 β -Hydroxysteroid dehydrogenase and 3 β -Hydroxysteroid dehydrogenase are involved in the metabolism of sex steroids.

Solution to Question 5:

Prednisone is a prodrug.

It is a synthetic glucocorticoid that is biologically inert until converted via hepatic metabolism (11-beta HSD1) to the active component, prednisolone.

Solution to Question 6:

Dexamethasone is the longest acting glucocorticoid. Its $t_{1/2}$ is >36 hours.

Solution to Question 7:

The highest mineralocorticoid activity is seen with aldosterone.

However, it cannot be used clinically. The highest mineralocorticoid activity in a drug that is used clinically is fludrocortisone.

	Glucocorticoid activity	Mineralocorticoid activity
Short-acting glucocorticoids		
CortisoneHydrocortisone	0.81	0.81
Intermediate-acting glucocorticoids		
Prednisone-prednisoloneMethylprednisolone	45	0.80.5
Long-acting glucocorticoids		
DexamethasoneBetamethasone	3030	00
Mineralocorticoids		
Deoxycorticosterone (DOCA) FludrocortisoneAldosterone	0.2150	100150200-500

Solution to Question 8:

Metyrapone is not a mineralocorticoid receptor antagonist.

It is a selective inhibitor of 11 β -hydroxylase and thereby reduces cortisol and corticosterone synthesis.

Options A and B - Spironolactone and eplerenone are aldosterone antagonist that is used in hypertension and chronic CHF for reversing the effects of aldosterone

Option C - Drospirenone is a 4th generation progestin used as an oral contraceptive having mineralocorticoid antagonist action

Solution to Question 9:

Glucocorticoids are administered before initiating treatment with thyroid hormones as administration of thyroxine can precipitate acute adrenal insufficiency.

In the case of combined adrenal insufficiency and hypothyroidism, as in panhypopituitarism (due to surgery or Sheehan's syndrome), adrenal insufficiency is generally treated first. This is because giving thyroxine first can enhance cortisol clearance and precipitate an adrenal crisis. Also, levothyroxine increases the basal metabolic rate (BMR) leading to an increased requirement of cortisol, which if not met can also precipitate an adrenal crisis.

Solution to Question 10:

The drug of choice for treating the fetus in-utero when suspected of having congenital adrenal hyperplasia is dexamethasone.

Dexamethasone readily crosses the placenta and suppresses the secretion of steroids and androgens by fetal adrenal glands. If started by 6 weeks of gestation, it reduces the virilization of external genitalia in affected females.

Chorionic villus biopsy is then performed to determine the sex and the genotype of the fetus and therapy is only continued if the fetus is an affected female.

This is not widely accepted currently due to insufficient data on long-term risks especially in unaffected females and males who derive no benefit from the treatment.

Solution to Question 11:

The replacement dose of hydrocortisone for chronic adrenal insufficiency is 20 mg/day.

The treatment of chronic adrenal insufficiency (not in crisis) requires glucocorticoid replacement at a dose that replaces the physiologic daily cortisol production. This is achieved by administering an oral replacement dose of 15-25mg Hydrocortisone.

A midpoint dose of 20mg is given in 2-3 divided doses: 10mg + 5mg + 5mg.

A higher morning dose is given at 8:30 am (usually 1/2 to 2/3 of the total dose). The next dose is given at 1:30 pm and in the evening (4:30-5:00 pm). This is to follow the normal circadian rhythm.

Solution to Question 12:

Betamethasone is indicated for the stimulation of fetal lung maturation in preterm labor.

As compared to cortisol, betamethasone binds to maternal proteins less and has less placental metabolism. This allows increased transfer to the fetus. Additionally, infants exposed to betamethasone are less likely to require postnatal resuscitation. 2 doses of 12 mg each are administered 18-24 hours apart when the delivery is anticipated before 34 weeks of gestation.

Dexamethasone is another long-acting glucocorticoid that can be given. It has benefits such as less risk of intraventricular hemorrhage and a shorter length of stay in the NICU.

Solution to Question 13:

Corticosteroid therapy does not cause muscular hypertrophy and the patient is unlikely to experience it as a side effect.

Excess glucocorticoid activity results in muscle wasting and myopathy while excess mineralocorticoid activity results in hypokalemia and causes muscle weakness.

Options B and C: Peptic ulceration and psychosis are known adverse effects of corticosteroids.

Option D: Glucocorticoids with some mineralocorticoid activity will result in sodium and water retention and may cause hypertension.

Solution to Question 14:

Eczematous skin diseases are not a contraindication for the use of oral corticosteroids. In fact, they are highly effective and are indicated in this condition.

All the contraindications to steroids are relative because of the life-saving nature of steroids.

Risk-benefit should be assessed cautiously when initiating steroids in the following conditions:

- Diabetes mellitus
- Hypertension
- Peptic ulcer
- Viral and fungal infections
- Tuberculosis and other infections
- Osteoporosis
- Herpes simplex keratitis
- Epilepsy
- CHF
- Renal failure

Solution to Question 15:

Chronic back pain following long term corticosteroid therapy is suggestive of osteoporosis. Glucocorticoids do not inhibit the action of osteoclasts.

It causes stimulation of osteoclasts and increases bone resorption.

Mechanism of glucocorticoids induced osteoporosis:

- Inhibition of gonadal steroidal hormones
- Reduced gastrointestinal absorption of calcium
- Secondary effect of decreased calcium absorption: increased parathyroid hormone leading to further resorption
- Suppressive action on osteoblasts
- Stimulation of osteoclasts

Solution to Question 16:

The femoral head is the most frequently affected site by osteonecrosis due to glucocorticoid therapy.

The earliest symptoms are joint pain and stiffness. When patients on steroid therapy develop hip, shoulder or knee pain, osteonecrosis must be considered and ruled out.

Options B and D - It can also affect the distal femur and humeral head

Option C - Osteonecrosis of the jaw is associated with the use of bisphosphonates

Solution to Question 17:

Corticotropin-releasing hormone (CRH) preparation used in CRH stimulation test which helps to differentiate between a pituitary source and an ectopic source for ACTH-dependent hypercortisolism.

Pituitary tumors tend to be sensitive to CRH stimulation and responds with a rise in cortisol while ectopic tumors do not usually respond. Ectopic sources of ACTH include carcinoid, medullary thyroid carcinoma, and pheochromocytoma.

It can also be used to test the HPA axis.

Option A - Gonadorelin is a recombinant GnRH.

Option C - Ganirelix is a GnRH antagonist.

Option D - Cosyntropin is a synthetic preparation of ACTH.

Solution to Question 18:

Mifepristone is a steroid receptor antagonist that can be used in Cushing's syndrome.

It is used in Cushing's syndrome especially in inoperable cases with ectopic ACTH secretion or adrenal carcinoma.

Options A, B, and D - Ketoconazole, metyrapone, and aminoglutethimide reduce steroid synthesis by enzyme inhibition.

Solution to Question 19:

The intravenous anesthetic that can be used for acute control of hypercortisolism is etomidate.

Etomidate at sub-hypnotic doses inhibits cortisol secretion by selectively inhibiting 11-beta-hydroxylase activity.

It is used off-label to treat hypercortisolism when rapid control is needed and the patient is not able to take oral medication.

Solution to Question 20:

Mitotane is used for medical adrenalectomy in this patient.

It has a non-selective toxic action on the adrenal cortex. It causes suppression of adrenal cortex hormone secretion comparable to that of adrenalectomy.

Solution to Question 21:

Ketoconazole can be used in the treatment of Cushing syndrome for symptomatic control.

It is a non-selective inhibitor of adrenal and gonadal steroid synthesis. These actions are more apparent at higher doses than those used for fungal infections.

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If you purchased this from someone else,
you may have been scammed.

Osteoporosis and Calcium Metabolism

Question 1:

Which of the following decreases both serum calcium and phosphate levels?

- a) Vitamin D
- b) Parathyroid hormone
- c) Calcitonin
- d) Vitamin A

Question 2:

Which of the following interferes with the absorption of calcium?

- a) 1,2 and 3
- b) 4 and 5
- c) 1,2,3 and 4
- d) 1,2,3,4 and 5

Question 3:

Following a road traffic accident, a patient underwent transfusion with 5 units of packed red blood cells. Soon after, he developed tetany. His serum calcium level was 7.5 mg/dl on lab reports. Which of the following is preferred as the treatment of choice for the given condition?

- a) Calcium chloride
- b) Calcium gluconate
- c) Calcium carbonate
- d) Calcium lactate

Question 4:

Which of the following is true regarding plasma calcium?

- a) Ca^{2+} sensing receptor is a voltage gated ion channel

- b) Albumin bound calcium is physiologically most important
- c) Calcium is mainly absorbed from large intestine
- d) Acidosis favours ionization of calcium

Question 5:

A 57-year-old post menopausal lady has been on alendronate for the past two years. By which of the following mechanisms does this drug act?

- a) Increasing the osteoid formation
- b) Increasing the mineralization of osteoid
- c) Decreasing the osteoclast mediated resorption of bone
- d) Decreasing the activity of osteoprotegerin

Question 6:

Which of the following is not an indication to use bisphosphonates?

- a) Paget's disease
- b) Vitamin D excess
- c) Postmenopausal osteoporosis
- d) Hypercalcemia of malignancy

Question 7:

A 65-year-old lady has been taking zoledronate for the treatment of osteoporosis. She is at a high risk of developing which of the following side effects?

- a) Acute renal failure
- b) Ventricular fibrillation
- c) Peptic ulcer
- d) Anterior uveitis

Question 8:

Which of the following is used in the first-line management of postmenopausal osteoporosis?

- a) Estrogen
- b) Bisphosphonates
- c) Raloxifene
- d) Combined oral contraceptives

Question 9:

You start a patient on alendronate for osteoporosis. The patient enquires about the side effects and their management. Which of the following measures would you not recommend to decrease oesophageal irritation?

- a) Remaining upright for 30 minutes
- b) Use of proton pump inhibitor at bedtime
- c) Shifting to intravenous formulation
- d) Administration with food

Question 10:

In which of the following conditions is teriparatide indicated?

- a) Severe osteoporosis
- b) Paget disease
- c) Hypercalcemia
- d) Osteosarcoma

Question 11:

Which of the following drugs promotes bone formation?

- a) Calcitonin
- b) Raloxifene
- c) Bisphosphonates
- d) Teriparatide

Question 12:

Which of the following acts by causing both decreased bone resorption and increased bone formation?

- a) Strontium ranelate
- b) Ibandronate
- c) Teriparatide
- d) Calcitonin

Question 13:

A 70-year-old patient with severe backache has a T-score of -3. He is started on a monoclonal antibody against RANKL. Identify the drug.

- a) Romosozumab
- b) Denosumab
- c) Odanacatib
- d) Blosozumab

Question 14:

A post-menopausal lady who is taking estrogen as part of her hormone replacement therapy reads about raloxifene on the internet and wants to change her current treatment. Which of the following is not an advantage of using raloxifene over estrogen?

- a) Reduces fracture rates
- b) Avoids endometrial hyperplasia
- c) Reduces incidence of venous thrombosis
- d) No increase in incidence of breast carcinoma

Question 15:

A 45-year-old patient with chronic kidney disease developed secondary hyperparathyroidism for which he was treated appropriately, and was started on cinacalcet. Which of the following best describes the mechanism of action of this drug?

- a) Calcitonin preparation
- b) PTH preparation
- c) Calcimimetic

d) Vitamin D3 analog

Question 16:

A 40-year-old male patient who requires weekly hemodialysis is started on sevelamer. Which of the following best describes the mechanism of action of this drug?

- a) Vitamin D analog
- b) Calcium-sensing receptor mimetic
- c) Phosphate binder
- d) Potassium binder

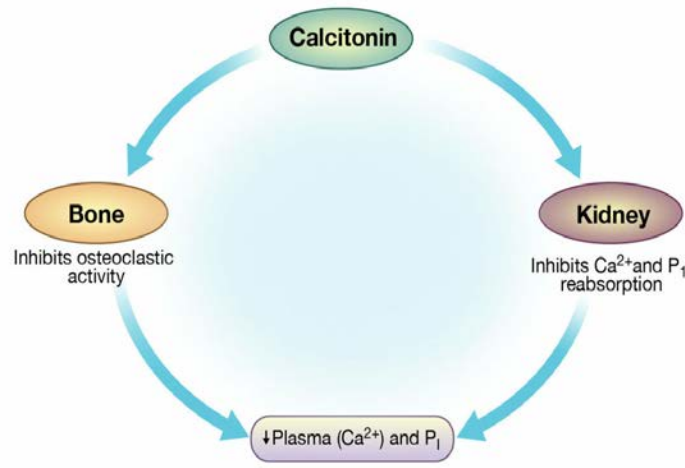
Answer Key

Question No.	Correct Option
1	c
2	d
3	b
4	d
5	c
6	b
7	a
8	b
9	d
10	a
11	d
12	a
13	b
14	c
15	c
16	c

Detailed Explanations

Solution to Question 1:

Calcitonin decreases both serum calcium and phosphate levels.



Solution to Question 2:

Glucocorticoids, phenytoin, tetracyclines, phytates, and oxalates all interfere with the absorption of calcium.

Glucocorticoids and phenytoin depress intestinal calcium transport whereas, tetracyclines, phytates, phosphates, and oxalates form insoluble complexes with calcium and thereby interfere with calcium absorption.

Solution to Question 3:

This is a case of hypocalcemia following a massive blood transfusion and IV calcium gluconate is the preferred treatment.

The hypocalcemia results from Ca²⁺ chelation by citrate (added to the blood product as preservative and anticoagulant). Calcium gluconate is the preferred over calcium chloride for the emergency treatment of hypocalcemia because it is less irritating and better tolerated.

However, calcium chloride may be preferable to calcium gluconate in the presence of abnormal liver function, because decreased citrate metabolism results in a slower release of ionized calcium. It is generally reserved for severe symptomatic hypocalcemia, (<4 mg/dL) and when given, should be administered via a central venous access device to avoid extravasation and tissue necrosis.

Calcium carbonate and calcium lactate are oral preparations of calcium.

Solution to Question 4:

Acidosis favors the ionization of calcium.

Hyperventilation by causing alkalosis favors the deionization of calcium. Alkalosis result in precipitation of tetany and laryngospasm in calcium-deficient patients

Option A - Ca^{2+} sensing receptor is a type of G protein coupled receptor.

Option B - 50% of the Ca^{2+} is in ionized form and is physiologically most important.

Option C - Calcium is mainly absorbed from the small intestine.

Solution to Question 5:

Alendronate is a bisphosphonate which acts by decreasing osteoclast mediated bone resorption.

Osteoclasts internalize bisphosphonates as they are released along with calcium from the bone surface and then reduce osteoclast activity by:

- Promoting apoptosis of osteoclasts
- Disrupting the cytoskeleton and ruffled border of osteoclasts
- Inhibiting the differentiation of osteoclasts

Solution to Question 6:

Bisphosphonates are not indicated in the treatment of vitamin D excess.

Indications for the use of bisphosphonates:

- Osteoporosis
- Paget's disease
- Hypercalcemia of malignancy
- Osteolytic bone metastasis

Solution to Question 7:

Zoledronate use has been associated with acute renal failure.

It is nephrotoxic. So baseline renal function and monitoring of renal function during therapy with zoledronate is recommended.

Long term complications seen with bisphosphonate therapy are osteonecrosis of jaw and subtrochanteric fracture of femur.

Solution to Question 8:

Bisphosphonates are used in the first-line management of postmenopausal osteoporosis.

Bisphosphonates are analogs of pyrophosphates. They specifically inhibit bone resorption and thus are very effective for the treatment of postmenopausal osteoporosis.

Bisphosphonates concentrate at sites of active remodeling. They remain in the matrix until the bone is remodeled. Then, they are released in the acid environment of the resorption lacunae and induce apoptosis in osteoclasts.

Classification of bisphosphonates (mainly based on potency):

First-generation:

- Etidronate (least potent)
- Clodronate
- Tiludronate

Second-generation: 10-100 times more potent than the first generation

- Alendronate
- Pamidronate
- Ibadronate

Third-generation: 10,000 times more potent than the first generation

- Risedronate
- Zoledronate

Solution to Question 9:

Bisphosphonates are not recommended to be administered with food to minimize esophageal irritation.

Oral bisphosphonates have very limited bioavailability (<1-10%). Bisphosphonates are recommended to be administered after an overnight fast with a full glass of water at least 30 minutes before breakfast.

Options A and B- Remaining upright for 30 minutes after dosing and taking proton pump inhibitors at bedtime can decrease the symptoms.

Option C- As esophageal irritation is due to the local direct effect of the oral formulation, shifting to intravenous formulations reduces the symptoms.

Note- The prescribing information of all the oral bisphosphonates states that it has to be taken along with tap or filtered water, and not mineral water as it contains higher amounts of calcium which can interfere with the absorption of bisphosphonates. It is prudent to inform patients to not have the medications with water having higher calcium levels.

Solution to Question 10:

Teriparatide is indicated in severe osteoporosis.

Teriparatide is a recombinant human parathyroid preparation. It increases bone formation especially when given in low doses and intermittently.

Although it may be more effective than bisphosphonates and estrogen, it has to be administered subcutaneously every day. Hence it is indicated mostly in severe osteoporosis.

Teriparatide increases the risk of osteosarcoma and it is contraindicated in conditions that cause high risk of osteosarcoma such as:

- Paget disease
- Prior h/o of radiotherapy
- Young adults with open epiphysis
- Unexplained elevated ALP
- Hypercalcemia

Abaloparatide is also a PTHrP preparation available.

Solution to Question 11:

Teriparatide increases bone formation especially when given in low doses and intermittently.

Teriparatide is a recombinant preparation of parathyroid hormone.

The pharmacologic activity of teriparatide, which is similar to the physiologic activity of PTH, includes stimulating osteoblast function, increasing gastrointestinal calcium absorption, and increasing renal tubular reabsorption of calcium.

Although it may be more effective than bisphosphonates and estrogen, it has to be administered subcutaneously every day. Hence it is indicated mostly in severe osteoporosis.

Teriparatide increases the risk of osteosarcoma and it is contraindicated in conditions that cause high risk of osteosarcoma such as:

- Paget disease
- Prior h/o of radiotherapy
- Young adults with open epiphysis
- Unexplained elevated ALP
- Hypercalcemia

Option A: Calcitonin decreases bone resorption.

Option B: Raloxifene is a selective estrogen receptor modulator, which decreases bone resorption. It is used to prevent and treat osteoporosis in postmenopausal women.

Option C: Bisphosphonates like alendronate and etidronate decrease bone resorption.

Solution to Question 12:

Strontium ranelate acts by both decreased bone resorption and increased bone formation.

It is generally reserved for severe osteoporosis. Due to an increased incidence of myocardial infarction and other cardiovascular events, the use is restricted in all the approved regions.

Option B- Ibandronate which is a bisphosphonate, decreases bone resorption only.

Option C- The pharmacologic activity of teriparatide, which is similar to the physiologic activity of PTH, includes stimulating osteoblast function, increasing gastrointestinal calcium absorption, and increasing renal tubular reabsorption of calcium.

Option D- Calcitonin also decreases bone resorption.

Solution to Question 13:

The given clinical history depicts a case of osteoporosis. Denosumab is a monoclonal antibody against RANKL that is indicated in the treatment of osteoporosis.

It inhibits the binding of RANKL to RANK receptors on osteoclasts and thereby inhibiting the activation of osteoclast and also promotes their apoptosis.

Options A and D: Romosozumab and blosozumab are monoclonal antibodies against sclerostin (sclerostin blocks osteoblast proliferation).

Option C: Odanacatib is an inhibitor of cathepsin K which is an enzyme in osteoclast that facilitates bone resorption.

Solution to Question 14:

Raloxifene causes a 3-fold increase in deep vein thrombosis and pulmonary embolism.

It is a selective estrogen receptor modulator (SERM) with the following actions:

- Agonist actions:
 - Bone - increases bone mineral density and reduces the risk of vertebral fracture
 - Lipids - increases the risk of deep vein thrombosis and pulmonary embolism.
- Antagonist actions:
 - Breast - incidence of breast carcinoma is reduced
 - Uterus - endometrial hyperplasia is avoided

Solution to Question 15:

Cinacalcet is a calcimimetic agent.

It activates the calcium-sensing receptor (CaSR) in the parathyroids and blocks PTH secretion.

Cinacalcet is a type II calcimimetic agent, it means that it needs either Ca^{2+} or some other agonist to activate CaSR.

It is indicated for secondary hyperparathyroidism due to chronic kidney disease and tumor of parathyroid gland.

Type I calcimimetic agents are those which can activate CaSR directly such as aminoglycoside and polylysine.

Solution to Question 16:

Sevelamer is a phosphate binder indicated in chronic kidney disease.

Sevelamer is a nonabsorbable polymer that acts as a nonselective anion exchanger that lowers serum phosphate concentration.

It is indicated to lower phosphate levels in patients with chronic kidney disease who are on dialysis.

Note: Patiromer is a potassium binder approved for use in hyperkalemia.

Anti-Diabetic Drugs - Oral

Question 1:

Which of the following does not cause insulin release?

- a) Rosiglitazone
- b) Nateglinide
- c) Glimepiride
- d) Tolbutamide

Question 2:

Insulin release due to the closure of K⁺ channels is seen with?

- a) Repaglinide
- b) Metformin
- c) Exenatide
- d) Sitagliptin

Question 3:

A diabetic patient being treated with an oral hypoglycemic agent develops dilutional hyponatremia. Which one of the following could be responsible for the effect?

- a) Tolbutamide
- b) Chlorpropamide
- c) Glyburide
- d) Glimepiride

Question 4:

A 62-year-old woman presents with complaints of fatigue. She is a known case of type 2 diabetes mellitus on treatment. On examination, there is pallor and knuckle pigmentation. Which of the following drugs could have caused her condition?

- a) Gliclazide

- b) Acarbose
- c) Metformin
- d) Rosiglitazone

Question 5:

Which of the following is incorrect regarding metformin?

- a) Causes little or no hypoglycemia in non-diabetic patients
- b) Acts by increased insulin secretion
- c) Increases peripheral utilization of glucose and decreases absorption of glucose from the intestine
- d) Increase intake of calcium may prevent metformin induced B₁₂ malabsorption

Question 6:

What is the central pharmacological action of metformin in type 2 diabetes?

- a) Reduction in glucose absorption
- b) Reduction in insulin resistance
- c) Reduction in hepatic glucose production
- d) Reduction in effects of glucagon

Question 7:

A 58-year-old woman with chronic kidney disease with recently diagnosed Type 2 DM was started on metformin therapy. Her repeat eGFR was 40/min/1.73 m². Which of the following is not true regarding the further management of this lady?

- a) Metformin dose has to be decreased.
- b) Metformin should be discontinued and replaced with insulin.
- c) eGFR should be measured at least annually in her.
- d) eGFR is preferred over serum creatinine concentration to assess renal function for metformin therapy.

Question 8:

A 13-year-old boy is diagnosed with type 2 diabetes mellitus. Which of the following drugs can be prescribed?

- a) Glimepiride
- b) Metformin
- c) Sitagliptin
- d) Pioglitazone

Question 9:

Which of the following is incorrect regarding pioglitazone?

- a) Metabolised in the liver by CYP3A4
- b) Selective agonist for the nuclear peroxisome proliferator activated receptor
- c) It can modulate the transcription of genes for carbohydrate and fat metabolism in the absence of insulin
- d) It should be avoided in any patient with cardiovascular disease

Question 10:

A 55-year-old diabetic man on treatment presents with complaints of difficulty reading. Fundoscopy shows the following findings. Which antidiabetic drug class is likely to cause this?



- a) Thiazolidinediones

- b) DPP-4 inhibitors
- c) SGLT-2 inhibitors
- d) Sulfonylureas

Question 11:

An elderly man was diagnosed with hypertension 2 years ago and has been on losartan. During a routine health checkup, his HbA1c was found to be 7.3%. Which of the following is a better alternative to losartan for this patient?

- a) Irbesartan
- b) Telmisartan
- c) Candesartan
- d) Olmesartan

Question 12:

A 60-year-old woman with a known history of hypertension and dyslipidemia was admitted for evaluation of recurrent TIA. On workup, she was diagnosed with diabetes mellitus. Which of the following drugs should be avoided in the management of this patient?

- a) Pioglitazone
- b) Rosiglitazone
- c) Metformin
- d) Troglitazone

Question 13:

A professional boxer is diagnosed with diabetes mellitus. He is concerned about weight changes as a side effect of anti-diabetic drugs and explains that he has to maintain his current weight category. Which of the following will you prescribe?

- a) Pioglitazone
- b) Saxagliptin
- c) Liraglutide
- d) Gliclazide

Question 14:

In a middle-aged diabetic patient with chronic kidney disease stage 5, which of the following drugs can be given safely without altering the dose?

- a) Vildagliptin
- b) Linagliptin
- c) Saxagliptin
- d) Sitagliptin

Question 15:

Which of the following is not true about dapagliflozin?

- a) It is indicated in heart failure
- b) It causes weight gain
- c) It is effective orally
- d) It will increase the risk of fractures

Question 16:

A patient with diabetes mellitus came to the OPD with a history suggestive of recurrent episodes of hypoglycemia. She was recently put on combination therapy. Her symptoms are least likely to be caused by which of the following drugs?

- a) Acarbose
- b) Insulin
- c) Gliclazide
- d) Repaglinide

Question 17:

While at a bakery, a man complains of giddiness and palpitations. As the only doctor there, you're asked for help. The man says he had just taken his diabetic medications and shows a prescription listing insulin and acarbose. What would you give for immediate management?

- a) Honey
- b) Bread

- c) Sugar
- d) Cake

Question 18:

A 52-year-old diabetic patient presented to the ED with complaints of chest pain on exertion. Laboratory tests showed dyslipidemia and features of metabolic syndrome. Which of the following drugs was recently approved for this condition?

- a) Alogliptazar
- b) Saroglitazar
- c) Muraglitazar
- d) Rosiglitazone

Question 19:

Which of the following is approved as an adjunctive treatment for type 2 diabetes mellitus?

- a) Bromocriptine
- b) Phenytoin
- c) Ropinirole
- d) Metoclopramide

Question 20:

A 54-year-old woman comes to the diabetic clinic for follow up. Her lipid profile shows high LDL cholesterol. Which bile acid sequestrant would you prescribe for her?

- a) Colesevelam
- b) Colestipol
- c) Cholestyramine
- d) Ezetimibe

Answer Key

Question No.	Correct Option
1	a
2	a
3	b
4	c
5	b
6	c
7	b
8	b
9	d
10	a
11	b
12	b
13	b
14	b
15	b
16	a
17	a
18	b
19	a
20	a

Detailed Explanations

Solution to Question 1:

Rosiglitazone does not increase insulin release from the beta cells of the pancreas.

Rosiglitazone is a thiazolidinedione that acts by activating PPAR-gamma (nuclear receptor) which induces several genes involved in lipid and glucose metabolism and thereby, decreases insulin resistance.

Nateglinide (a meglitinide) and glimepiride, tolbutamide (both sulfonylureas) are insulin secretagogues that act by inhibiting the ATP-sensitive K⁺ channels of the beta cells of the pancreas to cause insulin release.

Solution to Question 2:

Repaglinide acts by increasing insulin secretion by blockage of ATP-sensitive K^+ channels of the beta cells of the pancreas.

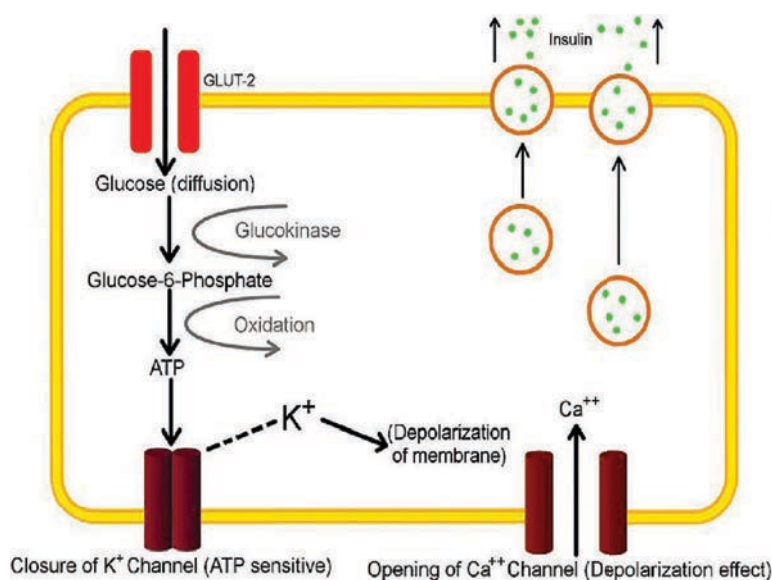
When ATP-sensitive K^+ channels are blocked, it will lead to depolarization of beta cells and the release of insulin.

Sulfonylureas also act by this similar mechanism.

Option B - Metformin is an insulin sensitizer and also inhibits gluconeogenesis and increases hepatic uptake of glucose.

Option C - Exenatide is a GLP-1 analog.

Option D - Sitagliptin is a DPP-4 inhibitor.



Solution to Question 3:

Chlorpropamide use is associated with dilutional hyponatremia.

It is a first generation sulfonylurea and can lead to SIADH.

Along with alcohol use, it can also lead to disulfiram-like reaction.

Solution to Question 4:

Metformin can cause vitamin B12 deficiency which manifests as megaloblastic anemia (pallor, fatigue) with knuckle pigmentation.

The most common side effect of metformin is GI intolerance as nausea, bloating, abdominal cramps and diarrhoea.

Note: The exact reason why metformin causes vitamin B12 deficiency is not known.

The proposed mechanisms include:

- Alterations in small bowel motility, which stimulates bacterial overgrowth and consequential vitamin B12 deficiency
- Competitive inhibition or inactivation of vitamin B12 absorption
- Alterations in intrinsic factor (IF) levels and interaction with the cubilin endocytic receptor.
- Inhibition of the calcium-dependent absorption of the vitamin B12-IF complex at the terminal ileum.

Solution to Question 5:

Metformin does not act by increasing insulin secretion.

It decreases insulin resistance and increases the peripheral utilization of glucose.

Metformin is a biguanide that acts by activation of AMP-dependent protein kinase activation and thereby mediates a series of actions that helps in reducing hepatic glucose production.

Option A - Metformin has minimal to nil effect on insulin secretion and as a result, has minimal or no hypoglycemic effect in non-diabetic patients.

Option C - Metformin reduces hepatic gluconeogenesis, increases the insulin-mediated peripheral glucose uptake and also decreases the absorption of glucose from the gut.

Option D - Increase intake of calcium will decrease the B12 malabsorption because metformin interferes with Ca²⁺ dependent B12 absorption.

Solution to Question 6:

The central pharmacological action of metformin is to reduce hepatic glucose production by inhibiting gluconeogenesis. Metformin effect is mainly mediated through activation of AMPK and altering the energy dynamics at mitochondrial respiration.

Option A - The reduction in glucose absorption can also occur but the contribution to the antihyperglycemic action is not significant

Option B - Metformin increases peripheral glucose utilization by both insulin-dependent (reducing insulin resistance) and independent ways. But the current evidence does not support the reduction in insulin resistance as the primary mechanism of metformin's action.

Option D - Recent evidence shows blunting of glucagon effects as one of the mechanisms of metformin's action

Note: All are mechanisms of metformin, however, the key action is Option C and it is the most appropriate answer.

Solution to Question 7:

According to current guidelines (KDIGO 2020), metformin should be discontinued only in patients with eGFR less than 30 ml/min/1.73 m² or kidney failure. Metformin dose should be reduced when the eGFR is less than 45 ml/min/1.73 m².

Metformin is excreted by the kidneys, and accumulation with reduced kidney function may increase the risk of lactic acidosis (due to inhibition of hepatic gluconeogenesis). It is contraindicated below eGFR < 30.

eGFR is obtained at baseline and at least annually in all patients taking metformin. It is recommended that the eligibility of CKD patients for metformin administration be changed from one based on a single laboratory parameter (blood creatinine concentration) to one that provides a better estimate of kidney function in patients with kidney disease i.e., glomerular filtration rate estimating equation (eGFR).

Solution to Question 8:

Metformin can be prescribed as it is FDA approved for medical treatment of type 2 diabetes in pediatric population as young as 10 years of age.

It is available as a liquid oral formulation (100mg/ml) for pediatric use.

Solution to Question 9:

Pioglitazone need not be avoided in all patients with cardiovascular disease.

Pioglitazone is known to cause fluid retention and edema and is therefore avoided in patients who have heart failure. It is not avoided in all cardiovascular diseases.

Option A - Pioglitazone is metabolized by CYP3A4 and CYP2C8.

Option B - Pioglitazone is a selective agonist of nuclear peroxisome proliferator-activated receptor (PPAR). It modulates the expression of genes involved in lipid and glucose metabolism, insulin signal transduction and adipocyte differentiation. These effects increase the sensitivity of the tissues to insulin.

Option C - Pioglitazone does not increase insulin levels in the body. It sensitizes the tissues to the action of insulin. Hence, insulin is needed for the action of pioglitazone. However, it does not need insulin to modulate the expression of genes.

Solution to Question 10:

The image shows diabetic macular edema, with hard exudates surrounding the blood vessels. It is a rare adverse effect seen with thiazolidinediones, and it resolves when the drug is discontinued.

Solution to Question 11:

This patient has hypertension and diabetes mellitus (HbA1c >6.5% is diagnostic for diabetes). Telmisartan would be a better alternative to losartan in patients with both hypertension and diabetes, due to its additional PPAR gamma agonist action.

PPAR gamma agonists act by inducing several insulins sensitizing genes and thereby reducing the overall insulin resistance.

While other angiotensin receptor blockers have been found to have PPAR gamma action, telmisartan has the highest and has been extensively studied in diabetes.

Solution to Question 12:

Rosiglitazone should be avoided as it increases the risk of cardiovascular events like myocardial infarction and stroke.

It remains banned in Europe, India, and other countries. It has not been withdrawn in the US and is marketed with a black box warning of increased risk of myocardial infarction and stroke.

Option A: Several studies have shown relative cardiovascular safety of pioglitazone over rosiglitazone.

Option C: Metformin use is associated with decreased risk of microvascular complications in diabetes.

Option D: Troglitazone was withdrawn due to increased hepatotoxicity.

Solution to Question 13:

Saxagliptin, a DPP-4 inhibitor, can be prescribed for this patient as it has a weight-neutral effect.

Note: *Metformin classification is a bit difficult. It can cause mild weight loss in some patients but few authorities do consider metformin as weight neutral.

EFFECT	DRUGS
Weight gain	Sulfonylureas, Insulin, Thiazolidinediones
Weight loss	GLP-1 analogues, Pramlintide, SGLT-2 inhibitors, Metformin*
Weight neutral	Alpha-glucosidase inhibitors, DPP-4 inhibitors

Solution to Question 14:

Linagliptin can be given safely without altering the dose for this patient.

Linagliptin is a dipeptidyl-peptidase-4 (DPP-4) inhibitor and is eliminated by the hepatobiliary route. Hence no dosage adjustment is needed in renal failure.

It is used in type 2 diabetes as monotherapy or as combination therapy with other drugs when metformin is not tolerated or contraindicated.

Other DPP-4 inhibitors like sitagliptin, vildagliptin, saxagliptin, alogliptin show renal elimination.

Solution to Question 15:

Dapagliflozin does not cause weight gain. It results in weight loss.

It acts by inhibiting the SGLT2 (sodium glucose-2) transporter, which is present on the proximal segment of the renal tubule and responsible for the reabsorption of 80-90% of filtered glucose. It inhibits glucose reabsorption resulting in glycosuria and a decrease in blood glucose level. It is effective orally.

The common adverse effects include urinary tract infections and genital mycotic infections due to the high glucose content of the urine and also the high risk of fractures.

Other SGLT2 inhibitors include empagliflozin, canagliflozin, and ertugliflozin.

Note: Empagliflozin has been found to decrease the risk of mortality from cardiovascular causes in diabetic patients. Dapagliflozin has been recently approved by the FDA for the treatment of heart failure with reduced ejection fraction (HFrEF) in both diabetic and non-diabetic patients.

Solution to Question 16:

Acarbose is least likely to cause hypoglycemia among the given options.

It is an alpha-glucosidase inhibitor. It inhibits the intestinal degradation and absorption of starch and disaccharides and thereby lowers the postprandial rise in glucose levels.

It has no role in stimulating insulin secretion and is therefore, least likely to cause hypoglycemia during monotherapy.

Insulin and insulin secretagogues like sulfonylureas (gliclazide) and meglitinides (repaglinide) can result in hypoglycemia.

Solution to Question 17:

Giddiness and palpitations after taking insulin and acarbose are suggestive of hypoglycemia. Honey is the most suitable agent among the given options for the management of hypoglycemia in a patient who is on acarbose therapy.

Acarbose is an alpha-glucosidase inhibitor. It inhibits the intestinal degradation of starch and disaccharides into glucose. Honey contains monosaccharides (fructose and glucose) that do not need to be broken down and are directly absorbed and would be beneficial in hypoglycemia.

caused by alpha-glucosidase inhibitors. However, glucose or dextrose tablets is recommended when available.

While alpha-glucosidase inhibitors themselves do not cause hypoglycemia, the risk of hypoglycemia is increased when they are taken along with insulin or insulin secretagogues.

Disaccharides and complex carbohydrates will not be completely broken down and absorbed in a patient who is on alpha-glucosidase inhibitors.

Option B - Bread predominantly contains complex carbohydrates like starch

Option C - Sugar contains sucrose, a disaccharide

Option D - Cake contains sugar (disaccharide) and other complex carbohydrates

Solution to Question 18:

Saroglitazar is a dual PPAR alpha/gamma agonist and is approved for diabetic dyslipidemia only in India.

Peroxisome proliferator-activated receptors (PPARs) are a superfamily of nuclear receptors involved in the regulation of energy homeostasis and metabolic function. It has three subtypes:

- PPAR α - reduces triglyceride level
- PPAR γ - causes insulin sensitization
- PPAR δ - enhances fatty acids metabolism

Of the three PPAR isoforms, synthetic pharmaceutical ligands that activate PPARalpha (the antidyslipidemic fibrates) and PPARgamma (the antidiabetic thiazolidinediones) have been studied extensively. Recently developed dual PPARalpha/gamma agonists combine the therapeutic effects of these drugs to be used in type 2 diabetes with dyslipidemia.

Options A and C - Aloglitazar and muraglitazar are also dual PPAR alpha/gamma agonists but their development was halted due to increased risk of cardiovascular mortality.

Option D - Rosiglitazone is a PPAR gamma agonist indicated in type 2 diabetes mellitus. It has been withdrawn from many countries including India due to increased cardiovascular mortality.

Solution to Question 19:

Bromocriptine is a dopamine agonist that has been approved as an adjunctive treatment for diabetes mellitus.

The exact mechanism is unknown but may involve alteration in circadian (hypothalamic) control of blood glucose levels. A fast-release, low-dose formulation of bromocriptine is approved for type 2 diabetes that has to be administered within 3 hours of awakening.

Option B - Phenytoin is a Na⁺ channel blocker and is an antiepileptic drug

Option C - Ropinirole is an antiparkinsonian drug that is a dopamine agonist and is also approved for treatment of restless leg syndrome

Option D - Metoclopramide is a 5HT₄ agonist and D₂ blocker and used as an antiemetic agent

Solution to Question 20:

Colesevelam can be prescribed for her as it is the only bile acid sequestrant that is specifically approved for the treatment of type 2 diabetes mellitus.

It is used to reduce LDL cholesterol and improve glycemic control in diabetics.

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Anti-Diabetic Drugs - Injectable

Question 1:

What is the half-life of endogenously secreted insulin?

- a) 1-2 minutes
- b) 5-6 minutes
- c) 20-30 minutes
- d) 1-2 hours

Question 2:

A 28-year-old nurse was brought to the casualty with seizures and altered mental status. Her plasma glucose was 35 mg/dl. Which test would help to rule out the possibility of exogenous insulin administration?

- a) Glucagon
- b) C-peptide
- c) Pancreatic polypeptide
- d) Preproinsulin

Question 3:

A patient requires the shortest acting formulation of insulin to prevent postprandial hyperglycemia. Which of the following formulations would you choose?

- a) Insulin lispro
- b) Insulin aspart
- c) Regular insulin
- d) Inhaled insulin

Question 4:

A patient on basal-bolus regimen for type 1 diabetes wants an alternative to multiple daily injections. She read about the new drug Afrezza and wants to know more about it. Which of

the following is not true about this formulation?

- a) Inhaled insulin
- b) Can be used as single or multiple pre meal doses
- c) If multiple doses used then no need of basal insulin
- d) Avoided in active smokers

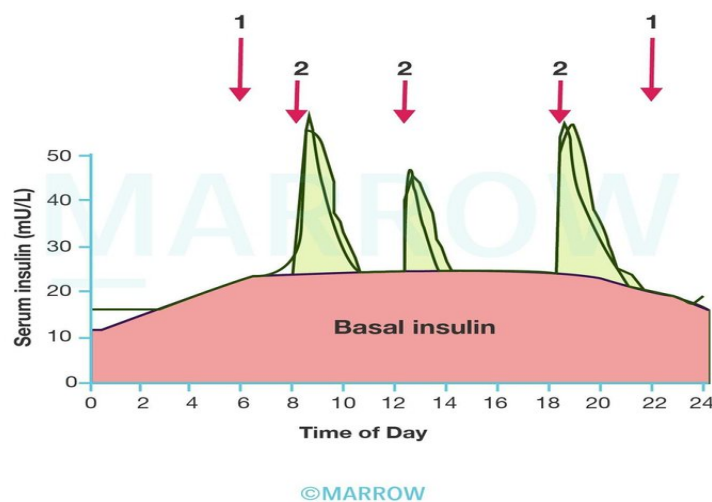
Question 5:

Which of the following preparations of insulin is not rapid-acting?

- a) Glargine
- b) Lispro
- c) Glulisine
- d) Aspart

Question 6:

Identify the correct insulin combination option.



- a) 1 - Glargine, 2 - Detemir
- b) 1 - Glargine, 2 - Degludec
- c) 1 - Detemir, 2 - Aspart
- d) 1 - Glulisine, 2 - Glargine

Question 7:

A 30-year-old patient with type 1 diabetes presented with complaints of vomiting and abdominal pain. On examination, she had fast, deep breathing with a fruity odor. Her capillary blood glucose was 450mg/dL. What is the clinical concern associated with administering insulin alone?

- a) Hypokalemia
- b) Hypernatremia
- c) Hyperkalemia
- d) Hyponatremia

Question 8:

You see a nurse administering an insulin preparation that appears cloudy white. Which formulation of insulin could it be?

- a) Insulin glargine
- b) Insulin glulisine
- c) Insulin detemir
- d) Insulin NPH

Question 9:

A patient was recently diagnosed with diabetes. She was prescribed the longest-acting insulin as it has the least risk of hypoglycemia. Which insulin preparation is she on?

- a) Insulin NPH
- b) Insulin glargine
- c) Insulin detemir
- d) Insulin degludec

Question 10:

A diabetic patient is in the postoperative area following cardiac surgery. He is to be given intravenous insulin for optimal glycemic control and to avoid sternal wound infection. Which of the following should not be administered?

- a) Insulin glulisine

- b) Insulin detemir
- c) Regular insulin
- d) Insulin aspart

Question 11:

Which of the following insulin precipitates into a cloudy white precipitate when mixed with short acting insulins?

- a) Insulin detemir
- b) Insulin degludec
- c) Insulin glargine
- d) Insulin glulisine

Question 12:

Composition of Lente insulin is:

- a) 70% amorphous 30% crystalline
- b) 70% crystalline 30% amorphous
- c) Only 70% amorphous
- d) Other name of NPH insulin

Question 13:

You are counseling a diabetic patient and teaching her the technique to self-administer insulin at home every day. Which of the following sites would you not suggest as the preferred site for insulin administration?

- a) Anterior part of thigh
- b) Lateral part of thigh
- c) Buttocks
- d) Medial part of thigh

Question 14:

Which of the following is incorrect regarding exenatide?

- a) It is a GLP- 1 analogue
- b) It can be used for the treatment of type 1 diabetes mellitus
- c) It is given subcutaneously
- d) It decreases glucagon

Question 15:

Which of the following is not a GLP-1 receptor agonist?

- a) Liraglutide
- b) Semaglutide
- c) Pramlintide
- d) Exenatide

Question 16:

Which of the following statements is not true about pramlintide?

- a) It can be used in both type 1 and type 2 diabetes mellitus
- b) It has got weight reducing property by causing a central anorectic effect
- c) It should not be administered with insulin in the same syringe
- d) It is given orally

Question 17:

A 40-year-old diabetic woman comes for follow up. Her BMI is 32 kg/m² despite several months of diet and exercise. Which of the following is the drug preferred for her condition?

- a) Liraglutide
- b) Linagliptin
- c) Insulin lispro
- d) Pioglitazone

Question 18:

A 45-year-old man comes for a routine health check-up. He has a family history of medullary thyroid cancer. His thyroid examination is normal but his HbA1c is found to be 7.5%. In view

of his history, which of the following should not be prescribed for him?

- a) SGLT 2 inhibitors
- b) DPP4 inhibitors
- c) GLP 1 agonists
- d) Sulfonylureas

Question 19:

A 50-year-old man with superior mesenteric artery occlusion underwent extensive bowel resection. Following surgery, he developed short bowel syndrome and has been on parenteral nutrition. Which of the following drugs would be beneficial for his condition?

- a) Liraglutide
- b) Lixisenatide
- c) Teduglutide
- d) Semaglutide

Question 20:

Identify the true statement regarding teduglutide.

- a) It is not recommended for patients with colorectal cancer
- b) It's half life is 6 - 8 hours
- c) It can be administered orally
- d) It promotes gastric acid secretion and gastric emptying

Answer Key

Question No.	Correct Option
1	b
2	b
3	d
4	c
5	a

6	c
7	a
8	d
9	d
10	b
11	c
12	b
13	d
14	b
15	c
16	d
17	a
18	c
19	c
20	a

Detailed Explanations

Solution to Question 1:

Endogenously secreted insulin has a half-life of only 5-6 minutes due to its extensive hepatic clearance.

Solution to Question 2:

C-peptide can be used to rule out the possibility of factitious hypoglycemia resulting from exogenously administered insulin.

Endogenous insulin is initially synthesized as preproinsulin that is processed to proinsulin inside beta cells. Proinsulin is released by secretory granules after processing it to form insulin and C-peptide.

Synthetic insulin usually lacks C-peptide, so in factitious hypoglycemia secondary to exogenous insulin administration, plasma insulin will be elevated, but there will be suppression of C-peptide and proinsulin, and insulin to C-peptide ratio will be greater than one.

On the other hand, insulinomas and insulin secretagogues like sulfonylureas stimulate endogenous insulin production. As a result, plasma insulin, C-peptide, and proinsulin levels will be all elevated. The only way to differentiate between insulinoma and insulin secretagogues-induced hypoglycemia is by detecting the drug, e.g., sulfonylurea, in the blood or urine.

C-peptide has no known physiological function or receptor. It has a half-life of about 30 min.

Solution to Question 3:

Inhaled insulin is the shortest acting formulation of insulin.

Solution to Question 4:

Afrezza should be used in combination with basal long-acting insulin to supplement it.

Option A - Afrezza is an inhaled insulin that is available in dry powder form. It is the shortest acting insulin (2-3 hours) and it reduces the use of short-acting injectable insulin. Adverse events include cough and throat irritation.

Option B - It is taken pre-meal to reduce postprandial hyperglycemia.

Option D - Afrezza is contraindicated in active smokers, asthmatics and COPD patients.

Solution to Question 5:

All of the following preparations of insulin are rapid-acting except glargine.

Insulin preparations and duration of action

Short-acting:

- Insulin lispro, Insulin aspart, Insulin glulisine: 3-4 hours
- Regular insulin: 4-6 hours

Long-acting:

- Insulin NPH: 10-16 hours
- Insulin detemir: 12-24 hours
- Insulin glargine: 12-24 hours
- Insulin degludec: 24-42 hours

Solution to Question 6:

Detemir is a long-acting insulin and aspart is a rapid-acting insulin and matches with 1 and 2 respectively in the image showing basal-bolus regimen.

Solution to Question 7:

Deep breathing (Kussmaul respiration) and fruity breath (acetone odor) in a symptomatic patient with type 1 diabetes are suggestive of diabetic ketoacidosis. Insulin therapy alone in a patient with diabetic ketoacidosis (DKA) may result in hypokalemia.

DKA is usually associated with a major K⁺ deficit. Potassium loss is caused by an extracellular shift of potassium (in exchange with hydrogen ions) due to acidosis. Much of the shifted extracellular potassium is lost due to emesis and osmotic diuresis.

When insulin therapy is initiated for DKA, the serum K⁺ is shifted intracellularly. Dangerous hypokalemia might occur if K⁺ supplementation is not initiated subsequently. Insulin should be withheld if the serum level of potassium is below 3.3 mmol/L.

Solution to Question 8:

The formulation is most likely NPH insulin as it produces a cloudy or whitish solution in contrast to the clear appearance of other insulin solutions.

NPH or isophane insulin is a suspension of native insulin complexed with zinc and protamine in phosphate buffer.

Solution to Question 9:

Insulin degludec has the maximum duration of action (24-42 hours) among long-acting insulins. It is sometimes considered an ultra long-acting insulin.

Solution to Question 10:

Insulin detemir is a long-acting insulin and should not be given intravenously or intramuscularly. It is only given via subcutaneous route.

Options A, C, and D: Regular insulin and short-acting insulins (glulisine and aspart) can be given intravenously. While short-acting insulins are approved for intravenous administration under supervision, regular insulin is preferred in perioperative glycemic control due to the lesser risk of hypoglycemia.

Intravenous insulin is used in diabetic ketoacidosis and other conditions where insulin requirements change drastically like perioperative period, labor, and in intensive care situations.

Solution to Question 11:

Insulin glargine forms a cloudy white precipitate when mixed with short-acting insulins.

Due to the acidic pH (4.0) of insulin glargine, it cannot be mixed with short-acting insulin preparations that are formulated at a neutral pH as it aggregates at neutral pH into a cloudy white

precipitate. This reduces the effectiveness of the insulin dose.

Insulin degludec and detemir have a pH of 7.4 that is very near to neutral pH and they do not form such aggregates when mixed with short-acting insulins.

But the manufacturers do not advise manual mixing with short-acting insulin as it may alter the activity profile of the combination that can turn unpredictable. Degludec is available as an approved fixed-dose combination with aspart that can be used with predictable effects.

Solution to Question 12:

Lente insulin is an intermediate-acting insulin and is a combination of Ultralente and Semilente in the ratio of 7:3

Two types of insulin zinc suspensions have been produced. The one with large particles is crystalline and practically insoluble in water (Ultralente). It is long-acting. The other has smaller particles and is amorphous (Semilente), is short-acting. Their 7:3 ratio mixture is called 'Lente insulin' and is intermediate-acting.

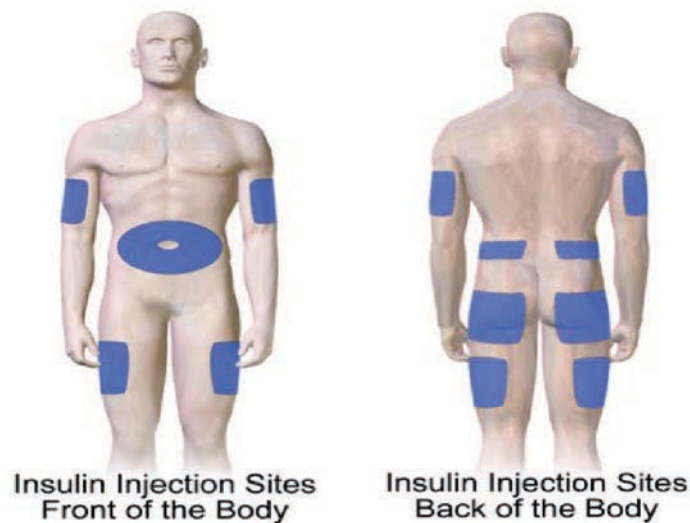
Solution to Question 13:

The medial part of the thigh is not a preferred site for insulin administration in patients. This is because of the relatively higher density of blood vessels and nerves in this area. Another consideration is that the inner thighs rub together while walking and may cause soreness at the injection sites.

The preferred sites for insulin administration are:

- Upper arm
- Anterior and lateral aspects of the thigh
- Buttocks
- Abdomen (with the exception of a circle with a 2-inch radius around the navel)

Rotation of the injection site is important to prevent lipodystrophy. However, rotating within one area is recommended (e.g., rotating injections systematically within the abdomen) rather than rotating to a different area with each injection. This helps decrease variability in absorption from day to day.



Solution to Question 14:

Exenatide is not indicated in type 1 diabetes mellitus.

Exenatide is a GLP-1 analog that acts as an insulin secretagogue. As the beta cells are already destroyed and do not secrete insulin in type 1 diabetes, there is no role of drugs that enhance insulin secretion.

Option A - Exenatide is a GLP-1 analog and it is synthesized from exendin-4

Option C - It is a peptide and hence has to be administered subcutaneously

Option D - It increases insulin secretion, decreases glucagon secretion, delays gastric emptying, and reduces food intake

Solution to Question 15:

Pramlintide is an islet amyloid polypeptide (IAPP) or amylin analogue.

Amylin analogues reduce glucagon secretion, slows gastric emptying, and centrally decreases appetite.

Options A, B and D - Liraglutide, semaglutide and exenatide are GLP-1 agonists.

Solution to Question 16:

Pramlintide is administered as a subcutaneous injection.

Option A - It is a synthetic analogue of amylin and is approved for both type 1 and type 2 diabetes mellitus

Option B - It has weight reducing property by causing a central anorectic effect. It acts by reducing glucagon secretion, delaying gastric emptying and decreasing appetite.

Option C - It should not be administered with insulin in the same syringe because of a difference in pH

Solution to Question 17:

Liraglutide is approved for the management of obesity.

GLP-1 is a physiological regulator of appetite and calorie intake, and the GLP-1 receptor is present in several areas of the brain involved in appetite regulation. Liraglutide is a GLP-1 analog and lowers body weight by decreasing calorie intake.

Note: FDA has recently approved semaglutide subcutaneous injection once weekly for chronic weight management. This is for use in patients with obesity and overweight individuals with at least one weight-related condition such as type 2 diabetes, hypertension or dyslipidemia.

Solution to Question 18:

GLP-1 analogues are contraindicated in patients with a personal or family history of medullary thyroid cancer and in patients with MEN 2 syndrome.

GLP-1 analogues cause thyroid C-cell tumours at clinically relevant exposures in rodents. However, studies have not yet shown if they cause thyroid C-cell tumours like medullary thyroid carcinoma in humans.

Solution to Question 19:

Teduglutide is a GLP-2 analog and is approved for the treatment of short bowel syndrome.

GLP-2 binds to enteric neurons and endocrine cells, stimulating the release of a number of hormones such as insulin-like growth factor. This stimulates mucosal epithelial growth and enhances fluid absorption.

GLP-1 analogs:

- Exenatide
- Liraglutide
- Albiglutide
- Dulaglutide
- Lixisenatide
- Semaglutide

Note: GLP-1 analogs are used for the treatment of Type II diabetes mellitus.

Solution to Question 20:

Teduglutide can potentially cause bowel cancer, therefore it is not recommended for patients with active malignancies.

It is a GLP-2 analog that has been recently approved for the treatment of short bowel syndrome. It inhibits gastric acid secretion and gastric emptying, stimulates mucosal epithelial growth and intestinal blood flow, and enhances nutrient and fluid absorption

The drug has a $t_{1/2}$ of 1–2 hours and is administered subcutaneously once daily.

Note: GLP-1 analogs are used for the treatment of Type II diabetes mellitus.

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OCPs, Estrogens and Progestins

Question 1:

Which of the following is not a natural estrogen?

- a) Estradiol
- b) Ethinylestradiol
- c) Estriol
- d) Estrone

Question 2:

A 55-year-old postmenopausal lady is on estrogen as part of her hormone replacement therapy. Which of the following statements about estrogen is not true?

- a) Decreases HDL
- b) Increases triglycerides
- c) Increases turnover of LDL receptors
- d) Increases apolipoprotein level

Question 3:

A post-menopausal lady who is on estrogen therapy for hot flashes is concerned about its adverse effects. She is not at an increased risk of developing which of the following cancers?

- a) Breast cancer
- b) Endometrial cancer
- c) Carcinoma of the gall bladder
- d) Hepatocellular carcinoma

Question 4:

In which of the following conditions is hormone replacement therapy not helpful?

- a) Vaginal atrophy
- b) Flushing

- c) Coronary heart disease
- d) Osteoporosis

Question 5:

Progestin dose remains constant throughout the cycle in which of the following type of combined contraceptive pills?

- a) Monophasic pills
- b) Biphasic pills
- c) Triphasic pills
- d) Multiphasic pills

Question 6:

A young lady who is on oral contraceptive pills refrains from taking any drug even for minor ailments due to the fear of contraceptive failure. Which of the following drugs can be taken safely?

- a) Aspirin
- b) Tetracycline
- c) Phenytoin
- d) Rifampicin

Question 7:

A 23-year-old lady has been on anti-epileptic therapy since childhood and now comes to the OPD seeking advice on a suitable method of contraception. Which contraceptive method should be avoided in her?

- a) Oral contraceptives
- b) Condoms
- c) Intrauterine contraceptive devices
- d) Post-coital pills

Question 8:

Which of the following statements about oral contraceptive pills is true?

- a) Progesterone is responsible for thromboembolism
- b) It is protective against ovarian cancer
- c) OCP are used in treatment mastalgia
- d) Rifampicin decreases the metabolism of OCP

Question 9:

A 23-year-old lady on oral combined contraceptive pills misses a single dose and reports to you in the OPD the next morning. What will be the most appropriate advice to this patient?

- a) Continue taking the next dose as scheduled
- b) Double the next dose to make up for the missed dose
- c) Interrupt the course and start taking the medications again on 5th day of bleeding
- d) Start a fresh course of OCP again

Question 10:

In which of the following patients is combined oral contraceptive pills not absolutely contraindicated?

- a) A perimenopausal woman with coronary heart disease
- b) A 23-year-old lady with a past history of hepatic adenoma
- c) A multiparous lady with abnormal undiagnosed vaginal bleeding
- d) A 40-year-old woman with migraine headache

Question 11:

Progestin-only pills do not act by which of the following mechanisms?

- a) Inhibition of ovulation
- b) Decreasing sperm penetration
- c) Impairing implantation
- d) Eliciting an inflammatory action in the endometrium

Question 12:

A 35-year-old lady, P2L2, has been taking depot medroxy progesterone acetate as a contraceptive. Which of the following best describes the formulation?

- a) Long acting progesterone preparation
- b) Long acting estrogen preparation
- c) Long acting anti-progestin preparation
- d) Short acting anti-progestin preparation

Question 13:

Which of the following progesterone is used in emergency contraception?

- a) Levonorgestrel
- b) Micronised progesterone
- c) Norgesterone
- d) Depot medroxyprogesterone acetate

Question 14:

A newly married couple came to the OPD saying that they had unprotected intercourse two days back but are not desiring a baby at present. Which of the following drugs can be prescribed in this case?

- a) Nomegestrel
- b) Tamoxifen
- c) Ulipristal
- d) Toremifene

Question 15:

The standard emergency contraceptive pill regimen, levonorgestrel can be taken up to how many hours after unprotected intercourse?

- a) 24 hours
- b) 48 hours
- c) 72 hours
- d) 120 hours

Question 16:

A 20-year old lady came to the OPD with complaints of missed period. You diagnose her to be 5 weeks pregnant but she is not willing to continue the pregnancy. You decide to prescribe her a drug that has anti-progestational activity. Which of the following drugs would you choose?

- a) Levonorgestrel
- b) Desogestrel
- c) Misoprostol
- d) Mifepristone

Question 17:

In the treatment of which of the following patients will you not consider clomiphene citrate?

- a) A 24-year-old lady with PCOS
- b) A 48-year old perimenopausal woman with hot flashes
- c) A 32-year-old man with oligospermia
- d) A 30-year old patient with SUNCT

Question 18:

A 45-year-old lady, diagnosed with breast cancer, is started on a selective estrogen receptor modulator. Identify the drug.

- a) Raloxifene
- b) Mifepristone
- c) Danazol
- d) Anastrozole

Question 19:

A 60-year-old lady who is on tamoxifen, for the treatment of breast cancer is worried about its side effects. Which of the following side effects of tamoxifen is least likely to occur in this patient?

- a) Increased risk of pulmonary embolism
- b) Endometrial carcinoma

- c) Cataract
- d) Cancer in opposite breast

Question 20:

Which of the following is a selective estrogen receptor down regulator?

- a) Tamoxifen
- b) Toremifene
- c) Fulvestrant
- d) Raloxifene

Question 21:

Which of the following is an aromatase inhibitor?

- a) Tamoxifen
- b) Danazol
- c) Ospemifene
- d) Exemestane

Question 22:

You decide to change the chemotherapeutic regimen of a patient with breast cancer from tamoxifen to an aromatase inhibitor. Which of the following is not an advantage of the latter in the treatment of breast cancer?

- a) No increase in risk of venous thromboembolism
- b) No increase in risk of endometrial cancer
- c) Can be used in advanced cases which are tamoxifen-resistant
- d) They maintain bone mineral density

Question 23:

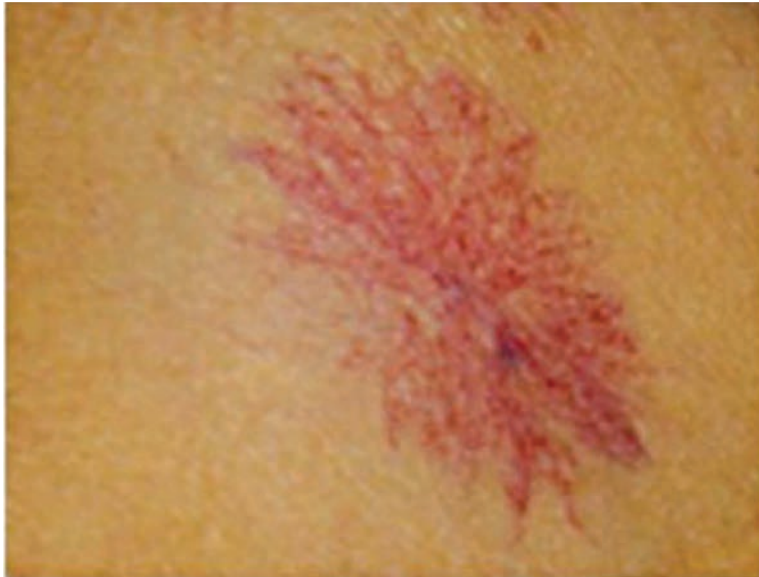
Yuzpe's regimen of emergency contraception consists of which of the following?

- a) Etonogestrel + ethinylestradiol

- b) Levonorgestrel + ethinylestradiol
- c) Levonorgestrel + desogestrel
- d) Norethisterone enanthate + estradiol valerate

Question 24:

You are an intern posted in the medicine ward and you notice the given dermatological finding in a patient with abdominal distension. Which of the following hormones is implicated in this finding?



- a) Testosterone
- b) Estrogen
- c) Hepatotoxins
- d) FSH

Question 25:

A 28-year-old man presents to the OPD with the following condition. Which of the following drugs can cause this adverse effect?



- a) 1 and 2
- b) 1,2 and 3
- c) 3,4 and 5
- d) Only 3

Answer Key

Question No.	Correct Option
1	b
2	a
3	c
4	c
5	a
6	a
7	a
8	b
9	b
10	d
11	d
12	a
13	a
14	c

15	c
16	d
17	b
18	a
19	d
20	c
21	d
22	d
23	b
24	b
25	c

Detailed Explanations

Solution to Question 1:

Ethinylestradiol is a synthetic estrogen.

Natural estrogens are:

- Estradiol
- Estriol
- Estrone

Solution to Question 2:

Estrogen causes an increase in plasma HDL levels.

Estrogen also causes an increase in triglycerides, apolipoprotein levels, and an increase in the turnover of LDL receptors.

Solution to Question 3:

Women receiving estrogen therapy do not have an increased risk of developing gall bladder carcinoma.

Long-term estrogen therapy can increase the risk of gall stones. Although gall stones can be associated with the development of gall bladder carcinoma, there is no direct correlation between long term estrogen therapy and gall bladder carcinoma.

Estrogen therapy has been associated with breast, endometrial and hepatocellular carcinoma.

Solution to Question 4:

Hormone replacement therapy(HRT) is not helpful in coronary heart disease.

Progesterone used in the HRT is associated with an increased risk of coronary artery disease, thromboembolism, and related mortality. Estrogens promote coagulation and thromboembolic episodes.

Options A and B: Estrogen is helpful in flushing and local preparations are preferred for vaginal atrophy.

Option D: Estrogen is beneficial for osteoporosis if it is started before the onset of significant osteoporosis.

Note: The use of HRT for cardiovascular benefits is quite controversial. The current consensus is that the HRT should be started only after assessing the benefit/risk profile individually and is not indicated for the prevention of cardiovascular events in all menopausal women as it was previously advocated.

Solution to Question 5:

Monophasic pills contain constant amounts of progesterone and estrogen that is taken daily for 21 days, followed by a 7-day pill-free period.

Biphasic and triphasic pills have variable amounts of active ingredients (especially progesterone) that mimic the natural changes in the hormonal levels in the menstrual cycle. These pills reduce the total amount of steroids that needs to be administered.

Solution to Question 6:

Among the given drugs, aspirin use is least likely to cause contraceptive failure.

Option B: Broad-spectrum antibiotics like tetracycline suppress the intestinal flora and interfere with the enterohepatic circulation of estrogen. This leads to lowering of plasma estrogen levels and can result in contraceptive failure.

Options C and D: Enzyme inducers like phenytoin and rifampicin increase the metabolism of oral contraceptives and can cause contraceptive failure.

Solution to Question 7:

Oral contraceptives must be avoided or used cautiously in epilepsy.

The most important aspect to consider is the drug interactions between anti-epileptic drugs and contraceptives. There is no conclusive evidence on whether estrogen is beneficial or harmful for

seizure control.

Note: Convulsive disorder is not an absolute contraindication; it is safe to avoid but can be used with caution.

Solution to Question 8:

Oral contraceptive pills (OCP) protect against ovarian cancer.

This benefit is thought to be due to the prolonged suppression of gonadotropin stimulation of the ovary on long-term use of OCPs.

Other benefits of oral contraceptive pills:

- Lesser incidence of endometrial carcinoma due to progestin component in OCP
- Regularisation of menstrual cycles results in relief of dysmenorrhea and menorrhagia
- Reduced incidence of fibrocystic breast disease and ovarian cysts
- Endometriosis and pelvic inflammatory disease are improved

Option A: The estrogen component of combined oral contraceptive pills is mainly responsible for venous thromboembolism.

Option C: OCP is not used in the treatment of mastalgia. Mastalgia, chloasma, and breakthrough bleeding are known side effects of oral contraceptive pills.

Option D: OCPs are metabolized by CYP3A4 enzymes. Rifampicin is a strong inducer of CYP3A4. Hence, if administered together, rifampicin can cause enhanced metabolism of OCPs and may lead to contraceptive failure.

Solution to Question 9:

The most appropriate advice for this patient who has missed a single dose of the oral contraceptive pill (OCP) is to double the next dose to make up for the missed dose.

Missed dose considerations for oral combined contraceptive pills:

- If a single dose is missed - a double dose can be taken on the next day to make up for the missed dose.
- If two or more doses are missed on subsequent days - the last missed pill has to be taken along with the present pill and alternative methods of contraceptive must be used for the next 7 days.
- If the pills were missed on the last week (days 15-21) - omit the hormone-free interval by finishing the hormonal pills in the current pack and starting the new pack from the next day.

Solution to Question 10:

A migraine headache is a relative contraindication for the use of combined oral contraceptives. Migraine with focal neurological symptoms is an absolute contraindication to combined oral contraceptive use.

Absolute contraindications to combined oral contraceptive use:

- Presence or history of thromboembolic disease, cerebrovascular disease, myocardial infarction, coronary artery disease
- Congenital hyperlipidemia
- Known or suspected carcinoma of the breast, female reproductive tract
- Known or suspected pregnancy
- Past or present liver tumors or impaired liver function
- Undiagnosed vaginal bleeding.
- Migraine with focal neurological symptoms

Relative contraindications to combined oral contraceptive use:

- Migraine headaches in ≥ 35 yrs female
- Hypertension
- Diabetes mellitus
- Obstructive jaundice of pregnancy
- Prior oral contraceptive use
- Gallbladder disease

Solution to Question 11:

Copper intrauterine devices elicit an inflammatory reaction within the endometrium. This impairs sperm viability, motility, and fertilization.

All the following actions contribute to the contraceptive action of progestin-only pills:

- The effectiveness of this drug is largely due to the thickening of cervical mucus which in turn impair sperm penetration.
- Inhibits ovulation via exerting negative feedback on the hypothalamus (GnRH). This leads to the absence of the FSH and LH surge, which is essential to mature the dominant follicle and trigger ovulation.
- Interfere with the implantation of a zygote by altering the endometrial environment.

The major reason women discontinue the use of progestin-only contraceptives is breakthrough bleeding.

Solution to Question 12:

Depot medroxyprogesterone acetate (DMPA) is a long-acting progestin-only injection formulation. DMPA is administered every 3 months intramuscularly as a contraceptive.

Solution to Question 13:

Levonorgestrel is the progesterone that is used in emergency contraception. Levonorgestrel-containing regimens are considered the standard regimen now.

Option B: Micronised progesterone is indicated for amenorrhea and to counteract estrogen effects on the uterus.

Option C: Norgestrel is usually used along with ethinyl estradiol for regular contraception.

Option D: Depot medroxyprogesterone acetate is a long-acting depot preparation indicated for long-term contraception.

Solution to Question 14:

Ulipristal is a selective progesterone receptor modulator (SPRM) that can be given as an emergency contraceptive in this case.

It should be administered within 5 days of unprotected intercourse.

It inhibits ovulation by suppressing LH surge as well as by direct effect on follicular rupture. In addition, its action on the endometrium can interfere with implantation.

Solution to Question 15:

Levonorgestrel regimen can be taken within 72 hours after unprotected intercourse but is most effective when taken as early as possible.

Levonorgestrel for emergency contraception can be taken as either of the following:

- 2 tablets of 0.75 mg 12 hours apart
- 1.5 mg single-dose

Solution to Question 16:

Mifepristone (RU-486) has potent anti-progestational (abortive activity) and significant anti-glucocorticoid and antiandrogenic activity.

Options A and B: Levonorgestrel and desogestrel are progesterone preparations.

Option C: Misoprostol is a synthetic prostaglandin (PGE₁) preparation.

Solution to Question 17:

Clomiphene citrate is not used in the treatment of hot flashes.

Option A- Clomiphene increases gonadotropin secretion and stimulates ovulation. It mainly acts by blocking the inhibitory actions of estrogen on gonadotropin release from the pituitary gland. It is primarily indicated in the management of infertility.

Option C - In men also, clomiphene increases gonadotropin secretion and promotes spermatogenesis. However, success rates are low.

Option D - Studies have shown clomiphene citrate to be effective in treating and preventing short-lasting unilateral neuralgiform headache attacks (SUNCT). It has not yet been FDA-approved for the treatment of SUNCT.

Solution to Question 18:

Raloxifene is a selective estrogen receptor modulator.

It has agonist action on bone and antagonist action on the breast and uterus.

It is used in the treatment and prevention of osteoporosis and in breast cancer prevention in high-risk postmenopausal women.

Solution to Question 19:

Tamoxifen acts as an estrogen receptor antagonist in breast tissues. The use of tamoxifen in carcinoma of breast patients decreases the incidence of carcinoma in opposite breasts.

The estrogen agonistic action of tamoxifen and its consequences are:

- Uterine endometrium - endometrial hypertrophy, vaginal bleeding, and endometrial cancer
- Coagulation system - thromboembolism leading to pulmonary embolism (tamoxifen decreases the level of antithrombin III and protein C in the blood)
- Bone metabolism - increase in bone mineral density
- Liver - alterations of blood lipid profile
- Cataract

Solution to Question 20:

Fulvestrant is a selective estrogen receptor down-regulator. It does not have estrogen agonistic activity for any tissue and is regarded as a pure estrogen antagonist.

Tamoxifen, toremifene, and raloxifene are selective estrogen receptor modulators with the agonist or antagonist action at different tissues.

Solution to Question 21:

Exemestane is an aromatase inhibitor.

Estrogen is locally produced from androgen by the action of the aromatase enzyme. Locally synthesized estrogens play an important role in the development of breast cancer because they are estrogen-dependent for proliferation.

Aromatase inhibitors decrease estrogen synthesis and are widely used now in the treatment of breast cancer in postmenopausal women.

Aromatase inhibitors are:

- Letrozole
- Anastrozole
- Exemestane (Irreversible inhibitor)

Solution to Question 22:

Aromatase inhibitors lack the property of maintaining bone mineral density when compared to tamoxifen.

Aromatase inhibitors dramatically reduce estrogen levels that result in a decrease in the risk of venous thromboembolism and also decreases the risk of endometrial hyperplasia/cancer. It is also indicated in tamoxifen-resistant advanced breast cancer.

Solution to Question 23:

Yuzpe's regimen of emergency contraception consists of a combination of levonorgestrel and ethinylestradiol.

The regimen includes two doses at 12 hours intervals within 72 hours of unprotected intercourse. It has a higher incidence of nausea and vomiting and is less efficacious than progesterone only standard regimen.

Methods of Emergency contraception:

Drug	Dose
Levonorgestrel	0.75mg stat and after 12 hours
Ethinyl estradiol 50 µg + Nor gestrel 0.25 mg	2 tab stat and 2 tab after 12 hours

Drug	Dose
Conjugated estrogen	15 mg BD x 5 days
Ethinyl Estradiol	2.5 mg BD X 5 days
Copper IUDs	Insertion within 5-7 days after accidental unprotected exposure. Not suitable for rape victims
Centchroman	2 tablets (60 mg) to be taken twice at an interval of 12 hours within 24 hours of intercourse.
Ulipristal	30 mg within 120 hours

Solution to Question 24:

The image shown is of spider nevus and the hormone implicated in its development is estrogen.

It is commonly seen in hyperestrogenic states such as pregnancy and in those taking estrogen-containing contraceptives. It can also be seen in male patients with cirrhosis due to the decreased ability of the liver to metabolize circulating estrogens, specifically estrone, which derives from the androgen androstenedione. Estrogens cause blood vessels to enlarge and dilate.

Solution to Question 25:

This is a case of gynecomastia which can occur as an adverse effect of spironolactone or ketoconazole or chlorpromazine.

Spironolactone - decreases testosterone production, increases peripheral conversion of testosterone to estradiol, and displaces estradiol from their plasma-protein-binding sites

Ketoconazole - decreases testosterone production and displaces testosterone from their binding sites

Chlorpromazine (neuroleptics) - increase prolactin release by blocking the inhibitory action of dopamine on prolactin

Clomiphene and tamoxifen are used mainly in reducing male gynecomastia.

Androgens and Drugs for Erectile Dysfunction

Question 1:

Which of the following is the most potent androgen?

- a) Dehydroepiandrostenidione
- b) Dihydrotestosterone
- c) Androstenedione
- d) Testosterone

Question 2:

Aromatase converts testosterone into _____

- a) Dihydrotestosterone
- b) Estradiol
- c) Epiandrosterone
- d) Androstenedione

Question 3:

A 59-year-old man with metastatic prostate cancer is started on flutamide therapy. What is the mechanism of action of this drug?

- a) 5-alpha reductase inhibitor
- b) Non-competitive androgen receptor inhibitor
- c) Competitive androgen receptor inhibitor
- d) Selective Androgen Receptor Modulator (SARM)

Question 4:

A 25-year-old medical student is worried about his receding hairline and is thinking about taking finasteride. The drug enhances hair growth by acting on which of the following?

- a) Aromatase enzyme
- b) 5 alpha reductase enzyme

- c) Androgen receptors
- d) PDE-5 enzyme

Question 5:

Which of the following inhibits spermatogenesis?

- a) Gemfibrosil
- b) Gemcadiol
- c) Gestodine
- d) Gossypol

Question 6:

A 30-year-old male patient presented to the urology OPD with complaints of erectile dysfunction and was prescribed a PDE-5 inhibitor. Identify the drug.

- a) Podofilox
- b) Finasteride
- c) Apalutamide
- d) Tadalafil

Question 7:

A 40-year-old diabetic patient was prescribed sildenafil for his erectile dysfunction. Following this, he develops a blurring of vision and complains that things seem to appear with a bluish-green hue. Inhibition of which of the following enzymes is likely to have caused his visual side effects?

- a) Phosphodiesterase 3
- b) Phosphodiesterase 5
- c) Phosphodiesterase 6
- d) Phosphodiesterase 2

Question 8:

Which of the following is a prostaglandin that can be administered as an intracavernosal injection for erectile dysfunction?

- a) Dinoprost
- b) Dinoprostone
- c) Alprostadil
- d) Misoprostol

Question 9:

A 35-year-old physician presented to the ED with a prolonged painful erection induced by intracorporeal injection of an alprostadil+phentolamine combination. He had been using this technique to overcome occasional performance anxiety. Corporeal aspiration reveals dark coloured blood and based on further workup, a diagnosis of low flow priapism is made. Which of the following is the drug of choice to treat his condition?

- a) Finasteride
- b) Phenoxybenzamine
- c) Phenylephrine
- d) Prazosin

Question 10:

A 19-year-old male athlete seeks your opinion on the usage of anabolic steroids to improve his performance. What would be your advice?

- a) Reassure the patient that all anabolic steroids are safe to use
- b) Explain clearly that only newer anabolic steroids are safe to use as they have negligible androgenic action
- c) Advice against use and sensitise the patient that all anabolic steroids have significant androgenic action
- d) Explain that older testosterone preparations are more appropriate for use in male athletes

Answer Key

Question No.	Correct Option
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1	b
2	b
3	c
4	b
5	d
6	d
7	c
8	c
9	c
10	c

Detailed Explanations

Solution to Question 1:

Dihydrotestosterone is the most potent androgen.

Solution to Question 2:

Aromatase (CYP19) converts testosterone to estradiol. Certain effects of testosterone are mediated by estradiol.

In men with deficient CYP19 or ER receptors, the epiphyses do not fuse, the long bones grow indefinitely and such patients are osteoporotic. Administration of estradiol can correct the bone abnormalities in CYP19 deficiency. However, this cannot be used in those with ER receptor defects.

Testosterone is converted to dihydrotestosterone (DHT) by 5-alpha-reductase.

Androstanedione and epiandrosterone are metabolites of DHT due to the action of 17-beta-HSD.

Solution to Question 3:

Flutamide acts by competitive inhibition of androgen receptors and is used to treat prostate cancer.

Flutamide is a non-steroidal antiandrogen that competes with androgens such as testosterone and dihydrotestosterone for binding to the androgen receptors in the prostate gland. By doing so, it prevents them from stimulating the prostate cancer cells to grow.

In addition to its use in prostate cancer, flutamide has been used to treat hyperandrogenism (excess androgen levels) in women, such as in those with polycystic ovary syndrome (PCOS) and

Hirsutism.

Other anti-androgens are:

- Bicalutamide, apalutamide, and nilutamide
- Spironolactone (aldosterone antagonist with antiandrogen activity)
- Cyproterone acetate (progestin with antiandrogen activity)
- Cimetidine (H₂ blocker with anti-androgen activity)

Selective Androgen Receptor Modulators (SARMs) are a class of androgen receptor ligands that have similar anabolic properties to anabolic steroids, but with reduced androgenic side effects. There are no FDA-approved SARMs at present.

Solution to Question 4:

Finasteride is a competitive inhibitor of 5 α -reductase enzyme and thereby inhibits the conversion of testosterone to dihydrotestosterone.

Finasteride selectively inhibits the Type 2 isoenzyme of 5 α -reductase. It is indicated for use in:

- Benign prostatic hyperplasia
- Male pattern baldness
- Hirsutism.

Solution to Question 5:

Gossypol is a non-steroidal agent that inhibits spermatogenesis.

Gossypol is obtained from cottonseed. It inhibits spermatogenesis by exerting toxic effects on seminiferous tubules but has low efficacy. Libido and potency are not significantly affected and fertility usually returns after a few months of discontinuation of the drug. But, around 10% of the patients may remain infertile. Hypokalemia is a major adverse effect and may even lead to transient paralysis.

Low efficacy and significant toxicity have caused gossypol to be abandoned as a candidate male contraceptive.

Solution to Question 6:

Tadalafil is a PDE-5 inhibitor used in the treatment of patients with erectile dysfunction.

Penile erection is mediated by NO and cGMP. During arousal, the corpus cavernosum produces nitric oxide which in turn stimulates the formation of cGMP. This leads to relaxation of the smooth muscle of corpus cavernosum and penile arteries, causing engorgement of corpus cavernosum producing an erection. The biological actions of cGMP are terminated by

phosphodiesterase enzymes and PDE-5 is the major cGMP metabolizing enzyme in the penile tissue.

Phosphodiesterase-5-inhibitors include:

- Sildenafil
- Tadalafil - Longest acting (24-36 hrs), bioavailability is not affected by food
- Vardenafil
- Avanafil - Most Rapid-acting (within 15 -30 mins)

Option A: Podofilox topical cream is used in the treatment of genital warts and molluscum contagiosum lesions.

Option B: Finasteride is a 5 α -reductase inhibitor.

Option C: Apalutamide is an androgen receptor antagonist.

Solution to Question 7:

Visual disturbance seen with sildenafil is due to PDE-6 inhibition.

Sildenafil is used for the treatment of erectile dysfunction principally due to its inhibitory actions on PDE-5 (Phosphodiesterase 5) in the vascular tissues of the corpus cavernosa. However, it also exerts minor inhibitory action against PDE-6, which is present exclusively in rod and cone photoreceptors and results in blurring and blue-green tinting of vision.

Tadalafil and avanafil have better PDE-5/PDE-6 selectivity, compared to sildenafil and vardenafil and hence have lesser visual side effects.

Solution to Question 8:

Alprostadil can be given as a direct intracavernosal injection for erectile dysfunction.

Alprostadil is a natural PGE₁ preparation. It is usually used in patients who are not responding to PDE-5 inhibitors. It is also available as an intraurethral suppository.

Solution to Question 9:

Phenylephrine is the drug of choice for the treatment of low flow priapism.

Characteristics of low-flow priapism include painful rigid erection and ischemic corpora, indicated by dark blood upon corporeal aspiration.

Prazosin, and, phenoxybenzamine are alpha-blockers that cause vasodilation and can worsen priapism.

Solution to Question 10:

All anabolic steroids have significant androgenic action and the use of anabolic steroids by athletes is not recommended.

Testosterone and its derivatives are used for their anabolic effects but they have significant androgenic action. Some of them (17 methyl derivatives) are also hepatotoxic. The current consensus is to advise against the use of such anabolic steroids.

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Drugs Acting on Uterus

Question 1:

A patient is administered oxytocin as part of the active management of the third stage of labor (AMTSL) protocol. Which of the following statements is correct regarding this drug?

- a) It is responsible for galactokinesis
- b) It can be given by oral route
- c) It is secreted by anterior pituitary
- d) At higher doses, it exerts diuretic like action

Question 2:

Which of the following is not an oxytocic?

- a) Ethacridine
- b) Ergometrine
- c) Pitocin
- d) Orciprenaline

Question 3:

A 32-year-old multipara at 40 weeks gestation developed water intoxication and hyponatremia following prolonged IV infusion with oxytocin by an inexperienced junior resident. This adverse effect is due to oxytocin's action on which of the following sites?

- a) Vasopressin V2 receptors on serosal surface of renal tubular cells
- b) Vasopressin V2 receptors on luminal surface of renal tubular cells
- c) Vasopressin V1 receptors on serosal surface of renal tubular cells
- d) Vasopressin V1 receptors on luminal surface of renal tubular cells

Question 4:

In which of the following patients is oxytocin not indicated?

- a) A primiparous woman with spontaneous premature labor

- b) A multiparous woman with postpartum haemorrhage
- c) A primiparous woman with uterine inertia
- d) A patient with breast engorgement due to inefficient milk ejection reflex

Question 5:

A 33-year old lady with an obstetric score of G4P3L3 delivered a baby by cesarean section. She is at risk of developing uterine atony and the resident asks you to bring a long-acting analog of oxytocin. Which drug will you get?

- a) Desamino-oxytocin
- b) Carbetocin
- c) Solifenacin
- d) Darifenacin

Question 6:

Which of the following is a synthetic PGF₂ alpha derivative used as an abortifacient?

- a) Dinoprost
- b) Dinoprostone
- c) Misoprostol
- d) Carboprost

Question 7:

Which of the following is/are the primary indication(s) of ergot alkaloids in obstetrics?

- a) 2,3
- b) 1,2,3
- c) Only 3
- d) 1,2,3,4,5

Question 8:

You administer methyl ergometrine to a patient with postpartum hemorrhage in the labor ward. Which of the following adverse effects will you not expect to occur in this patient?

- a) Nausea and vomiting
- b) Hypotension
- c) Decrease in pain threshold
- d) Decrease milk secretion postpartum

Question 9:

A 23-year old woman, G2P1L1 at 33 weeks of gestation is diagnosed with spontaneous preterm labor. The senior resident instructed you to give the patient antenatal steroids and a tocolytic. Which of the following drugs will you not choose?

- a) Atosiban
- b) Dopamine
- c) Nifedipine
- d) Isoxsuprine

Question 10:

A 30-year-old pregnant woman has an obstetric history of preterm labor and 2 spontaneous abortions. She is very anxious about her current pregnancy and enquires if she can take any medication to prevent preterm labor. Which of the following drugs can be given to her?

- a) Ritodrine
- b) Progesterone
- c) Terbutaline
- d) Nifedipine

Answer Key

Question No.	Correct Option
1	a
2	d
3	a
4	a
5	b

6	d
7	c
8	b
9	b
10	b

Detailed Explanations

Solution to Question 1:

Oxytocin is responsible for galactokinesis (milk ejection reflex).

Option B: It is a peptide and inactive orally.

Option C: It is secreted by posterior pituitary along with vasopressin.

Option D: At higher doses, it can exert vasopressin like action and cause water retention.

Solution to Question 2:

Orciprenaline is a beta 2 receptor agonist that has uterine relaxing property.

Oxytocics are uterine stimulants that increase uterine motility, especially at term.

Uterine stimulants are:

- Posterior pituitary hormones - oxytocin/pitocin
- Ergot derivatives - ergometrine, methylergometrine
- Prostaglandins - PGE₂, PGF₂ alpha, and others
- Others - quinine, ethacridine

Ethacridine is used as an abortifacient.

Solution to Question 3:

Oxytocin can bind to V₂ receptors on the serosal surface of renal tubular cells resulting in reduced free water clearance.

This is because oxytocin and vasopressin have structural similarities. Oxytocin can have vasopressin (ADH) like activity at V₂ receptors. The increase in water permeability occurs through the insertion of aquaporins (water channels) in collecting tubules.

Prolonged i.v. administration at high doses in conjunction with large volumes of hypotonic fluid may cause water intoxication associated with hyponatremia.

To avoid this complication, whenever high doses of oxytocin are administered over a long time:

- An electrolyte-containing diluent must be used (not dextrose)
- The volume of infused fluid should be kept low (by infusing oxytocin at a higher concentration)
- Fluid intake must be restricted, fluid balance chart should be kept and serum electrolytes should be monitored

Solution to Question 4:

Spontaneous premature labor is not an indication for oxytocics. Infact, it is an indication for tocolytics.

Uses of oxytocin are:

- Induction of labor
- Postpartum hemorrhage
- Uterine inertia
- Breast engorgement due to insufficient milk ejection reflex

Solution to Question 5:

Carbetocin is a long-acting analog of oxytocin.

It is indicated for the prevention of uterine atony following caesarian section and for prevention of PPH.

Option A: Desamino-oxytocin is a buccal formulation of oxytocin that is similar to oxytocin injection.

Options C and D: Solifenacin and darifenacin are synthetic tertiary amine antimuscarinic agents that are indicated in overactive bladder.

Solution to Question 6:

Carboprost is a 15-methyl synthetic derivative of PGF₂ alpha. It is used as an abortifacient and in refractory PPH.

Options A: Dinoprost is a natural PGF₂ alpha preparation.

Option B: Dinoprostone is natural PGE₂ preparation.

Option C: Misoprostol is a synthetic PGE₁ analog. Gemeprost is also a synthetic PGE₁ analog.

Solution to Question 7:

Ergot derivatives are indicated for the prevention and/or treatment of postpartum hemorrhage.

Ergot alkaloids markedly increase the motor activity of the uterus. The capacity to induce sustained uterine tetany and fetal distress exclude their use in the initiation or facilitation of labor.

Due to the high incidence of postoperative vomiting and lack of evidence in reducing bleeding, routine or prophylactic use of uterotonic medications at the time of MTP is not recommended.

Ergot alkaloids used in PPH include ergometrine and methylergometrine.

Solution to Question 8:

Administration of Ergot alkaloids (such as methyl ergotamine) result in hypertension.

The use of ergot alkaloids is preferably avoided in hypertensive patients.

Adverse effects of ergometrine/methylergometrine are:

- Nausea and vomiting
- Hypertension
- Decrease in pain threshold but analgesia might be required
- Decrease in milk secretion postpartum (due to dopaminergic action especially at higher doses)

Solution to Question 9:

Dopamine does not have any significant activity on the tone of the uterus, since uterine relaxation is mediated by beta-2 receptors.

Specific tocolytic agents are:

- Beta 2 agonists - ritodrine, isoxsuprine, terbutaline
- Calcium channel blockers - nifedipine
- Oxytocin receptor antagonist - atosiban
- Magnesium sulfate
- Indomethacin

Solution to Question 10:

Progesterone can be given prophylactically to reduce the risk of preterm labor.

It inhibits the secretion of pro-inflammatory cytokines and delays cervical ripening. Therefore, progesterone and its derivatives have been advocated to diminish the onset of preterm labor in women at high risk due to previous preterm delivery.

Intramuscular hydroxyprogesterone administered weekly as well as vaginal administration has both been shown to reduce the incidence of preterm delivery.

Ritodrine, terbutaline, and nifedipine are tocolytics that are indicated for inhibition of uterine contractions of preterm labor and not for prophylaxis against preterm labor.

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NSAIDs

Question 1:

Which of the following statements is true?

- a) NSAIDs act by inhibiting lipooxygenase.
- b) NSAIDs act by inhibiting the phospholipases.
- c) Cyclooxygenase-1 is a housekeeping enzyme.
- d) Phospholipase-A2 is a housekeeping enzyme.

Question 2:

A patient with a history of myocardial infarction asks you why he is being prescribed aspirin. You tell him it prevents the formation of clots which reduces the risk of MI. Formation of which of the following eicosanoids is blocked by aspirin for this purpose?

- a) PGI₂
- b) PGD₂
- c) Thromboxane A₂
- d) All of the above

Question 3:

Which of the following helps in maintaining the patency of ductus arteriosus?

- a) PGF₂^o
- b) PGE₁
- c) PGH₂
- d) PGI₂

Question 4:

A patient of CREST syndrome has developed pulmonary hypertension. Which of the following drugs will you not prescribe to treat this?

- a) Epoprostenol

- b) Iloprost
- c) Treprostinil
- d) Rioprostil

Question 5:

A 65-year-old man has presented to the OPD with a complaint of erectile dysfunction not responding to oral medication. The issue is determined to be of vascular origin. Which prostaglandin preparation can be given as a local injection to treat this?

- a) Dinoprost
- b) Epoprostenol
- c) Enprosnil
- d) Alprostadil

Question 6:

Which of the following statements is false about aspirin?

- a) It is an irreversible inhibitor of the cyclooxygenase enzyme.
- b) Its clearance is independent of plasma concentration at low dose.
- c) Alkalinization of urine increases its rate of excretion.
- d) Its antiplatelet effect at low dose is related to presystemic COX inhibition.

Question 7:

A 12-year-old child developed fever during an epidemic of influenza. Which of the following drugs is contraindicated in this child?

- a) Acetaminophen
- b) Aspirin
- c) Oseltamivir
- d) Baloxavir

Question 8:

A 30-year-old pilot with low back pain requires a painkiller that is effective at once-daily dosing. You decide to prescribe the NSAID with the longest plasma half-life. Which drug will you choose?

- a) Aspirin
- b) Phenylbutazone
- c) Piroxicam
- d) Ibuprofen

Question 9:

A patient with a history of gastritis complains of joint pain. You diagnose it to be a case of osteoarthritis and prescribe celecoxib. Which of the following statements is true regarding this drug?

- a) It causes renal toxicity
- b) It is more likely to cause gastric ulcers compared to ibuprofen
- c) It is not associated with cardiotoxicity
- d) It has antiplatelet action

Question 10:

Which of the following drugs has the least COX-2 selectivity?

- a) Lumiracoxib
- b) Etoricoxib
- c) Valdecoxib
- d) Celecoxib

Question 11:

Your senior asks you to administer a selective COX-2 inhibitor intravenously to a patient for post-op analgesia. Which of the following will you choose?

- a) Celecoxib
- b) Etoricoxib
- c) Rofecoxib
- d) Parecoxib

Question 12:

Which of the following is a true statement about acetaminophen?

- a) It has significant analgesic, antiinflammatory and antipyretic action.
- b) Acetaminophen toxicity causes periportal necrosis.
- c) It does not antagonize effects of uricosuric drugs.
- d) It has significant anti-platelet effect.

Question 13:

A 25-year-old patient of major depression presented to the ER after having consumed 20 tablets of acetaminophen (500 mg each) 1 hour ago in a self-harm attempt. She does not have any symptoms at the moment. Which of the following should you administer to her?

- a) Flumazenil
- b) N-acetylcysteine
- c) Methylene blue
- d) Nothing as she is asymptomatic

Question 14:

Which of the following drugs is not considered a hit and run drug?

- a) Phenelzine
- b) Omeprazole
- c) Piroxicam
- d) Aspirin

Question 15:

Which of the following patients should be prescribed NSAIDs for pain relief?

- a) A hypertensive patient on chlorthalidone having osteoarthritis
- b) Patient suffering from primary dysmenorrhea
- c) Patient complaining of low back pain who is a known case of CKD

d) Patient having painful diabetic neuropathy

Question 16:

A patient of chronic low back pain has presented to your OPD with deranged LFT reports. He tells you he has been on long term treatment with a painkiller for his condition. Which NSAID has he most likely been taking?

- a)** Acetaminophen
- b)** Diclofenac
- c)** Piroxicam
- d)** Indomethacin

Question 17:

A 30-year-old male on multiple drugs presents with a round macule on his right leg for the last 7 days. The patient gives a history of similar episodes in the past which resolved spontaneously. Which of the following drugs is the most likely cause?



- a)** Ibuprofen
- b)** Phenytoin
- c)** Captopril
- d)** Carbamazepine

Answer Key

Question No.	Correct Option
1	c
2	c
3	b
4	d
5	d
6	b
7	b
8	c
9	a
10	d
11	d
12	c
13	b
14	c
15	b
16	b
17	a

Detailed Explanations

Solution to Question 1:

Cyclooxygenase-1 (COX-1) is primarily a housekeeping enzyme.

NSAIDs act by inhibiting both COX-1 and COX-2.

- COX-1 is constitutively expressed in most cells. It performs housekeeping functions like gastrointestinal cytoprotection, fluid and electrolyte balance in the kidney.
- COX-2 is normally present in insignificant amounts in most of the cells. It is induced by cytokines, growth factors and other stimuli during inflammation.

Phospholipase-A2 is the initial enzyme that acts on the membrane phospholipids to generate arachidonic acid. Arachidonic acid produces prostaglandins by the action of the cyclooxygenase enzymes (COX).

Solution to Question 2:

Thromboxane A₂ is a pro-aggregatory eicosanoid whose formation requires the activity of COX enzyme. Aspirin blocks this step, thus reducing the chances of clot formation.

Thromboxane A₂ is an inducer of platelet aggregation and amplifies the signal of more potent platelet agonists, such as thrombin and ADP.

Solution to Question 3:

Prostaglandin E₁ (PGE₁) maintains the patency of ductus arteriosus (PDA). This is used in term neonates with underlying cyanotic congenital heart disease awaiting surgical intervention as they have duct-dependant circulation.

Indomethacin and Ibuprofen are the NSAID indicated for closing the ductus arteriosus in premature infants and neonates.

Solution to Question 4:

Rioprostil is a PGE₁ analog and is not indicated in the treatment of pulmonary hypertension.

Long term therapy with PGI₂ analogs can delay or preclude the need for lung or heart-lung transplantation. PGI₂ analogs used for the treatment of pulmonary hypertension are:

- Epoprostenol (IV)
- Iloprost (IV)
- Treprostinil (IV or SC)

PGI₂ (Epoprostenol) can also be used to prevent platelet aggregation and damage during haemodialysis or cardiopulmonary bypass. It also improves the harvest of platelets for transfusion.

Solution to Question 5:

Alprostadil (PGE₁) is given as an intracavernosal injection or urethral suppository for erectile dysfunction.

It is considered as a second-line therapy. Its use has been largely replaced by phosphodiesterase-5 inhibitors like sildenafil and tadalafil which are the drugs of choice.

Solution to Question 6:

Aspirin clearance is dependent on plasma concentration at a lower dose.

Aspirin shows saturation kinetics. This means that at lower plasma concentrations it has first-order kinetics where the rate of elimination is proportional to the plasma

concentration but at higher plasma concentrations it has zero-order kinetics where the rate of elimination is independent of its plasma concentration.

Around 80-90% of aspirin (salicylates) are bound to plasma proteins, especially albumin.

Low dose aspirin acetylates the COX enzyme of platelets in the portal circulation (pre-systemic) and deactivates it which is responsible for antiplatelet action.

Aspirin and salicylates are acidic in nature. Alkalinization of the urine increases the rate of excretion of aspirin, as ionised form of aspirin increases which is not reabsorbed from the tubular fluid in kidney.

Solution to Question 7:

Aspirin is contraindicated in this child due to the risk of development of Reye syndrome.

Children and young adults less than 20 years of age with viral fever have a risk of developing Reye's syndrome on administration of aspirin or other salicylates.

Features of Reye syndrome include:

- Acute onset of encephalopathy
- Liver dysfunction
- Fatty infiltration of the liver

Option A - Acetaminophen is considered the antipyretic of choice in children with viral infections.

Options C and D - Oseltamivir and baloxavir are antiviral drugs used in influenza.

Solution to Question 8:

Piroxicam is chosen as it has the longest plasma half-life among NSAIDs.

Piroxicam is absorbed completely and undergoes enterohepatic circulation. Due to the long half-life, once-daily administration of the drug is sufficient for the therapeutic response.

Solution to Question 9:

Celecoxib is a selective COX-2 inhibitor. As COX-2 enzyme is constitutively active within the kidney, recommended doses of selective COX-2 inhibitors cause renal toxicities similar to those associated with traditional NSAIDs.

Option B - It is less likely to cause gastric ulcers. Celecoxib has analgesic, antipyretic, and anti-inflammatory effects similar to those of non-selective NSAIDs but with better gastrointestinal tolerability.

Option C - It inhibits COX-2-mediated prostacyclin synthesis in the vascular endothelium. As a result, use of celecoxib is associated with increased cardiovascular risk.

Option D - Celecoxib at usual doses does not inhibit platelet aggregation. This is because platelet aggregation is mediated by thromboxane produced by the COX-1.

These points are also true for other selective COX-2 inhibitors.

Solution to Question 10:

Celecoxib has the least COX-2 selectivity among the following.

Some selective COX-2 inhibitors are withdrawn or have restricted use because of an increase in the risk of cardiovascular events like myocardial infarction, serious skin reactions (valdecoxib), fatal hepatic failure (lumiracoxib).

Nowadays, dual inhibitors inhibiting 5-LOX and COX like licofelone and tepoxalin are also used for their anti-inflammatory property.

Solution to Question 11:

Parecoxib is chosen as it is the only injectable COX-2 selective drug among the given options.

It is a water-soluble prodrug of valdecoxib. It is effective in the perioperative period when oral medications are not indicated.

Solution to Question 12:

Acetaminophen does not antagonize effects of uricosuric agents. It may be used concomitantly with probenecid in the treatment of gout as it does not have any effect on uric acid levels.

Option A - Acetaminophen has analgesic and antipyretic action equivalent to aspirin but lacks a significant anti-inflammatory effect.

Option B - Acetaminophen toxicity causes centrilobular necrosis.

Option D - It is a weak inhibitor of both COX-1 and COX-2 and has minimal or no effect on platelet activity.

Solution to Question 13:

The patient in the given clinical scenario has ingested a potentially toxic dose of acetaminophen. N-acetylcysteine is considered as a definitive antidote for acetaminophen poisoning to prevent hepatic injury.

N-acetyl-p-benzoquinone imine (NAPQI) is a toxic metabolite of acetaminophen which causes hepatotoxicity. In adults, hepatotoxicity may occur after ingestion of a single dose of 10–15 g (150–250 mg/kg) of acetaminophen; doses of 20–25 g or more are potentially fatal.

In acetaminophen poisoning, due to increased NAPQI, there is utilization of glutathione (GSH). This in turn leads to depletion of GSH leading to oxidative cellular damage of the hepatocytes. N-acetylcysteine replenishes GSH stores and also conjugates with NAPQI and prevents oxidative damage to the cell. It also protects against extrahepatic injury due to its antioxidant and anti-inflammatory properties.

Option A: Flumazenil is an antidote for benzodiazepine toxicity.

Option C: Methylene blue is used in methemoglobinemia.

Option D: Symptoms may take up to 2 days to develop in a case of acetaminophen poisoning by which time the hepatic damage may become irreversible. Thus, early treatment is initiated in all suspected cases.

Solution to Question 14:

Piroxicam is a reversible inhibitor of cyclooxygenase enzyme and cannot be considered as a hit and run drug.

Hit and run drugs are a group of drugs whose half-life is short, but cause irreversible inhibition of a particular enzyme. So, the effect of the drug is present even after the drug is cleared from the system. Examples include:

- MAO inhibitors (e.g.: Phenelzine)
- PPI (e.g.: omeprazole)
- Guanethidine
- Reserpine
- Aspirin

Solution to Question 15:

NSAIDs can be used in the treatment of primary dysmenorrhea. Prostaglandin release from endometrium causes cramps and other symptoms. NSAIDs inhibit the synthesis of prostaglandin and cause symptomatic relief.

Option A: NSAIDs interfere with the antihypertensive effect of diuretics. The exact mechanism is not yet known. So it should be avoided in such cases.

Option C: NSAIDs are generally avoided in renal diseases as inhibition of prostaglandin synthesis worsens renal function.

Option D: NSAIDs generally lack efficacy in cases of neuropathic pain like diabetic neuropathy.

Solution to Question 16:

Diclofenac is the NSAID that has the highest risk of hepatotoxicity within the therapeutic dose.

Liver injury occurs in acetaminophen overdose but it is extremely rare in therapeutic dose. NSAIDs are not recommended in advanced hepatic and renal failure.

Solution to Question 17:

This is a case of fixed drug eruption and the most likely cause is ibuprofen.

Fixed drug eruptions are characterized by sharply demarcated dull red to brown lesions. They occur in lips, hands, legs, face, genitalia, and oral mucosa, and cause a burning sensation. The lesion recurs in the same locations upon re-exposure to the offending drug.

Some of the drugs commonly known to cause fixed drug eruptions are Sulphonamides, NSAIDs, Analgesics, and Hypnotics.

Diagnosis is based on history and lesion morphology. Discontinuation of the offending drug is the first step in management.

Phenytoin and carbamazepine can lead to Stevens-Johnson syndrome (SJS)/toxic epidermal necrolysis (TEN) and drug reaction with eosinophilia and systemic symptoms (DRESS syndrome).

Antimigraine and Antigout drugs

Question 1:

Which of the following receptors is a ligand-gated ion channel?

- a) 5-HT 1 receptor
- b) 5-HT 2 receptor
- c) 5-HT 3 receptor
- d) 5-HT 4-7 receptors

Question 2:

Which of the following statements is false regarding triptans?

- a) They are 5HT 1B/1D agonists
- b) No action on adrenergic or cholinergic or dopaminergic receptors
- c) Sumatriptan is safe in pregnancy
- d) They are used for prophylaxis of migraine

Question 3:

A 25-year-old patient presented to the ER with an acute attack of migraine. You decide to administer a long-acting triptan. Which among the following do you choose?

- a) Sumatriptan
- b) Rizatriptan
- c) Frovatriptan
- d) Zolmitriptan

Question 4:

A patient of migraine has come to the OPD for follow up. She tells you that she gets very nauseous during the attacks and is not being able to take the oral drug. You decide to prescribe an intranasal formulation instead. Which of the following antimigraine agents will you not choose?

- a) Dihydroergotamine
- b) Sumatriptan
- c) Zolmitriptan
- d) Naratriptan

Question 5:

After the administration of a drug for an acute attack of migraine, a patient developed nausea and vomiting. She also developed tingling and numbness at the tips of her fingers that eventually turned blue. Which of the following is the most likely drug implicated?

- a) Dihydroergotamine
- b) Sumatriptan
- c) Butorphanol
- d) Aspirin

Question 6:

A patient has just been diagnosed with migraine. Which of the following drugs can you prescribe for prophylaxis?

- a) Sumatriptan
- b) Ergotamine
- c) Domperidone
- d) Topiramate

Question 7:

What is the mechanism of action of colchicine?

- a) Inhibits synthesis of uric acid
- b) Promotes excretion of uric acid
- c) Inhibits xanthine oxidase enzyme
- d) Binds to tubulin and inhibits granulocyte migration

Question 8:

Which one of the following drugs inhibits neutrophil recruitment?

- a) Montelukast
- b) Sodium cromolyn
- c) Colchicine
- d) Febuxostat

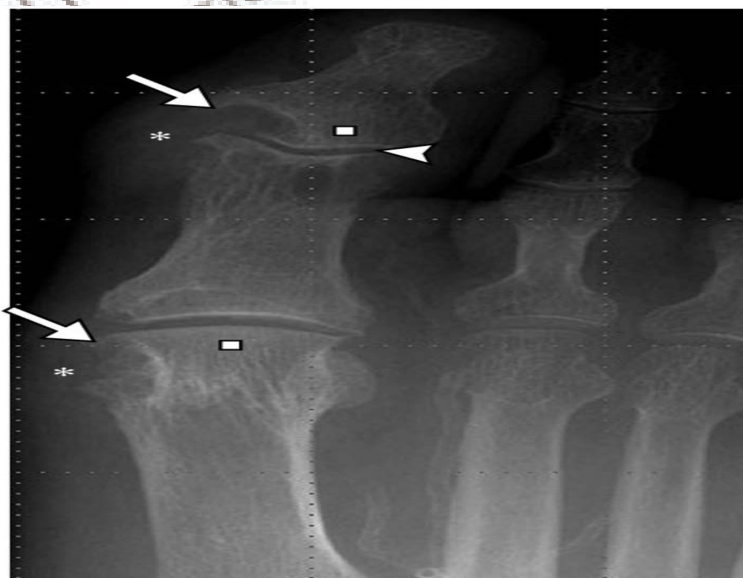
Question 9:

A patient with chronic tophaceous gout was prescribed sulfinpyrazone. What is the mechanism of action of this drug?

- a) Inhibits formation of uric acid
- b) Promotes excretion of uric acid
- c) Inhibits xanthine oxidase enzyme
- d) Inhibits cyclooxygenase enzyme

Question 10:

A middle-aged obese man has presented to the OPD with acute pain around his left first metatarsophalangeal joint. He has a history of similar episodes in the past. X-ray is given below. Which drug is commonly used as initial treatment for this condition?



- a) Aspirin
- b) Indomethacin

- c) Allopurinol
- d) Colchicine

Question 11:

A patient undergoing chemotherapy was diagnosed with tumour lysis syndrome. She was administered rasburicase. Which group does this drug belong to?

- a) Recombinant xanthine oxidase
- b) Recombinant urate oxidase
- c) Inhibitor of xanthine oxidase
- d) Inhibitor of uric acid oxidase

Question 12:

You have prescribed the first-line urate-lowering drug to a patient with chronic gout. Which of the following is true about this drug?

- a) Pyrimidine antimetabolite
- b) Inhibitor of neutrophil recruitment
- c) Uricosuric drug
- d) Xanthine oxidase inhibitor

Question 13:

A patient was on treatment with indomethacin for gout. You decide to add a non-purine xanthine oxidase inhibitor to his treatment plan. Which of the following will you prescribe?

- a) Allopurinol
- b) Febuxostat
- c) Either a or b
- d) Sulfipyrazone

Question 14:

A patient, who is a known case of gout on treatment with allopurinol, comes to the OPD for follow up. His serum uric acid level is 11 mg/dl. You add lesinurad to his treatment regime for

adequate disease control. Which group does this drug belong to?

- a) Uric acid synthesis inhibitor
- b) Uricosuric agent
- c) Xanthine oxidase inhibitor
- d) Recombinant xanthine oxidase

Question 15:

What is pegloticase used for?

- a) Chronic gout
- b) Rheumatoid arthritis
- c) Psoriatic arthritis
- d) Paralytic ileus

Question 16:

An elderly woman on treatment for migraine presents with chest pain. She has a previous history of angina/MI. Which of the following mechanism of the prescribed drug for migraine might have caused this presentation?

- a) 5-HT antagonist
- b) 5-HT agonist
- c) Calcium Channel Blocker action
- d) Nitric oxide mediated vasodilation

Question 17:

Which of the following is correct about the drugs used in gout?

- a) 1, 2 & 3
- b) 2, 3 & 4
- c) 1, 2 & 5
- d) 3 & 4

Question 18:

A patient was prescribed erenumab-aooe for the prophylaxis of migraine. What is the mechanism of action of this drug?

- a) Parathyroid hormone related protein (PTHrP) receptor agonist
- b) Parathyroid hormone related protein (PTHrP) receptor antagonist
- c) Calcitonin gene related protein (CGRP) receptor agonist
- d) Calcitonin gene related protein (CGRP) receptor antagonist

Answer Key

Question No.	Correct Option
1	c
2	d
3	c
4	d
5	a
6	d
7	d
8	c
9	b
10	b
11	b
12	d
13	b
14	b
15	a
16	b
17	a
18	d

Detailed Explanations

Solution to Question 1:

The 5-HT₃ receptor is a ligand-gated ion channel receptor. It rapidly depolarizes nerve endings by opening the cation channels located within the receptor.

The 5-HT₃ receptors are located in the GI tract (vagal & splanchnic afferents) and the brain (nucleus tractus solitarius & area postrema) and are involved in the emetic response. Hence 5-HT₃ receptor antagonists like ondansetron, dolasetron etc are used as antiemetics.

All other 5-HT receptors are G-protein-coupled receptors.

Solution to Question 2:

Triptans are not used in prophylaxis of migraine. They are indicated in the management of acute attacks of migraine.

Drugs used in prophylaxis of migraine are :

- Beta-blocker - Propranolol, metoprolol
- Antidepressant - Amitriptyline, nortriptyline
- Anticonvulsant - Topiramate, valproate
- Serotonergic drug - Pizotifen
- Antihistamine - Flunarizine
- Angiotensin receptor blocker - candesartan

Solution to Question 3:

Frovatriptan is a long-acting triptan among other triptans.

Headache recurrences in an attack are less with longer-acting triptans like frovatriptan and naratriptan but onset of action may be slow.

Solution to Question 4:

Naratriptan is not available for intranasal administration. It is available only for oral administration.

Solution to Question 5:

Nausea and vomiting, as well as tingling and numbness (paresthesia), are common adverse effects of dihydroergotamine.

The numbness progressing to bluish discolouration of the fingers are the initial signs of occlusion. It can eventually lead to gangrene if left untreated. This is a rare adverse effect of

dihydroergotamine.

Contraindications for ergotamine:

- Peripheral vascular disease
- Coronary artery disease
- Uncontrolled hypertension
- Fever and/or sepsis
- Pregnancy
- Within 24 hours of using triptans

Solution to Question 6:

Topiramate is used in the prophylaxis of migraine. Long-term use of topiramate can cause weight loss.

The drugs that have been approved by the FDA for the preventive treatment of migraine include:

- Propranolol
- Timolol
- Sodium valproate
- Topiramate

Options A and B - Sumatriptan, a 5HT_{1B/1D} agonist is a first-line agent for the termination of acute attacks. Ergotamines are also used in the termination of acute attacks.

Option C - Domperidone is indicated as an adjuvant in acute attacks. Acute attacks are associated with gastric stasis and an increased incidence of nausea and vomiting. It is helpful in reducing gastrointestinal effects. It can also increase gastrointestinal absorption of other antimigraine agents that are not absorbed due to gastric stasis.

Solution to Question 7:

Colchicine acts by binding to tubulin and inhibits granulocyte migration.

Colchicine produces its anti-inflammatory effects by binding to the intracellular protein tubulin. This prevents the polymerization of tubulin into microtubules which is necessary for leukocyte migration and phagocytosis. Inhibition of microtubule formation prevents granulocyte migration and also arrests cell division at G₁ phase.

Solution to Question 8:

Colchicine inhibits neutrophil (granulocyte) recruitment.

Colchicine acts by preventing the polymerisation of microtubules by binding to tubulin. This reduces neutrophil recruitment to inflamed tissue and also decreases the secretion of chemotactic factors by activated neutrophils, thus reducing the inflammatory response overall.

Colchicine is used in gout and Mediterranean fever.

Solution to Question 9:

Sulfinpyrazone is a uricosuric agent that promotes excretion of uric acid. It is used to decrease the body pool of urate in patients with gout. They act by inhibiting the organic anion transporter that is involved in uric acid reabsorption in renal tubules.

Uricosuric agents used in gout are:

- Probenecid
- Sulfinpyrazone
- Benzbromarone
- Lesinurad

Uricosuric therapy should be initiated in gouty patients with underexcretion of uric acid and when allopurinol or febuxostat is contraindicated or when tophi are present. Therapy should not be started until 2–3 weeks after an acute attack.

Solution to Question 10:

The given scenario and the X-ray showing Martel's sign point towards a diagnosis of acute on chronic presentation of gout. Indomethacin is commonly used in the initial treatment of gout.

In addition to inhibiting prostaglandin synthesis, indomethacin and other NSAIDs also inhibit urate crystal phagocytosis.

Option A - Aspirin is not used because it causes renal retention of uric acid at low doses. It is uricosuric only at higher doses that are poorly tolerated.

Option C - Allopurinol is used for prophylaxis of gout.

Option D - NSAIDs are selected over colchicine because of troublesome diarrhea associated with colchicine.

Solution to Question 11:

Rasburicase is a recombinant urate oxidase which converts uric acid into soluble and inactive allantoin. It is more effective in reducing the urate level than allopurinol.

Recombinant uric acid oxidase is available as rasburicase and pegloticase. It is produced by genetically modified *Saccharomyces cerevisiae* strain. They are indicated in gout only for chronic refractory gout. It is primarily indicated for hyperuricemia associated with tumor lysis syndrome.

Pegloticase is a recombinant pegylated uric acid oxidase that is indicated for treatment of severe, treatment-refractory, chronic gout and also prophylaxis of gout.

Solution to Question 12:

Allopurinol is recommended as the first-line urate-lowering agent in all patients with gout. It is a purine antimetabolite and an inhibitor of xanthine oxidase enzyme.

It prevents the conversion of hypoxanthine and xanthine to uric acid reducing plasma uric acid levels. Allopurinol is also a substrate for xanthine oxidase. The metabolite of allopurinol called oxypurinol is a non-competitive inhibitor of xanthine oxidase.

Solution to Question 13:

Febuxostat is a non-purine xanthine oxidase inhibitor that is indicated for reducing hyperuricemia in chronic gout. Both allopurinol and febuxostat can be used as first line agents but since the question mentions a non-purine inhibitor is chosen, febuxostat is the answer.

Option A - Allopurinol is a purine xanthine oxidase inhibitor

Option D - Sulfipyrazone is a uricosuric agent.

Solution to Question 14:

Lesinurad is a uricosuric agent.

It is a URAT1 inhibitor indicated in combination with a xanthine oxidase inhibitor for the treatment of hyperuricemia associated with gout. By inhibiting URAT1, it increases the urinary excretion of uric acid.

Solution to Question 15:

Pegloticase is used in the treatment of chronic gout.

Pegloticase is a pegylated uricase (urate oxidase). It catalyzes the enzymatic oxidation of uric acid into allantoin, which is a more soluble and inactive metabolite.

It is used in the treatment of severe, refractory chronic gout or in cases where other urate-lowering therapies are contraindicated.

Solution to Question 16:

The patient may have been given sumatriptan, a 5-HT_{1b/1d} receptor agonist which causes vasoconstriction and used in migraine. Thus, sumatriptan will worsen angina by causing coronary vasoconstriction.

Angina will be worsened if the O₂ demand of the heart is increased or there is reduced blood flow to the already ischemic areas.

Solution to Question 17:

Gout is a metabolic disease associated with hyperuricemia.

The drug of choice in an acute attack of gout is NSAIDs (non-steroidal anti-inflammatory drugs) like indomethacin or the drug colchicine. Colchicine is poorly tolerated and should not be used in elderly patients with chronic kidney disease.

Indomethacin is commonly used in the initial treatment of gout. In addition to inhibiting prostaglandin synthase, indomethacin and other NSAIDs also inhibit urate crystal phagocytosis

Intraarticular steroids like triamcinolone or methylprednisolone can also be given.

Aspirin is not used because it causes renal retention of uric acid at low doses. It is uricosuric only at higher doses that are poorly tolerated.

Solution to Question 18:

Erenumab is a calcitonin gene related protein (CGRP) receptor antagonist approved for the prophylaxis of migraine.

The pathology of migraine is not completely understood but it includes activation of trigeminal ganglion nociceptive neurons and the subsequent release of CGRP from fibers associated with meningeal vessels. This is believed to mediate vasodilation and mast cell degranulation. Thus, CGRP is involved in the development and maintenance of persistent pain, central sensitization and allodynia, which are events characteristic of migraine pathology.

Blockage of CGRP receptors by CGRP receptor antagonists, causes inhibition of the inflammatory and nociceptive effects of CGRP at multiple sites within the peripheral and central nervous systems that are implicated in the underlying pathology of migraine.

Monoclonal antibodies that directly inhibit CGRP have also been developed. These include:

- Fremanezumab
- Galcanezumab

Note:

The FDA has recently approved Atogepant, a calcitonin gene-related peptide (CGRP) receptor antagonist for the preventive treatment of episodic migraine in adults. The recommended dosage is 10 mg, 30 mg, or 60 mg orally once daily. In severe renal impairment or end-stage renal disease it is 10 mg once daily.

Anti-rheumatoid drugs

Question 1:

A patient with rheumatoid arthritis has been on treatment for the past 15 years. During an annual eye check-up, her fundus appeared as shown in the picture given. Which of the following anti-rheumatoid drugs likely led to this side effect?



- a) Cyclophosphamide
- b) Methotrexate
- c) Hydroxychloroquine
- d) Leflunomide

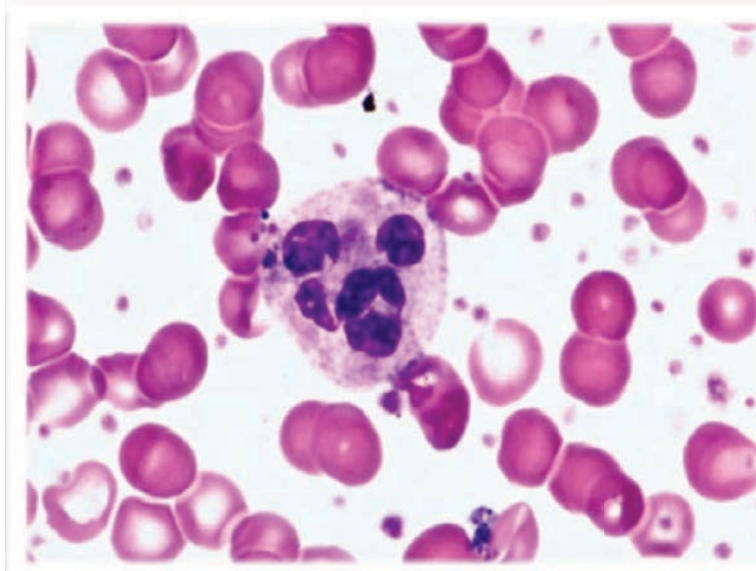
Question 2:

A middle-aged woman with rheumatoid arthritis was on DMARD therapy with a drug that increases extracellular adenosine. Which drug was she likely on?

- a) Methotrexate
- b) Lefunamide
- c) Sulphasalazine
- d) Hydroxychloroquine

Question 3:

A patient with rheumatoid arthritis on treatment with methotrexate came for a follow-up visit. Her peripheral blood smear shows the following finding. What drug can be administered to alleviate this side effect?



- a) Ascorbic acid
- b) Cyanocobalamin
- c) Thiamine
- d) Folinic acid

Question 4:

Which of the following drugs can be used for the treatment of a patient with rheumatoid arthritis?

- a) Sulfacetamide
- b) Sulfasalazine
- c) Silver sulfadiazine
- d) Mafenide

Question 5:

How does etanercept act?

- a) Interleukin 1 inhibition
- b) Interleukin 6 inhibition
- c) TNF alpha inhibition
- d) Janus kinase inhibition

Question 6:

You are using TNF alpha inhibitors in the treatment of a patient with rheumatoid arthritis. Which of the following is not true regarding these drugs?

- a) They can also be used in the treatment of plaque psoriasis
- b) They can be used in the treatment of psoriatic arthritis
- c) They can be used in the treatment of lymphomas
- d) They can be used in the treatment of ulcerative colitis

Question 7:

A middle-aged man with rheumatoid arthritis on treatment with methotrexate presents to your OPD complaining of no improvement in his condition. Which of the following is an IL-1 receptor antagonist that can you prescribe for him?

- a) Anakinra
- b) Rituximab
- c) Tofacitinib
- d) Golimumab

Question 8:

Which is the monoclonal antibody approved for the treatment of rheumatoid arthritis?

- a) Durvalumab
- b) Sarilumab
- c) Nivolumab
- d) Pembrolizumab

Question 9:

Which of the following is a Janus kinase inhibitor approved for use in rheumatoid arthritis?

- a) Ruxolitinib
- b) Baricitinib
- c) Denosumab
- d) Dasatinib

Question 10:

Which among the following is not an indication for the use of corticosteroids in the treatment of rheumatoid arthritis?

- a) Management of acute disease flares while on therapy with DMARDs
- b) As bridging therapy initially, till DMARDs take effect
- c) Chronic therapy in cases that cannot be controlled with DMARDs
- d) As monotherapy for the treatment of rheumatoid arthritis

Answer Key

Question No.	Correct Option
1	c
2	a
3	d
4	b
5	c
6	c
7	a
8	b
9	b
10	d

Detailed Explanations

Solution to Question 1:

The image shows bull's eye maculopathy.

Hydroxychloroquine use is associated with this side effect.

Solution to Question 2:

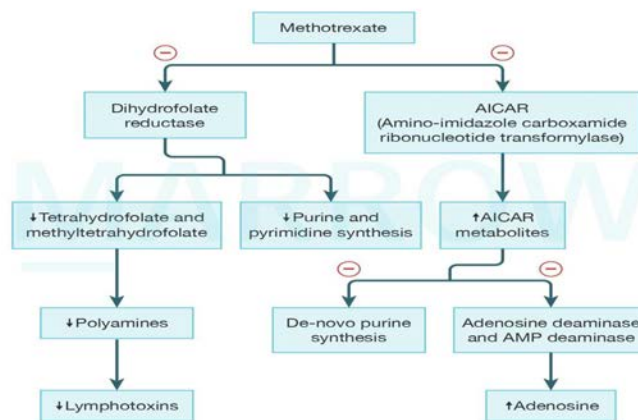
Methotrexate is the DMARD that increases extracellular adenosine.

It is a dihydrofolate reductase (DHFR) inhibitor that leads to purine synthesis inhibition and this is cytotoxic to lymphocytes.

It also inhibits amino-imidazole carboxamide ribonucleotide (AICAR) transformylase and thymidylate synthetase. This leads to increased extracellular adenosine which is anti-inflammatory.

The increased extracellular adenosine is hepatotoxic and hence monitoring with LFT is needed every 2-3 months.

Mechanism of action of methotrexate



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Solution to Question 3:

The peripheral smear findings point to a diagnosis of megaloblastic anemia. Folinic acid is given with methotrexate to decrease its side effects.

Methotrexate is a dihydrofolate reductase (DHFR) inhibitor and thus can lead to a deficiency of folic acid which may present as megaloblastic anemia.

Folinic acid/leucovorin calcium/citrovorum factor is the 5-formyl derivative of tetrahydrofolic acid. Thus, it decreases the side effects caused due to folate deficiency.

Solution to Question 4:

Sulfasalazine can be used in the treatment of rheumatoid arthritis, as a conventional DMARD. It is also used in the management of ulcerative colitis and regional enteritis.

Sulfasalazine is metabolized into the following:

- 5-amino salicylic acid (5-ASA) - it is not absorbed and is used to treat ulcerative colitis.
- Sulfapyridine - it is absorbed and is used to treat rheumatoid arthritis.

It is used as monotherapy in mild cases and as an add-on drug in moderate to severe cases.

Solution to Question 5:

Etanercept acts by TNF alpha inhibition. It is a biological DMARD (disease-modifying anti-rheumatoid drug).

Solution to Question 6:

TNF- α inhibitors are not used in the treatment of lymphomas.

Despite their benefits, multiple potential adverse effects following the use of TNF-alpha inhibitors have been reported. These include:

- Malignancy (lymphoma)
- Injection site reactions
- Infusion reactions
- Opportunistic infections.

TNF- α inhibitors can be used in the treatment of:

- Rheumatoid arthritis
- Ankylosing spondylitis
- Crohn's disease
- Ulcerative colitis
- Plaque psoriasis
- Psoriatic arthritis.

Solution to Question 7:

The IL-1 receptor antagonist that is used in the treatment of rheumatoid arthritis is anakinra.

Anakinra is a biological DMARD (Disease Modifying Anti-Rheumatoid Drug) used in the treatment of rheumatoid arthritis. All biological DMARDs downregulate the immune system and can lead to adverse effects like opportunistic infections and flaring of latent TB.

Solution to Question 8:

The monoclonal antibody approved for the treatment of rheumatoid arthritis is sarilumab.

It is an IL-6 receptor antagonist indicated for the treatment of adults with moderate to severe rheumatoid arthritis who show inadequate response or intolerance to one or more disease-modifying antirheumatic drugs (DMARDs).

Option A: Durvalumab is an anti PD-L1 antibody approved for metastatic urothelial cancer.

Option C: Nivolumab is a fully human monoclonal IgG4 antibody that blocks the interaction between PD-1 (Programmed cell Death-1) receptor and its ligands.

Option D: Pembrolizumab (formerly called lambrolizumab or MK-3475) is a humanized monoclonal IgG4- κ isotype antibody that blocks the interaction between PD-1 and its ligands. Nivolumab and pembrolizumab are used to treat advanced melanoma.

Solution to Question 9:

Baricitinib was approved for use in patients with rheumatoid arthritis. It is a Janus kinase inhibitor.

Oral Janus kinase inhibitors for rheumatoid arthritis and upadacitinib and tofacitinib.

Option A: Ruxolitinib is a Janus kinase inhibitor, approved for use in various forms of myelofibrosis, polycythemia vera, and acute graft vs host disease.

Option C: Denosumab is a RANK-L inhibitor that is approved for the treatment of osteoporosis.

Option D: Dasatinib and ponatinib are BCR-ABL tyrosine kinase inhibitors indicated for the treatment of CML.

Solution to Question 10:

Glucocorticoids are not used as monotherapy for the treatment of rheumatoid arthritis (RA).

High doses of glucocorticoids may be needed to treat severe extraarticular manifestations of RA.

Intraarticular injection of triamcinolone can be considered for one/few actively inflamed joints.

Option A: A burst of glucocorticoids for 1-2 weeks may be prescribed to manage acute disease flares.

Option B: They may be administered as bridging therapy initially to achieve rapid disease control till DMARDs take effect, which may take several weeks or even months.

Option C: Chronic therapy with low doses may be needed to control the disease in patients with an inadequate response to DMARD therapy.

Anti-histamines

Question 1:

Which of the following histamine receptors are presynaptic autoreceptors?

- a) H₁
- b) H₂
- c) H₃
- d) H₄

Question 2:

Which of the following will not be seen on the administration of an H₃ receptor agonist?

- a) Inhibition of H₁ receptor induced wakefulness
- b) Increase in H₂ mediated gastric secretion
- c) Inhibition of H₁ mediated bronchoconstriction
- d) Negative chronotropic effect on atria

Question 3:

You are about to prescribe an antihistamine to a patient with allergic rhinitis. You wish to prescribe a second-generation drug as they cause fewer side effects. Which of the following will you not choose?

- a) Olopatidine
- b) Loratidine
- c) Cetirizine
- d) Clemastine

Question 4:

A highway truck driver has presented with profuse rhinorrhea and sneezing. Which of the following drugs will you prefer to prescribe to him?

- a) Pheniramine

- b) Promethazine
- c) Hydroxyzine
- d) Fexofenadine

Question 5:

A patient presents to you with difficulty in writing and performing fine movements. On examination, he has a pill-rolling resting tremor. He has been on treatment with haloperidol for the past 12 years. Which of the following antihistamines would you prefer to prescribe to him?

- a) Chlorpheniramine
- b) Pheniramine
- c) Promethazine
- d) Hydroxyzine

Question 6:

A patient having persistent allergic rhinitis is on treatment with intranasal steroids. You wish to add an antihistamine to his treatment regimen. He requests an intranasal preparation. Which of the following will you prescribe?

- a) Azelastine
- b) Ebastine
- c) Fexofenadine
- d) Mizolastine

Question 7:

Which among the following antihistaminic drugs has additional platelet-activating factor antagonistic property?

- a) Fexofenadine
- b) Cetirizine
- c) Ebastine
- d) Rupatadine

Question 8:

A patient with cystic fibrosis was found to be underweight during his regular check-up. He complains of low appetite and you decide to prescribe an appetite-stimulating agent. Which antihistaminic drug can you prescribe to him?

- a) Cyclizine
- b) Azelastine
- c) Olapatidine
- d) Cyproheptadine

Question 9:

A patient has presented to your OPD complaining of hearing loss following an attack of vertigo. He says he has had similar episodes in the past. You decide to prescribe him betahistine. What is the mechanism of action of this drug?

- a) H₁ agonist
- b) H₁ antagonist
- c) H₂ agonist
- d) H₂ antagonist

Question 10:

A patient with narcolepsy has been prescribed tiprolisant. How does this drug act?

- a) H₁ receptor agonism
- b) H₁ receptor inverse agonism
- c) H₃ receptor agonism
- d) H₃ receptor inverse agonism

Question 11:

Which of the following drugs is used for the treatment of motion sickness?

- a) Cetrizine
- b) Promethazine
- c) Misolastin

d) Loratidine

Answer Key

Question No.	Correct Option
1	c
2	b
3	d
4	d
5	c
6	a
7	d
8	d
9	a
10	d
11	b

Detailed Explanations

Solution to Question 1:

H₃ histamine receptor is a presynaptic autoreceptor.

H₃ receptors are predominantly presynaptic and are present in basal ganglia, hippocampus, and cerebral cortex. They function as autoreceptors inhibiting histamine release from the neuron. H₃ receptors are also found postsynaptically but their function is not known.

Histamine receptor	G-protein coupling	Distribution
H ₁	G _q	Smooth muscle, endothelial cells, CNS
H ₂	G _s	Gastric parietal cells, cardiac muscle, mast cells, CNS
H ₃	G _{i/o}	CNS: predominantly presynaptic but also found postsynaptically
H ₄	G _{i/o}	Cells of hematopoietic origin

Solution to Question 2:

H₃ receptor agonist decreases H₂ mediated gastric secretion.

H₃ receptors are presynaptic inhibitory autoreceptors on histaminergic neurons. H₃ receptor agonists decrease the release of histamine and have actions similar to that of antihistamines.

Normally, histamine increases gastric secretion by acting through the H₂ receptor in the stomach. H₃ receptor agonist decreases histamine release and hence, decreases H₂ receptor-mediated gastric secretion.

Options A and C: H₁ receptor stimulation by histamine in the CNS causes wakefulness, and at the bronchus causes bronchoconstriction. Therefore H₃ receptor agonists by decreasing histamine release inhibit H₁ receptor-induced wakefulness and bronchoconstriction.

Option D: H₃ receptors are also presynaptic heteroreceptors on various neurons and their activation can decrease noradrenergic, serotonergic, cholinergic and glutaminergic transmission. A decrease in noradrenergic transmission in atria can result in a negative chronotropic effect.

Solution to Question 3:

Clemastine is a first-generation antihistamine.

H₁ antagonists are classified into 2 generations:

First-generation antihistamines - They have a strong sedative property and also have an antimuscarinic function. They include:

- Doxepin
- Clemastine, Diphenhydramine, Dimenhydrinate
- Promethazine
- Pheniramine, Chlorpheniramine, Dexchlorpheniramine
- Cyproheptadine
- Hydroxyzine, Cyclizine, Meclizine, Buclizine, Cinnarizine

Second-generation antihistamines - They are less sedative because of less central nervous system penetration and do not block autonomic receptors. They include:

- Olopatadine
- Fexofenadine, Loratadine, Desloratadine
- Cetirizine, Levocetirizine
- Azelastine
- Mizolastine
- Ebastine

- Rupatadine

Solution to Question 4:

The truck driver is suffering from allergic rhinitis and fexofenadine can be prescribed to him. This is preferred as it is a second-generation antihistamine having minimal/no sedative effects as it cannot cross the blood-brain barrier easily.

Pheniramine, promethazine, and hydroxyzine are first-generation antihistamines which should not be prescribed to drivers and heavy machine workers because they are lipophilic and cross the blood-brain barrier easily and cause sedation.

Highly sedative first-generation antihistamines:

- Diphenhydramine
- Dimenhydrinate
- Promethazine
- Hydroxyzine

Solution to Question 5:

The scenario points towards a diagnosis of drug-induced Parkinsonism due to long-term intake of haloperidol. Promethazine is the preferred antihistamine as it has the strongest anticholinergic (muscarinic blocking) activity among the given options.

Drug-induced Parkinsonism is caused by drugs that have D2 dopamine antagonistic action like typical antipsychotics (e.g. haloperidol). The decrease in dopaminergic activity results in an unbalanced cholinergic activity. Anticholinergics act by restoring balance.

First-generation H₁ antagonists have anticholinergic properties, while second-generation H₁ antagonists do not have anticholinergic properties.

H₁ antagonists having high anticholinergic activity are as follows:

- Promethazine
- Diphenhydramine
- Dimenhydrinate
- Pheniramine

Solution to Question 6:

Azelastine is an antihistamine which has an intranasal preparation for allergic rhinitis.

Olopatadine and azelastine are the second generations antihistaminics available as intranasal (nasal spray) preparations for allergic rhinitis. They have good topical activity.

Intranasal corticosteroids (INCS) remain the preferred monotherapy for persistent allergic rhinitis (AR). However, additional studies support the additive benefit of combination treatment with INCS and intranasal antihistamines in AR. For patients receiving INCS, an intranasal decongestant may be added to the treatment for upto 4 weeks.

Solution to Question 7:

Rupatadine is an antihistaminic drug that has additional platelet activating factor antagonistic property and is indicated in allergic rhinitis.

Platelet-activating factor (PAF) is involved in the pathophysiology of allergic rhinitis, asthma, and anaphylaxis.

Solution to Question 8:

Cyproheptadine has an appetite-stimulating effect and has been used to promote weight gain in poor eaters.

The appetite-stimulating effect may be due to the H₁ antihistaminic effect and an effect on growth hormone secretion.

Solution to Question 9:

The scenario given in the question likely points to a diagnosis of Meniere's disease. Betahistine is a selective H₁ agonist used in this condition.

It causes vasodilation in the inner ear which improves circulation. Hence, it is used to treat vertigo in Meniere's disease.

Solution to Question 10:

Tiprolisant is a selective H₃ receptor - inverse agonist.

Tiprolisant promotes wakefulness, improves cognitive functions (e.g., enhances memory, learning, and attention), and reduces food intake. It is also known as pitolisant, and it reduces excessive daytime sleepiness and cataplexy in adults with narcolepsy.

Solution to Question 11:

Promethazine is used for motion sickness. It is an antihistaminic agent that acts as an antagonist of H₁ receptors. It also has central anticholinergic activity that is responsible for its antiemetic

action.

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Antiplatelets, Fibrinolytics and Antifibrinolytics

Question 1:

A patient with a history of myocardial infarction has been on 75 mg aspirin OD for two years. The drug primarily acts by inhibiting which enzyme?

- a) Cyclooxygenase 1
- b) Cyclooxygenase 2
- c) Both a and b
- d) Lipoxygenase

Question 2:

A 55-year-old underwent coronary stent implantation. He was prescribed dual antiplatelet therapy for 6 months in which clopidogrel is one of the components. Which of the following is not true about this drug?

- a) It inhibits Gp IIb/IIIa receptors
- b) The onset of action is slow
- c) It is activated by the hepatic CYP2C19
- d) It has a long duration of action

Question 3:

The efficacy of which of the following antiplatelet drugs will be reduced, if omeprazole is used concurrently?

- a) Clopidogrel
- b) Prasugrel
- c) Ticagrelor
- d) Vorapaxar

Question 4:

Which of the following is not a prodrug?

- a) Clopidogrel
- b) Prasugrel
- c) Ticagrelor
- d) Ticlopidine

Question 5:

Which of the following statements is true about prasugrel?

- a) It is a reversible inhibitor of ADP receptor
- b) Has a slower onset of action than clopidogrel
- c) It is contraindicated in cerebrovascular accidents
- d) Dose adjustment is needed in renal impairment

Question 6:

Which of the following drugs does not act by blocking Gp IIb/ IIIa receptors?

- a) Abciximab
- b) Eptifibatide
- c) Tirofiban
- d) Vorapaxar

Question 7:

A patient with a urine output of 0.4ml/Kg per hour in the last 6 hours can be safely administered which of the following GP IIb/IIIa antagonists?

- a) Abciximab
- b) Eptifibatide
- c) Tirofiban
- d) All of the above

Question 8:

A 60-year-old male patient requires percutaneous coronary intervention. He was given an IV infusion of a reversible ADP receptor inhibitor. Which of the following drugs was used?

- a) Abciximab
- b) Eptifibatide
- c) Cangrelor
- d) Tirofiban

Question 9:

After suffering from a myocardial infarction, a patient was started on two drugs X and Y. The drug X inhibits the COX-1 enzyme and drug Y inhibits thrombin-induced platelet aggregation. Identify the drugs.

- a) X- Aspirin, Y- Clopidogrel
- b) X- Clopidogrel, Y- Vorapaxar
- c) X- Aspirin, Y- Vorapaxar
- d) X- Aspirin, Y- Cangrelor

Question 10:

How is alteplase differentiated from streptokinase?

- a) It is longer acting
- b) It is cheap
- c) It is derived from human kidney
- d) It has more affinity towards plasminogen bound to fibrin

Question 11:

In patients with which of the following conditions would fibrinolytic therapy be an absolute contraindication?

- a) 1, 3 and 5
- b) 2 and 4
- c) 3 and 4
- d) 1 and 2

Question 12:

Which of the following is not a recombinant tissue-type plasminogen activator?

- a) Alteplase
- b) Anistreplase
- c) Tenecteplase
- d) Reteplase

Question 13:

Which of the following plasminogen activator can be administered as a single bolus dose for a patient who requires coronary thrombolysis?

- a) Alteplase
- b) Tenecteplase
- c) Reteplase
- d) Anistreplase

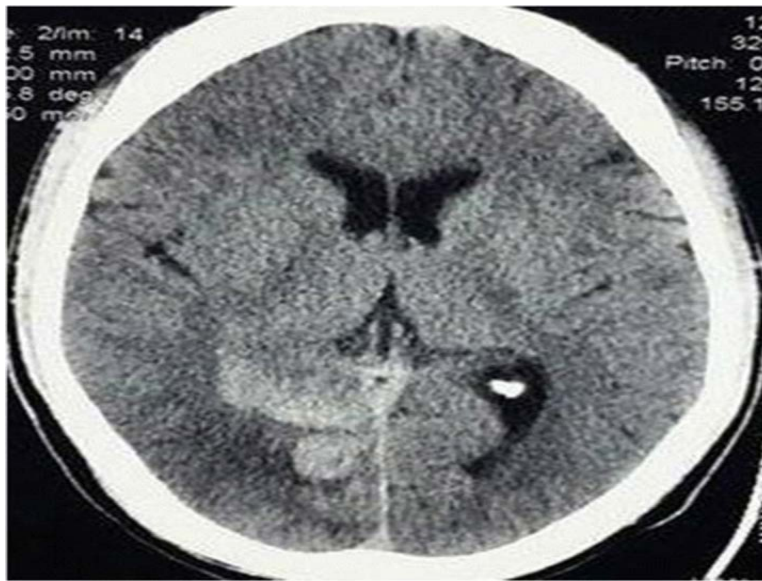
Question 14:

A 35-year-old woman came to the gynecology OPD with a history of heavy menstrual bleeding. She complained of having to change 4-5 pads in a day and an increase in bleeding to more than 7 days since the last few months. Which of the following is the preferred drug that can be given to this patient?

- a) Mefenamic acid
- b) Tranexamic acid
- c) Epsilon amino caproic acid
- d) Acetylsalicylic acid

Question 15:

A 55-year-old man admitted with ischemic stroke was given IV alteplase. After 8 hours, the ICU staff noticed decreasing mental status and acute hypertension in the patient. A repeat CT revealed the following, Which of the following drugs can be used to manage this patient?



- a) Protamine
- b) Streptokinase
- c) Heparin
- d) ϵ -Aminocaproic acid

Question 16:

Which of the following statements is true about prasugrel?

- a) It acts as an P2 Y₁₂ agonist
- b) It is 10 times less potent than clopidogrel
- c) It is a pro drug
- d) Omeprazole interferes in its activation to an active metabolite

Answer Key

Question No.	Correct Option
1	a
2	a
3	a
4	c
5	c

6	d
7	a
8	c
9	c
10	d
11	b
12	b
13	b
14	b
15	d
16	c

Detailed Explanations

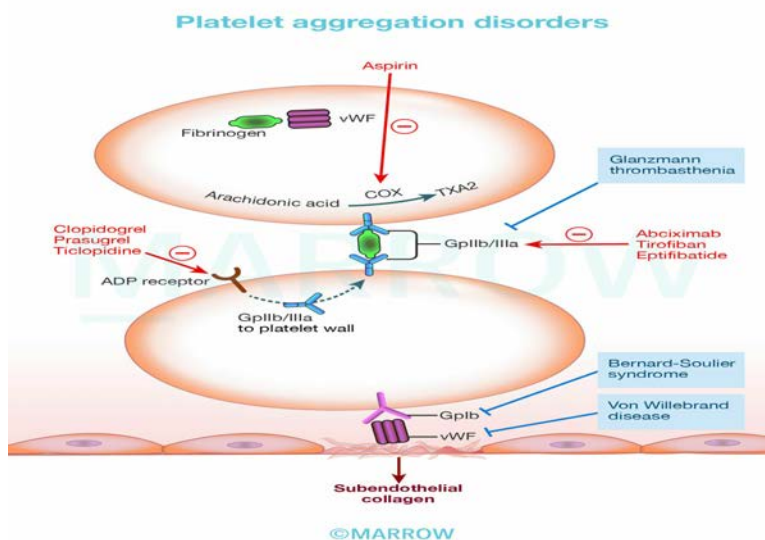
Solution to Question 1:

Aspirin at low doses (75mg) primarily acts by inhibiting cyclooxygenase 1 (COX-1).

Low-dose aspirin, used for antiplatelet action, results in selective inhibition of platelet COX-1 (thromboxane synthase) inhibiting its ability to form thromboxane A₂ (needed for platelet aggregation).

As platelets lack nuclei, new platelets have to be formed (life span of 7-10 days) for the return of COX-1 activity.

At higher doses (~1g/day), aspirin also inhibits COX-2 and results in decreased production of prostacyclin from endothelium. This may nullify the beneficial anti-aggregatory effect of aspirin.

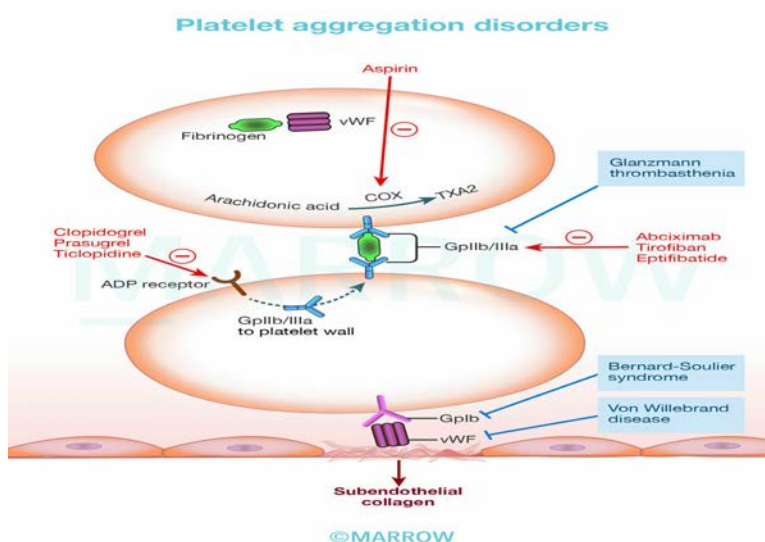


Solution to Question 2:

Clopidogrel acts by irreversible inhibition of P2Y₁₂ receptors of the platelets.

Options B and C: The onset of action is slow (starts after 4 hours) as it requires metabolic activation by the hepatic CYP2C19 enzyme. Patients with a loss of function mutation of the CYP2C19 allele exhibit reduced platelet inhibition and experience a higher rate of cardiovascular events.

Option D: Due to irreversible inhibition of platelet P2Y₁₂ receptors the duration of action is long and lasts for 5-7 days.



Solution to Question 3:

Clopidogrel is a prodrug that is converted into an active metabolite by the CYP2C19 enzyme and omeprazole can reduce the efficacy of clopidogrel due to its CYP2C19 inhibitory activity.

Option B: Prasugrel is converted to its active metabolite mainly by other CYP enzymes and therefore omeprazole does not interfere in its activation.

Option C: Ticagrelor does not require activation and is metabolized by the CYP3A4 enzyme.

Option D: Vorapaxar is a thrombin receptor antagonist that is also metabolized by the CYP3A4 enzyme.

Solution to Question 4:

Ticagrelor is not a prodrug and does not require metabolic activation to exert its antiplatelet action.

It is an orally active, reversible antagonist of P2Y₁₂ subtype of ADP receptors. Since the drug does not require hepatic activation, it can be used in patients with CYP2C19 mutations. Ticagrelor is metabolized into another active metabolite by CYP3A4 and is finally excreted in the urine.

Clopidogrel, prasugrel, and ticlopidine are irreversible antagonists of P2Y₁₂ and are prodrugs requiring metabolic activation.

Solution to Question 5:

Prasugrel is contraindicated in cerebrovascular accidents as it is associated with higher rates of fatal and life-threatening bleeding.

Option A: It is an irreversible inhibitor of the ADP (P2Y₁₂) receptor of platelets.

Option B: It has a faster onset of action and produces greater inhibition of platelets than clopidogrel.

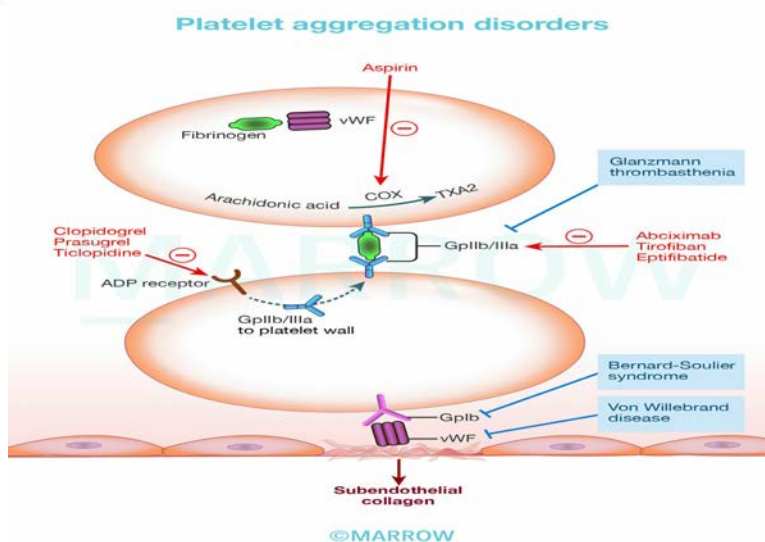
Option D: Dose adjustment is not needed in moderate hepatic and renal impairment.

Solution to Question 6:

Vorapaxar is a competitive antagonist of PAR-1 (thrombin receptor) and inhibits thrombin-induced platelet aggregation.

Abciximab, tirofiban, and eptifibatide act by blocking Gp IIb/IIIa receptors. They are used in patients undergoing percutaneous coronary intervention (PCI) for acute coronary syndrome. They carry a high risk of bleeding and can cause thrombocytopenia.

Eptifibatide is contraindicated in renal failure and tirofiban needs dose adjustment in cases of renal impairment.



Solution to Question 7:

Abciximab can safely be administered to this patient who is likely suffering from acute kidney injury (urine volume less than 0.5 mL/kg per hour for 6 hours).

Abciximab, eptifibatide, and tirofiban are parenteral GpIIb/IIIa receptor antagonists used in the treatment of acute coronary syndromes.

Eptifibatide is contraindicated in renal failure and tirofiban needs dose adjustment in cases of renal impairment. Both these drugs undergo renal clearance and used cautiously in renal impairment.

Solution to Question 8:

Cangrelor is a parenteral, reversible ADP receptor (P2Y₁₂ receptor) inhibitor, administered intravenously.

Abciximab, eptifibatide, and tirofiban are parenteral agents too but, they are GPIIb/IIIa inhibitors.

Solution to Question 9:

X- Aspirin and Y- Vorapaxar are the drugs administered to the patient.

Aspirin reduces thromboxane A₂ production by inhibiting the COX-1 enzyme.

Vorapaxar is a competitive antagonist of PAR-1 and inhibits thrombin-induced platelet aggregation. Vorapaxar is indicated in combination with aspirin and/or clopidogrel to reduce thromboembolic events in patients with MI or peripheral vascular disease.

Solution to Question 10:

Alteplase differs from streptokinase as it has more affinity towards plasminogen bound to fibrin.

It activates fibrin-bound plasminogen more rapidly than it activates plasminogen in the circulation. Streptokinase activates both free and fibrin bound plasminogen.

Option A: Alteplase is rapidly cleared by the liver and inactivated by plasminogen activator inhibitor-1 making it shorter-acting than streptokinase.

Option B: It is also more expensive than streptokinase

Option C: Alteplase is the recombinant tissue-type plasminogen activator derived from human tissue (endothelial cells) culture.

Solution to Question 11:

Among the given options, ischaemic stroke in the past 3 months and suspected aortic dissection are absolute contraindications to fibrinolytic therapy.

Active peptic ulcer and uncontrolled hypertension are relative contraindications for fibrinolytic therapy. Menstruation is not a contraindication to fibrinolytic therapy.

Note: History of a hemorrhagic stroke at any time is an absolute contraindication for thrombolytic therapy.

Solution to Question 12:

Anistreplase is not a recombinant tissue plasminogen activator (rtPA).

Anistreplase (anisoylated plasminogen streptokinase activator complex) is a preformed complex of human plasminogen and bacterial streptokinase which is used as a thrombolytic agent.

Solution to Question 13:

Tenecteplase can be administered as a single bolus dose for patients requiring coronary thrombolysis.

Options A and D: Alteplase and anistreplase are administered as intravenous infusion.

Option C: Reteplase can also be given as two IV boluses, given 30 minutes apart.

Solution to Question 14:

Tranexamic acid is the drug approved for treating heavy menstrual bleeding.

It is a lysine analogue that binds to the active site of plasmin and prevents the interaction of fibrin with plasmin. This prevents the dissolution of clots and decreases bleeding.

Option A: Mefenamic acid is an NSAID and indicated in primary dysmenorrhea.

Option C: ϵ -Aminocaproic acid is also an antifibrinolytic agent like tranexamic acid. However, it is not approved for heavy menstrual bleeding.

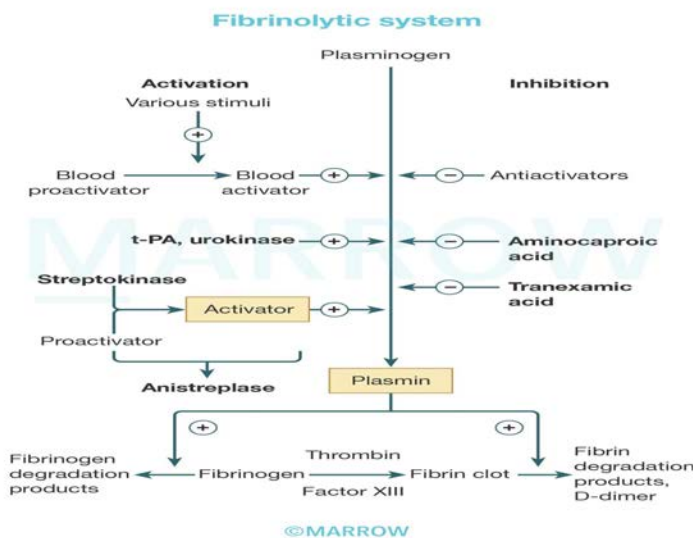
Option D: Acetylsalicylic acid is aspirin, an anti-platelet drug.

Solution to Question 15:

The given clinical scenario is suggestive of post-thrombolytic hemorrhagic transformation of an ischemic stroke. ϵ -Aminocaproic acid can be used as an antidote for fibrinolytic therapy.

It is a lysine analog that binds competitively to plasminogen, blocking plasminogen from binding to fibrin and the subsequent conversion to plasmin. This activity subsequently results in the inhibition of fibrinolysis.

It should not be given unless hemorrhage is life-threatening because it inhibits intrinsic fibrinolytic activity and can precipitate DIC. Fresh frozen plasma and cryoprecipitate may also be used to replenish fibrin and clotting factors.



Solution to Question 16:

Prasugrel is a prodrug.

It acts as an irreversible ADP (P2 Y12) antagonist and is converted into its active metabolite by CYP3A4 and CYP2B6. Since omeprazole is an inhibitor of CYP2C19, it does not interfere with its activation.

It is 10-times more potent than clopidogrel and has a more rapid onset of action because of better absorption and more predictable metabolic activation.

Hematinics

Question 1:

A newly married couple planning for pregnancy has come to your clinic for preconception counselling. Which of the following drugs will you prescribe to the patient to decrease the chances of having neural tube defects in the baby?

- a) Vitamin B9
- b) Vitamin B12
- c) Vitamin C
- d) Vitamin B6

Question 2:

A 25-year-old patient has subclinical folate deficiency. Which of the following drugs is least likely to precipitate megaloblastic anaemia in this patient?

- a) Alcohol
- b) Phenytoin
- c) Chloroquine
- d) Sulfasalazine

Question 3:

A 25-year-old patient presented with complaints of burning sensations of the palms and soles. History revealed a predominantly vegetarian diet and workup showed low levels of serum B12. Treatment for this patient must not be administered through which of the following routes?

- a) Oral route
- b) Deep subcutaneous route
- c) Intravenous route
- d) Intramuscular route

Question 4:

Absorption of iron is impeded by which of the following?

- a) Meat based foods
- b) Ascorbic acid
- c) Amino acids containing –SH groups
- d) Egg based foods

Question 5:

When treating an anemic patient with a history of inflammatory bowel disease, which of the following iron formulations will you prefer?

- a) Ferrous sulphate
- b) Ferrous gluconate
- c) Ferrous sucrose
- d) Ferrous fumarate

Question 6:

Which of the following statements regarding erythropoietin is false?

- a) 3, 4 and 5
- b) 3 and 5
- c) 1, 2 and 5
- d) 2 only

Question 7:

In a patient with anaemia due to chronic renal failure, which of the following is the long-acting derivative of erythropoietin suitable for treatment?

- a) Oprelvekin
- b) Darbepoetin alfa
- c) Dornase alfa
- d) Filgastrim

Question 8:

Sargramostim is a recombinantly engineered analogue of which of the following?

- a) G-CSF
- b) GM-CSF
- c) Erythropoietin
- d) IL-11

Question 9:

Which of the following is true about pegylated-filgrastim?

- a) Duration of action is shorter than normal filgrastim
- b) Added advantage is that it can be taken orally
- c) It should not be administered in sickle cell patients
- d) It is not effective in the treatment of severe neutropenia

Question 10:

Which of the following is not a thrombopoietin receptor agonist?

- a) Lusutrombopag
- b) Romiplostim
- c) Eltrombopag
- d) Oprelvekin

Question 11:

Which of the following agents is used in the treatment of patients who develop severe thrombocytopenia following anti-cancer therapy?

- a) Filgrastim
- b) Oprelvekin
- c) Erythropoietin
- d) Iron dextran

Question 12:

A patient requires autologous stem cell transplantation due to multiple myeloma. A CXCR-4 antagonist which mobilizes stem cells to the peripheral blood is administered few weeks before the procedure. Identify the drug.

- a) Maraviroc
- b) Sargramostim
- c) Filgrastim
- d) Plerixafor

Question 13:

A patient having chronic liver disease is set to undergo a tooth extraction. During pre-op evaluation, his platelet count is found to be $60 \times 10^9/L$. Which thrombopoietin receptor agonist can you prescribe to treat this?

- a) Eltrombopag
- b) Romiplostim
- c) Oprelvekin
- d) Avatrombopag

Question 14:

A 26-year-old woman presented to the OPD with multiple purpuric lesions. Further workup revealed isolated thrombocytopenia with a normal peripheral smear. Which tyrosine kinase inhibitor is indicated in her treatment?

- a) Romiplostim
- b) Bosutinib
- c) Fostamatinib
- d) Oprelvekin

Answer Key

Question No.	Correct Option
1	a
2	c

3	c
4	d
5	c
6	b
7	b
8	b
9	c
10	d
11	b
12	d
13	d
14	c

Detailed Explanations

Solution to Question 1:

Preconceptional intake of vitamin B₉ (folic acid) results in decreased incidence of neural tube defects like spina bifida, encephalocele, and anencephaly.

All women of reproductive age should be given 400 µg of folic acid daily. Women, who had fetal neural tube defects in the previous pregnancy, should be given 4 mg of folic acid daily beginning 1 month before conception to about 12 weeks of pregnancy.

Solution to Question 2:

Among the following, chloroquine is the least likely drug to precipitate folic acid deficiency.

The drugs that can result in the deficiency of folic acid include anticonvulsants (phenytoin, primidone), oral contraceptives, and dihydrofolate reductase inhibitors (trimethoprim, sulfasalazine, methotrexate). Megaloblastic anemia can be caused by a deficiency of folic acid or vitamin B₁₂.

A mnemonic for the causes of megaloblastic anemia is: Prime BATSMAN

- Prime – phenytoin, primidone
- B – B₁₂ and folate deficiency
- A – alcohol
- T – Trimethoprim
- S – Sulfasalazine
- M – Metformin

- A – Antifolate (methotrexate)
- N – N₂O

Solution to Question 3:

Cyanocobalamin (Vitamin B₁₂) should never be administered through the intravenous route. This is because IV administration will result in almost all of the vitamin being lost in the urine.

Cyanocobalamin is usually administered by the intramuscular or deep subcutaneous route. In patients with a previous history of sensitivity to vitamin B₁₂ injections, an intradermal skin test of cyanocobalamin should be done.

Solution to Question 4:

The absorption of iron is impeded by egg-based foods. They contain a high amount of phosphates (egg yolk) that form complexes with iron and impede absorption.

Heme-based iron, present in meat products, is easily absorbed without the aid of a carrier and is not much affected by other foods. Inorganic iron is affected by several factors.

Factors increasing iron absorption are:

- Reducing substances such as ascorbic acid and amino acids containing –SH groups - reduce ferric iron to the soluble ferrous form
- Meat-based foods - enhance heme absorption and increase acid secretion.

Factors decreasing iron absorption are:

- Phosphates, phytates, and tetracyclines - by forming complexes
- Alkalies - prevents the reduction of ferric iron to the soluble ferrous form
- Presence of other food in the stomach - by increasing pH

Solution to Question 5:

In an anemic patient with IBD, parenteral iron is preferred over oral formulations. The parenteral formulation of iron among the given options is ferrous sucrose.

On intravenous administration, it is taken up by reticuloendothelial cells where it dissociates and iron is utilized.

The formulations used for parenteral iron therapy are:

- Iron dextran
- Iron - sorbitol - citric acid
- Ferrous sucrose

- Ferric carboxy maltose

Ferrous sulfate, ferrous gluconate, and ferrous fumarate are oral iron preparations.

Solution to Question 6:

Erythropoietin decreases the need for blood transfusions and increases reticulocyte counts.

Therapeutic uses and effects of erythropoietin:

- The primary indication of erythropoietin is anaemia due to chronic renal failure
- It causes functional iron deficiency due to the inability to mobilize iron stores rapidly enough to support the increased erythropoiesis (low transferrin saturation)
- It increases the hematocrit, haemoglobin and reduces the need for transfusions
- It aggravates hypertension due to an increase in hematocrit and leads to thrombus formation
- Release of reticulocyte into circulation from the marrow and results in an increase in reticulocyte count.

Solution to Question 7:

Darbepoetin alpha is the recombinant derivative of erythropoietin that is long-acting and approved for anaemia due to chronic renal failure.

Option A: Oprelvekin is an IL-11 analog that stimulates megakaryocyte maturation.

Option C: Dornase alfa is a recombinant human DNase 1 that is used in cystic fibrosis to reduce the viscosity of mucosal secretions.

Option D: Filgrastim is a recombinant human G-CSF, indicated in the treatment of severe neutropenia.

Solution to Question 8:

Sargramostim is a recombinant human GM-CSF (Granulocyte-macrophage colony-stimulating factor).

Indications for use:

- Autologous bone marrow transplant - reduces the duration of neutropenia
- Cancer chemotherapy - reduces the duration of neutropenia
- Peripheral blood stem cell collection - to mobilize CD34 positive progenitor cells

It is administered subcutaneously or by slow intravenous infusion.

Solution to Question 9:

Patients with sickle cell anemia should not receive pegylated filgrastim as it can trigger a sickle cell crisis.

Filgrastim is a recombinant human G-CSF (granulocyte colony-stimulating factor). The pegylated forms of filgrastim, pegfilgrastim, and lipegfilgrastim, are longer-acting forms. It is administered subcutaneously or by slow intravenous infusion only.

Indications of recombinant G-CSF:

- Severe neutropenia after autologous hematopoietic stem cell transplantation
- Severe neutropenia after high dose cancer chemotherapy
- Severe congenital neutropenias
- Neutropenia of patients with AIDS receiving zidovudine

Solution to Question 10:

Oprelvekin is an IL-11 analogue that stimulates megakaryocyte maturation and is not a thrombopoietin receptor agonist.

Romiplostim, eltrombopag and lusutrombopag are thrombopoietin receptor agonists. They are indicated in patients with immune thrombocytopenic purpura (ITP) who have failed to respond to the conventional treatments.

Solution to Question 11:

Oprelvekin is used in the treatment of thrombocytopenia secondary to anti-cancer therapy.

It is a recombinant human interleukin-11. Its predominant hemopoietic activity is the stimulation of megakaryocytopoiesis. Oprelvekin is indicated for the prevention and treatment of severe thrombocytopenia and the reduction of platelet transfusion requirements following myelosuppressive chemotherapy in patients with non-myeloid malignancies.

Solution to Question 12:

Plerixafor is the CXCR-4 antagonist indicated for peripheral blood stem cell mobilization before autologous stem cell transplantation.

It is indicated for peripheral blood stem cell mobilization in patients with Non-Hodgkin's lymphoma and multiple myeloma having inadequate response to G-CSF. It mobilizes the release and entry of HSCs from the bone marrow into circulation. Peripheral blood stem cell mobilization/collection is followed by autologous transplantation.

Option A: Maraviroc is a CCR-5 receptor inhibitor indicated in HIV infections.

Options B and C: Sargramostim (GM-CSF) and filgrastim (G-CSF) are used for peripheral blood stem cell mobilization.

Solution to Question 13:

Avatrombopag is a thrombopoietin receptor agonist approved for the treatment of thrombocytopenia with chronic liver disease in patients who are scheduled to undergo a medical or dental procedure.

Thrombopoietin receptor agonist stimulates proliferation and differentiation of megakaryocytes from bone marrow progenitor cells resulting in increased production of platelets. Lusutrombopag is a drug with similar action and function.

Options A & B: Eltrombopag and romiplostim are thrombopoietin receptor agonists indicated in chronic immune thrombocytopenia.

Option C: Oprelvekin is a recombinant IL-11 that is used in the prophylaxis of thrombocytopenia during cancer chemotherapy.

Solution to Question 14:

The scenario in the question depicts a case of chronic immune thrombocytopenia which can be treated with a tyrosine kinase inhibitor called fostamatinib.

Fostamatinib has activity against spleen tyrosine kinase (SYK). The major metabolite of fostamatinib, R406, inhibits signal transduction of Fc-activating receptors and B-cell receptors. The fostamatinib metabolite R406 reduces antibody-mediated destruction of platelets.

Option A: Romiplostim is a TPO receptor agonist indicated in chronic immune thrombocytopenia.

Option B: Bosutinib is a BCR-ABL tyrosine kinase inhibitor indicated in CML.

Option D: Oprelvekin is a recombinant IL-11 that is used in the prophylaxis of thrombocytopenia during cancer chemotherapy.

Anticoagulants

Question 1:

A patient with sustained atrial fibrillation was given heparin cover. What is the primary mechanism of action of the drug?

- a) Activation of thrombin
- b) Activation of antithrombin
- c) Inhibition of antithrombin
- d) Direct inhibition of thrombin

Question 2:

Which of the following is not true about heparin?

- a) It is the weakest acid found in the body
- b) It is commonly obtained from pig intestine
- c) It acts by activation of antithrombin
- d) It can lead to thrombocytopenia

Question 3:

A patient is receiving low dose heparin infusions after hip replacement surgery to prevent thromboembolism. Which of the following abnormalities are least likely to be expected in this patient?

- a) Prolonged aPTT
- b) Alopecia
- c) Hypokalemia
- d) Thrombocytopenia

Question 4:

For the action of heparin as an anticoagulant, availability of which of the following chemical substances is mandatory?

- a) Thrombin
- b) Plasmin
- c) Antithrombin-III
- d) Thrombomodulin

Question 5:

A patient scheduled for a CABG procedure receives preoperative heparin for anticoagulation. During the surgery, the patient starts bleeding massively and there is a drop in BP. Which of the following would you use to manage this patient?

- a) Protamine sulfate
- b) Prothrombin complex concentrate
- c) Aminocaproic acid
- d) Tranexamic acid

Question 6:

Life-threatening bleeding in a patient receiving heparin is treated immediately with a drug X that reverses its effects. Identify the type of antagonism between X and heparin.

- a) Chemical
- b) Physical
- c) Physiological
- d) Competitive

Question 7:

Predominant activity of low-molecular-weight heparin is seen against which clotting factor?

- a) IIa
- b) IXa
- c) XIIa
- d) Xa

Question 8:

LMWH is given by which route?

- a) Intravenous
- b) Subcutaneous
- c) Intramuscular
- d) Intradermal

Question 9:

As compared to unfractionated heparin, which of the following is true about enoxaparin?

- a) Is absorbed more uniformly when given subcutaneously
- b) Requires more frequent laboratory monitoring
- c) Can be given to patients with heparin induced thrombocytopenia
- d) Predisposes to a higher risk of osteopenia

Question 10:

Which is the anticoagulant of choice for the management of cancer-associated thromboembolism?

- a) Unfractionated Heparin
- b) Warfarin
- c) Low molecular weight heparin
- d) Dabigatran

Question 11:

Fondaparinux can be best described as which of the following?

- a) Low molecular weight heparin
- b) Antithrombin activator
- c) Direct thrombin inhibitor
- d) Direct factor Xa inhibitor

Question 12:

A 65-year-old patient has been scheduled for cardiopulmonary bypass surgery. Which is the anticoagulant of choice for this patient?

- a) Unfractionated heparin
- b) LMWH
- c) Fondaparinux
- d) Warfarin

Question 13:

Lab reports of a 45-year-old female patient, who was recently started on heparin, showed rapidly falling platelet counts. She was suspected to have heparin-induced thrombocytopenia. What is the most common presentation seen in this condition?

- a) Bleeding
- b) Venous thrombosis
- c) Skin necrosis
- d) Arterial thrombosis

Question 14:

A 64-year-old man was started on heparin postoperatively for thromboprophylaxis. After a week he developed sudden difficulty in breathing. His platelet count was found to be 70,000/microliter. Which of the following is incorrect regarding the management of this patient?

- a) Heparin should be discontinued immediately
- b) Alternative anticoagulant such as argatroban should be administered
- c) Low molecular weight heparins should be avoided
- d) Heparin should be replaced with warfarin

Question 15:

A patient on anticoagulant therapy presented to the casualty with an uncontrolled bleeding episode and was given IV protamine sulfate. However, the patient did not respond to the treatment. Which of the following anticoagulant agents was most likely being used by the patient?

- a) Unfractionated heparin

- b) Enoxaparin
- c) Dalteparin
- d) Fondaparinux

Question 16:

A clinical trial is done to compare the safety profile of a certain direct thrombin inhibitor (drug X) with that of heparin in patients undergoing coronary angioplasty. Identify drug X.

- a) Enoxaparin
- b) Bivalirudin
- c) Fondaparinux
- d) Dalteparin

Question 17:

The oral thrombin inhibitor which can be used for prevention of stroke in patients with non-valvular atrial fibrillation is?

- a) Dabigatran
- b) Ximelgatran
- c) Lepirudin
- d) Saxagliptin

Question 18:

Which of the following is a parenteral direct thrombin inhibitor?

- a) Apixaban
- b) Argatroban
- c) Dabigatran
- d) Enoxaparin

Question 19:

Which of the following is an oral factor Xa inhibitor?

- a) Bivalirudin
- b) Dabigatran
- c) Rivaroxaban
- d) Enoxaparin

Question 20:

A patient on apixaban for nonvalvular atrial fibrillation requires urgent aortic surgery for a pseudoaneurysm. Identify the decoy factor Xa used for reversal of apixaban action before the surgery.

- a) Dornase alfa
- b) Andexanet alfa
- c) Idarucizumab
- d) Protamine sulfate

Question 21:

A patient with a history of regular oral anticoagulant therapy suffers a traumatic injury. He receives idarucizumab before damage control surgery to reverse the anticoagulant effect. Identify the oral anticoagulant that was being used by the patient.

- a) Heparin
- b) Warfarin
- c) Dabigatran
- d) Rivaroxaban

Question 22:

Which of the following drugs is under clinical trials and shows a reversal in the action of all direct anticoagulants?

- a) Protamine
- b) Idarucizumab
- c) Andexanet alfa
- d) Ciraparantag

Question 23:

Coagulation profile monitoring is needed in therapy with which of the following drugs?

- a) Fondaparinux
- b) Enoxaparin
- c) Dabigatran
- d) Lepirudin

Question 24:

Which of the following anticoagulants can be safely used in a pregnant patient?

- a) Heparin
- b) Warfarin
- c) Dicumarol
- d) Phenindione

Question 25:

A patient is on DVT prophylaxis with warfarin. It has an inhibitory action on which of the following enzymes?

- a) Gamma glutamyl carboxylase
- b) Vitamin K epoxide reductase
- c) Thrombin
- d) HMG-CoA reductase

Question 26:

Which of the following is vitamin K dependent clotting factor?

- a) Factor I
- b) Factor VII
- c) Factor XI
- d) Factor XII

Question 27:

A patient suffering from non-valvular atrial fibrillation is prescribed warfarin tablets. Which of the following clotting factors would you expect to decline the earliest in this patient after drug administration?

- a) Factor II
- b) Factor VII
- c) Factor IX
- d) Factor X

Question 28:

In a patient receiving warfarin tablets, which enzyme primarily metabolizes the drug?

- a) CYP2C19
- b) CYP2C9
- c) CYP3A4
- d) CYP1A2

Question 29:

Which of the following drugs does not increase the risk of bleeding in patients undergoing warfarin therapy?

- a) Cimetidine
- b) Carbamazepine
- c) Isoniazid
- d) Amiodarone

Question 30:

While prescribing warfarin, a decreased effect of the drug is expected in patients with which of the following conditions?

- a) Nephrotic syndrome
- b) Hepatic impairment
- c) Gastroenteritis on treatment with metronidazole

d) Congestive heart failure

Question 31:

A patient with splanchnic vein thrombosis has been started on warfarin therapy. Which of the following can be used to monitor the treatment in this patient?

- a) Prothrombin time**
- b) Activated partial thromboplastin time**
- c) Bleeding time**
- d) Clotting time**

Question 32:

A 50-year-old patient admitted with atrial fibrillation was started on bisoprolol and warfarin. After 4 days of admission, she started experiencing painful ecchymoses on her right lower limb. It progressed to the condition shown in the image over the next few days. Deficiency in which of the following can lead to this condition?



- a) Protein C**
- b) Protein S**
- c) Either a or b**
- d) Thrombin**

Question 33:

A 27-year-old male patient on anticoagulant therapy for his DVT started coughing up blood. He has a history of irregular warfarin intake. Workup revealed elevated INR and diffuse alveolar hemorrhage. Which of the following agents would be the best choice for immediate reversal of bleeding?

- a) Cryoprecipitate
- b) Fresh frozen plasma
- c) Prothrombin complex concentrates
- d) Vitamin K1

Question 34:

Which among the following is not a fetal abnormality expected after warfarin administration to a pregnant patient?

- a) Nasal hypoplasia
- b) Fetal haemorrhage
- c) Stippled epiphyses
- d) Macrocephaly

Question 35:

In patients with indications for warfarin usage, what is the recommended range of INR necessary for safe administration of the drug?

- a) 1.0 – 1.5
- b) 2.0 – 3.0
- c) 2.5 – 3.5
- d) 3.5 – 4.5

Question 36:

Apixaban belongs to which of the following class of drugs?

- a) Antifibrinolytic
- b) Factor Xa inhibitor

- c) Oral direct thrombin inhibitor
- d) Parenteral direct thrombin inhibitor

Answer Key

Question No.	Correct Option
1	b
2	a
3	c
4	c
5	a
6	a
7	d
8	b
9	a
10	c
11	b
12	a
13	b
14	d
15	d
16	b
17	a
18	b
19	c
20	b
21	c
22	d
23	d
24	a
25	b
26	b
27	b
28	b
29	b

30	a
31	a
32	c
33	c
34	d
35	b
36	b

Detailed Explanations

Solution to Question 1:

The primary mechanism of action of heparin is by activation of antithrombin.

In patients with atrial fibrillation that has persisted for more than 48 hours, heparin can be used to reduce the risk of thrombus formation and embolization until the warfarin level is therapeutic or cardioversion is performed.

Heparin, low molecular weight heparin (LMWHs), and fondaparinux have no intrinsic anticoagulant activity. They bind to antithrombin and accelerates the rate at which it inhibits various coagulation proteases, especially thrombin (factor IIa) and factor Xa.

LMWH has some action against thrombin. However, fondaparinux has no action against thrombin.

Antithrombin inhibits these factors by serving as a “suicide substrate”.

Solution to Question 2:

Heparin is the strongest organic acid present in the body.

Option B: Most commercial preparations of heparin are extracted from the porcine intestinal mucosa.

Option C: Heparin acts indirectly through the activation of antithrombin and inhibition of factors IIa and Xa.

Option D: Heparin use can result in thrombocytopenia due to the development of antibodies to heparin-platelet factor 4 complex.

Solution to Question 3:

Hypokalemia is least likely seen in this patient because heparin-induced hypoaldosteronism causes hyperkalemia.

aPTT is prolonged in heparin therapy and is used to monitor its anticoagulant effect.

The other side-effects of heparin are (OATH):

O - Osteoporosis

A - Alopecia (reversible)

T - Thrombocytopenia (Heparin-Induced Thrombocytopenia - HIT)

H - Hemorrhage, Hyperkalemia

Solution to Question 4:

Antithrombin-III is mandatory for the action of heparin as an anticoagulant. Reduced antithrombin levels can lead to resistance.

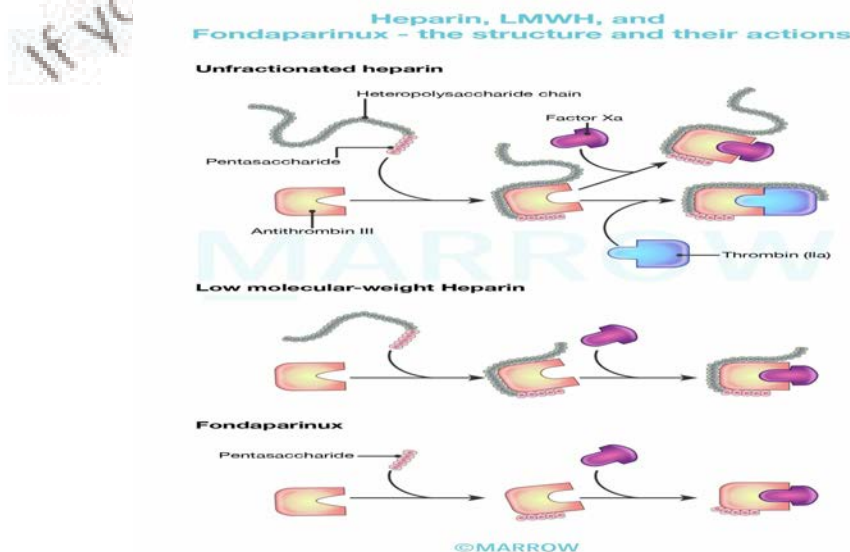
Heparin resistance refers to failure to achieve a therapeutic aPTT with daily doses of heparin of 35,000 units or more.

In true heparin resistance, there is an actual blunting of heparin's anticoagulant effect. This occurs due to:

- Low ATIII levels (most common cause)
- Increased heparin-binding plasma proteins (acute phase reactants)
- Increased heparin clearance (e.g. due to splenomegaly in liver disease)

Low ATIII levels could be due to decreased synthesis (hereditary, hepatic failure), increased consumption (DIC, MAHA, or in a bypass circuit) or increased loss (nephrotic syndrome).

The dose of heparin has to be increased to achieve therapeutic aPTT. Severe antithrombin deficiency may require antithrombin concentrate as well.



Solution to Question 5:

Protamine sulfate is the drug used to treat life-threatening, heparin-induced bleeding.

It is a mixture of basic polypeptides isolated from salmon sperm. It binds to acidic heparin with high affinity and neutralizes anticoagulant activity. The resultant protamine-heparin complexes are cleared. Protamine binds only to long heparin molecules (UFH). Therefore, protamine only partially reverses the activity of LMWH and has no effect on fondaparinux.

It is most commonly used to neutralize heparin-induced anticoagulation after separation from cardiopulmonary bypass. It is also utilized in the setting of dialysis, invasive vascular procedures, and acute ischemic strokes.

Note: Protamine sulfate has anticoagulant activity at higher doses and hence, only small amounts of protamine (1mg/100 units of heparin) are used for heparin-induced bleeding.

Solution to Question 6:

The drug antagonism between heparin and protamine sulfate (Drug X) is an example of chemical antagonism.

Protamine sulfate is a mixture of basic polypeptides that are positively charged. They bind to the highly acidic heparin with a strong negative charge and form protamine-heparin complexes that are cleared.

Solution to Question 7:

Low-molecular-weight heparin (LMWH) has predominant activity against factor Xa.

LMWH has an anti-factor activity against both Xa and IIa with a greater effect on Xa.

Solution to Question 8:

LMWH is given by the subcutaneous route.

Heparin, low molecular weight heparin (LMWH), and fondaparinux are not absorbed through GI mucosa and must be given parenterally.

LMWH and fondaparinux are given subcutaneously.

Heparin can be given through intravenous infusion and subcutaneous routes.

Solution to Question 9:

Enoxaparin (LMWH) has a very high bioavailability and is absorbed more uniformly when compared to unfractionated heparin. Hence, it does not require frequent monitoring.

LMWH can also lead to heparin-induced thrombocytopenia (HIT) and hence, it cannot be given in patients with HIT.

LMWH preparations:

- Enoxaparin
- Dalteparin
- Tinzaparin.

Solution to Question 10:

The anticoagulant of choice for the management of cancer-associated thromboembolism is low-molecular-weight heparin (LMWH).

It is the preferred agent because of more bioavailability, lack of need for anticoagulant monitoring, and a long $t_{1/2}$.

Venous thromboembolism (including deep vein thrombosis, pulmonary embolism, and splanchnic vein thrombosis), is a frequent complication of cancer and is the second-leading cause of death in cancer patients receiving chemotherapy.

Based on the results from the Comparison of Low-molecular-weight heparin vs. Oral anticoagulants for long-term anticoagulation in cancer patients with venous Thromboembolism (CLOT) Trial, all recent guidelines now advise LMWH for cancer-associated thromboembolism.

However, unfractionated heparin (UFH) can be used in patients with renal failure, and fondaparinux can be considered in patients with heparin-induced thrombocytopenia (HIT).

Solution to Question 11:

Fondaparinux is an antithrombin activator.

It is a synthetic analog of the pentasaccharide unit of heparin that binds and activates antithrombin. Activation of antithrombin causes the inactivation of factor Xa. So it is also described as an indirect factor Xa inhibitor.

It is not a direct factor Xa inhibitor. Direct factor Xa inhibitors include rivaroxaban, apixaban, and edoxaban.

Solution to Question 12:

Unfractionated heparin is the anticoagulant of choice for patients requiring cardiopulmonary bypass.

Cardiopulmonary bypass circuits activate factor XII which can lead to clotting in the oxygenator. This process is called catheter-induced thrombosis.

Unfractionated heparin is better than LMWH or fondaparinux for blocking it. Also, reversal of anticoagulation using protamine sulfate is easier for UFH compared to the other heparins.

Solution to Question 13:

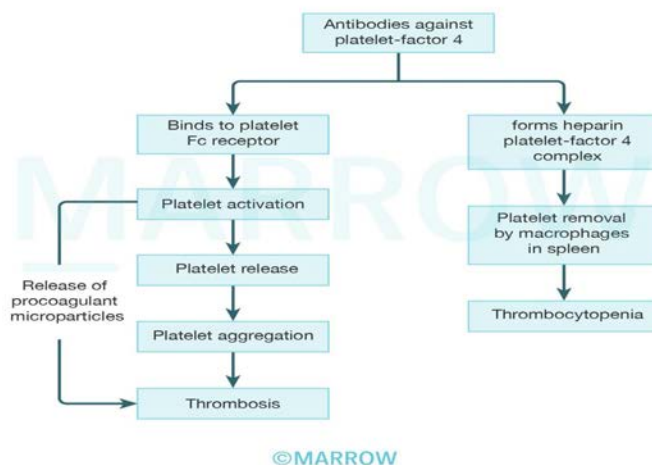
The most common presentation of heparin-induced thrombocytopenia is venous thrombosis.

HIT is an immune complication of heparin therapy caused by IgG antibodies against the heparin-platelet factor 4 complex. This results in the activation of platelets to propagate a hypercoagulable state culminating in life-threatening thrombosis.

It is characterized by thrombocytopenia (Platelet count $<1,00,000/\mu\text{L}$ or reduced by over 50% from pre-treatment value) and clinically presents as venous thrombosis (in the form of DVT or pulmonary embolism)

It occurs around 5-10 days after starting heparin. It is more common with UFH when compared to LMWH. It is also more common in women surgical patients. Arterial thrombosis (resulting in MI or stroke), as well as bilateral adrenal hemorrhage, can rarely occur.

Mechanism of heparin-induced thrombocytopenia



Solution to Question 14:

The patient in the above scenario has heparin-induced thrombocytopenia (HIT). Direct thrombin inhibitors, such as argatroban is the anticoagulant of choice in HIT.

Management of HIT:

- Stop heparin/LMWH
- Give an alternative anticoagulant, such as lepirudin, argatroban, bivalirudin, fondaparinux or rivaroxaban
- Do not give platelet transfusions (it only worsens the consumptive coagulopathy)

- Do not give warfarin until the platelet count returns to its baseline level. If warfarin was administered, give vitamin K to restore the INR to normal.

Increased thrombin generation associated with HIT can cause consumption of protein C. Warfarin administration can further decrease protein C levels and cause skin necrosis. Therefore warfarin is avoided in HIT.

Solution to Question 15:

Since no effect was noted with protamine sulfate, the likely anticoagulant in use must be fondaparinux. This is because protamine sulfate fails to bind the drug.

Protamine sulfate binds tightly to heparin and neutralizes its anticoagulant effect. It partially reverses the anticoagulant activity of LMWH also. But it has no effect on fondaparinux.

There is no known specific antidote approved for use in fondaparinux overdose.

Fondaparinux-associated bleeding complications are managed by treatment discontinuation and search for the primary cause. Surgical hemostasis, blood replacements, fresh plasma transfusion, plasmapheresis should be considered.

Solution to Question 16:

Among the given options, the direct thrombin inhibitor is bivalirudin.

Bivalirudin is a parenteral direct thrombin inhibitor (DTI) frequently used for anticoagulation in invasive cardiology, particularly percutaneous coronary intervention (PCI). Unlike other DTIs, it undergoes predominant non-organ elimination (proteolysis) and has the shortest half-life (approximately 25 min).

It can be used in patients with, or at risk for, heparin-induced thrombocytopenia.

Other options:

Options A and D: Enoxaparin and dalteparin are low-molecular-weight-heparin (LMWH) preparations.

Option C: Fondaparinux is a synthetic analog of the pentasaccharide unit of heparin.

Heparins and their analogs produce anticoagulant effects indirectly by activating antithrombin-III.

Solution to Question 17:

The oral thrombin inhibitor which can be used for the prevention of stroke in patients with non-valvular atrial fibrillation is dabigatran. However, it is contraindicated in patients with mechanical heart valves.

Option B: Ximelagatran is also an oral thrombin inhibitor, but it has been discontinued due to the risk of hepatotoxicity.

Option C: Lepirudin is a parenteral direct thrombin inhibitor used mainly in heparin-induced thrombocytopenia.

Option D: Saxagliptin is a DPP-4 inhibitor used in the management of type 2 diabetes mellitus.

Solution to Question 18:

The parenteral direct thrombin inhibitor is argatroban.

Argatroban binds reversibly to thrombin and inhibits it. It is metabolized in the liver and excreted in bile, so it can be safely used in patients with renal impairment.

It can be used for prophylaxis and treatment of heparin-induced thrombocytopenia (HIT).

Other parenteral direct thrombin inhibitors are their indications:

- Lepirudin - was indicated in HIT but is discontinued now
- Desirudin - indicated for thromboprophylaxis in hip replacement surgery
- Bivalirudin - used in procedures like coronary angioplasty or cardiopulmonary bypass surgery and HIT.

Solution to Question 19:

Rivaroxaban is an oral direct factor Xa inhibitor. They represent a new class of oral anticoagulant drugs that do not require monitoring.

This group of drugs inhibit factor Xa in the final common pathway of clotting [conversion of Prothrombin (Factor II) to thrombin (Factor IIa)]. They are given as fixed doses and have a rapid onset of action.

Oral direct factor Xa inhibitors are indicated in:

- Prevention of embolic stroke in patients with atrial fibrillation (without valvular heart disease)
- Venous thromboembolic disease (VTE)
- Postoperative thromboprophylaxis in patients undergoing hip and knee arthroplasty

Solution to Question 20:

Andexanet alfa is the decoy factor Xa approved for reversal of oral Xa inhibitors. It was approved in May 2018.

Option A: Dornase alfa is a recombinant human DNase 1 that is used in cystic fibrosis to reduce the viscosity of mucosal secretions.

Option C: Idarucizumab is a chimeric monoclonal antibody against dabigatran. It has a higher affinity towards dabigatran than thrombin and is approved as a reversal agent for dabigatran.

Option D: Protamine sulfate is an antidote for heparin overdose.

Solution to Question 21:

Idarucizumab is a reversal agent for dabigatran. It is a humanized mouse monoclonal antibody fragment.

It binds irreversibly to dabigatran and forms a complex that is cleared by the kidneys.

Solution to Question 22:

Ciraparantag is a synthetic cationic molecule that binds to all direct anticoagulants (dabigatran and other oral factor Xa inhibitors) as well as heparin and LMWH and reverses their actions. It is still under trial.

Option A: Protamine sulfate reverses the action of heparin and LMWH.

Option B: Idarucizumab is a specific antidote for dabigatran overdose.

Option C: Andexanet alfa is a decoy for oral factor Xa inhibitors.

Solution to Question 23:

Coagulation profile monitoring is needed for lepirudin.

Lepirudin is a direct irreversible thrombin inhibitor and its most adverse reaction is bleeding. The risk of bleeding can be reduced by implementing an appropriate monitoring and dose adjustment strategy. Thus, coagulation profile monitoring is required for lepirudin.

However, enoxaparin, fondaparinux, and dabigatran do not bind to endothelium or many plasma proteins and thus, the response to these drugs at the usual doses is consistent from patient to patient. Hence, monitoring is not required.

Solution to Question 24:

Heparin is the anticoagulant that can be safely used in pregnancy.

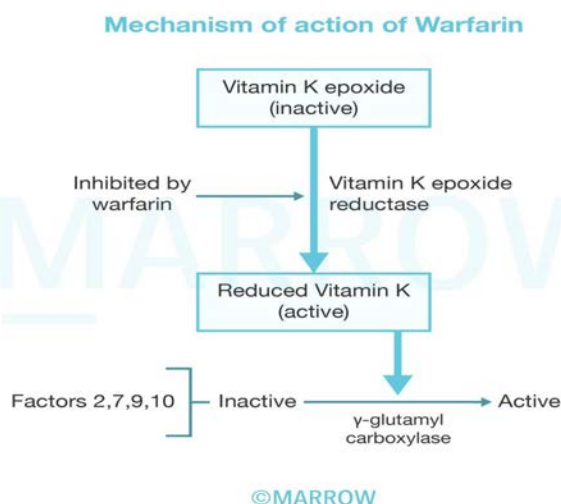
Heparin, LMWH, and fondaparinux do not cross the blood-brain barrier or placenta and hence they are the anticoagulants preferred in pregnancy.

Warfarin and other vitamin K antagonists are contraindicated during pregnancy due to the risk of teratogenicity.

Solution to Question 25:

Warfarin directly inhibits the enzyme Vitamin K epoxide reductase.

It inhibits the conversion of vitamin K epoxide to its reduced form - vitamin K hydroquinone which is a co-factor for the enzyme gamma-glutamyl carboxylase. This enzyme catalyzes the γ -carboxylation process of vitamin K dependent clotting factors.



Solution to Question 26:

Factor VII is a vitamin K dependant clotting factor.

The vitamin K dependent factors:

- Clotting factors: Factors II, VII, IX, X
- Anticoagulant factors: Protein C and Protein S

Solution to Question 27:

Factor VII activity declines most rapidly when warfarin is administered. It has the shortest plasma half-life among the vitamin K dependent clotting factors.

Clinical importance: Because of the long half-life of the other clotting factors, especially factor II, the full anticoagulant effect of warfarin is not seen for several days. Initially, parenteral anticoagulants are used along with warfarin until the full effect of warfarin is achieved.

Clotting factors	Plasma half-life (hours)
II	50
VII	6
IX	24
X	30

Solution to Question 28:

Warfarin is primarily metabolized by CYP2C9.

Genetic polymorphisms of CYP2C9 and drugs inhibiting or inducing CYP2C9 can result in variable response to warfarin administration.

Solution to Question 29:

Carbamazepine does not increase the risk of bleeding in patients undergoing warfarin therapy.

Warfarin is mainly metabolized by the CYP2C9 enzyme. Drugs that inhibit CYP2C9 enzyme reduce the metabolism of warfarin and thus, enhance the anticoagulant effect of warfarin and increase the risk of bleeding.

Drugs causing CYP2C9 inhibition:

- Amiodarone
- Azole antifungals
- Cimetidine
- Clopidogrel
- Cotrimoxazole
- Disulfiram
- Isoniazid
- Metronidazole

Solution to Question 30:

Decreased effect of warfarin is seen in patients with nephrotic syndrome.

Warfarin is highly bound to plasma protein and in nephrotic syndrome there is proteinuria. Drugs that are bound to plasma proteins are lost in the urine. Similarly, warfarin is lost in urine in nephrotic syndrome, resulting in a decreased effect.

Option C: Warfarin is mainly metabolized by the CYP2C9 enzyme. In patients being treated with metronidazole (a CYP2C9 inhibitor), the metabolism of warfarin is decreased. Hence, the effect of warfarin is increased.

Options B and D: Hepatic impairment, congestive heart failure, and hyperthyroidism lead to low concentrations of clotting factors and thus the effect of warfarin is enhanced in these conditions.

Solution to Question 31:

Warfarin therapy is usually monitored by measuring prothrombin time.

Prothrombin time is sensitive to the reduction in levels of prothrombin, factor VII, and factor X. The test is performed by adding thromboplastin, which is a reagent containing tissue factor, phospholipid and calcium to citrated plasma and determining the time to clot formation.

Due to the variable sensitivity of thromboplastin, the International normalized ratio (INR) was developed to standardize the reporting of prothrombin time. In INR, the patient's prothrombin time is divided by mean normal prothrombin time and this ratio is multiplied by the International Sensitivity Index (ISI) of the particular thromboplastin reagent.

Warfarin therapy should be monitored with INR every 3–4 weeks. The target INR is 2.0–3.0. In patients with mechanical heart valves, the target INR is 2.5–3.5.

Solution to Question 32:

The clinical features and the image given are suggestive of warfarin-induced skin necrosis seen due to a congenital or acquired deficiency in either protein C or protein S.

Warfarin therapy in such patients leads to a rapid fall in protein C or protein S, which are vitamin K dependent anticoagulants. This occurs before warfarin can exert an antithrombotic effect.

Treatment: Warfarin should be discontinued and vitamin K (the antidote for warfarin) should be administered.

Solution to Question 33:

Immediate or urgent reversal of warfarin-induced bleeding is best achieved by the administration of prothrombin complex concentrates (PCC).

These are highly purified concentrates prepared from pooled plasma and contain factors II, IX, and X. Factor VII may be included or produced separately.

Option D: Vitamin K₁ is the specific antidote for warfarin-induced bleeding, but it takes 6–24 hours for the clotting factors to be resynthesized and released in blood after Vitamin K administration and hence its action is delayed for several hours.

Option A: Cryoprecipitate mainly contains factor VIII and fibrinogen and therefore is not suitable for warfarin-induced bleeding, as warfarin inhibits factors II, VII, IX, X.

Option B: Fresh frozen plasma can also be used but prothrombin complex concentrate normalizes the INR more rapidly.

Solution to Question 34:

Macrocephaly is not a fetal abnormality caused by warfarin administered during pregnancy.

Warfarin crosses the placenta and can cause fetal abnormalities which include nasal hypoplasia and stippled epiphyses leading to abnormal bone formation. Fetal or neonatal hemorrhage is also seen.

The risk of embryopathy is highest with exposure in the first trimester of pregnancy. CNS abnormalities have also been reported following exposure during the second and third trimesters.

Solution to Question 35:

The INR ratio recommended for the administration of warfarin in most of the indications is in the range of 2.0-3.0.

Option A: The range of 1.0-1.5 is too low for beneficial effects.

Option C: In selected patients with mechanical heart valves, the INR target is around 2.5-3.5.

Option D: INR ≥ 3.5 can result in increased adverse effects and is not recommended for any condition.

Solution to Question 36:

Apixaban, a novel oral anticoagulant (NOAC), is a factor Xa inhibitor that is used in the treatment of VTE (venous thrombo embolism)/DVT and in its prophylaxis after major orthopedic surgeries.

Direct factor Xa inhibitors:

- Rivaroxaban
- Apixaban
- Edoxaban
- Betrixaban

Parenteral direct thrombin inhibitors:

- Bivalirudin
- Argatroban
- Desirudin

Oral direct thrombin inhibitor—Dabigatran is the only oral direct thrombin inhibitor available for clinical use.

The FDA recently approved the drug Apixaban for a new indication. The drug is now used to reduce the risk of stroke and systemic embolism in patients with nonvalvular atrial fibrillation.

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Respiratory System

Question 1:

Which of the following is a long-acting β -2 agonist that can be prescribed in a patient with moderately severe bronchial asthma?

- a) Salbutamol
- b) Indacaterol
- c) Terbutaline
- d) Formoterol

Question 2:

What is the current GINA recommendation for the prescription of long-acting β -2 agonists in patients suffering from bronchial asthma?

- a) As monotherapy
- b) In combination with an inhaled anticholinergic
- c) In combination with an inhaled corticosteroid
- d) In combination with a short acting β -2 agonist

Question 3:

Which among the following is the most appropriate therapy for bronchodilation in acute severe asthma?

- a) Intravenous aminophylline
- b) Inhalational anticholinergics
- c) Nebulized SABA
- d) Intravenous SABA

Question 4:

Which of the following is not a side effect of salbutamol?

- a) Hypoglycemia

- b) Hypokalemia
- c) Tremors
- d) Tachycardia

Question 5:

An asthmatic patient complains of wheezing and shortness of breath most days of the week. Which of the following therapies is preferred in him?

- a) Inhaled SABA
- b) Inhaled LABA
- c) Inhaled LABA + inhaled corticosteroids
- d) Inhaled SABA + inhaled corticosteroids

Question 6:

A 5-year-old child requires bronchodilator therapy. You attach the device shown below to the pressurized metered-dose inhaler for ease of use. Which of the following is false about this device?



- a) Decreases the deposition of drug on the oropharynx
- b) Increases the delivery of drug to lower airways
- c) Increases the incidence of oropharyngeal candidiasis
- d) Systemic absorption of drugs is unaffected

Question 7:

An asthmatic patient has been taking theophylline tablets for symptomatic relief. On which of the following receptors does this drug have an antagonistic action?

- a) Histamine receptors
- b) Bradykinin receptors
- c) Adenosine receptors
- d) Kallikrein receptors

Question 8:

Which of the following statements about theophylline is not correct?

- a) Increase in dose is required in cardiopulmonary disease
- b) Decreases the breakdown of cAMP in smooth muscles
- c) Increase in dose is required in smokers
- d) Is a non-selective phosphodiesterase inhibitor

Question 9:

Which of the following is an indication for theophylline?

- a) Supraventricular tachycardia
- b) Epilepsy
- c) Acute severe asthma
- d) Apnea of prematurity

Question 10:

A patient with severe persistent asthma continues to have symptoms despite being on high-dose ICS-LABA therapy. Which of the following is a leukotriene receptor antagonist, appropriate as an add-on therapy for this patient?

- a) Dupilumab
- b) Roflumilast
- c) Zafirlukast

d) Zileuton

Question 11:

The use of which of the following therapies for acute asthma requires serial monitoring of liver function tests?

- a) 3 and 4
- b) 2 and 3
- c) 1, 2 and 5
- d) 1, 4 and 5

Question 12:

An elderly asthmatic on treatment presented with fever, malaise, and lower limb sensory deficit. Significant investigations are given below. Which of the following drugs is associated with these findings?

- a) Montelukast
- b) Cromolyn sodium
- c) Theophylline
- d) Ketotifen

Question 13:

An asthmatic patient was brought to the ER in an agitated state. His respiratory rate was 32/min and pulse was 124/min. Pulse oximetry revealed a SpO₂ of 88% on room air. Which of the following drugs is not useful in this scenario?

- a) Salbutamol
- b) Hydrocortisone
- c) Cromolyn sodium
- d) Ipratropium bromide

Question 14:

Omalizumab is added as a step-up therapy in a child with severe asthma, refractory to treatment. Which of the following is true about this drug?

- a) Is administered intravenously
- b) Has anti interleukin-5 action
- c) Is indicated in acute severe asthma
- d) Is a monoclonal antibody against IgE

Question 15:

A patient diagnosed with Churg-Strauss syndrome is started on Mepolizumab. This drug acts against:

- a) IgE
- b) IgA
- c) IL-5
- d) IL-12

Question 16:

What is the thromboxane receptor antagonist used in the treatment of asthma?

- a) Seratrodast
- b) Rimanobant
- c) Sacubitril
- d) Roflumilast

Question 17:

In a case of acute exacerbation of asthma, salbutamol was given, however, no improvement was noticed. Intravenous corticosteroids and aminophylline were given and the condition improved. What is the probable mechanism by which the corticosteroids would have improved the condition?

- a) Corticosteroids increase bronchial responsiveness to salbutamol
- b) Corticosteroids cause direct bronchodilatation when used with xanthines
- c) Corticosteroids indirectly increase the effect of xanthines on adenosine receptor
- d) Corticosteroids increase mucociliary clearance

Question 18:

A patient has had 2 acute exacerbations of asthma over the span of 1-month despite being on therapy. Which of the following is a systemic steroid that can be used to treat her?

- a) Ciclesonide
- b) Beclomethasone
- c) Prednisolone
- d) Budesonide

Question 19:

An asthmatic patient presented to the ER with dyspnea and wheezing. Intake of which of the following is unlikely to have precipitated this?

- a) Phenylbutazone
- b) Naproxen
- c) Caffeine
- d) Aspirin

Question 20:

A patient with mild episodic asthma is prescribed daily low-dose inhaled budesonide therapy. Which of the following conditions is he not at increased risk of developing?

- a) Dysphonia
- b) Pulmonary TB
- c) Oropharyngeal candidiasis
- d) Carcinoma of lung

Question 21:

An elderly man with a 30-year history of smoking complains of breathing difficulty. A diagnosis of COPD is made, and treatment with a long-acting muscarinic antagonist is initiated. Which of the following will you not prescribe?

- a) Glycopyrronium

- b) Umeclidinium
- c) Ipratropium
- d) Tiotropium

Question 22:

Inhalational anticholinergics the mainstay of treatment in:

- a) Asthma
- b) COPD
- c) Both a and b
- d) Status asthmaticus

Question 23:

A chronic obstructive pulmonary disease (COPD) patient requires a step-up in therapy due to symptom persistence. Which PDE-4 inhibitor can you prescribe as an add-on drug to his treatment regimen?

- a) Seratrodist
- b) Roflumilast
- c) Zileuton
- d) Nedocromil

Question 24:

Which of the following drugs are antitussives?

- a) 1, 2 and 4
- b) 1, 2 and 5
- c) 3 and 5
- d) 3 and 4

Question 25:

A known case of CREST syndrome presents with dyspnea. On auscultation, a Graham-Steele murmur is heard. On further investigation, her pulmonary artery pressure is found to be 28

mm Hg. Which drug will you use to manage her condition?

- a) Non selective PDE inhibitors
- b) Lipoxygenase inhibitors
- c) Selective PDE-5 inhibitors
- d) Cyclo-oxygenase inhibitors

Question 26:

Which of the following groups does riociguat belong to?

- a) Soluble guanylate cyclase stimulator
- b) Soluble guanylate cyclase inhibitor
- c) Soluble adenylate cyclase stimulator
- d) Soluble adenylate cyclase inhibitor

Question 27:

Which of the following drugs is a prostacyclin analog that can be added to the treatment regimen of a patient with pulmonary artery hypertension?

- a) Sildenafil
- b) Misoprostol
- c) Dinoprost
- d) Iloprost

Question 28:

Which is a selective endothelin receptor A antagonist used in the management of pulmonary artery hypertension?

- a) Bosentan
- b) Ambrisentan
- c) Tezosentan
- d) Macitentan

Question 29:

Dornase alfa is indicated in which of the following patients?

- a) Only 3
- b) 2, 4 and 5
- c) 3 and 5
- d) 1, 2 and 3

Question 30:

Which of the following should not be prescribed with theophylline?

- a) Erythromycin
- b) Cefotaxime
- c) Cotrimoxazole
- d) Amoxicillin

Question 31:

If codiene is coming to the market in combination with dextromethorphan as an anti-tussive, this combination _____

- a) Should not be used due to risk of addiction
- b) Can be used as both have different mechanism of action
- c) Should not be used as it is irrational
- d) Can be used to treat cough effectively

Answer Key

Question No.	Correct Option
1	d
2	c
3	c
4	a
5	c
6	c

7	c
8	a
9	d
10	c
11	d
12	a
13	c
14	d
15	c
16	a
17	a
18	c
19	c
20	d
21	c
22	b
23	b
24	b
25	c
26	a
27	d
28	b
29	c
30	a
31	c

Detailed Explanations

Solution to Question 1:

Formoterol is a long-acting β -2 agonist (LABA).

Beta 2 agonists are classified based on their duration of action into:

- Short-acting β -2 agonists (SABA):
- Salbutamol
- Terbutaline
- Pirbuterol

- Long-acting β -2 agonists (LABA):
- Salmeterol
- Formoterol
- Very long-acting β -2 agonists (V-LABA):
- Indacaterol
- Olodaterol
- Vilanterol

Solution to Question 2:

LABA is used in combination with inhalational corticosteroids (ICS), preferably in the form of a combination inhaler.

LABA is never taken alone as it does not treat the underlying chronic inflammation of asthma. Corticosteroids will decrease the underlying inflammation and halt the progression of the disease.

Short-acting β -2 agonists (SABA) can be given as an add-on therapy to the LABA-ICS combination (as-needed therapy).

Solution to Question 3:

Nebulized short-acting beta-2 agonists (SABAs) are the most appropriate therapy in acute severe asthma among the given options.

Nebulized route is preferred over the intravenous route as it is easier and safer with equivalent efficacy.

Intravenous aminophylline is less effective than SABA and is reserved for those who are resistant or intolerant to beta-2 agonists with acute severe asthma.

Inhalational anticholinergics are not indicated in acute severe asthma as they are less effective than SABAs.

Solution to Question 4:

Hypoglycemia is not a side effect of salbutamol. It causes hyperglycemia.

The side effects of salbutamol are due to the stimulation of extrapulmonary β 2 receptors. They are more common after oral or intravenous administration. The side effects seen are:

- Muscle tremor
- Tachycardia
- Hypokalemia

- Restlessness
- Hypoxemia
- Increase in free fatty acid, insulin, glucose, pyruvate, and lactate

Solution to Question 5:

This patient has persistent asthma for which the preferred therapy is LABA with ICS available as a combination inhaler.

ICS should be started in any patient who needs to use a short-acting beta-2 agonist inhaler for symptomatic control more than twice a week (persistent asthma).

Solution to Question 6:

The given image is of a spacer device. It decreases the incidence of oropharyngeal candidiasis and other propellant-mediated oropharyngeal adverse effects like dysphonia that occur with the use of a pMDI.

Spacers reduce the velocity and size of particles entering the upper airways. This reduces the amount of drug that impinges on the oropharynx and increases the proportion of drug inhaled into lower airways. The systemic absorption is unchanged by the use of a spacer.

Spacer devices are useful in delivering inhaled drugs to children who are not able to use a pMDI alone but can also be used by adults for better drug delivery. The image below shows the use of a spacer device with a face mask which is preferred in children younger than 3 years of age.



The various devices used for the delivery of inhalational drugs are:

- Pressurized metered-dose inhalers - Drugs are propelled from a canister in the pMDI with the aid of a propellant. These devices are convenient, portable, and typically deliver 50–200 doses of a

drug



- Dry powder inhalers - these devices scatter the drug into a fine powder that is dispersed by air turbulence on inhalation. DPIs have been developed to deliver drugs such as insulin, but the dosing is inconsistent. Children <7 years find it difficult to use a DPI.



- Nebulizers: Two types of nebulizer are available -
- Jet nebulizers that are driven by a stream of gas (air or oxygen)
- Ultrasonic nebulizers that use a rapidly vibrating piezoelectric crystal and thus do not require a source of compressed gas.

The nebulizers deliver much higher drug doses as compared pMDIs. These are useful in:

- Treating acute exacerbations of asthma and COPD
- Extreme airway obstruction (severe COPD)

- Delivering inhaled drugs to infants and small children who cannot use the other inhalation devices
- Administering drugs such as antibiotics in relatively high doses



Solution to Question 7:

Theophylline has an antagonistic action on adenosine receptors.

Theophylline is a methylxanthine. It acts as a non-selective PDE inhibitor. Its other actions include:

- Increased IL-10 release
- Decrease in gene transcription of pro-inflammatory cytokines
- Promotes apoptosis of granulocytes
- Histone deacetylase activation. The recruitment of histone deacetylase-2 by glucocorticoid receptors switches off inflammatory genes. Through this mechanism, theophylline enhances the anti-inflammatory effects of corticosteroids.

Theophylline can be used as a bronchodilator in an asthmatic patient on ICS as an add-on drug. LABAs are more effective but theophylline is a cheaper option.

Solution to Question 8:

A decrease in dose of theophylline is required in cardiopulmonary disease. This is because reduced clearance of theophylline is seen in liver disease, pneumonia, and heart failure.

In smokers, clearance of theophylline is increased due to the induction of CYP1A2 enzyme.

Theophylline is a non-selective phosphodiesterase inhibitor that decreases the breakdown of cAMP levels in smooth muscles resulting in bronchodilation.

Solution to Question 9:

Theophylline is indicated in the treatment of apnea of prematurity.

In apnea of prematurity, methylxanthines such as theophylline and caffeine increase the central respiratory drive by lowering the threshold of response to hypercapnia. They also enhance the contractility of the diaphragm and prevent diaphragmatic fatigue.

Options A & B - The adenosine receptor antagonism property of theophylline precipitates cardiac arrhythmias and epileptic seizures.

Option C - Theophylline is no longer recommended as a treatment for acute severe asthma by the latest guidelines (GINA 2021) because of its poor efficacy and safety profile.

Solution to Question 10:

Zafirlukast and montelukast are leukotriene receptor antagonists. They are competitive antagonists of the cysLT₁ receptors. These drugs act by:

- Decreasing bronchoconstriction
- Reducing airway mucus secretion
- Decreasing vascular permeability and
- Reducing eosinophil chemotaxis

Option A: Dupilumab is an interleukin-4 receptor antagonist used in severe asthma.

Option B: Roflumilast is a PDE-4 selective inhibitor used in the treatment of COPD.

Option D: Zileuton is a lipoxygenase inhibitor that can be used in the treatment of asthma.

Solution to Question 11:

Zileuton, montelukast, and zafirlukast are all associated with rare instances of hepatic dysfunction. Liver function tests should be monitored during the course of their therapy.

Therapy with selective β -2 agonists (salbutamol, salmeterol) does not require serial monitoring of LFTs.

Solution to Question 12:

The given scenario points towards a diagnosis of Churg-Strauss syndrome. It is an eosinophilic vasculitis associated with the use of montelukast and zafirlukast. This is a very rarely reported side effect of these drugs.

These agents block the effects of the cysLTs (LTC₄, LTD₄, and LTE₄) but they do not block the effect of LTB₄, which is a chemoattractant for eosinophils and neutrophils. This can lead to eosinophilia.

Solution to Question 13:

The given clinical scenario is suggestive of acute severe asthma and cromolyn sodium is not useful in its treatment. It is indicated only for chronic prophylaxis.

Short-acting beta 2 agonists (SABA) and intravenous hydrocortisone are used in the treatment of acute severe asthma.

Ipratropium bromide can also be given in severe asthma exacerbations. It should be given after beta 2-agonists as they have a slower onset of bronchodilatory action.

Solution to Question 14:

Omalizumab is a monoclonal antibody against IgE.

It is used only in patients with very severe asthma (not acute severe asthma) who show poor symptom control even with step 4/5 measures as per GINA 2021 guidelines.

It can only be given to patients aged ≥ 6 years. It neutralizes circulating IgE and inhibits mast cell degranulation caused by IgE. It is administered subcutaneously.

Another drug used in severe asthma is mepolizumab. It is a monoclonal antibody acting against IL-5.

Solution to Question 15:

Mepolizumab is a monoclonal antibody against IL-5. It is indicated in:

- Selected (eosinophilic phenotype) patients with severe asthma
- Churg-Strauss syndrome
- Hypereosinophilic syndrome

IL-5 plays a pivotal role in eosinophilic inflammation and is also involved in eosinophil survival and priming.

Monoclonal antibodies against IL-5 approved for asthma are:

- Mepolizumab
- Reslizumab

- Benralizumab

Solution to Question 16:

Seratrodist is a thromboxane receptor antagonist indicated for use in long term prophylaxis of asthma. It is available and approved in India.

Option B- Rimonabant is a cannabinoid receptor antagonist that was investigated for weight gain but now withdrawn due to adverse psychological effects

Option C- Sacubitril is a neprilysin inhibitor approved in fixed-dose combination with valsartan for heart failure patients.

Option D- Roflumilast is a phosphodiesterase 4 inhibitor which reduces eosinophil infiltration and inhibits inflammatory cell ,effective against smoke induced inflammation and emphysema.

Solution to Question 17:

Corticosteroids improved the condition by increasing the airway responsiveness to salbutamol.

Mechanism of action of corticosteroids in an acute asthma attack:

- Increases responsiveness to β -2 agonists
- Decreases mucous secretion
- Decreases inflammatory cell response.

Steroids increase the transcription of β -2 receptor genes and hence increases responsiveness to β -2 agonists.

β -2 agonists increase the effect of steroids by enhancing the action of glucocorticoid receptors.

Theophylline increases the effects of steroids by activating the enzyme histone deacetylase which is the target for steroids as well.

Note: As per recent GINA 2021 guidelines, aminophyllines are no longer preferred in the management of acute severe asthma due to their low efficacy and poor safety profile.

Solution to Question 18:

Prednisolone is a systemic steroid used for maintenance therapy after an acute exacerbation of asthma.

Intravenous corticosteroids such as hydrocortisone are indicated if the lung function is predicted to be $<30\%$ or if there's poor response to nebulized β 2 agonists. After the exacerbation, oral prednisolone can be substituted with maximal beneficial effect at 30-40mg daily.

Inhaled corticosteroids commonly used in asthma are:

- Beclomethasone dipropionate

- Budesonide
- Ciclesonide
- Fluticasone
- Mometasone

Solution to Question 19:

The given clinical scenario is suggestive of an acute exacerbation of asthma. Caffeine does not precipitate this.

Caffeine is a dietary xanthine similar in structure to methylxanthines (theophylline) which are used as adjuvants in the treatment of asthma.

NSAIDs (like phenylbutazone, naproxen, and aspirin) inhibit the cyclooxygenase pathway thereby diverting arachidonic acid towards the lipoxygenase pathway. This increases the level of pro-inflammatory leukotrienes and can precipitate an acute asthma attack.

Solution to Question 20:

Inhaled corticosteroids (ICS) like budesonide do not increase the risk of developing carcinoma of the lung.

Common problems with the use of ICS:

- Hoarseness of voice
- Weakness of voice (dysphonia)
- Increased risk of bacterial and mycobacterial lung infection
- Oropharyngeal candidiasis

Solution to Question 21:

Ipratropium is a short-acting muscarinic antagonist which is not used for the long-term management of COPD. It can be used to treat acute exacerbations.

The long-acting muscarinic antagonists (LAMAs) are:

- Tiotropium
- Glycopyrronium
- Umeclidinium
- Aclidinium

Solution to Question 22:

Inhalational anticholinergics are the mainstay for the treatment of COPD.

Ipratropium is as effective as albuterol in patients with COPD. Longer-acting antimuscarinic agents, including tiotropium, aclidinium, and umeclidinium, are approved for maintenance therapy of COPD.

Anticholinergics are less effective than β -2 agonists as bronchodilators in asthma and are not the mainstay in the treatment of asthma.

Solution to Question 23:

Roflumilast is a PDE-4 selective inhibitor approved for use in COPD.

PDE-4 is the major isoform present in inflammatory cells and its inhibition results in higher anti-inflammatory effects.

Option A: Seratrodist is a thromboxane receptor antagonist used in the prophylaxis of asthma. It decreases the bronchial hypersensitivity caused by thromboxane.

Option C: Zileuton is a lipoxygenase inhibitor that can be used in the treatment of asthma.

Option D: Nedocromil is a mast cell stabilizer indicated in the prophylaxis of asthma.

Solution to Question 24:

Dextromethorphan, morphine, and gabapentin are all antitussives.

Dextromethorphan is an NMDA receptor antagonist that suppresses the cough center.

Opiates like morphine have a central mechanism of action on the medullary cough center and are indicated for intractable cough associated with bronchial carcinoma.

Gabapentin and pregabalin are GABA analogs that have been shown to benefit chronic idiopathic cough involving neural hypersensitivity.

Guaifenesin is an expectorant that is taken orally.

Carbocysteine is a mucolytic that reduces the disulfide bridges that bind glycoproteins to other proteins, such as albumin and secretory IgA. It is also taken orally and helps reduce the viscosity of sputum.

Solution to Question 25:

The clinical scenario points to a diagnosis of pulmonary artery hypertension (pulmonary artery pressure > 20 mmHg) which can be managed using selective PDE-5 inhibitors.

Inhibition of PDE-5 in the pulmonary bed leads to a reduction in the hydrolysis of cGMP which causes pulmonary vasodilation.

Sildenafil and tadalafil are the PDE-5 inhibitors approved for use in PAH.

Solution to Question 26:

Riociguat is a soluble guanylate cyclase stimulator (sGC), indicated in the treatment of pulmonary artery hypertension.

Riociguat increases the sensitivity of nitric oxide to sGC and also directly stimulates the sGC enzyme resulting in increased formation of cGMP. This leads to relaxation in the pulmonary vasculature.

Solution to Question 27:

Iloprost is a prostacyclin analog indicated in the management of pulmonary artery hypertension. It falls under the category of IP receptor (prostacyclin receptor) agonists.

Other drugs falling under the category of IP receptor agonists are:

-

Epoprostenol

-

Treprostinil

-

Iloprost

-

Selexipag

Option A: Sildenafil is a PDE-5 inhibitor.

Option B: Misoprostol is a synthetic PGE₁ analog.

Option C: Dinoprost is a natural PGF₂-alpha preparation. It should not be confused with dinoprostone which is a synthetic analogue of PGE₂.

Solution to Question 28:

Ambrisentan is a selective endothelin receptor A (ET-A) antagonist which is used in the management of pulmonary artery hypertension (PAH).

Bosentan, tezosentan, and macitantan are non-selective (ET-A and ET-B) endothelin receptor antagonists.

Solution to Question 29:

Dornase alfa is indicated in patients with cystic fibrosis and as a rescue treatment for refractory atelectasis in newborn and pediatric populations.

Dornase alfa is a recombinant DNase 1 that digests neutrophil DNA deposited in the pulmonary mucus thereby reducing mucosal viscoelasticity.

Solution to Question 30:

Erythromycin should not be prescribed with theophylline.

Theophylline is mainly metabolized by CYP1A2 in the liver. Erythromycin inhibits CYP1A2 and thereby decreases the clearance of theophylline.

Factors decreasing the clearance of theophylline are:

- Drugs - erythromycin, cimetidine, ciprofloxacin, allopurinol, fluvoxamine, zileuton, zafirlukast
- Congestive heart failure
- Liver disease
- Pneumonia
- Viral infection and vaccination
- High-carbohydrate diet
- Old age

Solution to Question 31:

It is an irrational drug combination and should not be given.

Codeine is an opioid analgesic having specific antitussive action while dextromethorphan is a centrally active NMDA receptor antagonist that may be effective as an antitussive. Both are centrally acting antitussives, and do not provide any added advantage when combined. Hence it is an irrational drug combination.

Rational fixed-dose combination drugs for the treatment of cough usually contain:

- A centrally acting anti-tussive (either codeine or dextromethorphan; not both), with
- Antihistamines, decongestants and expectorants.

Option A - Both of these drugs carry a risk of addiction, but it is not the reason to not give the drug. The main reason is that the combination is irrational.

Options B & D - Both codeine and dextromethorphan have different mechanisms of action and can treat cough effectively. However, the combination of two centrally acting antitussives does not provide any added advantage compared to using one drug alone. Hence, the combination is irrational.

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Acid Peptic Disorders and Inflammatory Bowel Disease

Question 1:

The final step in gastric acid production is inhibited by:

- a) H₂ blockers
- b) Proton pump inhibitors
- c) Octreotide
- d) Antimuscarinics

Question 2:

Despite their short half-lives, how do proton-pump inhibitors cause prolonged suppression of acid secretion for up to 48 hours?

- a) Prodrugs and undergo activation gradually
- b) Exit from plasma and enter acid secretory canaliculi, blocking the secretion of acid for a long time
- c) Irreversibly inhibit proton pump molecule, acid secretion requires synthesis of new proton pumps
- d) Available as enteric coated capsules from which drug is gradually released

Question 3:

A patient with gastroesophageal reflux disease was prescribed an oral PPI preparation that causes the fastest acid suppression. Identify the drug.

- a) Dexlansoprazole
- b) Esomeprazole
- c) Pantoprazole
- d) Rabeprazole

Question 4:

A neighbor approaches you for advice on the best time to take pantoprazole for heartburn. What would you recommend?

- a) After meals
- b) Shortly before meals
- c) Along with milk
- d) During meals

Question 5:

A patient undergoing evaluation for multiple gastroduodenal ulcers was diagnosed with Zollinger-Ellison syndrome. What would be the drug of choice for this patient?

- a) Proton pump inhibitors
- b) H₂ blockers
- c) Sucralfate
- d) Prostaglandin analogs

Question 6:

A 60-year-old woman with osteoarthritis developed gastric ulcers following long-term use of NSAIDs. Of the following, which drug is FDA-approved for her condition?

- a) PGE-1 analogue
- b) PGD-2 antagonist
- c) PGD-2 agonist
- d) PGF-2 analogue

Question 7:

What is the most effective therapy for NSAID-induced ulcers?

- a) Antacids
- b) H₂ blockers
- c) Misoprostol
- d) Proton pump inhibitors

Question 8:

Which proton pump inhibitor should be avoided in a patient on warfarin anticoagulation?

- a) Rabeprazole
- b) Pantoprazole
- c) Lansoprazole
- d) Omeprazole

Question 9:

You are counseling a patient with gastroesophageal reflux disease about the need for long-term therapy with proton-pump inhibitors. All of the following are complications of this therapy, except:

- a) Decreased vitamin B12 absorption
- b) Increased susceptibility to infections
- c) Hypogastrinemia
- d) Increased risk of bone fracture

Question 10:

Gynecomastia can occur as a side effect of:

- a) Aluminium hydroxide
- b) Cimetidine
- c) Pirenzepine
- d) Lansoprazole

Question 11:

A patient with peptic ulcer disease was prescribed ranitidine and sucralfate to be taken together. Why is this combination incorrect?

- a) Ranitidine combines with sucralfate and forms a toxic metabolite
- b) Sucralfate competes with ranitidine for H₂ receptor
- c) Ranitidine increases the gastric pH which reduces sucralfate efficacy
- d) Sucralfate inhibits the absorption of ranitidine

Question 12:

Gastric bezoars are commonly associated with _____.

- a) Proton pump inhibitors
- b) H₂ blockers
- c) Misoprostol
- d) Sucralfate

Question 13:

Which of the following is not a characteristic of milk-alkali syndrome?

- a) Alkalosis
- b) Renal insufficiency
- c) Hypercalcemia
- d) Hepatotoxicity

Question 14:

Which of the following can cause diarrhea?

- a) Sodium bicarbonate
- b) Magnesium hydroxide
- c) Calcium bicarbonate
- d) Aluminium hydroxide

Question 15:

Which of the following antacids is more likely to induce rebound acid secretion?

- a) Calcium carbonate
- b) Magnesium hydroxide
- c) Aluminium hydroxide
- d) Sodium bicarbonate

Question 16:

A patient presents with complaints of black stools and black coated tongue as shown in the image. Which of the following drugs is implicated?



- a) Sucralfate
- b) Rasburicase
- c) Colloidal bismuth subsalicylate
- d) Simethicone

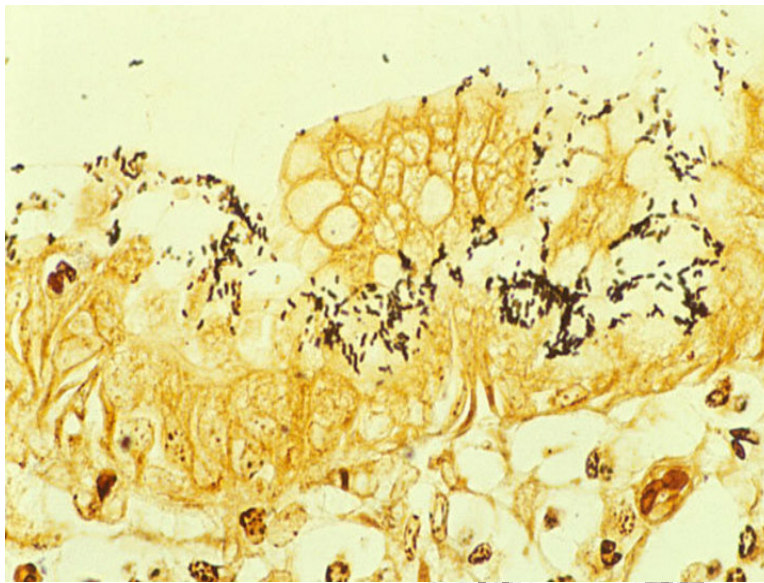
Question 17:

Which of the following is an antifoaming agent?

- a) Sucralfate
- b) Simethicone
- c) Colloidal bismuth subcitrate
- d) Rebamipide

Question 18:

A patient with a history of peptic ulcer underwent endoscopic evaluation. The biopsy image is shown below. Which of the following drugs is not used in treating this condition?



- a) Metronidazole
- b) Proton pump inhibitors
- c) Sucralfate
- d) Bismuth compounds

Question 19:

A 30-year-old man presented with bloody diarrhea and abdominal pain for the past week. Pseudopolyps were noted on colonoscopy. Which of the following can be used in managing his condition?

- a) Sulfasalazine
- b) Sulfamethoxazole
- c) Sulfinpyrazone
- d) Sulphadoxine

Question 20:

Which of the following is not a prodrug?

- a) Sulfasalazine
- b) Mesalamine
- c) Balsalazide
- d) Olsalazine

Question 21:

A patient with Crohn's disease, refractory to steroid therapy is planned to be started on an anti-TNF alpha monoclonal antibody. Which of the following drugs does not belong in this class?

- a) Infliximab
- b) Adalimumab
- c) Certolizumab pegol
- d) Vedolizumab

Question 22:

Which is not given in Wilson's disease?

- a) Zinc
- b) Penicillamine
- c) Calcium citrate
- d) Trientine

Answer Key

Question No.	Correct Option
1	b
2	c
3	d
4	b
5	a
6	a
7	d
8	d
9	c
10	b
11	c

12	d
13	d
14	b
15	a
16	c
17	b
18	c
19	a
20	b
21	d
22	c

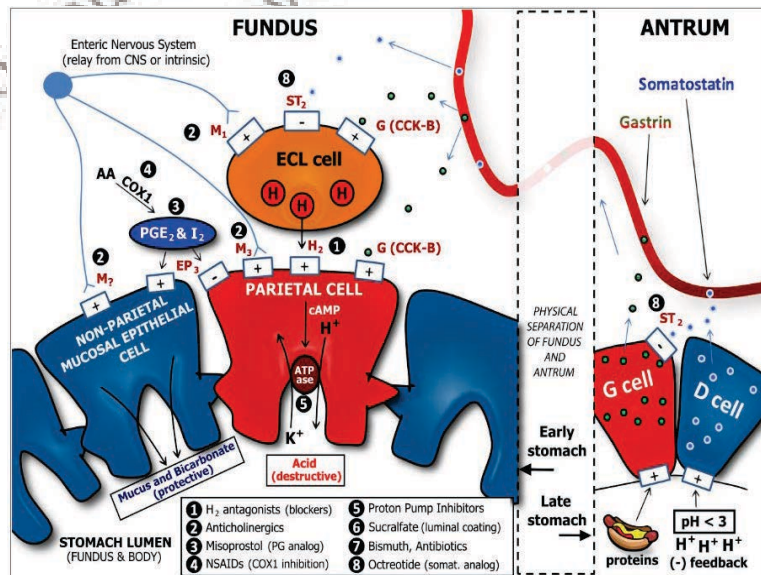
Detailed Explanations

Solution to Question 1:

The final step in gastric acid production is inhibited by the proton pump inhibitors (PPIs).

The secretion of protons by the activated H^+/K^+ -ATPase (proton pump) in the parietal cells into the gastric lumen is the final step involved in the production of gastric acid.

PPIs cause maximum suppression of gastric acid and also reduces basal acid secretion. Suppression of gastric acid occurs regardless of the stimulus present.



Solution to Question 2:

PPIs irreversibly inhibit the proton pump molecule and hence, acid secretion requires the synthesis of new proton pumps.

PPIs are prodrugs, when ingested, get absorbed into the systemic circulation. It then diffuses into the parietal cells of the stomach and accumulates in the acidic secretory canaliculi. Acidic pH in the parietal cell canaliculi causes activation of the PPIs. These activated PPIs irreversibly inhibit H^+/K^+ ATPase (proton pump) in the parietal cell.

The maximum proton pump activation corresponds to the peak concentration of PPI resulting in maximum efficacy.

The secretion resumes gradually over 3-4 days of stopping the drug when new proton pumps are synthesized.

Solution to Question 3:

Rabeprazole is the oral PPI preparation that causes the fastest acid suppression. It has a higher pKa that causes more rapid conversion to active species in acidic canaliculi.

The efficacy and potency are similar to that of omeprazole.

Solution to Question 4:

Proton-pump inhibitors (pantoprazole) should be ideally given shortly (20-30 minutes) before food.

PPIs are usually given before breakfast. Food hinders the absorption of PPIs and decreases their bioavailability.

PPIs are prodrugs, when ingested, get absorbed into the systemic circulation. It then diffuses into the parietal cells of the stomach and accumulates in the acidic secretory canaliculi. Acidic pH in the parietal cell canaliculi causes activation of the PPIs. The maximum proton pump activation corresponds to the peak concentration of PPI resulting in maximum efficacy.

Solution to Question 5:

Proton-pump inhibitors are the drug of choice in Zollinger–Ellison syndrome.

Zollinger Ellison syndrome includes pancreatic or duodenal gastrinoma which leads to the secretion of large amounts of gastrin. This, in turn, increases gastric acid leading to severe gastroduodenal ulceration and other effects of hyperchlorhydria.

Note: Netazepide is a gastrin/cholecystokinin-2 receptor antagonist used in the treatment of gastrinoma. It is also an antisecretory agent.

Solution to Question 6:

PGE1 analog (misoprostol) is approved by the FDA for NSAID-induced gastric ulcers. It reduces gastric acid secretion in a dose-dependent manner.

Prostaglandins inhibit gastric acid secretion. Misoprostol also increases mucosal blood flow and stimulates bicarbonate and mucin secretion which contribute to its cytoprotective action. However, the reduction in acid secretion is less than that of H₂ blockers.

Solution to Question 7:

The most effective therapy for NSAID-induced ulcers is proton pump inhibitors.

They are proven to be more effective than misoprostol and H₂ receptor antagonists.

Solution to Question 8:

Omeprazole should not be used in a patient on warfarin therapy as it decreases the metabolism of warfarin.

All PPIs are metabolized by hepatic P450 cytochromes, including CYP2C19 and CYP3A4, and may interfere with the metabolism of the other drugs. It inhibits the metabolism of clopidogrel, diazepam, and phenytoin.

Note: There is disagreement among different authors regarding the PPIs and their interactions with the metabolism of other drugs.

- Goodman and Gilman mention that lansoprazole and rabeprazole along with omeprazole and esomeprazole interact with the metabolism of warfarin.
- Katzung and various online sources agree that lansoprazole and rabeprazole do not or minimally interact with warfarin metabolism.
- All of them agree that pantoprazole does not or minimally interacts with warfarin metabolism.

Solution to Question 9:

Long-term PPI therapy is not associated with hypogastrinemia but is associated with hypergastrinemia. This hypergastrinemia contributes to rebound gastric acid hypersecretion if PPI therapy is discontinued.

Chronic PPI therapy is associated with:

- Decreased vitamin B12 absorption
- Increased susceptibility to infections- hospital-acquired pneumonia, pseudomembranous enterocolitis
- Reduced calcium absorption increases the risk of fractures

Solution to Question 10:

Gynecomastia can occur as a side effect of cimetidine. It inhibits the binding of dihydrotestosterone to androgen receptors, increases the serum prolactin level and inhibits the metabolism of estradiol by the liver. Cimetidine causes maximum hepatic inhibition.

The specific side effect of high doses of cimetidine given for a long time is gynecomastia or impotence in males and galactorrhea in females. This property is not shared by other H₂ blockers.

Drugs causing gynecomastia:

- Alcohol, marijuana, heroin
- Antiretroviral therapy
- Anabolic steroids
- Digoxin
- Risperidone
- Ketoconazole
- Spironolactone
- α -methyldopa
- Bromocriptine
- Cimetidine
- Phenytoin therapy.
- Androgens- testosterone and its esters (except dihydrotestosterone).

Solution to Question 11:

Ranitidine increases the gastric pH (i.e. less acidic) which reduces sucralfate efficacy; hence the combination is incorrect.

Polymerization of sucralfate occurs only at an acidic pH by crosslinking of the molecules which forms a sticky gel-like consistency. This adheres to the ulcer base predominantly and remains there for almost 6 hours, preventing the ulcer base from coming in contact with the acid.

Concurrent use of H₂ blockers like ranitidine increases the gastric pH and thus hinders the polymerization and reduces the efficacy of sucralfate.

Sucralfate should be taken at least 2 hours after the administration of any other drug.

Solution to Question 12:

Gastric bezoars are commonly associated with sucralfate.

Sucralfate produces a viscous gel that has a 'sticky nature'. The sticky nature may be responsible for the formation of gastric bezoars, particularly in those with gastroparesis.

Solution to Question 13:

Milk alkali syndrome is not characterized by hepatotoxicity.

Characteristics of the milk-alkali syndrome include:

- Alkalosis
- Hypercalcemia
- Renal insufficiency

Milk alkali syndrome used to occur in patients who were administered high doses of NaHCO_3 and CaCO_3 along with milk or cream for the management of peptic ulcers.

The syndrome is rare now and generally occurs with a large intake of calcium supplements along with milk.

Solution to Question 14:

Magnesium salts have a laxative property and can cause diarrhoea.

Aluminium salts can cause constipation by relaxing smooth muscles.

The combination of aluminium and magnesium salts can neutralize each other's action, thus the combination is preferred.

Solution to Question 15:

Calcium carbonate is an antacid that is more likely to induce rebound acid secretion.

Calcium containing antacids like calcium carbonate rapidly and effectively neutralizes gastric acid. Calcium in antacids causes rebound acid secretion, hence it is given at frequent intervals to reduce the incidence of rebound acid secretion.

Magnesium, aluminium and sodium-containing antacids are less likely to cause this phenomenon.

Solution to Question 16:

Colloidal bismuth subsalicylate is known to cause blackening of the tongue, dentures and stools.

It combines with trace amounts of sulfur in saliva and colon to form bismuth sulfide. Bismuth sulfide is a highly insoluble black salt. The adverse effect is claimed to be temporary and harmless.

Solution to Question 17:

Simethicone is a surfactant that decreases foaming. Due to this antifoaming action, it is included in many antacid preparations to reduce gastroesophageal reflux.

Option A: Sucralfate acts as an ulcer protective agent by forming a viscous sticky polymer that binds to the ulcer base.

Option D: Rebamipide is also a cytoprotective agent that acts by increasing the production of prostaglandin in gastric mucosa and by scavenging reactive oxygen species.

Option C: Bismuth compounds bind to the base of the ulcer, promote mucin and bicarbonate production. They have significant antibacterial action and are part of many anti-H.pylori regimens.

Solution to Question 18:

The image shows a Warthin silver starry stained biopsy specimen, diagnostic for *Helicobacter pylori*. Sucralfate does not have any anti-H. pylori action and is not used in the treatment of H. pylori gastritis.

Option A: Metronidazole is used in the triple, quadruple therapy, and sequential therapy regimens.

Option B: Proton pump inhibitors significantly enhance the effectiveness of regimens containing amoxicillin or clarithromycin.

Option D: Bismuth compounds have significant anti-bacterial action and are given in quadruple therapy.

Two or three antimicrobials are combined in regimens that range from 10-14 days. Monotherapy is not recommended due to the development of resistance.

Solution to Question 19:

Bloody diarrhea and colonoscopy findings of pseudopolyps are suggestive of ulcerative colitis. The sulfa drug indicated is sulfasalazine.

It is indicated mainly in mild-moderate ulcerative colitis and as an adjunctive in severe cases. Efficacy in Crohn's disease is less.

Sulfasalazine consists of 5-ASA (mesalamine or 5-amino salicylic acid) linked to sulfapyridine with an azo bond. Individually, 5-ASA and sulfapyridine would be absorbed in the upper gastrointestinal tract, the azo bond prevents such absorption. Colonic bacteria break the azo bond to liberate sulfapyridine and 5-ASA. 5-ASA is considered to be the active moiety.

Solution to Question 20:

Mesalamine is not a prodrug. Mesalamine or 5-ASA is the active component that is responsible for the beneficial effects of inflammatory bowel disease.

Cleavage of the bond of prodrugs that deliver mesalamine to the colon occurs by colonic bacteria.

Option A: Sulfasalazine - 5-ASA + sulfapyridine

Option C: Balsalazide - 5-ASA + an inert compound

Option D: Olsalazine - 5-ASA + 5-ASA

Solution to Question 21:

Vedolizumab is a humanized anti-alpha-4-beta-7 integrin monoclonal antibody that prevents lymphocyte recruitment to the intestinal mucosa.

It is approved for use in moderate-severe Crohn's disease and ulcerative colitis.

Anti-TNF alpha monoclonal antibodies indicated in IBD are:

- Infliximab
- Adalimumab
- Certolizumab pegol

Solution to Question 22:

Calcium citrate is not used in the treatment of Wilson's disease.

The drugs used in the management of Wilson's disease are:

Chelating agents: These drugs bind to copper to form complexes that get excreted in the urine. D-penicillamine and trientine are first-line drugs in symptomatic patients with hepatic diseases.

- D-penicillamine: It is the recommended drug due to its availability and cost-effectiveness as compared to trientine. Since it is an antagonist to pyridoxine, supplementation of pyridoxine is required. It can precipitate various adverse effects including aplastic anemia, myasthenia gravis, and membranous nephropathy. It can also worsen neurological symptoms

- Trientine: It is administered in those who cannot tolerate D-penicillamine

- Tetrathiomolybdate: It has been identified to be effective in patients with neuropsychiatric manifestation

Elemental zinc:

- Zinc blocks the intestinal absorption of copper by inducing metallothionein (endogenous chelator in enterocytes)

- It is commonly used as a maintenance therapy along with chelating agents. It can also be used as first-line therapy in pregnancy

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Anti-emetics and Drugs Affecting Gastrointestinal Motility

Question 1:

A 23-year-old man came to the OPD with complaints of vomiting for which he was prescribed metoclopramide. What is the primary mechanism of action of this drug?

- a) D2 antagonism
- b) Sensitization of muscarinic receptors on smooth muscles
- c) 5HT₃ antagonism
- d) 5HT₃ agonism

Question 2:

Which of the following actions is exhibited by metoclopramide?

- a) Inhibits cholinergic smooth muscle stimulation in the gastrointestinal tract
- b) Decreases the lower esophageal sphincter pressure
- c) Stimulates D2 receptor
- d) Enhances gastrointestinal motility

Question 3:

A patient presented to the casualty with signs of dehydration due to severe vomiting. The intern gave the patient a parenteral medication, following which the patient developed grimacing and abnormal posturing. What could have been the inciting drug for this reaction?

- a) Atropine
- b) Ranitidine
- c) Metoclopramide
- d) Domperidone

Question 4:

A patient developed abnormal posturing and grimacing following the administration of metoclopramide. Which drug will you administer to manage this patient?

- a) Phenelzine
- b) Promethazine
- c) Chlorpromazine
- d) Prochlorperazine

Question 5:

Which of the following drugs is implicated in prolonging QT interval and causing serious arrhythmias?

- a) Domperidone
- b) Metoclopramide
- c) Cisapride
- d) Ondansetron

Question 6:

Which of the following drugs do not act as a prokinetic agent by 5HT₄ agonism?

- a) Renzapride
- b) Metoclopramide
- c) Domperidone
- d) Cisapride

Question 7:

Which of the following is not a prokinetic?

- a) 5HT₄ agonist
- b) Dopamine antagonist
- c) Motilin
- d) Diphenylmethane

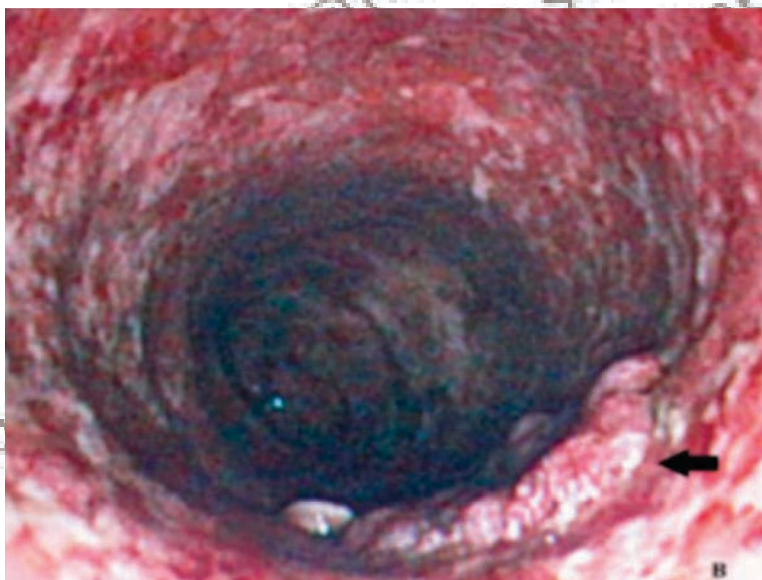
Question 8:

A 45-year-old male with unhealthy eating practices now complains of constipation. He took bisacodyl, following which he had symptomatic relief. Which group of drugs does this belong to?

- a) Bulk forming laxatives
- b) Stool softeners
- c) Stimulant laxatives
- d) Chloride secretory agents

Question 9:

A colonoscopy of an elderly patient revealed brownish pigmented lesions as shown below. He admitted to chronic laxative abuse. Which of the following laxatives is most likely to cause this condition?



- a) Bran
- b) Psyllium
- c) Senna
- d) Methylcellulose

Question 10:

You want to prescribe a bulk-forming laxative to an elderly patient with constipation. Which among the following would you not prescribe?

- a) Bran
- b) Psyllium
- c) Liquid paraffin
- d) Methyl cellulose

Question 11:

A 3-year-old child is brought to the pediatric OPD by his mother with complaints of acute watery diarrhea for 2 days. Oral rehydration solution, zinc supplement, and a peripheral enkephalinase inhibitor were prescribed. Which of the following drugs did he prescribe?

- a) Racecadotril
- b) Diphenoxylate
- c) Loperamide
- d) Linaclotide

Question 12:

A 30-year-old male who is on anti-retroviral therapy presents with complaints of diarrhea for three days. An infectious etiology is ruled out on work-up. Which of the following drugs is preferred to manage his diarrhea?

- a) Racecadotril
- b) Crofelemer
- c) Ondansetron
- d) Cisapride

Question 13:

A 38-year-old woman complains of recurrent episodes of constipation and crampy abdominal pain for the last 4 months. The pain is relieved on defecation. After investigations, a diagnosis of irritable bowel syndrome with constipation is considered. Which of the following drugs would you prescribe?

- a) Lubiprostone
- b) Loperamide
- c) Alosetron
- d) Eluxadoline

Question 14:

The use of which of the following drugs is associated with ischemic colitis?

- a) Ondansetron
- b) Alosetron
- c) Palonosetron
- d) Granisetron

Question 15:

A patient who is suffering from chronic constipation is prescribed a guanylcyclase C agonist. Identify the drug.

- a) Lubiprostone
- b) Loperamide
- c) Linaclotide
- d) Tegaserod

Question 16:

Prucalopride is a _____.

- a) 5HT_{2a} agonist
- b) 5HT_{2b} agonist
- c) 5HT₄ antagonist
- d) 5HT₄ agonist

Question 17:

A 40-year-old woman suffering from erratic bowel habits for the past few months is diagnosed with diarrhea-predominant irritable bowel syndrome. Which antibiotic is specifically approved for the treatment of her condition?

- a) Rifabutin
- b) Rifampicin
- c) Rifaximin

d) Rifapentine

Question 18:

A 20-year-old primiparous woman came to the OPD with complaints of nausea and vomiting at 8 weeks of gestation. Which of the following is specifically indicated to provide symptomatic relief in this patient?

- a) Domperidone with doxylamine**
- b) Doxylamine with pyridoxine**
- c) Dolasetron with pyridoxine**
- d) Dolasetron with metoclopramide**

Question 19:

What is the drug of choice in a patient complaining of nausea and vomiting, 2 days after commencement of a cisplatin-based chemotherapeutic regimen?

- a) 5HT₃ antagonist + NK-1 antagonist**
- b) NK-1 antagonist + 5HT₃ antagonist + Corticosteroid**
- c) NK-1 antagonist + Corticosteroid**
- d) 5HT₃ antagonist + Corticosteroid**

Question 20:

An AIDS patient presents with anorexia and weight loss. Which of the following is a naturally extracted cannabinoid that is approved for his condition?

- a) Dronabinol**
- b) Nabilone**
- c) Tibolone**
- d) Cannabidiol**

Question 21:

Which of the following statements about aprepitant is false?

- a) Agonist at NK₁ receptor**

- b) Crosses blood brain barrier
- c) Ameliorate nausea and vomiting induced by chemotherapy
- d) Metabolised by the CYP450 enzymes

Question 22:

Which of the following antiemetic drugs also decreases acid secretion due to its action on H₁ receptors?

- a) Promethazine
- b) Domperidone
- c) Palonosetron
- d) Ondansetron

Question 23:

The 5HT₃ antagonist having maximum potency in chemotherapy-induced vomiting is:

- a) Ondansetron
- b) Granisetron
- c) Cilansetron
- d) Palonosetron

Question 24:

Most effective antiemetic for chemotherapy-induced vomiting is_____.

- a) Domperidone
- b) Tegaserod
- c) Granisetron
- d) Doxylamine

Question 25:

QT interval prolongation is minimal with which of the following 5-HT₃ antagonists?

- a) Ondansetron

- b) Palonosetron
- c) Dolasetron
- d) Granisetron

Question 26:

Which of the following NK1 receptor antagonists has the longest half-life?

- a) Netupitant
- b) Rolapitant
- c) Aprepitant
- d) Fosaprepitant

Question 27:

A young male developed 5 episodes of loose stools 2 hours after eating in a restaurant. He is afebrile and has mild dehydration. What should be the treatment?

- a) Ciprofloxacin and tinidazole
- b) Only ciprofloxacin
- c) Only ORS
- d) Ciprofloxacin, tinidazole and ORS

Question 28:

You and your friend plan for a mountain trip for the day after. What medication would you advise him to help with his motion sickness?

- a) Ranitidine tablet before start of trip
- b) Dimenhydrinate before start of trip
- c) Scopolamine patch the night before trip
- d) Omeprazole the night before and before start of trip

Answer Key

Question No.	Correct Option
1	a
2	d
3	c
4	b
5	c
6	c
7	d
8	c
9	c
10	c
11	a
12	b
13	a
14	b
15	c
16	d
17	c
18	b
19	b
20	a
21	a
22	a
23	d
24	c
25	b
26	b
27	c
28	c

Detailed Explanations

Solution to Question 1:

The mechanism of action of metoclopramide is primarily D2 antagonism in the chemoreceptor trigger zone.

Metoclopramide has the following actions:

- 5HT₄ receptor agonism - prokinetic action
- 5HT₃ receptor antagonism - anti-emetic action at higher doses
- Peripheral D₂ antagonism - prokinetic action
- Sensitization of muscarinic receptors on smooth muscles

Solution to Question 2:

Metoclopramide enhances gastrointestinal motility.

Acetylcholine increases gastrointestinal motility and lower esophageal sphincter (LES) tone. Dopamine which is an inhibitory neurotransmitter suppresses ACh release. This decreases gastrointestinal motility and LES tone. Therefore, D₂ receptor inhibition augments ACh release and enhances GI motility and LES tone.

The action of metoclopramide is generally confined to the upper digestive tract and an increase in colonic motility is clinically insignificant.

Solution to Question 3:

The given clinical scenario describes the adverse effects characteristic of metoclopramide. The most common adverse effects of metoclopramide involve the central nervous system because it crosses the blood-brain barrier (BBB) and blocks the central D₂ receptors.

The adverse effects include:

- Restlessness
- Drowsiness
- Insomnia
- Anxiety
- Agitation
- Extrapyrimal effects - acute reaction with high doses and long-term therapy. (due to central dopamine receptor blockade)
- Dystonias
- Akathisia
- Parkinsonian features
- Tardive dyskinesia

Adverse events are more common among the elderly. Long-term use should be avoided unless absolutely necessary.

Solution to Question 4:

The given scenario is suggestive of dystonia associated with D2 blocker (metoclopramide) use. Promethazine is the drug of choice.

It is a first-generation antihistaminic having significant central anticholinergic activity.

Solution to Question 5:

Cisapride is associated with prolongation of QT interval and causation of serious ventricular arrhythmias.

Prolongation of QT interval is due to blockade of rapid delayed rectifier potassium channels. It is banned or has restricted use in most of the countries.

Ondansetron is associated with asymptomatic electrocardiogram changes, including prolongation of PT and QTc intervals.

Solution to Question 6:

Domperidone does not have any 5HT₄ agonist action. It mainly acts by D₂ antagonist action.

5HT₄ agonism and D₂ antagonism can result in prokinetic action by increasing the cholinergic activity of the gut.

The 5HT₄ agonist action is shown by:

- Metoclopramide
- Cisapride
- Mosapride
- Itopride
- Renzapride

Solution to Question 7:

Diphenylmethane is not a prokinetic agent. Its derivative (Bisacodyl) is used as a laxative.

Prokinetic agents are the drugs that increase GI motility and transit of material in the GI tract. These agents include:

- Dopamine receptor antagonists- metoclopramide and domperidone
- Serotonin receptor agonists- cisapride and prucalopride
- Motilin
- Macrolide antibiotics mimicking motilin - erythromycin

- Miscellaneous agents:
- Sincalide (C-terminal octapeptide of CCK)
- Camicinal (motilin receptor agonist)
- Relamorelin (ghrelin receptor agonist)
- Velusetrag and naronapride (5HT₄ agonists)

Solution to Question 8:

Bisacodyl is a stimulant laxative.

Stimulant laxatives act by irritation of the intestinal mucosa and direct action on myenteric plexus neurons stimulating motility and secretion.

Examples of stimulant/irritant laxatives are:

- Bisacodyl
- Sodium picosulfate
- Anthraquinones like senna, cascara sagrada
- Castor oil

Solution to Question 9:

The above colonoscopy image depicts a brownish moiré pattern that is indicative of melanosis coli. Chronic use of anthraquinone laxatives like senna and cascara sagrada can cause melanosis coli.

Bisacodyl is also implicated in causing melanosis coli.

Solution to Question 10:

Liquid paraffin is not a bulk-forming laxative. It is a mineral oil and acts as a stool softener.

It is rarely used in clinical practice today because of its drawbacks, such as:

- Bland and unpleasant taste
- Risk of lipoid pneumonia
- Anal leakage

Bulk-forming laxatives are:

- Bran
- Psyllium
- Methylcellulose

- Polycarbophil

Solution to Question 11:

Racecadotril is a peripheral enkephalinase inhibitor.

Enkephalins are endogenous delta-opioid receptor agonists that decrease intestinal secretions without affecting motility. Enkephalinase is the enzyme that degrades enkephalin.

Racecadotril, by inhibiting the enkephalinase enzyme increases enkephalin level and acts as an antisecretory agent in diarrhea. It has minimal/no effect on motility. Racecadotril is indicated for symptomatic treatment of diarrhea in infants and children.

Options B and C: Diphenoxylate and loperamide are μ -opioid receptor agonists.

Option D: Linaclotide is a guanylate cyclase agonist.

Solution to Question 12:

This is a case of non-infectious diarrhea associated with antiretroviral therapy (ART). Crofelemer is indicated for non-infectious diarrhea associated with ART in HIV/AIDS.

Crofelemer has minimal systemic absorption. It acts by inhibiting the cyclic AMP-stimulated CFTR chloride channel and calcium-activated chloride channels on the luminal side of the intestine.

Solution to Question 13:

The drug used in irritable bowel syndrome with constipation is lubiprostone.

It is approved for chronic idiopathic constipation and constipation-predominant irritable bowel syndrome (in female patients ≥ 18 years). It acts by stimulating Cl-channels in the intestine and increasing fluid secretion in the gut. This decreases the transit time.

Eluxadoline is an FDA-approved opioid drug for the treatment of diarrhea-predominant IBS in adults.

Solution to Question 14:

Ischemic colitis is associated with the use of alosetron. It is a 5HT₃ antagonist that may suppress intestinal relaxation leading to ischemic colitis in susceptible individuals.

It is approved for resistant diarrhea-predominant irritable bowel syndrome in female patients.

It is either withdrawn/ has restricted use because of the serious side effect.

Solution to Question 15:

The guanylcyclase-C (GC-C) agonist indicated in chronic constipation is linaclotide.

Linaclotide is a peptide agonist that activates GC-C which leads to increased synthesis of cyclic-GMP. This results in enhanced chloride and bicarbonate secretion into the intestinal lumen leading to increased water secretion and enhanced motility.

It is indicated in constipation-predominant IBS in females and in chronic idiopathic constipation in adults.

Solution to Question 16:

Prucalopride is a specific 5HT₄ receptor agonist which increases bowel movements.

It is approved for use in women with chronic constipation where other laxatives have failed to provide relief.

Solution to Question 17:

The antibiotic that is specifically approved for diarrhea-predominant IBS is rifaximin. It is an RNA synthesis inhibitor. It may provide benefit by altering the gastrointestinal flora favorably.

Rifabutin, rifampicin and rifapentine are antitubercular agents.

Solution to Question 18:

Doxylamine is indicated with pyridoxine to reduce mild to moderate nausea and vomiting in the early stages of pregnancy.

It is a histamine H₁ antagonist and its efficacy is improved when combined with pyridoxine.

Combinations of domperidone, dolasetron and metoclopramide with pyridoxine are not specifically indicated in pregnancy.

Solution to Question 19:

This patient is most likely suffering from delayed emesis associated with cancer chemotherapy.

Combined therapy with aprepitant (NK-1/ neurokinin-1 receptor antagonist), a 5-HT₃-receptor antagonist, and dexamethasone is the best treatment in delayed emesis associated with cancer chemotherapy.

Delayed chemotherapy-induced vomiting is any emesis occurring >24 hours after chemotherapy. If NK-1 receptor antagonists are not used in the combination, delayed emesis is prevented only in 30% of the patients. Usage of 5HT₃ antagonists alone has no role in the treatment of delayed emesis.

Solution to Question 20:

Dronabinol or delta-9-tetrahydrocannabinol is a cannabinoid that can be extracted naturally.

It is the only cannabinoid that is approved for anorexia and weight loss associated with AIDS. It is obtained from the marijuana plant or can be synthesized chemically.

Option B: Nabilone is a synthetic cannabinoid.

Dronabinol and nabilone are approved for the treatment of refractory cases of chemotherapy-induced nausea and vomiting.

Option C: Tibolone is a 19-norsteroid derivative specifically to be used for hormone replacement therapy.

Option D: Cannabidiol is a cannabinoid approved for the treatment of seizures associated with Lennox-Gastaut syndrome, Dravet syndrome, or tuberous sclerosis complex.

Solution to Question 21:

Aprepitant is a high-affinity NK₁ receptor antagonist that blocks the emetic action of substance P.

Aprepitant is well absorbed orally and crosses the blood-brain barrier.

Co-administration of oral aprepitant with dexamethasone and ondansetron significantly enhances the antiemetic efficacy against the highly emetogenic side effects of chemotherapeutic drugs like cisplatin and cyclophosphamide.

It is metabolized in the liver by the CYP450 enzyme.

Solution to Question 22:

An antiemetic drug that also decreases acid secretion due to its action on H₁ histamine receptors is promethazine.

The central anticholinergic activity of promethazine is responsible for antiemetic action.

Solution to Question 23:

The 5HT₃ antagonist having maximum potency in chemotherapy-induced vomiting is palonosetron.

It has a high affinity for 5HT₃ and is also the longest-acting 5HT₃ antagonist. This makes it more effective in delayed emesis associated with chemotherapy.

During chemotherapy, there is cell lysis and 5HT (serotonin) is released from the gut mucosa. The released serotonin acts on the 5HT₃ receptors, activating the vagal afferents in the gut that carry the emetogenic impulses to the nucleus tractus solitarius and chemoreceptor trigger zone. The selective blockade of 5HT₃ receptors reduces the emetogenic impulses.

Granisetron, dolasetron, and ondansetron are also effective for chemotherapy-induced vomiting.

Cilansetron is a new 5HT₃ antagonist indicated for irritable bowel syndrome with diarrhea.

Solution to Question 24:

Among the given options, the most effective antiemetic for chemotherapy-induced vomiting is granisetron.

Granisetron is a 5HT₃ receptor antagonist. They are the most effective drugs for the treatment of chemotherapy-induced and post-operative nausea and vomiting.

Palonosetron is a second-generation 5HT₃ antagonist that has the highest affinity for 5HT₃ receptors and has the maximum potency. It is also the longest acting that makes it effective in delayed emesis associated with chemotherapy. Palonosetron has minimal/no propensity to cause QT interval prolongation.

Solution to Question 25:

Palonosetron is a second-generation 5-HT₃ antagonist and has minimal/no propensity to cause QT prolongation.

First-generation 5-HT₃ antagonists have an increased risk of QT interval prolongation.

Intravenous dolasetron is contraindicated for prophylaxis of chemotherapy-induced nausea and vomiting for the same reason.

Solution to Question 26:

Rolapitant has the longest plasma half-life of 180 hours. It is a highly selective NK₁ receptor antagonist.

Solution to Question 27:

It is a case of acute diarrhea in an afebrile, stable patient which can be managed with ORS alone.

Management of diarrhea:

- Mild-moderate dehydration - ORS.
- Severe dehydration- IV fluids are given.
- Afebrile patient - No antibiotics are given.
- Febrile patient - empirical antibiotics. If only bacterial etiology is suspected, ciprofloxacin is given. If giardiasis is suspected, metronidazole or tinidazole are recommended.

Solution to Question 28:

In the above case, a scopolamine patch the night before the trip should be advised.

Treatment for motion sickness :

- Scopolamine patch- preferred, most effective.
- Do not give to children <10 years of age.
- Place patch behind the ears at least 4 hours before travel and replace every 72 hours as needed.
- Transdermal patch – Most effective
- Oral dose - 0.4–0.6 mg one hour before travel and then TID may be recommended.
- Cinnarizine – 30mg, 2 hours prior to travel, and 15mg TID as required. Not as effective as scopolamine. Moderately effective.
- Promethazine
- Other drugs (minimally effective):
- Cyclizine
- Dimenhydrinate
- Diphenhydramine
- Meclizine

PPI's / ranitidine / traditional ginger – all are ineffective.

Cell Cycle Specific Cytotoxic Drugs

Question 1:

The Nobel Prize in Physiology or Medicine, 2018 was awarded to James P. Allison and Tasuku Honjo for which of the following discoveries?

- a) Discovery of mechanisms of autophagy
- b) Discoveries of molecular mechanisms controlling the circadian rhythm
- c) Discovery of cancer therapy by inhibition of negative immune regulation
- d) Discovery that mature cells can be reprogrammed to become pluripotent

Question 2:

Which of the following is a cell-cycle specific agent?

- a) Ifosfamide
- b) Melphalan
- c) Vinblastine
- d) Cyclophosphamide

Question 3:

Which of the following is an antimetabolite?

- a) Cyclosporine
- b) Methotrexate
- c) Etoposide
- d) Vinblastine

Question 4:

Which of the following disease-modifying anti rheumatoid drug increases extracellular adenosine?

- a) Methotrexate
- b) Lefunamide

- c) Sulphasalazine
- d) Hydroxychloroquine

Question 5:

In which of the following conditions is methotrexate not used?

- a) Giant cell arteritis
- b) Psoriasis
- c) Sickle cell anemia
- d) Organ transplantation

Question 6:

In which of the following conditions is high dose methotrexate indicated in?

- a) Osteosarcoma
- b) Rhabdomyosarcoma
- c) Retinoblastoma
- d) Ewing's sarcoma

Question 7:

A 32-year-old lady was diagnosed with gestational trophoblastic neoplasia and started on methotrexate. However the hCG levels of the patient kept rising. What could be the cause of this phenomenon?

- a) Depletion of folate
- b) Overproduction of DHFR
- c) Overproduction of thymidylate kinase
- d) Decreased DHFR

Question 8:

A patient with impaired renal function, undergoing chemotherapy, develops methotrexate toxicity. Which of the following drugs is a methotrexate cleaving enzyme that can be used to manage this condition?

- a) Adenyl cyclase
- b) Glucarpidase
- c) Rasburicase
- d) Serine protease

Question 9:

Which of the following statements about methotrexate is false?

- a) Folinic acid enhances its action
- b) It inhibits dihydrofolate reductase
- c) Non-proliferative cells are resistant to it
- d) It targets the S-phase of the cell cycle

Question 10:

A patient is on methotrexate therapy as part of his chemotherapy regimen. Which of the following tests is not indicated for monitoring in this patient?

- a) Liver function test
- b) Lung function test
- c) Fundoscopy
- d) Hemogram

Question 11:

During rounds, one of your colleagues suggested alkalization of urine with sodium bicarbonate for a patient, to reduce the toxicity of a chemotherapeutic agent. Which is the chemotherapeutic drug being discussed here?

- a) Arabinoside-cytosine
- b) Ifosfamide
- c) Cisplatin
- d) Methotrexate

Question 12:

Which of the following is an inhibitor of DNA synthesis?

- a) 6-mercaptopurine
- b) Mitomycin
- c) Actinomycin
- d) Asparaginase

Question 13:

What should be done to the dose of intravenous 6-mercaptopurine when it is administered in a patient taking allopurinol?

- a) Reduce by 75%
- b) Reduce by 50%
- c) Increase by 50%
- d) No dose adjustment necessary

Question 14:

A patient developed the following lesions after 10 days of initiation of chemotherapy. The patient also complains of similar lesions over soles and they are sensitive to touch. Which is the most likely agent to cause this adverse effect?



- a) Cisplatin
- b) Vincristine

- c) Capecitabine
- d) Mitomycin-C

Question 15:

In which of the following patients is pemetrexed the drug of choice?

- a) A patient with malignant mesothelioma
- b) A patient with small cell carcinoma
- c) A patient with stage II non-small cell lung carcinoma
- d) A patient with inflammatory breast cancer

Question 16:

A patient who is undergoing chemotherapy is to be started on allopurinol. Which of these drugs might require dose reduction?

- a) 6-mercaptopurine
- b) Cyclophosphamide
- c) 6-thioguanine
- d) Methotrexate

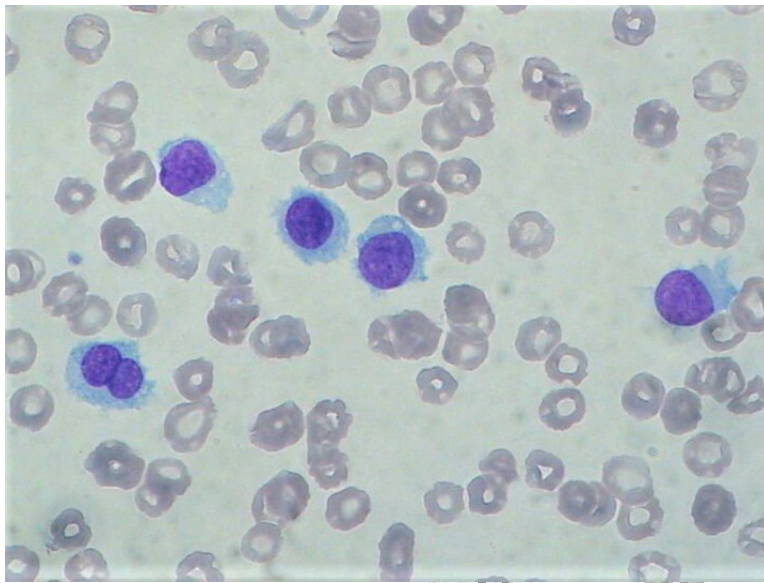
Question 17:

Which of the following enzymes is inhibited by pentostatin?

- a) Adenosine deaminase
- b) Aldolase
- c) Adenosine aminase
- d) Adenylyl cyclase

Question 18:

A 50-year-old patient presented with weakness, fatigue, and abdominal discomfort. On examination, splenomegaly was present. Investigations revealed pancytopenia, and his peripheral blood smear is shown below. Immunophenotype testing revealed CD11c, CD25, and CD103 positive. What is the drug of choice for this condition?



- a) Pentastatin
- b) Cladribine
- c) Fludarabine
- d) Clofarabine

Question 19:

Which of the following drugs is a topoisomerase I inhibitor?

- a) Doxorubicin
- b) Irinotecan
- c) Etoposide
- d) Vincristine

Question 20:

Which of the following is a mitotic polymerization inhibitor?

- a) Vinblastine
- b) Paclitaxel
- c) Dactinomycin
- d) Gemcitabine

Question 21:

A patient who was diagnosed with myocardial infarction underwent PCI and stenting with paclitaxel coated drug-eluting stent. What is the mechanism of action of this drug?

- a) Topoisomerase inhibition
- b) Increases the polymerisation of tubulin
- c) Inhibits protein synthesis
- d) Alkylation of DNA

Question 22:

Bone marrow suppression is not seen with which of the following drugs?

- a) Vincristine
- b) Vinblastine
- c) Vinorelbine
- d) Methotrexate

Question 23:

A 60-year-old man was diagnosed with AML and initiated on chemotherapy. During the course of his treatment, he develops ataxia and slurred speech. Which of the following drugs could have caused this?

- a) Cisplatin
- b) Cytarabine
- c) Bleomycin
- d) Actinomycin D

Question 24:

Which of the following statements is false regarding vincristine?

- a) It is an alkaloid
- b) Its use is associated with neurotoxicity
- c) It does not cause alopecia
- d) It is a useful drug for induction of remission in ALL

Question 25:

In which of the following conditions can all-trans-retinoic acid be used for treatment?

- a) Acute promyelocytic leukemia
- b) Acute lymphoblastic leukemia
- c) Chronic myelogenous leukemia
- d) Chronic lymphocytic leukemia

Question 26:

Which of the following antineoplastic agents is extracted from the plant shown below?



- a) Epothilones
- b) Camptothecin analogs
- c) Vinca alkaloids
- d) Taxanes

Question 27:

A 60-year-old woman is diagnosed with melanoma. On examination, a palpable lymph node was present in the posterior triangle. Biopsy revealed presence of tumour cells. Which of the following drugs used to treat this condition is a MEK inhibitor?

- a) Dabrafenib
- b) Vemurafenib
- c) Trametinib
- d) Ipilimumab

Question 28:

A 55-year-old woman had undergone modified radical mastectomy followed by chemotherapy with trastuzumab. She now presents with liver metastases. Which of the following drugs could be used in this condition?

- a) Lapatinib
- b) Panitumumab
- c) Erlotinib
- d) Imatinib

Answer Key

Question No.	Correct Option
1	c
2	c
3	b
4	a
5	c
6	a
7	b
8	b
9	a
10	c
11	d
12	a
13	d
14	c
15	a
16	a

17	a
18	b
19	b
20	a
21	b
22	a
23	b
24	c
25	a
26	c
27	c
28	a

Detailed Explanations

Solution to Question 1:

The Nobel prize in Physiology or Medicine, 2018 was awarded to James P. Allison and Tasuku Honjo for the discovery of cancer therapy by inhibition of negative immune regulation.

Immune checkpoint inhibitors inhibit proteins that keep immune responses to cancer cells in check. When these proteins are blocked, the brake on the immune system is released and T cells are able to kill cancer cells more efficiently.

The Nobel prize 2018 in Physiology or Medicine was awarded to James P. Allison and Tasuku Honjo for their breakthrough work on unleashing the body's immune system to attack cancer, that has led to an entirely new class of these drugs and brought lasting remissions to many patients who had run out of options.

Solution to Question 2:

Vinblastine is a cell cycle-specific agent and a vinca alkaloid. It arrests mitosis and acts specifically on M-phase.

Cyclophosphamide, melphalan, and ifosfamide are nitrogen mustards and are cell-cycle non-specific agents.

Cell cycle specific agents:

- Antimetabolites: S phase
- Camptothecin analogs: S Phase
- Taxanes, Vinca alkaloids: M phase

- Etoposide: G1-S phase
- Bleomycin: G2-M phase

Cell cycle non-specific agents:

- Alkylating agents
- Platinum analogs
- Anthracyclines
- Mitomycin, dactinomycin

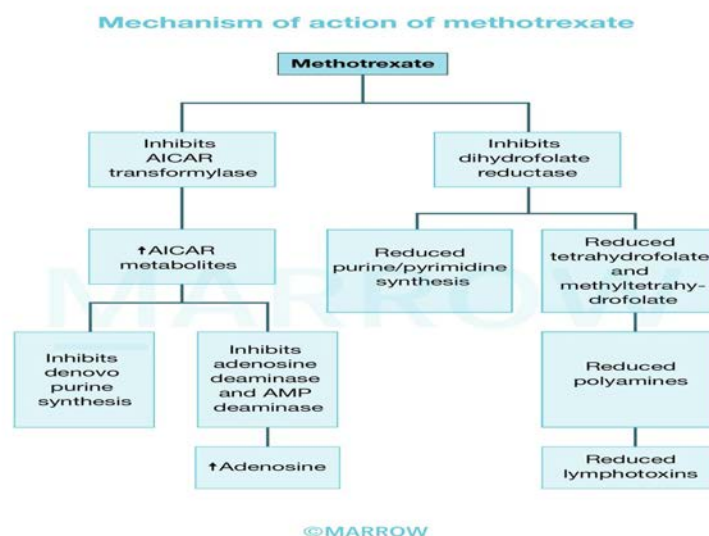
Solution to Question 3:

Methotrexate is an antimetabolite. It is a folic acid analog that inhibits the enzyme dihydrofolate reductase (DHFR). Methotrexate and its polyglutamate also inhibit the enzymes involved in de novo purine and thymidylate synthesis.

Solution to Question 4:

Methotrexate is the disease-modifying anti rheumatoid drug that increases extracellular adenosine. Methotrexate's principal mechanism of action relates to inhibition of amino-imidazole carboxamide ribonucleotide (AICAR) transformylase and thymidylate synthetase.

AICAR accumulates intracellularly, competitively inhibits AMP deaminase, leading to an accumulation of AMP. The AMP is released and converted extracellularly to adenosine, which is a potent inhibitor of inflammation. As a result, the inflammatory functions of neutrophils, macrophages, dendritic cells, and lymphocytes are suppressed.



Solution to Question 5:

Indications for methotrexate (inflammatory conditions) include:

- Rheumatoid arthritis
- Psoriasis
- Psoriatic arthritis
- Ankylosing spondylitis
- Polymyositis
- Dermatomyositis
- Wegener's granulomatosis
- Giant cell arteritis
- Systemic lupus erythematosus
- Other systemic vasculitis
- Graft versus host disease associated with organ transplantation.

Sickle cell anemia appears to be the most appropriate answer among the options. Though small trials have shown positive results in sickle cell disease, it is neither extensively used nor approved for this indication.

Solution to Question 6:

High-dose methotrexate with leucovorin rescue (HDM-L) is a standard agent for adjuvant therapy of osteosarcoma. HDM is critical in the management of childhood acute lymphocytic leukemia (ALL). HDM-L also produces a high response rate in CNS lymphomas.

Solution to Question 7:

The scenario depicts a case of Methotrexate resistance. This happens due to the overproduction of DHFR (dihydrofolate reductase).

Mechanism of resistance to methotrexate includes:

- Impaired transport of methotrexate into cells
- Production of altered forms of DHFR that have decreased affinity for methotrexate
- The increased concentration of intracellular DHFR through gene amplification or gene regulation
- Decreased ability to form methotrexate polyglutamates
- Increased expression of drug efflux transporter

Solution to Question 8:

Glucarpidase, which is a methotrexate cleaving enzyme, can be used in this condition. Glucarpidase is an FDA approved intravenous drug for the treatment of methotrexate toxicity in the treatment of cancer patients who have impaired kidney function.

Glucarpidase is a recombinant bacterial enzyme that breaks down methotrexate into inactive metabolites rapidly after injection. These inactive metabolites are then eliminated by the liver.

Note: Leucovorin is also a substrate of glucarpidase. Leucovorin is avoided 2 hours before and after the administration of glucarpidase to avoid its breakdown.

Solution to Question 9:

Folinic acid, also known as leucovorin, the active form of folic acid, is administered with methotrexate to reduce the toxicity associated with higher doses. It does not enhance the action of methotrexate.

Methotrexate is an antimetabolite that inhibits dihydrofolate reductase. It is a cell cycle (S-phase) specific antineoplastic agent and non-proliferative cells are relatively resistant to it.

Bone marrow suppression and gastrointestinal mucosa inflammation (mucositis) are the common side effects seen due to inhibition of DNA synthesis.

Methotrexate toxicity:

- Prevention - administration of leucovorin/folinic acid.
- Treatment - maintaining adequate hydration and forced alkaline diuresis in patients with normal renal function.
- Glucarpidase is an intravenous drug administered in patients who have impaired renal function.

Solution to Question 10:

Fundoscopy is not monitored in a patient on methotrexate therapy. This is because no significant eye-related adverse effect is associated with either low, normal or high-dose methotrexate use, mandating frequent monitoring.

Option A - Hepatic fibrosis or cirrhosis is a significant side effect of long-term therapy of methotrexate that requires regular hepatic function monitoring.

Option B - Allergic interstitial pneumonitis and other pulmonary-related adverse effects are associated with methotrexate.

Option D - Myelosuppression is a common and a serious adverse effect that mandates routine monitoring of hemogram.

Solution to Question 11:

The drug being discussed here is methotrexate. Alkalization of urine ameliorates the toxicity of methotrexate. Methotrexate is an acidic drug and the excretion is increased by alkalization of urine.

High-dose methotrexate with leucovorin (HDM-L) used in some conditions has a higher incidence of nephrotoxicity. It is attributed to the precipitation of methotrexate in the acidic tubular fluid. Vigorous hydration and alkalization of urine prior to HDM-L administration is required to minimize the chance of nephrotoxicity.

Solution to Question 12:

6-Mercaptopurine is an inhibitor of DNA synthesis.

6-Mercaptopurine is inactive in its parent form. It must be metabolized by hypoxanthine-guanine phosphoribosyltransferase to form monophosphate nucleotide 6-thioinosinic acid. This inhibits several enzymes of de novo purine nucleotide synthesis.

Option B- Mitomycin acts as an alkylating agent and crosslinks DNA. It also generates free radicals which damage DNA.

Option C- Actinomycin causes single-strand breaks in DNA. It blocks RNA transcription as well.

Option D- L-Asparaginase causes the breakdown of asparagine and thus interferes with protein synthesis.

Solution to Question 13:

Intravenous dosing of 6-mercaptopurine (6MP) does not require any dosing adjustment when administered with allopurinol.

6MP undergoes the first-pass metabolism by xanthine oxidase in the liver and intestine. Since allopurinol inhibits xanthine oxidase, it causes elevated levels of 6-MP. Hence, the oral dose of 6MP needs to be reduced by 75 % if allopurinol is given.

In the case of intravenous administration, the dosage need not be changed as it escapes the first-pass metabolism.

Solution to Question 14:

The given image depicts classical findings of hand-foot syndrome and capecitabine is commonly associated with this prominent side effect.

Hand foot syndrome or palmar-plantar erythrodysesthesia consists of erythema, desquamation, pain and sensitivity to touch of the palms and soles.

Drugs commonly implicated are:

- Capecitabine

- 5-Fluorouracil
- Cytarabine
- Pemetrexed
- Doxorubicin

Solution to Question 15:

Pemetrexed is the drug of choice for the treatment of malignant mesothelioma. It is an antifolate drug that belongs to the antimetabolite group.

It acts by inhibiting thymidylate synthase and dihydrofolate reductase (DHFR) and prevents the formation of DNA and RNA, which are required for the growth and survival of both normal cells and cancer cells.

Pemetrexed is used for the treatment of stage IV non-small cell carcinoma along with cisplatin.

Solution to Question 16:

A patient receiving allopurinol requires a dose reduction of 6-mercaptopurine.

Allopurinol is a potent xanthine oxidase inhibitor that is often used in acute leukemias as supportive care to prevent the development of hyperuricemia due to tumor lysis syndrome.

6-mercaptopurine is converted to an inactive metabolite (6-thiouric acid) by xanthine oxidase in the liver and intestine. Therefore, the oral bioavailability of the 6-mercaptopurine is reduced significantly due to high first-pass metabolism.

Concomitant allopurinol use increases 6-mercaptopurine availability which leads to toxicity, thereby mandating oral dose reduction.

Note: Intravenous dosing of 6-mercaptopurine need not require any dosing adjustment when administered with allopurinol.

Solution to Question 17:

Pentostatin mimics adenosine and inhibits the enzyme adenosine deaminase. Adenosine deaminase inhibition leads to the accumulation of adenosine and deoxyadenosine nucleotides that blocks DNA synthesis by inhibiting RNR (ribonucleotide reductase).

Pentostatin also inhibits RNA synthesis, its triphosphate derivative is incorporated into DNA causing strand breakage.

It is indicated in the treatment of hairy cell leukemia.

Solution to Question 18:

The clinical scenario, characteristic immunophenotype, and the presence of hairy cells on the Wright-stained peripheral blood smear point to a diagnosis of hairy cell leukaemia. Cladribine is considered the drug of choice for hairy cell leukemia.

It is administered daily by continuous infusion for 7 days. A single course of therapy achieves a complete response in about 80% of the patients.

Solution to Question 19:

Irinotecan is a topoisomerase I inhibitor.

Doxorubicin and etoposide inhibit topoisomerase II.

Vincristine is a vinca alkaloid derivative that inhibits microtubule polymerization.

Solution to Question 20:

Vinca alkaloids like vinblastine and vincristine bind to beta-tubulin and inhibit its polymerization with alpha-tubulin into microtubules.

Paclitaxel does not inhibit polymerization, rather it stabilizes polymerization and inhibits the breakdown of microtubules.

Gemcitabine is an antimetabolite and a pyrimidine analog.

Dactinomycin or actinomycin D is an actinomycin.

Solution to Question 21:

Paclitaxel binds specifically to beta-tubulin and promotes the polymerization and inhibits the disassembly of this cytoskeletal protein, resulting in the arrest of the cell cycle in M-phase.

The first generation drug eluting stents used in PCI were coated with paclitaxel or sirolimus as an anti-proliferative agent. Second-generation drug-eluting stents use newer agents such as everolimus, biolimus, and zotarolimus.

Solution to Question 22:

Bone marrow suppression is not seen with vincristine.

Vinblastine and vinorelbine have prominent bone marrow depression.

Bone marrow sparing cytotoxic drugs are:

- Vincristine

- Bleomycin
- L-asparaginase

Solution to Question 23:

This scenario describes a case of cerebellar toxicity and cytarabine is associated with this condition.

It is seen especially with intrathecal administration or high dose systemic use. Cerebellar toxicity manifests as ataxia and slurred speech. Cerebral toxicity (seizures, dementia, and coma) is also seen.

Solution to Question 24:

Vincristine use is associated with alopecia. It is usually reversible frequently without discontinuation of the drug.

Vincristine is a vinca alkaloid derivative. It is mainly indicated for induction of remission in acute leukemias including ALL. Neurotoxicity is one of the main side effects.

Solution to Question 25:

All-trans-retinoic acid (tretinoin) is used in the treatment of acute promyelocytic leukemia (APL). RAR-alpha and retinoid X receptor dimer binds tretinoin tightly and causes differentiation of promyelocytes resulting in decreased proliferation.

As monotherapy, tretinoin induces a high rate of complete remission and in combination with anthracyclines cures most of the patients.

Solution to Question 26:

The image shown is of the periwinkle plant (*Catharanthus roseus*). Vinca alkaloids like vinblastine and vincristine are extracted from this plant.

Sources of few other drugs are:

- Epothilones - *Myxobacterium Sorangium cellulosum*
- Camptothecin analogs - *Camptotheca acuminata* tree
- Taxanes - bark of the western yew tree.

Solution to Question 27:

The scenario is suggestive of metastatic melanoma and trametinib is used in this condition. It is a reversible inhibitor of mitogen-activated extracellular signal regulated kinase (MEK). It inhibits the serine-threonine kinase domain of BRAF V600E mutant of MEK.

It is widely used in the treatment of metastatic or unresectable melanoma. It can be used as a single agent but it is more effective when used with BRAF inhibitors like dabrafenib.

Adverse effects of this drug include rash, diarrhea, increased ALT, abdominal pain, stomatitis, increased AST, anemia, etc.

Dabrafenib and vemurafenib are BRAF inhibitors used in metastatic melanoma.

Ipilimumab is a human cytotoxic T-lymphocyte antigen 4 (CTLA-4) blocking antibody used in metastatic or unresectable melanoma.

Recent update: The FDA has approved the combination of the BRAF inhibitor encorafenib and the MEK inhibitor binimetinib for the treatment of patients with BRAF-mutant unresectable or metastatic melanoma.

Solution to Question 28:

The given scenario depicts a case of trastuzumab-refractory breast cancer and lapatinib is used in this condition.

Drug	Use
Lapatinib	Refractory and metastatic breast carcinoma
Panitumumab	Colorectal carcinoma
Erlotinib	Non small cell lung carcinoma Pancreatic carcinoma
Imatinib	Chronic myeloid leukemia Gastrointestinal stromal tumour

Non Cell Cycle Specific Cytotoxic Drugs

Question 1:

Which of the following is more likely to be susceptible to alkylation with a bifunctional alkylating agent?

- a) N3 of guanine
- b) N1 of adenosine
- c) N3 of adenosine
- d) N7 of guanine

Question 2:

During a study on the cytotoxic effects of alkylating agents, you observe that cyclophosphamide has a higher cytotoxic effect compared to procarbazine. Which property of cyclophosphamide is the reason for this?

- a) Cross-linking of DNA
- b) Methylation of DNA
- c) Additional inhibition of topoisomerase I
- d) Additional inhibition of topoisomerase II

Question 3:

Which among the following is not an alkylating agent?

- a) Carmustine
- b) Melphalan
- c) Chlorambucil
- d) 5-Flourouracil

Question 4:

Which of the following is not a nitrogen mustard?

- a) Mechlorethamine

- b) Melphalan
- c) Cyclophosphamide
- d) Carmustine

Question 5:

Temozolomide is used as a first-line agent for the management of which of the following conditions?

- a) Anaplastic astrocytoma
- b) Glioblastoma multiforme
- c) Acoustic neuroma
- d) Craniopharyngioma

Question 6:

A patient is diagnosed with astrocytoma. Which of the following drugs crosses the blood brain barrier and hence can be used for the treatment of this condition?

- a) Cisplatin
- b) Lomustine
- c) Vincristine
- d) Vinblastine

Question 7:

Which of the following drugs is not an anthracycline antibiotic?

- a) Doxorubicin
- b) Dactinomycin
- c) Mitoxantrone
- d) Epirubicin

Question 8:

Topical mitomycin C is used in which of the following patients?

- a) A patient with Sturge-Weber syndrome
- b) A patient with laryngotracheal stenosis
- c) A patient undergoing endoscopic treatment of angiofibroma
- d) A patient with hand-foot syndrome

Question 9:

A patient with superficial bladder carcinoma is being treated with a drug by intravesical therapy. Identify this drug.

- a) Actinomycin
- b) Mitomycin
- c) Cisplatin
- d) Vincristine

Question 10:

A patient was diagnosed with acute promyelocytic leukemia and chemotherapy was initiated. The patient achieved complete remission. However, after a few months molecular relapse was detected. Which of the following drugs would be highly effective in this case?

- a) Tretinoin
- b) Arsenic trioxide
- c) Idarubicin
- d) Cytarabine

Question 11:

A patient with osteosarcoma was initiated on chemotherapy. After a few days, he developed dysuria and hematuria. Which group of drugs among the following is most likely to have caused this complication?

- a) Alkylating agents
- b) Antimetabolites
- c) Mitotic inhibitors
- d) Topoisomerase inhibitors

Question 12:

Hemorrhagic cystitis is a characteristic side effect of which of the following drugs?

- a) Melphalan
- b) Busulfan
- c) Cyclophosphamide
- d) Procarbazine

Question 13:

A 62-year-old woman with breast carcinoma is started on cyclophosphamide. What is the rationale behind using sodium 2-mercaptoethanesulfonate along with this drug?

- a) Increase absorption
- b) Decrease excretion
- c) Ameliorate hemorrhagic cystitis
- d) Decrease metabolism

Question 14:

Which of the following statements about ifosfamide is false?

- a) Metabolised by cytochrome P450 enzymes
- b) Less neurotoxic than cyclophosphamide
- c) Chloroacetaldehyde is the metabolite of ifosfamide
- d) It is a nitrogen mustard

Question 15:

A 67 year old woman with a malignant lump in the breast who is undergoing chemotherapy, presents with shortness of breath, tachycardia and hypertension. ECG shows ST and T wave changes. Which among the following drugs could have led to this?

- a) Doxorubicin
- b) Mitomycin-C
- c) Bleomycin
- d) Actinomycin-D

Question 16:

A 54-year-old man is diagnosed with Kaposi sarcoma. You are planning to start chemotherapy with doxorubicin, which can cause cardiotoxicity. Which of the following drugs would you prescribe to prevent this?

- a) Leucovorin
- b) Mesna
- c) Dexrazoxane
- d) Dexamethasone

Question 17:

Least cardiotoxicity is seen with which of the following anthracycline antibiotic?

- a) Doxorubicin
- b) Daunorubicin
- c) Idarubicin
- d) Epirubicin

Question 18:

While reviewing the drug chart of a patient undergoing radiotherapy, which of the following drugs would require a dose adjustment to prevent radiation toxicity?

- a) Vincristine
- b) Doxorubicin
- c) Cyclophosphamide
- d) 6-mercaptopurine

Question 19:

A patient is undergoing chemotherapy for glioblastoma multiforme. You notice that his serial blood investigations reveal sustained neutropenia. Which of the following drugs could have caused this?

- a) Carmustine

- b) Cisplatin
- c) Vinblastine
- d) Cyclophosphamide

Question 20:

A patient with Hodgkin's disease on chemotherapy with an alkylating agent develops behavioral disturbances and experiences sedation. He presented to the emergency room with a disulfiram-like reaction. What is the drug being described here?

- a) Dacarbazine
- b) Procarbazine
- c) Carmustine
- d) Lomustine

Question 21:

A 50 year old man who presented with dysphagia was diagnosed with esophageal carcinoma and underwent multiple cycles of chemotherapy. During the cycles, he had developed tingling and numbness in the lower limbs and the symptoms progressed even after cessation of the cycles. Which drug could have led to this?

- a) Cisplatin
- b) Paclitaxel
- c) Mitomycin
- d) Vincristine

Question 22:

A 65 year old man who was diagnosed with small cell carcinoma of the lung had undergone chemotherapy about 5 years ago. Now he presents with features suggestive of AML. Which of the following drugs could have caused this condition?

- a) Vinblastine
- b) Paclitaxel
- c) Cisplatin
- d) Bleomycin

Question 23:

A patient who was diagnosed with gastric carcinoma is undergoing chemotherapy. His serum creatinine levels are rising and you suspect renal tubular damage caused by one of his chemotherapeutic drugs. Which drug could it be?

- a) Cisplatin
- b) Streptozocin
- c) Methysergide
- d) Cyclophosphamide

Question 24:

Which among the following is a highly emetogenic anticancer drug?

- a) Paclitaxel
- b) Vincristine
- c) Cisplatin
- d) 5-Fluorouracil

Question 25:

A patient who is undergoing chemotherapy for testicular cancer, now presents with dyspnea and dry cough. Investigations suggest pulmonary fibrosis. This condition is commonly associated with which of the following drugs?

- a) Bleomycin
- b) Cisplatin
- c) Methotrexate
- d) Actinomycin D

Question 26:

A patient who is undergoing chemotherapy, now presents with weight gain, ascites, hepatomegaly and hyperbilirubinemia. You suspect sinusoidal obstruction syndrome. Which drug is the most common cause of this condition?

- a) Ifosfamide
- b) Cyclophosphamide

- c) Bleomycin
- d) Busulfan

Question 27:

A patient undergoing chemotherapy was brought to the ER with seizures. He is diagnosed with cerebrovascular accident which you suspect to be due to a drug-induced hypercoagulable state. Which of the following drugs is implicated?

- a) 5-FU
- b) L-asparaginase
- c) Melphalan
- d) Carmustine

Question 28:

Which of the following drugs causes flagellate pigmentation?

- a) Bleomycin
- b) Doxorubicin
- c) Vincristine
- d) Minocycline

Question 29:

Which of the following drugs cause carcinoma bladder?

- a) Cyclophosphamide
- b) Cisplatin
- c) Taxane
- d) Tamoxifen

Answer Key

Question No.	Correct Option
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1	d
2	a
3	d
4	d
5	b
6	b
7	b
8	b
9	b
10	b
11	a
12	c
13	c
14	b
15	a
16	c
17	c
18	b
19	a
20	b
21	a
22	c
23	a
24	c
25	a
26	d
27	b
28	a
29	a

Detailed Explanations

Solution to Question 1:

The N7 atom of guanine is particularly susceptible to alkylation with bifunctional alkylating agents. It leads to the formation of a covalent bond with bifunctional alkylating agents and represents the key target in nucleic acids for alkylating agents. Alkylation results in cross-linking/abnormal base pairing of DNA which inhibits DNA replication.

Other sites that can be involved are:

- N1 and N3 of adenosine
- N3 of cytosine
- O6 of guanine

Solution to Question 2:

Cyclophosphamide (nitrogen mustard) is a bifunctional alkylating agent and it results in cross-linking of DNA that causes a major disruption in nucleic acid function and increased cytotoxicity.

Procarbazine is a monofunctional alkylating agent.

Differences between bifunctional and monofunctional alkylating agents are:

Bifunctional alkylating agents	Monofunctional alkylating agents
Usually transfers ethyl group to DNA resulting in cross-linking	Usually transfers a methyl group resulting in simple methylation. No cross-linking of DNA
Higher cytotoxicity because of cross-linking results in a greater threat to cellular survival	Higher carcinogenesis risk as simple methylation can be bypassed by DNA polymerases resulting in permanent modification of DNA sequence

Solution to Question 3:

5-Fluorouracil is a pyrimidine analogue and is an antimetabolite.

Solution to Question 4:

Carmustine belongs to the nitrosourea class of alkylating agents.

Solution to Question 5:

Temozolomide is used as a first-line agent for the management of glioblastoma multiforme. It is used for newly diagnosed cases along with radiotherapy and also used for maintenance therapy.

Temozolomide belongs to the triazene group of alkylating agents and is administered orally. The oral bioavailability of temozolomide is nearly 100%.

In anaplastic astrocytoma, refractory cases with disease progression are treated with a regimen containing nitrosourea and procarbazine.

Solution to Question 6:

Lomustine can cross the blood-brain barrier and is used in the management of brain tumours like astrocytoma.

Nitrosoureas like lomustine and carmustine are highly lipophilic drugs and can easily cross the blood-brain barrier. Hence, they are indicated in the management of brain tumors.

Solution to Question 7:

Dactinomycin or actinomycin D is an actinomycin and not an anthracycline antibiotic.

Solution to Question 8:

Topical mitomycin C is used in laryngotracheal stenosis.

Mitomycin C is an antitumor antibiotic that inhibits cell division, protein synthesis and fibroblast proliferation. Topical mitomycin C has been shown to cause symptomatic improvement when applied to patients with laryngotracheal stenosis.

Solution to Question 9:

Mitomycin C is used in the treatment of superficial transitional cell carcinoma by intravesical instillation. It is also used in upper gastrointestinal cancers (e.g. esophageal carcinoma), anal cancers, and breast cancers.

Topical mitomycin C is used in:

- Superficial bladder cancer
- Intraperitoneal tumors
- Strabismus surgery to reduce scarring
- To prevent tracheal and esophageal stenosis
- Post nasal surgery synechae formation

Solution to Question 10:

Arsenic trioxide (ATO) is highly effective in relapsed or refractory acute promyelocytic leukemia (APL). It acts by enhancing reactive oxygen free radical generation in APL cells.

Tretinoin and idarubicin (anthracycline) are usually first-line agents for acute promyelocytic leukemia.

Solution to Question 11:

This scenario depicts haemorrhagic cystitis and the possible cause could be alkylating agents like ifosfamide.

Chemotherapeutic agents used for the treatment of osteosarcoma include ifosfamide, which is one of the commonest drugs causing haemorrhagic cystitis.

Ifosfamide is a nitrogen mustard and belongs to the alkylating agents.

Solution to Question 12:

Hemorrhagic cystitis is a characteristic side effect of cyclophosphamide. The metabolite responsible for hemorrhagic cystitis is acrolein. This can be decreased by vigorous hydration and by the use of mercapto ethane sulfonic acid (mesna).

Solution to Question 13:

Sodium 2-mercaptoethanesulfonate (MESNA) is given along with cyclophosphamide to reduce the incidence of hemorrhagic cystitis.

MESNA concentrates in the bladder where acrolein accumulates after administration of cyclophosphamide and ifosfamide. It forms a conjugate with acrolein and other urotoxic metabolites. This conjugation reaction inactivates the urotoxic compounds to harmless metabolites which are excreted in the urine.

Solution to Question 14:

Ifosfamide is more neurotoxic than cyclophosphamide. It causes greater platelet suppression, nephrotoxicity, and urothelial damage when compared to cyclophosphamide.

Ifosfamide is a nitrogen mustard (alkylating agent). It is metabolized by hydroxylation by cytochrome P450 enzymes. Chloroacetaldehyde is a metabolite of ifosfamide that is responsible for severe neurological adverse effects.

Solution to Question 15:

This scenario depicts Doxorubicin induced cardiotoxicity. It is a unique adverse effect of doxorubicin. It is an anthracycline group of anticancer antibiotics.

Doxorubicin-induced cardiomyopathy takes two forms:

- Acute: pericarditis-myocarditis syndrome, arrhythmias, ST and T wave changes in ECG.
- Chronic: dilated cardiomyopathy.

Note: Dexrazoxane is a cardioprotective iron-chelating agent that is indicated for use along with high dose doxorubicin to avoid cardiotoxicity.

Solution to Question 16:

Dexrazoxane is a cardioprotective iron-chelating agent that is indicated for use along with high dose doxorubicin to avoid cardiotoxicity. The exact cardioprotective mechanism of dexrazoxane is unknown and it appears not to compromise with anticancer activity of doxorubicin.

Solution to Question 17:

Idarubicin has least cardiotoxicity when compared to other anthracyclines.

It is approved for use in acute myeloid leukemia along with cytarabine. Doxorubicin, daunorubicin and epirubicin share similar cardiotoxicity profile.

Solution to Question 18:

Doxorubicin requires dose adjustment during or after radiotherapy in order to prevent radiation toxicity.

Anthracyclines like doxorubicin can result in a radiation recall reaction, where erythema and desquamation of skin are observed at sites of previous radiation injury. The risk of cardiotoxicity also increases with previous irradiation. Therefore, careful monitoring and dosage adjustments are required for doxorubicin.

Solution to Question 19:

Sustained neutropenia is seen with nitrosoureas. Nitrosoureas like carmustine and lomustine cause profound and delayed myelosuppression with recovery 4-6 weeks after a single dose.

Solution to Question 20:

The most likely alkylating agent being described here is procarbazine. It is associated with behavioural disturbances and sedation. Concomitant CNS depressants should be avoided.

Procarbazine is a weak MAO inhibitor and can potentiate tyramine and catecholamines action. It also has disulfiram-like actions and alcohol should be avoided.

Solution to Question 21:

The scenario depicts coasting effect neuropathy, which is seen with cisplatin.

The coasting effect means the progression of symptoms even after the cessation of the drug. Neurotoxicity is usually dependent on the cumulative dose of anticancer drugs. The severity of neuropathy increases with the duration of treatment and progression stops once drug treatment is completed.

Platinum compounds are an exception, where sensory loss progresses for several months after cessation of treatment. The coasting effect is more common with cisplatin and oxaliplatin than with carboplatin.

Solution to Question 22:

The scenario is suggestive of secondary leukemia which is associated with cisplatin. Cisplatin is a platinum-containing anticancer agent that forms DNA adduct. It is associated with the development of AML usually more than 4 years after treatment.

Procarbazine, a methylhydralazine (alkylating agent), is most commonly associated with the development of leukemias with a 5-10% risk of acute leukemia in patients treated with the MOPP regimen.

Cisplatin produces responses in cancers of the bladder, head and neck, cervix, and endometrium; all forms of carcinoma of the lung; anal and rectal carcinomas; and neoplasms of childhood.

Solution to Question 23:

The given scenario is suggestive of nephrotoxicity and it is one of the dose-limiting toxicity of cisplatin therapy. Cisplatin achieves a high concentration in nephrons. Both direct toxicity and inflammation are responsible for nephrotoxicity. A high concentration of chloride stabilizes cisplatin and reduces nephrotoxicity. Therefore, chloride diuresis is used to reduce nephrotoxicity.

Option B - Streptozotocin use can result in diabetes mellitus due to the destruction of beta cells of the pancreas.

Option C - Methysergide use is associated with retroperitoneal fibrosis.

Option D - Cyclophosphamide causes hemorrhagic cystitis as one of its characteristic side effect.

Solution to Question 24:

Cisplatin is a highly emetogenic anticancer drug. Cisplatin causes marked nausea and vomiting in almost all cases and is usually administered along with anti-emetic regimens.

This can usually be controlled with 5HT₃ receptor antagonists, neurokinin (NK₁) receptor antagonists (e.g., aprepitant), and high-dose corticosteroids.

Solution to Question 25:

Pulmonary fibrosis is commonly seen with bleomycin. The main abnormality in bleomycin toxicity occurs in endothelial and epithelial cells. This leads to the destruction and desquamation of type I pneumocytes and the resultant hyperplasia of type II pneumocytes.

Pulmonary toxicity begins with a dry cough, fine rales, and diffuse basal infiltrates on X-ray and may progress to life-threatening pulmonary fibrosis. No specific treatment is available to ameliorate the toxicity and managed by general symptomatic care.

Note: Few case reports of methotrexate causing pulmonary fibrosis are reported. The association is still uncertain as the underlying disease process like RA can also contribute to fibrosis.

Solution to Question 26:

Sinusoidal obstruction syndrome is most commonly associated with high-dose busulfan. Incidence increases when coadministered with CYP inhibitors.

Sinusoidal obstruction syndrome, previously known as veno-occlusive hepatic disease (VOD), involves primary hepatic sinusoidal injury that progresses to diffuse hepatocyte involvement and panvasculitis. This leads to multi-organ failure, and can be life-threatening.

Solution to Question 27:

L-asparaginase causes a hypercoagulable state. It is an enzyme and toxicity results due to the inhibition of protein synthesis in normal tissues.

Side effects of L-asparaginase:

- Thrombosis and increased coagulation - physiological anticlotting factors are impaired
- Hemorrhage and bleeding - clotting factors are impaired
- Hyperglycemia - insulin deficiency.

Solution to Question 28:

Flagellate hyperpigmentation/flagellate erythema is a characteristic cutaneous reaction to treatment with bleomycin.

Clinical features include multiple linear streaks (erythematous or hyperpigmented) at sites of trauma and generalized pruritus.



Solution to Question 29:

The only chemotherapeutic agent that has been proven to cause bladder cancer is cyclophosphamide.

The risk of the bladder cancer formation is linearly related to the duration and intensity of cyclophosphamide treatment, supporting a causative role. Phosphoramidate mustard is the primary mutagenic metabolite that causes bladder cancer in patients exposed to cyclophosphamide.

Monoclonal Antibodies

Question 1:

Cetuximab is approved for use in which of the following conditions?

- a) Palliation in head and neck cancer
- b) Anal canal carcinoma
- c) Gastric carcinoma
- d) Lung carcinoma

Question 2:

A patient diagnosed with metastatic colorectal carcinoma is being treated with panitumumab. This drug acts against which of the following receptors?

- a) PDGFR
- b) EGFR
- c) HER2/neu receptor
- d) VEGF receptor

Question 3:

A 57-year-old woman is diagnosed with HER 2/neu positive breast cancer. Which of the following monoclonal antibodies is indicated for the management of this patient?

- a) Panitumumab
- b) Cetuximab
- c) Trastuzumab
- d) Bevacizumab

Question 4:

A 57-year-old woman diagnosed with breast carcinoma was prescribed trastuzumab. Which of the following statements regarding this drug is true?

- a) It is an antibody produced entirely from mouse containing no human component

- b) It is a monoclonal antibody produced by injecting HER-2 antigen
- c) It is a polyclonal antibody
- d) It is a monoclonal antibody containing only human component

Question 5:

A patient was diagnosed with HER2/neu positive breast cancer and was started on adjuvant therapy with trastuzumab. Which of the following is the most serious and dose-limiting toxicity of this drug?

- a) Nephrotoxicity
- b) Neurotoxicity
- c) Cardiotoxicity
- d) Ototoxicity

Question 6:

A patient diagnosed with wet age-related macular degeneration is being administered ranibizumab intravitreally. All of the following are known side effects of the drug, except _____.

- a) Eye pain
- b) Fall in IOP
- c) Conjunctival hemorrhage
- d) Floaters

Question 7:

Which among the following is the monoclonal antibody that is targeted specifically against VEGFR?

- a) Bevacizumab
- b) Ranibizumab
- c) Ramucirumab
- d) Aflibercept

Question 8:

Rituximab is not used in which of the following conditions?

- a) Non-Hodgkin lymphoma
- b) Paroxysmal nocturnal hemoglobinuria
- c) Rheumatoid arthritis
- d) Systemic lupus erythematosus

Question 9:

A 70-year-old man is found to have leukocytosis on his yearly health check-up. Further evaluation led to a diagnosis of chronic lymphocytic leukemia. Immunophenotyping was positive for CD52. Which is the monoclonal antibody indicated in this condition?

- a) Rituximab
- b) Alemtuzumab
- c) Cetuximab
- d) Obinutuzumab

Question 10:

Which of the following monoclonal antibodies is not a HER1 inhibitor?

- a) Pertuzumab
- b) Cetuximab
- c) Necitumumab
- d) Panitumumab

Question 11:

Which of the following is a monoclonal antibody directed against PDGFR-alpha?

- a) Panitumumab
- b) Pertuzumab
- c) Cetuximab
- d) Olaratumab

Question 12:

Which of the following is a bispecific monoclonal antibody?

- a) Daratumumab
- b) Alemtuzumab
- c) Blinatumomab
- d) Panitumumab

Question 13:

A 40-year-old man who presented with fever, night sweats and weight loss was evaluated and diagnosed with diffuse large B-cell lymphoma. Which of the following is a radioimmunoconjugate that can be used in this condition?

- a) Gemtuzumab ozogamicin
- b) Brentuximab vedotin
- c) Denileukin deftitox
- d) Ibritumomab tiuxetan

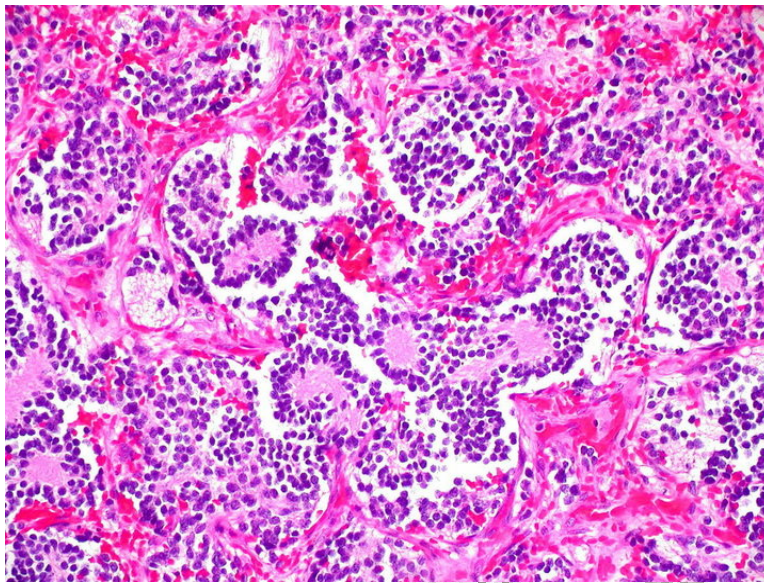
Question 14:

What is basiliximab?

- a) IL-1 receptor antagonist
- b) Anti-CD3 antibody
- c) IL-2 receptor antagonist
- d) TNF inhibitor

Question 15:

A 2-year-old girl was brought to the hospital by her mother with complaints of abdominal pain and diarrhea. Examination revealed a palpable abdominal mass. Investigations showed elevated levels of urinary homovanillic acid and vanillyl mandelic acid. The lesion could not be completely resected during surgery and histopathology findings are shown in the image. Which of the following is a monoclonal antibody approved for use in this condition?



- a) Alirocumab
- b) Alemtuzumab
- c) Dinutuximab
- d) Denosumab

Question 16:

Which of the following drugs is recently approved for the treatment of colorectal cancer arising through Lynch Syndrome?

- a) Ipilimumab
- b) Nivolumab
- c) Pembrolizumab
- d) Atezolimumab

Question 17:

Pembrolizumab is not approved for use in which of the following malignancies?

- a) Non-metastatic small cell lung cancer
- b) Metastatic squamous cell lung cancer
- c) Refractory classical hodgkin's lymphoma
- d) Urothelial cancer

Question 18:

Which among the following is the Interleukin-5 antagonist indicated in the management of asthma?

- a) Omalizumab
- b) Lebrikizumab
- c) Reslizumab
- d) Atezolimumab

Question 19:

Which of the following is the monoclonal antibody developed against Bacillus anthracis?

- a) Alemtuzumab
- b) Omalizumab
- c) Obiltoxaximab
- d) Abciximab

Question 20:

Which of the following is an immune checkpoint inhibitor?

- a) Alemtuzumab
- b) Denosumab
- c) Durvalumab
- d) Abciximab

Question 21:

A patient with rheumatoid arthritis has inadequate disease control with methotrexate. Which Janus kinase inhibitor can you add to his treatment regimen?

- a) Ruxolitinib
- b) Baricitinib
- c) Dasatinib
- d) Ponatinib

Question 22:

A patient has presented to the OPD with raised erythematous plaques on his arms and torso covered in silvery scales. You decide to prescribe a monoclonal antibody targeting interleukin-23 for treatment. Identify this drug.

- a) Secukinumab
- b) Ixekizumab
- c) Brodalumab
- d) Tildrakizumab

Question 23:

Dupilumab is indicated in the management of _____.

- a) Resistant osteoporosis
- b) Moderate atopic dermatitis
- c) Guttate psoriasis
- d) Severe rheumatoid arthritis

Answer Key

Question No.	Correct Option
1	a
2	b
3	c
4	b
5	c
6	b
7	c
8	b
9	b
10	a
11	d
12	c
13	d

14	c
15	c
16	c
17	a
18	c
19	c
20	c
21	b
22	d
23	b

Detailed Explanations

Solution to Question 1:

Cetuximab is used for palliation in head and neck cancer.

Indications for use of cetuximab are:

- Advanced squamous cell carcinoma of the head and neck in combination with radiotherapy for locally or regionally
- Metastatic colorectal cancer

Cetuximab is a recombinant chimeric monoclonal antibody against the external domain of the epidermal growth factor receptor (EGFR).

Solution to Question 2:

Panitumumab is a recombinant fully-humanized monoclonal antibody that binds specifically to the external domain of the epidermal growth factor receptor (EGFR).

It is approved for use in patients with metastatic colorectal carcinoma.

Solution to Question 3:

Trastuzumab is indicated for adjuvant therapy in early and metastatic HER2 positive breast cancer.

Trastuzumab is a humanized monoclonal antibody that binds to the extracellular domain of HER2/Neu.

Human epidermal growth factor receptor 2 tyrosine kinase receptor (HER-2-TK) inhibitors are:

- Lapatinib
- Neratinib

Anti-Her-2 monoclonal antibodies are:

- Trastuzumab
- Pertuzumab

Solution to Question 4:

Trastuzumab is a humanized monoclonal antibody against the external domain of HER2/neu. In humanized antibodies, all components are identical to human antibodies (95% human component), except the complementary region of the variable chain that contains sequences from other species.

It is indicated in the management of adjuvant therapy in early and metastatic HER-2 positive breast cancer.

Human epidermal growth factor receptor 2 tyrosine kinase receptor (HER-2-TK) inhibitors are:

- Lapatinib
- Neratinib

Anti-Her-2 monoclonal antibodies are:

- Trastuzumab
- Pertuzumab

Solution to Question 5:

The most serious toxicity of trastuzumab is cardiac failure. Cardiac failure is a potentially disabling or fatal side effect unless recognized early and the drug is discontinued.

Cardiotoxicity is caused by the interruption of HER2/4 heterodimer signaling in cardiomyocytes, which is a type of signaling that is essential for the maintenance of contractile function.

Clinical correlation:

- Baseline ECG and cardiac ejection fraction should be monitored to rule out underlying heart disease
- Careful clinical follow-up for signs and symptoms of CHF

Other side effects include fever, chills, nausea, dyspnea, and rashes.

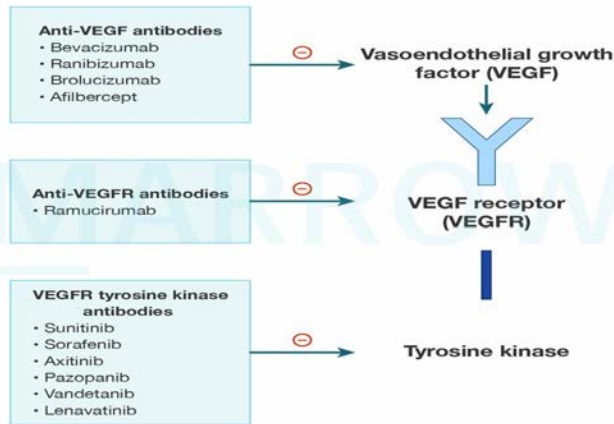
Dose-limiting toxicities are side effects of a drug or other treatment that are serious enough to prevent an increase in dose or level of that treatment.

Solution to Question 6:

Ranibizumab is associated with a raise in IOP, not a fall.

Ranibizumab is a monoclonal antibody directed against VEGF-A (vascular endothelial growth factor-A). Ranibizumab is indicated in wet age-related macular degeneration and is used intravitreally where it inhibits VEGF-A induced ocular neovascularization.

Inhibitors of VEGF and VEGFR antibodies



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Solution to Question 7:

Ramucirumab is a monoclonal antibody that binds to VEGFR-2 and inhibits the binding of VEGFR ligands. It is indicated in NSCLC, gastric cancer and metastatic colorectal cancer.

Bevacizumab, aflibercept and ranibizumab are monoclonal antibodies against VEGF ligands and not the receptor.

Solution to Question 8:

Rituximab is used in all conditions, except paroxysmal nocturnal hemoglobinuria.

Indications for rituximab include: (mnemonic- CANT MIS Ritu):

- CLL
- Autoimmune hemolytic anemia
- Non-Hodgkin lymphoma
- TTP
- Marginal zone lymphomas, mantle cell lymphoma, myasthenia gravis
- ITP

- SLE
- Rheumatoid arthritis

It is a chimeric monoclonal antibody targeted against the CD20 receptor that is present on the surface of B lymphocytes.

Ofatumumab and obinutuzumab are the other monoclonal antibodies directed against CD20 receptors. They are also approved for use in CLL.

Solution to Question 9:

Alemtuzumab is a humanized monoclonal antibody against CD52 that is indicated in the management of chronic lymphocytic leukemia (CLL).

Rituximab and obinutuzumab are monoclonal antibodies against CD20 that are also indicated in the treatment of chronic lymphocytic leukemia.

Cetuximab is a chimeric human-mouse anti-epidermal growth factor receptor monoclonal antibody against the extracellular domain of EGFR.

Solution to Question 10:

Pertuzumab is a monoclonal antibody against the HER2 receptor.

Pertuzumab is indicated in HER2-positive breast cancer in combination with trastuzumab and docetaxel.

Monoclonal antibodies against EGFR (HER1) are:

- Panitumumab
- Cetuximab
- Necitumumab

Solution to Question 11:

Olaratumab is a monoclonal antibody against PDGFR-alpha and blocks ligand-mediated receptor activation. It is indicated along with doxorubicin for adult patients with soft tissue sarcoma.

Panitumumab and cetuximab are monoclonal antibodies against EGFR (HER1).

Pertuzumab is a monoclonal antibody against HER2.

Solution to Question 12:

Blinatumomab is a bispecific monoclonal antibody.

Bispecific monoclonal antibody simultaneously binds to two different types of antigen. It binds simultaneously to CD3 on T cells and to CD19 on B cells. This links the two cells and activates T cells to exert cytotoxic activity against CD19 B cells.

Blinatumomab is approved for the treatment of Philadelphia chromosome-negative relapsed or refractory B-cell precursor ALL.

Option A: Daratumumab is a monoclonal antibody against CD-38.

Option B: Alemtuzumab is a monoclonal antibody against CD-52.

Option D: Panitumumab is a monoclonal antibody against EGFR.

Solution to Question 13:

Ibritumomab tiuxetan is a radioimmunoconjugate that is approved for use in non-Hodgkin's lymphoma. Diffuse large B-cell lymphoma is a subtype of non-Hodgkin's lymphoma.

Ibritumomab is a monoclonal antibody against CD20 which is attached to the radioactive isotope yttrium-90 using the chelator tiuxetan. Beta emission from Y-90 destroys the surrounding tissues.

Radioimmunoconjugates provide targeted delivery of radionuclides to tumor cells. It is made by attaching a radioactive molecule to a monoclonal antibody.

Option A: Gemtuzumab ozogamicin is a monoclonal antibody against CD33 covalently linked to a semisynthetic derivative of calicheamicin which is a potent antitumor antibiotic.

Option B: Brentuximab vedotin is a monoclonal antibody against CD30 linked with microtubule disrupting agent MMAE.

Option C: Denileukin diftitox is an immunotoxin produced by genetic recombination of IL-2 and the catalytically active fragment of diphtheria toxin.

Solution to Question 14:

Basiliximab exhibits a high affinity towards IL-2 receptor and inhibits it. It is an anti- CD-25 monoclonal antibody.

It has a short plasma half-life of 1 week. It is used to prevent transplant rejection reactions. Its adverse effects are anaphylactic reactions & opportunistic infections.

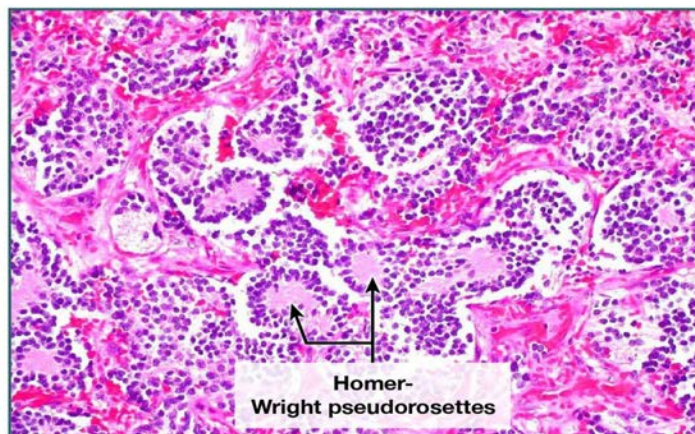
Solution to Question 15:

The given scenario and the histopathology (image) showing small round blue tumor cells with Homer-Wright pseudorosettes, point towards a diagnosis of paediatric neuroblastoma.

Dinutuximab is the monoclonal antibody against glycolipid GD2 which is indicated in this condition.

GD2 glycolipid is expressed on neuroblastoma cells and on normal cells of neuroectodermal origin, including the central nervous system and peripheral nerves.

Neuroblastoma



Solution to Question 16:

Pembrolizumab is the drug recently approved for the treatment of colorectal cancer arising through Lynch syndrome. The increased risk for colorectal cancer in Lynch Syndrome is due to inherited mutations that impair DNA mismatch repair.

Immune checkpoint inhibitors inhibit proteins that help to keep immune responses to cancer cells in check. When these proteins are blocked, the brake on the immune system is released and T cells are able to kill cancer cells more efficiently.

Pembrolizumab is an immune checkpoint inhibitor. It is a humanized monoclonal IgG4- κ isotype antibody that blocks the interaction between programmed cell death protein (PD-1) and its ligands.

The FDA has approved pembrolizumab for patients with recurrent, locally advanced or metastatic, squamous cell carcinoma of the esophagus and head and neck squamous cell cancer in combination with platin agents and 5-FU for the first-line treatment.

Solution to Question 17:

Pembrolizumab is not approved for use in non-metastatic small-cell lung cancer. It is approved for metastatic small cell lung cancer (SCLC).

Pembrolizumab is a monoclonal antibody (IgG4) against PD-1 and blocks interaction between PD-1 and its ligands PD-L1 and PD-L2. It is used in the following conditions:

- Metastatic SCLC
- Hodgkin's lymphoma

- Metastatic urothelial cancer
- Metastatic non SCLC
- Merkel cell carcinoma
- Chemotherapy-refractory head and neck cancers
- Refractory melanoma
- Triple-Negative Breast Cancer
- Esophageal Cancer
- Colorectal cancer due to Lynch syndrome

Solution to Question 18:

Reslizumab is a monoclonal antibody against interleukin-5 that is approved for use in severe asthma.

Option A & B: Omalizumab is an anti-IgE antibody and lebrikizumab is an antibody against interleukin 13, which are also indicated in the management of asthma.

Option D: Atezolizumab is an immune checkpoint modulator that inhibits the interaction of PD-L1 with PD-1 and B7-H1. It is approved for use in metastatic urothelial cancer and for NSCLC.

Solution to Question 19:

Obiltoxaximab is a monoclonal antibody against the protective antigen of Bacillus anthracis.

It is indicated in the treatment of inhalational anthrax due to B. anthracis in combination with appropriate antibacterial drugs. It is also indicated for prophylaxis of inhalational anthrax.

Solution to Question 20:

Durvalumab is an immune checkpoint inhibitor. It is an antagonist of PD-L1 which is a protein-ligand that is present on antigen-presenting cells, T cells, B cells, and some cancer cells.

Immune checkpoint inhibitors inhibit proteins that help the cancer cells evade the immune response. When these proteins are blocked, the brake on the immune system is released and T cells are able to kill cancer cells more efficiently.

Durvalumab is approved for use in metastatic urothelial and Merkel cell cancer. The FDA approved Durvalumab in combination with etoposide and either carboplatin or cisplatin as first-line treatment of patients with extensive-stage small-cell lung cancer.

Other PD-L1 inhibitors include:

- Atezolizumab

- Avelumab
- Durvalumab

Solution to Question 21:

Baricitinib is a Janus kinase inhibitor approved for use in rheumatoid arthritis.

Oral Janus kinase inhibitors for rheumatoid arthritis are:

- Baricitinib
- Tofacitinib

Option A: Ruxolitinib is a Janus kinase inhibitor approved for use in various forms of myelofibrosis, polycythemia vera and acute graft vs host disease.

Options C and D: Dasatinib and ponatinib are BCR-ABL tyrosine kinase inhibitors indicated for CML.

Solution to Question 22:

The scenario described in the question points towards a diagnosis of chronic plaque psoriasis. Tildrakizumab is a monoclonal antibody directed against IL-23 used in the treatment of this condition.

IL-23 is a naturally occurring cytokine that is involved in inflammatory and immune responses. Tildrakizumab prevents the interaction of IL-23 with its receptors. It is indicated in moderate-to-severe plaque psoriasis.

Secukinumab, ixekizumab, brodalumab are other monoclonal antibodies directed against IL-17A and indicated in moderate-to-severe plaque psoriasis.

Note: Ustekinumab is a monoclonal antibody against both IL-12 and IL-23. It is used for the treatment of Crohn's disease, ulcerative colitis, plaque psoriasis and psoriatic arthritis.

Solution to Question 23:

Dupilumab is indicated in the treatment of moderate-to-severe atopic dermatitis.

Dupilumab inhibits interleukins IL-4 and IL-13. When these interleukins are inhibited, they are unable to form complexes with their receptors. This results in a decrease in the immune response including the release of proinflammatory cytokines, chemokines, and IgE.

Interleukins, Growth Factors and Targeted Therapies

Question 1:

What is the mechanism of action of imatinib mesylate?

- a) Increase in metabolism of P glycoprotein
- b) Blocking the action of P glycoprotein
- c) Blocks the action of chimeric fusion protein of BCR-ABL
- d) Non-competitive inhibition of ATP binding site

Question 2:

Which of the following is not a BCR-ABL tyrosine kinase inhibitor?

- a) Bosutinib
- b) Dasatinib
- c) Erlotinib
- d) Nilotinib

Question 3:

A patient presented with abdominal pain. Imaging revealed a mass in the small intestine and a lesion in the liver. A diagnosis of a c-KIT positive gastrointestinal stromal tumor with liver metastasis was made on further evaluation. What is the drug of choice?

- a) Imatinib
- b) Bevacizumab
- c) Denosumab
- d) Rituximab

Question 4:

A patient with chronic myeloid leukemia was switched to nilotinib as he showed only partial response to imatinib therapy. What is the mechanism resulting in the loss of response?

- a) Amplification of wild-type kinase gene
- b) Increased efflux
- c) Point mutations in kinase domain
- d) Overexpression of MDR gene

Question 5:

Imatinib is indicated in which of the following patients?

- a) 2 and 4 only
- b) 1, 2 and 4
- c) 1 and 5 only
- d) 1, 3 and 5

Question 6:

Which of the following is a third generation BCR-ABL kinase inhibitor?

- a) Nilotinib
- b) Bosutinib
- c) Ponatinib
- d) Dasatinib

Question 7:

A patient with trastuzumab-refractory breast cancer is started on lapatinib. Which of the following statements is true regarding this drug?

- a) Dual inhibitor of VEGFR and HER 2 tyrosine kinase
- b) Dual inhibitor of MET and HER 2 tyrosine kinase
- c) Dual inhibitor of VEGFR and MET tyrosine kinase
- d) Dual inhibitor of EGFR and HER 2 tyrosine kinase

Question 8:

Which among the following is an irreversible inhibitor of EGFR tyrosine kinase?

- a) Gefitinib
- b) Erlotinib
- c) Trametinib
- d) Afatinib

Question 9:

A patient undergoing treatment for colon cancer develops hypertension. Which of the following drugs is most commonly implicated?

- a) Trastuzumab
- b) Bevacizumab
- c) Panitumumab
- d) Cetuximab

Question 10:

A patient with uncontrolled diabetes is diagnosed to have diabetic macular edema. Which of the following anti-VEGF therapies is effective in the treatment of this condition?

- a) Bevacizumab
- b) Ranibuzumab
- c) Aflibercept
- d) Ramucirumab

Question 11:

A patient is referred to your center for the management of hepatocellular carcinoma. Imaging reveals a large lesion in the liver with multiple intrahepatic metastases. The tumor is found to be unresectable. Which of the following is considered as first-line treatment for this patient?

- a) Sunitinib
- b) Sorafenib
- c) Regorafenib
- d) Thalidomide

Question 12:

Thalidomide should not be used in which of the following conditions?

- a) HIV-associated peripheral neuropathy
- b) HIV-associated aphthous ulcers
- c) Behcet syndrome
- d) Erythema nodosum leprosum

Question 13:

Use of lenalidomide over thalidomide would lead to the reduction of incidence of the following side effects, except _____.

- a) Constipation
- b) Sedation
- c) Neuropathy
- d) Bone marrow suppression

Question 14:

Which of the following is a proteasome inhibitor?

- a) Paclitaxel
- b) Bortezomib
- c) Cetuximab
- d) Irinotecan

Question 15:

Which of the following drugs can be used for the treatment of a patient with hormone-responsive breast cancer?

- a) Adriamycin
- b) Clomiphene citrate
- c) Diethylstilbestrol
- d) Tamoxifen

Question 16:

A patient undergoing adjuvant radiotherapy for an oral malignancy is prescribed a radioprotective drug. Identify this drug.

- a) Cetuximab
- b) Amifostine
- c) Cisplatin
- d) Gemcitabine

Question 17:

Which among the following is a cyclin-dependent kinase (CDK) 4/6 inhibitor?

- a) Ruxolitinib
- b) Palbociclib
- c) Cobimetinib
- d) Trametinib

Question 18:

Which of the following inhibits V600E mutated BRAF?

- a) Vemurafenib
- b) Dabrafenib
- c) Both A and B
- d) Ibrutinib

Question 19:

Which among the following is a hedgehog pathway inhibitor?

- a) Vemurafenib
- b) Vismodegib
- c) Olaparib
- d) Neratinib

Question 20:

Evaluation of a patient with CLL showed evidence of 17p deletion. Which of the following is a BCL-2 inhibitor approved for use in this condition?

- a) Olaparib
- b) Rucaparib
- c) Venetoclax
- d) Vismodegib

Question 21:

Sipuleucel-T is the first vaccine approved for the treatment of cancer. This vaccine induces an immune response against which of the following?

- a) HER2 receptors
- b) Prostatic acid phosphatase
- c) Vascular endothelial growth factor
- d) Vascular endothelial growth factor receptor

Question 22:

Which of the following is not a histone deacetylase inhibitor?

- a) Vorinostat
- b) Romidepsin
- c) Panabino
- d) Febuxostat

Question 23:

A 60-year-old woman was diagnosed with melanoma. Sentinel lymph node biopsy showed malignant cells. Which of the following drugs used in this condition is a MEK inhibitor?

- a) Dabrafenib
- b) Vemurafenib
- c) Trametinib
- d) Ipilimumab

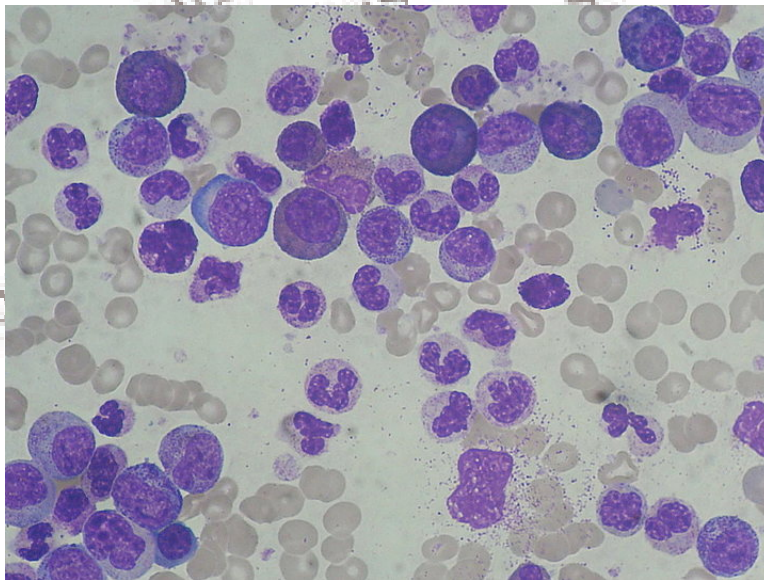
Question 24:

A 4-year-old child who was being evaluated for pancytopenia is diagnosed with acute lymphoblastic leukemia. Which of the following is the first gene therapy-based treatment approved for use in this condition?

- a) Trastuzumab
- b) Tisagenlecleucel
- c) Riociguat
- d) Dinutuximab

Question 25:

A 67-year-old man presented with fatigue and weight loss. On examination, pallor and splenomegaly were noted. Peripheral blood smear examination findings are shown in the image. Which of the following BCR-ABL tyrosine kinase inhibitors is not a first-line agent for this condition?



- a) Imatinib
- b) Dasatinib
- c) Nilotinib
- d) Bosutinib

Question 26:

A 63-year-old postmenopausal woman came to the OPD with a 12 cm adnexal mass. Her CA-125 levels were raised and her risk of malignancy index was calculated to be 250. PET scan revealed hepatic metastases. Which PARP inhibitor can you use in her treatment?

- a) Palbociclib
- b) Rucaparib
- c) Vorinostat
- d) Romidepsin

Question 27:

Which of the following conditions has hydrogen peroxide been recently approved for?

- a) Psoriasis
- b) Rosacea
- c) Atopic eczema
- d) Seborrheic keratoses

Question 28:

Which of the following conditions is apalutamide approved for?

- a) Breast cancer
- b) Colon cancer
- c) Prostate cancer
- d) Gastric cancer

Question 29:

A 60-year-old woman was diagnosed with melanoma. Sentinel lymph node biopsy showed malignant cells. Which among the following is a recombinant interleukin-2 preparation approved for use in this condition?

- a) Denileukin diftitox
- b) Aldesleukin
- c) Imatinib
- d) Nilotinib

Question 30:

Denileukin diftitox, a drug approved for the treatment of recurrent/refractory cutaneous T-cell lymphomas, is an immunotoxin made of:

- a) Fusion protein of IL-2 and pertussis toxin
- b) Fusion protein of IL-2 and diphtheria toxin
- c) Monoclonal antibody against CD33
- d) Monoclonal antibody against CD30

Answer Key

Question No.	Correct Option
1	c
2	c
3	a
4	c
5	d
6	c
7	d
8	d
9	b
10	c
11	b
12	a
13	d
14	b
15	d
16	b
17	b
18	c
19	b
20	c
21	b
22	d
23	c

24	b
25	d
26	b
27	d
28	c
29	b
30	b

Detailed Explanations

Solution to Question 1:

Imatinib mesylate blocks the action of the chimeric fusion protein of BCR-ABL. It targets the BCR-ABL tyrosine kinase and inhibits the activity of the fusion protein.

A single molecular event, 9:22 translocation, leads to the expression of the Abelson proto-oncogene kinase (ABL) fused to BCR (breakpoint cluster region) yielding a constitutively activated protein kinase in chronic myelogenous leukemia. Imatinib causes remission in >90% of patients in the chronic phase of CML.

Solution to Question 2:

Erlotinib is an epidermal growth factor receptor (EGFR) inhibitor.

Classification of BCR-ABL kinase inhibitors:

- First-generation: most likely to develop resistance.
- Imatinib
- Second generation: active against imatinib-resistant CML cases but not against T315I mutant cases.
- Nilotinib
- Dasatinib
- Bosutinib
- Third generation: active against T315I mutant CML cases and other known mutant cases.
- Ponatinib

Solution to Question 3:

The drug of choice for the management of a patient with a metastatic gastrointestinal stromal tumor is imatinib.

It is an oral anticancer drug that acts by inhibiting tyrosine kinase activated due to BCR-ABL fusion. It also inhibits other related tyrosine kinases and hence is used in the treatment of conditions like gastrointestinal stromal tumor (driven by a c-KIT mutation).

Imatinib is the agent of choice for GIST patients with metastatic disease and as adjuvant therapy for c-kit positive GIST.

Solution to Question 4:

The given scenario is suggestive of the development of resistance to imatinib, which is a tyrosine kinase inhibitor. This condition arises due to point mutations in three separate regions of the kinase domain.

The contact points between imatinib and the enzyme become sites of mutations in drug-resistant leukemic cells and hence these mutations prevent tight binding of the drug and lock the enzyme in its open configuration (the configuration in which it has access to the substrate and is enzymatically active).

Mechanisms other than BCR-ABL kinase mutations play a minor role in resistance to imatinib. They include:

- Amplification of the wild-type kinase gene has been identified in some cases.
- Overexpression of MDR-gene leading to increased drug efflux transporter has been shown experimentally but not implicated in clinical resistance.

Nilotinib retains inhibitory activity in the presence of most point mutations that confer resistance to imatinib.

Solution to Question 5:

Imatinib is the agent of choice in patients with metastatic GIST and as adjuvant therapy for c-kit positive GIST. Imatinib also shows therapeutic benefits in patients with hypereosinophilia syndrome and dermatofibrosarcoma protuberans (driven by activating mutations in the PDGFR).

Imatinib is an oral anticancer drug that acts by inhibiting tyrosine kinase activated due to ABL-BCR fusion. Imatinib also inhibits other kinases and hence is used in the above-mentioned conditions.

Indications for imatinib:

- BCR-ABL kinase mutations:
- Chronic myelogenous leukemia (CML)
- Acute lymphoblastic leukemia (ALL)
- C-KIT mutations - gastrointestinal stromal tumors (GIST)
- PDGFR mutations:

- Dermatofibrosarcoma protuberans
- Hypereosinophilia syndrome
- Chronic myelomonocytic leukemia (CML)

Solution to Question 6:

Ponatinib is a third-generation BCR-ABL kinase inhibitor.

Classification of BCR-ABL kinase inhibitors:

- First-generation: most likely to develop resistance.
- Imatinib
- Second-generation: active against imatinib-resistant CML cases but not against T315I mutant cases.
- Nilotinib
- Dasatinib
- Bosutinib
- Third-generation: active against T315I mutant CML cases and other known mutant cases.
- Ponatinib

Solution to Question 7:

Lapatinib is a dual inhibitor of epidermal growth factor (EGFR) and human epidermal growth factor 2 (HER2) tyrosine kinase.

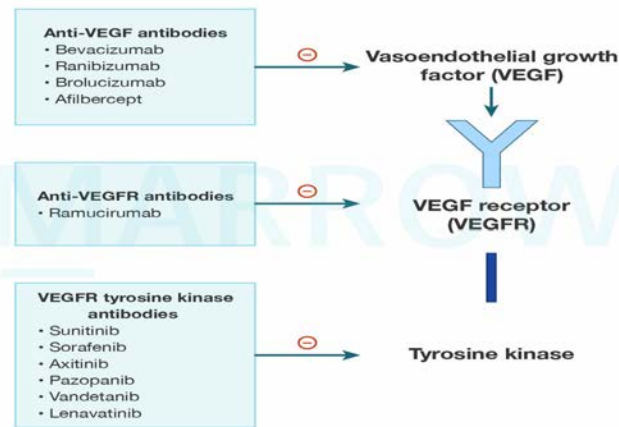
Lapatinib in combination with capecitabine is approved for the treatment of patients with metastatic HER2-positive and trastuzumab-refractory breast cancer.

Solution to Question 8:

Afatinib binds covalently with epidermal growth factor receptor tyrosine kinase (EGFR-TK) also known as Her-1 receptor and inhibits it irreversibly. It is approved for the use of metastatic non-small cell lung cancer.

Gefitinib and erlotinib compete with ATP binding to EGFR and results in competitive inhibition, which is reversible.

Inhibitors of VEGF and VEGFR antibodies



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Solution to Question 9:

Hypertension is one of the characteristic side effects of angiogenesis inhibitors like bevacizumab.

The mechanism is not clear but may be related to decreased endothelial nitric oxide production. Patients receiving higher doses and prolonged treatment might require antihypertensive medications.

Bevacizumab can be used in the treatment of metastatic colon cancer, in combination with FOLFOX or FOLFIRI regimen. It can also be used in the treatment of non-small cell lung cancer.

Solution to Question 10:

Aflibercept is more effective in diabetic macular edema in patients with worse baseline values of visual impairment when compared to bevacizumab and ranibizumab.

All of them are administered intravitreally. Clinical trials have shown the advantage of aflibercept over bevacizumab and ranibizumab in visual and anatomic terms at the end of one year.

Ramucirumab which is a monoclonal antibody against VEGFR is not indicated for diabetic macular edema.

Solution to Question 11:

Sorafenib is indicated as a first-line treatment for unresectable hepatocellular carcinoma. It is an angiogenesis inhibitor.

Hepatocellular carcinoma management:

- Resection, ablation, and other physical measures are the mainstay.
- Only two drugs (angiogenesis inhibitors) are approved in this condition:
- Sorafenib (first-line)
- Regorafenib (second-line).

Sunitinib and thalidomide are not indicated in hepatocellular carcinoma.

Solution to Question 12:

Thalidomide use is associated with peripheral neuropathy and is usually avoided in HIV-associated peripheral neuropathy as it can worsen the condition.

It is being investigated as an orphan drug for:

- Recurrent aphthous ulcers
- GVHD
- Behcet syndrome
- primary brain malignancies
- Kaposi's sarcoma
- Crohn's disease.

It is approved for use in multiple myeloma and erythema nodosum leprosum.

Solution to Question 13:

Lenalidomide, which is a congener of thalidomide, suppresses bone marrow and is associated with significant leukopenia.

Lenalidomide use is associated with a lesser incidence of sedation, constipation, and neuropathy when compared to thalidomide.

Solution to Question 14:

Bortezomib is a proteasome inhibitor.

It is a boron-containing anticancer drug that covalently binds to the proteasome and inhibits its proteolytic activity, thereby disrupting many intracellular signaling pathways. The most important of these is nuclear factor- κ B (NF- κ B) mediated signaling. It is indicated in the management of multiple myeloma.

Option A: Paclitaxel is a taxane.

Option C: Cetuximab is a monoclonal antibody against the extracellular domain of EGFR.

Option D: Irinotecan is a topoisomerase I inhibitor.

Solution to Question 15:

Tamoxifen is a selective estrogen receptor modulator that is indicated for hormone-responsive breast cancer.

It is approved for endocrine treatment of women with ER+ metastatic breast cancer or following primary tumor excision as adjuvant therapy.

Solution to Question 16:

Amifostine is the radioprotective drug that has been prescribed to this patient.

It is an organic thiophosphate cytoprotective agent that is indicated for the following conditions:

- Reduction of nephrotoxicity associated with cisplatin.
- Reduction of xerostomia in patients with head and neck cancer undergoing postoperative irradiation.

Solution to Question 17:

Palbociclib is a cyclin-dependent kinase (CDK) 4/6 inhibitor. It is approved for use in advanced or metastatic ER-positive HER2-negative breast cancer.

Ruxolitinib is an inhibitor of Janus associated kinases: JAK1 and JAK2.

Trametinib and cobimetinib are inhibitors of serine-threonine kinases.

Solution to Question 18:

Vemurafenib and dabrafenib inhibit V600E mutated BRAF present in melanoma.

Both are indicated in the management of melanoma.

The name vemurafenib is derived from the target V600E mutated BRAF.

Ibrutinib is a Bruton tyrosine kinase inhibitor.

Solution to Question 19:

Vismodegib acts as a competitive antagonist of the SMO (smoothened hedgehog) receptor in the hedgehog signaling pathway.

Hedgehog signaling plays a key role in stem cell regulation and tissue regeneration. Basal cell carcinoma (BCC) is associated with dysregulation of this pathway. Vismodegib is indicated in patients with metastatic or relapsed BCC.

Option A: Vemurafenib is a BRAF kinase inhibitor.

Option C: Olaparib is an inhibitor of PARP [Poly (ADP-ribose) polymerase] enzyme.

Option D: Neratinib is an irreversible inhibitor of HER-2 and EGFR.

Solution to Question 20:

Venetoclax is a small-molecule inhibitor of BCL-2 that is approved for use in CLL with a 17p deletion.

Overexpression of BCL-2 has been demonstrated in CLL cells where it mediates tumor cell survival. Venetoclax helps restore the process of apoptosis by binding directly to the BCL-2 protein, displacing proapoptotic proteins like BIM, triggering mitochondrial outer membrane permeabilization and the activation of caspases.

Olaparib and rucaparib are inhibitors of PARP [Poly (ADP-ribose) polymerase] enzymes.

Solution to Question 21:

Sipuleucel-T is a cell-based approach to induce an immune response against prostatic acid phosphatase (PAP) that is commonly expressed in prostatic cancer. It is approved for use in minimally symptomatic, hormone-refractory, metastatic prostate cancer.

Sipuleucel-T preparation is done by isolating a patient's peripheral blood cells by leukapheresis and then exposing them to PAP-GM-CSF (which is a human recombinant protein). When these treated cells are infused to a patient, the APCs present in the blood produces antibodies directed against PAP.

Prostate-specific membrane antigen (PSMA) is a specific prostate epithelial cell membrane antigen, showing increased expression in prostate cancers, which is also a potential target for monoclonal antibody therapy.

Solution to Question 22:

Febuxostat is a uric acid synthesis inhibitor that acts by inhibiting the enzyme xanthine oxidase and it is indicated in gout management.

Examples of histone deacetylase inhibitors are:

- Vorinostat - cutaneous T cell lymphoma
- Belinostat - peripheral T cell lymphoma
- Romidepsin - cutaneous/peripheral T cell lymphoma

- Panobinostat - multiple myeloma

Solution to Question 23:

The scenario is suggestive of metastatic melanoma and trametinib is used in this condition. It is a reversible inhibitor of the mitogen-activated extracellular signal-regulated kinase (MEK). It inhibits the serine-threonine kinase domain of BRAF V600E mutant of MEK.

It can be used as a single agent but it is more effective when used with BRAF inhibitors like dabrafenib.

Adverse effects of this drug include:

- Cutaneous rash
- Acneiform dermatitis
- Diarrhea
- Fatigue
- Nausea
- Lymphedema.
- Cardiomyopathy
- Hypertension
- Hemorrhage
- Interstitial lung disease
- Ocular toxic effects.

Options A and B: Dabrafenib and vemurafenib are BRAF inhibitors used in metastatic melanoma.

Option D: Ipilimumab is a human cytotoxic T-lymphocyte antigen 4 (CTLA-4) blocking antibody used in metastatic or unresectable melanoma.

Solution to Question 24:

Tisagenlecleucel is the first gene therapy-based treatment that is approved for use in acute lymphoblastic leukemia (ALL).

The therapy involves reprogramming a patient's own T-cells with a transgene encoding a chimeric antigen receptor (CAR) to identify and eliminate CD19-expressing malignant and normal cells.

Solution to Question 25:

The given scenario and the peripheral smear findings (increased WBCs with myelocytes, metamyelocytes, and band forms), is suggestive of chronic myelogenous leukemia

(CML). Bosutinib is approved for use in this condition only for resistant and/or intolerant cases and not in newly diagnosed cases.

Agents that can be used as first-line for CML are:

- Imatinib
- Dasatinib
- Nilotinib

Dasatinib and nilotinib can be used in newly diagnosed cases of CML, that are found to be imatinib-resistant.

Solution to Question 26:

The clinical scenario points towards a diagnosis of metastatic ovarian cancer. Rucaparib is a poly ADP-ribose polymerase/ PARP inhibitor approved for the treatment of advanced ovarian cancer.

PARP is a nuclear protein that transfers ADP-ribose from NAD⁺ to target proteins and this plays an important role in the DNA damage response. Rucaparib and olaparib are PARP inhibitors.

Option A: Palbociclib is a selective inhibitor of the cyclin-dependent kinases CDK4 and CDK6. It is used in the treatment of metastatic breast cancer.

Options C & D: Vorinostat and romidepsin are histone deacetylase inhibitors.

Solution to Question 27:

A topical solution of hydrogen peroxide has been approved for the treatment of raised seborrheic keratoses.

Hydrogen peroxide is a keratolytic agent. Its exact mechanism of action in the treatment of seborrheic keratoses is not fully understood. Supraphysiologic concentrations may cause oxidative damage and death to seborrheic keratosis cells.

Hydrogen peroxide is commonly used as a disinfectant to clean wounds. It can also be used as a mouthwash.

Solution to Question 28:

Apalutamide is an androgen receptor inhibitor indicated for the treatment of patients with non-metastatic castration-resistant prostate cancer.

Solution to Question 29:

The clinical scenario is suggestive of metastatic melanoma. The recombinant interleukin-2 (IL-2) preparation that is approved for use in metastatic melanoma is aldesleukin. It acts as an IL-2 agonist and is also approved for use in metastatic renal cell cancer.

Denileukin diftitox is an immunotoxin made from the recombinant combination of IL-2 and the catalytically active fragment of diphtheria toxin. It is approved for use in cutaneous T cell lymphoma.

Imatinib and nilotinib are a BCR/ABL tyrosine kinase inhibitor.

Solution to Question 30:

Denileukin diftitox is an immunotoxin made from the genetic recombination of IL-2 and the catalytically active fragment of diphtheria toxin.

It is indicated for advanced cutaneous T-cell lymphoma in patients with $\geq 20\%$ of T cells expressing the surface marker CD25. After binding to the IL-2 receptors, it is internalized by endocytosis. The active fragment of diphtheria toxin then is released into the cytosol, where it inhibits protein synthesis through ADP ribosylation, leading to cell death.

Immunotoxins are proteins that contain a toxin along with an antibody that binds specifically to target cells.

Antibody-drug conjugates are complex molecules composed of an antibody linked to a biologically active cytotoxic (anticancer) drug. These antibody-targeted drug therapies are cell-specific to kill efficiently the target cells that bind and internalize the drug conjugate. Examples are:

- Gemtuzumab ozogamicin - Monoclonal antibody against CD33 linked to a semisynthetic derivative of calicheamicin, a potent antitumor antibiotic. It was approved for the treatment of acute myeloid leukemia.
- Brentuximab vedotin - Anti-CD30 IgG1 monoclonal antibody linked with the microtubule-disrupting agent MMAE. It is used for the treatment of Hodgkin and anaplastic lymphoma.
- Ado-trastuzumab emtansine - It combines the HER2-targeted properties of trastuzumab with the antimicrotubule agent DM1 (derived from maytansine), allowing preferential intracellular drug delivery to HER2+ cells. It is used in the treatment of HER+ breast cancer.